The Loop Atlas: the record

Every prediction a run judged, every other check a run made, how each came out, and every miss with its reason

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Preprint. The companion volume to The Loop Atlas*, made by script from the runs that travel with it, rebuilt at every build so that it cannot fall behind them. TOY MODEL throughout: every number here was printed by a small computer model with values we chose. CC BY 4.0.*

How to read this record

This volume is made for checking rather than for reading through. *The Loop Atlas* tells the story and counts the results; this record holds the receipts. Open it to find exactly what a run was asked and what it answered, and, in Part 4, every run file that travels with the book, in the archive deposited with it.

Every row names its run as the book’s notes do, by the tool in capitals and the run’s date and time: TRAIN_PARTING_EDGE 20261029_070800 is the tool TRAIN_PARTING_EDGE, run on 29 October 2026 at 07:08:00. Each tool and each run’s own file travel with the book (Part 4 lists the run files, and names the few runs cited here that do not travel), the tool’s header carrying every prediction it was built to judge and the run file every verdict it printed. The label is the prediction’s name in that run. The words are the run’s own, as it printed them, bar and all; where a run prints only a label, the words are in the tool’s header. In a few rows the run’s own words carry a working name of the project, given here as the book gives it: a card, the project’s name for one registered test, by the tool that ran it; the second computer, on which the long runs were flown, by that name; and the companion book by its title. The verdict is the run’s, with the run’s own number beside it where it printed one.

A prediction judged again in a later run appears again there, so the counts are of judgements. Runs made before 5 October print no verdict line in a form the script can read, and are not in these tables.

The misses are printed whole in Part 3, each under the number the tables give it, in the misses file’s own words and its own four headings: what happened; what kind of miss it was; the lesson, with the redesign where one was named; and whether the lesson has been proved, and by what. Their working names are given as above, a part of the working draft by its chapter in *The Loop Atlas*, and a lesson sent to the Field Guide without its number. Where a row quoted a word typed in conversation, named one of the project’s working records, or dated itself by a working session’s handle, it was read by hand and the place given in plain words, the handle as its date, up to row 540 or so; the later rows, written in the last weeks, keep some of the working words, and these are what they mean. *The crate* is the folder of every file the work made, the archive that travels with the book; a *manifest*, named by two letters, is the list of those files with their checksums, sent after each round of work. A *session*, named by a letter, is one long working conversation, and its *exchanges* are numbered within it; a *sitting*, in the earliest rows, is one of the first four working conversations, on 2 and 3 October; *session K’s exchange 30* is a date, 8 November 2026, the last of them. The *proposed list* is the list of tests the surprises raised and that were not run as cards, numbered from P1, kept with the working files; the *backward pass* is the reading of both editions and this record against their runs by readers who had not seen the work, done before the upload. A miss that names another by its row, as in row 292, means that miss in Part 3.

Part 1. Every prediction, and how it came out

1,104 predictions about the toys, judged in the runs of record: 706 held, 369 missed and 29 were left undecided; day by day, in the order the runs were made. A prediction that missed or was left undecided names its row in Part 3 where that row cites the run and the label together, and shows a dash where none does.

5 October

Run	Label	The prediction, as the run prints it	Verdict	Misses row
HIDDEN_SCISSORS 20261005_064106	F0	0.557209 against 0.526206	Missed	—
HIDDEN_SCISSORS 20261005_064106	H1a	needles still slipping at 0.9120: 16 of 64 (needs 48)	Missed	—

Run	Label	The prediction, as the run prints it	Verdict	Misses row
HIDDEN_SCISSORS 20261005_064106	H1b	median re-lock at least 0.002 below 0.91233 (0.9103 or below)	Missed	—
HIDDEN_SCISSORS 20261005_064106	H2	re-lock paces 10th to 90th percentile 0.9115 to 0.9120, spread 0.0005 (needs under 0.003)	Held	
HIDDEN_SCISSORS 20261005_064106	H3	fresh needle’s angle at phase zero, locked at 0.850: 5.33784165; re-locked needles’ largest difference 2.83e-09 rad	Held	
HIDDEN_SCISSORS 20261005_064106	H4	median time per slip 7.74 cycles at 0.9120 against 7.74 at 0.9120 (last pace with at least 8 slipping)	Missed	—
HIDDEN_SCISSORS_ SLIVER 20261005_145137	S1	targets below the edge where the share still slipping falls during the hold (or is already zero by cycle 100): 12 of 12	Held	
HIDDEN_SCISSORS_ SLIVER 20261005_145137	S2	mean lifetime 11.9 cycles at 0.9122 against 0.0 at 0.9116, ratio nan (needs 5 or more)	Missed or not read	—
HIDDEN_SCISSORS_ SLIVER 20261005_145137	S3	still slipping in the last 100 cycles at 0.9112 to 0.9121: 0 needles	Held	
HIDDEN_SCISSORS_ SLIVER 20261005_145137	C0	still slipping in the last 100 cycles at 0.9124: 64 of 64	Held	
PULSED_SCISSORS 20261005_155747	F0	a = 0 edges Omega 0.5: fall 0.99875, climb-back 0.96625; Omega 1.0: fall 0.99875, climb-back 0.96625	Held	
PULSED_SCISSORS 20261005_155747	F1	a = 0 spin exponents +0.0262, +0.0131 per drive period	Missed	—
PULSED_SCISSORS 20261005_155747	H1	cells against it: [(1.0, 0.05, ‘chaotic spin, gap open’), (1.0, 0.2, ‘chaotic spin, gap open’), (1.0, 0.3, ‘chaotic spin, gap open’)]	Missed	—
DIVER 20261005_160158	A0	3.32e-10 rad	Held	
DIVER 20261005_160158	A1	forward plus backward +1.34e-10 rad	Held	
DIVER 20261005_160158	A2	ratio 0.4902	Missed	—
DIVER 20261005_160158	A3	difference 6.2e-11 rad	Held	
DIVER 20261005_160158	A4	at A 0.6, 0.0 degrees in 1.43 s and 0.0 in 2.35 s (needs both under 180)	Held	
DIVER 20261005_160158	B0	0.0e+00 turns	Held	
DIVER 20261005_160158	B1	largest drift 1.4e-14	Held	
DIVER 20261005_160158	B2	mean spin over prediction minus 1 at 2, 5, 10 degrees: 0.944, 0.280, 0.050	Missed	—
DIVER 20261005_160158	B3	ratios 1.747, 1.617, 1.642 against 1.643	Missed	—
DIVER_B 20261005_160338	A5	ratios 0.9988, 1.0041	Held	
DIVER_B 20261005_160338	A6	pike at A 0.6 turns 1.40 degrees per loop; loops for half a turn 128.67	Missed	—
DIVER_B 20261005_160338	B4	largest below 4.412 degrees 0.1779 turns; smallest above 4.5 degrees 1.8660	Missed	—
PULSED_SCISSORS_LONG 20261005_170134	LH1	Omega 1.0 a 0.0: share 1.000, gap +0.0325 -> +0.0325; Omega 0.5 a 0.05: share 1.000, gap +0.0250 -> +0.0250; Omega 0.5 a 0.1: share 1.000, gap +0.0350 -> +0.0350; Omega 1.0 a 0.1: share 1.000, gap +0.0225 -> +0.0225	Held	
PULSED_SCISSORS_LONG 20261005_170134	LH2	Omega 1.0 a 0.05: share 1.000, gap +0.0275 -> +0.0275; Omega 1.0 a 0.2: share 1.000, gap +0.0125 -> +0.0125; Omega 1.0 a 0.3: share 1.000, gap +0.0050 -> +0.0050	Held ; rival (share under 0.5) in: none	
GAP_DIAL 20261005_ 170519	GD0	gap 90 fall 0.99875, climb-back 0.96625; gap 45 gap +0.00250, fall 0.98875 against 0.987488	Missed	—
GAP_DIAL 20261005_ 170519	GD1	gap 85 gap +0.02750 (at least 0.0225)	Held	

Run	Label	The prediction, as the run prints it	Verdict	Misses row
GAP_DIAL 20261005_170519	GD2	steps in the gap as c grows: -0.0050, -0.0075, -0.0025, -0.0050, -0.0025, -0.0050, -0.0025, +0.0000, +0.0000	Held	
GAP_DIAL 20261005_170519	GD3	gaps against it: [(90, ‘chaotic spin, gap open’), (85, ‘chaotic spin, gap open’), (80, ‘chaotic spin, gap open’), (75, ‘chaotic spin, gap open’), (70, ‘regular spin, gap shut’), (65, ‘chaotic spin, gap open’), (60, ‘chaotic spin, gap open’)]	Missed	—
GAP_DIAL_2 20261005_172939	GD2-0	gap 90 edge 0.96475 (0.96475)	Held	
GAP_DIAL_2 20261005_172939	GD2-1	G 88: -0.00013; G 86: -0.00019; G 84: -0.00075; G 82: -0.00231; G 80: -0.00288; P 0.01: +0.00026; P 0.02: +0.00010; P 0.03: -0.00057; P 0.05: +0.00010; P 0.07: -0.00072; P 0.1: -0.00171	Missed	—
GAP_DIAL_2 20261005_172939	GD2-2	largest edge minus line +0.00026 (at most +0.0005)	Held	
POCKET_HOLD 20261005_195333	PK-1	at 50000: [6, 3]; stopped after 10000: [0, 0]	Held	
POCKET_HOLD 20261005_195333	PK-2	median exponent of the slipping pocket needles -0.2890 per field turn (at most +0.02)	Held	

6 October

Run	Label	The prediction, as the run prints it	Verdict	Misses row
HITCH_KICK 20261006_011635	HK0	rigid trunk turn -60.000000 degrees (-60 to 1e-6)	Held	
HITCH_KICK 20261006_011635	HK1	one arm circle at 1, 2, 3 per second: +13.865584, +13.865584, +13.865584 degrees; spread 2.13e-10	Held	
HITCH_KICK 20261006_011635	HK2	one forward arm circle turns the trunk +13.866 degrees (backward, 15 to 45); legs over arms 12.476 (2.5 to 5)	Missed	—
HITCH_KICK 20261006_011635	HK3	arms at 2 per second hold off 63.3 per cent (40 to 100); trunk lands -21.99 degrees	Held	
HITCH_KICK 20261006_011635	HK4	largest relative change of the whole body’s turning momentum 7.11e-15 (1e-9 relative, 1e-12 absolute at L0 = 0)	Held	
ROOM_NEEDLE 20261006_013956	RN-0	carried at tau 456.7 in 60 of 60 cells (at 1184.6: 60); upstream never slips	Held	
ROOM_NEEDLE_2 20261006_014758	RN2-C	control needles carried 120 of 120 (largest advance over the read +3.40e-01 rad)	Held	
ROOM_NEEDLE_2 20261006_014758	RN2-W	cells whose slipping needles first turn at or after 3,500: [(18, 3848.0000000839927), (38, 3996.0000000872233), (56, 3837.0000000837526), (58, 3538.000000077226), (117, 3952.000000086263)]	Missed (window flagged short there)	—
ROOM_NEEDLE_2 20261006_014758	RN-0	carried at tau 456.7 in 120 of 120 cells (at 1184.6: 120); upstream never slips	Held	
ROOM_NEEDLE_2 20261006_014758	RN-1	eps 0.0, rho 1.0113: slope -0.580 (-1.15 to -0.85)	Missed	—
ROOM_NEEDLE_2 20261006_014758	RN-2	eps 0.0, rho 1.25: slope -0.479 (-0.80 to -0.40)	Held	
ROOM_NEEDLE_2 20261006_014758	RN-3	eps 0.0, rho 1.0113: five smallest uncensored rooms -0.385, five largest -0.770 (smallest shallower)	Held	
ROOM_NEEDLE_2 20261006_014758	RN2-L	eps 0.0, rho 1.0113: five smallest uncensored rooms -0.385 (-0.80 to -0.40)	Missed	—
ROOM_NEEDLE_2 20261006_014758	RN-4	steepest slope -0.600 on 6 of 6 readable cells (none steeper than -1.15)	Held	

Run	Label	The prediction, as the run prints it	Verdict	Misses row
ROOM_NEEDLE 20261006_014807	RN-0	carried at tau 456.7 in 60 of 60 cells (at 1184.6: 60); upstream never slips	Held	
SWAP_RIDER_BAND 20261006_041001	SW-S	largest change of the band per unit A when the step halves 1.80e-09 (below 1e-4)	Held	
SWAP_RIDER_BAND 20261006_041001	SW-1	the bound first turns positive at mass 0.75 (between 1.0 and 2.0)	Missed	—
SWAP_RIDER_BAND 20261006_041001	SW-2	folds [np.float64(1.0143), np.float64(1.0133), np.float64(1.012), np.float64(1.0104), np.float64(1.0089), np.float64(1.0077), np.float64(1.0056), np.float64(1.0041), np.float64(1.0031)]; largest distance from 1 0.0143 (within 0.03)	Held	
FERRY_ROPE 20261006_045826	FR-1	reversed wave, A 0.3: [np.float64(0.8735), np.float64(0.9018), np.float64(0.9672)] (below 0.95, below 0.95, within 0.015 of 0.9672)	Held ; gravel shoulder (any above 1): loses	
FERRY_ROPE 20261006_045826	FR-2	rise ratio A 0.3 over A 0.15 at masses 0.25 and 1: [np.float64(4.33), np.float64(4.16)] (3 to 5)	Held ; reversed A 0.15: [np.float64(0.9679), np.float64(0.9758), np.float64(0.9924)] (within 0.01 of 0.9661, 0.9749, 0.9924): HELD	
FERRY_ROPE 20261006_045826	FR-3	largest mean speed 6.22e-01 (within 0.001 of 0)	Missed	—
NEVER_BLIND 20261006_045833	NB-1	margin 0.2 up at every delay to 0.8: False; limits 0.6 (margin 0.1) and 0.5 (margin 0.2), the same within 0.1: True	Missed	—
NEVER_BLIND 20261006_045833	NB-2	limit with margin 0.2 0.5 (between 1.0 and 1.5)	Missed	—
NEVER_BLIND 20261006_045833	NB-3	rest held at 0.5, fewer than half up at every delay from 0.6	Held	
NEVER_BLIND_2 20261006_050123	NB2-1	each change alone falls at 0.8 or later	Missed	—
NEVER_BLIND_2 20261006_050123	NB2-2	both together up at every delay to 1.0	Missed	—
NEVER_BLIND_2 20261006_050123	NB2-3	no version up at 1.5 (past 1.414)	Held	
POCKET_GEARS 20261006_052012	PG-1	every pocket needle repeats within 1e-6 with q up to 12, one ratio per pocket	Held	
POCKET_GEARS 20261006_052012	PG-2	ratios 1/2 and 1/2 (the same)	Held	
SWAP_STATE 20261006_052545	SS-1	mass 1.0 lands in gear 2 and holds it below the fold coming down	Missed	—
SWAP_STATE 20261006_052545	SS-2	masses 2.0 to 3.0 running (not gear 2) on leaving gear 1 and at the top	Held	
SWAP_STATE 20261006_052545	SS-3	masses 2.0 to 3.0 come straight back to gear 1, no gear 2 held, opening at least A7's bound	Missed	—
SWAP_STATE_2 20261006_052914	SS2-0	half rate, every mass leaves gear 1 within 0.003 above its held fold	Held (+0.0017, +0.0017, +0.0020, +0.0016, +0.0021, +0.0023, +0.0024, +0.0019, +0.0019)	
SWAP_STATE_2 20261006_052914	SS2-1	half rate, mass 1.0, first window wholly after leaving is gear 2 and gear 2 holds below the fold coming down	Held	

Run	Label	The prediction, as the run prints it	Verdict	Misses row
SWAP_STATE_2 20261006_052914	SS2-2	gear 2 caught coming down at every mass 1.0 to 3.0: True; its last window within 0.004 of SWAP_STATE's: True; every opening from 0.75 at least A7's bound: True	Held	
SWAP_STATE_2 20261006_052914	SS2-3	gear 1 and gear 2 held at one pace at every mass 1.0 to 3.0	Held	
SIZED_TAP 20261006_060952	ST-1	best held-rest setting up at every delay to 1.0	Missed	—
SIZED_TAP 20261006_060952	ST-2	best held-rest fall 1.0 (between 1.0 and 1.3)	Held	
SIZED_TAP 20261006_060952	ST-3	best never-blind fall 1.3 (at or before 1.0, and at least 0.2 before the held rest's 1.0)	Missed	—
SIZED_TAP 20261006_060952	ST-4	no setting half up at 1.5	Held	
SIZED_TAP_2 20261006_061606	ST2-1	some setting half up at 1.3	Held	
SIZED_TAP_2 20261006_061606	ST2-2	no setting half up at 1.5	Missed	—
SIZED_TAP_2 20261006_061606	ST2-3	best new setting c 2.5 look-ahead 1.0 (fall None): wobble from 0.3 to 1.4: 0.0123, 0.0143, 0.0169, 0.0201, 0.0247, 0.0295, 0.0363, 0.0441, 0.0555, 0.0690, 0.0857, 0.1126; rising at every step	Held	
SIZED_TAP_3 20261006_061940	ST3-1	some setting half up at 2.0	Missed	—
SIZED_TAP_3 20261006_061940	ST3-2	no setting half up at 3.0	Held	
SIZED_TAP_3 20261006_061940	ST3-3	longest-holding setting c 2.5 look-ahead 1.0 (fall 1.6): wobble from 1.4 to 1.4: 0.1126; rising at every step	Missed	—
CLIMB_HOLD 20261006_065510	CH-F	mass 0.75, drive 1.6400, all 8 running at the end	Held	
CLIMB_HOLD 20261006_065510	CH-1	mass 0.75, no rider started running still running at any drive 1.6300 to 1.6371	Held	
CLIMB_HOLD 20261006_065510	CH-2	mass 0.75, every rider started in the lock in gear 1 at every drive 1.6300 to 1.6371	Held	
CLIMB_HOLD 20261006_065510	CH-3	masses 1.0 and 1.5, no rider started running above the band's top ends in gear 1	Missed	—
RUN_MAP 20261006_071157	RM-1	its words in the tool's own header	Held	
SLIP_BEND 20261006_071248	SB-1	T x sqrt(p^2-1) = 2 pi within 1e-6 at every delta	Held	
SLIP_BEND 20261006_071248	SB-2	slopes within 0.02 of -1/2 for delta at most 1e-3 (-0.5000, -0.5001) and within 0.03 of -1 at the largest pair (-0.9820)	Held	
RUN_STROBE 20261006_170149	RS-1	its words in the tool's own header	Held	
RUN_STROBE 20261006_170149	RS-2	its words in the tool's own header	Missed	—
RUN_STROBE 20261006_170149	RS-4	its words in the tool's own header	Held	
RUN_STROBE 20261006_170149	RS-3	14 landed riders off the gears, 2 distinct (mass, drive, slips, period); periods off the address rule: none	Held	
RUN_STROBE_2 20261006_174220	RS2-b	periods [8]; largest distance of a last strobe from the orbit 8.5e-08	Held	

Run	Label	The prediction, as the run prints it	Verdict	Misses row
RUN_STROBE_2 20261006_174220	RS2-a	slopes over the five smallest distances, mass 2.0 0.2124, mass 3.0 0.2925	Missed	—
RUN_STROBE_2 20261006_174220	RS2-a2	slopes over the five smallest distances, mass 2.0 0.4835, mass 3.0 0.4818	Held	
RUN_STROBE_2 20261006_174220	RS2-c	0 runs	Missed	—
STROBE_NEWTON 20261006_184350	N2	its words in the tool’s own header	Missed	—
STROBE_NEWTON 20261006_184350	N1	its words in the tool’s own header	Held	
STROBE_NEWTON 20261006_184350	N3	its words in the tool’s own header	Held	
STROBE_NEWTON 20261006_184350	N4	its words in the tool’s own header	Missed	—
STROBE_NEWTON 20261006_184350	N5	its words in the tool’s own header	Missed	—
GROOVES 20261006_193519	G1	5/4 width at mass 3.0 5.528e-05 against the 4/3’s 6.860e-05	Missed	—
GROOVES 20261006_193519	G2	widths 4.183e-05 to 5.528e-05, ratio 1.321 (at most 1.2 registered), found at 5 of 5 masses	Missed	—
GROOVES 20261006_193519	G3	its words in the tool’s own header	Held	
GROOVES 20261006_193519	G4a	its words in the tool’s own header	Held	
GROOVES 20261006_193519	G4b	groove from 1.423013597 to 1.423128081, width 1.145e-04 against the golden clock’s 3.877e-05	Held	
GROOVES 20261006_193519	G4	its words in the tool’s own header	Held	
CAT_3D 20261006_201322	C1	|net turn| at 90 degrees 57.16 deg, 41 times the planar pike’s 1.40	Missed	—
CAT_3D 20261006_201322	C2	strictly growing True; every axis within 10 degrees of the end-to-end line False	Missed	—
CAT_3D 20261006_201322	C3	its words in the tool’s own header	Missed	—
CAT_3D_2 20261006_202140	K1	at radius 0.3 the legs turn 158.93 and 158.93 degrees (at least 150 registered)	Held	
CAT_3D_2 20261006_202140	K2	legs nearer 180 as the halves thin, both halves: True; the V moves less as the halves thin: True	Held	
GROOVES_2 20261006_202753	H1	its words in the tool’s own header	Held	
GROOVES_2 20261006_202753	H2	its words in the tool’s own header	Held	
CAT_3D_2_READER 20261006_202842	K3	its words in the tool’s own header	Missed	—
FUNNEL_CURLS 20261006_203001	F1	its words in the tool’s own header	Held	
FUNNEL_CURLS 20261006_203001	F2	its words in the tool’s own header	Held	

Run	Label	The prediction, as the run prints it	Verdict	Misses row
SPOKES 20261006_204038	Q1	its words in the tool’s own header	Missed (a period-16 orbit was not found, not stable, or a drive lies above the refined doubling)	—
SPOKES 20261006_204038	Q2	at the smallest distance the strands sit on opposite sides of the period-8 point, distances differing by 25.134 per cent	Missed	—
GROOVES_3 20261006_204201	J1	width in drive falls strictly with weight	Held	
GROOVES_3 20261006_204201	J2	width in turns per passing rises strictly with weight	Held	
CASCADE 20261006_211853	S1	the inequality’s log-log slope 0.4978; at 1e-6 5.983 per cent	Held	
CASCADE 20261006_211853	S2	separation over the square root within 0.5 per cent of 1.662 at each [np.True_, np.True_, np.True_]; stable at each [True, True, True]	Held	
CASCADE 20261006_211853	R1	the period-16 doubling is 1.2016e-04 below the first	Held ; the period-8 band 5.5460e-04 is 4.6155 times the period-16 band (printed)	
CASCADE 20261006_211853	R2	the period-32 band is 2.5803e-05; the period-16 band over it 4.6570 (Feigenbaum’s 4.669; 3.97 to 5.37 registered)	Held	
GROOVES_4 20261006_212231	V1	in turns per passing 3/2 > 11/7 > 8/5	Missed	—
GROOVES_4 20261006_212231	V2	in drive 3/2 > 11/7 > 8/5	Missed	—
CASCADE_2 20261006_213511	S1	the inequality’s log-log slope 0.4978; at 1e-6 5.983 per cent	Held	
CASCADE_2 20261006_213511	S2	separation over the square root within 0.5 per cent of 1.662 at each [np.True_, np.True_, np.True_]; stable at each [True, True, True]	Held	
CASCADE_2 20261006_213511	R1	the period-16 doubling is 1.2016e-04 below the first	Held ; the period-8 band 5.5460e-04 is 4.6155 times the period-16 band (printed)	
CASCADE_2 20261006_213511	R2	the period-32 band is 2.5803e-05; the period-16 band over it 4.6570 (Feigenbaum’s 4.669; 3.97 to 5.37 registered)	Held	
CASCADE_2 20261006_213511	T3	log-log slope 0.5000	Held	
CASCADE_2 20261006_213511	T1	the period-64 doubling at 1.6893109278, -2.54e-09 from 1.6893109303	Held ; band 5.5291e-06; the period-32 band over it 4.6667 (printed)	
CASCADE_2 20261006_213511	T2	its words in the tool’s own header	Held	
GROOVES_5 20261006_213610	W1	8/5 wider than 11/7 in both units, each at least ten times its resolution	Missed	—
GROOVES_5 20261006_213610	W2	7/5 wider than 10/7 in drive	Held	

Run	Label	The prediction, as the run prints it	Verdict	Misses row
GROOVES_5 20261006_213610	W3	8/5 wider than 10/7 in drive	Missed	—
GROOVES_6 20261006_215909	X1	both widths resolved ten times over and under 1e-12	Held	
GROOVES_6 20261006_215909	X2	8/5 narrower than 10/7 (2.3819e-12)	Held	
GROOVES_6 20261006_215909	X3	11/7 narrower than 8/5	Held	
CASCADE_3 20261006_220635	S1	the inequality’s log-log slope 0.4978; at 1e-6 5.983 per cent	Held	
CASCADE_3 20261006_220635	S2	separation over the square root within 0.5 per cent of 1.662 at each [np.True_, np.True_, np.True_]; stable at each [True, True, True]	Held	
CASCADE_3 20261006_220635	R1	the period-16 doubling is 1.2016e-04 below the first	Held ; the period-8 band 5.5460e-04 is 4.6155 times the period-16 band (printed)	
CASCADE_3 20261006_220635	R2	the period-32 band is 2.5803e-05; the period-16 band over it 4.6570 (Feigenbaum’s 4.669; 3.97 to 5.37 registered)	Held	
CASCADE_3 20261006_220635	G0	doublings 1.6893422593 and 1.6893164568 against CASCADE’s 1.6893422593 and 1.6893164568	Held	
CASCADE_3 20261006_220635	T3	log-log slope 0.5000	Held	
CASCADE_3 20261006_220635	T1	the period-64 doubling at 1.6893109278, -2.54e-09 from 1.6893109303	Held ; band 5.5291e-06; the period-32 band over it 4.6667 (printed)	
CASCADE_3 20261006_220635	T2	its words in the tool’s own header	Held	
CASCADE_3 20261006_220635	U2	slope 0.4993; coefficients 0.56558, 0.56542, 0.56454 against 1.2588	Missed	—
CASCADE_3 20261006_220635	U3	1.66206/1.45803 = 1.13994; 1.45803/0.56558 = 2.57795; against 1.1583	Missed	—
CASCADE_3 20261006_220635	U4	same class at 6 of 16 points	Missed	—
CASCADE_3 20261006_220635	U1	the period-128 doubling at 1.6893097435, -1.33e-10 from 1.6893097436	Held ; band 1.1843e-06; the period-64 band over it 4.6687 (printed)	
CASCADE_4 20261006_221804	S1	the inequality’s log-log slope 0.4978; at 1e-6 5.983 per cent	Held	
CASCADE_4 20261006_221804	S2	separation over the square root within 0.5 per cent of 1.662 at each [np.True_, np.True_, np.True_]; stable at each [True, True, True]	Held	
CASCADE_4 20261006_221804	R1	the period-16 doubling is 1.2016e-04 below the first	Held ; the period-8 band 5.5460e-04 is 4.6155 times the period-16 band (printed)	
CASCADE_4 20261006_221804	R2	the period-32 band is 2.5803e-05; the period-16 band over it 4.6570 (Feigenbaum’s 4.669; 3.97 to 5.37 registered)	Held	

Run	Label	The prediction, as the run prints it	Verdict	Misses row
CASCADE_4 20261006_221804	G0	doublings 1.6893422593 and 1.6893164568 against CASCADE’s 1.6893422593 and 1.6893164568	Held	
CASCADE_4 20261006_221804	T3	log-log slope 0.5000	Held	
CASCADE_4 20261006_221804	T1	the period-64 doubling at 1.6893109278, -2.54e-09 from 1.6893109303	Held ; band 5.5291e-06; the period-32 band over it 4.6667 (printed)	
CASCADE_4 20261006_221804	T2	its words in the tool’s own header	Held	
CASCADE_4 20261006_221804	U2	slope 0.4993; coefficients 0.56558, 0.56542, 0.56454 against 1.2588	Missed	—
CASCADE_4 20261006_221804	U3	1.66206/1.45803 = 1.13994; 1.45803/0.56558 = 2.57795; against 1.1583	Missed	—
CASCADE_4 20261006_221804	U4	same class at 6 of 16 points	Missed	—
CASCADE_4 20261006_221804	U1	the period-128 doubling at 1.6893097435, -1.33e-10 from 1.6893097436	Held ; band 1.1843e-06; the period-64 band over it 4.6687 (printed)	
CASCADE_4 20261006_221804	Z1	at the critical point 2.8231 and 2.3951 (2.2 to 2.8 registered)	Missed	—
CASCADE_4 20261006_221804	Z2	alpha-squared class at m = [1, 2, 3, 4, 5, 6, 7, 8] (8 consecutive: True) and m = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15] (8 consecutive: False); the same: False	Missed	—
CASCADE_4 20261006_221804	Z3	coefficient ratios 2.8080 (against 2.8174) and 1.1235 (against 1.1056)	Held	
GROOVES_7 20261006_222345	E1	its words in the tool’s own header	Missed	—
GROOVES_8 20261006_225037	P1	9/7 width 5.7740e-08 against 1.0937e-07 (factor 0.528)	Held	
GROOVES_8 20261006_225037	P2	the period-8 power of the gap 17.7337 against 22.36	Missed	—
CASCADE_5 20261006_225126	S1	the inequality’s log-log slope 0.4978; at 1e-6 5.983 per cent	Held	
CASCADE_5 20261006_225126	S2	separation over the square root within 0.5 per cent of 1.662 at each [np.True_, np.True_, np.True_]; stable at each [True, True, True]	Held	
CASCADE_5 20261006_225126	R1	the period-16 doubling is 1.2016e-04 below the first	Held ; the period-8 band 5.5460e-04 is 4.6155 times the period-16 band (printed)	
CASCADE_5 20261006_225126	R2	the period-32 band is 2.5803e-05; the period-16 band over it 4.6570 (Feigenbaum’s 4.669; 3.97 to 5.37 registered)	Held	
CASCADE_5 20261006_225126	G0	doublings 1.6893422593 and 1.6893164568 against CASCADE’s 1.6893422593 and 1.6893164568	Held	
CASCADE_5 20261006_225126	T3	log-log slope 0.5000	Held	

Run	Label	The prediction, as the run prints it	Verdict	Misses row
CASCADE_5 20261006_225126	T1	the period-64 doubling at 1.6893109278, -2.54e-09 from 1.6893109303	Held ; band 5.5291e-06; the period-32 band over it 4.6667 (printed)	
CASCADE_5 20261006_225126	T2	its words in the tool’s own header	Held	
CASCADE_5 20261006_225126	U2	slope 0.4993; coefficients 0.56558, 0.56542, 0.56454 against 1.2588	Missed	—
CASCADE_5 20261006_225126	U3	1.66206/1.45803 = 1.13994; 1.45803/0.56558 = 2.57795; against 1.1583	Missed	—
CASCADE_5 20261006_225126	U4	same class at 6 of 16 points	Missed	—
CASCADE_5 20261006_225126	U1	the period-128 doubling at 1.6893097435, -1.33e-10 from 1.6893097436	Held ; band 1.1843e-06; the period-64 band over it 4.6687 (printed)	
CASCADE_5 20261006_225126	Z1	at the critical point 2.8231 and 2.3951 (2.2 to 2.8 registered)	Missed	—
CASCADE_5 20261006_225126	Z2	alpha-squared class at m = [1, 2, 3, 4, 5, 6, 7, 8] (8 consecutive: True) and m = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15] (8 consecutive: False); the same: False	Missed	—
CASCADE_5 20261006_225126	Z3	coefficient ratios 2.8080 (against 2.8174) and 1.1235 (against 1.1056)	Held	
CASCADE_5 20261006_225126	K1	the shrink from 64 to 128 at the critical point 2.5491 (2.40 to 2.60 registered; the sequence 2.8231, 2.3951, 2.5491)	Held	
CASCADE_5 20261006_225126	K2	its words in the tool’s own header	Held	
CASCADE_5 20261006_225126	K3	the period-256 doubling at 1.6893094898, -7.97e-11 from 1.6893094899	Held ; band 2.5365e-07; the period-128 band over it 4.6691 (printed)	

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Run	Label	The prediction, as the run prints it	Verdict	Misses row
GROOVES_10 20261007_002739	V0	6/5 upper edge 1.20188238090441145687354605478 against version 1’s 1.2018823809044119304974610543 5 (within 1.64e-15; differs by 4.74e-16)	Held	
GROOVES_10 20261007_002739	R3	7/5 from its loop 1.4889793e-8 against 1.4890e-08 (ratio 0.99998608)	Held	
GROOVES_10 20261007_002739	R2	11/7 from its loop 2.8068117e-22 against 7.5780578e-24 (ratio 37.0387)	Missed	—
GROOVES_10 20261007_002739	R1	8/5 from its loop 2.4662389e-20 against 1.2532901e-21 (ratio 19.6781)	Missed	—
GROOVES_10 20261007_002739	T1	8/5 at N 341 and 2N: widths 2.4662389e-20 and 2.4662385e-20, ratio 0.99999984	Held	
GROOVES_9 20261007_011618	F0	10/7 from its loop 2.6116567e-12 and 9/8 1.4576737e-05 against 2.6116567e-12 and 1.4576737e-05	Held	
GROOVES_9 20261007_011618	L1	11/8 from its loop 1.191112e-11 against GROOVES_8’s 5.196122e-11	Missed	—

Run	Label	The prediction, as the run prints it	Verdict	Misses row
GROOVES_9 20261007_011618	L2	the period-9 powers of the gap 23.0608 (11/9 to 13/9) and 21.7910 (13/9 to 14/9)	Missed	—
SWING_LISTEN 20261007_013201	S1	its words in the tool’s own header	Held	
SWING_LISTEN 20261007_013201	S2	its words in the tool’s own header	Held	
SWING_LISTEN 20261007_013201	S3	its words in the tool’s own header	Held	
SWING_LISTEN_2 20261007_024253	T1	its words in the tool’s own header	Held	
SWING_LISTEN_2 20261007_024253	T2	its words in the tool’s own header	Held	
FOLD_OR_FLIP 20261007_050721	F1	its words in the tool’s own header	Held	
FOLD_OR_FLIP 20261007_050721	F2	its words in the tool’s own header	Held	
FOLD_OR_FLIP 20261007_050721	F3	its words in the tool’s own header	Held	
FOLD_OR_FLIP 20261007_050721	F4	its words in the tool’s own header	Held	
FOLD_OR_FLIP 20261007_050721	F5	its words in the tool’s own header	Held	
FOLD_OR_FLIP 20261007_050721	F6	its words in the tool’s own header	Held	
FOLD_OR_FLIP_2 20261007_053443	S1	its words in the tool’s own header	Held	
FOLD_OR_FLIP_2 20261007_053443	S2	its words in the tool’s own header	Held	
FOLD_OR_FLIP_3 20261007_054521	V1	its words in the tool’s own header	Held ; V1b: HELD	
FOLD_OR_FLIP_3 20261007_054521	V2	its words in the tool’s own header	Held	
FOLD_OR_FLIP_3 20261007_054521	V1b	its words in the tool’s own header	Held	
FOLD_OR_FLIP_4 20261007_061947	W0	largest spread at period 32 4.58e-13; largest of all 4.58e-13	Missed	—
FOLD_OR_FLIP_4 20261007_061947	W1a	its words in the tool’s own header	Missed ; W1: MISSED	—
FOLD_OR_FLIP_4 20261007_061947	W1	its words in the tool’s own header	Missed	—
FOLD_OR_FLIP_4 20261007_061947	W2	its words in the tool’s own header	Missed	—
GROOVES_10 20261007_065508	R3	7/5 from its loop 1.4889793e-8 against 1.4890e-08 (ratio 0.99998608)	Held	
GROOVES_10 20261007_065508	R2	11/7 from its loop 2.8068117e-22 against 7.5780578e-24 (ratio 37.0387)	Missed	—
GROOVES_10 20261007_065508	R1	8/5 from its loop 2.4662389e-20 against 1.2532901e-21 (ratio 19.6781)	Missed	—
GROOVES_10 20261007_065508	T1	8/5 at N 341 and 2N: widths 2.4662389e-20 and 2.4662385e-20, ratio 0.99999984	Held	

Run	Label	The prediction, as the run prints it	Verdict	Misses row
RUNG_ABOVE 20261007_071341	U1	the clock of *A Rider in Two Waves* 1.692490282152 against B11's 1.6924902823 (differs -1.48e-10)	Held	
RUNG_ABOVE 20261007_071341	U2	golden less the clock of *A Rider in Two Waves* +5.8412e-09	Held	
RUNG_ABOVE 20261007_071341	U3	golden clock (D2 - D4) / (D4 - D8) = 4.45954 against 4.4595	Held	
RUNG_ABOVE 20261007_071341	U4	its words in the tool's own header	Held	
RUNG_ABOVE_2 20261007_073220	V2	the period-2 clock shift at 487 steps +5.8412e-09 (at 486 +5.8412e-09; differs -6.66e-16) against +5.8412e-9 within 1e-11	Held	
POLARISATION_LOOP_2 20261007_074423	W1	largest difference from 2 pi a J1(2a) sin(gap) 2.51e-07 (bar 1e-6)	Held	
POLARISATION_LOOP_2 20261007_074423	W2	plates' intensity off 1 by at most 2.5e-14, home in its own state within 1.2e-14 (bars 1e-12); light's phase, plates less filters, at most 4.44e-15 (bar 1e-6)	Held	
POLARISATION_LOOP_2 20261007_074423	W3	its words in the tool's own header	Held	
GROOVES_11 20261007_080706	E1	inside each edge the residual changes sign, outside it keeps one sign	Held	
GROOVES_11 20261007_080706	E2	inside each edge the residual changes sign, outside it keeps one sign	Missed	—
GROOVES_12 20261007_083300	G1	the highest edge round the circle above the old upper edge by more than a thousandth of the old width, at an angle between 4.4 and 5.1	Missed	—
GROOVES_12 20261007_083300	G2	the width round the circle 2.8068117e-22 against 2.8068117e-22 (ratio 1.0), between 1.5 and 3	Missed	—
GROOVES_12 20261007_083648	G0	8/5 round the whole circle 2.4662389e-20 against one period's 2.4662389e-20	Held	
GROOVES_12 20261007_083648	G3	7/5 round the whole circle 1.4889793e-8 against one period's 1.4889793e-8	Held	
GROOVES_13 20261007_090124	H1	a closed orbit whose leading multiplier is more than 0.01 from 1	Undecided	—
GROOVES_13 20261007_090124	H2	its leading multiplier larger than 1 in size	Undecided	—
GROOVES_15 20261007_164232	S3	upper edge after eight rounds 9.44e-24 from the fold (bar 1e-18), moved 4.76e-16 from round four (more than 1e-16)	Held	
GROOVES_15 20261007_164232	S4	lower edge after eight rounds 2.5e-23 from the fold (bar 1e-19)	Held	
GROOVES_15 20261007_165418	S5	died in the second edge with 'matrix is numerically singular', the centre then 2.16387 radian from the lowest sample (more than 0.2)	Held	
GROOVES_14 20261007_170021	K2	an orbit within 0.05 of each edge's angle inside, none anywhere outside	Held	
GROOVES_14 20261007_170021	K3	8/5): both edges from the dips within 1e-4 of the width of GROOVES_10's (off by 1.25e-10, 1.82e-10 of the width	Held	
GROOVES_14 20261007_170021	K3_8/5	its words in the tool's own header	Held	
GROOVES_14 20261007_170022	K1	an orbit within 0.05 of each edge's angle inside, none anywhere outside	Held	

Run	Label	The prediction, as the run prints it	Verdict	Misses row
GROOVES_14 20261007_170022	K3	11/7): both edges from the dips within 1e-4 of the width of GROOVES_10’s (off by 1.36e-8, 1.94e-8 of the width	Held	
GROOVES_14 20261007_170022	K3_11/7	its words in the tool’s own header	Held	
GROOVES_16 20261007_182009	Q1	11/7	Held	
GROOVES_16 20261007_182009	Q2	1 - d changes sign 2 times; crossings at 4.51392830281, 4.9292888282; within 1e-6 of 4.5139283028: yes, 6.07e-14; between 4.9166 and 5.0051: yes; 1 - d at the sample nearest 4.75 (4.7640) -0.02329	Held	
GROOVES_16 20261007_182009	Q4	the circle at the far point’s angle (landed 3.29e-28 from it in 3 landings): speed 1.2182148520294294798, h 7.25124e-22; GROOVES_13’s far point 1.2182148521805400114, off the circle by 1.51111e-10 (window 1e-12 to 1e-8); sqrt of h there 2.693e-11	Held	
GROOVES_16 20261007_182009	Q3	(8/5 control): 1 - d changes sign 0 times round the circle; the closed-loop law held at every sample	Held	
GROOVES_16 20261007_182009	Q1_11/7	its words in the tool’s own header	Held	
GROOVES_16 20261007_182010	Q5	its words in the tool’s own header	Held ; Q6: HELD; Q7 (every estimate from round two on inside the lock): HELD	
GROOVES_16 20261007_182010	Q6	its words in the tool’s own header	Held	
GROOVES_16 20261007_182010	Q7	its words in the tool’s own header	Held	
GROOVES_16 20261007_183554	Q8	5 dips, extremes -7.901e-32, -8.525e-31, -2.902e-30, -4.833e-30, -2.908e-30 (bar 1e-28 in size)	Held	
GROOVES_17 20261007_234942	T1	the 8/5’s five implied edges with 2 pi agree within 8.23e-38 (bar 1e-32; with TWOM 4.756e-30)	Held	
GROOVES_17 20261007_234943	T2	the 11/7’s seven implied edges with 2 pi agree within 7.883e-38 (bar 1e-32); with TWOM within 1.788e-30 (more than 1e-31)	Held	
CONE_TOMOGRAPHY 20261007_235426	C1	largest difference 8.44e-15 (bar 1e-8)	Held	
CONE_TOMOGRAPHY 20261007_235426	C2	outside the foot (radii 2.0368, 2.1579, 2.2789, 2.4000) the largest turn 0.00e+00 (bar 1e-8); the largest turn at radius 0.9474, the radius nearest the steepest ring (0.8944) is 0.9474	Held	
CONE_TOMOGRAPHY 20261007_235444	C3	centres within 1.29e-08 (bar 0.01), slopes within 3.11e-15 radian (bar 1e-6)	Held	
CONE_TOMOGRAPHY 20261007_235444	C4	the big circle round all three feet (centre (3, 1.5), radius 8) turns the frame by +0.00e+00 (bar 1e-8)	Held	

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Run	Label	The prediction, as the run prints it	Verdict	Misses row
CONE_TOMOGRAPHY 20261008_001220	C5	largest difference 1.82e-14 (bar 1e-8)	Held	

Run	Label	The prediction, as the run prints it	Verdict	Misses row
CONE_TOMOGRAPHY 20261008_001220	C6	largest difference 4.80e-14 (bar 1e-8)	Held	
CONE_TOMOGRAPHY_2 20261008_002937	E1	the twins’ turns on their steepest rings against 1.516187431740: largest difference 4.33e-13 (bar 1e-8)	Held	
CONE_TOMOGRAPHY_2 20261008_002937	E2	the three mountains of one shape at the ten radii u R: largest difference 0.00e+00 (bar 1e-8)	Held	
CONE_TOMOGRAPHY_2 20261008_002937	E3	difference +8.88e-16 (bar 1e-8)	Held ; the half-height slice over the full one 0.336108398 (29.198120 degrees of paper against 86.871141)	
CONE_TOMOGRAPHY_2 20261008_002937	E4	largest difference 1.60e-14 (bar 1e-8)	Held ; the quarter-circle mountain is 2.7097 percent taller than the mountain at (0, 0)	
CONE_TOMOGRAPHY_2 20261008_002939	D1	best radius 3.00 (predicted 3.00), 2 feet reached (predicted 2), centre outside every foot True	Held	
CONE_TOMOGRAPHY_2 20261008_002939	D2	largest difference 4.88e-11 (bar 1e-8)	Held	
CONE_TOMOGRAPHY_2 20261008_002939	D3	radius above 3.0 True, converged True, more than 0.01 from every centre True, outside every foot True	Held	
CONE_TOMOGRAPHY_3 20261008_011149	N2	the two-mountain circle third order False; mountain 1’s ring fourth order True	Missed	—
CONE_TOMOGRAPHY_3 20261008_011150	N1	centres (mountains above 1e-3, the fourth within 1e-12 of 0) True; along the best circles (mountains under 1e-5, the fourth above 1e-4 somewhere) True	Held	
CONE_TOMOGRAPHY_3 20261008_011157	N3	by 4 and by 1/4 exactly 0 (True, True); by 3 nonzero somewhere and under 1e-12 (True)	Held	
CONE_TOMOGRAPHY_4 20261008_012042	K2	error ratios per doubling 2.000, 0.250 (each 13 to 19); signs +, -, +	Missed	—
CONE_TOMOGRAPHY_4 20261008_012043	K1	changes sign True; scaled error spans a factor 15.2 (more than 3)	Held	
CONE_TOMOGRAPHY_5 20261008_012723	K3a	n = 5 ratios each 13 to 19	Held	
CONE_TOMOGRAPHY_5 20261008_012723	K3b	n = 3’s error over n = 5’s at 500, 1,000 and 2,000 steps: 46.8, 4.47, 2.27e+03 (each more than 100)	Missed	—
DELTA_BY_TOP 20261008_034013	D1	nu = 2’s last doubling ratio less 4.669201609: -2.15e-8 (bar 1e-5)	Held	
DELTA_BY_TOP 20261008_034013	D2	the quartic’s last doubling ratio 7.284725495 against 7.28 (bar 0.01)	Held	
DELTA_BY_TOP 20261008_034013	D3	nu = 2’s last width ratio -2.502907898 against -2.5029 (bar 1e-3)	Held	
DELTA_BY_TOP 20261008_034013	D4	last doubling ratios by order nu 2: 4.6692016, nu 4: 7.2847255, nu 6: 9.2976544, nu 8: 10.962189; rising and under 29.8	Held	

Run	Label	The prediction, as the run prints it	Verdict	Misses row
PEAK_RING 20261008_040518	R1	the scan’s largest total 11.928858947 at radius 2.75 (interior, between 2.5 and 3.5: True); above radius 3.5 the largest total 0.00e+00 (bar 1e-8); a circle of radius 0.02 at the middle, all seven peaks: turn 0.000e+00	Held	
PEAK_RING 20261008_040518	R2	refined best radius 2.7328845743; peak 0’s share there at 40,000 steps 1.707734186512 (0.413074 of its best slice); total 11.954139305584 radian = 1.902560361 turns; above one peak’s best slice (4.134209) and above 2 pi	Held	
PEAK_RING 20261008_040518	R3	the frame round the best circle over all seven peaks, 40,000 steps, reads -0.612225683983; the total less 2 whole turns is -0.612231308776; difference +5.62e-06 (bar 1e-6)	Missed	—
PEAK_RING 20261008_040518	R4	shares agree within 1e-8 False; every move lowers the total True (the middle’s total at 4,000 steps 11.954172508142)	Missed	—
PEAK_RING_2 20261008_040854	Q0	peak 0 alone on its steepest ring (radius 0.3333333), 40,000 steps, reads -2.148975939303; the slice less a whole turn -2.148975939303; difference +1.15e-14 (bar 1e-8)	Held	
PEAK_RING_2 20261008_040854	Q1	best radius 2.6751533997; the seven shares, each peak alone, 40,000 steps: 1.709559085452, 1.709559085451, 1.709559085451, 1.709559085451, 1.709559085451, 1.709559085451, 1.709559085451; spread 2.97e-13 (bar 1e-8)	Held	
PEAK_RING_2 20261008_040854	Q2	the frame over all seven peaks reads -0.599457016199; the sum of the shares 11.966913598160 less 2 whole turns is -0.599457016199 (difference +2.68e-13); seven times peak 0’s share 11.966913598161 (difference -1.07e-12); bar 1e-8 both	Held	
PEAK_RING_2 20261008_040854	Q3	the best total 11.966913598160 radian = 1.904593453 turns, above 2 pi	Held ; each share is 0.413515 of a peak’s best slice	
PENTA_SHARE 20261008_060159	S1a	radius 20 against 50, by slope: 30: -2.25e-03, 50: -2.42e-03, 70: -2.77e-03 (each under 1e-3)	Missed	—
PENTA_SHARE 20261008_060159	S1b	the radius-50 fractions 0.329420, 0.346712, 0.379270, spread 4.99e-02 (bar 1e-3)	Missed	—
PENTA_SHARE 20261008_060200	W1	pentagram less (inner pentagon + outline), modulo a turn, +4.44e-15 (bar 1e-8)	Held	
PENTA_SHARE 20261008_060200	W2	outer pentagon +0.00e+00 (within 1e-8 of 0), pentagram -0.930998 (more than 0.1 from 0)	Held	
PENTA_SHARE_2 20261008_062403	S2	its words in the tool’s own header	Held	
PENTA_SHARE_2 20261008_062420	W3	pentagram less (inner pentagon + outline), modulo a turn, -5.72e-13 (bar 1e-8)	Held	
DELTA_AREA 20261008_062647	A1	the last ratio 8.720271433 against 8.72 (bar 0.01)	Held	
DELTA_AREA 20261008_062647	A2	the last ratio above 7	Held	
DELTA_BY_TOP_2 20261008_063744	E2	nu 12: 0.9619, nu 16: 0.9690, nu 24: 0.9756 after nu 8’s 0.9465; rising and below 1	Held	
DELTA_BY_TOP_2 20261008_063744	E3	the last, 2.549064, less 2.5029: +0.0462 (bar 1e-2)	Missed	—
GROW_LOOP 20261008_133646	G1	at radius 2.6751533997 the raw reading -0.599457015845, unwrapped +11.966913598514 (1.904593 turns; whole turns added 2), against the seven shares’ sum 11.966913598160: +3.54e-10 (bar 1e-6)	Held	

Run	Label	The prediction, as the run prints it	Verdict	Misses row
GROW_LOOP 20261008_133646	G2	at 50 radii from 0.002 to 3.528, unwrapped reading against seven times peak 0’s share: largest difference 2.07e-09 at radius 2.450 (bar 1e-6)	Held	
GROW_LOOP 20261008_133646	G3	its words in the tool’s own header	Held	
DRAFT_TEST 20261008_134707	F1	read jelly_viscous_grid_rung6_animal_v24_rope1_ringA_tether_passive_2mm_3s_20260918_container.txt: J_total -6.538, +3.878, -0.759 (summed -3.419 mN s); jelly_viscous_grid_rung6_animal_v24_rope1_ringB_free_passive_2mm_3s_20260918_container.txt: distances -0.118, -0.050, -0.063 (last row dist/R -0.231); against -6.538, +3.878, -0.759 and -0.118, -0.050, -0.063	Held	
DRAFT_TEST 20261008_134707	D1	closed and held, J_total summed over three cycles +51.671 mN s (positive registered)	Held	
DRAFT_TEST 20261008_134707	D2	closed and free, distance at 3 s +1.337 R (positive registered)	Held	
DRAFT_TEST 20261008_134707	D3	open and held, summed J_total -1609.346 against closed +51.671 (below registered)	Held	
DRAFT_TEST 20261008_134707	D4	open and free, distance at 3 s +nan against closed +1.337 (below registered)	Missed	—
BENT_PAPER 20261008_164513	P1	largest |reading| over 30 loops 0.00e+00 (bar 1e-8)	Held	
BENT_PAPER 20261008_164513	P2	its words in the tool’s own header	Held (bar 1e-8)	
BENT_PAPER 20261008_164513	P3a	its words in the tool’s own header	Held (bar 1e-8)	
BENT_PAPER 20261008_164513	P3b	radius R/3 round the top: -2.709729640646, -0.560754 from the plain’s -2.148975939303 (registered: more than 1e-3 apart)	Held	
BENT_PAPER_2 20261008_164721	P1b	largest |reading| over 30 loops 3.11e-12 (bar 1e-8)	Held	
BENT_PAPER_2 20261008_164721	S1	its words in the tool’s own header	Held (bar 1e-10)	
GROW_SLIP 20261008_195205	S1	(arithmetic): k = 1 to 10 all keep +2	Held	
GROW_SLIP 20261008_195205	S2	k = 20, count -1 (not +2 registered)	Held	

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Run	Label	The prediction, as the run prints it	Verdict	Misses row
COUNTER_FLOW 20261009_002621	X1	over 201 NTU from 0.01 to 100: parallel at most 0.500000000001 (bar 0.5 + 1e-9); counter from NTU 9 up at least 0.901187 (bar 0.9)	Held	
COUNTER_FLOW 20261009_002621	X2	counter over parallel 1.000033 at NTU 0.01 (within 1 per cent of 1) and 1.818182 at NTU 10 (1.8 to 2)	Held	
SOUP_JUMP 20261009_045400	J0	mean of 1/crest over the 8 soups 0.3760 (range 0.3548 to 0.3895); band 0.30 to 0.40	Held	
SOUP_JUMP 20261009_045400	J3	onset over 1/crest per soup [‘1.0898’, ‘1.0461’, ‘1.0481’, ‘1.0429’, ‘1.0818’, ‘1.0744’, ‘1.0436’, ‘1.1295’]; band 0.97 to 1.06	Missed	—
SOUP_JUMP_2 20261009_051426	K1	ratios [‘1.0187’, ‘0.9825’, ‘1.0287’, ‘1.0089’, ‘1.0113’, ‘1.0183’, ‘1.0002’, ‘1.0079’]; band 1.00 to 1.04	Missed	—
SOUP_JUMP_2 20261009_051426	K4	its words in the tool’s own header	Held	

Run	Label	The prediction, as the run prints it	Verdict	Misses row
SOUP_JUMP_2 20261009_052404	K2	mean B 1.0636 against mean A 1.0096	Held	
SOUP_JUMP_2 20261009_052404	K3	mean C 1.1434 against mean B 1.0636	Held	
LATTICE_READER 20261009_092420	LR1	columns 300 to 780 clean with exactly 8 runs; exceptions 481 [300, 301, 302, 303, 304, 305, 306, 307, 308, 309]	Missed	108
LATTICE_READER 20261009_092420	LR2	its words in the tool’s own header	Undecided	108
LATTICE_READER 20261009_092420	LR3	its words in the tool’s own header	Missed	108
LATTICE_READER 20261009_092420	LR4	fork medians [np.float64(780.0), None, None]; (fork 1 - fork 2)/(fork 2 - fork 3) = nan (record 4.6155; band 4.2 to 5.1)	Undecided	108
LATTICE_READER 20261009_092420	LR5	median slope 0.2308; power best in 0 of 4	Undecided	108
LATTICE_READER 20261009_092420	LR6	the clean run holding column 780 ends at column 900 (drive 1.690100; +83.0 columns from the fold of record) and begins at column 265 (drive 1.689465)	Missed	110
LATTICE_READER 20261009_092420	LR7	0 of 2468 (family, column) cases have every run’s revisit interval 8 x runs (0.0 per cent); first exceptions [(0, 270, np.int64(2), 8), (0, 271, np.int64(2), 8), (0, 272, np.int64(2), 8), (0, 273, np.int64(2), 8), (0, 274, np.int64(2), 8), (0, 275, np.int64(2), 8), (0, 276, np.int64(2), 8), (0, 277, np.int64(2), 8)]	Missed	108
LATTICE_READER 20261009_092420	LR8	two largest peaks of the change of change per family [(815, 761), (815, 766), (815, 761), (815, 766)] (doubling 262.4, fold 817.0)	Undecided	108
LATTICE_READER_2 20261009_093040	LR2-2	its words in the tool’s own header	Missed	—
LATTICE_READER_2 20261009_093040	LR3-2	its words in the tool’s own header	Missed	—
LATTICE_READER_2 20261009_093040	LR4-2	its words in the tool’s own header	Held	
LATTICE_READER_2 20261009_093040	LR5-2	(f_top - f1)/(f1 - f2) = 4.6333, (f1 - f2)/(f2 - f3) = 4.6154 (record 4.6155, 4.6570; band 4.2 to 5.1)	Held	
LATTICE_READER_2 20261009_093040	LR6-2	(a) median slope 0.5115, power best in 4 of 4; (b) median slope 0.4959, power best in 8 of 8	Held	
LATTICE_READER_2 20261009_093040	LR7-2	the two largest peaks per family [(815, 260), (815, 260), (815, 260), (815, 260)] (doubling 262.4, top end 817.0)	Held	
LATTICE_READER_2 20261009_093040	LR8-2	columns 112 to 900 (outside the windows) with a single-landing pad: 0 []; columns 0 to 100 with one: 89.1 per cent	Held	
LATTICE_READER_2 20261009_093040	LT1	drive 1.690050, period 4 with 2 turns: converged True, moved 9.9e-10; multipliers -0.972272, +0.000000	Held	
LATTICE_READER_2 20261009_093040	LT2	the period-4 orbit’s leading multiplier reaches -1 at drive 1.690017021430 (+4.30e-10 from 1.690017021)	Held	
HISTORY_SETTLE 20261009_093424	H1	first column going up from 812 with history period 1, per family: [822, 821, 820, 822] (819 or 820)	Missed	—

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Run	Label	The prediction, as the run prints it	Verdict	Misses row
LIFT_3D 20261010_031527	P1	holes 1e-4, columns 122 to 137 against distinct points	Missed	—
LIFT_3D 20261010_031527	P2	holes 1e-4, columns 111 to 115 against distinct points	Missed	—
LIFT_3D 20261010_031527	P3	the touching strands more than 1e-4 apart in speed	Missed	—

Run	Label	The prediction, as the run prints it	Verdict	Misses row
LIFT_3D 20261010_031527	P1b	holes 1e-5, columns 122 to 140 against distinct points	Missed	—
LIFT_3D 20261010_031527	P2b	holes 1e-5, columns 111 to 115 against distinct points	Missed	—
LOBE_TWIN 20261010_043453	T1	the two lobes opposite in sign, every lane, both events	Held	
LOBE_TWIN 20261010_043453	T2	their midpoint within one column of the event of record, every lane, both events	Held	
ZERO_CROSS 20261010_043523	Z1	its words in the tool’s own header	Held	
ZERO_CROSS 20261010_043523	Z2	its words in the tool’s own header	Held	
CORNER_PRINT 20261010_061715	C1	each lane’s pair correlates with a pure corner’s echo at 0.95 or more, three doublings	Missed	—
CORNER_PRINT 20261010_061715	C2	the pure corner’s echo alone crosses zero within 0.01 of a column of the event, three doublings	Held	
SHUFFLE_NULL 20261010_190812	N1	ordered bands: share in strings of 20 or more, picture 0.9997 at least 0.9, shuffles’ mean 0.0003 at most 0.05	Held	
SHUFFLE_NULL 20261010_190812	N2	ordered bands: shuffles’ mean count of strings of 4 or more, 209.8, above the picture’s 17	Held	
SHUFFLE_NULL 20261010_190812	N3	cloud: picture’s share in strings of 20 or more, 0.3512, within a factor of 2 of the shuffles’ mean 0.2659	Held	
SHUFFLE_NULL_2 20261010_190944	M1	cloud: picture at least 3 spreads above the column shuffles, here +1.8	Missed	626
DELTA_TAIL 20261010_205937	T1	order 8’s delta_8 between 10.951 and 10.957, here 10.95436008	Held	
DELTA_TAIL 20261010_205937	T2	order 8’s step ratio between 0.40 and 0.47, here 0.42182	Held	
DELTA_TAIL 20261010_205937	T3	order 8’s Aitken from delta_7 to delta_9 within 3e-4 of 10.9486, here 10.948788, off by 0.000188	Held	
DELTA_TAIL 20261010_205937	T4	order 6’s Aitken from delta_8 to delta_10 within 5e-5 of 9.2962, here 9.2962197, off by 1.97e-5	Held	

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Run	Label	The prediction, as the run prints it	Verdict	Misses row
ULP_NUDGE 20261011_044600	V1	every fold’s step count unchanged under all four nudges, here 32 of 32	Held	
ULP_NUDGE 20261011_044600	V2	at least three of the four doubling refinements change their count under some nudge, here 4 of 4	Held	
ULP_NUDGE 20261011_044600	V3	every fold’s drive moves by less than 5e-11, here at most 5.36e-11	Missed	—
DELTA_TAIL_2 20261011_051600	U1	order 8’s Aitken from delta_9 to delta_11 above 10.94875, here 10.9486659	Missed	—
DELTA_TAIL_2 20261011_051600	U2	order 6’s Aitken from delta_9 to delta_11 within 2e-5 of 9.2962468327, here 9.29625632, off by 9.48e-6	Held	
FOLD_TIGHT 20261011_051636	T1	with the 1e-12 bar every fold’s drive moves by less than 1e-12 under every nudge, here at most 1.24e-14	Held	
FOLD_TIGHT 20261011_051636	T2	every fold reaches the 1e-12 bar within two steps more than its 1e-9 count, under every nudge	Missed	—
DAMPER_LAWS 20261011_051848	A1	every law ends home within 1e-5 m and slower than 1e-9 m/s	Held	

Run	Label	The prediction, as the run prints it	Verdict	Misses row
DAMPER_LAWS 20261011_051848	A2b	the stepped pay-out reaches the three formulas' values to four figures	Held	
DAMPER_LAWS 20261011_051848	B1	his speed is the momentum share and the damper takes half mu V^2, to six figures	Held	
TURNING_TILT 20261011_055620	T1	above pace 1 the count is within one of 200 sqrt(pace^2 - 1) / pace	Held	
TURNING_TILT 20261011_055620	T2	below pace 1 no slip	Held	
SPIN_OR_SIT 20261011_055807	L1	top-timed successive gaps correlate +0.11 to +0.21, here +0.2082	Held	
SPIN_OR_SIT 20261011_055807	L2	landing-timed successive gaps within 0.05 of zero, here +0.0568	Missed	128
SPIN_OR_SIT 20261011_055807	L3	the tie across the landing rises with memory 0, 1, 2, 5: -0.4450, +0.0043, +0.0683, +0.5082	Undecided	129
SPIN_OR_SIT 20261011_055807	R1	spins and locks: median / mean between 0.6 and 0.8, here 0.708 and 0.443	Missed	130
SPIN_OR_SIT 20261011_055807	R2	spinning share x running rate gives the slip rate within 2 per cent	Held	
SPIN_OR_SIT 20261011_055807	C1	fast climbs peak between 0.10 and 0.30 periods	Held	
SPIN_OR_SIT 20261011_055807	C2	climbs over two periods are at least 1 per cent	Missed	131
SPIN_OR_SIT 20261011_055807	C3	climbs of 0.5 to 2 periods fewer than climbs over 2	Held	
SLIP_SENSE 20261011_060035	P1	every judged row reads the pace back within 0.01	Held	
CHAOS_SETTLES 20261011_060320	E1	each needle's exponent drops from positive to negative once and ends negative	Held	
CHAOS_SETTLES 20261011_060320	E2	the drop times spread like a constant chance per unit time	Held	
CHAOS_SETTLES 20261011_060320	N1	no needle locks: every needle slips in every window	Held	
LOCK_FROM_CHAOS 20261011_060551	C0	all 256 needles still slipping at 0.9126	Held	
LOCK_FROM_CHAOS 20261011_060551	X1	a single exponential at every pace: median/mean 0.6 to 0.8, and 0.090 to 0.203 left at twice the mean	Missed	132
LOCK_FROM_CHAOS 20261011_060551	X2	slope between -0.65 and -0.35, here -0.3089	Missed	133
SHOVE_SKATER 20261011_070846	S0	no drag: the three speeds agree to 1e-10	Held	
SHOVE_SKATER 20261011_070846	S1	as registered: drag set by distance, the square law, three speeds within 0.005 m/s	Missed	134
SHOVE_SKATER 20261011_070846	L1	Claude's lean: S1 misses and the speed rises with the rate	Held	
SHOVE_SKATER 20261011_070846	S2	drag charged per swizzle: three speeds agree to 1e-10 and read 3.0951	Held	
SHOVE_SKATER 20261011_070846	S3	the record's linear drag: speed after 10 swizzles rises more than 5 per cent from 0.5 to 2	Held (ratio 1.3404)	
WADDLE_PRICE 20261011_071302	W1	same pads within 1e-3 L, about 1 L per landing	Held	

Run	Label	The prediction, as the run prints it	Verdict	Misses row
WADDLE_PRICE 20261011_071302	C1	as registered: positive work, waddle 0.75 of shuvit within 0.01	Missed	136
WADDLE_PRICE 20261011_071302	C2	net work equal within 1 per cent	Held	
WADDLE_PRICE 20261011_071302	C3	positive plus braking: 0.60 within 0.01	Missed	136
WADDLE_PRICE 20261011_071302	P1	as registered: steady energy 4/3 of the pop within 1 per cent	Held	
WADDLE_PRICE 20261011_071302	P2	the gap shrinks by 0.25 per landing within 0.01	Missed	137
WADDLE_PRICE 20261011_071302	P3	a fixed kick settles at 2 J / I_end within 1 per cent	Held	
WADDLE_PRICE 20261011_071302	F1	excess within 15 per cent of (1.5 tau)^2 / k at each couple	Missed	—
WADDLE_PRICE 20261011_071302	F2	couple 1: ratios 0.75 and 0.60 within 0.002	Held	
METER_CLOCKS 20261011_072646	a	relative difference under 0.10	Held	
METER_CLOCKS 20261011_072646	b	hour-wear fleet: lowest waste at k between 1.25 and 1.75	Missed	140
METER_CLOCKS 20261011_072646	c	tach fleet: loud catch 0.85 to 0.95 with more waste; quiet catch and waste each move under 0.05	Missed	141
TWIST_LONG 20261011_072842	W1	below: the 10 s angle stays inside its first 2.35 s range, widened by 1 per cent	Held	
TWIST_LONG 20261011_072842	W2	above: turns gained 5-10 s are 0.9 to 1.1 times those of 0-5 s	Held	
TWIST_LONG 20261011_072842	T1	the jump within 0.25 degrees of the formula at every cartwheel moment	Held	
METER_K_SPREAD 20261011_080641	i	in each fleet the least-spread k within 0.5 of the lowest-waste k	Held	
METER_K_SPREAD 20261011_080641	ii	tach-wear fleet: lowest-waste k above 2	Held	
EXP_SHAPE 20261011_080708	E1	median over mean of R between 0.6 and 0.8 at every pace	Missed	142
EXP_SHAPE 20261011_080708	E2	R’s distance under the null’s 95th percentile at every pace	Missed	142
EXP_SHAPE 20261011_080708	E3	the excess of early locks is largest at the farthest pace, 0.91200	Held (largest at 0.91200)	
WADDLE_FOLLOW 20261011_080808	G1	the gap from the walker’s own limit shrinks by 0.25 within 0.01, landings 1 to 7, every pop	Held	
WADDLE_FOLLOW 20261011_080808	G2	the floor halves within 20 per cent when the pin is twice as stiff	Missed	143
MEMORY_SWITCH 20261011_084432	S1	the switch between 0.70 and 0.71	Undecided	144
MEMORY_SWITCH 20261011_084432	S2	memory 0.9 keeps slipping, memory 0.6 stops	Held	
MEMORY_SWITCH_2 20261011_085331	R1	the three readings agree within 1e-6	Held	
MEMORY_SWITCH_2 20261011_085331	R2	within 0.001 of 0.7026	Held	

Run	Label	The prediction, as the run prints it	Verdict	Misses row
LOCK_LANDING 20261011_170402	K1	locks under 1 period: at least 80 per cent in the fall at the middle, here 1.0	Held	
LOCK_LANDING 20261011_170402	K2	locks over 2 periods: at least 80 per cent in the well at the middle, here 0.786	Missed	146
LOCK_LANDING 20261011_170402	G1	the fall-with-next-fall term is the largest and more than half the total, here 0.749 of it	Held	
LOCK_LANDING 20261011_170402	T1	memory 0's all-turns tie within two standard errors of zero, here -0.0209 against 0.0211	Held	
LOCK_LANDING 20261011_170402	T2	the spinning tie rises at memories 1, 1.5, 2, 3, 5: -0.0014, +0.0442, +0.0683, +0.2617, +0.5082	Held	
EXP_SHAPE_2 20261011_194909	P1	both bars passed at both paces with 1,024 needles	Held	
LOCK_MATCH 20261011_195010	A1	at least 90 per cent of the long locks overlap a caught climb, here 0.821	Missed	147
LOCK_MATCH 20261011_195010	A2	at least 90 per cent of the caught climbs overlap a long lock, here 1.000	Held	
LOCK_MATCH 20261011_195010	B1	the median long lock spends at least 0.8 of its recorded steps short of the top, here 0.784	Missed	147
FALL_CHAIN 20261011_200714	C1	a chain, not a slow drift: r_2 within two errors of r_1 squared and below half of r_1	Held	
FALL_CHAIN 20261011_200714	C2	r_2 more than two errors above zero, *A Rider in Two Waves*'s question; near a coin flip at this size	Missed	148
CATCH_TRAP 20261011_202500	a	every caught climb overlaps a long lock, and the matched long locks number the catches within 2: 64 of 64, 64 locks	Held	
CATCH_TRAP 20261011_202500	b	the catches' lengths less 2 periods have median over mean 0.6 to 0.8, here 0.800	Missed	149
SETTLED_COPIES 20261011_212014	S1	every needle at 0.98375 repeats its state after 4 drive periods to 1e-6, here 1.672e-09	Held	
SETTLED_COPIES 20261011_212014	S2	4 clusters, the npz's groups, carried round one cycle by the drive period	Missed	150
SETTLED_COPIES_2 20261011_212705	a	at 0.98375 the clusters are the groups read on settled windows	Held	
SETTLED_COPIES_2 20261011_212705	b	at 0.99375 every needle repeats within 200 drive periods	Held	
SETTLED_GROOVES 20261011_223349	W1	the 1/4 groove is lost on both sides between 0.98 and 0.99	Held	
SETTLED_GROOVES 20261011_223349	O1	at a 1e-9 sieve all 16 needles at 0.99375 share one repeat period within 400	Missed	152
SETTLED_GROOVES 20261011_223349	O2	that period is a multiple of 19, with 9 turns per 19	Undecided	—

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Run	Label	The prediction, as the run prints it	Verdict	Misses row
FALL_CHAIN_2 20261012_041438	a	r_2 more than two standard errors above zero	Missed	154
FALL_CHAIN_2 20261012_041438	b	r_2 within two standard errors of r_1 squared	Missed	154
SETTLED_GROOVES_2 20261012_041558	W2	the groove is lost within 25 steps of 0.00001 on each side	Held	

Run	Label	The prediction, as the run prints it	Verdict	Misses row
SETTLED_GROOVES_2 20261012_041558	a	every needle at 0.99375 repeats within 4,000 drive periods at 1e-9	Missed	153
SETTLED_GROOVES_2 20261012_041558	b	every period a multiple of 19, with 9 turns per 19	Undecided	—
SETTLED_GROOVES_2 20261012_041558	c	at least three different periods among the 16	Undecided	—
FALL_SHARE 20261012_075804	a	given the landing speed, the fall-to-fall correlation drops under half its size	Held	
FALL_SHARE 20261012_075804	b	the landing speed correlates above 0.1 in size with both falls it touches	Held	
WINDOW_EDGE 20261012_080030	E1	the bisection keeps the groove at its lower end and none at its upper	Held	
WINDOW_EDGE 20261012_080030	E2	the mean calm stretch falls like d to a power between -0.65 and -0.35	Missed	155
WINDOW_CASCADE 20261012_081249	a	the doublings 4 to 8, 8 to 16, 16 to 32 and 32 to 64 all lie between 0.98371 and 0.98373	Missed	157
WINDOW_CASCADE 20261012_081249	b	the second ratio of gaps lies within 0.5 of 4.669	Held	
WINDOW_CASCADE_2 20261012_171509	R0	its words in the tool’s own header	Held	
WINDOW_CASCADE_2 20261012_171509	a	the third ratio within 0.1 of 4.6692 and nearer than the second	Held	
WINDOW_CASCADE_2 20261012_171509	b	the fourth ratio within 0.05 of 4.6692	Held	
WINDOW_CASCADE_2 20261012_171509	c	held 4,000 drive periods at 0.98371, the walker repeats after 8 with 2 turns	Held	
WINDOW_EDGE_2 20261012_171509	a	the fold lies above the bisected edge by between 1e-8 and 1e-6 of pace	Missed	158
WINDOW_EDGE_2 20261012_171509	b	one minus the larger multiplier falls like the square root of the distance, slope 0.45 to 0.55	Held	
WINDOW_EDGE_2 20261012_171509	c	the first passage from the window’s lower end falls like d to a power between -0.6 and -0.4	Held	
SPEED_CHAIN 20261012_171949	a	the landing speed forgets in one step, |corr(s k, s k+1)| under 0.05	Held	
SPEED_CHAIN 20261012_171949	b	given the top speed, climb k and fall k+1 keep under a tenth of their correlation	Missed	159

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Run	Label	The prediction, as the run prints it	Verdict	Misses row
TOP_SECTION 20261013_005355	a	with 200 bins the remainders keep under a tenth of the plain correlation	Held	
INNER_OUTER 20261013_005545	a	a return at every one of the seven distances: an inner fold	Held	
INNER_OUTER 20261013_005545	b	the mean return grows like d to a power between -0.65 and -0.35	Held	
GHOST_TRIAL 20261013_005637	a	every edge above p_f, each fourfold lengthening cutting the distance to under a quarter	Missed	160
SADDLE_LOG 20261013_012417	a	the fall’s slope within 30 per cent of -0.695, between -0.90 and -0.49	Missed	161

Run	Label	The prediction, as the run prints it	Verdict	Misses row
GHOST_TRIAL_2 20261013_015743	R1	its words in the tool’s own header	Held	
GHOST_TRIAL_2 20261013_015743	a	the answer switches at least 5 times along the scan	Held	
GHOST_TRIAL_2 20261013_015743	b	at least 5 paces more than 1e-8 past the fold answer on	Held	
REINJECT 20261013_015743	R1	its words in the tool’s own header	Held	
REINJECT 20261013_015743	a	mean return over first passage above 0.92 at 1/64 and 0.95 at 1/256	Held	
REINJECT 20261013_015743	b	the ratio rises from 1/16 to 1/64 to 1/256	Held	
WINDOW_CASCADE_3 20261013_015743	R0	its words in the tool’s own header	Held	
WINDOW_CASCADE_3 20261013_015743	a	the fifth ratio short of 4.669202 by 0.0005 to 0.0025	Held	
WINDOW_CASCADE_3 20261013_015743	b	the fifth ratio exceeds the recomputed fourth	Held	
REINJECT_CLOCK 20261013_024653	R1	its words in the tool’s own header	Held	
REINJECT_CLOCK 20261013_024653	a	at 1/1024 the time outside returns per return between 500 and 750	Held	
REINJECT_CLOCK 20261013_024653	b	at 1/1024 the share of time in returns above 0.70	Held	
GHOST_DEPTH 20261013_024747	a	the depth grows in proportion to the distance, slope 0.9 to 1.1	Held	
GHOST_DEPTH 20261013_024747	b	the depth reaches 1e-6 between 1.85e-8 and 1e-7 past the fold	Held	
SADDLE_NOISE 20261013_024850	a	at noise 0 the slope within 15 per cent of -0.6949	Missed	162
SADDLE_NOISE 20261013_024850	c	the slope’s size falls at each step of the noise	Missed	162
SADDLE_ASYMPTOTE 20261013_025252	a	over 1e-4 to 1e-3 the slope within 3 per cent of -0.6949	Held	
SADDLE_ASYMPTOTE 20261013_025252	b	the slope’s size grows decade by decade toward the law	Held	
CATCH_TRAP_2 20261013_030643	a	not one exponential: Kolmogorov distance above the 95th percentile	Held	
CATCH_TRAP_2 20261013_030643	b	median over mean above ln 2 by more than two standard errors	Missed	163
ORBIT_919_3 20261013_031807	a	every needle shrinks onto needle 30’s orbit, log distance straight against time, r squared above 0.9	Missed	164
ORBIT_919_3 20261013_031807	b	the orbit attracts, its multiplier’s rate within a factor 2 of the needles’ mean slope	Missed	164
ORBIT_919_4 20261013_040039	a	somewhere along the orbit a deviation grows more than a million-fold	Missed	165
ORBIT_919_4 20261013_040039	b	extended: all 13 within 1e-11 of the reference at every reading from 1,000 to 2,000	Missed	165
ORBIT_919_4 20261013_040039	c	double: at least one beyond 1e-9 at some reading from 1,000 to 2,000	Held	
ORBIT_919_4 20261013_040039	d	folded: all 13 within 1e-11 at every reading from 1,000 to 2,000	Missed	165

Run	Label	The prediction, as the run prints it	Verdict	Misses row
CATCH_TRAP_3 20261013_040120	a	at c = 3 and at c = 4 the Kolmogorov distance below the reference’s 95th percentile	Held	
CATCH_TRAP_3 20261013_040120	b	the mean excess at c = 4 within two standard errors of that at c = 3	Held	
ORBIT_19 20261013_041840	a	needle 30 repeats after 19 drive periods and no fewer, 9 turns	Missed	166
ORBIT_19 20261013_041840	b	each of the 13 needles within 1e-12 in median of needle 30’s 19 readings	Missed	166
ORBIT_19 20261013_041840	c	the 19-period pull between 0.105 and 0.135, its 23rd power within a factor 3 of 4.93e-22	Held	
STRETCH_WHERE 20261013_045856	a	growth and share rank together above 0.8 across the 437 drive periods	Missed	167
STRETCH_WHERE 20261013_045856	b	the largest growth’s period among the 22 with the largest share	Held	
STRETCH_DEPTH 20261013_055240	a	at every counted pace, growth ranks with depth above 0.9	Held	
STRETCH_DEPTH 20261013_055240	b	pooled, growth ranks with depth above 0.9 and at least 0.1 above share	Held	
LOCK_EDGE_EXACT 20261013_063227	F0	a multiplier passes +1 at the turning point	Held	
LOCK_EDGE_EXACT 20261013_063227	a	the fold between 0.91231 and 0.9124, here 0.9123313911	Held	
LOCK_EDGE_EXACT 20261013_063227	b	LOCK_FROM_CHAOS’s slope against the fold between -0.65 and -0.35	Held	
WADDLE_BOOKS 20261013_063353	a	the books close within 1e-9 in every top-up at k = 1e5, here 1.107e-08	Missed	168
WADDLE_BOOKS 20261013_063353	b	the damper part over 0.8 of the floor at couple 1, here 0.0059	Missed	168
WADDLE_STEP 20261013_073935	a	the no-step floor goes as 1/k within 20 per cent at 2e5 and 4e5	Missed	169
WADDLE_STEP 20261013_073935	b	the spring’s part at 4e5 within 10 per cent of the centrifugal stretch	Held	
QUANT_ZERO 20261013_081905	a	at memory 0 no pace locks at q of 2 to 40	Held	
WADDLE_CATCH 20261013_082013	a	the catch’s shortfall is half to all of the floor at the stiffest pin and finest step	Missed	171
QUANT_STEP 20261013_082454	a	each halving cuts the miss by 8 or more at each pace	Held	
QUANT_STEP 20261013_082454	b	under 1e-10 at 640 steps at each pace	Held	
HALF_BUBBLE 20261013_083609	a	a 2-period orbit of one turn under the 4-period band, its leading multiplier real and below -1, at 0.9995, 1.0005, 1.0015	Held	
HALF_BUBBLE 20261013_083609	b	each end of the 4-period band a doubling, within 2e-4 of 0.9993 and of 1.0019	Held	
BUBBLE_FAR 20261013_084037	a	the multiplier comes back up through -1 between 1.0090 and 1.0100, no Newton failure	Held	
BUBBLE_FAR 20261013_084037	b	the 2-period orbit stays below -1 from 1.0015 to 1.0105	Held	
SIZED_TAP_4 20261013_161615	a	halving the jostling moves the edge out by at least 0.05 at c 2.5 and 3.0	Held	
SPIN_START 20261013_161916	a	started spinning, every pace from 0.9800 to 0.9950 still turning	Missed	172

Run	Label	The prediction, as the run prints it	Verdict	Misses row
SPIN_START 20261013_161916	b	started spinning, near 1/4 at 0.9837 and 0.9838 and near 9/19 at 0.9937 and 0.9938, within 0.01	Missed	172
HALF_PARENT 20261013_163612	a	Claude’s parent: multiplier above 0.9 and split below 1e-4 within 1e-3 of p2	Held	
HALF_PARENT 20261013_163612	b	p2 between 1.0150 and 1.0200	Held	
HALF_PARENT 20261013_163612	B	Bear’s tooth: a peak below 0.9 and a fall of 0.2 or more after it	Missed	174
HALF_PARENT 20261013_163612	c	from rest, every pace 1.0101 to 1.0300 locks at q 2 or 4, and at q 2 above p2 + 5e-4	Missed	174
SPIN_WINDOWS 20261013_163655	a	inside each window at least 8 of 16 turning at the window’s rate	Held	
SPIN_WINDOWS 20261013_163655	b	just outside the 1/4 window all 16 sit	Missed	173
GEAR_TRAIN 20261013_170243	a	a band with p/q between 2/3 and 1	Held	
GEAR_TRAIN 20261013_170243	b	the bands’ p/q rise with the pace	Held	
EAR_COST 20261013_183245	a	the steepest pull held falls as the pack grows, at every look interval	Missed	175
EAR_COST 20261013_183245	b	each pack has a best look interval inside the grid, and looking every half fall time costs at least 0.01	Missed	175
HALF_TOP 20261013_183449	F0	the multiplier nearest +1 passes 1 at the turning point	Held	
HALF_TOP 20261013_183449	a	the fold between 1.0225 and 1.0230, here 1.0229104794, and the square-root law’s slope between 0.45 and 0.55, here +0.4999	Held	
HALF_TOP 20261013_183449	b	the exponent above +0.01 at four or more of 1.0235, 1.0245, 1.0255, 1.0265, 1.0270	Held	
EAR_COST_2 20261013_183913	a	the loss is the looks’ average pull, slope 0.75 to 1.25; here +1.0405	Held	
EAR_COST_2 20261013_183913	b	a whole-fall-time look every 4 costs at least 1.5 times a quarter look every 1	Missed	176
STAIR_WIDTHS 20261013_184033	a	each of 3/4, 4/5, 5/6, 6/7 between 0.35 and 0.6 of the width before it	Held	
STAIR_WIDTHS 20261013_184033	b	1/1 reached by 1.2000, and no step back	Missed	177
WOBBLE_SOFT 20261013_192713	a	the gauge reads the softened pull, within 0.03 at every no-slip setting	Missed	178
WOBBLE_SOFT 20261013_192713	b	at each fresh noise the gauge reaches 0.95 before slips pass 1 per 100	Held	
EAR_COST_3 20261013_192738	a	looks of 4 fall times every 16 lose at least 0.0375	Held	
SECOND_GEAR 20261013_192828	a	at every pace at least two locked fractions, one the step-back’s	Missed	179
CHAOS_RAMP 20261013_192906	a	the rotation number never falls by more than 0.003 from one pace to the next	Missed	180
CHAOS_RAMP 20261013_192906	b	at 1.0272 the rotation number within 0.02 of 2/3	Missed	180
SLIP_PROOF 20261013_194935	a	the slip-proof gauge reads the softened pull wherever slips are under 1 per 100	Missed	181
SLIP_PROOF 20261013_194935	b	without the slips, no early warning at the fresh noises	Missed	182

Run	Label	The prediction, as the run prints it	Verdict	Misses row
STEP_BACK_RAMP 20261013_195010	a	somewhere the rotation number falls by more than 0.003 from one pace to the next	Held	
RAMP_EDGE 20261013_195233	F0	the multiplier nearest +1 passes 1 at the turning point	Missed	183
RAMP_EDGE 20261013_195233	a	a millionth below the edge, from rest, more than 0.02 short of 2/3	Undecided	183
QUIET_TAP 20261013_195754	a	no jostling: c 2.5 and 3.0 keep every rider up to 1.80	Missed	187
QUIET_TAP 20261013_195754	b	no jostling: c 3.5 drops its riders somewhere in 1.20 to 1.65 and holds them somewhere in 1.70 to 2.20	Held	
SLIP_PROOF_2 20261013_200028	a	the slip-proof gauge reads the softened pull wherever slips are under 1 per 100	Missed	185
SLIP_PROOF_2 20261013_200028	b	without the slips, no early warning at the fresh noises	Missed	185
RAMP_EDGE_2 20261013_200133	F0	the multiplier nearest +1 passes 1 at the turning point	Held	
RAMP_EDGE_2 20261013_200133	a	a millionth below the edge, from rest, more than 0.02 short of 2/3	Missed	186
COLUMN_ACCORDION 20261013_222625	a	at time gap 1.0 s, the stumble shrinks with no delay and grows at some delay	Held	
COLUMN_ACCORDION 20261013_222625	b	the shortest safe time gap rises with the delay	Missed	189
FIVE_SEVENTHS 20261013_222744	a	every pace from 1.05775 to 1.05820 locks at 5/7 with least repeat 7	Missed	190
CHAOS_ORDER 20261013_222835	a	each chaotic pace’s spread changes by less than a factor of 2 from no noise to sigma 0.03	Missed	191
CHAOS_ORDER 20261013_222835	b	at sigma 0.03 both locked paces spread more than every chaotic pace	Missed	191
COLUMN_ACCORDION_2 20261013_222950	a	delays to 0.6 s: a time gap of 1.5 s or more keeps the stumble from growing; a fixed distance makes it grow at every delay	Held	
COLUMN_ACCORDION_2 20261013_222950	b	at delays of 0.8 and 1.0 s the stumble grows at every time gap	Held	
MORTAR_SPACING 20261013_224843	a	fixed 5 m: some walker stops, and some 10 m of road holds 5 or more	Held	
MORTAR_SPACING 20261013_224843	b	opening gap: no walker stops, never more than 3 in 10 m, the tail’s range no wider than the leader’s	Held	
BANDED_FIVE 20261013_224945	a	the excursion from 5/7 stays within 2 turns over 30,000 periods at every pace	Held	
BANDED_FIVE 20261013_224945	b	the exponent is above 0.005 per period at every pace	Missed	192
CHAOS_ORDER_2 20261013_225021	a	from sigma 0.001, the nearest locks’ rates move more than every chaotic pace’s	Held	
CHAOS_ORDER_2 20261013_225021	b	a lock breaks at no more noise nearer its edge	Held	
SLIP_PROOF_3 20261013_225419	a	the slip-proof gauge reads the softened pull wherever slips are under 1 per 100	Missed	193
SLIP_PROOF_3 20261013_225419	b	without the slips, no early warning at the fresh noises	Held	
NIGHT_SPACING 20261013_231417	a	by sight at 1 m, some walker stops	Held	
NIGHT_SPACING 20261013_231417	b	by touch at 1 m, no walker stops, none pressed to the floor, the tail’s range within 1.6 m/s	Held	

Run	Label	The prediction, as the run prints it	Verdict	Misses row
ZIGZAG_GAUGE 20261013_231629	a	where the slope is above 0.325 and the fall line steady, slope within 10 per cent and fall line within 5 degrees	Undecided	195
ZIGZAG_GAUGE_2 20261013_231751	a	where the slope is above 0.325 and the fall line steady, slope within 10 per cent and fall line within 5 degrees	Missed	196

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Run	Label	The prediction, as the run prints it	Verdict	Misses row
SPRING_STALL 20261014_063112	a	exactly zero at and above critical damping, above two standard errors at 0.90 and 0.95, at every tune	Held	
CATS_EYES 20261014_063138	a	T 0 with the reading error: none stops, none at the floor, tail within 1.6 m/s	Held	
CATS_EYES 20261014_063138	b	T 0.3 s: some walker stops, with the error and without it	Held	
QUICK_READ 20261014_065420	a	no stop at 0.05 to 0.07 s; some stop at every look-time from 0.11 to 0.15 s	Held	
QUICK_READ 20261014_065420	b	at 0.12 s, walker 15’s strongest wobble within 15 per cent of 2.93 rad/s	Held	
WORD_BACK 20261014_065420	a	fixed distance, the word obeyed at once: the tail dips the leader’s 0.5 m/s within 1 per cent	Held	
WORD_BACK 20261014_065420	b	fixed distance, the word obeyed after the delay: the tail dips more than 100 times the leader’s	Held	
WINDOW_TOUCH_2 20261014_074538	a	each window within 0.2 of a touch by its own image	Undecided	—
WINDOW_TOUCH_2 20261014_074538	b	cycle = 4 x pieces x 3 or 5 for the windows of 12, 20, 24 and 48	Missed	202
WINDOW_TOUCH_2 20261014_074538	c	crossings lock at their least repeat, 0.8 or more; controls 0.2 or less	Undecided	201
WINDOW_TOUCH_3 20261014_080520	a	each window within 0.2 of a touch by its own image	Undecided	—
WINDOW_TOUCH_3 20261014_080520	b	cycle = 4 x pieces x 3 or 5 for the windows of 12, 20, 24 and 48	Missed	—
WINDOW_TOUCH_3 20261014_080520	b2	the band in 4 pieces beside the window of 48	Missed	204
WINDOW_TOUCH_3 20261014_080520	c	touches lock at their least repeat, 0.8 or more; controls 0.2 or less	Missed	203
SMOOTH_DIRT 20261014_084856	a	the extra buzz is the dry law’s; within 10 per cent of 2.1044	Missed	207
SEAM 20261014_090006	s1	top edge minus image 3 positive; reaches nought between -12.0 and -11.9	Held	
SEAM 20261014_090006	s2	the partner above the top edge, meeting it within 0.05 of (s1)’s pace	Held	
SEAM 20261014_090006	s3	Bear’s inner arc on image 3, half or more, twice the shuffle	Missed	205
WINDOW_TOUCH_4 20261014_090037	c4	scanned places lock at their cycle, 0.8 or more; controls 0.2 or less	Missed	208
SEAM_2 20261014_090552	s1	top edge minus image 3 positive; reaches nought between -12.0 and -11.9	Held	
SEAM_2 20261014_090552	s2	the partner above the top edge, meeting it within 0.05 of (s1)’s pace	Held	
SEAM_2 20261014_090552	s3	Bear’s inner arc on image 3, half or more, twice the shuffle	Held	

Run	Label	The prediction, as the run prints it	Verdict	Misses row
CRISIS_BURSTS 20261014_091405	t1	slope between -0.65 and -0.45	Held	
WINDOW_TOUCH_5 20261014_092920	c5	a window at 0.8 or more of the places, 0.2 or less of the controls	Held	
COLUMN_TAIL 20261014_145058	g	every ratio from walker 15 to 29 within 5 per cent of the largest gain, at all three settings	Missed	210
HEAVY_SHAPE 20261014_155553	h1	C/S within 2 per cent of 9/4 at M = 1.0309, 1.0879, 1.1662 and 1.2030	Held	
COLUMN_STEADY 20261014_164637	k1	at each judged setting the 30th walker’s largest swing is more than 100 times the first’s	Held	
COLUMN_STEADY 20261014_164637	k2	at each judged setting every walker’s swing in the last 20 s is under 1e-4 of its own largest	Held	
HEAVY_SHAPE_2 20261014_164913	b1	S within 3 per cent of Bear’s purple at the three judged weights	Missed	211
HEAVY_SHAPE_2 20261014_164913	b2	C/S above 2.35 at M 1.254018 and 1.272465, as Bear’s cyan	Undecided	212
HEAVY_SHAPE_2 20261014_164913	h2	C/S within 2 per cent of 9/4 at the three judged weights	Undecided	212
HEAVY_SHAPE_3 20261014_170837	b2	C/S above 2.35 at M 1.254018 and 1.272465, as Bear’s cyan	Missed	213
HEAVY_SHAPE_3 20261014_170837	h2	C/S within 2 per cent of 9/4 at M 1.254018 and 1.272465	Held	
TYRE_KNEE 20261014_174309	a	the loop’s width at the 25- and 32-degree lanes is their band, within 0.05 m/s	Held	
TYRE_KNEE 20261014_174309	b	one sweep leaves the car at the down-edge lane for 12 m/s; the second leaves no more	Held	
TYRE_KNEE 20261014_174309	c	a 5-second sweep widens the 30-degree loop by more than 0.5 m/s	Held	
STOP_WALL 20261014_180544	w1	the gap at memory 1.2725 is less than half the gap at memory 1	Held	
STOP_WALL 20261014_180544	w2	the line through the four heaviest gaps reaches nought below memory 1.6	Held	
STOP_WALL 20261014_180544	w3	every needle walked past the edge and back ends stopped, at every weight	Held	
RING_JAM 20261014_224934	f	the ring grows exactly where the open column’s largest gain exceeds 1.001	Held	
STOP_WALL_2 20261014_225838	z1	at memory 1.6 the gap is above 0.0005	Missed	214
STOP_WALL_2 20261014_225838	z2	read to 1e-5 the gap falls at every step from memory 1 to 1.2725	Held	
NOISE_EDGE 20261014_232230	n1	the slope within 25 per cent of the barrier	Held	
NOISE_EDGE 20261014_232230	n2	at memory 5 every fallen needle still running at the end	Missed	215

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Run	Label	The prediction, as the run prints it	Verdict	Misses row
STOP_WALL_3 20261015_000651	B1	the gap at memory 1.7998, 1.9998 and 2.1998 between 270 and 345 window widths	Missed	216

Run	Label	The prediction, as the run prints it	Verdict	Misses row
STOP_WALL_3 20261015_000651	B2	the gap still falls from 1.6 to 1.9998, by less than the 0.00077 it fell from 1.2030 to 1.6	Missed	216
STOP_WALL_3 20261015_000651	z3	the gap at memory 1.9998 is more than 0.01: the stop has left the window	Held	
STOP_WALL_3 20261015_000651	h3	C/S at memory 1.9998 within 2 per cent of 9/4	Held	
ROOT_JUMPS 20261015_031511	j1	every jump across Kp = 1 away from the foot is by exactly 1	Held	
ROOT_JUMPS 20261015_031511	j2	every other jump is by an even number	Held	
HEAVY_WINDOW_4 20261015_032553	D1	S at 2.22, 2.37 and 2.52 within 10 per cent of S at 1.62, 1.47 and 1.32	Missed	217
HEAVY_WINDOW_4 20261015_032553	D2	S at 2.52 more than 10 per cent below S at 1.32	Held	
HEAVY_WINDOW_4 20261015_032553	D3	the window there at memory 3.0, C/S within 2 per cent of 9/4	Held	
BELL_LOG 20261015_033341	L1	against ln M, S at 0.3 and 0.4 either side of the least within 5 per cent of each other	Missed	218
HALT_ORDER 20261015_052122	j1	passed man to man: the 30th halted more than 13 s, at least 3 s longer than the head	Held	
HALT_ORDER 20261015_052122	j2	ordered at once: every walker halted 10 s within 0.1 s, no gap below 0.9 m	Held	
STOP_LEAVES 20261015_052943	J1	a jump: some judged gap beyond 0.012, none between 0.006 and 0.012	Held	
STOP_LEAVES 20261015_052943	J2	at memory 1.7598 no edge inside the 0.02 cap	Held	
GAPS_READ 20261015_061859	G1	Bear’s dip: at step 95, memory 1.7598, a stable cycle with q over 2 dividing 200; no return at steps 94 and 96	Held	
GAPS_READ 20261015_061859	G2	Bear’s flickers: the half-turn cycle unstable at steps 164 (1.7248), 147 and 194 (1.7598), stable at the steps before	Missed	219
HALVINGS 20261015_064027	H1	every edge a doubling: the largest multiplier real and negative there	Missed	220
HALVINGS 20261015_064027	H2	both width ratios within 25 per cent of 4.669	Missed	220
HALVINGS 20261015_064027	B1	Bear’s rungs: the two ratios within 25 per cent of each other	Missed	220
HALVINGS_2 20261015_064427	H1	every edge a doubling: the largest multiplier real and negative there	Held	
HALVINGS_2 20261015_064427	H2	both width ratios within 25 per cent of 4.669	Missed	221
HALVINGS_2 20261015_064427	B1	Bear’s rungs: the two ratios within 25 per cent of each other	Held	
BANK_BUMP 20261015_064644	K1	stable minus unstable, the wall counted, is 1 at every speed	Held	
BANK_BUMP 20261015_064644	K2	three seats exactly from 23.80 to 28.69 m/s, one elsewhere	Held	
HALVINGS_3 20261015_064938	H3	the third ratio within 10 per cent of 4.669 and above 4.2353	Held	
GAPS_REREAD 20261015_071452	G1	Bear’s dip: at step 95, memory 1.7598, a stable cycle with q over 2 dividing 200; no return at steps 94 and 96	Held	
GAPS_REREAD 20261015_071452	G2	Bear’s flickers: the half-turn cycle unstable at steps 164 (1.7248), 147 and 194 (1.7598), stable at the steps before	Missed	—

Run	Label	The prediction, as the run prints it	Verdict	Misses row
LADDER_LIGHT 20261015_071735	L1	every edge a doubling, multiplier within 1e-3 of -1	Held	
LADDER_LIGHT 20261015_071735	L2	both ratios within 10 per cent of 3.4280 and 4.2353	Held	
LADDER_LIGHT 20261015_071735	L3	the 4-to-8 edge more than 0.001 deeper than 0.014758	Held	
LEAK_CLOSE 20261015_073039	C1	a jump: some weight between with no stop inside the cap, none with its gap between 0.004 and 0.012	Held	
SNAKE_LONG 20261015_073142	S1	96 links between 3.76 and 4.5; 192 between 4.18 and 5.0	Missed	223
SNAKE_LONG 20261015_073142	S2	both below 6.2832	Held	
SNAKE_LONG 20261015_073142	S3	the gap shrinks by 0.7 to 0.95 a doubling	Held	
WOBBLY_SHUVIT 20261015_074302	W1	the mean reach within 3 per cent of 1/(1 - c)	Held	
WOBBLY_SHUVIT 20261015_074302	W2	the mean squared distance within 5 per cent of the formula	Held	
SWITCH_END 20261015_074716	E1	the switch at or beyond C/S and within 0.03 of it, both weights	Held	
SONG_GAPS 20261015_074750	G1	1 m against 0.5 m, Kolmogorov-Smirnov distance under 0.03	Missed	224
SONG_GAPS 20261015_074750	G2	the first 50 gaps differ for at least 12 of 16 riders	Missed	225
SONG_GAPS_2 20261015_074852	G1b	in whole steps, 1 m against 0.5 m, Kolmogorov-Smirnov distance under 0.03	Held	
SONG_GAPS_2 20261015_074852	G2b	gaps 101 to 150 differ for at least 12 of 16 riders	Held	
LEAK_RATE 20261015_080150	R1	the share fallen never rises with weight	Held	
LEAK_RATE 20261015_080150	R2	some weight from 1.7204 to 1.7219 with none fallen	Held	
LEAK_RATE 20261015_080150	R3	exponential survival, share 0.95 or more, where 16 or more fall in the hold	Held	
SNAKE_768 20261015_151934	T1	at 384 links the shared setting is best, 4.4117 within 1e-4	Missed	226
SNAKE_768 20261015_151934	T2	768 links below 6.2832, the shrink from 384 between 0.888 and 0.96	Missed	227
SNAKE_EXACT 20261015_154247	X1	rises at every doubling, below 6.2832	Held	
SNAKE_EXACT 20261015_154247	X2	the gap’s four log slopes within 0.15 of their mean	Held	
SNAKE_EXACT 20261015_154247	X3	the best start drifts one way only	Held	
RING_KNOBS 20261015_165858	K1	the pair’s advantage peaks at a gap from 0.5 to 1.5, smaller at 0.25 and 3.0, 3.0 between 0.95 and 1.10	Held	
RING_KNOBS 20261015_165858	K2	at gap 1.0 the advantage grows with the core, 0.02 < 0.05 < 0.1 < 0.2	Missed	228
RING_KNOBS 20261015_165858	K3	three rings beat their budget ring by more than two at gap 1.0, a0 0.05	Missed	229

Run	Label	The prediction, as the run prints it	Verdict	Misses row
TRAIN_HOLD 20261015_171853	H1	a clean edge between 0.5 and 0.75	Held	
TRAIN_HOLD 20261015_171853	H2	a holding trio’s advantage rises with the gap up to the edge	Missed	230
TRAIN_HOLD 20261015_171853	H3	every broken trio leaves a pair within 3 radii	Held	
TRAIN_HOLD 20261015_171853	H4	four rings hold only at a smaller gap than three	Held	
SLOPE_EAR 20261015_190935	S1	noise, every 1/4 cycle: the ideal rate sensor at least 0.0816	Held	
SLOPE_EAR 20261015_190935	S2	noise, every 1/16 cycle: the two-look slope below the plain ear	Missed	231
SLOPE_EAR 20261015_190935	S3	noise, every 1/4 cycle: the two-look slope above the plain ear	Held	
FILTER_SLOPE 20261015_195645	F1	1/16 cycle: the four-look fit drifts at least 0.10314 and switches at most 1.815 a cycle	Missed	232
FILTER_SLOPE 20261015_195645	F2	1/4 cycle: the four-look fit drifts below 0.07895	Held	
FLIP_MISTAKES 20261015_195739	D1	DWELL’s share of mistaken flips above SAMPLED’s at every tau	Held	
FLIP_MISTAKES 20261015_195739	D2	DWELL’s wrong-way time above SAMPLED’s at every tau	Held	
FLIP_MISTAKES 20261015_195739	D3	DWELL’s wrong-way time grows with tau	Held	
PILOT_BEAT 20261015_203855	B1	a soft beat locks at 0.9 to 1.1 and not at 0.5 or 2.0	Missed	233
PILOT_BEAT 20261015_203855	B2	a loud beat locks at every pitch from 0.7 to 1.5	Missed	234
PILOT_BEAT 20261015_203855	B3	the loud beat costs net push at every pitch	Held	
BEAT_SWING 20261015_204207	C1	halving the step nearly doubles the switch count	Held	
BEAT_SWING 20261015_204207	C2	by the swing a soft beat locks at 0.9 to 1.1, not at 0.5 or 2.0	Missed	235
BEAT_SWING 20261015_204207	C3	by the swing a loud beat locks from 0.7 to 1.5	Missed	236
FIT_KNOBS 20261015_204319	G1	some setting drifts at least 0.10314 and switches at most 1.815	Missed	237
FIT_KNOBS 20261015_204319	G2	the switching falls as the knee widens, at every number of looks	Held	
PILOT_PRICE 20261015_220321	P1	at the tune, the push per pilot power falls with every step up in loudness	Held	
PILOT_PRICE 20261015_220321	P2	at 0.03 the pilot’s power is greatest at the tune	Missed	238
PILOT_PRICE 20261015_220321	P3	at 0.01 an off-tune whisper, 2.0, wakes more push per power than one at the tune	Held	
PILOT_PRICE 20261015_220321	T1	the lighter bead’s working swing within 10 per cent of its tune	Held	
PILOT_PRICE 20261015_220321	T2	a loud beat sets the lighter bead’s count at the tune	Missed	239

Run	Label	The prediction, as the run prints it	Verdict	Misses row
PAUSE_BEST 20261015_220655	Q1	bill at half the step within 0.9 to 1.15 of the bill, every step, h 0.5	Held	
PAUSE_BEST 20261015_220655	Q2	best price looking every step at h 0.5 to 1.0	Missed	240
PAUSE_BEST 20261015_220655	Q3	best h at 1/16 cycle within one grid step of every step’s	Held	
PAUSE_WIDE 20261015_223744	Z1	the price turns over, its best at h 1.75 to 2.5	Missed	244
PAUSE_WIDE 20261015_223744	Z2	drift at the best price at least 0.08	Missed	245
JUMP_STORE 20261015_223918	J1	clean, the skate pushes more than every jump	Held	
JUMP_STORE 20261015_223918	J2	the triggered jump’s impulse per event rises with h, and the listener’s crest-firing’s falls	Missed	242
JUMP_STORE 20261015_223918	J3	the best jump carries more impulse per event than the noisy listener with the knee	Missed	243
LATCH_EVENT 20261015_232643	L1	the clean latch’s impulse per event rises at every step of h from 0 to 3	Held	
LATCH_EVENT 20261015_232643	L2	at h 2 the latch carries at least 1.3 times the crest-firer’s impulse per event	Held (0.5508 against 0.3637)	
LATCH_EVENT 20261015_232643	L3	where it fires, the spring’s latch carries less than half of 0.3776 per event	Undecided	246
KNEE_CREEP 20261015_232920	K1	noise relaxing over 0.1: last halving within 0.97 to 1.03	Held	
KNEE_CREEP 20261015_232920	K2	noise relaxing over 0.03: last halving within 0.97 to 1.03	Missed	247
KNEE_CREEP 20261015_232920	K3	noise relaxing over 0.03: the first halving’s ratio above the last’s	Missed	248

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Run	Label	The prediction, as the run prints it	Verdict	Misses row
CREEP_LAW 20261016_035058	T1	the traders’ law within 25 per cent at noise times 0.1, 0.03 and 0.01, steps 0.00125 to 0.04	Missed (0.1: True, 0.03: False, 0.01: False)	249
CREEP_LAW 20261016_035058	T2	the halvings’ change falls as the step to a power from 0.4 to 0.6	Missed (power 0.332)	250
CREEP_LAW 20261016_035058	T3	the fall grows with roughness at every noise time	Held	
BRIDGE_COUNT 20261016_044439	B1	within 5 per cent: noise 1.0 every step, noise 0.1 to step 0.01; the plain count short	Missed	251
BRIDGE_COUNT 20261016_044439	B2	noise 0.01 at steps 0.01 to 0.08: more than 10 per cent short	Held	
BRIDGE_COUNT 20261016_044439	B3	the bridge recovers at least half the missed pokes everywhere	Missed	252
RIDER_FLEET 20261016_044707	M4	slowing reaction: no policy beats the calendar by more than 0.02	Missed	253
RIDER_FLEET 20261016_044707	M5	weakening tap: the taps rise before the fall, and the learned power reminder beats the calendar by more than 0.02	Missed	254

Run	Label	The prediction, as the run prints it	Verdict	Misses row
WORN_RIDER 20261016_065645	W1	the effort rises above 1.5 times before the fall	Held	
WORN_RIDER 20261016_065645	W2	the fixed effort level beats the calendar by more than 0.02	Held	
WORN_RIDER 20261016_065645	W3	the riders who work harder live at least 10 per cent longer	Held	
HITCH_LEGS 20261016_065650	a	one stride turns the trunk back 15 to 60 degrees	Missed (+6.573)	255
HITCH_LEGS 20261016_065650	b	whole arm circles hold off within 5 per cent of circles times 13.87/60	Missed	256
BELL_CLOCK 20261016_072241	C1	the water’s three clocks restore the distance per squeeze within 5 per cent	Missed	257
BELL_CLOCK 20261016_072241	C2	the three clocks come closer than period and memory alone	Held	
BAND_BRIDGE 20261016_072612	F1	within 3 per cent: noise 1.0 to step 0.08, noise 0.1 to step 0.01; the plain flips short by more than 3 per cent	Missed	258
BAND_BRIDGE 20261016_072612	F2	noise 0.03 and 0.01, steps up to the noise’s time: at least half the missed flips recovered	Held	
BAND_BRIDGE 20261016_072612	F3	the fastest watcher’s band bridge within 1 per cent of its flips, noise 1.0 and 0.1	Missed	259
NEVER_BLINK 20261016_090509	N1	the traders’ law from the reference predicts each slower watcher’s fall within 10 per cent, steps to a tenth of the noise’s time	Held	
NEVER_BLINK 20261016_090509	N2	the band bridge within 1 per cent of the never-blinking estimate: noise 1.0 to step 0.08, noise 0.1 to step 0.01	Held	
NEVER_BLINK 20261016_090509	N3	the two never-blinking estimates agree within 0.3 per cent	Held	
TRUCK_ROAD 20261016_090656	c1	the slow lane runs to the held fold at every mass	Held	
TRUCK_ROAD 20261016_090656	c2	masses 1.5 to 3.0: the passing lane only below the held fold	Held	
TRUCK_ROAD 20261016_090656	c3	mass 1.0: the passing lane above the held fold	Held	
TRUCK_ROAD 20261016_090656	c4	mass 0.75: no passing lane	Missed	261
TRUCK_ROAD 20261016_090656	c5	masses 1.0 to 3.0: more runners caught by gear 2 than by gear 1 where both lanes hold	Missed	262
HEAT_RING 20261016_174342	h1	the brick stays above zero at L/2 pi 1.05, 1.10, 1.20	Held	
HEAT_RING 20261016_174342	h1-b	the brick dips below zero at L/2 pi 0.25	Held	
HEAT_RING 20261016_174342	h2	the brick’s deepest dip never grows with L	Held	
HEAT_RING 20261016_174342	h3	heat’s speed limit: below 1e-9 until d - 1.5, above 1e-6 by d + 0.5, Fourier’s above 1e-6 by d - 1.5	Held	
LIGHT_LANES 20261016_174648	P1	Bear’s prediction: at mass 1.0 the passing lane fills the road from 1.00 to 1.80, no runner keeps running	Missed	263
LIGHT_LANES 20261016_174648	P2	mass 0.75: a passing band at least 0.05 wide, starting at or below 1.06, ending below 1.80	Held	

Run	Label	The prediction, as the run prints it	Verdict	Misses row
LIGHT_LANES 20261016_174648	P3	the passing lane climbs as the rider gets lighter: lowest at 0.5 above lowest at 0.75, above 0.985	Held	
LIGHT_LANES 20261016_174648	P4	mass 1.25: open road at some pace	Held	
ARM_SHARE 20261016_182910	A1	one circle turns the trunk back 15.439 degrees within 0.001	Held (15.438876)	
ARM_SHARE 20261016_182910	A2	the reading gives the three shares within 1e-6	Held	
ARM_SHARE 20261016_182910	A3	the cruder form within 5 per cent of the fixed share	Missed (1.0748)	265
ARM_SHARE 20261016_182910	A4	half the arm mass: turn and fixed share each 0.45 to 0.60 of themselves	Missed	266
BELL_RATIO 20261016_183422	B1	S 1.41 and 2: the gain within 5 per cent of 1 at every memory	Missed	267
BELL_RATIO 20261016_183422	B2	the gain at S 0.5 above that at S 2, at every memory	Missed	268
BELL_RATIO 20261016_183422	B3	S sets the gain more than the memory does	Missed	269
EQUAL_CATCH 20261016_192548	Q1	the learned wobble reminder’s gain within twice its sd of nought	Held	
EQUAL_CATCH 20261016_192548	Q2	the learned power reminder’s gain within twice its sd of nought	Held	
EQUAL_CATCH 20261016_192548	Q3	the fixed power level wastes more than the calendar at every seed	Held	
BELL_BALANCE 20261016_193628	T1	Bear’s prediction: the period’s spread exceeds the memory’s, both above 0.05	Held	
BELL_BALANCE 20261016_193628	T2	the gain at period 6 above that at 1.5, at every memory	Undecided	—
BELL_BALANCE 20261016_193628	T3	memory 12 further from 1 than memory 3, at every period	Held	
BELL_BALANCE 20261016_193628	T4	the reopening kicks above plain water’s at both balance points	Held	
LAG_METER 20261016_203014	W1	the learned lag reminder beats the calendar by more than twice its sd	Missed	273
LAG_METER 20261016_203014	W2	its mean gain at least 0.10 of a life	Missed	274
LAG_METER 20261016_203014	W3	the fixed lag level beats the calendar by more than twice its sd	Missed	275
SWING_GRID 20261016_203014	G1	push per watt x pace = 1 within 1e-4 everywhere	Held	
SWING_GRID 20261016_203014	G2	locked push = pace; above the edge below 0.99: the most push at the edge	Held	
SWING_GRID 20261016_203014	G3	the running branch buys less than the locked, inside the band	Held	
SWING_GRID 20261016_203014	G4	at memory 5, less than half	Held	
BRICK_LINE 20261016_205112	B1	the brick dips wherever its first mode does	Held	
BRICK_LINE 20261016_205112	B2	from 0.80 to 0.95, within 35 per cent of 4/pi times the first mode’s dip	Missed	278

Run	Label	The prediction, as the run prints it	Verdict	Misses row
RESTED_LAG 20261016_205504	R1	the learned rested-lag reminder beats the calendar by more than twice its sd	Held	
RESTED_LAG 20261016_205504	R2	its mean gain at least 0.10 of a life	Held	
RESTED_LAG 20261016_205504	R3	the fixed rested-lag level beats the calendar by more than twice its sd	Held	
RING_WIDTH 20261016_213858	R1	first order: the width doubles with A within 20 per cent	Held	
RING_WIDTH 20261016_213858	R2	the lane’s middle moves less than 0.02 from A 0.075 to 0.3	Held	
STRIDE_MAP 20261016_213858	S1	hip 1.40 turns 1.6 to 2.4 times hip 0.70, knee 1.57	Held	
STRIDE_MAP 20261016_213858	S2	the biggest stride turns less than one arm circle, 15.44	Missed	279
STRIDE_MAP 20261016_213858	S3	three human strides hold off under 0.35; none with hip up to 1.05 holds off half	Missed	280
BELL_CORNER 20261016_224449	C1	every crowded point settles	Missed	282
BELL_CORNER 20261016_224449	C2	every settled crowded gain between 0 and 1	Undecided	—
BELL_CORNER 20261016_224449	C3	the gain at period 6 above that at 1.5, at every memory	Undecided	—
BELL_CORNER 20261016_224449	C4	memory 12 further from 1 than memory 3, at periods 1.5 and 2.12	Undecided	—
SOUP_JUMP_3 20261016_225440	Q1	the knock-off falls with the margin, by at least 0.5 from 0.02 to 0.4	Held	
SOUP_JUMP_3 20261016_225440	Q2	the stronger left comb knocks at least as many off	Held	
SOUP_JUMP_3 20261016_225440	Q3	by 500, above half at margin 0.02 and below half at 0.2	Held	

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Run	Label	The prediction, as the run prints it	Verdict	Misses row
ARMS_LEGS 20261017_012927	W1	together within 10 per cent of the two alone, added	Held	
ARMS_LEGS 20261017_012927	W2	the shape part at least 0.7 of the share at every pairing	Held	
ROUGH_FLOOR 20261017_013514	F1	a trap’s spread, median 0.5 to 0.9 of the mean, on every floor	Undecided	287
ROUGH_FLOOR 20261017_013514	F2	runs past three means at most a tenth	Held	
FACE_ALONG 20261017_070629	F1	broad ends farther than broad middle	Held	
FACE_ALONG 20261017_070629	F2	broad head and broad tail the same, to 1e-4 of the travel	Missed	294
FACE_ALONG 20261017_070629	F3	every body the same way as the round body	Held	
FACE_ALONG 20261017_070629	F4	broad ends buy at least half of what all broad buys	Missed	295
DIVE_CUE 20261017_082454	D1	the event lands nearer at every spin error	Held	
DIVE_CUE 20261017_082454	D2	both errors in proportion, and R = B/K to 1e-4	Held	
DIVE_CUE 20261017_082454	D3	R between 0.25 and 0.5	Held	

Run	Label	The prediction, as the run prints it	Verdict	Misses row
SNAKE_LONGEST 20261017_155052	X4	the best within 0.03 of 5.60	Held	
SNAKE_LONGEST 20261017_155052	X5	the gap’s slope between -0.50 and -0.45	Held	
SNAKE_LONGEST 20261017_155052	X6	the start drifts on toward 0 and the width narrows	Held	
SPRING_FLOOR 20261017_160023	S1	the floored latch stays awake to the end	Missed	298
SPRING_FLOOR 20261017_160023	S2	push per event under half of 0.3776	Held	
LANE_EDGES 20261017_164501	L1	width x sqrt(mass) within 10 per cent of its mean at every mass	Held	
LANE_EDGES 20261017_164501	L1b	the width’s log slope between -0.6 and -0.4	Held (slope -0.5135)	
LANE_EDGES 20261017_164501	L2	both edges fall at every step of mass	Held	
LANE_EDGES 20261017_164501	L3	the strip straddles the slow lane’s top at 1.0 and 1.25 only	Held	
TONG_DULLIN 20261017_173718	T1	21.0647 and 5.2888 within 0.001	Missed	306
TONG_DULLIN 20261017_173718	T2	the extremes where theirs are	Missed	307
TONG_DULLIN 20261017_173718	T3	B’s largest size and place	Missed	—
TONG_DULLIN 20261017_173718	T4	spin 120: {1, 0.8593}, 0.0189 rad	Missed	—
TONG_DULLIN 20261017_173718	T5	spin 30: {0.9818, 0.3158}, 0.2327 rad	Missed	—
TONG_DULLIN 20261017_173718	T6	the corner from 193.65	Missed	307
TONG_DULLIN 20261017_173718	T7	no spin: {0, 0}	Held	
TONG_DULLIN_READINGS 20261017_174547	R1	exactly one reading gives both inertias	Missed	309
TONG_DULLIN_READINGS 20261017_174547	R2	its words in the tool’s own header	Undecided (no single matching reading)	—
TONG_DULLIN_SCAN 20261017_181953	S1	exactly one entry brings both inertias	Missed	312
TONG_DULLIN_SCAN 20261017_181953	S2	that entry also brings the extremes and the field	Undecided	—
TONG_DULLIN_SCAN 20261017_181953	S3	the torso’s moment moves the field’s peak by more than 1 per cent	Held	
TONG_DULLIN_SCAN 20261017_181953	S4	s_in leaves 1 between spins 30.5 and 31.5	Held	
TONG_DULLIN_KNEE 20261017_182629	K1	both inertias within 0.01	Held	
TONG_DULLIN_KNEE 20261017_182629	K2	the extremes’ places within 0.005	Held	
TONG_DULLIN_KNEE 20261017_182629	K3	the field’s peak and place	Held	
TONG_DULLIN_KNEE 20261017_182629	K4	spin 120	Held	
TONG_DULLIN_KNEE 20261017_182629	K5	spin 30	Held	
TONG_DULLIN_KNEE 20261017_182629	K6	the corner’s spin within 2	Held	
TONG_DULLIN_KNEE 20261017_182629	K7	no spin: (0, 0)	Held	

Run	Label	The prediction, as the run prints it	Verdict	Misses row
HALT_THRESHOLD 20261017_184055	l1	a valley: the earliest arrival strictly inside the read	Missed	318
HALT_THRESHOLD 20261017_184055	l2	the best threshold below 6 m	Missed	319
HALT_THRESHOLD 20261017_184055	l3	below the start-up stretch no arrival; 1 m above it, arrival	Missed	321
LIMIT_AUDIT 20261017_190938	A1	some number moves by more than 1 per cent	Held	
LIMIT_AUDIT 20261017_190938	A2	HALT_ORDER moves most; the most: HALT_ORDER	Held	
LIMIT_AUDIT 20261017_190938	A3	no verdict changes	Missed	323
TWO_HANDS 20261017_203811	W1	test 1: flat within 3 per cent across the sweep	Held	
TWO_HANDS 20261017_203811	W2	test 2: the drift shrinks as the waves multiply	Held	
TWO_HANDS 20261017_203811	W3	test 3: the half-turn separates the hands on a skewed soup, not on an even one	Missed	329
TWO_HANDS 20261017_203811	W4	test 4: each hand moves, the product holds	Held	
TWO_HANDS 20261017_203811	W5	the noise check	Held	
TWO_HANDS 20261017_203811	W6	Rice at 96 waves within 1.5 per cent	Held	
COIL_SNAKE 20261017_204030	C1	the closed form to 1e-6	Held	
COIL_SNAKE 20261017_204030	C2	the best pitch angles 54.7 and 67.8	Held	
COIL_SNAKE 20261017_204030	C3	the log coil as an even coil of its mean radius, within 2 per cent	Held	
TWO_HANDS_2 20261017_204131	V1	the half-turn separates the hands on a skewed soup	Missed	330
TWO_HANDS_2 20261017_204131	V2	the even soup’s mirror agrees at 192 soups	Held	
TWO_HANDS_BAND 20261017_210543	B1	the count barely moves with the band	Missed	334
CAT_TWIST 20261017_231928	L1	the twist-free trick does not land it on its feet	Held	
CAT_TWIST 20261017_231928	L2	some constant bend from 90 to 135 turns a twist-free cat over in one full circuit	Held	
SLING_RUNDOWN 20261017_232234	S1	the share crosses one half between 70 and 80 degrees	Held	
SLING_RUNDOWN 20261017_232234	S2	the best cut in degrees rises at every degree and stays below 45	Held	
SLING_RUNDOWN 20261017_232234	S3	as an angle, the best cut lies between 25 and 45 degrees from 40 to 90	Held	

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Run	Label	The prediction, as the run prints it	Verdict	Misses row
SAGNAC_LOOP 20261018_032138	G1	the shape does not matter	Held	
SAGNAC_LOOP 20261018_032138	G2	the axis’s place does not matter	Held	
SAGNAC_LOOP 20261018_032138	G3	a figure-eight counts its lobes with signs	Held	
SAGNAC_LOOP 20261018_032138	G4	a tilt of 60 degrees halves it	Held	
TRAIN_LIFETIME 20261018_034211	T1	a sharp cliff: all hold to 0.650, all break from 0.720	Missed	347
TRAIN_LIFETIME 20261018_034211	T2	a third of neighbours differ by more than a factor of 2	Held	
TRAIN_LIFETIME 20261018_034211	T3	an exponential lifetime, p above 0.05	Held	
TRAIN_LIFETIME 20261018_034211	T4	every break leaves a pair	Held	

Run	Label	The prediction, as the run prints it	Verdict	Misses row
CAT_SWEET 20261018_034953	K1	within 0.01 degrees of straight down at some bend from 75 to 85, radius 0.3	Held	
CAT_SWEET 20261018_034953	K2	on its feet at some bend at each radius	Held	
CAT_SWEET 20261018_034953	K3	the best bend moves more than 5 degrees from 0.15 to 0.45	Held	
CAT_SWEET 20261018_034953	U1	unequal halves: the toy’s half turn more than 1 degree off	Held	
CAT_SWEET 20261018_034953	U2	unequal halves: the twist-free circuit tilted more than 0.1 degrees	Held	
CAT_MIRROR 20261018_041123	Z1	the pairing exact on the 1-degree grid	Held	
CAT_MIRROR 20261018_041123	Z2	a law of thickness: the radius found at 90 matches at 60 and 120	Held	
CAT_MIRROR 20261018_041123	Z3	the law halves the radius	Held	
CAT_MIRROR 20261018_041123	Z4	no clock	Held	
CAT_MIRROR 20261018_041123	Z5	the same angle about different axes	Held	
TRAIN_TIMING 20261018_082139	D1	between 1.1 and 1.6	Held	
TRAIN_TIMING 20261018_082139	D2	the long shot, above 2	Missed	361
COIL_PUMP 20261018_082424	P1	a loop moves it, and the reversed loop moves it back	Missed	358
COIL_PUMP 20261018_082424	P2	halving the loop quarters the travel	Missed	359
COIL_PUMP 20261018_082424	P3	the long shot: no spin left per cycle	Held	
COIL_WAVE 20261018_082544	W1	the wave moves it, the reversed wave moves it back	Held	
COIL_WAVE 20261018_082544	W2	the travel goes as the square of the wave	Held	
COIL_WAVE 20261018_082544	W3	it spins as it goes	Held	
COIL_WAVE 20261018_082544	W4	the long shot: it swims the way its wave runs	Missed	362
TRAIN_CLOSE 20261018_103739	C1	0.25 to 0.40 all hold	Held	
TRAIN_CLOSE 20261018_103739	C2	some start from 0.41 to 0.59 breaks	Held	
TRAIN_CLOSE 20261018_103739	C3	one of 0.60, 0.62, 0.63 breaks	Held	
TRAIN_CLOSE 20261018_103739	C4	every break leaves a pair	Held	
CAT_MIRROR_PROOF 20261018_174959	X1	the half trick is not paired	Held	
CAT_MIRROR_PROOF 20261018_174959	X2	the long shot: unequal halves, both halved, still paired	Missed	363
CAT_MIRROR_PROOF 20261018_174959	X3	the second long shot: another thickness map with the hinge at 0.3 L	Missed	364
TRAIN_EDGE 20261018_181641	E1	the edge is no wall: some gap both holds and breaks	Held	
TRAIN_EDGE 20261018_181641	E2	more break in the upper half than the lower	Held	
TRAIN_EDGE 20261018_181641	E3	the throws at 0.65 differ by a median factor from 1.33 to 3	Held	
TRAIN_EDGE 20261018_181641	E4	the long shot: their lifetime within 20 per cent of 862.03	Missed	380
TRAIN_NUDGE 20261018_192138	N1	a 1e-12 nudge moves two of three breaks by over a tenth	Held	
TRAIN_NUDGE 20261018_192138	N2	so does a 1e-15 nudge	Held	
TRAIN_NUDGE 20261018_192138	N3	the long shot: at 1e-15 the break at 0.70 moves under 1 per cent	Missed	370
SAIL 20261018_202451	S1	faster than the true wind on some heading	Missed	373
SAIL 20261018_202451	S2	fastest between 90 and 135 degrees	Missed	374
SAIL 20261018_202451	S3	best upwind progress between 40 and 55 degrees	Held	

Run	Label	The prediction, as the run prints it	Verdict	Misses row
SAIL 20261018_202451	S4	the long shot: best downwind progress beats the wind	Missed	375
SAIL_HULL 20261018_202650	H1	a clean dinghy’s hull beats the wind	Held	
SAIL_HULL 20261018_202650	H2	a cleaner hull sails fastest further off the wind, at every step	Missed	376
SAIL_HULL 20261018_202650	H3	the long shot: the cleanest hull’s best downwind progress beats the wind	Missed	377

19 October

Run	Label	The prediction, as the run prints it	Verdict	Misses row
TRAIN_DRIFT 20261019_034330	D1	the integrator’s: tighter cuts the drift by 5 or more at both gaps	Held	
TRAIN_DRIFT 20261019_034330	D2	a creep at both gaps	Held	
TRAIN_DRIFT 20261019_034330	D3	within a factor of 3 of the second computer’s at 0.25	Held	
TRAIN_DRIFT 20261019_034330	D4	the long shot: a cut of more than 30 at 0.25	Missed	392
KNEE_SIZE 20261019_035236	K1	the square root, power 0.4 to 0.6 at noise times 1.0 and 0.1	Held	
KNEE_SIZE 20261019_035236	K2	the law within 15 per cent wherever the step is at most a tenth of the memory	Missed	389
KNEE_SIZE 20261019_035236	K3	the long shot: within 15 per cent at noise time 0.03, step 0.01	Missed	390
KNEE_FINE 20261019_040109	J1	the ruler’s: within 15 per cent at 0.00125 and 0.0025 on the finer grid	Missed	393
KNEE_FINE 20261019_040109	J2	the long shot: every step to 0.04 within 5 per cent	Missed	394
KNEE_REF 20261019_042520	R1	the scatter’s: under two standard errors above the law at 0.00125 and 0.0025, old reference	Held	
KNEE_REF 20261019_042520	R2	the long shot: every step 0.000625 to 0.04 within 5 per cent, new reference	Missed	397
ERROR_BARS 20261019_162750	B1	at most one claim besides KNEE_FINE’s changes its wording	Missed (6 change)	400
ERROR_BARS 20261019_162750	B2	Kramers’ four first ratios each within two standard errors of 1	Held (+0.00, -0.11, -0.02, -0.92)	
ERROR_BARS 20261019_162750	B3	the long shot: every noisy number within two standard errors of its law or bar	Missed	401
KNEE_READER 20261019_163948	Q1	the fitted reader within two bootstrap standard errors of the law at every step, both references	Held	
KNEE_READER 20261019_163948	Q2	the staircase reader’s bootstrap under 0.7 of its jackknife at 0.0025	Held	
KNEE_READER 20261019_163948	Q3	the long shot: fitted, new reference, every step within 5 per cent	Missed	406

20 October

Run	Label	The prediction, as the run prints it	Verdict	Misses row
CAT_MIRROR_PATHS 20261020_034021	P1	Claude’s lean: the pairing breaks at b0 90, a 10, by more than 1e-6	Missed	416
CAT_MIRROR_PATHS 20261020_034021	P2	the gap grows more than fourfold from a 5 to a 20	Undecided	—
CAT_MONTGOMERY 20261020_035025	B1	the twist-free cat is Montgomery’s, within 1e-8	Held	
CAT_MONTGOMERY 20261020_035025	B2	the toy is Montgomery’s with its halves spinning alike, within 1e-8	Held	
CAT_MONTGOMERY 20261020_035025	B3	hinge 0.3, the matched radius pairs them within 1e-9	Held	
CAT_MONTGOMERY 20261020_035025	B4	the flip bend is CAT_SWEET’s sweet spot	Held	
CAT_MIRROR_RATES 20261020_041152	R1	hinge at the ends, three sizes, within 1e-13	Held	
CAT_MIRROR_RATES 20261020_041152	R2	hinge moved in, matched radius, within 1e-13	Held	
ZETA_FIT 20261020_055322	Z1	BETA read back within two standard errors of 0.5826	Missed	421
ZETA_FIT 20261020_055322	Z2	0.5 more than two standard errors away	Held	
ZETA_LIMIT 20261020_055829	L1	the limit within two standard errors of 0.5826	Held	
ZETA_LIMIT 20261020_055829	L2	BETA rises with r	Held	
ZETA_LIMIT 20261020_060256	L1	the limit within two standard errors of 0.5826	Held	
ZETA_LIMIT 20261020_060256	L2	BETA rises with r	Held	
CAT_RTB 20261020_063617	R1	the flown cat turns their formula’s turn within 1e-8, both hinges	Held	
CAT_RTB 20261020_063617	R2	their printed figures are their formula rounded	Held	
CAT_RTB 20261020_063617	R3	their formula is G at bend 180 - a within 1e-12	Held	
KRAMERS_EXACT 20261020_073520	K1	pooled within two standard errors of 1 against the exact rate	Held	
KRAMERS_EXACT 20261020_073520	K1b	at most 2 of 15 rows two or more errors from 1	Missed	425
KRAMERS_EXACT 20261020_073520	K2	exact over Kramers averages 0.85 to 0.95	Held	
SLIP_SPREAD 20261020_075155	F1	the Fano factor from 0.9 to 1.3 in at least 15 of 18 rows	Held	
SLIP_SPREAD 20261020_075635	F1	the Fano factor from 0.9 to 1.3 in at least 15 of 18 rows	Held	
SLIP_SPREAD 20261020_075635	K1bp	its words in the tool’s own header	Missed	—
SLIP_SPREAD 20261020_075635	K1p	its words in the tool’s own header	Held	
BENCH_LOOP 20261020_081001	L1	E’s four runs home in the other motion, N’s four and W’s two in the same	Missed	—
BENCH_LOOP 20261020_081001	L3	jumps + winding odd exactly when home changed; continuous changes only on pace 1 below the tip	Held	
SLIP_RHYTHM 20261020_083426	R1	long-run Fano within two errors of the exact one, and below 1, in each row	Missed	430
SLIP_RHYTHM 20261020_083426	R2	exact Fano lower at sigma 0.1 than 0.2 than 0.3	Held	

Run	Label	The prediction, as the run prints it	Verdict	Misses row
SLIP_RHYTHM 20261020_083426	R3	long-run Fano within two errors of the intervals’ squared spread	Held	
SLIP_RHYTHM 20261020_083426	R4	under half of 0.0952 of intervals shorter than a tenth of the mean	Missed	431
SLIP_RECHECK 20261020_083908	Q1	at most 2 of 15 rows two or more errors out	Held	
SLIP_RECHECK 20261020_083908	Q2	the three far rows within two errors of 1	Held	
SLIP_RECHECK 20261020_083908	Q3	pooled within two errors of 1	Missed	432
SLIP_RECHECK 20261020_083908	Q4	pooled less 0.9885 within two combined errors of zero	Held	
SLIP_RHYTHM_2 20261020_084919	S1	intervals’ squared spread within two errors of the exact Fano factor in each row	Missed	433
SLIP_RHYTHM_2 20261020_084919	S2	long-run Fano within two errors of the exact one in each row	Held	
SLIP_RHYTHM_2 20261020_084919	S3	at one mean interval more than two errors below the exact one in each row	Missed	434
SLIP_START 20261020_090154	T1	from 50 to 500 within two errors of the exact rate	Held	
SLIP_START 20261020_090154	T2	the first 50 more than two errors below it	Held	
SLIP_START 20261020_090154	T3	lateness 2 to 20 time units	Held	
SLIP_RHYTHM_3 20261020_090717	V1	the reader within two errors of the exact Fano factor in each row	Held	
SLIP_RHYTHM_3 20261020_090717	V2	the window pool below the reader by 0.0005 to 0.005 in each row	Missed	435
RHYTHM_BAND 20261020_154009	N1	at 3 units the Fano factor rises with the noise at every step	Held	
RHYTHM_BAND 20261020_154009	N2	noise 0.025 and 0.05 within 0.001	Held	
RHYTHM_BAND 20261020_154009	N3	noise 0.5 more than 0.005 above noise 0.05	Held	
RHYTHM_BAND 20261020_154009	G1	gamma within 0.01 of -2.476 at noise 0.025	Held	
RHYTHM_BAND 20261020_154009	C1	pace 0.95: near 1, a minimum below 0.9, above 1 at the strongest noise	Held	
SLIP_BAND 20261020_161039	H1	each noise within two errors of the exact Fano factor	Missed	437
SLIP_BAND 20261020_161039	H2	noise 0.7 below 0.3 and 1.5 by more than two combined errors	Held	
SLIP_BAND_2 20261020_161539	K1	noise 1.5 within two errors of the exact 0.50967	Held	
BELL_SWITCH 20261020_161902	D1	at each memory and step the eight starts do not all agree	Held	
BELL_SWITCH 20261020_161902	D2	the share forward moves by at most 2 of 8 between the steps	Missed	438
BELL_SWITCH 20261020_161902	D3	softened core: both answers at each memory	Missed	439
TRAIN_FLOOR 20261020_224239	F1	the edge thins downward: fewer than 8 of 10 break at 0.550 and 0.555	Missed	444
TRAIN_FLOOR 20261020_224239	F2	the breaking reaches below 0.55	Held	
TRAIN_FLOOR 20261020_224239	F3	the long shot: none break at 0.540 or tighter, and 6 or more of 10 at 0.550 and 0.555	Missed	445

21 October

Run	Label	The prediction, as the run prints it	Verdict	Misses row
BELL_SOFT_MAP 20261021_000736	F1	all 25 points settle	Missed	440
BELL_SOFT_MAP 20261021_000736	F2	at the 20 thin-settled points the softened gain within 0.05	Missed	441

Run	Label	The prediction, as the run prints it	Verdict	Misses row
BELL_SOFT_MAP 20261021_000736	F3	the gain crosses 1 at every memory, between periods 2.5 and 3.5	Missed	442
BELL_SOFT_MAP 20261021_000736	F4	Bear's (T1): the period's spread exceeds the memory's, both above 0.05	Held	
BELL_SOFT_MAP 20261021_000736	F5	the gain at period 6 above that at 1.5, at every memory	Held	
BELL_SOFT_SCALE 20261021_054731	S3	judged at widths [0.01, 0.05]: 0.01	Held ; 0.05: HELD	
BELL_SOFT_SCALE 20261021_054731	S1	the width a knob: more than half move by more than 0.05 from 0.0225 to 0.05	Held	
BELL_SOFT_SCALE 20261021_054731	S2	more than half closer to one at 0.05	Missed	443
BELL_SOFT_SCALE 20261021_054731	S4	the corner settles at 0.05	Held	
TRAIN_FLOOR_2 20261021_083613	G1	the floor is real: none of the ten long flights breaks by 100,000	Missed	447
TRAIN_FLOOR_2 20261021_083613	G2	the breaking reaches below 0.530: at least one of the twenty from 0.526 to 0.529 breaks	Held	
TRAIN_FLOOR_2 20261021_083613	G3	the long shot: none at 0.526 to 0.528, at least 2 of 5 at 0.529	Missed	448
BELL_WIDTH_LADDER 20261021_151833	W1	the fast point's peak at a length from 0.125 to 0.175	Held	
BELL_WIDTH_LADDER 20261021_151833	W2	at memory 12 slow beats fast at every width where both settle, at least eight	Missed	446
BELL_WIDTH_LADDER 20261021_151833	W3	the slow point within 0.10 of 1.30 from length 0.05 to 0.30	Held	

22 October

Run	Label	The prediction, as the run prints it	Verdict	Misses row
BELL_SPIKES 20261022_053740	P1	the fast spike a narrow hill, no neighbours more than 0.08 apart	Undecided	—
BELL_SPIKES 20261022_053740	P2	the memory-12 spike a narrow hill too	Held	
BELL_SPIKES 20261022_053740	P3	both spikes broad, within 0.05 of their own value	Undecided	—
TRAIN_DOOR 20261022_055043	D1	the door moves with the core, by more than 0.01 each way	Missed	449
TRAIN_DOOR 20261022_055043	D2	at each core the door's core over radius within 0.05 of 0.52	Missed	450
TRAIN_DOOR 20261022_055043	D3	the long shot: at core 0.05 every break through 0.200 to 0.220	Missed	451
TRAIN_EDGE_MAP 20261022_060414	E1	the door holds at core 0.06, 0.205 to 0.215	Missed	452
TRAIN_EDGE_MAP 20261022_060414	E2	some core-0.04 ring below 0.215	Missed	453
TRAIN_EDGE_MAP 20261022_060414	E3	each core's edge inside the grid	Held	
TRAIN_EDGE_MAP 20261022_060414	E4	the long shot: the same first breaking gap at all three cores	Missed	454
TRAIN_EDGE_DOOR 20261022_070632	K1	every core-0.06 break through a squeeze to 0.200 to 0.220	Missed	455
TRAIN_EDGE_DOOR 20261022_070632	K2	core 0.04 reaches the door, some ring below 0.215	Missed	456

Run	Label	The prediction, as the run prints it	Verdict	Misses row
TRAIN_EDGE_DOOR 20261022_070632	K3	the two edges within 0.015	Held	
TRAIN_EDGE_DOOR 20261022_070632	K4	the long shot: core 0.04’s breaks through a door of their own, within 0.02	Missed	457
TRAIN_LONG_EDGE 20261022_175421	L1	every late core-0.06 break through a squeeze to 0.200 to 0.220	Missed	463
TRAIN_LONG_EDGE 20261022_175421	L2	a thin-core holder squeezes as deep as a late thin-core breaker	Held	
TRAIN_LONG_EDGE 20261022_175421	L3	the three cores’ lifetimes at 0.530 within a factor of 2	Missed	464
TRAIN_LONG_EDGE 20261022_175421	L4	the long shot: core 0.04 outlives core 0.06 more than twice	Held	
TRAIN_DOOR_LINE 20261022_222935	D1	at 0.045 and 0.055 every break within 0.004 of the line’s door	Held	
TRAIN_DOOR_LINE 20261022_222935	D2	every core-0.04 break within 200 of a scaled deep squeeze	Held	
TRAIN_DOOR_LINE 20261022_222935	D3	core 0.04 pooled outlives TRAIN_LONG_EDGE’s core 0.06 more than twice	Held	
TRAIN_DOOR_LINE 20261022_222935	D4	the long shot: every break 120 to 190 after its last scaled squeeze, at all three cores	Held	

23 October

Run	Label	The prediction, as the run prints it	Verdict	Misses row
CAT_MIRROR_FAMILY 20261023_001232	F1	the map fails on the third direction, by more than 1e-6	Held	
CAT_MIRROR_FAMILY 20261023_001232	F2	the matching rate varies by more than 1e-6	Held	
CAT_MIRROR_FAMILY 20261023_001232	F3	the long shot: the same at every shape within 1e-12	Missed	460
CAT_MIRROR_CENTRES 20261023_001618	C1	at the centres the whole family mirrors, within 1e-13	Held	
CAT_MIRROR_CENTRES 20261023_001618	C2	the failure shrinks steadily as the hinge moves in	Held	
CAT_MIRROR_CENTRES 20261023_001618	C3	the long shot: the failure goes as A squared, within 10 per cent	Missed	462

28 October

Run	Label	The prediction, as the run prints it	Verdict	Misses row
TRAIN_FUSE_CLOCK 20261028_101647	F1	most of the fuse is the trip: median parting over fuse below 0.4	Missed	467
TRAIN_FUSE_CLOCK 20261028_101647	F2	at 0.035 and 0.065 every break within 0.004 of the line’s door	Held	
TRAIN_FUSE_CLOCK 20261028_101647	F3	the long shot: the passages from ts to the parting within 10 per cent of their median, at all five cores	Missed	469
TRAIN_FUSE_CLOCK 20261028_101647	F4	at 0.035 and 0.065 every break 120 to 190 after its last scaled squeeze	Held	
TRAIN_BEAT 20261028_212034	B1	every deep squeeze is a threading: every scaled squeeze within 0.5 of one	Held	
TRAIN_BEAT 20261028_212034	B2	exactly one threading between the last scaled squeeze and the break	Missed	471
TRAIN_BEAT 20261028_212034	B3	the beat after ts within 1 of the beat before it	Held	
TRAIN_BEAT 20261028_212034	B4	the long shot: the break 35 to 70 after its last threading	Missed	472
TRAIN_BEAT 20261028_212034	B5	a break is for good: no third ring back within 3 radii in 1,000	Missed	473

29 October

Run	Label	The prediction, as the run prints it	Verdict	Misses row
TRAIN_PARTING_EDGE 20261029_070800	P1	read to the parting for good, the lifetime under 1.10 times that to the first 5-radius spread	Missed	480
TRAIN_PARTING_EDGE 20261029_070800	P2	no ring comes back from 20 radii or more	Held	
TRAIN_PARTING_EDGE 20261029_070800	P3	the first two returns’ farthest reaches within 2 per cent, in every train with two or more	Held	
TRAIN_PARTING_EDGE 20261029_070800	P4	the long shot: an even number of returns before every parting that had any	Held	
TRAIN_PARTING 20261029_112955	G1	no ring comes back from 20 radii: every returning swing’s farthest reach under 20	Held	
TRAIN_PARTING 20261029_112955	G2	read to the parting of a ring that is gone, the lifetime at least 1.3 times the lifetime to the first 5-radius spread	Missed	477
TRAIN_PARTING 20261029_112955	G3	the long shot: some ring comes back twice or more	Held	
TRAIN_PARTING_EDGE_2 20261029_211416	Q1	read to the parting for good, the lifetime at least 1.3 times that to the first 5-radius spread	Missed	485
TRAIN_PARTING_EDGE_2 20261029_211416	Q2	no ring comes back from 14 radii or more	Missed	486
TRAIN_PARTING_EDGE_2 20261029_211416	Q3	the first two returns’ farthest reaches within 2 per cent, in every train with two or more	Held	
TRAIN_PARTING_EDGE_2 20261029_211416	Q4	an even number of returns before every parting that had any	Held	

31 October

Run	Label	The prediction, as the run prints it	Verdict	Misses row
RAMP_SPEED 20261031_232517	S1	a runner coming in at rest is caught by gear 1 at every push of every mass	Missed	551
RAMP_SPEED 20261031_232517	S2	at every mass, gear 2 catches a larger share at 0.6 times the pace than at 1.0	Missed	552
RAMP_SPEED 20261031_232517	S3	at every mass, gear 2’s share at 1.1 to 1.6 times the pace within 0.10 of its share at 1.0	Missed	552
RAMP_SPEED 20261031_232517	S4	at mass 3.0, some speed coming in has more than half caught by gear 2	Held	
HALT_TRADE 20261031_235932	T1	one edge: never arriving up to some threshold, arriving above it	Held	
HALT_TRADE 20261031_235932	T2	the highest threshold that never arrives is 2.5 m or more	Missed	554
HALT_TRADE 20261031_235932	T3	between 3 and 4 m the longest gap jumps from under 6 m to over 20 m between neighbours	Missed	555
HALT_TRADE 20261031_235932	T4	a threshold keeps contact and brings the tail home within 1,000 s	Missed	555

1 November

Run	Label	The prediction, as the run prints it	Verdict	Misses row
COMB_STRETCH 20261101_002907	W1	the rule foretells at least 90 per cent of the 5,120 fates	Missed	556
COMB_STRETCH 20261101_002907	W2	at least 80 per cent of the foretold falls come within 30 time units after the foretold time	Missed	556
COMB_STRETCH 20261101_002907	W3	every all-or-none soup foretold exactly	Missed	556
COMB_STRETCH_B 20261101_003130	B1	the twin misses no fall	Held	
COMB_STRETCH_B 20261101_003130	B2	the rule and its twin disagree on at most 20 per cent	Missed	557
COMB_STRETCH_B 20261101_003130	B3	the midpoint foretells at least 95 per cent	Missed	557
SNAKE_RACE 20261101_175747	R1	the hardest pull uses at least twice the thriftiest torque, every length	Held	
SNAKE_BEAT 20261101_190003	B1	a quarter off, one beat longer the same and half a beat different, sixteen comparisons at 0.75 and 1.25	Held	
SNAKE_BEAT 20261101_190003	B2	the beat is 1/|1 - k| at 0.875 and 0.9, eight comparisons	Held	
SNAKE_BEAT 20261101_190003	B3	in tune, a straight line in the length within 1e-9, at 12.0, 13.6, 16.0 and 20.8	Held	
FREE_SWIM 20261101_222356	E2	the hardest start plus the summed lateness is arctan(W_e), within 1e-6 degrees, all 45; largest apart 5.3e+01	Missed	574
FREE_SWIM 20261101_222356	E3	the held pull falls as W_e rises, every length	Held	
FREE_SWIM 20261101_222356	E4	the re-tuned best falls as W_e rises, every length	Held	
FREE_SWIM_2 20261101_222740	Q1	the body’s answer a quarter turn behind the water at the hardest pull, within 1e-6 degrees, all 30; largest apart 1.0e-10	Held	

2 November

Run	Label	The prediction, as the run prints it	Verdict	Misses row
RING_PATHS 20261102_013613	R2	every pair passes at least three times, its beat within 1 per cent	Held	
RING_PATHS 20261102_013613	R3	every ring of the two holding trios takes the lead	Held	
RING_PATHS 20261102_013613	R4	the pair a broken train leaves passes at least twice after the break	Held	
ROUGH_COUNT 20261102_173941	C1	the count levels off on every floor at both paces, F(1,024) over F(512) below 1.2	Missed	578
ROUGH_LONG 20261102_200436	L2	the count levels off given windows long enough, a slow trap	Held	

3 November

Run	Label	The prediction, as the run prints it	Verdict	Misses row
BRICK_QUARTER 20261103_180126	Q1	at the quarter the dip moves by more than 1 per cent from 400 to 800 cells	Held	
BRICK_QUARTER 20261103_180126	Q2	at the quarter the turning points grow by at least half with each doubling	Held	
BRICK_FINE 20261103_180438	E1	the echo between 1.45 and 1.52 lengths at every length to 0.88, both finer grids	Held	
BRICK_FINE 20261103_180438	E2	the echo’s end at 0.87, 0.88 or 0.89 on both finer grids	Held	
BRICK_FINE 20261103_180438	E3	the brick still stops at the line on both finer grids	Held	

4 November

Run	Label	The prediction, as the run prints it	Verdict	Misses row
BLOB_RING 20261104_170523	B1	one wave turns the ring but does not carry it, m = 2, 3, 4	Held	
BLOB_RING 20261104_170523	B2	the two waves together carry it, above 1e-4 radii a lap	Held	
BLOB_RING 20261104_170523	B3	the two waves’ travel as e squared, 3.8 to 4.2 times	Missed	590
BLOB_RING 20261104_170523	B4	farther at grip 5 than 2 than 1.5	Held	
BLOB_SMALL 20261104_170747	S1	from 0.0125 to 0.025 the travel grows 3.8 to 4.2 times	Held	
BLOB_SMALL 20261104_170747	S2	the growth climbs toward 4 as the height shrinks	Held	

5 November

Run	Label	The prediction, as the run prints it	Verdict	Misses row
LEAN_DRIFT 20261105_025051	D1	the late share above the early by more than two errors: guesses grown safer	Missed	592
RING_AREA 20261105_032546	A1	the excess ranks with the loop’s area at Spearman 0.9 or more	Missed	594
RING_AREA 20261105_032546	A2	the excess is the area times one number, every pair within 10 per cent of the median ratio	Missed	595
RING_SPLIT 20261105_050121	S1	within each core the own part rises and the partner’s part falls at every step of the gap	Held	
RING_SPLIT 20261105_050121	S2	the partner’s part ranks with the area swept per unit time at 0.9 or more	Held	
SWIM_RIDGE 20261105_145417	W1	C-ray laps at 4 to 12 all forward, S-ray laps at 4 and 6 backward	Held	
SWIM_RIDGE 20261105_145417	W2	C-ray travel falls at every step from 0 to 12, and at 12 below 0.9 of at 4	Held	
SWIM_RIDGE 20261105_145417	W3	the straight body’s lap at grip 1,000,000 between 795 and 815 forward	Missed	598

7 November

Run	Label	The prediction, as the run prints it	Verdict	Misses row
OWED_BARS 20261107_023832	O1	noise time 0.1: two standard errors of the agreement exceed 0.004 per cent	Held	
OWED_BARS 20261107_023832	O2	both agreements within two standard errors of zero	Held	
OWED_BARS 20261107_023832	B1	the rate sensor above the slope on every path	Held	
OWED_BARS 20261107_023832	B2	the noise costs the slope more than the sensor, by two standard errors	Held	
OWED_BARS 20261107_023832	Q1	the edge not the same at every draw	Held	
OWED_BARS 20261107_023832	Q2	the edge’s standard deviation under 0.01 s	Held	
OWED_BARS 20261107_023832	G1	two standard errors of the gap past 0.034 at a judged setting	Held	
OWED_BARS 20261107_023832	G2	noise 0.4: at most one judged gap two standard errors above zero	Missed	—
CUT_WIDE 20261107_033035	W1	noise 0.4: each judged gap moves by less than one of its standard errors when the cut doubles	Held	
CUT_WIDE 20261107_033035	W2	the largest judged gap stays above 0.03 when the cut doubles	Held	
COIL_CHEAP 20261107_060011	B1	rho 6, a body of gamma0: cheapest between 22.21 and 35.26	Held	
COIL_CHEAP 20261107_060011	B2	rho 2, a body of gamma0: cheapest within 0.5 of 35.26	Held	
COIL_CHEAP 20261107_060011	L1	the growing coil: cheapest within 1 degree of 35.26	Held	
QUADRATURE_ACROSS 20261107_080953	P1	the rider’s best within 3 degrees of 90	Held	
QUADRATURE_ACROSS 20261107_080953	W1	the wing’s best at twist 30 within 2 degrees of 90	Missed	616
QUADRATURE_ACROSS 20261107_080953	W2	farther from 90 at twist 45 than at 5	Held	
QUADRATURE_ACROSS 20261107_080953	Q1	all five within 5 degrees of the rule’s gap	Missed	616

8 November

Run	Label	The prediction, as the run prints it	Verdict	Misses row
COLUMN_WALK 20261108_234842	W1	all 10 lines agree at this tool’s first run of record	Held	
COLUMN_WALK 20261108_234842	W2	passed man to man at a fixed metre, the thirtieth man’s first halt more than 15 s after the head’s	Missed	628

9 November

Run	Label	The prediction, as the run prints it	Verdict	Misses row
CLOCK_SHIFT_30 20261109_034237	C1	the ratios climb to delta from below	Held	
CLOCK_SHIFT_30 20261109_034237	C2	r(16-32-64) within 0.01 of delta	Missed	632
THIRD_AXIS 20261109_170732	T1	its words in the tool’s own header	Missed	—
THIRD_AXIS 20261109_170732	T1b	its words in the tool’s own header	Held	
THIRD_AXIS 20261109_170732	T2	its words in the tool’s own header	Held	
THIRD_AXIS 20261109_170732	Z1b	its words in the tool’s own header	Held	
THIRD_AXIS 20261109_170732	Z2	its words in the tool’s own header	Missed	—
THIRD_AXIS_2 20261109_181451	F	its words in the tool’s own header	Held	
THIRD_AXIS_2 20261109_181451	P127	its words in the tool’s own header	Missed	—
THIRD_AXIS_2 20261109_181451	D	its words in the tool’s own header	Missed	640
THIRD_AXIS_3 20261109_191727	P	its words in the tool’s own header	Missed	—
THIRD_AXIS_3 20261109_191727	W	its words in the tool’s own header	Missed	—
THIRD_AXIS_3 20261109_191727	P126	its words in the tool’s own header	Missed	—
THIRD_AXIS_3 20261109_191727	P128	its words in the tool’s own header	Held	
THIRD_AXIS_3 20261109_191727	P129	its words in the tool’s own header	Missed	—

Part 2. Every other check a run made

Beside its predictions, a run printed checks of other kinds, 1,015 lines in all: the rivals written to win if a prediction was wrong, the checks planted in each tool, the toy pages’ engine checked against the runs, and the book builder’s own checks. Each kind has its table, in the order the runs were made.

The rivals

183 lines: written to win if the prediction beside it was wrong: a rival that held means its prediction missed, and a rival can miss beside a prediction that also missed, when neither idea was right.

Run	Label	The check, as the run prints it	Verdict	Misses row
HEAT_RING 20261016_174342	h1r	the rival: it dips below zero there	Missed	
LIGHT_LANES 20261016_174648	P1r	the rival: at mass 1.0 the passing lane ends below 1.80 and open road begins above it	Held	
BELL_RATIO 20261016_183422	B3r	the memory sets it at least as much	Held	
EQUAL_CATCH 20261016_192548	Q1r	more than twice its sd above nought	Missed	
BELL_BALANCE 20261016_193628	T1r	the memory’s spread at least the period’s	Missed	
LAG_METER 20261016_203014	W1r	within twice its sd of nought	Held	
BRICK_LINE 20261016_205112	B1r	the brick stops sooner	Missed	
RESTED_LAG 20261016_205504	R1r	within twice its sd of nought	Missed	
RING_WIDTH 20261016_213858	R1r	second order: both ratios 3 or more	Missed	

Run	Label	The check, as the run prints it	Verdict	Misses row
STRIDE_MAP 20261016_213858	S3r	some stride with hip up to 1.05 holds off half	Held	
BELL_CORNER 20261016_224449	C1r	at least one does not	Held	
SOUP_JUMP_3 20261016_225440	Q1r	the same within 0.1 at every margin	Missed	
ARMS_LEGS 20261017_012927	W1r	some pairing departs by more than 25 per cent	Missed	
ROUGH_FLOOR 20261017_013514	F1r	Bear’s sandpile: a rough floor’s median below 0.4 of its mean	Missed	
EXPLORER_ENGINE_CHECK 20261017_063123	W1r	over 500 ms	Missed	
EXPLORER_ENGINE_CHECK 20261017_063123	W2r	at 1,500 steps at least one leaves it	Missed	
FACE_ALONG 20261017_070629	F1r	broad middle farther than broad ends	Missed	
EXPLORER_ENGINE_CHECK 20261017_070749	W1r	over 500 ms	Missed	
EXPLORER_ENGINE_CHECK 20261017_070749	W2r	at 1,500 steps at least one leaves it	Missed	
DIVE_CUE 20261017_082454	D1r	the clock lands nearer at some spin error	Missed	
SNAKE_LONGEST 20261017_155052	X5r	it flattens, above -0.43	Missed	
EXPLORER_DIVE_CHECK 20261017_155902	W3r	over 500 ms	Missed	
EXPLORER_ENGINE_CHECK 20261017_155903	W1r	over 500 ms	Missed	
EXPLORER_ENGINE_CHECK 20261017_155903	W2r	at 1,500 steps at least one leaves it	Missed	
SPRING_FLOOR 20261017_160023	S2r	some setting carries at least half	Missed	
EXPLORER_ROUTINE_CHECK 20261017_164432	G1r	the rival: more than 30 degrees	Missed	
EXPLORER_ROUTINE_CHECK 20261017_164432	W4r	over 500 ms	Missed	
LANE_EDGES 20261017_164501	L1r	the rival: outside	Missed	
EXPLORER_ENGINE_CHECK 20261017_165820	W1r	over 500 ms	Missed	
EXPLORER_ENGINE_CHECK 20261017_165820	W2r	at 1,500 steps at least one leaves it	Missed	
EXPLORER_DIVE_CHECK 20261017_165821	W3r	over 500 ms	Missed	
TONG_DULLIN 20261017_173718	T1r	either more than 0.1 away	Held	
TONG_DULLIN_READINGS 20261017_174547	R1r	none does	Held	
TONG_DULLIN_SCAN 20261017_181953	S1r	none does	Missed	
TONG_DULLIN_KNEE 20261017_182629	K1r	either more than 0.05 away	Missed	
EXPLORER_CAT_CHECK 20261017_183736	W4r	more than 0.01 away	Missed	
EXPLORER_CAT_CHECK 20261017_183736	W5r	over 100 ms	Missed	
EXPLORER_ENGINE_CHECK 20261017_183749	W1r	over 500 ms	Missed	
EXPLORER_ENGINE_CHECK 20261017_183749	W2r	at 1,500 steps at least one leaves it	Missed	
EXPLORER_DIVE_CHECK 20261017_183750	W3r	over 500 ms	Missed	
EXPLORER_ROUTINE_CHECK 20261017_183751	G1r	the rival: more than 30 degrees	Missed	
EXPLORER_ROUTINE_CHECK 20261017_183751	W4r	over 500 ms	Missed	
HALT_THRESHOLD 20261017_184055	l1r	the lowest threshold best	Missed	
LIMIT_AUDIT 20261017_190938	A1r	none moves by more than 0.1 per cent	Missed	
LIMIT_AUDIT 20261017_190938	A3r	at least one changes	Held	
TWO_HANDS 20261017_203811	W1r	more than 10 per cent somewhere	Missed	

Run	Label	The check, as the run prints it	Verdict	Misses row
TWO_HANDS 20261017_203811	W2r	the three drifts agree	Missed	
COIL_SNAKE 20261017_204030	C3r	more than 10 per cent away	Missed	
TWO_HANDS_2 20261017_204131	V1r	they agree	Missed	
TWO_HANDS_2 20261017_204131	V2r	beyond three standard errors	Missed	
TWO_HANDS_BAND 20261017_210543	B1r	a small band cuts the count by over 1 per cent	Missed	
EXPLORER_CAT_ROOM_CHECK 20261017_211325	P5r	they differ by more than 0.01 somewhere	Missed	
EXPLORER_CAT_ROOM_CHECK 20261017_211325	W6r	over 150 ms	Missed	
EXPLORER_ENGINE_CHECK 20261017_211710	W1r	over 500 ms	Missed	
EXPLORER_ENGINE_CHECK 20261017_211710	W2r	at 1,500 steps at least one leaves it	Missed	
EXPLORER_DIVE_CHECK 20261017_211712	W3r	over 500 ms	Missed	
EXPLORER_ROUTINE_CHECK 20261017_211712	G1r	the rival: more than 30 degrees	Missed	
EXPLORER_ROUTINE_CHECK 20261017_211712	W4r	over 500 ms	Missed	
EXPLORER_CAT_CHECK 20261017_211715	W4r	more than 0.01 away	Missed	
EXPLORER_CAT_CHECK 20261017_211715	W5r	over 100 ms	Missed	
CAT_TWIST 20261017_231928	L1r	it does	Missed	
CAT_TWIST 20261017_231928	L2r	none does	Missed	
SLING_RUNDOWN 20261017_232234	S2r	it passes 45	Missed	
EXPLORER_CAT_TWISTFREE_CHECK 20261018_031532	Q1r	outside	Missed	
EXPLORER_CAT_TWISTFREE_CHECK 20261018_031532	W7r	over 150 ms	Missed	
EXPLORER_ENGINE_CHECK 20261018_031535	W1r	over 500 ms	Missed	
EXPLORER_ENGINE_CHECK 20261018_031535	W2r	at 1,500 steps at least one leaves it	Missed	
EXPLORER_DIVE_CHECK 20261018_031538	W3r	over 500 ms	Missed	
EXPLORER_ROUTINE_CHECK 20261018_031539	G1r	the rival: more than 30 degrees	Missed	
EXPLORER_ROUTINE_CHECK 20261018_031539	W4r	over 500 ms	Missed	
EXPLORER_CAT_CHECK 20261018_031543	W4r	more than 0.01 away	Missed	
EXPLORER_CAT_CHECK 20261018_031543	W5r	over 100 ms	Missed	
EXPLORER_CAT_ROOM_CHECK 20261018_031545	P5r	they differ by more than 0.01 somewhere	Missed	
EXPLORER_CAT_ROOM_CHECK 20261018_031545	W6r	over 150 ms	Missed	
SAGNAC_LOOP 20261018_032138	G1r	shapes differ by over 1 per cent	Missed	
TRAIN_LIFETIME 20261018_034211	T1r	some start at 0.750 or above holds	Missed	
TRAIN_LIFETIME 20261018_034211	T3r	p below 0.01	Missed	
CAT_SWEET 20261018_034953	K1r	the nearest more than 0.1 away	Missed	
CAT_SWEET 20261018_034953	K2r	radius 0.15 at none	Missed	
CAT_SWEET 20261018_034953	U1r	within 1e-6	Missed	

Run	Label	The check, as the run prints it	Verdict	Misses row
CAT_SWEET 20261018_034953	U2r	level within 1e-6	Missed	
CAT_MIRROR 20261018_041123	Z1r	parts by more than 1e-6	Missed	
CAT_MIRROR 20261018_041123	Z2r	no such radius	Missed	
CAT_MIRROR 20261018_041123	Z3r	off by more than 1e-3	Missed	
TRAIN_TIMING 20261018_082139	D1r	below 1.05	Missed	
TRAIN_CLOSE 20261018_103739	C1r	one of them breaks	Missed	
TRAIN_CLOSE 20261018_103739	C2r	all hold	Missed	
TRAIN_CLOSE 20261018_103739	C3r	all three hold	Missed	
TRAIN_DRIFT 20261019_034330	D1r	the rival: the rings’ own, a cut of less than 2 at 0.25	Missed	
KNEE_SIZE 20261019_035236	K1r	the rival: first order, power 0.8 or more at 1.0	Missed	
KNEE_FINE 20261019_040109	J1r	the rival, the law’s: 15 per cent or more above at both	Held	
KNEE_REF 20261019_042520	R1r	the rival, real and the law’s: two or more at both, against both references	Missed	
ERROR_BARS 20261019_162750	B1r	the rival: two or more change	Held	
KNEE_READER 20261019_163948	Q1r	the rival: the excess survives, new reference, 0.0025, two or more	Missed	
CAT_MIRROR_PATHS 20261020_034021	P1r	the rival: within 1e-9 on every path	Held	
CAT_MONTGOMERY 20261020_035025	B2r	it misses by more than 1e-6	Missed	
CAT_MONTGOMERY 20261020_035025	B3r	they part by more than 1e-6	Missed	
CAT_MIRROR_RATES 20261020_041152	R2r	fails by more than 1e-6	Missed	
ZETA_FIT 20261020_055322	Z1r	BETA more than two standard errors from 0.5826	Held	
ZETA_LIMIT 20261020_055829	L1r	more than two away	Missed	
ZETA_LIMIT 20261020_060256	L1r	more than two away	Missed	
KRAMERS_EXACT 20261020_073520	K1r	a shortfall remains	Missed	
SLIP_SPREAD 20261020_075635	K1bpr	its words in the tool’s own header	Held	
BENCH_LOOP 20261020_081001	L3r	some run breaks the count	Missed	
SLIP_RHYTHM 20261020_083426	R1r	within two errors of 1 in at least two rows	Missed	
SLIP_RECHECK 20261020_083908	Q1r	3 or more	Missed	
SLIP_RHYTHM_2 20261020_084919	S1r	more than two errors in at least one row	Held	
SLIP_START 20261020_090154	T1r	more than two errors below it	Missed	
SLIP_RHYTHM_3 20261020_090717	V1r	more than two errors in at least one row	Missed	
RHYTHM_BAND 20261020_154009	N3r	within 0.005	Missed	
RHYTHM_BAND 20261020_154009	C1r	no minimum below 0.9	Missed	
SLIP_BAND 20261020_161039	H1r	at least one more than two errors from it	Held	
SLIP_BAND_2 20261020_161539	K1r	more than two errors from it	Missed	
BELL_SWITCH 20261020_161902	D1r	at each memory and step all eight agree	Missed	
BELL_SWITCH 20261020_161902	D3r	softened core: all eight agree at both memories	Held	
TRAIN_FLOOR 20261020_224239	F1r	the rival: the edge is a comb, a breaking gap tighter than an all-hold gap	Missed	
BELL_SOFT_MAP 20261021_000736	F1r	at least one does not	Held	

Run	Label	The check, as the run prints it	Verdict	Misses row
BELL_SOFT_SCALE 20261021_054731	S1r	a plateau: half or fewer	Missed	
TRAIN_FLOOR_2 20261021_083613	G1r	the rival: at least one breaks	Held	
BELL_WIDTH_LADDER 20261021_151833	W1r	the rival: the peak outside that band	Missed	
BELL_SPIKES 20261022_053740	P1r	the rival: a switch	Undecided	
TRAIN_DOOR 20261022_055043	D1r	the rival: the door is the train’s	Held	
TRAIN_EDGE_MAP 20261022_060414	E2r	the rival: none	Held	
TRAIN_EDGE_DOOR 20261022_070632	K2r	the rival: none does	Held	
TRAIN_LONG_EDGE 20261022_175421	L1r	the rival: one lies outside	Held	
TRAIN_DOOR_LINE 20261022_222935	D1r	the rival: some break more than 0.008 off the line	Missed	
CAT_MIRROR_FAMILY 20261023_001232	F1r	the rival: it holds within 1e-13	Missed	
CAT_MIRROR_CENTRES 20261023_001618	C1r	the rival: it fails there by more than 1e-6	Missed	
TRAIN_FUSE_CLOCK 20261028_101647	F1r	the rival: the train waits whole, that median above 0.7	Held	
TRAIN_FUSE_CLOCK 20261028_101647	F2r	the rival: some break more than 0.008 off the line	Missed	
TRAIN_BEAT 20261028_212034	B1r	the rival: some scaled squeeze more than 2 from every threading	Missed	
TRAIN_BEAT 20261028_212034	B5r	the rival: some third ring comes back within 3 radii	Held	
TRAIN_PARTING_EDGE 20261029_070800	P1r	the rival: at least 1.3 times	Held	
TRAIN_PARTING 20261029_112955	G1r	the rival: some ring comes back from 20 radii or more	Missed	
TRAIN_PARTING_EDGE_2 20261029_211416	Q1r	the rival: under 1.15 times	Held	
COMB_STRETCH 20261101_002907	W1r	the pointwise rule foretells at least as many	Missed	
EXPLORER_TRUCK_CHECK 20261101_015149	W8r	over 300 ms	Missed	
EXPLORER_LATCH_CHECK 20261101_042950	W9r	at least one does not	Missed	
SNAKE_RACE 20261101_175747	R1r	somewhere less than twice	Missed	
SNAKE_BEAT 20261101_190003	B1r	one fails	Missed	
SNAKE_BEAT 20261101_190003	B2r	one fails	Missed	
SNAKE_BEAT 20261101_190003	B3r	one does not	Missed	
FREE_SWIM 20261101_222356	E2r	somewhere not	Held	
FREE_SWIM 20261101_222356	E3r	somewhere it rises	Missed	
FREE_SWIM 20261101_222356	E4r	somewhere it rises	Missed	
FREE_SWIM_2 20261101_222740	Q1r	one case off by more	Missed	
RING_PATHS 20261102_013613	R2r	its words in the tool’s own header	Missed	
RING_PATHS 20261102_013613	R3r	its words in the tool’s own header	Missed	
RING_PATHS 20261102_013613	R4r	its words in the tool’s own header	Missed	
ROUGH_COUNT 20261102_173941	C1r	Bear’s sandpile: on some rough floor it is 1.2 or more	Held	
ROUGH_LONG 20261102_200436	L2r	Bear’s sandpile: it keeps growing	Missed	
BRICK_QUARTER 20261103_180126	Q1r	by 1 per cent or less	Missed	
BRICK_QUARTER 20261103_180126	Q2r	they do not	Missed	
BRICK_FINE 20261103_180438	E1r	it leaves the band	Missed	

Run	Label	The check, as the run prints it	Verdict	Misses row
BRICK_FINE 20261103_180438	E3r	it rings past the line	Missed	
BLOB_RING 20261104_170523	B1r	one wave carries it	Missed	
BLOB_RING 20261104_170523	B2r	they do not	Missed	
BLOB_SMALL 20261104_170747	S1r	it does not	Missed	
LEAN_DRIFT 20261105_025051	D1r	the late share below the early by more than two errors: the bench grew bolder	Missed	
RING_AREA 20261105_032546	A1r	it ranks higher with the loop’s length than with its area	Missed	
RING_SPLIT 20261105_050121	S1r	in at least three cores the partner’s part is largest away from the smallest gap; it is in 0	Missed	
RING_SPLIT 20261105_050121	S2r	it ranks higher with the pass rate	Held	
SWIM_RIDGE 20261105_145417	W1r	C-ray laps at 8, 10 and 12 backward	Missed	
SWIM_RIDGE 20261105_145417	W2r	travel at 12 at least 0.9 of at 4	Missed	
OWED_BARS 20261107_023832	O2r	noise time 1.0: the agreement two or more standard errors from zero, a bias	Missed	
OWED_BARS 20261107_023832	B2r	the noise costs both alike, within two standard errors	Missed	
OWED_BARS 20261107_023832	Q1r	the edge 0.13 s at all eight draws	Missed	
OWED_BARS 20261107_023832	G2r	noise 0.4: three or more judged gaps two standard errors above zero, a bias	Held	
CUT_WIDE 20261107_033035	W1r	noise 0.4, pace 0.65: the gap falls by two or more of its standard errors, the cut’s bias	Missed	
COIL_CHEAP 20261107_060011	B1r	rho 6, a body of gamma0: cheapest above 35.26	Missed	
COIL_CHEAP 20261107_060011	L1r	the growing coil: cheapest more than 3 degrees from 35.26	Missed	
QUADRATURE_ACROSS 20261107_080953	W1r	the rival: more than 5 degrees from 90	Held	
COLUMN_WALK 20261108_234842	W1r	at least one line differs at its printed digits	Missed	
COLUMN_WALK 20261108_234842	W2r	within 15 s	Held	
CLOCK_SHIFT_30 20261109_034237	C1r	the ratios alternate about delta	Missed	
THIRD_AXIS 20261109_170732	T1r	its words in the tool’s own header	Held	
THIRD_AXIS 20261109_170732	T2r	its words in the tool’s own header	Missed	
THIRD_AXIS 20261109_170732	Z2r	its words in the tool’s own header	Held	
THIRD_AXIS_2 20261109_181451	Fr	its words in the tool’s own header	Missed	
THIRD_AXIS_2 20261109_181451	P127r	its words in the tool’s own header	Held	
THIRD_AXIS_2 20261109_181451	Dr	its words in the tool’s own header	Missed	
THIRD_AXIS_3 20261109_191727	Pr	its words in the tool’s own header	Held	
THIRD_AXIS_3 20261109_191727	Wr	its words in the tool’s own header	Held	

The planted checks

572 lines: the checks planted in each tool, which it had to pass before its numbers were used, with the checks each tool’s own docstring calls fidelity or rig checks; every one that failed was a fault or a limit of the tool, caught.

Run	Label	The check, as the run prints it	Verdict	Misses row
PULSED_SCISSORS_LONG 20261005_170134	LH0	slip equal True, largest exponent difference 0.000e+00	Held	

Run	Label	The check, as the run prints it	Verdict	Misses row
GAP_DIAL_2 20261005_172939	GD2-F	40 parent paces compared; differing: none	Held	
POCKET_HOLD 20261005_195333	PK-F	counts at 700 and 10,000: [(6, 6), (3, 3)] (6, 6) and (3, 3)	Held	
ROOM_NEEDLE_2 20261006_014758	RN2-F	onsets at rooms 0.00625 and 0.00442, ROOM-NEEDLE against this run (grid index; None = none): [(3, 3), (6, 6), (0, 0), (2, 2), (0, 0), (0, 0), (4, 4), (6, 6), (1, 1), (3, 3), (0, 0), (0, 0)]	Held	
SWAP_RIDER_BAND 20261006_041001	SW-F	p_h(1) 0.9643271245 (0.964327165); gap-dial band per unit c 0.17495 (0.17495)	Missed	—
FERRY_ROPE 20261006_045826	FR-0	[np.float64(1.0143), np.float64(1.0104), np.float64(1.0031)] against [1.0143, 1.0104, 1.0031]	Held	
NEVER_BLIND 20261006_045833	NB-0	its words in the tool’s own header	Held	
NEVER_BLIND_2 20261006_050123	NB2-0	its words in the tool’s own header	Missed ([np.float64(1.0), np.float64(1.0), np.float64(1.0), np.float64(0.188), np.float64(0.0), np.float64(0.0), np.float64(0.0), np.float64(0.0), np.float64(0.0), np.float64(0.0), np.float64(0.0)])	—
POCKET_GEARS 20261006_052012	PG-F	slipping at 10,000: 6 and 3 (6 and 3)	Held	
SWAP_STATE 20261006_052545	SS-0	every needle leaves gear 1 within 0.003 above its held fold	Missed	—
SWAP_STATE_2 20261006_052914	SS2-F	F repeats SWAP_STATE’s runs for masses 1.0 and 2.0	Missed	—
SIZED_TAP 20261006_060952	ST-0	its words in the tool’s own header	Held (NB-0 line same; control series 13 of 13 delays the same)	
SIZED_TAP_2 20261006_061606	ST2-F	c 2.0, look-ahead 1.0 repeats SIZED_TAP’s shares and wobbles at all 13 delays	Held	
SIZED_TAP_3 20261006_061940	ST3-F	c 2.5 and 3.0, look-ahead 1.0, delay 1.4 repeat SIZED_TAP_2	Held	
RUN_MAP 20261006_071157	RM-F	mass 0.75 at 1.64, all 8 running	Held	
RUN_STROBE 20261006_170149	RS-F	the table equals run 20261006_071157’s, 53 lines	Held	
RUN_STROBE 20261006_170149	RS-0	its words in the tool’s own header	Missed	—
RUN_STROBE_2 20261006_174220	RS2-F	fidelity: largest difference from RUN_STROBE’s saved strobes, L 1.2e-07, V 7.4e-09	Held	
STROBE_NEWTON 20261006_184350	N0	its words in the tool’s own header	Held	
GROOVES 20261006_193519	G0	its words in the tool’s own header	Held	
CAT_3D 20261006_201322	C0	its words in the tool’s own header	Held	
GROOVES_2 20261006_202753	H0	slopes 2.2017 and 1.2764 against the reader’s 2.2017 and 1.2764	Held	
GROOVES_3 20261006_204201	J0	its words in the tool’s own header	Held	
CASCADE 20261006_211853	F0	its words in the tool’s own header	Held	

Run	Label	The check, as the run prints it	Verdict	Misses row
CASCADE_2 20261006_213511	G0	doublings 1.6893422593 and 1.6893164568 against CASCADE’s 1.6893422593 and 1.6893164568	Held	
GROOVES_6 20261006_215909	Y0	every 30-digit fold within 1e-10 of its double fold	Held	
GROOVES_10 20261007_002739	F0	6/5 from its loop 3.996105e-05 and 10/7 2.6116567e-12 against 3.996105e-05 and 2.6116567e-12	Held	
SWING_LISTEN 20261007_013201	F0	its words in the tool’s own header	Held	
SWING_LISTEN_2 20261007_024253	T0	its words in the tool’s own header	Held	
FOLD_OR_FLIP 20261007_050721	R0	its words in the tool’s own header	Held	
FOLD_OR_FLIP 20261007_050721	R1	against SPOKES’ 1.6894624205: -1.29e-11	Held	
RUNG_ABOVE_2 20261007_073220	V1	at 486 steps 1.692490287993 (golden) and 1.692490282152 against RUNG_ABOVE’s 1.692490287993 and 1.692490282152	Held	
POLARISATION_LOOP_2 20261007_074423	W0	its words in the tool’s own header	Held	
GROOVES_13 20261007_090124	F0	the same sign changes as GROOVES_11 (4.5241, 5.0150)	Held	
GROOVES_15 20261007_163523	S1	the 9/7 from its loop 5.7739504e-08 against GROOVES_8’s 5.7739504e-08	Held	
GROOVES_15 20261007_165418	R0	spread 2.67729e-5 against 2.67729e-5; first edge 1.201882380904411930049746105435 against 1.20188238090441193049746105435	Held	
GROOVES_14 20261007_170021	F0	the loader’s two lines reading GROOVES_10’s edges against GROOVES_11 20261007_080706’s: 2 of 2 identical	Held	
GROOVES_14 20261007_170022	F0	the loader’s two lines reading GROOVES_10’s edges against GROOVES_11 20261007_080706’s: 2 of 2 identical	Held	
GROOVES_16 20261007_182009	P0	h range 1.04e-24 to 1.584e-21 and 7 dip extremes against GROOVES_14’s 1.04e-24 to 1.584e-21 and 7	Held	
GROOVES_16 20261007_182010	P0B	the r = 5 run’s fourth round 1.33684215441747836190584981 86 and 1.33684213952768564970768588 84 against GROOVES_10’s 1.33684215441747836190584981 86 and 1.33684213952768564970768588 84	Held	
GROOVES_17 20261007_234942	F0	the new closing function with TWOM against GROOVES_9’s close at three points: 3 of 3 identical to every digit (turn residuals there 1.47e-39, 2.64e-38, -1.54e-38)	Held	
GROOVES_17 20261007_234943	F0	the new closing function with TWOM against GROOVES_9’s close at three points: 3 of 3 identical to every digit (turn residuals there 1.47e-39, 2.64e-38, -1.54e-38)	Held	
CONE_TOMOGRAPHY_2 20261008_002939	D0	identical to run 20261007_235444’s line	Held	
CONE_TOMOGRAPHY_4 20261008_012043	F0	at 40,000 steps the generalised derivatives with n = 3 give +1.378589569923743, N2’s +1.378589569923743, difference +0.00e+00 (bar 1e-13)	Held	
DELTA_BY_TOP 20261008_034013	F0	c_1 less (1 + sqrt 5)/4 = 0.0	Held	

Run	Label	The check, as the run prints it	Verdict	Misses row
PEAK_RING 20261008_040518	P0	peak 0 alone on its steepest ring (radius 0.4472136), 40,000 steps: the frame reads -2.148975939303; 2 pi (1 - cos 70) = 4.134209367876, less a whole turn -2.148975939303; difference modulo a turn +8.88e-15 (bar 1e-8)	Held	
PENTA_SHARE 20261008_060159	F0	circle of radius 0.6 walked as a path +2.505485638101380, CONE_TOMOGRAPHY_4's turns +2.505485638101379, difference +4.44e-16 (bar 1e-10); arc and chord (radius 3, t 0.2) +1.682980939722595, the full circle at 400,000 steps +1.682980939730466, difference -7.87e-12 (bar 1e-9)	Held	
PENTA_SHARE 20261008_060200	F0	circle of radius 0.6 walked as a path +2.505485638101380, CONE_TOMOGRAPHY_4's turns +2.505485638101379, difference +4.44e-16 (bar 1e-10); arc and chord (radius 3, t 0.2) +1.682980939722595, the full circle at 400,000 steps +1.682980939730466, difference -7.87e-12 (bar 1e-9)	Held	
PENTA_SHARE_2 20261008_062403	F0	+2.505485638101380 and +1.682980939722595 against run 20261008_060200's +2.505485638101380 and +1.682980939722595: differences +0.0e+00, -2.2e-16 (bar 1e-14)	Held	
PENTA_SHARE_2 20261008_062420	F0	+2.505485638101380 and +1.682980939722595 against run 20261008_060200's +2.505485638101380 and +1.682980939722595: differences +0.0e+00, -2.2e-16 (bar 1e-14)	Held	
DELTA_AREA 20261008_062647	F0	version 3's first three rungs less version 1's: 3.94e-31, 8.48e-27, -1.98e-28 (bar 1e-20)	Held	
DELTA_AREA 20261008_062647	A0	its words in the tool's own header	Held	
DELTA_BY_TOP_2 20261008_063744	F0	against CASCADE_5's printed 2.8231, 2.3951, 2.5491 to four places	Held	
GROW_LOOP 20261008_133646	G0	reading at radius 2.6751533997, 40,000 steps, -0.599457016199 against -0.599457016199: -2.13e-13 (bar 1e-8)	Held	
DRAFT_TEST 20261008_134707	F0	its words in the tool's own header	Held	
BENT_PAPER 20261008_164513	F0	the 70-degree peak, circle at R/3 round its top: -2.148975939303 against -2.148975939303: -2.90e-13 (bar 1e-9)	Held	
GROW_SLIP 20261008_195205	S0	at the best radius, raw -0.599457015845 (+2.58e-13 from -0.599457015845), unwrapped +11.966913598514 (+4.32e-13 from +11.966913598514), bar 1e-12	Held	
COUNTER_FLOW 20261009_002621	X0	NTU 10, 20,000 cells: counter 0.909090909092 against 10/11 = 0.909090909091 (+7.04e-13); parallel 0.499999998970 against (1 - e^-20)/2 = 0.499999998969 (+3.14e-13); bar 1e-6	Held	
SOUP_JUMP 20261009_045400	F0	its words in the tool's own header	Held	
SOUP_JUMP 20261009_045400	R1	its words in the tool's own header	Held	
LATTICE_READER 20261009_092420	LF0	its words in the tool's own header	Held	
LATTICE_READER_2 20261009_093040	LT0	drive 1.690000, period 8 with 4 turns: converged True; leading multiplier +0.94260 (record +0.94260)	Held	
HISTORY_SETTLE 20261009_093424	H0	strobes at passings 8,001 to 8,256 against LOOP_ATLAS_LATTICE_READER_20261009_091531_partS2.npz: largest difference 0.0e+00	Held	

Run	Label	The check, as the run prints it	Verdict	Misses row
ZERO_CROSS 20261010_043523	Z0	recomputed D2 against the saved one, columns 156 to 894: largest difference 0.0e+00	Held	
SHUFFLE_NULL 20261010_190812	N0	pads equal the period in at least 99 per cent of three bands' columns	Held	
SHUFFLE_NULL_2 20261010_190944	M0	the picture's numbers equal SHUFFLE_NULL's	Held	
DELTA_TAIL 20261010_205937	M0	the record's c_n for orders 6 and 8 reproduced to 35 digits, 17 of 17	Held	
ULP_NUDGE 20261011_044600	M0	k = 0 gives FOLD_OR_FLIP_4's record: period-32 values ['1.6893164568304', '1.6893164568305', '1.6893164514289', '1.6893164514289'], counts [1, 6, 9, 40]; fold drives as recorded, counts [3, 2, 3, 3, 3, 7, 4, 6]	Held	
DELTA_TAIL_2 20261011_051600	M0	the record's c_n for orders 6 and 8 reproduced to 35 digits, 17 of 17	Held	
FOLD_TIGHT 20261011_051636	M0	the 1e-9 results at k = 0 are FOLD_OR_FLIP_4's record	Held	
DAMPER_LAWS 20261011_051848	A0	the linear law gives the record: 53.7024 J, home within 1e-6 m	Held	
SPIN_OR_SIT 20261011_055807	M0	memory 2 reproduces KNEE_RECOUNT_E part A	Held	
SLIP_SENSE 20261011_060035	M0	NOVIKOV_NEEDLE_B's function gives its first two rows of record	Held	
CHAOS_SETTLES 20261011_060320	M0	the protocol to 700 reproduces the parent's arrays exactly	Held	
LOCK_FROM_CHAOS 20261011_060551	M0	SLIVER's 0.9123 row reproduced	Held	
SHOVE_SKATER 20261011_070846	M0	the record's lines	Held	
SHOVE_SKATER 20261011_070846	M1	exactly the record	Held	
WADDLE_PRICE 20261011_071302	M0	its words in the tool's own header	Held	
WADDLE_PRICE 20261011_071302	M2	its words in the tool's own header	Held	
METER_CLOCKS 20261011_072646	M0	ten seed lines and six mean lines, character for character against the run of record	Held (16 of 16 identical)	
TWIST_LONG 20261011_072842	M0	its words in the tool's own header	Held	
TWIST_LONG 20261011_072842	M1	its words in the tool's own header	Held	
METER_K_SPREAD 20261011_080641	M0	its words in the tool's own header	Held	
EXP_SHAPE 20261011_080708	M0	its words in the tool's own header	Held	
WADDLE_FOLLOW 20261011_080808	M0	its words in the tool's own header	Held	
MEMORY_SWITCH 20261011_084432	M0	its words in the tool's own header	Held	
LOCK_LANDING 20261011_170402	M0	its words in the tool's own header	Held	
LOCK_LANDING 20261011_170402	G0	the four terms sum to the total within 1e-9 of its size	Held	
EXP_SHAPE_2 20261011_194909	M0	its words in the tool's own header	Held	

Run	Label	The check, as the run prints it	Verdict	Misses row
LOCK_MATCH 20261011_195010	M0	its words in the tool’s own header	Held	
FALL_CHAIN 20261011_200714	M0	its words in the tool’s own header	Held	
CATCH_TRAP 20261011_202500	M0	its words in the tool’s own header	Held	
SETTLED_COPIES 20261011_ 212014	M0	its words in the tool’s own header	Held	
SETTLED_COPIES_2 20261011_ 212705	M0	its words in the tool’s own header	Held	
SETTLED_GROOVES 20261011_ 223349	M0	its words in the tool’s own header	Held	
FALL_CHAIN_2 20261012_041438	M0	its words in the tool’s own header	Held	
SETTLED_GROOVES_2 20261012_ 041558	M0	its words in the tool’s own header	Held	
SETTLED_GROOVES_2 20261012_ 041558	R0	the walker held at 0.98375 stays on the 1/4 groove	Held	
FALL_SHARE 20261012_075804	M0	its words in the tool’s own header	Held	
WINDOW_EDGE 20261012_ 080030	M0	its words in the tool’s own header	Held	
WINDOW_CASCADE 20261012_ 081249	M0	its words in the tool’s own header	Held	
WINDOW_CASCADE 20261012_ 081249	R0	its words in the tool’s own header	Held	
WINDOW_CASCADE_2 20261012_ 171509	M0	its words in the tool’s own header	Held	
WINDOW_EDGE_2 20261012_ 171509	M0	its words in the tool’s own header	Held	
SPEED_CHAIN 20261012_171949	M0	its words in the tool’s own header	Held	
TOP_SECTION 20261013_005355	M0	its words in the tool’s own header	Held	
INNER_OUTER 20261013_005545	M0	its words in the tool’s own header	Held	
INNER_OUTER 20261013_005545	R1	its words in the tool’s own header	Held	
INNER_OUTER 20261013_005545	R2	first passages within 2 drive periods of WINDOW_ EDGE_2’s	Held	
GHOST_TRIAL 20261013_005637	M0	its words in the tool’s own header	Held	
GHOST_TRIAL 20261013_005637	R0	its words in the tool’s own header	Held	
SADDLE_LOG 20261013_012417	M0	its words in the tool’s own header	Held	
GHOST_TRIAL_2 20261013_ 015743	M0	its words in the tool’s own header	Held	
REINJECT 20261013_015743	M0	its words in the tool’s own header	Held	
REINJECT 20261013_015743	R0	the first passage at 1/16 millionth is 254	Held	
WINDOW_CASCADE_3 20261013_ 015743	M0	its words in the tool’s own header	Held	
REINJECT_CLOCK 20261013_ 024653	M0	its words in the tool’s own header	Held	
REINJECT_CLOCK 20261013_ 024653	R0	the first passage at 1/256 millionth is 1,058	Held	
GHOST_DEPTH 20261013_024747	M0	its words in the tool’s own header	Held	
SADDLE_NOISE 20261013_024850	M0	its words in the tool’s own header	Held	

Run	Label	The check, as the run prints it	Verdict	Misses row
SADDLE_NOISE 20261013_024850	b	at noise 0.2 within 0.1 of SADDLE_LOG’s -0.3231	Held	
SADDLE_ASYMPTOTE 20261013_025252	M0	its words in the tool’s own header	Held	
CATCH_TRAP_2 20261013_030643	M0	its words in the tool’s own header	Held	
ORBIT_919_3 20261013_031807	M0	its words in the tool’s own header	Held	
ORBIT_919_4 20261013_040039	M0	its words in the tool’s own header	Held	
CATCH_TRAP_3 20261013_040120	M0	its words in the tool’s own header	Held	
ORBIT_19 20261013_041840	M0	its words in the tool’s own header	Held	
STRETCH_WHERE 20261013_045856	M0	its words in the tool’s own header	Held	
STRETCH_DEPTH 20261013_055240	M0	its words in the tool’s own header	Held	
LOCK_EDGE_EXACT 20261013_063227	M0	SLIVER’s 0.9123 row reproduced	Held	
LOCK_EDGE_EXACT 20261013_063227	M1	LOCK_FROM_CHAOS’s six means from its npz	Held	
WADDLE_BOOKS 20261013_063353	M0	WADDLE_FOLLOW’s six rows character for character	Held	
WADDLE_BOOKS 20261013_063353	M1	the booking copy matches pinned number for number	Held	
WADDLE_STEP 20261013_073935	M0	WADDLE_BOOKS’s lines and its no-step floor reproduced	Held	
QUANT_ZERO 20261013_081905	M0	the memory-1 needle reproduces the saved 1/4 window orbit within 1e-9	Held	
QUANT_ZERO 20261013_081905	M1	memory 0’s drive-period map is a Möbius map to under 1e-9 radian	Missed	170
WADDLE_CATCH 20261013_082013	M0	WADDLE_STEP’s line at 4e5 and 2.5e-6 reproduced	Held	
QUANT_STEP 20261013_082454	M0	QUANT_ZERO’s 160-step misses reproduced	Held	
HALF_BUBBLE 20261013_083609	M0	Newton finds the saved 1/4 window orbit at 0.98375 within 1e-8	Held	
BUBBLE_FAR 20261013_084037	M0	the first crossing at HALF_BUBBLE’s 1.0019857	Held	
SIZED_TAP_4 20261013_161615	M0	SIZED_TAP_3’s shares and wobbles at 1.40 and 1.60 repeated	Held	
SPIN_START 20261013_161916	M0	QUANT_ZERO’s memory-1 count line from rest, character for character	Held	
HALF_PARENT 20261013_163612	M0	the 4-cycle walk crosses -1 upward at 1.0097823	Held	
SPIN_WINDOWS 20261013_163655	M0	QUANT_ZERO’s M0: the saved 1/4 window orbit within 1e-9 over 640 steps	Held	
GEAR_TRAIN 20261013_170243	M0	1.0300 locks at 2/3, as in HALF_PARENT’s sweep	Held	
EAR_COST 20261013_183245	M0	BALANCE_BOARD’s OWN 1 m, 0.1 s row repeated with the hook adding nothing	Held	
HALF_TOP 20261013_183449	M0	the 2-cycle orbit’s leading multiplier at 1.0225 is +0.5073	Held	
HALF_TOP 20261013_183449	M1	the exponent at 1.0225 is half the log of that multiplier, within 0.01	Held	
EAR_COST_2 20261013_183913	M0	BALANCE_BOARD’s OWN 1 m, 0.1 s row repeated	Held	

Run	Label	The check, as the run prints it	Verdict	Misses row
STAIR_WIDTHS 20261013_184033	M0	GEAR_TRAIN’s 701 paces repeated exactly	Held	
WOBBLE_SOFT 20261013_192713	M0	CLIFF_EDGE’s 30 recorded rows repeated exactly	Held	
EAR_COST_3 20261013_192738	M0	BALANCE_BOARD’s OWN 1 m, 0.1 s row repeated	Held	
EAR_COST_3 20261013_192738	M1	EAR_COST_2’s 0.1275 and 0.1025 repeated	Held	
SECOND_GEAR 20261013_192828	M0	the lag-0 needle takes STAIR_WIDTHS’s fraction at each pace	Held	
CHAOS_RAMP 20261013_192906	M0	HALF_TOP’s four judged rotation numbers repeated to five decimals	Held	
SLIP_PROOF 20261013_194935	M0	CLIFF_EDGE’s rows repeated; the two gauges agree where nothing slips	Held	
STEP_BACK_RAMP 20261013_195010	M0	the locked paces’ rotation numbers equal their fractions within 1e-6	Missed	184
RAMP_EDGE 20261013_195233	M0	1.0280 locks at 2/3 with least repeat 3, and the orbit there is stable	Held	
QUIET_TAP 20261013_195754	M0	SIZED_TAP_4’s shares and wobbles at jostling 0.02 repeated	Held	
SLIP_PROOF_2 20261013_200028	M0	CLIFF_EDGE’s rows repeated; the finder counts the master’s slips; the gauges agree where nothing slips	Missed	185
RAMP_EDGE_2 20261013_200133	M0	1.0280 locks at 2/3 with least repeat 3, the orbit there is stable, and the walk repeats RAMP_EDGE’s	Held	
COLUMN_ACCORDION 20261013_222625	M0	one walker passes on a gentle swing at its closed form’s gain, within 1 per cent	Held	
FIVE_SEVENTHS 20261013_222744	M0	STAIR_WIDTHS’s locks repeated, rotation numbers exact within 1e-9	Missed	190
CHAOS_ORDER 20261013_222835	M0	CHAOS_RAMP’s rotation number at 1.0250, and the two locks, repeated with no noise	Held	
COLUMN_ACCORDION_2 20261013_222950	M0	the gentle walker passes on a swing at its closed form’s gain within 1 per cent, and stays steady alone at every setting	Held	
MORTAR_SPACING 20261013_224843	M0	COLUMN_ACCORDION_2’s growth repeated with the limits off	Held	
BANDED_FIVE 20261013_224945	M0	FIVE_SEVENTHS’s rotation numbers repeated to ten decimals	Held	
CHAOS_ORDER_2 20261013_225021	M0	CHAOS_RAMP’s rotation number and the six locks, with no noise	Held	
SLIP_PROOF_3 20261013_225419	M0	CLIFF_EDGE’s rows repeated; the finder counts the master’s slips where slips are few; the gauges agree where nothing slips	Held	
SLIP_PROOF_3 20261013_225419	M1	the planted test counted exactly	Held	
QUARTER_STROBE 20261013_231124	M0	the orbit at 0.98374 steady with 1 turn in 4; exactly 4 readings at every pace inside the 4-cycle stretch	Missed	194
NIGHT_SPACING 20261013_231417	M0	MORTAR_SPACING’s growth repeated	Held	
QUARTER_STROBE_2 20261013_231511	M0	the orbit at 0.98374 steady with 1 turn in 4; exactly 4 readings at every pace inside the 4-cycle stretch	Held	
ZIGZAG_GAUGE 20261013_231629	M0	the plain slope read back exactly	Held	
ZIGZAG_GAUGE_2 20261013_231751	M0	the plain slope read back exactly	Held	

Run	Label	The check, as the run prints it	Verdict	Misses row
SPRING_STALL 20261014_063112	M0	the 2 October sweep repeated at tune 1.0, loss 1.9	Held	
CATS_EYES 20261014_063138	M0	T 0, no error, repeats NIGHT_SPACING’s touch: none stops, closest 0.825, tail 1.355	Held	
QUICK_READ 20261014_065420	M0	CATS_EYES’s numbers at 0.05, 0.10 and 0.15 s repeated	Held	
WORD_BACK 20261014_065420	M0	COLUMN_ACCORDION_2’s growth repeated	Held	
WINDOW_TOUCH_2 20261014_074538	M0	images 1 to 8 repeat QUARTER_TREE_CURVES’s, touches repeat WINDOW_TOUCH’s	Held	
WINDOW_TOUCH_3 20261014_080520	M0	images 1 to 8 repeat QUARTER_TREE_CURVES’s, touches repeat WINDOW_TOUCH’s	Held	
SMOOTH_DIRT 20261014_084856	M0	the dry law repeats EAR_CURES_B’s C1 rows	Held	
SEAM 20261014_090006	M0	images at -10.0 repeat QUARTER_TREE_CURVES’s	Missed	206
SEAM 20261014_090006	M1	the quantile carry agrees with the direct carry within 1e-4	Held	
WINDOW_TOUCH_4 20261014_090037	M0	the 19 places repeat WINDOW_TOUCH_3’s	Held	
SEAM_2 20261014_090552	M0	images at -10.0 repeat QUARTER_TREE_CURVES’s	Held	
SEAM_2 20261014_090552	M1	the quantile carry agrees with the direct carry within 1e-4	Held	
CRISIS_BURSTS 20261014_091405	M0	no burst inside the window	Held	
WINDOW_TOUCH_5 20261014_092920	M0	the places repeat; the eight judged are WINDOW_TOUCH_4’s	Held	
COLUMN_TAIL 20261014_145058	M0	the first walker’s dips repeat COLUMN_ACCORDION_2’s	Held	
COLUMN_TAIL 20261014_145058	M1	the planted case: no ratio above 1.001	Held	
HEAVY_SHAPE 20261014_155553	M0	at M = 1, C/S within 0.01 of 2.2508	Held	
COLUMN_STEADY 20261014_164637	M0	the first walker’s dip at a fixed distance and no delay is 0.2409 m/s	Held	
HEAVY_MARKS 20261014_164806	M0	all ten markers within 2 pixels of the scale	Held	
HEAVY_MARKS 20261014_164806	M1	his red and green each within 3 per cent of the data, median	Held	
HEAVY_SHAPE_2 20261014_164913	M0	at M 1.2030, S and C repeat HEAVY_SHAPE’s to 1e-4	Held	
HEAVY_SHAPE_3 20261014_170837	M0	at M 1.2030, S and C repeat HEAVY_SHAPE’s to 1e-4	Held	
HEAVY_SHAPE_3 20261014_170837	M1	at M 1.233497, S and C repeat HEAVY_SHAPE_2’s to 1e-2	Held	
TYRE_KNEE 20261014_174309	M0	BANKED_TRACK’s band repeated	Held	
TYRE_KNEE 20261014_174309	M1	no grip: no leftover	Held	
STOP_WALL 20261014_180544	M0	at memory 1 the running edge lies between 0.9795 and 0.9805	Held	
RING_JAM 20261014_224934	M0	the closed form’s 1.5862	Held	
RING_JAM 20261014_224934	M1	at 1.5 s the gain is not above 1.001 and the ring’s growth is below 1	Held	
STOP_WALL_2 20261014_225838	M0	at memory 1 the fine edge lies between 0.98040 and 0.98055	Held	

Run	Label	The check, as the run prints it	Verdict	Misses row
NOISE_EDGE 20261014_232230	M0	the planted holding time within 6 per cent of the exact one	Held	
STOP_WALL_3 20261015_000651	M0	at memory 1.6 the fine edge repeats 0.876292 within 2e-5	Held	
ROOT_JUMPS 20261015_031511	M0	SMOOTH_PUSH_EDGE's 1,699 repeated	Held	
HEAVY_WINDOW_4 20261015_032553	M0	at memory 2.2 the fold within 1e-12 of HEAVY_WINDOW_3's and S within 0.002	Held	
BELL_LOG 20261015_033341	M0	the least 1.1672 at memory 1.9198	Held	
HALT_ORDER 20261015_052122	M0	WORD_BACK's growth repeated	Held	
STOP_LEAVES 20261015_052943	M0	at memory 1.7198 the fine edge repeats STOP_WALL_3's 0.853272 within 2e-5	Held	
GAPS_READ 20261015_061859	M0	the window's 4-cycle found stable, its largest multiplier within 1e-6 of HEAVY_WINDOW's finder's	Held	
HALVINGS 20261015_064027	M0	the 4-cycle at step 160 stable, largest multiplier 0.5333 within 1e-3	Held	
HALVINGS_2 20261015_064427	M0	the 4-cycle at step 160 stable, largest multiplier 0.5333 within 1e-3	Held	
HALVINGS_2 20261015_064427	M1	the window's first doubling found at S below its top within 0.002 millionths, multiplier real and near -1	Held	
BANK_BUMP 20261015_064644	M0	the flat bank: seat at 22.18 degrees at 20 m/s, count 1 below 26.21 m/s	Held	
HALVINGS_3 20261015_064938	M0	the 4-cycle at step 160 stable, largest multiplier 0.5333 within 1e-3	Held	
HALVINGS_3 20261015_064938	M1	the window's first doubling found at S below its top within 0.002 millionths, multiplier real and near -1	Held	
GAPS_REREAD 20261015_071452	M0	the window's 4-cycle found stable, its largest multiplier within 1e-6 of HEAVY_WINDOW's finder's	Held	
LADDER_LIGHT 20261015_071735	M0	the 4-cycle at step 180, largest multiplier 0.3317 within 1e-3	Held	
LADDER_LIGHT 20261015_071735	M1	the planted edge at memory 1.7598	Held	
LEAK_CLOSE 20261015_073039	M0	at memory 1.7198 the fine edge repeats 0.853272 within 2e-5, and at 1.7248 no edge inside the cap	Held	
SNAKE_LONG 20261015_073142	M0	the 48-link body repeats DRIFT_D's 3.1086	Held	
WOBBLY_SHUVIT 20261015_074302	M0	no wobble: 1,000 L in 1,000 landings	Held	
WOBBLY_SHUVIT 20261015_074302	M1	the arrays repeat SHUVIT_WALK's walk for 20 walkers within n^2 1e-14 L	Held	
SWITCH_END 20261015_074716	M0	S by HEAVY_SHAPE's measure within 0.002 millionths of the table's at both weights	Held	
SONG_GAPS 20261015_074750	M0	1 m board: taps per fall time 1.912	Held	
SONG_GAPS_2 20261015_074852	M0	1 m board: taps per fall time 1.912	Held	
LEAK_RATE 20261015_080150	M0	at memory 1.7198 more than half the needles fallen by the end	Held	
SNAKE_768 20261015_151934	M0	the 192-link body repeats SNAKE_LONG's 4.1760	Held	
SNAKE_EXACT 20261015_154247	M0	the grid's values from the seven numbers: 384 and 768 links within 1e-6 of the saved values, 48 within the four places' rounding	Held	
SNAKE_EXACT 20261015_154247	M1	a direct solve at the exact best repeats it within 1e-6, every length	Held	

Run	Label	The check, as the run prints it	Verdict	Misses row
RING_KNOBS 20261015_165858	M0	the record’s pair and budget ring repeated to four places	Held	
RING_KNOBS 20261015_165858	M1	three rings conserve energy and impulse within 1e-8	Held	
TRAIN_HOLD 20261015_171853	M0	RING_KNOBS’s trio repeated over 400 time units	Held	
TRAIN_HOLD 20261015_171853	M1	energy and impulse within 1e-8 at the largest holding gap	Held	
SLOPE_EAR 20261015_190935	M0	the plain ear repeats RF_SKATER_COUNT’s three drifts	Held	
SLOPE_EAR 20261015_190935	M1	looking every step, the slope ears equal the plain ear	Held	
FILTER_SLOPE 20261015_195645	M0	plain, two-look slope and rate repeat SLOPE_EAR	Held	
FILTER_SLOPE 20261015_195645	M1	the fit over 2 looks repeats the two-look slope within 1e-6	Held	
FLIP_MISTAKES 20261015_195739	M0	the six drifts repeat EAR_CURES	Held	
PILOT_BEAT 20261015_203855	M0	PILOT_TONE’s four net pushes at the tune	Held	
PILOT_BEAT 20261015_203855	M1	one soup repeats PILOT_TONE’s run to 1e-9	Held	
BEAT_SWING 20261015_204207	M0	PILOT_BEAT repeated at all 50 settings	Held	
FIT_KNOBS 20261015_204319	M0	the three values at a knee of 0.5 repeat FILTER_SLOPE	Held	
PILOT_PRICE 20261015_220321	M0	BEAT_SWING repeated at its 50 settings	Held	
PAUSE_BEST 20261015_220655	M0	RF_SKATER_COUNT’s ear repeated at h 0.5	Held	
PAUSE_WIDE 20261015_223744	M0	PAUSE_BEST repeated at h 1.5	Held	
JUMP_STORE 20261015_223918	M0	SPRING_BEAD’s blade repeated at tune 1, losses 0.05 and 0.4	Held	
JUMP_STORE 20261015_223918	M1	the knee ear’s events free of the step at tune 1, loss 1	Held	
LATCH_EVENT 20261015_232643	M0	JUMP_STORE’s clean listener and crest-firer repeated	Held	
LATCH_EVENT 20261015_232643	M1	JUMP_STORE’s knee ear repeated	Held	
KNEE_CREEP 20261015_232920	M0	EAR_CURES_B repeated	Held	
CREEP_LAW 20261016_035058	M0	the exact recipe’s spread, and the counter	Held	
BRIDGE_COUNT 20261016_044439	M0	CREEP_LAW’s knee flips repeated	Held	
RIDER_FLEET 20261016_044707	M0	the fleet repeats ride() and FB3	Held	
WORN_RIDER 20261016_065645	M0	RIDER_FLEET’s weakening fleet repeated	Held	
WORN_RIDER 20261016_065645	M1	the taps settle as the step is halved twice	Held	
HITCH_LEGS 20261016_065650	M0	the even legs repeat HITCH_KICK	Held	
HITCH_LEGS 20261016_065650	M1	one stride’s turn the same at two paces	Held	
BELL_CLOCK 20261016_072241	M0	HOW_IT_COUNTS’s P7 repeated	Held	
BELL_CLOCK 20261016_072241	M1	every clock stretched gives the same distance per squeeze	Held	
BAND_BRIDGE 20261016_072612	M0	CREEP_LAW’s knee flips repeated	Held	
BAND_BRIDGE 20261016_072612	M1	the fastest watcher’s flips settle at noise time 1.0	Held	
NEVER_BLINK 20261016_090509	M0	BAND_BRIDGE’s path and counts repeated	Held	
NEVER_BLINK 20261016_090509	M1	the reference settles, the step test	Held	
NEVER_BLINK 20261016_090509	M2	a single line does not, the band test	Held	

Run	Label	The check, as the run prints it	Verdict	Misses row
NEVER_BLINK 20261016_090509	M3	the two counters agree	Held	
TRUCK_ROAD 20261016_090656	M0	SWAP_STATE’s ramp repeated	Held	
TRUCK_ROAD 20261016_090656	M1	the step test	Held	
TRUCK_ROAD 20261016_090656	M2	the band test, the read halved	Held	
HEAT_RING 20261016_174342	M0	the first mode against the exact curve	Held	
HEAT_RING 20261016_174342	M1	the first mode’s line is critical damping	Held	
HEAT_RING 20261016_174342	M2	the step test	Held	
HEAT_RING 20261016_174342	M3	the band test	Held	
LIGHT_LANES 20261016_174648	M0	TRUCK_ROAD repeated where the grids meet	Held	
LIGHT_LANES 20261016_174648	M1	the step test	Held	
LIGHT_LANES 20261016_174648	M2	the band test, the read halved	Held	
ARM_SHARE 20261016_182910	M0	HITCH_LEGS’s shares repeated	Held	
ARM_SHARE 20261016_182910	M1	the step test	Held	
BELL_RATIO 20261016_183422	M0	BELL_CLOCK repeated at three points	Held	
BELL_RATIO 20261016_183422	M1	the step test	Held	
BELL_RATIO 20261016_183422	M2	the band test, the quarters	Held	
BELL_RATIO 20261016_183422	M3	no rings: the balance	Held	
EQUAL_CATCH 20261016_192548	M0	RIDER_FLEET repeated at seed 11	Held	
BELL_BALANCE 20261016_193628	M0	BELL_RATIO repeated	Held	
BELL_BALANCE 20261016_193628	M1	the step test	Missed	271
BELL_BALANCE 20261016_193628	M2	the band test, the quarters	Missed	271
LAG_METER 20261016_203014	M0	EQUAL_CATCH repeated, the gauge added	Held	
SWING_GRID 20261016_203014	M0	NOVIKOV_NEEDLE’s part D repeated at memory 0	Held	
SWING_GRID 20261016_203014	M1	the step test	Held	
BRICK_LINE 20261016_205112	M0	HEAT_RING repeated	Held	
BRICK_LINE 20261016_205112	M1	the step test	Held	
BRICK_LINE 20261016_205112	M2	the band test	Held	
RESTED_LAG 20261016_205504	M0	EQUAL_CATCH repeated, the gauge added	Held	
RESTED_LAG 20261016_205504	M1	the gauge reads the reaction, not the rest	Held	
RING_WIDTH 20261016_213858	M0	TRUCK_ROAD repeated at A = 0.3	Held	
RING_WIDTH 20261016_213858	M1	the step test	Held	
RING_WIDTH 20261016_213858	M2	the band test, the read halved	Held	
STRIDE_MAP 20261016_213858	M0	HITCH_LEGS’s stride repeated	Held	
STRIDE_MAP 20261016_213858	M1	no knee bend, no turn	Held	
STRIDE_MAP 20261016_213858	M2	the exact form, flown	Held	
STRIDE_MAP 20261016_213858	M3	the step test	Held	
BELL_CORNER 20261016_224449	M0	BELL_BALANCE repeated	Held	
BELL_CORNER 20261016_224449	M1	the step test on a settled point	Held	

Run	Label	The check, as the run prints it	Verdict	Misses row
SOUP_JUMP_3 20261016_225440	M0	the riders’ code reprints run_beads	Held	
SOUP_JUMP_3 20261016_225440	M1	no knock-off without the left comb	Held	
SOUP_JUMP_3 20261016_225440	M2	the step test	Held	
SOUP_JUMP_3 20261016_225440	M3	the band test, distance 5 and 20	Held	
ARMS_LEGS 20261017_012927	M0	the limbs alone repeat their records	Held	
ARMS_LEGS 20261017_012927	M1	the shape part does not care how fast	Held	
ARMS_LEGS 20261017_012927	M2	share = count part + shape part	Held	
ARMS_LEGS 20261017_012927	M3	the step test	Held	
ROUGH_FLOOR 20261017_013514	M0	AVALANCHE repeated on the smooth floor	Held	
ROUGH_FLOOR 20261017_013514	M1	the step test	Held	
ROUGH_FLOOR 20261017_013514	M2	the band test	Missed	286
FACE_ALONG 20261017_070629	M0	the parent’s lap and the record repeated	Held	
FACE_ALONG 20261017_070629	M1	no grip, no travel	Held	
FACE_ALONG 20261017_070629	M2	the step test	Held	
FACE_ALONG 20261017_070629	M3	the work loop repeats the lap	Held	
DIVE_CUE 20261017_082454	M0	HITCH_LEGS’s record	Held	
DIVE_CUE 20261017_082454	M1	the loop repeats fly	Held	
DIVE_CUE 20261017_082454	M2	no area, and the clock’s exact error	Held	
DIVE_CUE 20261017_082454	M3	the step test	Held	
DIVE_CUE 20261017_082454	M4	the event at no error	Held	
SNAKE_LONGEST 20261017_155052	M0	SNAKE_EXACT repeated at 768 links	Held	
SNAKE_LONGEST 20261017_155052	M1	a direct solve repeats the best at 1,536 links	Held	
SPRING_FLOOR 20261017_160023	M0	floor 0 repeats LATCH_EVENT	Held	
SPRING_FLOOR 20261017_160023	M1	the step test	Held	
SPRING_FLOOR 20261017_160023	M2	the band test	Held	
LANE_EDGES 20261017_164501	M0	TRUCK_ROAD and LIGHT_LANES repeated where the grids meet	Held	
LANE_EDGES 20261017_164501	M1	the step test, mass 2.0	Held	
LANE_EDGES 20261017_164501	M2	the band test, the read halved	Held	
LANE_EDGES 20261017_164501	M3	each strip one unbroken run inside its window	Held	
TONG_DULLIN 20261017_173718	M0	the machinery repeats HITCH_LEGS	Held	
TONG_DULLIN 20261017_173718	M1	held: one over the inertia	Held	
TONG_DULLIN 20261017_173718	M2	Fig. 11’s out and back turns nothing	Held	
TONG_DULLIN 20261017_173718	M3	the step test	Held	
TONG_DULLIN 20261017_173718	M4	Green’s theorem	Held	
TONG_DULLIN 20261017_173718	M5	two parts against the flight	Held	
TONG_DULLIN_READINGS 20261017_174547	M0	the printed reading repeats TONG_DULLIN	Held	
TONG_DULLIN_SCAN 20261017_181953	M0	the printed table repeats TONG_DULLIN	Held	

Run	Label	The check, as the run prints it	Verdict	Misses row
TONG_DULLIN_SCAN 20261017_181953	M1	every root on 21.0647	Held	
TONG_DULLIN_SCAN 20261017_181953	M2	the fine grid at a spin of 30	Held	
TONG_DULLIN_KNEE 20261017_182629	M0	the printed table repeats TONG_DULLIN	Held	
HALT_THRESHOLD 20261017_184055	M0	HALT_ORDER’s growth repeated	Held	
HALT_THRESHOLD 20261017_184055	M1	the steady column’s known arrival	Held	
HALT_THRESHOLD 20261017_184055	M2	the step test	Held	
LIMIT_AUDIT 20261017_190938	A0	each copy as it stands repeats its run of record	Held	
TWO_HANDS 20261017_203811	M0	JUMP_STORE’s crest-firer repeated	Held	
TWO_HANDS 20261017_203811	M1	the mirror’s true nothing	Held	
TWO_HANDS 20261017_203811	M2	the step test	Held	
COIL_SNAKE 20261017_204030	M0	no grip ratio, no travel	Held	
COIL_SNAKE 20261017_204030	M1	the mirror travels back	Held	
COIL_SNAKE 20261017_204030	M2	the wire’s cutting	Held	
TWO_HANDS_2 20261017_204131	M0	TWO_HANDS’s even soup repeated	Held	
TWO_HANDS_BAND 20261017_210543	M0	TWO_HANDS repeated at band 0	Held	
CAT_TWIST 20261017_231928	M0	the port repeats the engine and CAT_3D	Held	
CAT_TWIST 20261017_231928	T1	each convention’s relative turning has no part along its own line	Held	
CAT_TWIST 20261017_231928	T2	the toy’s waist winds a full turn over the half swing; the twist-free one’s never	Held	
CAT_TWIST 20261017_231928	T3	the toy’s loop is the odd one, the twist-free one’s the even one	Held	
SLING_RUNDOWN 20261017_232234	M0	SWING_TWENTY part C repeated on its own grid	Held	
SAGNAC_LOOP 20261018_032138	S0	the delay is 4 A W / c^2	Held	
TRAIN_LIFETIME 20261018_034211	M0	TRAIN_HOLD’s breaks repeated	Held	
TRAIN_LIFETIME 20261018_034211	M1	energy and impulse within 1e-8	Held	
CAT_SWEET 20261018_034953	M0	the two-thickness port repeats CAT_TWIST	Held	
CAT_MIRROR 20261018_041123	M0	the seven pairs repeated	Held	
TRAIN_TIMING 20261018_082139	M0	the same break either way	Held (spread 0.0e+00)	
COIL_PUMP 20261018_082424	M0	isotropic drag goes nowhere	Held	
COIL_PUMP 20261018_082424	M1	one knob goes nowhere	Held	
COIL_PUMP 20261018_082424	M2	no clock	Held	
COIL_WAVE 20261018_082544	M0	isotropic drag goes nowhere	Held	
COIL_WAVE 20261018_082544	M1	a standing wave goes nowhere	Held	
COIL_WAVE 20261018_082544	M2	no clock	Held	

Run	Label	The check, as the run prints it	Verdict	Misses row
TRAIN_CLOSE 20261018_103739	M0	TRAIN_LIFETIME’s breaks repeated	Missed	366
TRAIN_CLOSE 20261018_103739	M1	energy and impulse within 1e-6 on every holding train	Held	
CAT_MIRROR_PROOF 20261018_174959	M0	the reduction repeats the flights	Held	
CAT_MIRROR_PROOF 20261018_174959	M1	the identity, hinge at the ends	Held	
CAT_MIRROR_PROOF 20261018_174959	M2	it fails with the hinge elsewhere	Held	
TRAIN_EDGE 20261018_181641	M0	energy and impulse within 1e-6 on every train that holds	Held	
TRAIN_EDGE 20261018_181641	M1	every break leaves a pair	Held	
TRAIN_NUDGE 20261018_192138	M0	no nudge repeats TRAIN_LIFETIME	Held	
SAIL 20261018_202451	M0	the course theorem at every heading	Held	
SAIL 20261018_202451	M1	pure drag downwind is slower than the wind	Held	
SAIL_HULL 20261018_202650	M0	the barge repeats SAIL’s run of record	Held	
SAIL_HULL 20261018_202650	M1	the course theorem at every heading of every hull	Held	
TRAIN_DRIFT 20261019_034330	M0	the reader: planted creep CREEP, planted wobble WOBBLE	Held	
TRAIN_DRIFT 20261019_034330	M1	every flight holds the whole flight	Held	
KNEE_SIZE 20261019_035236	M1	the reader on a straight line, planted shift 0.02: read 0.020000000	Held	
KNEE_SIZE 20261019_035236	M2	the reader on a bent curve, planted shift 0.0225: fine grid 0.0225032, grid of 0.05 0.0228173	Held	
KNEE_SIZE 20261019_035236	M0	flips at knee 1.0 repeat CREEP_LAW’s json, largest difference 2.22e-16	Held	
KNEE_FINE 20261019_040109	M1	the reader on a straight line, planted shift 0.0123: read 0.012300000	Held	
KNEE_FINE 20261019_040109	M0	the finest step’s flips at the 17 shared knees repeat KNEE_SIZE’s json, largest difference 0.00e+00	Held	
KNEE_REF 20261019_042520	M3	the reader on a straight line, planted shift 0.0123: read 0.012300000	Held	
KNEE_REF 20261019_042520	M1	soup 0’s first path, 2,559,996 fine innovations: their variance over the noise’s for one fine step 0.99888; straight-line filling 0.24950, judged to miss	Held	
KNEE_REF 20261019_042520	M0	the watchers 0.000625 to 0.08 on the finer path repeat KNEE_FINE’s flips at knee 1.0, largest difference 5.55e-17	Held	
KNEE_REF 20261019_042520	M2	against the old reference the grid of 0.001 gives KNEE_FINE’s extra sizes at 0.00125 to 0.04, 6 steps, largest difference 3.65e-04	Missed	395
ERROR_BARS 20261019_162750	M1	the jackknife: mean of 64 unit normals, 400 times, gives 0.1241 against 0.125; a ratio of two correlated means, over its delta-method value 1.0000	Held	
ERROR_BARS 20261019_162750	M0	every rerun repeats its run of record within 1e-9	Held	
KNEE_READER 20261019_163948	M1	the fitted reader on a planted line and a planted quadratic, shift 0.0123: 0.012300000000, 0.012300000000	Held	
KNEE_READER 20261019_163948	M2	the bootstrap over the jackknife for the mean of 64 unit normals, averaged over 200 draws: 0.9935	Held	

Run	Label	The check, as the run prints it	Verdict	Misses row
KNEE_READER 20261019_163948	M0	KNEE_REF’s flips at knee 1.0 and both references’ curves repeated, largest difference 2.78e-16	Held	
CAT_MIRROR_PATHS 20261020_034021	M0	a = 0 repeats CAT_MIRROR	Held	
CAT_MIRROR_PATHS 20261020_034021	M1	the hinge moved in breaks the steady pairing	Held	
CAT_MIRROR_PATHS 20261020_034021	M2	the path at a = 0 is CAT_TWIST’s	Held	
CAT_MONTGOMERY 20261020_035025	M0	the flights repeat CAT_MIRROR	Held	
CAT_MONTGOMERY 20261020_035025	M1	a wrong closed form fails	Held	
CAT_MONTGOMERY 20261020_035025	M2	the closed forms’ algebra	Held	
CAT_MONTGOMERY 20261020_035025	M3	the double mirror repeats CAT_MIRROR_PROOF’s X3	Held	
CAT_MIRROR_RATES 20261020_041152	M0	the six instants	Held	
CAT_MIRROR_RATES 20261020_041152	M1	the unmatched radius fails	Held	
ZETA_FIT 20261020_055322	M0	CREEP_LAW repeated	Held	
ZETA_FIT 20261020_055322	M1	a known BETA read back	Held	
ZETA_LIMIT 20261020_055829	M0	CREEP_LAW repeated	Held	
ZETA_LIMIT 20261020_055829	M1	a known BETA(r) read back	Held	
ZETA_LIMIT 20261020_060256	M0	CREEP_LAW repeated	Held	
ZETA_LIMIT 20261020_060256	M1	a known BETA(r) read back	Held	
CAT_RTB 20261020_063617	M0	straight gives no turn	Held	
CAT_RTB 20261020_063617	M1	a misreading fails	Held	
KRAMERS_EXACT 20261020_073520	M0	no noise: the noiseless rate	Held	
KRAMERS_EXACT 20261020_073520	M1	no washboard: the pace	Held	
KRAMERS_EXACT 20261020_073520	M2	Kramers recovered as the barrier grows	Held	
SLIP_SPREAD 20261020_075155	M0	SLIP_SENSE’s totals repeated	Held	
SLIP_SPREAD 20261020_075155	M1	the spread reader on Poisson draws	Held	
SLIP_SPREAD 20261020_075635	M0	SLIP_SENSE’s totals repeated	Held	
SLIP_SPREAD 20261020_075635	M1	the spread reader on Poisson draws	Held	
BENCH_LOOP 20261020_081001	M0	a loop of zero size changes nothing	Held	
BENCH_LOOP 20261020_081001	M1	the tip repeated	Held	
BENCH_LOOP 20261020_081001	M2	the jump reader on known cases	Held	
SLIP_RHYTHM 20261020_083426	M0	SLIP_SPREAD’s three rows repeated	Held	
SLIP_RHYTHM 20261020_083426	M1	no washboard: D0 and the pace	Held	
SLIP_RHYTHM 20261020_083426	M2	no tilt: the untilted spread	Held	
SLIP_RHYTHM 20261020_083426	M3	the mean speed is KRAMERS_EXACT’s	Missed	429

Run	Label	The check, as the run prints it	Verdict	Misses row
SLIP_RHYTHM 20261020_083426	M4	20 noise units: independent slips	Held	
SLIP_RECHECK 20261020_083908	M0	SLIP_SENSE’s row repeated	Held	
SLIP_RECHECK 20261020_083908	M1	KRAMERS_EXACT repeated at its printed paces	Held	
SLIP_RHYTHM_2 20261020_084919	M0	SLIP_RHYTHM’s row repeated	Held	
SLIP_RHYTHM_2 20261020_084919	M1	SLIP_RHYTHM’s exact Fano factors repeated	Held	
SLIP_START 20261020_090154	M0	SLIP_RECHECK’s row repeated	Held	
SLIP_START 20261020_090154	M1	SLIP_RECHECK’s exact rates repeated	Held	
SLIP_RHYTHM_3 20261020_090717	M0	the reader on synthetic renewal needles	Held	
SLIP_RHYTHM_3 20261020_090717	M1	the formula’s two sign versions agree	Held	
SLIP_RHYTHM_3 20261020_090717	M2	SLIP_RHYTHM’s row repeated	Held	
RHYTHM_BAND 20261020_154009	M0	SLIP_RHYTHM repeated	Held	
RHYTHM_BAND 20261020_154009	M1	20 noise units: independent slips	Held	
RHYTHM_BAND 20261020_154009	M2	no washboard: free diffusion	Held	
SLIP_BAND 20261020_161039	M0	SLIP_RHYTHM’s row repeated	Held	
SLIP_BAND 20261020_161039	M1	the reader on synthetic needles	Held	
SLIP_BAND_2 20261020_161539	M0	SLIP_BAND’s noise-0.3 row repeated by the record reader	Held	
BELL_SWITCH 20261020_161902	M0	BELL_BALANCE repeated at the gap 1.5	Held	
BELL_SWITCH 20261020_161902	M1	the softened equations at 1e-3 are MEMORY_BELL’s	Held	
TRAIN_FLOOR 20261020_224239	M3	planted step: thinning read True, comb read False	Held	
TRAIN_FLOOR 20261020_224239	M4	planted band: flat thinning False, flat comb False; 5,0,5,0 comb True	Held	
TRAIN_FLOOR 20261020_224239	M0	energy and impulse within 1e-6 on every train that holds	Held	
TRAIN_FLOOR 20261020_224239	M1	every break leaves a pair	Held	
TRAIN_FLOOR 20261020_224239	M2	the six unnudged copies hold, as TRAIN_CLOSE’s did on the second computer	Held	
BELL_SOFT_MAP 20261021_000736	M0	the copy at 1e-3 is BELL_BALANCE	Held	
BELL_SOFT_MAP 20261021_000736	M1	the plain stroke unchanged	Held	
BELL_SOFT_SCALE 20261021_054731	M0	the copy at 0.0225 is BELL_SOFT_MAP’s	Held	
TRAIN_FLOOR_2 20261021_083613	M0	energy and impulse within 1e-6 over 20,000 and 5e-6 over 100,000 on every train that holds	Held	
TRAIN_FLOOR_2 20261021_083613	M1	every break leaves a pair	Held	
TRAIN_FLOOR_2 20261021_083613	M2	the ten long flights pass 20,000 whole, as the same starts did on the second computer	Held	

Run	Label	The check, as the run prints it	Verdict	Misses row
BELL_WIDTH_LADDER 20261021_151833	M0	the copy at 0.05 is BELL_SOFT_SCALE’s	Held	
BELL_SPIKES 20261022_053449	M0	the copy at 0.04 is BELL_WIDTH_LADDER’s	Held	
BELL_SPIKES 20261022_053740	M0	the copy at 0.04 is BELL_WIDTH_LADDER’s	Held	
TRAIN_DOOR 20261022_055043	M0	this tool’s flight at core 0.05 is TRAIN_FLOOR_2’s	Held	
TRAIN_DOOR 20261022_055043	M1	every break leaves a pair	Held	
TRAIN_DOOR 20261022_055043	M2	energy and impulse within 1e-6 on every train that holds	Held	
TRAIN_EDGE_MAP 20261022_060414	M0	TRAIN_DOOR’s break repeated on this machine	Held	
TRAIN_EDGE_MAP 20261022_060414	M1	every break leaves a pair	Held	
TRAIN_EDGE_MAP 20261022_060414	M2	energy and impulse within 1e-6 on every train that holds	Held	
TRAIN_EDGE_DOOR 20261022_070632	M0	energy and impulse within 1e-6 on every train that holds	Held	
TRAIN_EDGE_DOOR 20261022_070632	M1	every break leaves a pair	Held	
TRAIN_EDGE_DOOR 20261022_070632	M2	TRAIN_FLOOR_2’s break repeated on the second computer	Held	
TRAIN_LONG_EDGE 20261022_175421	M0	energy and impulse within 5e-06 on every train that holds	Held	
TRAIN_LONG_EDGE 20261022_175421	M1	every break leaves a pair	Held	
TRAIN_LONG_EDGE 20261022_175421	M2	TRAIN_FLOOR_2’s break repeated on the second computer	Held	
TRAIN_LONG_EDGE 20261022_175421	M3	TRAIN_EDGE_DOOR’s ten copies repeat their first 20,000	Held	
TRAIN_DOOR_LINE 20261022_222935	M0	energy and impulse within 5e-06 on every train that holds	Held	
TRAIN_DOOR_LINE 20261022_222935	M1	every break leaves a pair	Held	
TRAIN_DOOR_LINE 20261022_222935	M2	TRAIN_FLOOR_2’s break repeated on the second computer	Held	
TRAIN_DOOR_LINE 20261022_222935	M3	TRAIN_LONG_EDGE’s break repeated on the second computer	Held	
CAT_MIRROR_FAMILY 20261023_001232	M0	the family’s slices are the two cats	Held	
CAT_MIRROR_FAMILY 20261023_001232	M0b	the rate ignores chi itself	Held	
CAT_MIRROR_FAMILY 20261023_001232	M1	CAT_MIRROR_RATES’s sweeps repeated	Held	
CAT_MIRROR_FAMILY 20261023_001232	M2	the unmatched radius fails	Held	
CAT_MIRROR_CENTRES 20261023_001618	N0	CAT_MIRROR_FAMILY repeated	Held	
CAT_MIRROR_CENTRES 20261023_001618	N1	the toy slice mirrors at the centres	Held	

Run	Label	The check, as the run prints it	Verdict	Misses row
TRAIN_FUSE_CLOCK 20261028_101647	M0	energy and impulse within 5e-06 on every train that holds	Held	
TRAIN_FUSE_CLOCK 20261028_101647	M1	every break leaves a pair	Held	
TRAIN_FUSE_CLOCK 20261028_101647	M2	TRAIN_DOOR_LINE’s 21 breaks again, to three decimals	Held (21 of 21)	
TRAIN_FUSE_CLOCK 20261028_101647	M3	TRAIN_DOOR_LINE’s scaled squeezes again, every one	Held (21 of 21)	
TRAIN_BEAT 20261028_212034	M2	this tool’s flight is TRAIN_FUSE_CLOCK’s, every field, core 0.05, gap 0.527, nudge 2e-15, to 500	Held (26 of 26 fields equal)	
TRAIN_BEAT 20261028_212034	M3	the reader gives back TRAIN_FUSE_CLOCK’s read	Held (35 breaks; 177 scaled squeezes, the largest distance from a threading 0.061; one threading after ts in 35; the beat after ts against the beat before, largest 0.116)	
TRAIN_BEAT 20261028_212034	M0	energy and impulse within 5e-06 on every train that holds	Held	
TRAIN_BEAT 20261028_212034	M1	every break leaves a pair	Held	
TRAIN_PARTING_EDGE 20261029_070800	M0	energy and impulse within 5e-06 on every train that stays whole	Held	
TRAIN_PARTING_EDGE 20261029_070800	M1	every train whose ring is gone keeps a pair within 3 radii	Held	
TRAIN_PARTING_EDGE 20261029_070800	M2	TRAIN_FLOOR_2’s train spreads 5 radii first at 11373.597 again	Held (got 11373.597400086826)	
TRAIN_PARTING_EDGE 20261029_070800	M3	TRAIN_LONG_EDGE’s train spreads 5 radii first at 4222.621 again	Held (got 4222.620609838368)	
TRAIN_PARTING 20261029_112955	M0	energy and impulse within 5e-06 on every train	Held	
TRAIN_PARTING 20261029_112955	M1	every train with a ring gone keeps a pair within 3 radii	Held	
TRAIN_PARTING 20261029_112955	M2	TRAIN_BEAT’s first partings again, 1846.427 and 1608.505	Held (got 1846.427 and 1608.505)	
TRAIN_PARTING_EDGE_2 20261029_211416	M0	energy and impulse within 5e-06 on every train that stays whole	Held	
TRAIN_PARTING_EDGE_2 20261029_211416	M1	every train whose ring is gone keeps a pair within 3 radii	Held	
TRAIN_PARTING_EDGE_2 20261029_211416	M2	TRAIN_FLOOR_2’s train spreads 5 radii first at 11373.597 again	Held (got 11373.597400086826)	
TRAIN_PARTING_EDGE_2 20261029_211416	M3	TRAIN_LONG_EDGE’s train spreads 5 radii first at 4222.621 again	Held (got 4222.620609838368)	
RAMP_SPEED 20261031_232517	M0	the runners at the pace repeat TRUCK_ROAD’s run of record	Held	
RAMP_SPEED 20261031_232517	M1	the step test	Held	
RAMP_SPEED 20261031_232517	M2	the band test, the read halved	Held	
HALT_TRADE 20261031_235932	M0	HALT_ORDER’s growth repeated	Held	

Run	Label	The check, as the run prints it	Verdict	Misses row
HALT_TRADE 20261031_235932	M1	the steady column’s known arrival	Held	
HALT_TRADE 20261031_235932	M2	the step test	Held	
HALT_TRADE 20261031_235932	M3	HALT_THRESHOLD repeated at 2, 3 and 4 m	Held	
COMB_STRETCH 20261101_002907	F0	the patterns add up to the ride’s velocity	Held	
COMB_STRETCH 20261101_002907	F1	the json gives the printed shares	Held	
COMB_STRETCH_B 20261101_003130	B0	f = 1 repeats COMB_STRETCH	Held	
SNAKE_RACE 20261101_175747	C0	the seven numbers repeat SNAKE_EXACT’s record at five lengths and three wave numbers within 1e-9; largest gap 0.0e+00	Held	
SNAKE_RACE 20261101_175747	C1	the motion carries the push at every length’s best setting of record within 1e-9; largest gap 1.2e-15	Held	
SNAKE_RACE 20261101_175747	C2	the joints’ heights are the solver’s body shape within 1e-12; largest gap 1.3e-13	Held	
SNAKE_BEAT 20261101_190003	B0	the seven numbers and f0 repeat SNAKE_RACE’s within 1e-9	Held	
FREE_SWIM 20261101_222356	E0	at W_e = 1 the seven numbers and f0 repeat SNAKE_RACE’s within 1e-9	Held	
FREE_SWIM 20261101_222356	E1	the motors alone push nothing at every encounter frequency, |F2| below 1e-7 of the cross term	Held	
FREE_SWIM_2 20261101_222740	Q0	at W_e 1.25 the seven numbers repeat FREE_SWIM’s within 1e-9	Held	
RING_PATHS 20261102_013613	R0	the six flights on RING_KNOBS’s grid repeat its json within 1e-9; largest gap 0.0e+00	Held	
RING_PATHS 20261102_013613	R1	the four on TRAIN_HOLD’s grid repeat its json, gains and distances within 1e-9, the rest exactly; largest gap 0.0e+00	Held	
ROUGH_COUNT 20261102_173941	M0	the counting repeats ROUGH_FLOOR’s json	Held	
ROUGH_COUNT 20261102_173941	M1	the count tells a made-up trap from a made-up heavy tail	Held	
ROUGH_LONG 20261102_200436	L0	the first 20,000 time units repeat ROUGH_COUNT’s F	Held	
ROUGH_LONG 20261102_200436	L1	the count tells a made-up trap from a made-up heavy tail at this length	Held	
BRICK_QUARTER 20261103_180126	M0	HEAT_RING repeated at 400 cells, to the last bit	Held	
BRICK_FINE 20261103_180438	M0	BRICK_LINE repeated at 400 cells, to the last bit	Held	
BLOB_RING 20261104_170523	M0	the toy repeats the first swimmer’s lap	Held	
BLOB_RING 20261104_170523	M1	the even breathing goes nowhere and does not turn	Held	
BLOB_RING 20261104_170523	M2	the step test	Held	
BLOB_RING 20261104_170523	M3	the node test	Held	
BLOB_RING 20261104_170523	M4	the time mirror	Held	
BLOB_SMALL 20261104_170747	M0	BLOB_RING repeated at e = 0.05 and 0.1, to the last bit	Held	
LEAN_DRIFT 20261105_025051	M0	the decided predictions counted here equal the register’s	Held	

Run	Label	The check, as the run prints it	Verdict	Misses row
RING_AREA 20261105_032546	M0	the 36 pairs repeat RING_KNOBS's json within 1e-9; largest gap 0.0e+00	Held	
RING_AREA 20261105_032546	M1	the four kept pairs repeat RING_PATHS's kept paths exactly	Held	
RING_AREA 20261105_032546	M2	the area at half the samples within 1e-6 of that at the full number; largest gap 9.0e-08	Held	
RING_SPLIT 20261105_050121	M0	the 36 pairs repeat RING_KNOBS's json within 1e-9; largest gap 0.0e+00	Held	
RING_SPLIT 20261105_050121	M1	the split adds up within 1e-6 over the first two periods; largest gap 6.7e-11	Held	
RING_SPLIT 20261105_050121	M2	the own part at half the samples within 1e-7; largest gap 2.8e-16	Held	
RING_SPLIT 20261105_050121	M3	the span's gain within 1 per cent of RING_KNOBS's; largest 0.053 per cent	Held	
SWIM_RIDGE 20261105_145417	M0	the follow-on's three laps at grips 10,000 and 100,000 repeat its printed travel	Held	
SWIM_RIDGE 20261105_145417	M1	every judged lap at 1,200 steps within 1 per cent of 600, same sign; largest 0.000 per cent	Held	
OWED_BARS 20261107_023832	M1	ERROR_BARS's jackknife on its planted cases: the mean of 64 unit normals, 400 times, 0.1241 against 0.125; a ratio of two means over its delta-method value 1.0000	Held	
OWED_BARS 20261107_023832	M0	every rerun repeats its run of record: A True, B True, C True, D True	Held	
OWED_BARS 20261107_023832	M2	part D's needle sums rebuild every reading	Held	
CUT_WIDE 20261107_033035	M0	the record's rows repeated at the cut of 15	Held	
CUT_WIDE 20261107_033035	M1	OWED_BARS's judged gaps and errors repeated at the cut of 15	Held	
COIL_CHEAP 20261107_060011	M0	the parent's coil repeated	Held	
COIL_CHEAP 20261107_060011	M1	the wire's cutting at the cheapest angles	Held	
COIL_CHEAP 20261107_060011	M2	ten turns held: the hand law, cheapest at 35.3	Held	
COIL_CHEAP 20261107_060011	M3	the wire's length held: cheapest at 45.0	Held	
QUADRATURE_ACROSS 20261107_080953	M0	the wing repeats toy run 5d	Held	
QUADRATURE_ACROSS 20261107_080953	M1	the skater repeats its part A	Held	
QUADRATURE_ACROSS 20261107_080953	M2	the rider repeats S1 at 75, 90, 105	Held	
QUADRATURE_ACROSS 20261107_080953	M3	the snake and the light loop, carried	Held	
QUADRATURE_ACROSS 20261107_080953	M4	the wing's best stands with the step halved and read finer	Held	
QUADRATURE_ACROSS 20261107_080953	M5	the skater's best within 0.5 degree of 0	Held	
QUADRATURE_ACROSS 20261107_080953	M6	the wing's two half circles mirror each other	Held	
COLUMN_WALK 20261108_234842	M1	the walk prints the room's 10 corrected lines at their printed digits; 10 of 10 the same	Held	

Run	Label	The check, as the run prints it	Verdict	Misses row
CLOCK_SHIFT_30 20261109_034237	M0	its words in the tool’s own header	Held	
CLOCK_SHIFT_30 20261109_034237	M1	its words in the tool’s own header	Held	
CLOCK_SHIFT_30 20261109_034237	M2	its words in the tool’s own header	Held	
THIRD_AXIS 20261109_170732	M0	its words in the tool’s own header	Held	
THIRD_AXIS 20261109_170732	Z1a	its words in the tool’s own header	Held	
THIRD_AXIS_2 20261109_181451	M0	its words in the tool’s own header	Held	
THIRD_AXIS_3 20261109_191727	M0	its words in the tool’s own header	Held	

The teaching pages’ engine

243 lines: the toy pages’ engine checked against the runs of record.

Run	Label	The check, as the run prints it	Verdict	Misses row
EXPLORER_ENGINE_CHECK 20261017_063123	E1	the rigid forward turn, human legs (degrees)	Held	
EXPLORER_ENGINE_CHECK 20261017_063123	E2	one arm circle, nothing carried (degrees)	Held	
EXPLORER_ENGINE_CHECK 20261017_063123	E3	one human stride, nothing carried (degrees)	Held	
EXPLORER_ENGINE_CHECK 20261017_063123	E4	the arms alone, 1, 2 and 3 circles (share)	Held	
EXPLORER_ENGINE_CHECK 20261017_063123	E5	the legs alone, 3 strides, knee 1.57 and 2.36 (share)	Held	
EXPLORER_ENGINE_CHECK 20261017_063123	E6	two circles with two human strides (share)	Held	
EXPLORER_ENGINE_CHECK 20261017_063123	E7	two circles with two deep strides (share)	Held	
EXPLORER_ENGINE_CHECK 20261017_063123	E8	the same routine, nothing carried, in 0.9 s and 0.45 s (degrees)	Held	
EXPLORER_ENGINE_CHECK 20261017_063123	E9	diver: pike loop at amplitude 0.6 (degrees a loop)	Held	
EXPLORER_ENGINE_CHECK 20261017_063123	E10	diver: zigzag loop at amplitude 0.6 (degrees a loop)	Held	
EXPLORER_ENGINE_CHECK 20261017_063123	E11	diver: fall from 3 m and 10 m (s)	Held	
EXPLORER_ENGINE_CHECK 20261017_063123	W1	a jumper flight at 6,000 steps under 100 ms	Held (median 46.19 ms)	
EXPLORER_ENGINE_CHECK 20261017_063123	W2	at 1,500 steps E1 to E8 all within tolerance	Held	
EXPLORER_ENGINE_CHECK 20261017_070749	E1	the rigid forward turn, human legs (degrees)	Held	
EXPLORER_ENGINE_CHECK 20261017_070749	E2	one arm circle, nothing carried (degrees)	Held	
EXPLORER_ENGINE_CHECK 20261017_070749	E3	one human stride, nothing carried (degrees)	Held	
EXPLORER_ENGINE_CHECK 20261017_070749	E4	the arms alone, 1, 2 and 3 circles (share)	Held	

Run	Label	The check, as the run prints it	Verdict	Misses row
EXPLORER_ENGINE_CHECK 20261017_070749	E5	the legs alone, 3 strides, knee 1.57 and 2.36 (share)	Held	
EXPLORER_ENGINE_CHECK 20261017_070749	E6	two circles with two human strides (share)	Held	
EXPLORER_ENGINE_CHECK 20261017_070749	E7	two circles with two deep strides (share)	Held	
EXPLORER_ENGINE_CHECK 20261017_070749	E8	the same routine, nothing carried, in 0.9 s and 0.45 s (degrees)	Held	
EXPLORER_ENGINE_CHECK 20261017_070749	E9	diver: pike loop at amplitude 0.6 (degrees a loop)	Held	
EXPLORER_ENGINE_CHECK 20261017_070749	E10	diver: zigzag loop at amplitude 0.6 (degrees a loop)	Held	
EXPLORER_ENGINE_CHECK 20261017_070749	E11	diver: fall from 3 m and 10 m (s)	Held	
EXPLORER_ENGINE_CHECK 20261017_070749	W1	a jumper flight at 6,000 steps under 100 ms	Held (median 60.19 ms)	
EXPLORER_ENGINE_CHECK 20261017_070749	W2	at 1,500 steps E1 to E8 all within tolerance	Held	
EXPLORER_DIVE_CHECK 20261017_155902	E12	the nominal spin (kg m^2/s)	Held	
EXPLORER_DIVE_CHECK 20261017_155902	E13	the cue’s angle at 1.00 s (degrees)	Held	
EXPLORER_DIVE_CHECK 20261017_155902	E14	the clock’s entry errors at -10 and +10 per cent (degrees)	Held	
EXPLORER_DIVE_CHECK 20261017_155902	E15	the cue’s entry errors at -10, -5, -2, +2, +5, +10 per cent (degrees)	Held	
EXPLORER_DIVE_CHECK 20261017_155902	E16	the cue’s moment at +10 per cent (s)	Held	
EXPLORER_DIVE_CHECK 20261017_155902	E17	the cued miss over the clock’s at +10 per cent	Held	
EXPLORER_DIVE_CHECK 20261017_155902	W3	a cued dive under 100 ms	Held (median 49.78 ms)	
EXPLORER_ENGINE_CHECK 20261017_155903	E1	the rigid forward turn, human legs (degrees)	Held	
EXPLORER_ENGINE_CHECK 20261017_155903	E2	one arm circle, nothing carried (degrees)	Held	
EXPLORER_ENGINE_CHECK 20261017_155903	E3	one human stride, nothing carried (degrees)	Held	
EXPLORER_ENGINE_CHECK 20261017_155903	E4	the arms alone, 1, 2 and 3 circles (share)	Held	
EXPLORER_ENGINE_CHECK 20261017_155903	E5	the legs alone, 3 strides, knee 1.57 and 2.36 (share)	Held	
EXPLORER_ENGINE_CHECK 20261017_155903	E6	two circles with two human strides (share)	Held	
EXPLORER_ENGINE_CHECK 20261017_155903	E7	two circles with two deep strides (share)	Held	
EXPLORER_ENGINE_CHECK 20261017_155903	E8	the same routine, nothing carried, in 0.9 s and 0.45 s (degrees)	Held	
EXPLORER_ENGINE_CHECK 20261017_155903	E9	diver: pike loop at amplitude 0.6 (degrees a loop)	Held	

Run	Label	The check, as the run prints it	Verdict	Misses row
EXPLORER_ENGINE_CHECK 20261017_155903	E10	diver: zigzag loop at amplitude 0.6 (degrees a loop)	Held	
EXPLORER_ENGINE_CHECK 20261017_155903	E11	diver: fall from 3 m and 10 m (s)	Held	
EXPLORER_ENGINE_CHECK 20261017_155903	W1	a jumper flight at 6,000 steps under 100 ms	Held (median 88.29 ms)	
EXPLORER_ENGINE_CHECK 20261017_155903	W2	at 1,500 steps E1 to E8 all within tolerance	Held	
EXPLORER_ROUTINE_CHECK 20261017_164432	R1	its words in the tool’s own header	Held	
EXPLORER_ROUTINE_CHECK 20261017_164432	R2	its words in the tool’s own header	Held	
EXPLORER_ROUTINE_CHECK 20261017_164432	R3	its words in the tool’s own header	Held	
EXPLORER_ROUTINE_CHECK 20261017_164432	G1	arms swung forward: within 10 degrees of the record’s entry	Held (5.2446)	
EXPLORER_ROUTINE_CHECK 20261017_164432	G2	the arm circle, nothing carried, turns the trunk forward 8 to 30 degrees	Held (-21.3929)	
EXPLORER_ROUTINE_CHECK 20261017_164432	G2b	at the nominal spin it enters 8 to 35 degrees over	Held (-18.7706)	
EXPLORER_ROUTINE_CHECK 20261017_164432	G3	the rip window 28.785 ms to 0.01 ms	Held (28.784943 ms)	
EXPLORER_ROUTINE_CHECK 20261017_164432	W4	a routine flight under 100 ms	Held (median 31.55 ms)	
EXPLORER_ENGINE_CHECK 20261017_165820	E1	the rigid forward turn, human legs (degrees)	Held	
EXPLORER_ENGINE_CHECK 20261017_165820	E2	one arm circle, nothing carried (degrees)	Held	
EXPLORER_ENGINE_CHECK 20261017_165820	E3	one human stride, nothing carried (degrees)	Held	
EXPLORER_ENGINE_CHECK 20261017_165820	E4	the arms alone, 1, 2 and 3 circles (share)	Held	
EXPLORER_ENGINE_CHECK 20261017_165820	E5	the legs alone, 3 strides, knee 1.57 and 2.36 (share)	Held	
EXPLORER_ENGINE_CHECK 20261017_165820	E6	two circles with two human strides (share)	Held	
EXPLORER_ENGINE_CHECK 20261017_165820	E7	two circles with two deep strides (share)	Held	
EXPLORER_ENGINE_CHECK 20261017_165820	E8	the same routine, nothing carried, in 0.9 s and 0.45 s (degrees)	Held	
EXPLORER_ENGINE_CHECK 20261017_165820	E9	diver: pike loop at amplitude 0.6 (degrees a loop)	Held	
EXPLORER_ENGINE_CHECK 20261017_165820	E10	diver: zigzag loop at amplitude 0.6 (degrees a loop)	Held	
EXPLORER_ENGINE_CHECK 20261017_165820	E11	diver: fall from 3 m and 10 m (s)	Held	
EXPLORER_ENGINE_CHECK 20261017_165820	W1	a jumper flight at 6,000 steps under 100 ms	Held (median 59.24 ms)	

Run	Label	The check, as the run prints it	Verdict	Misses row
EXPLORER_ENGINE_CHECK 20261017_165820	W2	at 1,500 steps E1 to E8 all within tolerance	Held	
EXPLORER_DIVE_CHECK 20261017_165821	E12	the nominal spin (kg m^2/s)	Held	
EXPLORER_DIVE_CHECK 20261017_165821	E13	the cue’s angle at 1.00 s (degrees)	Held	
EXPLORER_DIVE_CHECK 20261017_165821	E14	the clock’s entry errors at -10 and +10 per cent (degrees)	Held	
EXPLORER_DIVE_CHECK 20261017_165821	E15	the cue’s entry errors at -10, -5, -2, +2, +5, +10 per cent (degrees)	Held	
EXPLORER_DIVE_CHECK 20261017_165821	E16	the cue’s moment at +10 per cent (s)	Held	
EXPLORER_DIVE_CHECK 20261017_165821	E17	the cued miss over the clock’s at +10 per cent	Held	
EXPLORER_DIVE_CHECK 20261017_165821	W3	a cued dive under 100 ms	Held (median 28.24 ms)	
EXPLORER_CAT_CHECK 20261017_183736	E18	the full circuit	Held	
EXPLORER_CAT_CHECK 20261017_183736	E19	the bends	Held	
EXPLORER_CAT_CHECK 20261017_183736	E20	the radii	Held	
EXPLORER_CAT_CHECK 20261017_183736	E21	the half circuits	Held	
EXPLORER_CAT_CHECK 20261017_183736	E22	bend and straighten: no turn	Held	
EXPLORER_CAT_CHECK 20261017_183736	E23	backward, and no momentum	Held	
EXPLORER_CAT_CHECK 20261017_183736	W4	within 0.001 degrees at 1,000 steps	Held	
EXPLORER_CAT_CHECK 20261017_183736	W5	under 20 ms	Held	
EXPLORER_ENGINE_CHECK 20261017_183749	E1	the rigid forward turn, human legs (degrees)	Held	
EXPLORER_ENGINE_CHECK 20261017_183749	E2	one arm circle, nothing carried (degrees)	Held	
EXPLORER_ENGINE_CHECK 20261017_183749	E3	one human stride, nothing carried (degrees)	Held	
EXPLORER_ENGINE_CHECK 20261017_183749	E4	the arms alone, 1, 2 and 3 circles (share)	Held	
EXPLORER_ENGINE_CHECK 20261017_183749	E5	the legs alone, 3 strides, knee 1.57 and 2.36 (share)	Held	
EXPLORER_ENGINE_CHECK 20261017_183749	E6	two circles with two human strides (share)	Held	
EXPLORER_ENGINE_CHECK 20261017_183749	E7	two circles with two deep strides (share)	Held	
EXPLORER_ENGINE_CHECK 20261017_183749	E8	the same routine, nothing carried, in 0.9 s and 0.45 s (degrees)	Held	
EXPLORER_ENGINE_CHECK 20261017_183749	E9	diver: pike loop at amplitude 0.6 (degrees a loop)	Held	

Run	Label	The check, as the run prints it	Verdict	Misses row
EXPLORER_ENGINE_CHECK 20261017_183749	E10	diver: zigzag loop at amplitude 0.6 (degrees a loop)	Held	
EXPLORER_ENGINE_CHECK 20261017_183749	E11	diver: fall from 3 m and 10 m (s)	Held	
EXPLORER_ENGINE_CHECK 20261017_183749	W1	a jumper flight at 6,000 steps under 100 ms	Held (median 55.39 ms)	
EXPLORER_ENGINE_CHECK 20261017_183749	W2	at 1,500 steps E1 to E8 all within tolerance	Held	
EXPLORER_DIVE_CHECK 20261017_183750	E12	the nominal spin (kg m^2/s)	Held	
EXPLORER_DIVE_CHECK 20261017_183750	E13	the cue’s angle at 1.00 s (degrees)	Held	
EXPLORER_DIVE_CHECK 20261017_183750	E14	the clock’s entry errors at -10 and +10 per cent (degrees)	Held	
EXPLORER_DIVE_CHECK 20261017_183750	E15	the cue’s entry errors at -10, -5, -2, +2, +5, +10 per cent (degrees)	Held	
EXPLORER_DIVE_CHECK 20261017_183750	E16	the cue’s moment at +10 per cent (s)	Held	
EXPLORER_DIVE_CHECK 20261017_183750	E17	the cued miss over the clock’s at +10 per cent	Held	
EXPLORER_DIVE_CHECK 20261017_183750	W3	a cued dive under 100 ms	Held (median 27.86 ms)	
EXPLORER_ROUTINE_CHECK 20261017_183751	R1	its words in the tool’s own header	Held	
EXPLORER_ROUTINE_CHECK 20261017_183751	R2	its words in the tool’s own header	Held	
EXPLORER_ROUTINE_CHECK 20261017_183751	R3	its words in the tool’s own header	Held	
EXPLORER_ROUTINE_CHECK 20261017_183751	G1	arms swung forward: within 10 degrees of the record’s entry	Held (5.2446)	
EXPLORER_ROUTINE_CHECK 20261017_183751	G2	the arm circle, nothing carried, turns the trunk forward 8 to 30 degrees	Held (-21.3929)	
EXPLORER_ROUTINE_CHECK 20261017_183751	G2b	at the nominal spin it enters 8 to 35 degrees over	Held (-18.7706)	
EXPLORER_ROUTINE_CHECK 20261017_183751	G3	the rip window 28.785 ms to 0.01 ms	Held (28.784943 ms)	
EXPLORER_ROUTINE_CHECK 20261017_183751	W4	a routine flight under 100 ms	Held (median 32.61 ms)	
EXPLORER_CAT_ROOM_CHECK 20261017_211325	R1	the routine of record, a loop with no clock	Held	
EXPLORER_CAT_ROOM_CHECK 20261017_211325	R2	no swing, no turn	Held	
EXPLORER_CAT_ROOM_CHECK 20261017_211325	R3	the other way round, its mirror	Held	
EXPLORER_CAT_ROOM_CHECK 20261017_211325	P4	the tilt grows with the bend	Held	
EXPLORER_CAT_ROOM_CHECK 20261017_211325	P5	the tilt is the half circuit’s V movement, the body a half turn	Held	

Run	Label	The check, as the run prints it	Verdict	Misses row
EXPLORER_CAT_ROOM_CHECK 20261017_211325	W6	under 30 ms	Held	
EXPLORER_ENGINE_CHECK 20261017_211710	E1	the rigid forward turn, human legs (degrees)	Held	
EXPLORER_ENGINE_CHECK 20261017_211710	E2	one arm circle, nothing carried (degrees)	Held	
EXPLORER_ENGINE_CHECK 20261017_211710	E3	one human stride, nothing carried (degrees)	Held	
EXPLORER_ENGINE_CHECK 20261017_211710	E4	the arms alone, 1, 2 and 3 circles (share)	Held	
EXPLORER_ENGINE_CHECK 20261017_211710	E5	the legs alone, 3 strides, knee 1.57 and 2.36 (share)	Held	
EXPLORER_ENGINE_CHECK 20261017_211710	E6	two circles with two human strides (share)	Held	
EXPLORER_ENGINE_CHECK 20261017_211710	E7	two circles with two deep strides (share)	Held	
EXPLORER_ENGINE_CHECK 20261017_211710	E8	the same routine, nothing carried, in 0.9 s and 0.45 s (degrees)	Held	
EXPLORER_ENGINE_CHECK 20261017_211710	E9	diver: pike loop at amplitude 0.6 (degrees a loop)	Held	
EXPLORER_ENGINE_CHECK 20261017_211710	E10	diver: zigzag loop at amplitude 0.6 (degrees a loop)	Held	
EXPLORER_ENGINE_CHECK 20261017_211710	E11	diver: fall from 3 m and 10 m (s)	Held	
EXPLORER_ENGINE_CHECK 20261017_211710	W1	a jumper flight at 6,000 steps under 100 ms	Held (median 54.49 ms)	
EXPLORER_ENGINE_CHECK 20261017_211710	W2	at 1,500 steps E1 to E8 all within tolerance	Held	
EXPLORER_DIVE_CHECK 20261017_211712	E12	the nominal spin (kg m^2/s)	Held	
EXPLORER_DIVE_CHECK 20261017_211712	E13	the cue’s angle at 1.00 s (degrees)	Held	
EXPLORER_DIVE_CHECK 20261017_211712	E14	the clock’s entry errors at -10 and +10 per cent (degrees)	Held	
EXPLORER_DIVE_CHECK 20261017_211712	E15	the cue’s entry errors at -10, -5, -2, +2, +5, +10 per cent (degrees)	Held	
EXPLORER_DIVE_CHECK 20261017_211712	E16	the cue’s moment at +10 per cent (s)	Held	
EXPLORER_DIVE_CHECK 20261017_211712	E17	the cued miss over the clock’s at +10 per cent	Held	
EXPLORER_DIVE_CHECK 20261017_211712	W3	a cued dive under 100 ms	Held (median 27.15 ms)	
EXPLORER_ROUTINE_CHECK 20261017_211712	R1	its words in the tool’s own header	Held	
EXPLORER_ROUTINE_CHECK 20261017_211712	R2	its words in the tool’s own header	Held	
EXPLORER_ROUTINE_CHECK 20261017_211712	R3	its words in the tool’s own header	Held	
EXPLORER_ROUTINE_CHECK 20261017_211712	G1	arms swung forward: within 10 degrees of the record’s entry	Held (5.2446)	

Run	Label	The check, as the run prints it	Verdict	Misses row
EXPLORER_ROUTINE_CHECK 20261017_211712	G2	the arm circle, nothing carried, turns the trunk forward 8 to 30 degrees	Held (-21.3929)	
EXPLORER_ROUTINE_CHECK 20261017_211712	G2b	at the nominal spin it enters 8 to 35 degrees over	Held (-18.7706)	
EXPLORER_ROUTINE_CHECK 20261017_211712	G3	the rip window 28.785 ms to 0.01 ms	Held (28.784943 ms)	
EXPLORER_ROUTINE_CHECK 20261017_211712	W4	a routine flight under 100 ms	Held (median 32.55 ms)	
EXPLORER_CAT_CHECK 20261017_211715	E18	the full circuit	Held	
EXPLORER_CAT_CHECK 20261017_211715	E19	the bends	Held	
EXPLORER_CAT_CHECK 20261017_211715	E20	the radii	Held	
EXPLORER_CAT_CHECK 20261017_211715	E21	the half circuits	Held	
EXPLORER_CAT_CHECK 20261017_211715	E22	bend and straighten: no turn	Held	
EXPLORER_CAT_CHECK 20261017_211715	E23	backward, and no momentum	Held	
EXPLORER_CAT_CHECK 20261017_211715	W4	within 0.001 degrees at 1,000 steps	Held	
EXPLORER_CAT_CHECK 20261017_211715	W5	under 20 ms	Held	
EXPLORER_CAT_TWISTFREE_CHECK 20261018_031532	F1	the half trick repeats CAT_TWIST	Held	
EXPLORER_CAT_TWISTFREE_CHECK 20261018_031532	F2	the full circuits repeat CAT_TWIST	Held	
EXPLORER_CAT_TWISTFREE_CHECK 20261018_031532	F3	every twist-free landing level	Held	
EXPLORER_CAT_TWISTFREE_CHECK 20261018_031532	F4	the twist reading: a full turn for the toy, none twist-free	Held	
EXPLORER_CAT_TWISTFREE_CHECK 20261018_031532	F5	the toy unchanged	Held	
EXPLORER_CAT_TWISTFREE_CHECK 20261018_031532	F6	no clock	Held	
EXPLORER_CAT_TWISTFREE_CHECK 20261018_031532	Q1	the best bend for one full circuit lies between 70 and 80	Held	
EXPLORER_CAT_TWISTFREE_CHECK 20261018_031532	W7	under 30 ms	Held	
EXPLORER_ENGINE_CHECK 20261018_031535	E1	the rigid forward turn, human legs (degrees)	Held	
EXPLORER_ENGINE_CHECK 20261018_031535	E2	one arm circle, nothing carried (degrees)	Held	
EXPLORER_ENGINE_CHECK 20261018_031535	E3	one human stride, nothing carried (degrees)	Held	
EXPLORER_ENGINE_CHECK 20261018_031535	E4	the arms alone, 1, 2 and 3 circles (share)	Held	

Run	Label	The check, as the run prints it	Verdict	Misses row
EXPLORER_ENGINE_CHECK 20261018_031535	E5	the legs alone, 3 strides, knee 1.57 and 2.36 (share)	Held	
EXPLORER_ENGINE_CHECK 20261018_031535	E6	two circles with two human strides (share)	Held	
EXPLORER_ENGINE_CHECK 20261018_031535	E7	two circles with two deep strides (share)	Held	
EXPLORER_ENGINE_CHECK 20261018_031535	E8	the same routine, nothing carried, in 0.9 s and 0.45 s (degrees)	Held	
EXPLORER_ENGINE_CHECK 20261018_031535	E9	diver: pike loop at amplitude 0.6 (degrees a loop)	Held	
EXPLORER_ENGINE_CHECK 20261018_031535	E10	diver: zigzag loop at amplitude 0.6 (degrees a loop)	Held	
EXPLORER_ENGINE_CHECK 20261018_031535	E11	diver: fall from 3 m and 10 m (s)	Held	
EXPLORER_ENGINE_CHECK 20261018_031535	W1	a jumper flight at 6,000 steps under 100 ms	Held (median 89.59 ms)	
EXPLORER_ENGINE_CHECK 20261018_031535	W2	at 1,500 steps E1 to E8 all within tolerance	Held	
EXPLORER_DIVE_CHECK 20261018_031538	E12	the nominal spin (kg m^2/s)	Held	
EXPLORER_DIVE_CHECK 20261018_031538	E13	the cue’s angle at 1.00 s (degrees)	Held	
EXPLORER_DIVE_CHECK 20261018_031538	E14	the clock’s entry errors at -10 and +10 per cent (degrees)	Held	
EXPLORER_DIVE_CHECK 20261018_031538	E15	the cue’s entry errors at -10, -5, -2, +2, +5, +10 per cent (degrees)	Held	
EXPLORER_DIVE_CHECK 20261018_031538	E16	the cue’s moment at +10 per cent (s)	Held	
EXPLORER_DIVE_CHECK 20261018_031538	E17	the cued miss over the clock’s at +10 per cent	Held	
EXPLORER_DIVE_CHECK 20261018_031538	W3	a cued dive under 100 ms	Held (median 44.45 ms)	
EXPLORER_ROUTINE_CHECK 20261018_031539	R1	its words in the tool’s own header	Held	
EXPLORER_ROUTINE_CHECK 20261018_031539	R2	its words in the tool’s own header	Held	
EXPLORER_ROUTINE_CHECK 20261018_031539	R3	its words in the tool’s own header	Held	
EXPLORER_ROUTINE_CHECK 20261018_031539	G1	arms swung forward: within 10 degrees of the record’s entry	Held (5.2446)	
EXPLORER_ROUTINE_CHECK 20261018_031539	G2	the arm circle, nothing carried, turns the trunk forward 8 to 30 degrees	Held (-21.3929)	
EXPLORER_ROUTINE_CHECK 20261018_031539	G2b	at the nominal spin it enters 8 to 35 degrees over	Held (-18.7706)	
EXPLORER_ROUTINE_CHECK 20261018_031539	G3	the rip window 28.785 ms to 0.01 ms	Held (28.784943 ms)	
EXPLORER_ROUTINE_CHECK 20261018_031539	W4	a routine flight under 100 ms	Held (median 60.16 ms)	

Run	Label	The check, as the run prints it	Verdict	Misses row
EXPLORER_CAT_CHECK 20261018_031543	E18	the full circuit	Held	
EXPLORER_CAT_CHECK 20261018_031543	E19	the bends	Held	
EXPLORER_CAT_CHECK 20261018_031543	E20	the radii	Held	
EXPLORER_CAT_CHECK 20261018_031543	E21	the half circuits	Held	
EXPLORER_CAT_CHECK 20261018_031543	E22	bend and straighten: no turn	Held	
EXPLORER_CAT_CHECK 20261018_031543	E23	backward, and no momentum	Held	
EXPLORER_CAT_CHECK 20261018_031543	W4	within 0.001 degrees at 1,000 steps	Held	
EXPLORER_CAT_CHECK 20261018_031543	W5	under 20 ms	Held	
EXPLORER_CAT_ROOM_CHECK 20261018_031545	R1	the routine of record, a loop with no clock	Held	
EXPLORER_CAT_ROOM_CHECK 20261018_031545	R2	no swing, no turn	Held	
EXPLORER_CAT_ROOM_CHECK 20261018_031545	R3	the other way round, its mirror	Held	
EXPLORER_CAT_ROOM_CHECK 20261018_031545	P4	the tilt grows with the bend	Held	
EXPLORER_CAT_ROOM_CHECK 20261018_031545	P5	the tilt is the half circuit’s V movement, the body a half turn	Held	
EXPLORER_CAT_ROOM_CHECK 20261018_031545	W6	under 30 ms	Held	
EXPLORER_TRUCK_CHECK 20261101_015149	H1	the port repeats TRUCK_ROAD, 465 settings, 6,045 riders	Held	
EXPLORER_TRUCK_CHECK 20261101_015149	H2	the port repeats LIGHT_LANES, 324 settings	Held	
EXPLORER_TRUCK_CHECK 20261101_015149	H3	the port repeats RING_WIDTH, 564 settings, its widths, middles and ratios	Held	
EXPLORER_TRUCK_CHECK 20261101_015149	H4	the room’s data and printed numbers against the five runs	Held	
EXPLORER_TRUCK_CHECK 20261101_015149	H5	the step test, mass 2.0 at steps of 0.01	Held	
EXPLORER_TRUCK_CHECK 20261101_015149	W8	one flight under 60 ms	Held	
EXPLORER_LATCH_CHECK 20261101_042950	K1	the compact data equals the six runs’ json, value for value	Held	
EXPLORER_LATCH_CHECK 20261101_042950	K2	the six runs’ 99 table lines printed again from the data, the same	Held	
EXPLORER_LATCH_CHECK 20261101_042950	W9	every number in the six knob rows agrees with the data at its printed rounding	Held	
EXPLORER_SNAKE_CHECK 20261101_171827	K1	the compact data equals the two runs’ json, value for value	Held	
EXPLORER_SNAKE_CHECK 20261101_171827	K2	the two runs’ 13 result lines printed again from the data, the same	Held	

Run	Label	The check, as the run prints it	Verdict	Misses row
EXPLORER_SNAKE_CHECK 20261101_171827	K3	every claim in the three knob rows agrees with the data at its printed rounding; expected to miss on the 4.7	Missed	564
EXPLORER_RING_CHECK 20261102_045348	K1	the room’s data equals the five runs’ json, value for value	Held	
EXPLORER_RING_CHECK 20261102_045348	K2	the four runs’ 117 lines about the room’s cases printed again from the data, the same	Held	
EXPLORER_RING_CHECK 20261102_045348	K3	every claim in the four ring rows agrees with the data as the rows now stand; expected to miss on ‘over 400 time units’	Missed	576
EXPLORER_BELL_CHECK 20261103_025532	K1	the room’s data equals the seven runs’ json, value for value	Held	
EXPLORER_BELL_CHECK 20261103_025532	K2	the seven runs’ 134 lines about the room’s cases printed again from the data, the same	Held	
EXPLORER_BELL_CHECK 20261103_025532	K3	every claim in the bell row agrees with the data as the row now stands; expected to miss on ‘a longer memory to make them pay more’	Missed	580
EXPLORER_HEAT_CHECK 20261103_065559	K1	the room’s data equals the two runs’ json, value for value	Held	
EXPLORER_HEAT_CHECK 20261103_065559	K2	the two runs’ 44 lines about the room’s cases printed again from the data, the same	Held	
EXPLORER_HEAT_CHECK 20261103_065559	K4	the rod in JavaScript repeats every stored case: the brick’s to the last bit, the hump’s within 1e-12, times exactly	Held	
EXPLORER_HEAT_CHECK 20261103_065559	K3	every claim in the heat row agrees with the data as the row now stands; expected to miss on ‘1.485 lengths of heat’s travel after the quench’	Missed	584
EXPLORER_NEEDLE_CHECK 20261104_222610	K1	the room’s data equals the orbit file bit for bit; the page data of 13 October equals it at its own rounding	Held	
EXPLORER_NEEDLE_CHECK 20261104_222610	K2	STRETCH_READS’s 43 lines about the two orbits printed again from the data, with the allowance set before	Held	
EXPLORER_NEEDLE_CHECK 20261104_222610	K3	the book’s 20 claims on the needle’s stretch agree with the data; expected to miss on (1) and (5) and nowhere else	Missed	591
EXPLORER_GRID_CHECK 20261107_073202	K1	the room’s data equals the runs’ data	Held	
EXPLORER_GRID_CHECK 20261107_073202	K2	SWING_GRID’s 15 untimed lines printed again from the room’s data	Held	
EXPLORER_GRID_CHECK 20261107_073202	K4	the engine repeats every stored case; the relock lines within their brackets	Held	
EXPLORER_GRID_CHECK 20261107_073202	K3	every claim of the grid row and the two paragraphs agrees with the data; judged, not a lean	Missed	613
EXPLORER_SOUP_CHECK 20261108_070632	K1	the room’s data equals the runs’ data	Held	
EXPLORER_SOUP_CHECK 20261108_070632	K2	the runs’ 159 untimed lines printed again from the room’s data	Held	
EXPLORER_SOUP_CHECK 20261108_070632	K3	every claim of the seven rows and the three paragraphs agrees with the data; judged, not a lean	Missed	618
EXPLORER_METER_CHECK 20261108_182139	K1	the room’s data equals the runs’ data	Held	

Run	Label	The check, as the run prints it	Verdict	Misses row
EXPLORER_METER_CHECK 20261108_182139	K2	the runs’ 60 untimed lines printed again from the room’s data	Held	
EXPLORER_METER_CHECK 20261108_182139	K3	every claim of the three rows and the paragraphs agrees with the data; judged, not a lean	Missed	621
EXPLORER_COLUMN_CHECK 20261108_194717	K1	the room’s data equals the runs’ data	Held	
EXPLORER_COLUMN_CHECK 20261108_194717	K2	LIMIT_AUDIT’s 101 untimed lines printed again from the room’s data	Held	
EXPLORER_COLUMN_CHECK 20261108_194717	K3	every claim of the four rows and the paragraphs agrees with the data; judged, not a lean	Missed	622
EXPLORER_FACE_CHECK 20261108_202547	K1	the room’s data equals the run’s data	Held	
EXPLORER_FACE_CHECK 20261108_202547	K2	the run’s 11 untimed lines printed again from the room’s data	Held	
EXPLORER_FACE_CHECK 20261108_202547	K3	every claim of the row and the paragraph agrees with the data; judged, not a lean	Held	

The publishing tool’s checks

12 lines: the book builder’s own checks of what it prints.

Run	Label	The check, as the run prints it	Verdict	Misses row
BUILD_PAPER_CHECK 20261010_201918	B1	made-up case: plain short by more than 100, fixed by none	Missed	—
BUILD_PAPER_CHECK 20261010_201918	B1b	made-up case with 150-character names: plain short by more than 100, fixed by none	Held	
BUILD_PAPER_CHECK 20261010_201918	B2	data ledger: the same	Held	
V13_FIDELITY 20261013_202203	F1	the self-test passes	Held	
V13_FIDELITY 20261013_202203	F2	every file builds with exit 0	Missed (62 of 66)	188
V13_FIDELITY 20261013_202203	F3	no file loses letters it did not lose before	Missed (64 of 66)	—
V13_FIDELITY 20261013_203154	F1	the self-test passes	Held	
V13_FIDELITY 20261013_203154	F2	every file builds with exit 0	Missed (64 of 66)	188
V13_FIDELITY 20261013_203154	F3	no file loses letters it did not lose before	Missed (65 of 66)	—
V13_FIDELITY 20261013_204846	F1	the self-test passes	Held	
V13_FIDELITY 20261013_204846	F2	every file builds with exit 0	Missed (64 of 66)	188
V13_FIDELITY 20261013_204846	F3	no file loses letters it did not lose before	Missed (65 of 66)	—

The checks of the kit and the records

5 lines: checks of files and records rather than of a toy, read so by hand from each tool’s own words, since a run’s name and its first line could not say.

Run	Label	The check, as the run prints it	Verdict	Misses row
NEEDS_AUDIT 20261010_205839	N1	every file a fold tool opens is named in its launcher (its bat file)	Held	
NEEDS_AUDIT 20261010_205839	N2	at least one launcher (bat file) names a file its tool never opens	Missed	550
BENCHMARK_TAGS 20261020_055158	S1a	each reader's universal count from 23 to 31	Held	
BENCHMARK_TAGS 20261020_055158	S1b	the readers agree on at least 36 rows	Held	
BENCHMARK_TAGS 20261020_055158	S1r	either reader under 23	Missed	550

Part 3. Every miss, with its reason

656 misses, numbered in the order they were found, from the first sitting to the last run. Each says what happened, what kind of miss it was, what it taught and whether a redesign was named, and whether the lesson has been proved by a run. These are the misses file's rows: a missed prediction in Part 1 whose Misses row column shows a dash has no row here, and its reason, where one was written, is in the book's text.

Misses 1 to 50

1. An all-or-nothing blade stalled three times: the spring bead at rest, the very lossy bead, the turning blade. Once the charge was off, nothing turned it back on.

Kind: NATURE, first seen as a TOOL FAULT. *Lesson:* A blade should never go fully dark. A blade that keeps a fifth of its charge did not stall. This may be a real effect to look for: any harvester that needs to be moving before it can switch has a cold-start problem. Specimen (third sitting): the single-phase induction motor, which has no starting torque because at rest the field it sees is a standing wave pushing both ways equally, and is started by a second winding a quarter-cycle out of step, cut out once up to speed; a failed start capacitor leaves it humming and still, a blade gone fully dark. Bear reports the stalled, humming rotor as familiar from the field, not as expert testimony. The crystal-set detector tie is recall only. *Proved?* Stall seen in 3 runs; two cures proved by run: a blade that keeps a fifth of its charge, and Bear's pilot tone (row 17); the motor specimen is a familiar one pending a bench check; the detector tie is recall only.

2. The snake went 0.1508 forward and 0.1472 backward.

Kind: TOOL FAULT. *Lesson:* In a partner that forgets, backward must undo forward exactly. Using that as a free check exposed a drifting stepping rule. Corrected: 0.14898 both ways; the gripless swimmer gives exactly 0, not 0.002. The tool itself was replaced at the third sitting. The Field Guide already carries this check (mirror the pipeline). *Proved?* Proved by run; tool replaced (run_LOOP_ATLAS_SNAKE_20261002b).

3. Claude said a loop cares about speed only when its partner remembers.

Kind: WRONG GUESS. *Lesson:* Speed matters in two ways: memory, or a pace limit. Four forgetting-partner toys cared about speed through a pace limit. *Proved?* Shown by count over the tag table.

4. Claude expected a listener as noisy as the soup to catch nothing.

Kind: WRONG GUESS. *Lesson:* The catch fades gently (72% kept at equal noise) and being late hurts far more than being noisy. *Proved?* Proved by run.

5. Claude taught that a quarter cycle is the special gap.

Kind: WRONG GUESS. *Lesson:* True for two knobs. For a chain the best step is about half a wave along the body. *Proved?* Proved by run (worm chain).

6. An event trigger for the bell (squeeze when the flow has peaked) did no better than a clock.

Kind: WRONG GUESS. *Lesson:* One trigger rule failing does not settle event against clock. In the wave soup the event-driven blade won outright. The trigger has to listen to the right thing. *Proved?* Both seen in runs.

7. Two free bells never passed each other, so the leap-through was never tested.

Kind: TOY LIMIT. *Lesson:* Point bells with nothing holding them together drift apart. A tether or a controller is part of the idea, not an extra. *Proved?* Seen in run; the idea itself is still HELD.

8. The three bell toys rest on a rule Claude assumed (a kick is stronger into oncoming water).

Kind: TOY LIMIT. *Lesson:* Every bell result stands or falls with that rule. *Proved?* Not proved; needs Bear's solver.

9. The draft horse reached a speed of 9.7 where the waves travel at 1.

Kind: TOY LIMIT. *Lesson:* The toy lets the pull go on for ever and does not count what the soup can supply. A pull is a fair reading of that run; a speed is not. *Proved?* Stated.

10. In the wave soup the square rule held to a strength of 0.3 and then jumped at 0.4.

Kind: NATURE, probably. *Lesson:* At that strength some beads begin to be caught and carried by single waves. The rule changes where the bead can keep up with a wave. *Proved?* Seen in 1 run; cause not tested. [Settled in part, 9 October: not single waves (none comes near the wave speed) but the whole right-going comb, which moves as one because every wave travels at the same speed. The jump sits where the strength times the comb's tallest crest reaches the wave speed: 1 over the crest is 0.355 to 0.390 in the record's eight soups (SOUP_JUMP 20261009_045400, J0). In each soup the onset comes 4 to 13 per cent above that line, never below (row 105).]

11. The three-link swimmer fell below the square rule at big strokes.

Kind: NATURE, probably. *Lesson:* The room-inside-the-loop rule is a small-stroke rule. What limits a big loop is not known to us yet. *Proved?* Seen in run; cause not tested. [Settled, found 9 October: a run of 3 October answered it and nothing marked this row. The grip is a hill, highest at the straight body; a big lap encloses a disk that reaches down its slopes, and the grip summed over each disk matches the travel within 2 per cent to 60 degrees, where the square rule overshoots by 25 (BIG_LOOP 20261003_175803, chapter 3).]

12. Two rings merged into one travelled faster than two rings leapfrogging.

Kind: Not looked at. *Lesson:* It cuts against the leap-through idea and was reported once and dropped. *Proved?* Seen in 1 run.

13. The first swing run (20 swings) and the run of record (6 swings) gave very different numbers.

Kind: Not looked at. *Lesson:* Claude put it down to the swing drifting out of step as it grows and never tested that. *Proved?* Not proved.

14. Graphs of push against distance did not land for Bear, twice. The butter-knife picture did not land either.

Kind: TEACHING MISS. *Lesson:* Hands and moving pictures first. The rubber band against a sink of water worked; the wing snapshots worked. *Proved?* Bear's report.

15. Claude misread Bear's tesseract sketch twice before getting the two-jelly leap-through.

Kind: TEACHING MISS, the other way round. *Lesson:* Ask what a drawing shows before building on it. *Proved?* Bear's report.

16. A batch of files went out stale because an update script stopped partway and the send ran anyway.

Kind: TOOL FAULT, in the shop and not in a toy. *Lesson:* Check that every file changed before sending. Corrected copies were sent the same turn. *Proved?* Fixed.

17. Bear's cure for the stall: a small steady feed from the ship, a pilot tone.

Kind: NATURE, Bear's idea. *Lesson:* A pilot of one hundredth of the soup's push woke both stalled beads completely (0.000 to 0.145, and 0.000 to 0.225). A pilot as strong as the soup cut the catch (to 0.132 and 0.099), because the bead then follows the pilot's clock and stops listening to the soup. *Proved?* Proved by run.

18. The listening blade got a steady pull from an even soup with its charge appearing from nowhere.

Kind: TOY LIMIT, first read as a finding. *Lesson:* Charge is conserved. With the opposite charge on the ship, the pull is exactly zero at zero tether and grows only with the tether's length against the wave. The old listener's 0.124 was a free lunch the toy did not bill for. The Field Guide's common-mode rule names the principle; the toy gives the price. *Proved?* Proved by run (counter-charge toy).

19. Claude filed the Latch Atlas's powered lock and the Loop Atlas's ride on a difference as separate rows.

Kind: WRONG GUESS, a filing. *Lesson:* Both are driven from outside and last only while the drive runs; the one difference is whether a whole number is held. One family, two members. Bear's call. *Proved?* Settled by reading both books.

20. Bear tied the soup toy's local tilt to the polarity issue in the Sun work.

Kind: WRONG GUESS, Bear's, settled by reading. *Lesson:* The Sun paper's turning is the instrument's (about 3 degrees, in the instrument's frame, reproduced by a one-layer fit); the soup's lopsidedness is physical. Hunter and hunted. The paper's mirror test is kept instead. *Proved?* Settled by reading Dancing on the Sun in full.

21. Claude proposed the mirror audit as new after reading the Sun paper.

Kind: WRONG GUESS, about novelty. *Lesson:* It is a Field Guide rule already, with thresholds frozen. The Guide hands the test; we run it. *Proved?* Settled by reading the Guide in full.

22. The bell pair and bell tether quote single settings in a medium with memory.

Kind: TOOL FAULT, by the Guide's rule. *Lesson:* The Guide's memory-reading rule asks for at least five phase points before a per-cycle number is quoted. The memory bell met it; these two did not. Swept at the third sitting with ten points per line: the tethered pair at gaps 0.5 and 1.0 is exactly the case the rule is written for (a tenth of a period moves the speed by 0.037 to 0.049, a sixth of its value), so its one gain is reported as a slope and a crossing; at gaps 1.5 and beyond, and for the free pair, the lines are flat and the single values stand. *Proved?* Swept (run_LOOP_ATLAS_BELL_SWEEP_A/B_20261002); fence lifted.

23. The time mirror missed by half a turn on the locked rotor and conveyor (pace 0.5) and by 3.8 on the clock blade, while slipping cases retraced to 1e-13.

Kind: NATURE, first read as a possible TOOL FAULT. *Lesson:* A lock forgets. A caught bead or a locked needle is an attractor; run backward an attractor is a repeller, and no stepping rule retraces through it. So the time mirror is a stepping-rule test only where nothing is locked, and a lock detector where something is. Agrees with the Latch Atlas W8 (a stall has no memory of how it was reached). *Proved?* Proved by run: the clock blade's miss falls 3.83 -> 0.48 -> 0.12 -> 0.0067 as strength falls 0.3 -> 0.2 -> 0.15 -> 0.1; the plain bead misses 0.00000 at 0.3 and 50.5 at 0.6 (run_LOOP_ATLAS_MIRROR_AUDIT_20261002b).

24. The draft horse with no soup read +0.00048 and flipped sign under the rider mirror, which the parity test calls a ride.

Kind: TOY LIMIT, and a trap for the audit. *Lesson:* It is the starting nudge: 0.01 of momentum over an outfit of 21 is 0.000476. Parity cannot tell a reflected starting condition from a reflected engine; only the arithmetic can. Every blade toy now prints its nudge's momentum beside its result. *Proved?* Proved by arithmetic and run (run_LOOP_ATLAS_MIRROR_AUDIT_20261002b).

25. The first pass of the mirror audit read the turning blade's twist as red (eps 0.970).

Kind: WRONG GUESS by the auditor. *Lesson:* The written expectation said the twist should flip under a reflected soup. The twist is the blade's, so the reflected soup leaves it, and it did (+0.0332 -> +0.0312, eps 0.030). The auditor's expectation is a registered prediction too, written before the run, and the run corrects it. *Proved?* Both passes kept; every number identical by diff; only the expectation label changed.

26. The hinge-geometry tool gave a long-headed swimmer 0.078 and a long-tailed one 0.085 body lengths per lap, as if the body knew which end was which.

Kind: TOOL FAULT. *Lesson:* The two bodies are mirror images and must travel the same. On the two-slope rule both give 0.076. A stepping rule can invent an asymmetry, and a mirror pair inside the toy (head for tail) is a free check, like forward for backward. *Proved?* Proved by run (run_LOOP_ATLAS_HINGE_GEOMETRY_20261002b against run_LOOP_ATLAS_HINGE_GEOMETRY_20261002, the first version, retired and never kept by any crate; said so on 10 November, when P62 found it cited and absent, row 651).

27. The 11-hinge swimmer's best timing came out at 28 degrees, against Claude's guess of below 25; the 5-hinge shift at 7 degrees and 2.6 per cent, against under 7 and 2.

Kind: WRONG GUESS, narrow. *Lesson:* More hinges want more of a wave along the body: 0.42 of a wave at 2 hinges up to 0.86 at 11. The coarse grid had hidden no better timing. *Proved?* Proved by run (HINGE_FINE 20261004_080159).

28. Row 13 settled: the swing's numbers at 20 swings differ from those at 6 because a big swing is slower (its period 1.0019, 1.031 and 1.073 of the small-swing period at 10, 40 and 60 degrees), so a clocked rider falls out of step: 63.16 degrees at swing 10, back to 8.91 by swing 20. The first rider timed to the swing's own motion, with no lead, did nothing.

Kind: NATURE (row 13's excuse proved) and WRONG GUESS (the rider). *Lesson:* A rider who pumps on the swing's own motion never falls out of step, but only with a lead: at a lead of 0 or 180 degrees it is its own mirror image about the bottom and does exactly nothing. *Proved?* Proved by run (SWING_TWENTY 20261004_075715 and _B 20261004_075902).

29. The first polarisation run defined the light's phase with the opposite sign to the loop's orientation.

Kind: TOOL FAULT. *Lesson:* Fix every sign convention against the area's orientation before reading a sign. The magnitudes were identical; the run was retired. *Proved?* Fixed (POLARISATION_LOOP 20261004_071425; 071354 retired).

30. Row 12 reversed: the merged ring had carried 1.41 to 1.73 times the leapfrogging pair's energy for the same push. Given exactly the pair's budget, one ring travels at 0.84 to 0.88 of the pair's speed.

Kind: NATURE, a miss that reversed. *Lesson:* Compare on one budget before calling a winner. On one budget the leapfrogging pair is 14 to 19 per cent faster, which supports the leap-through idea it had seemed to cut against. *Proved?* Proved by run (RING_BUDGET 20261004_080107).

31. Row 8 settled: asked the solver's own question, the assumed kick rule moved the bell the right way but a tenth as far and by a different ring, and Claude's registered prediction missed in sign.

Kind: TOY LIMIT and WRONG GUESS. *Lesson:* In the solver more than half of the effect is the resting bell carried forward by its own wake, which a rule that acts only at kicks cannot do. The bell toys' numbers are numbers of the rule, not findings. *Proved?* Proved by run (KICK_RULE_CHECK 20261005_031525).

32. The leader's own ring, as the draft test seeded it, pushed a closed follower back (51.671 mN·s on the leash, 1.337 radii free); the open follower's one-way valve ran away within a tenth of a second.

Kind: NATURE (the push) and TOOL LIMIT (the valve). *Lesson:* A ring wider than the follower's door is a tow, not a draft. A valve that switches on the bare sign of the flow chatters under a strong ring; it needs a band. *Proved?* Run once (DRAFT_TEST 20261008_134707); parked by Bear's ruling, its two follow-ons named.

33. A fidelity check on a chaotic needle missed (0.557209 against 0.526206): one long call had rounded the drive's phase differently from the parent's sequence of calls, and chaos amplified the difference.

Kind: TOOL FAULT, of method. *Lesson:* On a chaotic state, fidelity needs the same sequence of calls, not only the same code. Repeating the parent's sequence reproduced 0.526206 to the digit. *Proved?* Proved by run (HIDDEN_SCISSORS_F0B 20261005_064308).

34. Needles slipping just below the lock's edge looked like a second state waiting beside the lock (registered guesses H1a and H1b missed).

Kind: WRONG GUESS. *Lesson:* It was a transient: held long, every needle below the edge sat down. A needle does not slow before it sits; the time per slip stretched only because the crowd thinned. *Proved?* Proved by run (HIDDEN_SCISSORS_SLIVER 20261005_145137).

35. A lifetime ratio could not be read: every needle had already locked on the walk down to its pace.

Kind: TOOL LIMIT, of design. *Lesson:* Read lifetimes from needles placed straight onto the pace, so no time is spent locking on the way. *Proved?* Follow-on registered, open.

36. Claude's guess that chaos closes scissors failed (three chaotic cells with the scissors open), and a threshold of 0.01 sat below the instrument's own floor.

Kind: WRONG GUESS and TOOL LIMIT. *Lesson:* Chaos is not enough: held 10,000 time units, nothing drained; the chaotic spins are attractors. A threshold must sit above the instrument's floor (a Field Guide rule). *Proved?* Proved by runs (PULSED_SCISSORS 20261005_155747, _LONG 20261005_170134).

37. A note this book sent to *A Rider in Two Waves* used the junction's retrapping law, 4/(pi times the square root of memory), at memory 1, where it gives 1.273, above the fall line.

Kind: WRONG GUESS, a law used outside its range. *Lesson:* That law is the large-memory limit. At memory 1 the needle's own climb-back line is 0.9643. *Proved?* Corrected by reading (a note from *A Rider in Two Waves*).

38. The record's climb-back line, 0.964327165, was 4.05e-8 high.

Kind: TOOL LIMIT. *Lesson:* A nine-digit figure from a fixed-step integrator is not nine digits. Four adaptive integrators from three starts agree on 0.96432712447 within 2.2e-12. *Proved?* Proved by run (CLIMB_BACK_LINE_CHECK 20261006_041633).

39. A registered rig check compared two machines that only looked alike (memory 1.0 against the co-turning wave in one, 0.924 in the other).

Kind: WRONG GUESS, in the registration. *Lesson:* Check that two machines are one equation before calling their agreement a rig check (a lesson sent to the Field Guide). *Proved?* Seen in run (GAP_DIAL 20261005_170519).

40. The climb-back blade fell short of the split loop's band at the larger clocks, and the shortfall survived a hold fifteen times longer.

Kind: NATURE. *Lesson:* The band's top is a ceiling the blade touches only while the band is narrow. *Proved?* Proved by runs (GAP_DIAL_2 20261005_172939, GAP_DIAL_3 20261005_194344).

41. A full run started by mistake after its smoke run had failed: the launch line, chained after a pipe, ran anyway.

Kind: TOOL FAULT, in the shop. *Lesson:* A pipe's exit status is its last command's. Never chain a launch after a pipe. The run was stopped within seconds, the smoke fixed and passed, and the run relaunched. *Proved?* Fixed.

42. The hitch-kick's arm circle turned the trunk 13.9 degrees, against a guessed 15 to 45, and its legs 12.5 times the arms, against 2.5 to 5.

Kind: WRONG GUESS and TOY LIMIT. *Lesson:* The arm guess was a hand estimate; the toy's leg circle, a straight leg swung full circle, is not a human motion. Redesigned with knees and two legs out of step. *Proved?* Seen in run (HITCH_KICK 20261006_011635); follow-on registered.

43. The climb-back reader, counting a pace as spinning only when 8 of 16 needles spun, missed pockets of 6 and 3 needles that kept spinning below the edge.

Kind: TOOL LIMIT of the reader, then NATURE. *Lesson:* The pockets are a 1:2 gear beside the 1:1 lock at one push: a remembering needle holds either, decided by where it came from. *Proved?* Proved by runs (POCKET_HOLD 20261005_195333, POCKET_GEARS 20261006_052012).

44. A relay test read nothing: its rooms were copied from *A Rider in Two Waves*'s machine, where our needles' onsets sat about a hundred times lower.

Kind: WRONG GUESS, in the registration. *Lesson:* Look where your own machine's onsets sit before borrowing another machine's grid. The redesign moved the rooms down and read. *Proved?* Proved by run (ROOM_NEEDLE 20261006_013956, ROOM_NEEDLE_2 20261006_014758).

45. *A Rider in Two Waves*'s room rule, slope −1, did not carry to our needle relay: slopes −0.48 to −0.60.

Kind: NATURE. *Lesson:* The rule is a law of *A Rider in Two Waves*'s stage. What our needles add is −1/2 off the beat, steepening toward −1 only where the downstream's own restoring rate outruns the beat. *Proved?* Proved by run (ROOM_NEEDLE_2).

46. A bound this book offered for the rider of *A Rider in Two Waves* assumed a running state to climb back from; at masses 1.0 and 1.5 there is none, and the bound failed at mass 0.75 in *A Rider in Two Waves*'s own run.

Kind: WRONG GUESS. *Lesson:* A bound built on a picture fails where the picture's object does not exist. Hold riders at one drive; a ramp cannot test what holding can. The bound is withdrawn. *Proved?* Proved by runs (CLIMB_HOLD 20261006_065510; a note from *A Rider in Two Waves*).

47. At one push, three of eight needles were caught by the passing wave in a running lock, where the registered clause expected all eight in the gear.

Kind: NATURE. *Lesson:* Both of Bear's pictures appear: the rope's pull helps the needle in the gear, and some needles are trapped by the wave. In *A Rider in Two Waves*'s frame the trap is its second gear. *Proved?* Proved by run (FERRY_ROPE 20261006_045826).

48. A ramped rider left the lock later than a tolerance allowed, and a slowing runner was caught by the second gear on the way down.

Kind: TOOL LIMIT (the tolerance) and NATURE (the catch). *Lesson:* A tolerance must leave room for a ramp's lag. A runner slowing through the clock's rate can be caught. *Proved?* Proved by runs (SWAP_STATE 20261006_052545, SWAP_STATE_2 20261006_052914).

49. A run's last window was filed under the hold: the tool's clock, summed in steps of 0.02, passed the ramp's end 4.7e-7 late.

Kind: TOOL FAULT. *Lesson:* Compare changes of state, not a run's last window; a clock summed in floating steps drifts. *Proved?* Seen in run (SWAP_STATE_2); nothing in the result moved.

50. Two riders failed a period-8 test at one drive; a spread test missed at three cells.

Kind: TOOL LIMIT, of hold and reader. *Lesson:* Riders near a doubling close on their orbit slowly and need longer holds (all eight repeat at 1,500 passings); turns per passing read end to end carries the spin's wobble, and read as a slope the riders agree. *Proved?* Proved by runs (RUN_STROBE 20261006_170149, RUN_STROBE_2 20261006_174220).

Misses 51 to 100

51. An edge read from a ramp gave the second gear's ghost a slope of 0.36 to 0.42, where a fold's law says one half.

Kind: TOOL LIMIT. *Lesson:* Find edges by holding, not ramping. From held edges the ghost obeys the square-root law, 0.48, over two and a half decades. *Proved?* Proved by run (RUN_STROBE_2).

52. No drive on a grid of 0.0005 repeated as the 3/2 tongue at heavy mass.

Kind: TOOL LIMIT, then a redesign. *Lesson:* The tongue is a few times 1e-5 wide and riders inside it settle too slowly to sort. Follow the orbit to its two folds by Newton's method instead. *Proved?* Seen in run (RUN_STROBE_2); redesigned as STROBE_NEWTON.

53. The first Newton follower of the strobe's orbits ran its secant away to negative drives and missed a fold's tolerance.

Kind: TOOL FAULT. *Lesson:* A search needs a bracket. The faulty version was retired with its readings disclosed, and its successor reproduced every shared number. *Proved?* Fixed (STROBE_NEWTON version 2, 20261006_184350).

54. A period-8 orbit's leading multiplier came out +0.94, not negative as guessed, and the end of its band was first called a fold.

Kind: WRONG GUESS, then a misreading. *Lesson:* It was the period-4 orbit's doubling, where the period-8 orbit is born with multiplier +1, the square of its parent's −1. Before calling a +1 a fold, look at the parent. *Proved?* Corrected by run (FOLD_OR_FLIP 20261007_050721, with a note from *A Rider in Two Waves*).

55. Claude guessed that a groove's width is set by its place, and that weight hardly moves it; read in drive, the order of the grooves and the effect of weight came out otherwise.

Kind: WRONG GUESS, and a ruler. *Lesson:* A width read in drive carries the staircase's slope. Read in the rider's own units, turns per passing, grooves lower on the record are wider and heavier riders hold wider grooves, as blind tests then confirmed. *Proved?* Proved by runs (GROOVES 20261006_193519, GROOVES_2 20261006_202753, GROOVES_3 20261006_204201).

56. An orbit follower carried its angle at about 2,360 radians, so its noise floor rose above its own tolerance and it failed on several grooves.

Kind: TOOL FAULT. *Lesson:* Reduce an angle to one turn at every entry. *Proved?* Fixed (GROOVES version 2).

57. Claude's bridge lean put *A Rider in Two Waves*'s hold time and our needle's slip period on one square-root law.

Kind: WRONG GUESS. *Lesson:* It merged two regimes: the loan (slope −1, far from the lock) and the ghost (slope −1/2, near it). One formula holds both, a slip period of 2 pi over the square root of the detuning squared less the band squared (textbook injection locking, from recall). *Proved?* Proved by run (SLIP_BEND 20261006_071248).

58. The beat reading said *A Rider in Two Waves*'s room slope should flatten toward −1/2 away from its watershed; on the edge reached by the gain it stayed at −0.985.

Kind: WRONG GUESS, weakened. *Lesson:* A reading that names no crossover number cannot be rescued by one run; it is weakened and left as registered. *Proved? A Rider in Two Waves*'s room-bottom run of 5 October (20261005_164542).

59. *A Rider in Two Waves*'s scissors (an opening between gripping and slipping) appeared on 20-passing readers and vanished on 100-passing ones.

Kind: TOOL LIMIT, *A Rider in Two Waves*'s, a shared lesson. *Lesson:* A reader's window shorter than the wander inside a gear invents structure. *A Rider in Two Waves* withdrew them; its headline became 'chaos needs a second axis: mass, or a partner that answers back'. *Proved?* Three notes from *A Rider in Two Waves*.

60. A registered recovery after a kick of 0.3 failed: near the edge the kick was bigger than the margin itself; and the wobble gauge, which needs no master, failed once slips began.

Kind: WRONG GUESS, of design. *Lesson:* Size a test's kick to the margin it probes. Redesigned with a kick of 0.01 and held. *Proved?* Proved by run (CLIFF_EDGE 20261004_062732 and _B 062930).

61. A first avalanche run counted noise as slipping; a registered spread of waiting times missed because its yardstick included the runs themselves.

Kind: TOOL FAULT and WRONG GUESS. *Lesson:* Measure a spread against a yardstick that excludes what it measures. And a nature lesson beside it: a needle with memory 5 at pace 0.75 under weak noise slips 99.8 per cent of the time at a pace it holds for ever when undisturbed. *Proved?* Proved by run (AVALANCHE 20261004_064914; 064551 retired).

62. The listening test's own prediction, that a noisy ear moves a cell down the map but not off it, failed for noise: the plain ear chatters, its bill doubling each time the step halves.

Kind: NATURE, of switches. *Lesson:* A switch fed fresh noise chatters; a knee about three times the noise brings the cell back onto the map, at about a sixth of the clean ear's drift. With noise drawn fresh every step the knee still leans slightly on the step; with slow noise it does not. *Proved?* Proved by runs (EAR_PRICE_CHECK, EAR_KNEE, EAR_CURES_B).

63. Of Bear's three cures for a chattering switch, the pause state cured nothing, and a pause after each flip capped the flips but still leaned on the step.

Kind: NATURE, Bear's cures sorted. *Lesson:* Pause the looking, not the acting: looking only every so often is free of the step exactly, and with a small knee its flips land on Rice's rate. *Proved?* Proved by runs (EAR_CURES 20261004_045151, EAR_CURES_B 045721).

64. Jumping (firing only on crests) never beat skating for the listening bead; output per event was flat or fell.

Kind: WRONG GUESS. *Lesson:* A real jump needs a store. A memoryless skater can only skate less often. *Proved?* Seen in run (RF_SKATER_COUNT 20261004_061647); the redesign is open.

65. A fourth local peak on the cone map was read as an artifact of the map's limits.

Kind: WRONG GUESS, then NATURE. *Lesson:* It is a true local maximum belonging to no mountain: a hilltop among circles with plain under it, held up by two mountains at once. *Proved?* Proved by run (CONE_TOMOGRAPHY_2 20261008_002939, CONE_TOMOGRAPHY_3 20261008_011150).

66. The error at a mountain's foot was counted as third order and missed; a control aimed at an error already at round-off; a single step count landed near a cancellation; a smoke run stopped on a missing name.

Kind: WRONG GUESS, of counting and design, and a TOOL FAULT. *Lesson:* A foot smooth only to the second derivative gives second order with scatter; smooth feet give clean fourth order. A test must aim where an error is still visible, and scatter is read by envelopes over many counts, never by one count. *Proved?* Proved by runs (CONE_TOMOGRAPHY_3 to _5, 8 October); the envelope test registered.

67. Round Bear's ring of steep peaks, the frame's reading and the seven shares missed their bars.

Kind: TOOL LIMIT. *Lesson:* The integrator stumbles at every foot, and a walk round the ring crosses fourteen. Smooth feet took the shares' spread from 9.76e-7 to 3e-13. *Proved?* Proved by run (PEAK_RING 20261008_040518, PEAK_RING_2 20261008_040854).

68. A share near two-fifths of a peak's slice, seen in four geometries, was taken for a constant.

Kind: WRONG GUESS. *Lesson:* It falls with the cutting circle's size to a limit set by the slope. Passing near the pentagon's 1/phi squared on the way down was not a resting place: a number that looks famous may only be passing through. *Proved?* Proved by runs (PENTA_SHARE 20261008_060159, PENTA_SHARE_2 20261008_062403).

69. Thirty loops on ridged paper printed exactly 0.000e+00.

Kind: TOOL LIMIT, of design. *Lesson:* The arrow had been started down the ridges, where it stays in their plane and reads zero by symmetry whatever the arithmetic does. Exact zeros are a warning. Started on a slant, the test could fail, and held. *Proved?* Proved by runs (BENT_PAPER 20261008_164513, BENT_PAPER_2 20261008_164721).

70. Bear's house in space: can a damper keep a share of the inertia? Three registered clauses missed as written, one of them on a tool fault that put the ball back on the station's mark.

Kind: NATURE, and TOOL FAULT. *Lesson:* A damper eats energy, never momentum: free fall lets you keep a chunk of position, never of motion, and the holder's mass sets its size. The fault was fixed and each clause redesigned aloud and held. *Proved?* Proved by runs (HOUSE_IN_SPACE 20261004_150225, _B 150847, _C 151017).

71. The divers: two clauses were read on a symmetry's zero and said nothing; the twist threshold was missed at small tilts; the cat's planar way turned a pike only 1.40 degrees per loop against a guess of 5 to 40; and a full run was started before its smoke had passed.

Kind: WRONG GUESS, TOOL LIMIT and a slip in the shop. *Lesson:* Below a threshold tilt a carried somersault only wobbles; above it, it twists. The planar cat's way is far weaker than guessed. A run is started only after its smoke passes. *Proved?* Proved by runs (DIVER 20261005_160158, DIVER_B 20261005_160338).

72. The three-dimensional cat turned 57 degrees per loop, not half a turn, and fatter halves turned more, against the guess.

Kind: WRONG GUESS, the wrong question. *Lesson:* A full circuit of the bend is what the move costs, not what it buys; what the cat buys is its legs turned over in half a circuit, 159 degrees. A first version counted the halves' own spin and was retired; an axis clause failed at the floor of an arccosine. *Proved?* Proved by runs (CAT_3D 20261006_201322, CAT_3D_2 20261006_202140); read small angles as atan2.

73. An orbit finder compared angles without reducing them to one turn and kept the first orbit Newton reached, one of them 51 turns away; then the period-16 strands sat 25 and 51 per cent unequal where equal parts were guessed.

Kind: TOOL FAULT, then NATURE. *Lesson:* The half-turn's two directions come into equal parts only in the limit: the inequality shrinks as the square root of the distance to the doubling. *Proved?* Proved by runs (SPOKES 20261006_204038, CASCADE 20261006_211853).

74. Strands counted from an arbitrary point changed class between generations, so three leans missed; a first version set its acceptance at its own arithmetic's floor and took the period-32 orbit for a period-64 one.

Kind: WRONG GUESS and TOOL FAULT. *Lesson:* Count from the orbit's critical point. Never set an acceptance at the floor of the arithmetic. *Proved?* Proved by runs (CASCADE_3 20261006_220635, CASCADE_4 20261006_221804).

75. The critical point's shrink missed its window by 0.023, and the alpha-squared block was half the orbit, not a fixed eight.

Kind: WRONG GUESS. *Lesson:* The shrinks close on 2.5029 from alternate sides, and the alpha-squared class runs exactly from the critical point's image for half the orbit: Feigenbaum's fine structure, read rightly. *Proved?* Proved by run (CASCADE_5 20261006_225126).

76. The high grooves' widths in double precision compared 8/5 and 11/7 in the wrong order.

Kind: TOOL LIMIT. *Lesson:* Those widths were rounding noise. In 30-digit arithmetic the 8/5 is 1.25e-21 of drive wide and the 11/7 7.6e-24. Never compare numbers under the instrument's floor. *Proved?* Proved by runs (GROOVES_4, _5, _6, 6 October).

77. The thin grooves' widths did not fall on a straight line in drive, and a first refinement carried both folds of one groove to the same point.

Kind: WRONG GUESS, of ruler, and TOOL FAULT. *Lesson:* The right ruler is how far the rider's fraction sits below the speed limit: the period-5 widths are one power, about the 12th, of that distance across sixteen orders of magnitude. Cap every Newton step. *Proved?* Proved by run (GROOVES_7 20261006_222345).

78. Two thin grooves' folds crept in 30 digits without converging, and a 'not converged' flag hid a 4.4-fold error; one power per period then held only to about 6 per cent.

Kind: TOOL LIMIT, then NATURE. *Lesson:* Read an exponentially thin groove from its loop, whose highest and lowest points are its edges, not from badly conditioned fold conditions; a flag that says 'not converged' is a number not yet measured. The law bends a little, both ways. *Proved?* Proved by runs (GROOVES_8 20261006_225037, GROOVES_9 20261007_011618).

79. The thin grooves read again from their loops came out 19.7 and 37 times wider than the 30-digit folds of record (the 8/5 and the 11/7 at mass 3.0), and Claude's lean, that the folds' small last steps meant only the arithmetic's floor, was wrong. A first version died on a singular matrix, and a restart of the cloud workspace killed three parts in flight.

Kind: TOOL FAULT, of the old readings, and WRONG GUESS. *Lesson:* The old folds had found one edge twice and never the other. On a flat condition a small step is not arrival. The new widths, 2.47e-20 and 2.81e-22 of drive, changed chapter 10's figure (c) and its notes. *Proved?* Proved by runs (GROOVES_10 20261007_065508 and the second computer's 20261007_002739; version 1 20261007_024105 retired).

80. A fold refined on 6 October turned out to be luck: at a doubling the fold refiner wandered for 17 to 21 steps, its residual rising and falling, until one step happened to land under the bar. A registered tell (a true fold snaps in within 6 steps) then missed on one fold of eight, which took 7.

Kind: NATURE, of the method, and WRONG GUESS, of a count. *Lesson:* At a true fold the refiner snaps in (2 to 7 steps at eight true folds); at a doubling the fold's equations hold all along the doubled parent, a corridor rather than a point, and it wanders. Snap or wander tells the two apart as a separation, not as a fixed count. *Proved?* Proved by runs (FOLD_OR_FLIP_2 20261007_053443, _3 20261007_054521, _4 20261007_061947).

81. The other three verdicts of the same follow-on printed MISSED, and none was about the needle: one compared fresh values with a printed rounding against a bar below the print's own resolution (Claude's registration), and two printed MISSED where an unconverged refinement left no value. An earlier HELD (a ratio of 4.689 against Feigenbaum's 4.669) had been judged inside its own floor. A first launch died on a mistyped file name.

Kind: WRONG GUESS, of registration, and TOOL FAULT. *Lesson:* Never set a bar below the resolution of the number it is judged against. A verdict whose reading failed is UNDECIDED, not MISSED. Past period 16 double precision cannot see the clock's slide (a ratio of 5.0, anywhere from 2.2 to 19), so the lean that the shifts alternate about 4.669 is withdrawn as a claim and kept as a question. *Proved?* Read by run (FOLD_OR_FLIP_4 20261007_061947); the question parked as CLOCK-SHIFT-30.

82. Halving the integrator's step moved the 8/5 groove's edges by 6.9e-9 of drive, some seventy times Claude's lean of about 1e-10, while its width changed by 1.6e-7 of itself.

Kind: WRONG GUESS, of size. *Lesson:* The prediction held: the width belongs to the rider, not the integrator. The step moves where a thin lock sits by about 2.8e11 times its width and leaves the width alone. The row first named the wrong step count (487, the doubling's), corrected before the run. *Proved?* Proved by run (GROOVES_10 version 2, 20261007_065508 part C, as THIN-LOCK-STEP).

83. The fixed-drive check of the 11/7 groove found sign changes far from either edge, outside the groove as well as inside (E2 missed). Claude read them after the data as the loop rising higher elsewhere; walked round the whole circle, it did not (G1 and G2 missed). Bisected sixty times, the far 'zero' stayed at 4.91e-10: not a zero at all. Two versions of the check were retired (one wandered, one lost its branch), and the next tool's first version died on a replacement that matched nothing.

Kind: TOOL FAULT, of the reading, then WRONG GUESS twice. *Lesson:* A sign change counts as an orbit only if it shrinks under bisection. The far sign changes belonged to the speed-only solve, which is a feedback loop of gain d: its reading is the true residual times (1 - lambda)/(1 - d), and where d crosses 1 the reading flips sign although the

residual does not (Black's formula; all eleven verdicts held). A reading after the data is a guess until a run. *Proved?* Proved by runs (GROOVES_11 20261007_080706, _12 083300 and 083648, _13 090124, _16 182009).

84. The fixed-drive reading, redone, printed two verdicts UNDECIDED on a condition stricter than the one registered (landed samples a step out of order). Then Claude reported that the five passings gave the same 8/5 edge to within 3e-30 of drive, a misread of printed digits.

Kind: TOOL FAULT, and a misreading. *Lesson:* A check in the code stricter than the registration is a fault too: version 2 re-aimed the stray samples and held all five verdicts, a fourth reading of both thin widths. Numbers drawn from printed digits are computed by script: the passings' edges sat 2.7e-30 to 2.77e-29 from the edge of record, 1.1e-9 of the width at most. *Proved?* Proved by runs (GROOVES_14 version 1 20261007_160414 and 160415, version 2 20261007_170021 and 170022).

85. Each edge's printed resolution, its parabola's last move, overstated its error about 625 times; and Claude's reading of version 1's death on the 6/5 groove (the loop folding back in angle) was wrong.

Kind: TOOL LIMIT, and WRONG GUESS, of a cause. *Lesson:* A parabola whose spacing shrinks r times a round cuts its error r to the fourth (81, 625 and 10,000 at r = 3, 5 and 10, all held), always toward the inside of the lock, so every edge is read slightly narrow. The death was replayed exactly: a round thrown 2.16 radian from its sample, where the chained Newton started from a stale point. A cause is a guess until it is replayed. *Proved?* Proved by runs (GROOVES_15 20261007_164232 and 165418, GROOVES_16 20261007_182010).

86. The five passings of one edge orbit, which must touch zero at one drive, touched it at drives up to 4.8e-30 apart.

Kind: TOOL LIMIT, of a constant. *Lesson:* The tools counted whole turns with a 2 pi typed to 30 digits, 3.39e-31 short. With 2 pi to 40 digits the passings agreed within 8e-38, some 58 million times tighter. Every edge of record carries that fingerprint at a few times 1e-30 of drive, a ten-billionth of the 8/5's width and a hundred-millionth of the 11/7's; the widths stand. *Proved?* Proved by runs (GROOVES_16 20261007_183554, GROOVES_17 20261007_234942 and 234943).

87. The needle's last width ratio at the critical point missed 2.5029 by 0.046 against a bar of 0.01; the verdict was judged on numbers already in print, and said so.

Kind: WRONG GUESS, of design. *Lesson:* One term cannot be held to 0.01 while the sequence still swings by more: the ratios swing about 2.5029 by +0.32, -0.11 and +0.046, each a third to two-fifths the size of the one before, and their extrapolation lands 0.005 above. Judge a swinging sequence by its swing. The needle stays in the rounded-top class; the next rung (period 256) is registered and waits. *Proved?* Proved by run (DELTA_BY_TOP_2 20261008_063744).

88. The area-keeping map's ladder stopped twice on the tool: the first version's last Newton step left its bracket and settled below the rung before, and the second picked up an older orbit of the same period farther out, giving a ratio of 50.

Kind: TOOL FAULT, twice. *Lesson:* Refuse any refined value outside the bracket it was found in, and tell a just-born orbit by its trace (between 0 and 2, on its way from +2 to -2). Version 3 held: the ratios climb toward 8.72 from below (8.72027 at period 32), as the needle's climb toward 4.669. *Proved?* Proved by runs (DELTA_AREA 20261008_062447 and 062624, retired; 062647).

89. The three falling stars' loop law seemed to leave a small offset, which Claude placed at the closest pair approach, and the three-body work's memory test read +0.87. Both were the reader's: all of the offset sat on 24 loops spanning one moment, when the shape raced across its sphere.

Kind: WRONG GUESS twice, on a TOOL LIMIT of the reader. *Lesson:* Read a fast moment finer before calling anything a law: at 400 points per sample those 24 loops fall to +0.00007, and all 58 hold to a mean of 0.00003. A correlation can be one moment: the memory test reads -0.22 with those loops set aside. The race's own speed never converged (379 to 1,149 and climbing); only the residual did. *Proved?* Proved by runs (TRIPLE_CARD_CHECK 20261005_055204; the three-body work's TRIPLE_READER_CHECKS 20261005_063526).

90. Under a uniform outside field the three stars and the baseline run shared only their first two loops; then their histories parted. And this book's first note to the three-body work carried 'worst residual 0.0011' where 0.0011 was the largest mean.

Kind: NATURE, of the arithmetic, and a slip in a note. *Lesson:* The stars obey the equivalence principle and the arithmetic does not; the loop law held in both runs. The expected cause, rounding that grows with distance from the origin, is not tested. A number travels with its name. *Proved?* Seen in runs (triple_fourth_body 20260930_173710, TRIPLE_CARD_CHECK 20261005_055204); cause not tested.

91. Bear's wave snake: more hinges were guessed to take less, and a many-link snake was guessed to have a best length. Neither: they reach a ceiling, and there is no best length.

Kind: WRONG GUESS, twice. *Lesson:* With enough hinges a longer body always takes more, about 0.07 to 0.09 per wavelength of body, and about three links per wavelength reach the ceiling. Two links take exactly nothing at two wavelengths, each link covering a whole wave. The toy's fence: its push and its drag share one footprint, so the best length cannot move with the wave's rhythm. *Proved?* Proved by runs (WAVE_SNAKE 20261004_073636, _B 073735).

92. The motor snake: a clock motor at torque 0.03 was too weak to push against the waves, many motors did not beat one motor's 2.5, and a quiet body did not push better. Two first runs were retired, one whose motor wave ran opposite to its description and one whose loader left a stray parent run file.

Kind: WRONG GUESS, three times, and TOOL FAULT. *Lesson:* Length is the lever. A motor can hand power to the wave only through body shapes that match the wave's shape, and a body shorter than a wave has none: the exchange rises with body length (0.25 at 1.2 wavelengths to 3.11 at 9.6), always below the wave's own 6.28. *Proved?* Proved by runs (WAVE_SNAKE_DRIFT 20261004_074634, _B 074716, _C 080331, _D 080506).

93. The listening swing's growth leaned off its formula by up to 0.46 per cent, in a pattern the formula's single sine does not carry; and the swing counter three tools share reads a growing swing at the end of each swing and a dying one at its start, half a swing apart.

Kind: NATURE (the lean), and TOOL LIMIT (the counter). *Lesson:* The lean is a second-order term: halving the rope's change cut it to a quarter (0.2469), with the same sign pattern. Counted in turning points the pump and the brake take the same 18 each way, and the brake is the pump's climb run backward to 8.5e-13 degree. No judged reading changed. *Proved?* Proved by runs (SWING_LISTEN 20261007_013201, the reader SWING_LISTEN_TURNS 014205, SWING_LISTEN_2 024253).

94. Bear heard the push price's 'light is the most expensive thing for sale' as the Floer market of his Crossover Atlas. Then a registered slip rate missed in one row (5 slips), and backward slips came only 7 to 15 times rarer than forward.

Kind: WRONG GUESS, a bridge that was our own wording; and TOOL LIMIT, of the counter. *Lesson:* Both markets were images Claude chose, and nothing numerical crosses. The chapter's real object was one aisle over, and exact: the needle is Novikov's circle-valued flow, its locks always in stable-unstable pairs. Kramers' rate needs enough events (296 slips held it, 0.949); a single-level counter logs jitter across its own line, and a knee found no backward slips at all. *Proved?* Proved by runs (NOVIKOV_NEEDLE 20261004_152937, _B 153129).

95. Recounted with the knee, the cliff edge's slips missed at the edge (0.265 against 0.25 per 100 cycles), and the avalanche's ratchet ran 6.6 per cent above the knee against 1. After two fixes the knee still leaned the same way at every watch length.

Kind: TOOL FAULT, of the counters. *Lesson:* The ratchet's excess is exactly the needles left hanging past the barrier when the watch ends, to the count in all five rows: an end effect. The lean was a starting rule that placed some needles one well too low. A counter with a knee must be running before the watch starts; a starting rule can only guess what its own history sets. No published count changed its story. *Proved?* Proved by runs (KNEE_RECOUNT 20261004_184524, _B 190250, _C 192352, _D 195102).

96. The remembering needle's run statistics moved with where a slip was timed, and five registered predictions about them missed (KB2, KB2b, KB2c, B1, B2). Claude's binning reading of the spread was wrong, and one registered test was found ill-posed by derivation before it ran.

Kind: WRONG GUESS, of cause and of design. *Lesson:* A run ends only at a long gap, and breaks are rare (about 200 to 250 of 15,000 gaps), so a run statistic is decided by a thin tail. The barrier top logs false starts; a climb and its own fall are tied (+0.590) while a fall and the next climb are nearly strangers (+0.068), so a landing-timed gap holds both halves of one slow turn. Time a slip at the bottom of the turn or the landing, and print a run statistic with its break line and the tail it rests on. *Proved?* Proved by runs (KNEE_RECOUNT_B 20261004_190250, _C 192352, _D 195102, _E 20261005_003928).

97. The forgetting needle was guessed to spend under a tenth of its time past the barrier top at pace 0.95; it spent 0.117.

Kind: WRONG GUESS, narrow. *Lesson:* At that pace the barrier stands only 1.06 noise units high, so 'past the barrier top' counted lingering over a barrier the noise barely notices. Swept over pace, the needle matches the tilted washboard's exact density within 0.0052 in every twelfth of the turn. A state is meaningful only once its barrier stands about three noise units high. *Proved?* Proved by runs (KNEE_RECOUNT_D 20261004_195102, _E 20261005_003928).

98. Bear's picture of the balance board's fall, the left-right taps and the wobble growing before it goes, missed: up to the cliff the taps and the wobble stay within 7 per cent of their no-delay values, and the wobble, if anything, shrinks. A first run was retired because its test could not tell a decaying run from one that had blown up.

Kind: NATURE, against Bear's picture; and TOOL FAULT. *Lesson:* Some cliffs are silent. A tapping rider falls exactly where its reaction delay reaches its own rest, with no warning in its motion, where the needle's fold hums from far away. Faster taps on a shorter board are the slide rule sliding, not a new song. *Proved?* Proved by runs (BALANCE_BOARD 20261004_161712, _B 162140; the silence measured again by SEAM_MAP 20261005_021349).

99. A grid could not find the edge of the smooth push's balancing gains near its delay limit (A1 and A1b missed); of two later attempts, one was too slow to finish and one used a grid coarser than the island its own first part had just printed.

Kind: TOOL LIMIT, of the grid. *Lesson:* An island narrower than any affordable grid is found from its equation, not a grid: a winding count of the characteristic equation put the island's top 1.2e-4 above Kp = 1 at delay 1.40 and found none from 1.415, and the textbook limit, the square root of two, stands. *Proved?* Proved by run (SMOOTH_PUSH_EDGE 20261004_170203).

100. The rider that never taps blind was guessed to hold to 1.0 to 1.5 fall times of delay; it fell at 0.5 to 0.6, where its rest reached 0.7 fall times. Neither fix tried next (a look-ahead, harder taps) helped, and one fidelity check printed MISSED on a value typed as its printed rounding.

Kind: WRONG GUESS, and TOOL FAULT, of typing. *Lesson:* Never tapping blind removes the first cliff and walks the rider into a second: a fixed tap given once per rest cannot bring back a board that falls for most of a fall time between taps. A fixed tap holds to about 0.55 fall times whatever its rest, look-ahead or kick. Take a fidelity value from the record's numbers, never from its printed rounding (0.188 for 3 of 16). *Proved?* Proved by runs (NEVER_BLIND 20261006_045833, NEVER_BLIND_2 20261006_050123).

Misses 101 to 150

101. A tap sized by the reading surprised three ways: with a held rest it fell one grid step short of the guess (ST-1); never tapping blind, guessed to cost it reach, gave it more (ST-3, missed in the good direction); and it held past the smooth push's limit of the square root of two (ST2-2: 14 of 16 riders up at 1.5 fall times). Its cliff then sat between 1.5 and 1.6, short of the 2.0 guessed (ST3-1).

Kind: WRONG GUESS, of a borrowed limit, and NATURE. *Lesson:* A rider that pulses and then waits out its own delay is a different controller from a push that acts all the time (act-and-wait, from recall; names owed): not bound by the square root of two, and not free of it either, it buys 0.1 to 0.2 fall times past it. And its cliff hums: the wobble rises at every step into the edge. *Proved?* Proved by runs (SIZED_TAP 20261006_060952, _2 061606, _3 061940).

102. Two of the seam map's gauges misread near the edges: the kick recovery ran 3 to 12 per cent slow near the fold, and the share of time at the wall came out 0.072 where more than 0.3 was predicted.

Kind: TOOL FAULT, of the gauges. *Lesson:* A kick toward the lip meets a softer slope, and a short settle had left the needle short of its seat: kick both ways and average, and settle many times 1/cos(lag) first. The wall's share counted the steps on which the clip acted, so it fell with the step: a gauge that changes with the step measures the integrator. Read rightly, the end warns only within about two wobble widths. *Proved?* Proved by run (SEAM_MAP_B 20261005_023746).

103. The avalanche's climb-back lines were slow-ramp readings, 0.0169 and 0.0153 low; slowing the ramp shrank the gap, but not by the steady factor per halving predicted. The tapping rider's memory time near its cliff dropped 16 to 23 per cent against a guess of within 10; read finer, the drop was smaller and differed from rest to rest, against the guess that it comes at the same share of the rest.

Kind: TOOL LIMIT, of the ramp and of a gauge, and WRONG GUESS. *Lesson:* Read a line from methods that stand still: two such methods agree within 2.1e-7 (0.7898648 and 0.5413103). One needle's ramp reading depends on where in its turn the ramp crosses the line, so an ensemble ramp is the redesign; the memory gauge can jump by a whole ring of its autocorrelation. Both parked: neither changes what the book says. *Proved?* Proved in part (SEAM_MAP_B 20261005_023746, against AVALANCHE 20261004_064914); parked.

104. The maintenance meter's control missed: in the fleet that wears by the hour, a calendar counted in running turns still threw away less life than one counted in hours (0.473 against 0.533). Then the learned reminder's waste in the two fleets missed agreeing within 10 per cent, twice and narrowly (17 and 14.9 per cent).

Kind: WRONG GUESS, then NATURE. *Lesson:* A machine asked to work harder meets its cliff at a stronger grip, and running turns count the work: turns beat hours as a calendar even when the wear is hourly, which is how a small aircraft is already logged, off tach time. A gauge that knows the room throws away about a tenth of a life where the calendar throws away about half. *Proved?* Proved by runs (MAINTENANCE_METER 20261005_012150, _B 012355).

105. The wave soup's jump was registered to start within 0.97 to 1.06 of each soup's crest line, the strength at which the right-going comb's tallest crest pushes as fast as the waves move. In all eight soups it started 1.04 to 1.13 above it, never below.

Kind: WRONG GUESS, of size; NATURE, the cause open. *Lesson:* The crest sets the jump and something taxes it, always upward. Two suspects: reach (a crest far down the beads' path is met too late to count) and the left-going comb knocking carried beads off (one soup carried 42 beads and then only 22 as the strength rose). Below each jump the drift climbs before it, as beads linger at a crest just short of holding them, the ghost of a fold. *Proved?* Seen in run (SOUP_JUMP 20261009_045400, J3); SOUP-JUMP-2 registered to split the two suspects.

106. The follow-on's one-way check (K1) missed on one soup of eight: soup 1 carried beads at 0.98 of its crest line. And Claude's first reading after the data, that soup 1's falling count (42, then 22) was the left comb knocking riders off, was not borne out as the only cause: the one-way soup, with no left comb, showed falling counts too.

Kind: WRONG GUESS, in the registration; and a reading after the data. *Lesson:* The crest window (x from -400 to 40) was copied from a 400-unit run into a 600-unit one, where beads slide farther and meet crests the window never held. Size a window to the run it serves. A count that falls as the drive rises can come from a crest just short of holding, which delays the riders past the counting window, as well as from a knock-off. *Proved?* Seen in runs (SOUP_JUMP_2 20261009_051426 part A, against SOUP_JUMP 20261009_045400 part B); K1's redesign goes into SOUP-JUMP-3.

107. A launch meant to run two parts side by side ran them one after another: the line put only the closing echo in the background.

Kind: TOOL FAULT, in the shop. *Lesson:* Group the whole command before backgrounding it, (python …; echo …) &, and count the running processes right after a launch. The first part was stopped within seconds by a script, its headers-only file set aside, and the parts relaunched. *Proved?* Fixed (SOUP_JUMP_2, 9 October).

108. Bear's lattice reader (LATTICE_READER, run 20261009_092420) was registered to find the light needle's eight strands as eight coarse strings. At half a degree the eight-step dance shows four, each the two strands of a tight pair, so the predictions built on eight (LR1, LR3, LR7) missed and four more (LR2, LR4, LR5, LR8) could not be read.

Kind: WRONG GUESS, in the registration (of scale). *Lesson:* A lattice's scale decides what counts as a string. Take the families from the picture's own side histogram, not from an expected count; and when one coarse string holds two strands, a pad's history says so, because the rider's visits alternate. *Proved?* Seen in runs (LATTICE_READER 20261009_092420; LATTICE_READER_2 20261009_093040 read the same picture with four families from the histogram and history periods 2, 4, 8 and 16).

109. The same reader called a pad crowded (more than 10 of the 24 holes within radius 2 filled) a sign of a cloud, so a string three rows thick below the first doubling ended its families' trace at column 265, three columns above the doubling it was meant to find.

Kind: TOOL FAULT, of design. *Lesson:* Thickness is not disorder. Bear's counter is the mark of a cloud: in a dance every pad holds many landings (512 over the period), in a cloud many pads hold one. The counter found the cloud below the cascade (89.1 per cent of columns 0 to 100) and nowhere else. *Proved?* Fixed (LATTICE_READER_2 20261009_093040, LR8-2 held).

110. The registration listed the top of the eight-step band as 'the fold of record' from STROBE_NEWTON (20261006_184350), though the record had corrected it on 7 October: FOLD_OR_FLIP (20261007_050721, after a note from *A Rider in Two Waves*) showed it is the four-step dance's doubling, at 1.690017021430, and the book's working draft says so. LATTICE_READER's LR6 (a ghost of the fold, then a cloud above it) was built on the old word and missed: above the top every column is the clean four-step dance. Claude then read the doubling off the reader's picture as if it were new, before finding it in the record.

Kind: WRONG GUESS, in the registration (a stale source). *Lesson:* Read an answer of record from the latest row that speaks of it, not the first: a row of the data ledger can be overtaken by a later one. The picture still told the truth from dots alone, and LATTICE_READER_2's Newton check landed on FOLD_OR_FLIP's 1.690017021430 to the twelve digits printed. *Proved?* Seen in runs (LATTICE_READER 20261009_092420, LR6; LATTICE_READER_2 20261009_093040, part T).

111. History saw every doubling going down inside its window (columns 264, 143, 116), but going up it first read the four-step dance's single visit at column 824 in three lanes and 823 in one, against a window that ended at 823 (LR2-2).

Kind: WRONG GUESS, narrow (of settling). *Lesson:* Beside a doubling the riders settle slowly: one column above the top the four-step dance's multiplier is -0.99918 (part T), so after 8,000 passings the riders still alternate by more than the history's 1e-7 for about six columns. A history's window has to allow for the settling on both sides of an event; the reading itself is right. *Proved?* Seen in runs (LATTICE_READER_2 20261009_093040); the follow-on HISTORY-SETTLE settles four times longer.

112. On the fine lattice the 32-step band held 32 runs in only 81.2 per cent of its columns (31 in columns 135 to 137), so LR3-2 missed, though its other three stretches held in every column and its four first columns all fell in their windows.

Kind: TOOL LIMIT, of scale. *Lesson:* Two strands that pass within a row of each other read as one run. The counter does not merge them: every column of that band holds 32 pads of 16 landings, and the 64-step band 63 or 64 pads of 8 while its runs read 58 to 61. Where strands crowd, count pads, not runs, or halve the row. *Proved?* Seen in runs (LATTICE_READER_2 20261009_093040; the pad counts read after the data).

113. HISTORY-SETTLE (run 20261009_093424) settled the riders four times longer at the top of the eight-step band; the history's first single-visit column came down by two or three columns in every lane (to 822, 821, 820 and 822), not to the registered 819 or 820.

Kind: WRONG GUESS, of size. *Lesson:* A multiplier gives the settling's rate only once the alternation is small. Claude's estimate took the multiplier from the first period, and the riders alternated longer. Measure the alternation's size against time before predicting where a history will read clean. *Proved?* Seen in runs (HISTORY_SETTLE 20261009_093424); SETTLE-SIZE registered.

114. The third practice in reading patterns lifted the light needle's strands into the rider's speed at each passing and predicted that the two strands touching in angle at columns 135 to 137 were an overpass, more than 1e-4 apart in speed, so that holes of 1e-4 in both numbers would count every strand (P1, P2, P3). They are 1.58e-4 to 1.84e-4 apart in angle and only 6.41e-5 to 7.51e-5 in speed: the lift read 31 strands where there are 32, as the angle alone had.

Kind: WRONG GUESS, of kind (a meeting taken for a shadow). *Lesson:* A touch the third number does not separate is a real meeting: here a fork's young daughters, 2 to 7 columns below the 16-to-32 doubling (column 142.3), still close in every number. Resolve it by scale (holes of 1e-5 separate them: 32 strands in every column from 122 to 140) or by history, not by another dimension. The prediction assumed the picture's shadow, the error Bear had named just before: lift a flat picture into a third dimension and you tend to see what you want to see. *Proved?* Seen in runs (LIFT_3D 20261010_031527, the run of record; first counted by two scratch scripts the same night); predictions written down before the counts.

115. The follow-on (P1b, P2b) predicted that holes of 1e-5 in both numbers count as many strands as there are distinct strobe points, with angle and speed rounded to 1e-6. The strands were right and the reference was not: the rounded count ran one high in columns 122, 123 and 127, and at column 111 the riders had not settled (no repeat within 128 passings).

Kind: TOOL LIMIT, of the reference count (rounding). *Lesson:* A rounded count has a scale of its own: a value on a rounding boundary rounds two ways. Count the truth by history, each rider's period (the smallest repeat of its passings within 1e-7 in angle and speed): the 1e-5 lift agrees with it in every column from 112 to 140 where both riders repeat with one period. *Proved?* Seen in runs (LIFT_3D 20261010_031527).

116. CORNER-PRINT's C1 asked that each lane's twin lobes correlate with a pure corner's echo at 0.95 or more. The sizes of the correlations were 0.993 to 1.00000 at all three doublings, but in half the lanes the correlation was negative, because there the corner bends the other way, so C1 missed as written.

Kind: REGISTRATION FAULT, of wording. *Lesson:* The same registration called the corner's sign free and printed it, then asked for a signed number. When a sign is free, register the size: 'correlation at least 0.95 in size'. Nothing was rerun or re-judged; the sizes stand printed beside the verdict. *Proved?* Seen in runs (CORNER_PRINT 20261010_061715).

117. SHUFFLE-NULL-2 bet that the needle picture's cloud (columns 0 to 100) holds real order between neighbouring columns, because the first null (rows permuted among pads) had put the picture +9.4 spreads above chance. With the order of the columns permuted instead, so that nothing merges, the excess fell to +1.8 spreads, under the registered 3.

Kind: WRONG GUESS, of kind; and TOOL FAULT, of the null. *Lesson:* The first null merged 11 per cent of the cloud's pads (two shuffled pads landing in one hole), thinning the shuffled picture and shortening its strings by itself; its N3 had held as written on a band (a factor of 2) too loose to decide what its own spread decided. A null must keep everything it is not testing; a shuffle that changes the counts is a test of its own. Choose the statistic before the shuffle, too: on the ordered bands the count of short strings pointed the wrong way (-20.7 spreads) while long reach separated (+895). *Proved?* Seen in runs (SHUFFLE_NULL 20261010_190812, SHUFFLE_NULL_2 20261010_190944).

118. The kit for the Field Guide's new session was staged with every tool the Guide named, each with its run of record and the files each tool names in its own text. Run in a clean folder holding only the kit, FOLD_OR_FLIP, FOLD_OR_FLIP_2 and FOLD_OR_FLIP_4 stopped on 'No such file' before printing a result: two strobe pictures (RUN_STROBE 20261006_170149 and RUN_STROBE_2 20261006_174220) were missing.

Kind: PACKING FAULT, of a kit's list of needs. *Lesson:* The tools load GROOVES.py by text, and the part they load opens the two pictures; the list of needs had been read from each tool's own text only. A tool that loads another tool by text inherits that tool's data files. Read the list of needs from every file each tool opens, through the loaded tools too, and run the kit in a clean folder before it is sent: the clean folder is what found the fault, at no cost to the Guide. With the pictures added, all nine tools reproduced their runs of record, every line (333). Sharpened the same day: the needs had been written down all along, in each fold tool's own bat ('the RUN_STROBE and RUN_STROBE_2 npz files beside it'); the staging read the tools' text and not their bats. The bat is the first place to read a tool's needs, and what a tool opens is the last word (NEEDS-AUDIT, registered). *Proved?* Seen in runs (the first clean-folder test, 10 October, which wrote no run files; the second, runs 20261010_202355 to 20261010_202803; LOOP_ATLAS_KIT_COMPARE).

119. NEEDS-AUDIT's N2 bet that at least one of the kit's bats names a file its tool never opens: that the bats, written by hand, would be generous. Logged by an audit hook, every bat's list is exact: the four fold tools open 7, 7, 8 and 10 files, every one named in their bats and nothing named left unopened; the shuffle tools and the build check likewise.

Kind: WRONG GUESS, of kind (a hand-written list taken for a loose one). *Lesson:* The bats were right and the reader was wrong: the needs had been written down exactly, and the kit's first staging missed two of them by reading the tools' text instead (row 118). Nothing to redesign in the audit; one limit to fix in a next version: the hook counted a folder opened by a cleanup (the build check's _build_tmp) as if it were a file. A bat's times are not as exact as its needs: 'a few minutes' was 2 s. *Proved?* Seen in runs (NEEDS_AUDIT 20261010_205839).

120. The wander test's first try on the second computer (run 20261010_144307) stopped at once: 'No such file: LOOP_ATLAS_GROOVES.py', the first companion FOLD_OR_FLIP_4 loads, though all 15 files had been sent loose with the instruction to put them in one folder and to take any '(1)' off a renamed download.

Kind: INSTRUCTION FAULT (Claude's), and TOOL LIMIT (a tool that opens its companions by exact name). *Lesson:* The instruction broke Bear's standing rule of 16 September: a '(n)' copy in Downloads is never renamed; a bat finds the right copy by its hash. The kit's tools were written for the cloud workspace, where names are exact. Which cause it was could not be seen from here, and Claude guessed a renamed copy. The launcher's census on the second computer (LAPTOP_F4, run 20261010_170450, 135 folders) showed the guess wrong: none of the ten files it found was a '(n)' copy; LOOP_ATLAS_GROOVES.py sat in another folder (Downloads\LOOP2), and the two strobe pictures and the comparison were not in any folder it looked in, under their names. So the version 2 launcher, LOOP_ATLAS_LAPTOP_F4_2, looks two folders deep, under any ending and inside zip files, the kit's zip holding all thirteen. LOOP_ATLAS_LAPTOP_F4 finds every needed file by md5 in its folder, the folder above and Downloads, copies the right copies (renaming nothing) into a fresh folder under their proper names and runs the unchanged tool there; tested on a made-up Downloads with '(1)' and '(2)' copies, a same-name file with other contents, a file one folder down, a missing file, and one full run. *Proved?* Seen in runs (Bear's runs 20261010_144307 and 20261010_170450, as he pasted them; LAPTOP_F4's and LAPTOP_F4_2's tests of 10 and 11 October).

121. The launcher's first full test, on a made-up Downloads folder with a space in its name, ran FOLD_OR_FLIP_4 to the end and reproduced every number, yet the comparison (version 2) called the run DIFFERENT on one line: the run-file line, which differs only in its folder.

Kind: TOOL FAULT, of the comparison (latent), caught before use. *Lesson:* The launcher had named its work folder with the run's own stamp (LOOP_ATLAS_LAPTOP_F4_20261010_214554), and the comparison sets the stamp aside before the folder, so the folder no longer matched and the path, holding a space, could not be set aside either. The launcher now names its folder with dashes (LOOP_ATLAS_LAPTOP_F4_work_2026-10-10_21-57-31), which no stamp matches. Owed in the comparison's next version: set the folder aside before the stamp. A made-up end-to-end run is what found it; a dry run would not have. *Proved?* Seen in runs (LAPTOP_F4's full tests of 10 October).

122. LAPTOP-WANDER's L1 bet that where the refiner settles, at the eight folds, the second computer would match the cloud workspace's record to the 12 printed decimals. Three of the eight drives differed, by 1.0e-12, 2.0e-12 and 2.8e-11; the grooves and every fold's step count matched.

Kind: REGISTRATION FAULT, of precision (a bar finer than the instrument's floor). *Lesson:* The record already showed the fold refiner's floor: its final residuals run up to 9.7e-10 and its distances from STROBE_NEWTON's values 2.1e-11 to 4.9e-10, so 12 decimals asked the arithmetic for more than it holds at a fold; the largest difference, 2.8e-11, sits inside that floor. Redesign, said aloud: register a cross-machine match at the method's own floor, read off its record before the run (second specimen of a Field Guide rule), and look for the fold-or-doubling difference where it lives, in the step counts (ULP-NUDGE, registered). Later the same day FOLD-TIGHT showed the 1e-9 bar's own error at these folds reaches 7.2e-11, which holds the second computer's 2.8e-11 inside it. *Proved?* Seen in runs (FOLD_OR_FLIP_4 20261007_061947 and 20261010_172149).

123. LAPTOP-WANDER's L3 borrowed the second computer's pace from a Field Guide lesson (3.46 to 3.51 times the cloud workspace's seconds) and bet on 3 to 4 times the record's 336.9 s. The second computer took 273.3 s, 0.81 times: it was faster.

Kind: WRONG GUESS, of a borrowed number. *Lesson:* The borrowed ratio was measured on other tools under other loads, which cannot be checked from here; the pace is the machine's and the tool's own. The irony is the lesson: that lesson's own sentence is that a borrowed number belongs to the machine it was made on, and its number did too. Register a pace only from a run of the same tool on the same machine, or print it unjudged. *Proved?* Seen in runs (FOLD_OR_FLIP_4 20261010_172149).

124. ULP-NUDGE's V3 bet that every fold's drive moves by less than 5e-11 when its start is nudged by one or two units in the last place. The largest move was 5.36e-11, at the 3/2 lower fold with k = -2; the other 31 moves were smaller, and every fold's step count was unchanged.

Kind: REGISTRATION FAULT, of margin (a bar set less than twice the largest difference already seen). *Lesson:* The bar came from the second computer's largest difference (2.8e-11, at the same fold) with too little room, and the fold that moves most is the one whose refinement stopped closest to its 1e-9 bar (last residual 9.7e-10 in the record).

Redesign, said aloud: the drive's spread may be set by where the refiner stops rather than by the arithmetic; FOLD-TIGHT, registered, refines the same folds to 1e-12 and asks whether one more snapping step removes it. *Proved?* Seen in runs (ULP_NUDGE 20261011_044600).

125. DELTA-TAIL-2's U1 bet that order 8's doubling constant, extrapolated from its last three rungs at rung 12, would lie above 10.94875, outside the rounding of the reference's 10.9486, because the extrapolated values had been rising (10.947846, 10.948648, 10.948788). At rung 12 it was 10.9486659.

Kind: WRONG GUESS, of kind (a zigzag read as a trend). *Lesson:* The rise was the second correction that changes sign every rung, which the step ratio had already shown zigzagging (and goes on doing: 0.4158, 0.4406, 0.4119); extrapolated twice, allowing for it, the value settles at 10.94862, on the reference. The Loop Atlas's own lesson of the day before (the lesson sent to the Field Guide: read the step ratio first) was not applied to its own guess. Nothing to redesign; the reference stands. *Proved?* Seen in runs (DELTA_TAIL_2 20261011_051600).

126. DAMPER_LAWS version 1's A2 bet that the constant brake's and the square law's extra reaches equal their formulas to four figures; both came out 0.000500 m long.

Kind: TOOL FAULT, two at once. *Lesson:* The flight was stepped, so the ribbon was 0.0005 m past taut (one step of the relative speed) when the damper took hold; and the pay-out was solved with the very formulas A2 tested, so A2 could fail only through the flight and could never test the law (a Field Guide rule). Version 2, the same hour, ends the flight exactly at taut and steps the pay-out with the force law; A2b, registered before it ran, held to six figures. *Proved?* Seen in runs (DAMPER_LAWS 20261011_051800, set aside, and 20261011_051848).

127. FOLD-TIGHT's T2 bet that every fold reaches a bar of 1e-12 within two steps more than its count at 1e-9, from every nudged start. Seven folds did; the 5/4 lower fold took 13, 8, 9, 20 (not reached) and 14 steps against its 6.

Kind: WRONG GUESS, of a floor. *Lesson:* That fold's residual cannot go much below 1e-12 (its stopping residuals run 7e-13 to 2.5e-12), so the tighter bar sits on its floor and the refiner wanders there as it does at a doubling, while the drive itself holds to 1e-14. A fold snaps only while its bar stands above its arithmetic's floor; the step-count test for fold against doubling needs the bar set above each fold's floor, which this run has now measured for all eight. *Proved?* Seen in runs (FOLD_TIGHT 20261011_051636).

128. SPIN_OR_SIT's L2 bet that at memory 2 successive landing-timed gaps on spinning windows correlate within 0.05 of zero. They correlated +0.0568.

Kind: REGISTRATION FAULT, of reading (the bet counted one of four terms). *Lesson:* The registration read the landing's forgetting from the tie across it alone (a fall with the next climb, covariance +0.0006), but successive landing-timed gaps also pair a climb with the next climb, a climb with the next fall and a fall with the next fall, and with memory the needle's speed carries across the landing in those. Redesign, said aloud: split the covariance into its four terms; GAP-TERMS registered. At memory 5 the same correlation is +0.3843. *Proved?* Seen in runs (SPIN_OR_SIT 20261011_055807).

129. SPIN_OR_SIT's L3 bet that the tie across the landing rises with memory 0, 1, 2 and 5, judged on spinning turns with at least 100 at each memory. The forgetting needle gave 12: UNDECIDED.

Kind: DESIGN FAULT, of sample (a test the run could not feed). *Lesson:* At pace 0.95 the forgetting needle seldom slips twice running, so its spinning turns are rare; and without memory a landing is a renewal point, so its tie is zero by theory and is better read on all turns. Redesign, said aloud: memory 0 on all turns against zero within its standard error, memories 1, 1.5, 2, 3 and 5 on spinning turns; LANDING-TIE-2 registered. *Proved?* Seen in runs (SPIN_OR_SIT 20261011_055807).

130. SPIN_OR_SIT's R1 bet that both kinds of spell, told apart by the needle's speed over the last field period with a knee at 0.35 and 0.65 of the pace, spread like a trap (median 0.6 to 0.8 of the mean). The spins did (0.708); the locks did not (0.443: 1,051 locks, median 0.70 periods, mean 1.58, the longest 47.3).

Kind: READER FAULT (the instrument defined what it counted). *Lesson:* The speed knee calls 1,051 stretches locks where the lag finds 69 climbs over two periods: most speed-knee locks are short slow-downs, a mixture with the true catches, not one trap. Redesign, said aloud: ask where along the turn each speed-knee lock sits; LOCK-READERS registered, with SPIN_OR_SIT's 69 catches as the place-reader's count. Followed up 11 October: LOCK_LANDING (20261011_170402) and LOCK_MATCH (20261011_195010) found every speed-knee lock under two periods to be a hesitation read late (all 808 under one period mid-fall at their middle), and the long locks to be the lag's 69 catches, one for one, plus 15 long hesitations past the top; whether the catches alone spread like a trap is CATCH-TRAP. *Proved?* Seen in runs (SPIN_OR_SIT 20261011_055807).

131. SPIN_OR_SIT's C2 bet that climbs over two periods are at least 1 per cent of all climbs. They were 0.46 per cent (69 of 14,969).

Kind: REGISTRATION FAULT, of a bar set without a number in hand. *Lesson:* The picture held (C1 and C3: a fast pass at 0.15 to 0.20 periods, 30 climbs of 0.5 to 2 periods against 69 over 2); the 1 per cent was a guess. Redesign, said aloud: none is owed to the picture; the catch rate, about one climb in 217, is the number to carry, and LOCK-

READERS sets it beside the speed knee's count. *Proved?* Seen in runs (SPIN_OR_SIT 20261011_055807).

132. LOCK_FROM_CHAOS's X1 bet that at every pace the lifetimes spread like a single exponential (median / mean 0.6 to 0.8, and 0.090 to 0.203 left at twice the mean). Three paces fell under 0.6 (0.586, 0.599, 0.584) and all six sat under the exponential's 0.69, while the tails matched (0.137 to 0.168).

Kind: MODEL FAULT, of shape (two drains, not one). *Lesson:* Not noise: for 256 exponential lifetimes the ratio's spread is 0.046 and it falls under 0.6 in 2.2 per cent of samples (measured by simulation after the run). More needles lock in the first cycles than a constant chance allows (at 0.91200, 139 of 256 still slipping at cycle 10, where the exponential with mean 22.27 leaves 163), and after that the drain is exponential. Redesign, said aloud: read the tail past cycle 20 as its own exponential from the saved lifetimes; EXP-SHAPE registered. *Proved?* Seen in runs (LOCK_FROM_CHAOS 20261011_060551).

133. LOCK_FROM_CHAOS's X2 bet that the mean lifetime against the distance below the edge 0.91232 has a log-log slope between -0.65 and -0.35. It was -0.3089.

Kind: REGISTRATION FAULT, of an inherited constant (the edge). *Lesson:* The three nearest paces flatten at 54 to 63 cycles; the farthest three alone give -0.391. A power law against an edge is only as good as the edge: against 0.91235 the slope would be -0.50 (read after the run), and 0.91232 came from a walk-down on a 1e-5 grid, not from the lock's own fold. Redesign, said aloud: find the fold from the lock itself, then refit; LOCK-EDGE-EXACT registered. *Proved?* Seen in runs (LOCK_FROM_CHAOS 20261011_060551).

134. SHOVE_SKATER's S1 (as registered) bet that a skater given a fixed shove per swizzle, with a drag set by distance (read as the square law, whose speed falls by e per fixed distance), reaches the same speed after 10 swizzles at 0.5, 1 and 2 swizzles per second, within 0.005 m/s. The speeds were 2.8634, 3.2530 and 3.5030 m/s.

Kind: WRONG GUESS, of kind (a length taken for a count). *Lesson:* A drag with a length constant still charges the skater for the distance coasted in each swizzle, and that distance is the speed over the rate, so the faster swizzler loses less per swizzle. Claude's lean, registered beside it (L1), said so and held. Redesign, said aloud: covered in the same run by S2, a drag charged per swizzle (the record's coefficient times the rate), under which the three speeds agree to the bit and equal the record's 3.0951 m/s. Read the same way, the record skater counts seconds only to about 1e-4 (3.0950 against 3.0951 m/s): an event counter can be exact where a time counter is only close. *Proved?* Seen in runs (SHOVE_SKATER 20261011_070846).

135. WADDLE_PRICE version 1 read the passive waddle's spin energy as half the board's turning heft times the spin, not the spin squared, so P1 and P2 printed 'energies' that changed sign at every landing.

Kind: TOOL FAULT, of a reader. *Lesson:* Caught by the sign: an energy that alternates in sign cannot be an energy. The run was retired and set aside; version 2 squares the spin, reproduces every line of version 1 on the shuvit walk's price (44 of 44, compared by script), and says in its header that version 1's printed spins had already shown the steady energy at 4/3 of the pop, so its P1 was not blind. *Proved?* Seen in runs (WADDLE_PRICE 20261011_071049, retired; 20261011_071302).

136. WADDLE_PRICE's C1 (as registered) bet that, with a dead catch and the rider's top-up, the waddle's positive work per landing is 0.75 of the shuvit walk's within 0.01; it was 0.76328. C3 (Claude's) bet the positive-plus-braking ratio at 0.60 within 0.01; it read 0.61734. C2 held: the net work per landing is the same for both walks (ratio 0.99958), so the shuvit walk's whole extra price is its brake.

Kind: TOY LIMIT (a springy pin), and NATURE (the price of a jerk on a joint). *Lesson:* Both walks paid about 0.0023 more per top-up than the rigid algebra, close to k delta^2 for the dead pin (stiffness 1e5) pushed by the couple of size 10 (delta = 1.5 tau / k, so 0.00225): a force switched on against a damped spring does F delta of work, half stored in the spring and half lost in the damper whatever the damping, as a capacitor charged from a battery through a resistor loses half the battery's work in the resistor whatever the resistance. Redesign, said aloud: shrink the couple; WADDLE-PIN registered and run as version 2's part F: at couple 1 the ratios are 0.75016 and 0.60020, the rigid algebra's answers. *Proved?* Seen in runs (WADDLE_PRICE 20261011_071302 and part F).

137. WADDLE_PRICE's P2 (Claude's) bet that the spin energy's distance from its steady value, 4/3 of the pop, shrinks by 0.25 within 0.01 from each landing to the next over the first six landings. It did for five steps (0.25000 to 0.24967) and then read 0.27506 at pops 1/6 and 0.5.

Kind: REGISTRATION FAULT, of a reference (the ideal limit taken for the toy's own). *Lesson:* By landing 7 the distance is about 2e-5 of the pop, and the toy's own steady value sits about 2e-6 of the pop below 4/3 (its late energies read 1.333331 and 1.333325 pops), because its catch keeps a quarter only to about five figures; measured against the ideal 4/3, that hair becomes the gap. Redesign, said aloud: measure the distance from the walker's own late energy; WADDLE-GAP registered. P1 held: from landing 12 the energy is 4/3 of the pop within 1e-5 of a pop. *Proved?* Seen in runs (WADDLE_PRICE 20261011_071302).

138. WADDLE-PIN's F1 bet that the waddle's extra positive work per landing over 1/8 is (1.5 tau)^2 / k within 15 per cent at couples 10, 3 and 1. It was at 10 and 3 (ratios 1.009 and 1.050) and missed at 1 (1.451: 3.265e-5 against 2.250e-5).

Kind: REGISTRATION FAULT, of a floor (a bar finer than the toy's own floor). *Lesson:* A second, smaller cost does not shrink with the couple: about 1.0e-5 in the waddle and 7.4e-6 in the shuvit walk's positive work, at every couple. Read after the run, not a verdict: a floor plus a square law through couples 10 and 1 gives the square law's coefficient 2.2610e-5 against (1.5)^2 / k = 2.2500e-5 and a floor of 1.004e-5, and predicts 2.135e-4 at couple 3 against the 2.127e-4 measured. Redesign, said aloud: find what the floor is before reading a law under it; WADDLE-FLOOR registered. F2 held all the same: at couple 1 the ratios are 0.75016 and 0.60020. *Proved?* Seen in runs (WADDLE_PRICE 20261011_071302, part F).

139. TWIST_LONG version 1 printed 'M1: UNDECIDED' on M1's own line while its summary printed 'M1: HELD'; the difference it judged was 0.00e+00 turns.

Kind: TOOL FAULT, of a printer. *Lesson:* The verdict was a numpy true, and the printer asks whether a verdict is Python's True, so a numpy true fell through to UNDECIDED; the summary converted it first and read HELD. A run file of record cannot say two things about one verdict, so the run was retired and set aside; version 2 converts every verdict and reproduces version 1's other 33 lines, compared by script. Convert a verdict to a plain true or false before printing it, everywhere it is printed. *Proved?* Seen in runs (TWIST_LONG 20261011_072737, retired; 20261011_072842).

140. METER_CLOCKS's (b) bet that, in the hour-wear fleet, the calendar counting hours times duty to a power k wastes least at k between 1.25 and 1.75, derived from the median hours at death of duties 0.45 and 0.65. The lowest mean waste was at k = 2 (0.413), and in the tach-wear fleet the waste was still falling at the scan's edge, k = 2.5 (0.424, against 0.510 for tach turns alone).

Kind: WRONG GUESS, of a derivation from two medians; and NATURE (duty moves the cliff as well as the wear). *Lesson:* Two medians set one ratio, but the waste depends on how spread the machines' ages at death are, within each duty as well as between duties; and the calendars caught unequally at the same training bar (0.911 to 0.939 in the hour fleet), so their wastes are not read at matched catch. The finding stands either way: even where wear is per turn, tach turns are not the best clock, because a machine asked to work harder meets its cliff at a stronger grip. Redesign, said aloud: ask which power makes age at death least spread, and scan past 2.5; METER-K-SPREAD registered. *Proved?* Seen in runs (METER_CLOCKS 20261011_072646).

141. METER_CLOCKS's (c) bet that thresholds set per room class would raise the loud rooms' catch to 0.85 to 0.95 at more waste there, and leave the quiet rooms' catch and waste each within 0.05. The loud half held (0.839 to 0.887, waste 0.157 to 0.195); the quiet rooms' catch fell from 0.963 to 0.891.

Kind: WRONG GUESS, of kind (a shared bar taken for a fair one). *Lesson:* One threshold set to catch 90 per cent of the whole fleet over-catches the quiet rooms and under-catches the loud ones; a threshold per class sets each near 90 per cent, so catch moves from the quiet rooms to the loud ones, and waste moves the other way (quiet 0.067 to 0.046). Redesign, said aloud: covered by the run's own numbers, which show the trade in both directions and in both fleets; it answers the open question of why the room input saved life in quiet rooms but caught no more loud-room machines: the shared bar did it, not the input. *Proved?* Seen in runs (METER_CLOCKS 20261011_072646).

142. EXP_SHAPE's E1 and E2 bet that, past their early rush, the lock-from-chaos lifetimes drain as one exponential at every pace: remaining lifetimes past cycle 20 with median over mean 0.6 to 0.8, and a Kolmogorov distance under the 95th percentile of whole-cycle exponential samples of the same size. Four paces passed both; 0.91230 (0.549; 0.0896 against 0.0840) and 0.91224 (0.599; 0.1038 against 0.0949) failed both.

Kind: WRONG GUESS, or sampling (not yet told apart). *Lesson:* Two misses in six tests at a 5 per cent bar happen by chance about one time in thirty, so the tail may hold a second, slower drain at some paces, or the 256 needles may simply have been too few. The two failing paces are not neighbours in distance below the edge, which leans toward sampling. E3 held: the early rush grows with the distance below the edge (excess locks by cycle 10, +3.2 nearest to +24.4 farthest). Redesign, said aloud: four times the needles at the two failing paces; EXP-SHAPE-2 registered, with Claude's lean that the misses were sampling. Told apart 11 October: sampling. With 1,024 needles at each of the two paces both bars pass (median over mean 0.658 and 0.634; distance 0.0404 against 0.0449 and 0.0465 against 0.0522; EXP_SHAPE_2 20261011_194909), with the 256 needles of record inside the larger samples, their lifetimes unchanged. *Proved?* Seen in runs (EXP_SHAPE 20261011_080708).

143. WADDLE_FOLLOW's G2 bet, on Claude's guess that the toll per top-up is the pin's give, that a pin twice as stiff halves the floor at couple 1 within 20 per cent. It fell to a quarter (0.2534), and at couple 3 it hardly moved (1.0162e-5 to 1.0077e-5).

Kind: WRONG GUESS, of a mechanism. *Lesson:* A static give would scale as one over the stiffness at every couple; this floor scales one way at couple 1 and not at all at couple 3, so it lives in the pin's motion, not its stretch: most likely the ring the catch leaves in the pin, whose phase at the moment the couple switches on decides what the couple pays.

It is under a ten-thousandth of the bill, 1.0e-5 against 1/8 a landing [corrected 8 November from ‘about a hundred-thousandth’], and changes nothing the book says. Redesign, said aloud: book every joule step by step in each top-up; WADDLE-BOOKS registered and kept for the working draft. G1 held beside it: measured from the walker’s own limit, the gap shrinks by 0.25000 per landing. *Proved?* Seen in runs (WADDLE_FOLLOW 20261011_080808).

144. MEMORY_SWITCH’s S1 bet that the memory at which our needle starts to remember, approached by climb-back lines ever nearer the fall edge (1 - p from 1e-2 to 1e-6), settles between 0.70 and 0.71, judged at 1e-6 only if the last two readings agreed within 0.002. The reading at 1e-6 was 0.7037, inside the band, but it had moved 0.0026 from 1e-5: UNDECIDED.

Kind: REGISTRATION FAULT, of a stopping rule (a bar on the step that the approach’s own law could not meet). *Lesson:* The readings approach their limit as the square root of the distance to the edge (steps shrinking by 0.2776, 0.3040, 0.3124 per decade, toward 0.3162), so each decade moves the reading by about 0.3 of the last move, and a step under 0.002 needed 1 - p near 1e-7. A walk toward an edge should be judged by the law of its approach, or done at the edge itself. Redesign, said aloud: shoot at pace exactly 1, from the point where the lock is born; MEMORY-SWITCH-2 registered and run. *Proved?* Seen in runs (MEMORY_SWITCH 20261011_084432).

145. MEMORY_SWITCH_2 took three versions to give its first reading. Version 1 shot each needle forward from the lock’s birthplace at pace 1 and called it caught if it never passed the next birthplace within 1e6 time units; version 2 compared speeds at a half-turn section, following the arriving path backward in time, and died when that path turned round before the section at memory 0.8.

Kind: TOOL FAULT, twice (a stopping rule that never stops; a path assumed to cross a line it need not cross). *Lesson:* A caught needle creeps up to the birthplace as one over time and never arrives, so ‘not yet passed’ is never decided and every caught shot ran to the limit, about 70 s each; ask a question whose answer arrives in finite time. Run backward in time the drag becomes a push, and the path into the birthplace may come from behind the section rather than across it; count that case as speed zero, since every forward needle crossing the section is then above it. Version 3 read the switch to eight figures in a minute. Neither faulty version printed a reading; a scratch timing check of single shots had shown 0.70 caught and 0.71 passing, disclosed in version 3’s header. *Proved?* Seen in runs (MEMORY_SWITCH_2 20261011_084809 and 20261011_085049, set aside; version 3’s run of record).

146. LOCK_LANDING’s K2 bet that of the speed knee’s whole lock spells over 2 periods (memory 2, pace 0.95), at least 80 per cent have the needle in the well, short of the barrier top, at the spell’s middle. 66 of 84 did (78.6 per cent).

Kind: WRONG GUESS, of a line (the barrier top drawn sharp on a barrier about one noise unit high). *Lesson:* The long locks sit where a lock should: 60 of 84 within a twelfth of the well floor and 6 more between there and the top. The other 18 lie just past the top (15 in its own twelfth, 3 in the next), on a barrier about one noise unit high at this pace (KNEE_RECOUNT_E’s 1.06), where a needle can cross the top and come back; Claude’s reading, not yet tested, is that these are still locks, lingering on the shoulder. The crossing flag, a memory of the last crossing rather than a place, would have done worse (83 per cent ‘past the top’). K1 held beside it: every one of the 808 locks under one period sits mid-fall at its middle, a hesitation read late through the one-period window. Redesign, said aloud: ask whether each long lock is one of the lag’s caught climbs, and read the share of its whole length spent short of the top rather than its place at one instant; LOCK-MATCH registered. *Proved?* Seen in runs (LOCK_LANDING 20261011_170402).

147. LOCK_MATCH’s A1 bet that at least 90 per cent of the speed knee’s 84 long locks (over two periods; memory 2, pace 0.95) overlap one of the lag’s caught climbs on the same needle, and its B1 bet that the median long lock spends at least 0.8 of its recorded steps short of the barrier top. 69 of 84 overlapped (0.821); the median share was 0.784. A2 held beside them: all 69 caught climbs overlap a long lock.

Kind: WRONG GUESS, of what a reader counts (a hesitation can outlast the line drawn for a lock). *Lesson:* The two readers agree on every catch, 69 and 69. The 15 long locks with no catch are hesitations past the top that outlasted the two-period line: all between 2.05 and 2.89 periods, no landing inside, almost no time short of the top (nine of them none). They also pull B1 under its bar; read after the run, the 69 matched locks have median share 0.832, each running a median 0.56 periods longer than its climb because the one-period window reads the fall late. Redesign, said aloud: the question asked is answered (the speed knee’s locks are the lag’s catches plus hesitations), and B1 needs no tool of its own, its shortfall being two effects already measured; what stays open is R1’s (row 130), whether the catches alone spread like a trap. CATCH-TRAP registered, on a fresh seed, since seed 1’s lengths have been seen. *Proved?* Seen in runs (LOCK_MATCH 20261011_195010).

148. FALL_CHAIN’s C2 bet, *A Rider in Two Waves*’s question, that two falls apart the remembering needle’s falls still correlate more than two standard errors above zero (memory 2, pace 0.95, 368,185 pairs over two seeds). It read +0.0022 against a standard error of 0.0016: 1.4 errors.

Kind: DESIGN FAULT, of power (registered knowing it was near a coin flip). *Lesson: A Rider in Two Waves*’s own arithmetic said a chain’s +0.0034 would sit only about two errors up at this size; it landed at 1.4, within 0.75 errors of the chain and 1.4 of no memory, so it tells neither. C1 held beside it: r_2 sits 35 errors below a drift’s +0.058, so the

falls forget within two turns. Redesign, said aloud: about four errors are needed, about 1.4 million pairs; FALL-CHAIN-2 registered, with eight fresh seeds, a run for the background or for the second computer. *Proved?* Seen in runs (FALL_CHAIN 20261011_200714).

149. CATCH_TRAP's (b) bet that on a fresh seed (seed 2) the remembering needle's caught climbs, less their first 2 periods, spread like a trap: median over mean between 0.6 and 0.8. It read 0.800, over the bar by less than 0.0002 (64 catches; median 8.704 periods, mean 10.879). (a) held beside it: all 64 catches overlap a long lock, and exactly 64 long locks overlap a catch.

Kind: REGISTRATION FAULT, of a bar narrower than its sample's scatter. *Lesson:* Read after the run in a scratch calculation (not a run of record): for 64 draws from one exponential, median over mean averages 0.701 with a spread of 0.088, and 13 per cent of samples reach 0.8, so the bar tested luck as much as shape. But both seeds sit above the exponential's 0.693 (0.800 here; 0.853 on seed 1, which 3.9 per cent of 69-draw samples reach), leaning, weakly, toward a catch less memoryless than a trap: fewer quick escapes and fewer very long holds than a fixed chance per period gives. Redesign, said aloud: judge the shape on thousands of catches by a Kolmogorov distance, as EXP_SHAPE did, from FALL_CHAIN_2's saved landings (eight fresh seeds), with no new replay; CATCH-TRAP-2 registered. *Proved?* Seen in runs (CATCH_TRAP 20261011_202500).

150. SETTLED_COPIES's S2 bet that at pace 0.98375 the 16 settled needles fall into exactly 4 clusters at one reading, that the clusters are the needles' groups in CHAOS_SETTLES's saved file, and that one drive period carries the clusters round one cycle. The 4 clusters and the one cycle came out (cluster 0 to 2 to 3 to 1 to 0); the groups, read on all 50 windows, came out 16 single needles.

Kind: REGISTRATION FAULT, of a definition (groups read over windows that hold each needle's own transient). *Lesson:* The groups the data ledger named (agreeing to 3.4e-8 inside a group) were read on settled windows; over all 50 windows each needle's early windows differ, since each settles at its own time, so no two needles agree within 1e-6. S1 held beside it: every needle repeats after 4 drive periods to 1.7e-9, gaining 1 turn, so the settled spin is a 1/4 groove. Redesign, said aloud: read the groups on windows 31 to 50, where every needle of both paces has settled; SETTLED-COPIES-2 registered and run, with the slower orbit at 0.99375 beside it. *Proved?* Seen in runs (SETTLED_COPIES 20261011_212014).

Misses 151 to 200

151. SETTLED_GROOVES version 1 stopped after its M0 line (HELD) and before any reading of its three bets, on a name it looked for in the wrong namespace: the needle's weight and the pulse's size live in the namespace SETTLED_COPIES builds for CHAOS_SETTLES's text, two layers down, not in SETTLED_COPIES's own.

Kind: TOOL FAULT (a name read one layer too shallow when loading a tool that loads a tool). *Lesson:* Loading by text nests namespaces: a tool loaded by text keeps what it loaded inside its own dictionary, and only the names it pulled out itself are at its top. Ask for every name the new code uses before the long run starts, or read it from the layer that made it. Version 2 reads the two names from the inner namespace; nothing else changed. Set aside with its run among the retired runs. *Proved?* Seen in runs (SETTLED_GROOVES 20261011_222608, set aside).

152. SETTLED_GROOVES's O1 bet, Claude's lean, that at pace 0.99375, settled 2,440 drive periods and sieved at 1e-9, all 16 needles share one repeat period of at most 400 drive periods. Only needles 17 and 25 repeat within 400 (after 399, with 189 turns); the other 14 do not.

Kind: WRONG GUESS, of how long the orbits are and how many. *Lesson:* At the 1e-6 sieve the needles had looked like 9/19 orbits of 19, 38, 57 or 95 periods; finer, each passes within a millionth of itself every 19 periods but closes only much later, and the two that close do so after 21 laps of 19. With negative exponents and long settling these are periodic orbits, most longer than 400 periods, and the 13 clusters at one reading say several coexist rather than one. Redesign, said aloud: read 4,001 times and ask which needles share an orbit; ORBIT-919-2 registered. *Proved?* Seen in runs (SETTLED_GROOVES 20261011_223349).

153. SETTLED_GROOVES_2's (a), Claude's lean, that read 4,000 drive periods at the 1e-9 sieve every needle at pace 0.99375 repeats. Only needle 30 did, after 437 (23 laps of 19, 207 turns).

Kind: WRONG GUESS, of what a long read would find (and a sieve asking for exact repeats of needles still settling). *Lesson:* The longer read sharpened the picture instead of closing it: 13 of the 16 needles lie within 1e-9 of needle 30's 437-period path at some reading, {23, 28} share another path and 22 a third, yet only 30 closes. Either the 13 creep onto 30's orbit very slowly, or the paths are longer still; a sieve asking for exact repeats cannot tell. Redesign, said aloud: measure each needle's distance from 30's orbit lap by lap, and the orbit's own pull (its Floquet multiplier); ORBIT-919-3 registered. *Proved?* Seen in runs (SETTLED_GROOVES_2 20261012_041558).

154. FALL_CHAIN_2's (a) and (b), the falls as a simple chain: with eight fresh seeds (1.48 million pairs), two falls apart the correlation would stand more than two errors above zero and within two errors of the square of one apart (+0.0036). It read +0.0012: 1.52 errors above zero and 2.86 below the chain.

Kind: WRONG GUESS, of a mechanism (a chain where the memory is one step deep). *Lesson:* The chain was the natural first picture, and the first two seeds could not tell it from no memory; with four times the pairs the falls forget faster than a chain does. One step apart they share +0.060; two apart, almost nothing. That is the signature of a moving average of one shared disturbance: each fall shares one kick with the next and none with the one after, which also fits FALL_CHAIN's finding that the halves do not pass their memory through the climb. Redesign, said aloud: find the shared kick; Claude's guess is the speed at landing; FALL-SHARE registered. *Proved?* Seen in runs (FALL_CHAIN_2 20261012_041438).

155. WINDOW_EDGE's E2, Claude's lean, that just past the 1/4 window's upper edge (p_c 0.983752845182936, bisected) the mean calm stretch grows like one over the square root of the distance, a log-log slope between -0.65 and -0.35 over d = 1 to 32 millionths. It read -0.298: the mean fell only from 5.1 to 2.0 drive periods.

Kind: REGISTRATION FAULT, of a definition (a calm test loose enough to be met by chance in the chaos). *Lesson:* The calm test (the state 4 drive periods on within 0.05, one turn gained) is met for a period or two at a time by the chaos itself: the median calm stretch is 1 or 2 drive periods at every pace and there are 500 to 600 of them in 20,000 periods, so the mean counted coincidences. The flicker the bet was about shows in the long stretches instead: the longest whole stretch 62, 42, 26, 18, 10 and 6, and the first passage from the old groove state 58, 38, 26, 18, 10 and 6 (least-squares slope -0.56 over 1 to 8 millionths, -0.65 over all six), and at the two nearest paces the needle came back into the channel for longer than its first passage. Redesign, said aloud: find the fold itself by Newton's method, test the square-root law on the orbit's multiplier, and time the first passage from the window's far end so the start does not sit inside the channel; WINDOW-EDGE-2 registered. *Proved?* Seen in runs (WINDOW_EDGE 20261012_080030).

156. WINDOW_CASCADE version 1 found the doublings 4 to 8, 8 to 16 and 16 to 32 and picked up the period-32 orbit; walking that orbit down in pace, Newton's method slid back onto the period-16 orbit, so no multiplier would ever pass -1 and the walk would have run on to its limit. Stopped before its verdicts.

Kind: TOOL FAULT (a continuation that can fall back onto the orbit it was born from). *Lesson:* A doubled orbit is born touching its parent, and the parent is a fixed point of the doubled map too: Newton started near both will take whichever is closer. The guard is cheap: count a solution only if it fails to repeat over half its period, and halve the step when it does. A stand-in run (the same equation written afresh) showed the slide while version 1 was still loading its parent, so version 1 was stopped there; version 2 adds the guard, the halving and a start carried along the last two solutions. Set aside with its run among the retired runs. *Proved?* Seen in runs (WINDOW_CASCADE 20261012_080715, set aside).

157. WINDOW_CASCADE's (a), Claude's lean, that the doublings 4 to 8, 8 to 16, 16 to 32 and 32 to 64 all lie between 0.98371 and 0.98373. Only the first does (0.98372406); the others lie at 0.98370962, 0.98370621 and 0.98370546.

Kind: WRONG GUESS, of where the cascade sits, read off a walker that had not settled. *Lesson:* The bracket came from SETTLED_GROOVES_2's walker, which found nothing repeating within 40 drive periods at 0.98371. There the period-8 orbit is stable, but its multiplier is near -1 (the next doubling is only 4e-7 of pace below), and 100 drive periods of settling were not enough to reach it to 1e-6. A walker's 'nothing repeats' near a doubling means 'not yet'. The ratio bet (b) held: 4.2365 then 4.5513, climbing toward 4.6692. Redesign, said aloud: hold the walker 4,000 drive periods at 0.98371, and carry the cascade two doublings deeper; CASCADE-2 registered. *Proved?* Seen in runs (WINDOW_CASCADE 20261012_081249).

158. WINDOW_EDGE_2's (a), Claude's lean, that Newton's fold lies above WINDOW_EDGE's bisected edge (0.983752845182936) by 1e-8 to 1e-6 of pace, because near the fold the orbit pulls too weakly for a 400-period trial to settle onto it. It lies below, by 1.593e-8 (p_f 0.983752829249151).

Kind: WRONG GUESS, of which way a finite trial errs. *Lesson:* Claude weighed the slow settling below the fold and forgot the slow leaving above it. Just past a fold the vanished orbit's ghost holds the needle for a passage that grows like one over the square root of the distance (about 510 drive periods at 1.6e-8 past it, by a fit read after the run), and inside the ghost the state barely moves in 4 drive periods; so a 400-period trial counted the ghost as the groove. The square-root law itself held twice over in the same run (the orbit's multiplier, slope +0.5015; the first passage, slope -0.5498). Redesign, said aloud: make the trial's length the knob, so the bisected edge should close on the fold as the trial lengthens; GHOST-TRIAL registered. *Proved?* Seen in runs (WINDOW_EDGE_2 20261012_171509).

159. SPEED_CHAIN's (b), Claude's lean, that the barrier top is a section like the landing: given the needle's speed as it crosses the top, the climb before it and the fall after it keep under a tenth of their correlation (+0.5865). They keep +0.0980, a sixth.

Kind: WRONG GUESS, of how clean a section the top is (or of what a straight-line partial can take out). *Lesson:* The top speed carries most of the tie (+0.7004 with the climb, +0.7741 with the fall) but not all, where the landing speed had carried all of the falls' memory. Two reasons fit, said aloud: the tie may run through the speed along a curve a straight line cannot follow (a crossing at almost no speed makes both the climb and the fall long); or the landed top, being the last crossing before the landing, is not a moment the needle can know it has reached, so what comes after it

is not cut off from what came before. Redesign: take the top speed out along its own curve, by binned means, from the seed files already written; TOP-SECTION registered. If it misses too, the last-crossing reason is next, with a replay that counts each top's crossings. *Proved?* Seen in runs (SPEED_CHAIN 20261012_171949).

160. GHOST_TRIAL's (a), Claude's lean, that each fourfold lengthening of the trial cuts the bisected edge's distance from Newton's fold to under a quarter (the passage's square-root law says to a sixteenth). From 400 to 1,600 drive periods it fell to 0.215; from 1,600 to 6,400 only to 0.458 (distances 1.59e-8, 3.43e-9, 1.57e-9).

Kind: WRONG GUESS, of what a long trial measures past an inner fold. *Lesson:* Claude treated the trial as timing the first passage. But the fold is inner (INNER_OUTER, the same batch): the chaos keeps sending the needle back into the ghost, a quarter of the time at the nearest distance. At 1.57e-9 past the fold the first passage lasts about 1,640 drive periods, far short of 6,400, so the trial that counted as on there caught the needle on a return. A long trial past an inner fold answers by where the needle happens to be when the trial ends. Redesign, said aloud: scan many paces past the fold with 6,400-period trials and look for answers that switch back and forth; GHOST-TRIAL-2 registered. *Proved?* Seen in runs (GHOST_TRIAL 20261013_005637).

161. SADDLE_LOG's (a), Claude's lean, that over the slowest crossings of the barrier top the mean fall grows against the log of the lag's crossing speed with a slope within 30 per cent of -0.695 periods, the saddle's unstable escape rate (0.2176) read as a log law. It grew with slope -0.3231 (R squared 0.94): a log law, half as steep.

Kind: WRONG GUESS, of a mechanism (a noiseless escape where the noise is as large as the crossing speeds). *Lesson:* The log law is there, but the noise on the needle's velocity (0.2, divided by its memory 2) accumulates about 0.2 in speed over the 4.6 time units of a slow escape, as much as the slowest crossings have, so it shakes the needle off the saddle sooner than the crossing speed alone would. The climb, printed beside, came within 6 per cent of the saddle's other rate (-0.223 against -0.211): the approach to a saddle is ruled by its stable rate and the leaving by its unstable one, and the approach was not shortened. Redesign, said aloud: launch needles from the top itself at chosen speeds and turn the noise down; SADDLE-NOISE registered. *Proved?* Seen in runs (SADDLE_LOG 20261013_012417).

162. SADDLE_NOISE's (a) and (c), Claude's leans, that launched without noise from the barrier top the fall keeps the saddle's full log law within 15 per cent (-0.6949), and that the law weakens at each step of the noise. Without noise the slope is -0.5457, 0.79 of the law; at noise 0.05 and 0.1 it is -0.5798 and -0.5727, a little steeper; only at 0.2 does it drop, to -0.3787.

Kind: WRONG GUESS, of the regime (a law that is an asymptote read as if it held across the range). *Lesson:* The saddle's log law is what the fall does as the crossing speed goes to zero; at crossing speeds of 0.09 to 0.39 the needle leaves the top before the escape is purely along the unstable direction, so the noiseless slope is a fifth short. The noise matters only when it is as large as the crossing speeds (0.2), where it takes the slope down to -0.38, close to the replays' -0.32 (rig check held). So SADDLE_LOG's halving was a fifth range and the rest noise, not noise alone. Redesign, said aloud: launch without noise at crossing speeds a hundred to a thousand times smaller, where the law should be whole; SADDLE-ASYMPTOTE registered. *Proved?* Seen in runs (SADDLE_NOISE 20261013_024850).

163. CATCH_TRAP_2's (b), Claude's lean, that on the 6,613 catches of eight seeds the excess over 2 periods has median over mean above ln 2 by more than two standard errors, as both small seeds leaned (0.853 and 0.800 on 69 and 64 catches). It came out 0.6782, 1.68 errors below ln 2 = 0.6931.

Kind: WRONG GUESS, from two small samples. *Lesson:* The two seeds' lean was a small sample's scatter (on 69 catches the standard error is about 0.09), and on a hundred times the catches it reversed. The catches are still not one exponential ((a) held: Kolmogorov distance 0.0240 against 0.0134), but the departure runs the other way: too many short excesses (1,612 under a quarter of the mean against the exponential's 1,463) and a slightly long tail (357 beyond three means against 329; mean squared over variance 0.97). My reading: the 2-period cut lets in climbs that are merely slow, not caught, and they pile up just past it; the saddle's log law (SADDLE_LOG, SADDLE_NOISE, GHOST_DEPTH, REINJECT_CLOCK and SADDLE_ASYMPTOTE) makes slow climbs common. Redesign, said aloud: raise the cut to 3 and 4 periods and ask whether what is left is one exponential and forgets; CATCH-TRAP-3 registered. *Proved?* Seen in runs (CATCH_TRAP_2 20261013_030643).

164. ORBIT_919_3's (a) and (b), Claude's leans, that the 12 needles on needle 30's path creep onto its 437-period orbit lap by lap on a straight line in log, and that the orbit's own pull matches the creep within a factor 2. Nothing crept: over 20 laps their distances wander from 3.7e-12 to 7.9e-8 with no trend, and the orbit's pull is about a thousand times stronger than any creep (largest multiplier 4.93e-22 over 437 drive periods).

Kind: WRONG GUESS, of a mechanism; and probably the instrument (the arithmetic). *Lesson:* I read 'on its path without closing' as 'arriving slowly'. But an orbit that squeezes a deviation by about 1e-21 every lap would pull any needle within 1e-8 onto itself to the last digit within a lap, and needle 30 itself strays from its own first lap by up to 1.1e-8. So what the 1e-9 sieve saw is not slow arrival but scatter about one orbit, at a size double precision can reach only if somewhere along its 437 drive periods the orbit stretches a deviation a million-fold before squeezing it. The 13 needles are probably on one orbit, and SETTLED_GROOVES_2's sieve (row 153) was finer than the arithmetic holds

there. Redesign, said aloud: measure the orbit's largest stretch, and carry the 13 needles in extended precision, where the scatter should shrink about two thousand-fold if it is the rounding; ORBIT-919-4 registered. *Proved?* Seen in runs (ORBIT_919_3 20261013_031807).

165. ORBIT_919_4's (a), (b) and (d), Claude's leans: that somewhere along needle 30's orbit a deviation grows over a million-fold; and that carried in extended precision, or in double precision with the lag folded into one turn and the time into one drive period, all 13 needles stay within 1e-11 of the orbit at every reading. The largest stretch is 1.09e5 (over 17 drive periods); the largest distances are 1.88e-9 extended and 7.17e-11 folded.

Kind: WRONG GUESS, of a size (bars set on the largest value of a scatter whose largest runs a thousand times its median). *Lesson:* The reading the card was built to test came through: the hovering is the arithmetic. The median distance falls from 3.82e-9 in double precision to 4.94e-13 extended and 9.95e-14 folded, the physics unchanged, and needle 30 scatters exactly as the other twelve do, so the 13 are one orbit. What I misjudged is the shape of the scatter: the stretch, 1e5 rather than 1e6, recurs at phases a multiple of 19 drive periods apart, and the largest distance sits a thousand or more times above the median, so a bar on the largest value fails while the median moves with the grain as the reading says. Redesign, said aloud: the question of one orbit is settled and needs no rerun; the size of the scatter against the grain could be priced by a fourth carry (folded and extended together), but it changes no claim. The live question is the stretch itself: STRETCH-WHERE registered. *Proved?* Seen in runs (ORBIT_919_4 20261013_040039).

166. ORBIT_19's (a) and (b), Claude's leans, that with the numbers kept small needle 30 repeats after 19 drive periods to below 1e-12 in median, and that each of the 13 needles lies within 1e-12 in median of its 19 readings. The repeat at 19 came out 1.04e-12 (at 38, 8.5e-13), and the needles' medians 4.3e-13 to 2.9e-12.

Kind: WRONG GUESS, of a size (a bar set on the arithmetic's floor rather than on the gap). *Lesson:* The orbit is one of 19 drive periods beyond any doubt: at every other q up to 40 the median is 0.72 to 1.62, twelve orders of magnitude above the repeat; the turns are 9; and the 19-period pull raised to the 23rd power gives the 437-period pull ((c) held). What missed is a bar placed at 1e-12, where the folded arithmetic's own floor sits. Redesign, said aloud: none needed for the question, which is answered; the lesson is to bet on the gap between the repeat and the rest, not on a floor the arithmetic sets. It is the third bar of this thread set on a largest or least value at the floor (rows 164 and 165). *Proved?* Seen in runs (ORBIT_19 20261013_041840).

167. STRETCH_WHERE's (a), Claude's lean, that across the 437 drive periods of the orbit at 0.99375 a period's growth ranks with the share of it spent on the top half of the turn (where the junction's inductance is negative) above 0.8. It came out +0.7094.

Kind: WRONG GUESS, of a measure (a measure that saturates: 138 of the 437 periods, 6 in every 19, are spent wholly on the top half). *Lesson:* The circuit picture held where it was tested hardest: the largest growth, 20.9 in one period, is in a period spent wholly on top ((b) held), its multiplier is real and nearly as large (20.5), so it is growth and not a shear, and the periods spent mostly on the bottom ring down at exactly e^(-pi) each, the size the drag sets. What missed is the measure. The share counts time on the top half but not how far over the top the needle climbs, and the frozen growth rate is largest at the very top, so a period that grazes the top half and one that crosses its crest score the same share. Six periods in every 19 tie at a share of 1, with growths from 4.9 to 20.9, and those ties cap the correlation. Redesign, said aloud: rank by the depth (the period's mean of -cos L), which keeps ranking where the share saturates. On STRETCH_WHERE's own orbit the depth ranks at 0.97, but that orbit was seen while building the Stretch and Squeeze Explorer, the book's companion page, so the test runs on eight fresh paces: STRETCH-DEPTH registered. *Proved?* Seen in runs (STRETCH_WHERE 20261013_045856).

168. WADDLE_BOOKS's (a) and (b): that in every top-up the couple's work equals the board's, the spring's and the damper's shares within 1e-9, and Claude's guess that over 80 per cent of the waddle's floor at couple 1 is the damper eating the catch's leftover ring. The books close to 3e-9 at couple 1 and 1.1e-8 at couple 10; the damper is 0.6 per cent of the floor, the board's own energy 88 per cent.

Kind: THE INSTRUMENT (the integrator's step), and a WRONG GUESS, of a mechanism. *Lesson:* Both legs point at the same culprit. The books' residual halves each time the step is halved, so it is the integrator's own first-order error; and the floor itself falls in proportion to the step (1.01e-5, 6.40e-6, 4.52e-6 at dt 1e-5, 5e-6, 2.5e-6), so about three quarters of it was never the walk: it was the arithmetic's grain, paid as extra spin energy at the end of each top-up. The damper guess was also wrong on its own terms, since the catch's leftover ring is paid for by the ring, not by the couple. What is left with no step, about 2.6e-6, looks like the pin stretching under the bigger spin's centrifugal pull (the spring's part 1.12e-6 against 1.17e-6 by formula, a reading after the run). Redesign, said aloud: take the step out by halving it twice and test that the step-free floor goes as one over the stiffness, the centrifugal stretch's law; WADDLE-STEP registered. The card changes nothing the book says; the lesson for the working draft is that a toll a hundred-thousandth of the bill must be checked against the step before it is given a mechanism. *Proved?* Seen in runs (WADDLE_BOOKS 20261013_063353).

169. WADDLE_STEP's (a), Claude's lean, that with the step taken out the waddle's floor at couple 1 goes as one over the pin's stiffness, the pin's centrifugal stretch, so that it halves at 2e5 and quarters at 4e5. It stayed near 2.5e-6: 2.646e-6, 2.572e-6 and 2.535e-6 at 1e5, 2e5 and 4e5.

Kind: WRONG GUESS, of a mechanism. *Lesson:* Half of the guess was right: the spring's share is the pin stretched by the spin's centrifugal pull and goes as one over the stiffness to within 2 per cent ((b) held). But the stretch was never most of the floor, and as the pin stiffens the board takes up what the spring gives back, so the sum tends to a floor of the rigid walk, about 2.5e-6, that no pin explains. A second thing came out beside it: the step's part of the floor belongs to stiffness 1e5 alone (7.5e-6 there, under 5e-10 at 2e5 and 4e5), so WADDLE_FOLLOW's floor falling to a quarter with a stiffer pin was the step's part disappearing, not the pin's give. Redesign, said aloud: look at the catch, where the rigid walk could leave the board with a little less than half its spin, so the top-up must pay a little more than 1/8; WADDLE-CATCH registered. Still a question for the working draft only: none of it changes what the book says. *Proved?* Seen in runs (WADDLE_STEP 20261013_073935).

170. QUANT_ZERO's M1, Claude's bar, that at memory 0 the pulsed needle's one-drive-period map fits a Möbius map to under 1e-9 radian at paces 0.98, 0.99 and 1.00. It came out 9.61e-9, 6.28e-9 and 8.47e-9.

Kind: THE INSTRUMENT (a bar set on the integrator's floor). *Lesson:* The structure the card leans on came through. Memory 1 misses the same fit by 2.78 radian, eight orders of magnitude above, and at memory 0 the scan found no lock at any q from 2 to 40, as the theorem says ((a) held). What missed is the bar: 1e-9 for a map carried at 160 RK4 steps a drive period, where the integrator's own error is about 1e-8. Redesign, said aloud: the lesson is the Field Guide's, to bet on the gap and not on a floor; the cheap check was to halve the step twice and watch the miss fall sixteen times at each halving, so QUANT-STEP was registered, run at once, and held (15.75 to 16.43 each halving, 2e-12 at 1,280 steps). It is the fourth bar of this thread set on a floor (rows 164 to 166). *Proved?* Seen in runs (QUANT_ZERO 20261013_081905, QUANT_STEP 20261013_082454).

171. WADDLE_CATCH's (a), Claude's guess, that the waddle's rigid floor at couple 1 (2.5352e-6 at stiffness 4e5 and step 2.5e-6) is half to all the catch keeping a little less than half the spin, so that the top-up must pay a little more than 1/8. The catch keeps 0.499999704 of the spin; its shortfall, 4.94e-8, is 0.0195 of the floor.

Kind: WRONG GUESS, of a mechanism. *Lesson:* The catch is nearly perfect, and its small loss goes as one over the stiffness (1.97e-7, 9.87e-8 and 4.94e-8 at 1e5, 2e5 and 4e5), a pin's effect like the spring's share, not a rigid one. So the floor that stays near 2.5e-6 as the pin stiffens is not the step, not the spring and not the catch. Redesign, said aloud: look next at where in the walk the extra positive work is paid, by booking the floor over time windows (during the catch, during the top-up, between them), and at the top-up's stopping rule, the spin it stops at and how finely that is read. Not registered as a card: the result changes nothing the book says, so under the rules of the time the test is laid down on the proposed list as a question for the working draft. *Proved?* Seen in runs (WADDLE_CATCH 20261013_082013).

172. SPIN_START's (a) and (b), Claude's leans: that a remembering pulsed needle started spinning (speed 1) is still turning after 2,000 drive periods at every pace from 0.9800 to 0.9950, and that at the grid paces nearest the 1/4 window and the 9/19 orbit (0.9837, 0.9838, 0.9937, 0.9938) its rotation number lies within 0.01 of 1/4 and of 9/19. It sat at every pace up to 0.9949, those four among them.

Kind: THE INSTRUMENT (a grid coarser than the windows it was looking for), and a WRONG GUESS, of a mechanism. *Lesson:* Leg (b) was set where it could not be met: the 1/4 window is 4.8e-5 wide, under half the sweep's step of 1e-4, and the four grid paces sit outside both windows; I knew the width and registered the leg anyway. Leg (a) found something true: outside the windows the spin on that side does not last. A needle started at a full turn a cycle tumbles and then sits, within 2,000 drive periods, at every pace below 0.995, so the windows are islands of lasting spin in a sea of spin that drains into the sitting lock, and the book's sentence that a needle has to be spinning already to find them is true but not enough: it must also be inside. Redesign, said aloud: start the spinning needles at the windows' own paces and just outside them, many starts and long enough (CHAOS_SETTLES saw settlings up to about 31,000 time units at 0.99375); SPIN-WINDOWS registered. *Proved?* Seen in runs (SPIN_START 20261013_161916) Corrected the same day (misses row 173): the 'true' finding of leg (a) was a misreading; the needles that sat had sat within two cycles, decided by their start, and from other starts the spin outside the window lasts.

173. SPIN_WINDOWS's (b), Claude's lean, that just outside the 1/4 window (0.98365 and 0.98380) all 16 needles started spinning end sitting within 8,000 drive periods; and with it the reading of SPIN_START written earlier that day, that outside the windows a spinning needle tumbles and then sits. Only 7 of 16 sat, the same 7 starting lags at each pace by the window, each within its first two cycles; the other 9 tumbled on for all 8,000 cycles.

Kind: WRONG GUESS, of a mechanism, resting on a MISREADING (a conclusion drawn from one start per pace, without the time it sat). *Lesson:* SPIN_START gave one needle per pace, started at lag 0, and recorded only that it sat by 2,000 cycles. I read that as a spin that drains, wrote it into the book, the Field Guide's lessons and a no-physics read, and registered (b) on it. With 16 starts and the sitting-down time printed, the picture is plain: where a spinning needle sits is decided in its first cycle by where it starts, and lag 0 is one of the starts that sit at once. Outside the window the spin is lasting chaos, at about 0.4 turn a cycle; inside, the same starts settle on the window's orbit ((a) held). Redesign, said aloud: none needed for the question, which is answered; the corrections are made where the misreading went (the book's paragraph, SPIN_START's row and result, the data ledger's row SPIN-START, the Field

Guide's lesson, and a note under the no-physics read written beside SPIN_START). The lesson: a sweep with one start per pace reads one basin, and a 'when' that was never measured must not be written as a 'how'. *Proved?* Seen in runs (SPIN_WINDOWS 20261013_163655; SPIN_START 20261013_161916).

174. HALF_PARENT's (B) and (c). (B), the rival picture registered from Bear's reading of the figure, that past 1.0105 the 4-cycle orbit's multiplier peaks below +1 and falls again, another tooth that might grow; it climbed without a pause to +0.9987 and the orbit ran into its parent at 1.01512. (c), Claude's lean, that from rest every pace from 1.0101 to 1.0300 locks at 2 or 4 cycles; 45 paces locked at nothing, 43 of them from 1.0230 to 1.0272, and 28 at 2/3 from 1.0273.

Kind: WRONG GUESS, of a shape ((B)), and WRONG GUESS, of an extent ((c)). *Lesson:* (B) read the climb correctly: the curve does keep rising past the figure's edge, and something does grow after it, the parent's own multiplier, which climbs from -1 at 1.01511 toward +1 and ends the half gear near 1.023. What it did not have is the parent: a doubled orbit climbing toward +1 is heading for the orbit it was born from. (c) assumed the half gear runs past 1.03; it ends near 1.023, the sweep's 'none' returns there, as Bear's look at the top panel guessed it might, and a third gear, 2/3, follows from 1.0273. Redesign, said aloud: for (B), none needed, (a) answers the question; for (c), where the half gear ends and what fills the gap to 2/3 is the next question: HALF-TOP registered. *Proved?* Seen in runs (HALF_PARENT 20261013_163612).

175. EAR_COST's (a) and (b), Claude's lean (registered 6 October): that the steepest pull held falls as the pack grows at every look interval (at least 0.01 lower at pack 0.4 than at 0.1), and that every pack has a best look interval inside the grid, looking every half fall time costing at least 0.01. At a look every 2 fall times the heaviest pack held only 0.005 less than the lightest, at 2.5 every pack held the same 0.020, and the lightest pack paid only 0.005 for looking every half fall time.

Kind: WRONG GUESS, of an extent (a law that holds where looks are frequent read as holding at every look rate), and THE INSTRUMENT (a grid step as large as the effect). *Lesson:* The lean's shape came through where it could: pack 0.4 held 0.090, 0.100 and 0.110 at a look every 0.5, 0.75 and 1 fall time, against 0.125 with no pack. What I missed is that the pack's cost is paid per look: where looks are rare it falls under the grid's step of 0.005, and past a look every 2 fall times the path decides everything (6, 31, 62 and 94 per cent lost on flat ground at 2, 2.5, 3 and 4). Read after the run, the losses sit within one step of the looks' average pull, the pack's shift (0.17 rad per unit of pack) times the share of time spent looking. Redesign, said aloud: turn the path off, halve the grid's step, double the hikers and test that law directly, with a leg of long, rare looks to see whether a peak costs more than its average; EAR-COST-2 registered. *Proved?* Seen in runs (EAR_COST 20261013_183245).

176. EAR_COST_2's (b), Claude's lean, that at pack 0.4 and the same share of time spent looking, a quarter, looks of a whole fall time every 4 fall times cost at least 1.5 times the loss of looks of a quarter fall time every fall time. They cost 1.2 times (0.0300 against 0.0250), and looks of half a fall time every 2 cost 0.0225.

Kind: WRONG GUESS, of a size. *Lesson:* I expected a look that outlasts the rider's reaction (a delay of 0.31 fall times, a rest of 0.5) to be paid at its peak, since for a whole fall time the pull is more than the rider can hold steadily. It is paid nearly at its average, and (a) held: the loss against the looks' average pull has slope 1.04 over 20 settings. A rider who falls only by gathering tilt can ride out a short excess. Redesign, said aloud: lengthen the looks at the same share, 2 and 4 fall times every 8 and 16, until the peak shows or the average holds to the end; EAR-COST-3 registered. *Proved?* Seen in runs (EAR_COST_2 20261013_183913).

177. STAIR_WIDTHS's (b), Claude's lean, that the locked fractions never step back along the sweep (with one turn a cycle reached by 1.20, which held: 1.14410). They step back at five single paces: 3/5 at 1.03280 and 1.03295 inside 2/3, 9/13 at 1.05575 after 5/7, 5/7 at 1.06025 after 8/11, 13/17 at 1.07620 after 10/13.

Kind: WRONG GUESS, of an extent (GEAR_TRAIN's bands, three paces or more, rise; single paces need not). *Lesson:* GEAR_TRAIN's (b) judged bands, and its lone 3/5 at 1.0328 was printed beside, one of the three surprises named after that run; the finer sweep finds it on three paces and four other step-backs, each on one or two paces at the edge of a step or inside the chaos beside one. A needle from rest reads one basin, so a step-back may be a second gear that a different start would not take. Redesign, said aloud: the test proposed that day already covers it (16 starts at the lone 3/5, as SPIN_WINDOWS did), widened to all five paces; SECOND-GEAR registered. *Proved?* Seen in runs (STAIR_WIDTHS 20261013_184033).

178. WOBBLE_SOFT's (a), Claude's lean (registered 4 October as an unrun guess): that wherever no needle slipped (slips under 0.05 per 100 field cycles) the wobble gauge reads within 0.03 of the pace a clean needle of the same mean stiffness would have. It did at 23 of 26 settings; at three it read 0.918, 0.952 and 0.984 against 0.589, 0.785 and 0.863.

Kind: THE INSTRUMENT (a no-slip bar coarser than one slip). *Lesson:* The guess's mechanism came through where it could be read: the gauge reads the noise-softened stiffness, sqrt(1 - mean cos(lag) squared), within 0.02, so a noisy needle really is nearer its edge than its pace says. The three misses are the settings where a few needles slipped once (0.01 to 0.03 per 100 field cycles), and one slip in the record destroys the gauge's correlation and pins it near 1. My

bar for no slip let them in. Redesign, said aloud: covered by SLIP_PROOF, the slip-proof gauge, which counts slips as events and leaves them out; it also tells whether the gauge's early warning (b, held) comes from the softened pull or from hearing the first rare slips. *Proved?* Seen in runs (WOBBLE_SOFT 20261013_192713).

179. SECOND_GEAR's (a), Claude's lean, that at every one of the five step-back paces 16 needles started round the circle take at least two locked fractions, one of them the step-back's. It held at 1.03280 and 1.03295 (3/5 for 2 starts, 2/3 for 14), and missed at 1.05575 (9/13 for 15, one tumbling at 0.6924), 1.06025 (5/7 for all 16) and 1.07620 (13/17 for 6, ten tumbling at 0.7692 to 0.7699).

Kind: WRONG GUESS, of a mechanism (one cause given to step-backs of two kinds). *Lesson:* I took every step-back for a second basin, as the lone 3/5 is. At 1.05575 and 1.06025 the lower fraction is the only state any start finds, so there the staircase itself runs back with the pace; at 1.07620 the second state is not a locked gear but tumbling at about 10/13. Redesign, said aloud: read the needle's own turn rate, over 4,000 cycles and on a fine grid, across the stretch between the two islands of 5/7, to see whether it falls with the pace there; STEP-BACK-RAMP registered. *Proved?* Seen in runs (SECOND_GEAR 20261013_192828).

180. CHAOS_RAMP's (a) and (b), Claude's leans, that on a grid of 1e-4 from 1.0230 to 1.0272 the turn rate never falls by more than 0.003 from one pace to the next, and that at 1.0272 it lies within 0.02 of 2/3. It fell by up to 0.00364, and at 1.0272 it was 0.63049, 0.036 short.

Kind: WRONG GUESS, of a size ((a)) and of a shape ((b)). *Lesson:* HALF_TOP's seven paces had climbed in order, and I read a smooth ramp into them. On a grid ten times finer the climb stumbles back by a few thousandths (one pace reads exactly 3/5), and it does not close on 2/3 as an intermittent approach to a fold would, with a gap shrinking as the square root of the distance: over the last ten paces the gap's slope is only +0.15, so the chaos meets the step in a jump. Redesign, said aloud: (a) is a size, answered; for (b) find the 2/3 step's lower edge from its own orbit, as the half gear's fold was found, and read the turn rate a millionth below it; RAMP-EDGE registered. *Proved?* Seen in runs (CHAOS_RAMP 20261013_192906).

181. SLIP_PROOF's (a), registered: that wherever slips are under 1 per 100 field cycles, a wobble gauge with its slips found and left out reads within 0.03 of the softened pull. It did at 23 of 33 settings.

Kind: THE INSTRUMENT (a slip finder that could not see slips). *Lesson:* I asked the finder for half a turn within 2 time units without checking that a slip makes that. It does not: a noiseless slip turns at most 0.78 of half a turn in any 2 time units at pace 0.50 and 0.98 at 0.95, so the finder caught only the slips noise hurried, and missed the rest at the very settings the leg was about. A ten-line run of one noiseless slip, before the registration, would have shown it. Redesign, said aloud: a finder that counts whole turns against the needle's own line, checked against the master's count in its fidelity leg; SLIP-PROOF-2 registered, run in the same batch. *Proved?* Seen in runs (SLIP_PROOF 20261013_194935).

182. SLIP_PROOF's (b), Claude's lean: that at the fresh noises 0.15, 0.25 and 0.4 the slip-proof gauge first reaches 0.95 no earlier than the slips pass 1 per 100. It reached 0.95 at the same paces as the plain gauge, 0.10 to 0.15 earlier.

Kind: THE INSTRUMENT (the same blind finder). *Lesson:* With slips left in, the slip-proof gauge is the plain gauge, so this leg could only repeat WOBBLE_SOFT's (b). Nothing is learned about the lean from this run. Redesign, said aloud: covered by row 181's redesign, which keeps this leg unchanged. *Proved?* Seen in runs (SLIP_PROOF 20261013_194935).

183. RAMP_EDGE's F0, a check: that the multiplier nearest +1 passes 1 at the walk's least pace. The walk used its 800 steps and stopped at 1.0273103676, the multiplier at 0.893, with no turn; (a) went undecided.

Kind: THE INSTRUMENT (a budget too small for the walk). *Lesson:* Near the fold the walk moves fast through the orbit's lag and speed and slowly in pace; once the step cap fell to 2e-5 each step lowered the pace by about 3e-8, and the walk needed some 1,600 such steps, not the 330 it had left. I set the budget in steps without asking how far a step moves the pace. Redesign, said aloud: the same walk with 4,000 steps; RAMP-EDGE-2 registered, run in the same batch, where the fold came at step 1,635. *Proved?* Seen in runs (RAMP_EDGE 20261013_195233).

184. STEP_BACK_RAMP's M0, a fidelity check: that at every pace STAIR_WIDTHS locked, the rotation number over 4,000 drive periods equals the fraction within 1e-6. It did at 6 of 33.

Kind: THE INSTRUMENT (a tolerance finer than the reading). *Lesson:* A rotation number read over 4,000 periods ends part-way round any cycle whose length does not divide 4,000, and so carries an error of up to about 1/4000 of a turn; 7, 11 and 13 do not divide it, and only the two 7/10 paces were exact. Every lock sat within 5.6e-5 of its fraction, so no pace changed its gear. The check could not pass as written. Redesign, said aloud: read over 10,010 periods, a whole number of cycles of 7, 10, 11 and 13, with the least repeat tested at each pace, which also asks the run's new question, whether 5/7 holds a third island; FIVE-SEVENTHS-ISLANDS registered. *Proved?* Seen in runs (STEP_BACK_RAMP 20261013_195010).

185. SLIP_PROOF_2's M0, (a) and (b): that its finder matches the master's slip count at 95 in 100 needles at every setting below pace 1, and SLIP_PROOF's two legs. It matched at 26 of 54 settings; (a) held at 24 of 33; (b) missed at all three noises.

Kind: THE INSTRUMENT (a second blind finder). *Lesson:* The finder measured the needle's slips against the needle's own straight line, and a single slip tilts that line: a whole turn lost in the middle of the record tilts the line by a turn and a half from end to end, so against it the slip shows as about half a turn, under the three-quarter trigger. The first round found nothing, and the refit, which only corrects slips already found, never began. A planted slip in a test record, before the registration, would have shown it; I wrote the finder from the picture of a level line. Beside the judged legs, at the one setting with a slip where the finder matched the master needle by needle (noise 0.15, pace 0.85), leaving the slip out took the gauge from 0.984 to 0.867, against a softened pull of 0.863. Redesign, said aloud: a finder whose line is fitted within the calm stretches only, checked on fresh needles against the master's count before it is registered exact; SLIP-PROOF-3 registered. *Proved?* Seen in runs (SLIP_PROOF_2 20261013_200028).

186. RAMP_EDGE_2's (a), RAMP_EDGE's lean kept as registered: that a millionth below the 2/3 step's edge, from rest, the turn rate lies more than 0.02 short of 2/3, a jump onto the step. It lies 0.009 short; 0.013 a hundred-thousandth below and 0.039 a ten-thousandth below.

Kind: WRONG GUESS, of a shape. *Lesson:* I read a jump from one gap, 0.036 a ten-thousandth below a step whose edge I had not found. Found exactly, at 1.027298976, the edge sits a millionth below the sweep's first 2/3 pace, and the gap closes toward it. Before this run my lean had already turned to a squeeze by intermittency, and I said so in the registration while keeping the old lean to be judged. The squeeze is there; its law is not yet the square root such a squeeze usually has (slope 0.31 on reads of 4,000 periods). Redesign, said aloud: covered for the question asked, jump or squeeze; the law of the closing gap is new, and goes to the proposed list (longer reads, nearer the edge). *Proved?* Seen in runs (RAMP_EDGE_2 20261013_200133).

187. QUIET_TAP's (a), Claude's lean: that with no jostling the never-blind sized tap keeps every rider up at c 2.5 and 3.0 at every delay up to 1.80 fall times. c 3.0 did, to 2.00; c 2.5 kept every rider to 1.75 and lost all at 1.80.

Kind: WRONG GUESS, of a size (by one grid step). *Lesson:* The lean's point, that the noise and not the controller sets the edge near 1.55 to 1.70, came through: without jostling the edges are 1.75 to 1.80 and 2.00 to 2.05. My bar sat on the gentler kick's own edge. Redesign, said aloud: covered; the run finds both edges on its grid. What it adds is new: just before its fall the gentler kick's hum turns into a swing, its wobble leaping from 0.049 to 0.21 and 0.34 with every rider still up; that goes to the proposed list. *Proved?* Seen in runs (QUIET_TAP 20261013_195754).

188. V13_FIDELITY's F2, said before both runs: that every markdown file in the project's folder that has a PDF builds with build_paper.py v1.3 at exit 0. The first run built 62 of 66; the second and the third, of record, after v1.3 was amended, 64 of 66.

Kind: THE INSTRUMENT, twice, then THE FILES. *Lesson:* The first run caught two faults in the builder before release: a fenced block of very long lines widened the page and shrank a paper to 5.0 pt (v1.3's own layout check refused it, which is the check working), and v1.2's read-back took a nested list's first item into the outer item's probe and called a whole item missing. Both were mended and given fixtures. The two files that still do not build are of 2 October and their shape, not the builder, stops them: a table before the first heading, set as the byline, and no second-level heading at all. Redesign, said aloud: covered; the builder is right to refuse them, and they keep their old PDFs. *Proved?* Seen in runs (V13_FIDELITY 20261013_202203, 20261013_203154 and 20261013_204846).

189. COLUMN_ACCORDION's (b), Claude's lean (one of the road questions of 6 October, on the column of Bear's mountain road), that the shortest time gap keeping a stumble from growing down a column rises with the reaction delay. It read 0.5, 0.5, none, none, none and 3.0 s at delays 0 to 1.0 s; (a) read HELD only because walkers fell over.

Kind: THE INSTRUMENT (a grid past the walker's own stability). *Lesson:* With the gains I chose, a lone walker's own loop rings up and falls over once its delay passes 0.34 to 0.73 s, by its closed form, which I did not work out before registering; from 0.4 s the run read the walker, not the column, and its growths are ratios of runaway numbers. The early lesson to work out the instrument before the registration was not applied to the toy's own parameters. Redesign, said aloud: gentler walkers that stay steady alone at every delay, with the closed form worked out first; COLUMN-ACCORDION-2 registered and run in the same batch, every leg held. *Proved?* Seen in runs (COLUMN_ACCORDION 20261013_222625).

190. FIVE_SEVENTHS's M0 and (a): that read over 10,010 periods, a whole number of cycles of 7, 10, 11 and 13, every lock's rotation number is exact within 1e-9, and that the paces 1.05775 to 1.05820 lock at 5/7 with least repeat 7. M0 held at 30 of 33; no pace in the middle locks at all.

Kind: THE INSTRUMENT (M0) and WRONG GUESS, of a mechanism ((a)). *Lesson:* For M0 I counted the reduced fractions' denominators and not the orbits' least repeats: the '5/7' islands repeat after 14 and 28 cycles and the '7/10' after 20, and 20 and 28 do not divide 10,010; the labels themselves all repeated exactly. It is row 184's lesson again, missed by reading the fraction instead of the orbit. For (a) I took a turn rate equal to 5/7 for a lock; it is a needle that repeats at no cycle up to 40 yet keeps 5/7 within 3e-6, which looks like chaos in bands that keep the gear. Redesign,

said aloud: M0 covered (every label repeated; a read over a multiple of 28 and 20 would make it exact); for (a), test the middle as chaos that keeps an exact gear, its exponent and its bounded count; BANDED-FIVE-SEVENTHS registered. *Proved?* Seen in runs (FIVE_SEVENTHS 20261013_222744).

191. CHAOS_ORDER's (a) and (b), Claude's leans (on Bear's chaos that keeps order): that the chaotic needle's spread changes by less than a factor of 2 from no noise to sigma 0.03, and that at sigma 0.03 locked needles near their edges spread more than chaotic ones. The spread rose 1.5 to 2.5 times; at 0.03 one lock tied the chaos (1.846e-2 against 1.853e-2).

Kind: WRONG GUESS, of a measure. *Lesson:* I measured order by the spread of the turn count, and the spread is the noise's as much as the chaos's. The order Bear named shows in the average: at noise 0.003 both locks near their edges lost their gear, every needle slipping 56 to 149 turns in 3,960 periods and the rates moving by 0.02 to 0.03, while the chaotic needles' averages moved by 0.0054 or less; the locks also spread 2 to 5 times more there. And 64 needles give the spread only to about 18 per cent, too coarse for a bar at a factor of 2 or a tie. Redesign, said aloud: judge the average rate on a ladder of noise, with 256 needles, and find the noise each lock takes at three distances from its edge; CHAOS-ORDER-2 registered. *Proved?* Seen in runs (CHAOS_ORDER 20261013_222835).

192. BANDED_FIVE's (b), Claude's lean, that the needle's largest Lyapunov exponent is above 0.005 a period at every pace from 1.05775 to 1.05820. It is +0.024 to +0.107 at nine, and -0.031 at 1.05805.

Kind: WRONG GUESS, of extent. *Lesson:* I took the whole middle of 5/7 for one chaos. Bands of chaos are threaded with windows, short stretches of pace where a long cycle takes over, and 1.05805 is one: a cycle longer than 40, which the earlier lock rule could not see. The main finding stands at all ten paces, the count kept within a quarter turn of five in seven. Redesign, said aloud: covered; the window's cycle length goes to the proposed list (a lock rule out to 400 cycles at 1.05805). *Proved?* Seen in runs (BANDED_FIVE 20261013_224945).

193. SLIP_PROOF_3's (a), SLIP_PROOF's lean kept unchanged, that with the slips left out the gauge reads sqrt(1 - mean cos(lag) squared) within 0.03 wherever slips are under 1 per 100. It did at 31 of 33; at noise 0.4 it read +0.033 and +0.034 over.

Kind: WRONG GUESS, of a size. *Lesson:* The softened-hold formula is a small-wobble reading; at noise 0.4 the lag swings wide and the gauge sits a few hundredths above it, as WOBBLE_SOFT's widest gaps already hinted. The question the card was built for, whether the warning is the slips, is answered by (b), which held. Redesign, said aloud: covered; the size at loud noise is a property of the formula, written into the book as 'within 0.034'. *Proved?* Seen in runs (SLIP_PROOF_3 20261013_225419).

194. QUARTER_STROBE's M0 (the data run for the 1/4 window's figure): that at every grid pace 0.4e-6 inside the first doubling and the fold, the 256 readings take exactly 4 values within 1e-6. They did at 137 of 140; the three just past the doubling showed 74, 24 and 9.

Kind: THE INSTRUMENT (the settle). *Lesson:* Next to a doubling the orbit's multiplier nears -1 (-0.981 at the nearest judged pace), so a stray is pulled in only slowly: some 1,460 periods for a start 0.001 off, more for a reading split in two by the alternation, against the 1,000 given. My check before registering counted the judged paces but did not ask the settle time at the one nearest the bifurcation, which its multiplier gives in a second. Redesign, said aloud: QUARTER-STROBE-2, carried 4,000 periods, run in the same batch, held; the check now names the settle time at a judged case near a bifurcation. *Proved?* Seen in runs (QUARTER_STROBE 20261013_231124).

195. ZIGZAG_GAUGE's (a), Claude's lean (leg (b) of the road questions of 6 October): at every turn where the slope is above 0.325 and the fall line turns less than 10 degrees between the legs' middles, the slope read within 10 per cent and the fall line within 5 degrees. No turn passed the gate: UNDECIDED.

Kind: THE INSTRUMENT (a test that could not decide). *Lesson:* The registered hill's base grade, 0.30, lies under the gate, and its steep places are on the bumps, where the fall line swings 10 to 26 degrees between legs; the walker turns every 45 m up the road against bumps 80 and 100 m across. My check planted the plain slope and did not count the turns the gate would judge. Redesign, said aloud: ZIGZAG-GAUGE-2, the base grade raised to 0.40 and its gate counted before registering (28 of 35), run in the same batch; the check now asks whether the test can decide, its judged cases counted without their verdicts. *Proved?* Seen in runs (ZIGZAG_GAUGE 20261013_231629).

196. ZIGZAG_GAUGE_2's (a), Claude's lean: the same, on the same bumps over a base grade of 0.40. 24 of the 28 judged turns were within; the fall line held within 4.38 degrees at all 28, but the slope was off by 21.5, 26.8, 23.8 and 11.5 per cent at four turns on the rise's flank and the hollow's far rim.

Kind: WRONG GUESS, of a mechanism. *Lesson:* I gated on the fall line's direction holding between the legs and took that for the hill being smooth over them. The slope's size can change along a leg while its direction holds, and the reader returns the legs' average, which I compared with the slope at the turn. The gauge reads direction truly and size as an average. Redesign, said aloud: compare the read slope with the true slope averaged along the two legs (ZIGZAG-AVERAGE, on the proposed list). *Proved?* Seen in runs (ZIGZAG_GAUGE_2 20261013_231751).

197. QUARTER_TREE_READ version 1's part (5) and QUARTER_TREE_MARKS's part (c), reading tools: densities of the chaos built by starting 256 needles from the strobe's 256 readings at a pace. Those readings are one needle's 256 consecutive readings, so the 256 needles retraced one path: about 356 and 556 distinct points a pace, each counted many times.

Kind: THE INSTRUMENT (the sample). *Lesson:* Found while building QUARTER_TREE_CURVES: in a scratch probe its fold finder saw only 112 and 224 distinct points at two paces that are plainly chaotic. What it changed: READ's 'emptiest middle hole, 26 readings, where the bands joined' was the single path's (with separate needles the middle holes hold 194 to 431 and the edges 1,112 and 449), and Claude had reported that reading; MARKS's gap walls stand, repeated to the hole with separate needles. A second fault in the redesign itself: needles settled only 200 cycles left a few crossing the gap (in a scratch probe, 5 of 256 at +3.6 and 2 at +4.0; none after 2,500 cycles). Redesign, said aloud: covered; separate needles spread across the band and settled 1,500 cycles, in READ version 2 and QUARTER_TREE_CURVES, and the report corrected. *Proved?* Seen in runs (QUARTER_TREE_READ 20261014_022707, retired; QUARTER_TREE_MARKS 20261014_033510).

198. QUARTER_TREE_MARKS's (a), said before the run: that the histogram's two peaks are the steady 4-cycle road and the first split's daughters. The upper peak is the steady road alone (768 of 834 readings); the lower is the lower daughter and its children.

Kind: WRONG GUESS, of a source. *Lesson:* The upper daughter curves and its children spread wide, so its readings pile nowhere; the lower daughter's children stay bunched, which is the lopsidedness of the split again. Redesign, said aloud: covered; described in the book. *Proved?* Seen in runs (QUARTER_TREE_MARKS 20261014_033510).

199. QUARTER_TREE_CURVES's (e), said before the run: that the void at +0.2 is a window turning once in every 4 cycles, found by the lock rule. No repeat within 1,000 cycles.

Kind: WRONG GUESS, of a kind. *Lesson:* QUARTER_TREE_VOID: a window gone on to its own chaos, 14 clusters in the strip, seven single points and seven thin bands up to 0.003 rad, as Bear's note on his drawing said: bands made of thin bands. Redesign, said aloud: covered; its cycle before the chaos (the window's own period) is on the proposed list. *Proved?* Seen in runs (QUARTER_TREE_CURVES 20261014_055131, QUARTER_TREE_VOID 20261014_060108).

200. WINDOW_TOUCH's (a) and (b), Bear's leans made exact: each of five windows within 0.2 millionths of a touch by the fold's image after its cycle plus 4; the cycle 4 times the band's pieces times 3 or 5. (a) UNDECIDED by its own check: the share of strobe paces near a touch by the windows' images was 0.026, 0.052, 0.169, 0.779 and 0.896, and the test decides only under one half for all. (b) printed 12 = 4 x 1 x 3, 20 = 4 x 1 x 5 and 24 = 4 x 2 x 3, then the tool stopped with an error at the window of 48 cycles, before its verdict lines.

Kind: THE INSTRUMENT (a test that could not decide, and a rule with a hole). *Lesson:* The decides-check added on 13 October did its work: the fold's later images (13 and 21, after 52 and 84 cycles) touch the edges 69 and 132 times between -10 and +5.2, so any pace lies near a touch, and nearness to a dense family of curves is no evidence. My registration measured each window from the pace where it was first seen locked, not from where it opens, and I checked the arithmetic of the images' numbers but never counted their touches before registering. And (b)'s rule asked for chaos on both sides of every window; above +4.8 the strobe holds no chaotic pace before the pile-up, a case I would have seen had I listed the paces. Redesign, said aloud: WINDOW-TOUCH-2, Bear's guess turned round, each crossing of the low edge by images 3 to 9 predicting a window of 4(m - 1) cycles, judged by locks against controls; (b) given a one-side rule; run in the same batch. *Proved?* Seen in runs (WINDOW_TOUCH 20261014_073237).

Misses 201 to 250

201. WINDOW_TOUCH_2's (c), Claude's lean: each crossing of the low edge by images 3 to 9 locks at its predicted cycle. No crossing exists: UNDECIDED, 0 of at least 10.

Kind: THE INSTRUMENT (a definition that contradicted its own theory). *Lesson:* The low edge is image 1, the lowest the band reaches, so image m minus image 1 is never below nought; at a window's heart the image comes down and touches the edge from above, as my said-before for WINDOW_TOUCH put it, and I then defined a crossing as a change of sign. The registration named a planted case, the 12-cycle window's crossing by image 4 near -4.95, but I did not run it before registering; it would have found no crossing. Redesign, said aloud: WINDOW-TOUCH-3, a touch found as a local minimum of image m minus image 1, refined, and judged exact when it comes within 1e-6 rad of nought; the planted case run in a scratch probe before registering. *Proved?* Seen in runs (WINDOW_TOUCH_2 20261014_074538).

202. WINDOW_TOUCH_2's (b), Bear's lean made exact, with the one-side rule: for the window of 48 cycles at +4.8, the cycle is 4 times the band's pieces times 3 or 5. The rule took +4.2, 2 pieces: 48 = 4 x 2 x 6. MISSED.

Kind: THE INSTRUMENT (the sample). *Lesson:* The strobe paces +4.4, +4.6, +5.0 and +5.2 are chaotic by eye (204 to 253 values) but fail the rule's 'under 0.6 of the readings in the 20 fullest holes', because there the band is in thin pieces that bunch the readings; so the nearest pace passing the rule lay across the place near +4.4 where 2 pieces become 4. By arithmetic from the doublings' last measured ratio, 4.668, and the pile-up at +5.2545, the band's

merges come at about +1.1 (2 into 1, measured), +4.36 (4 into 2) and +5.06 (8 into 4), which puts +4.8 in the 4-piece stretch and 48 = 4 x 4 x 3; that is the reading, not a result. Redesign, said aloud: count the pieces with separate needles at paces just beside +4.8 (WINDOW-TOUCH-3's (b2)). *Proved?* Seen in runs (WINDOW_TOUCH_2 20261014_074538).

203. WINDOW_TOUCH_3's (c), Claude's lean: at 0.8 or more of 19 places where an arch touches the band's low edge, the needle locks to 4(m - 1) cycles, and at 0.2 or less of 19 controls. 10 of 19 locked, exactly at their predicted cycles; 0 of 19 controls. MISSED.

Kind: OPEN: THE INSTRUMENT or the toy, not yet told apart; and one WRONG GUESS of mine. *Lesson:* Right of -6 millionths 8 of 9 places lock; left of it 2 of 10, and the one at -9.3026 locks only at the farthest offset, 3e-4 below the touch, so the touch's place there is off by about that much, and windows of 28 and 32 cycles nearer the crisis may be narrower still; or there may be no window. The place at -5.3100 is my error: it lies inside the 12-cycle window, where the arches are no longer the chaos's edges but its orbit settling, and its refined minimum, 2.1e-5 rad, was the largest of the twenty; the 1e-4 bar, set after counting, let it in. Redesign, said aloud: WINDOW-TOUCH-4, to run next: at every place that did not lock, the lock rule scanned across 1e-3 either side in steps of 2e-5, the controls scanned alike; each touch found from the fold at its own pace, not the nearest strobe pace's; and touches inside a window (where the needle at the strobe pace beside it repeats) set aside before the count. *Proved?* Seen in runs (WINDOW_TOUCH_3 20261014_080520).

204. WINDOW_TOUCH_3's (b2), Bear's lean made exact: beside the window of 48 at +4.8, the band in 4 pieces at every chaotic pace of six. Pieces 3, 3, 8, 3, 4 and 4. MISSED.

Kind: THE INSTRUMENT (a rule blind to gaps just opening). *Lesson:* The piece rule counts a gap only at 0.003 rad or more. The 4 pieces are born from 2 near +4.36, their inner gaps opening from nothing, and QUARTER_PHASES's saved density (holes of 0.0007 rad) still shows a single gap at +4.4; so at +4.65 to +4.85 one inner gap is under the rule's 0.003 (3 pieces), and by +4.90 both are over it (4). At +4.75, beside the window, the readings bunch round the window's ghost and leave extra empty runs (8). Redesign, said aloud: covered for now; 48 = 4 x 4 x 3 rests on +4.90 and +4.95 and on the merges' arithmetic, and a piece rule that scales with the gap's growth goes on the proposed list. *Proved?* Seen in runs (WINDOW_TOUCH_3 20261014_080520).

205. SEAM's (s3), Claude's lean on Bear's picture: his inner arc lies on the fold's image 3. Read 0.000 on image 3 against 0.085 shuffled: MISSED, and void.

Kind: THE INSTRUMENT (a ruler read at the wrong scale). *Lesson:* The panel's axes were read from matplotlib at its own 100 dots an inch while the PNG was saved at 110, so every stroke was placed about a tenth off; the pink rungs, which should sit on the windows, came out at -3.71 to +10.40. My check redrew the figure and found it equal to the PNG pixel for pixel, and took that for a check of the axes, which it was not; one look at where the spines sit in the PNG would have caught it. Redesign, said aloud: SEAM-2, the axes at 110 dots an inch, checked on the rungs first; run in the same batch. *Proved?* Seen in runs (SEAM 20261014_090006).

206. SEAM's M0, fidelity: at -10.0 the fold's images repeat QUARTER_TREE_CURVES's within 1e-6 rad. Largest difference 1.90e-05: MISSED.

Kind: THE INSTRUMENT (a parameter one rounding off). *Lesson:* The tool built its pace as 0.98370 - 10e-6, which differs from the strobe file's pace in the last binary digit; in the chaos that small a change gives a new sample of 256 needles after 1,500 cycles, and the fold, fitted to the sample, moves by its own scatter, about 2e-5 rad. A scratch test had shown the stepper gives the same numbers for an array of any length, so I expected an exact repeat, and did not check that the paces were bit for bit the same. Redesign, said aloud: SEAM-2 takes -10.0 from the strobe file; the lesson for fidelity checks of a fitted quantity in chaos is to match the parameter bit for bit or to set the tolerance from the fit's scatter. *Proved?* Seen in runs (SEAM 20261014_090006).

207. SMOOTH_DIRT's (a), Claude's lean (an older test, carried forward): the dirty spring bead's extra buzz is the dry law's, so a smooth friction law (eps 0.001) at friction 0.05 brings the flips back within 10 per cent of the clean bead's 2.1044. Read 2.3519, 11.8 per cent above: MISSED.

Kind: WRONG GUESS, of a size. *Lesson:* Most of the extra buzz is the law's: of the dry law's 0.904 extra flips a time unit, the smooth law keeps 0.248 at eps 0.001 and 0.070 at eps 0.003. What is left grows as the law sharpens toward a step at zero speed, so the buzz follows how sharply the friction turns over as the bead stops, and the dry law, which stops the bead dead, is that turn at its sharpest. I set the bar at one eps and should have registered the trend. Redesign, said aloud: covered for the card's question (the law, by the trend in eps); the trend to eps 0.0001 with a step that handles the stiff drag goes on the proposed list. *Proved?* Seen in runs (SMOOTH_DIRT 20261014_084856).

208. WINDOW_TOUCH_4's (c4), Claude's lean: scanned 1e-3 either side of the eight touches left of -6 in steps of 2e-5, at least 0.8 of them lock at their predicted cycle. None does, at any cycle; none of eight controls either. MISSED.

Kind: OPEN: THE INSTRUMENT or the toy, not yet told apart. *Lesson:* The settle-and-look lock rule can only find a window wider than its step, and the one window found near the crisis (24 cycles at -9.30) is 3.6e-4 wide or more, lying 2e-4 to 6e-4 below its touch. Windows of 28 and 32 cycles nearer the crisis, where the chaos stretches harder (the partner's multiplier is 4.0 there, 3.75 at 0), may be far narrower than 2e-5. Or they are not there. A rule that does not need to land inside the window is wanted. Redesign, said aloud: WINDOW-TOUCH-5, to run next: Newton's method on the predicted cycle, started from the fold's image at each touch and followed across the scan in pace, looking for the cycle's larger multiplier passing through nought, which marks a window however narrow; the controls alike; its probe first. [Settled the same night by WINDOW_TOUCH_5, run 20261014_092920: THE INSTRUMENT, and not by narrowness. All eight windows are there, lying 1.2e-3 to 3e-3 below their touches, beyond the scan's 1e-3; they are 2e-4 to 1.2e-3 wide. My reading above, that they might be narrower than the step, was wrong; the scan was too short.] *Proved?* Seen in runs (WINDOW_TOUCH_4 20261014_090037, WINDOW_TOUCH_5 20261014_092920).

209. CRISIS_BURSTS, said before the run by Claude, from recall: below the window's end the mean stay grows as d^-gamma with gamma = 1/2 + ln a1 / (-ln a2) = 0.553, the correction coming from the squeeze of the orbit that mediates the crisis (a1 = 4.05, a2 = exp(-8 pi)/a1). Measured 0.4982 with a standard error of 0.0011: 48 standard errors from 0.553 and 1.6 from the plain 1/2. The judged lean, a range from 0.45 to 0.65, held.

Kind: WRONG GUESS, of a law's reach (from recall). *Lesson:* The needle behaves here exactly as a map of one dimension. Either I recalled the formula wrongly, or it belongs to a different crisis from this one, where the band's own edge, the fold's image, runs into the orbit; the squeeze, 1e-11 a lap, may simply be too strong to leave a trace. Redesign, said aloud: covered for the window (the measurement answers); on the proposed list, the same count with the squeeze weakened (a heavier needle with less drag) to see whether gamma moves off 1/2, and the formula's exact form and reach for the literature check, names owed. *Proved?* Seen in runs (CRISIS_BURSTS 20261014_091405).

210. COLUMN_TAIL's (g), the lean of 6 October made exact by Claude: past the delay where it grows, the stumble is multiplied by the same factor at every link, every ratio of successive walkers' largest dips from walker 15 to 29 within 5 per cent of the closed form's largest gain. Up to 5.66, 5.26 and 6.99 per cent off at the three settings: MISSED.

Kind: THE INSTRUMENT (an extreme read on a carrier). *Lesson:* Far down the line the stumble is a wave packet: an envelope that grows by the largest gain a link, filled with crests. A walker's largest dip is the deepest crest, and as the packet slides the deepest crest hands over to the next one, so the ratio of extremes saws round the envelope's ratio, about 6 per cent either way. On average the lean holds exactly: the geometric mean ratio is 0.972 to 0.981 of the largest gain against the stationary phase's 0.977. Redesign, said aloud: covered for the lean (it holds on average, to the closed form's own factor); the envelope's size read by a smooth measure (the swing's root mean square, or the envelope of the speed) at every walker goes on the proposed list. Also wrong before registering: I said COLUMN_ACCORDION_2's 120-second run was cut off at the slowest setting; it was not, the growth over 300 s being the same to the last digit. *Proved?* Seen in runs (COLUMN_TAIL 20261014_145058).

211. HEAVY_SHAPE_2's (b1), Bear's purple stroke made exact by Claude: beyond memory 1.2 the window's first doubling lies within 3 per cent of his hand-drawn continuation, 4.876, 4.294 and 3.813 millionths at M 1.2335, 1.2540 and 1.2725. Measured 4.9459, 4.4188 and 4.0147, 1.4, 2.9 and 5.3 per cent above it: MISSED.

Kind: WRONG GUESS, of a shape (a curve carried on straight while it bends). *Lesson:* His purple runs within 0.7 per cent of the straight line in 1/M through the five measured S, so as a continuation it was as good as a straight line can be; the curve itself keeps bending, its slope against 1/M falling 9.48, 9.53, 9.35, 9.00, then 8.72, 8.50 and 8.30, so the window narrows more and more slowly. His word on the picture, that the curves bend, holds for S as well as for C. Redesign, said aloud: covered; the bend is now measured at eight weights, and how it goes on is on the proposed list with a finer follower. *Proved?* Seen in runs (HEAVY_SHAPE_2 20261014_164913).

212. HEAVY_SHAPE_2's (b2) and (h2), Bear's cyan stroke and Claude's lean on C over S at M 1.2540 and 1.2725. Neither decided: the window's end was not found at either weight: UNDECIDED.

Kind: THE INSTRUMENT (needles on a long detour, first read as outside the window's basin and corrected on 14 October; and a probe that did not touch the run's own input). *Lesson:* At these weights another motion lives beside the window, and its basin comes within 1e-4 rad of the window's cycle: of 128 needles started beside the cycle 1e-4 rad apart, 3 left it within 300 cycles even inside the stable stretch (a probe after the run), and wandering across the circle they were read as the window ending at the walk's first step. Started 1e-7 apart, all 128 stay. The blind probe before registering had said the test could decide at 1.2725, but it ran with a first guess of S (3.85) other than the one the run's chain gave (4.42), so it settled the needles at another pace and none strayed: the check touched a neighbour of the run's input, not the input itself, the lesson of the overnight runs earlier that day again. Redesign, said aloud: HEAVY_SHAPE_3, the same measure with the needles 1e-7 apart, its blind probe run on the very chain the tool runs, in the same batch (the chain's second run). Corrected 14 October: the reading of where the three went was wrong; followed for 6,000 cycles, all 128 needles started 1e-4 apart are back on the window, so the three were on a detour hundreds of cycles long, not in another motion's basin (STOP_WALL 20261014_180544); the motion that does live

beside the window at its pace is the needle standing still, which none of them reached. The redesign stands: started 1e-7 apart, none took the detour. *Proved?* Seen in runs (HEAVY_SHAPE_2 20261014_164913; STOP_WALL 20261014_180544).

213. HEAVY_SHAPE_3's (b2), Bear's cyan stroke made exact by Claude: beyond memory 1.2 the window's end shrinks more slowly than its first doubling, C over S above 2.35 at M 1.254 and 1.2725 (his strokes 2.472 and 2.559). Measured 2.2574 and 2.2560: MISSED.

Kind: WRONG GUESS, of a shape (two curves bending at different rates). *Lesson:* Both curves bend, as he saw, and they bend together: against 1/M the slopes of ln C and ln S between neighbouring weights are 8.62 and 8.72, 8.65 and 8.50, 8.35 and 8.30 at the three heavier steps, so their ratio stays within 0.4 per cent of 9/4 at all eight weights followed. The lean that held (h2) is the one-hump law from recall. Redesign, said aloud: covered for the curves; the question behind his stroke, where the one-hump picture gives way as the squeeze lightens, stays open, and the road to it, a larger window followed to memory 3 to 5, is on the proposed list. *Proved?* Seen in runs (HEAVY_SHAPE_3 20261014_170837).

214. STOP_WALL_2's (z1), Claude's lean: the gap from the 1/4 window down to the stop is still above 0.0005 at memory 1.6. Measured 0.00044: MISSED.

Kind: WRONG GUESS, of a size. *Lesson:* I leaned on the gap levelling off, read from STOP_WALL's last two equal gaps; read to 1e-5 it kept falling at every weight (0.00091, 0.00081, 0.00071, 0.00061, 0.00054, 0.00048, 0.00044 from memory 1.2725 to 1.6), more slowly, so my floor came too high and the straight line's nought at 1.48 too soon: both readings of the same few points missed, on opposite sides. Redesign, said aloud: covered by the follow-on already wanted, the window carried further (memory 2 and beyond) and the gap read there, with the gap in window widths, which levels near 300, as the gauge to watch. *Proved?* Seen in runs (STOP_WALL_2 20261014_225838).

215. NOISE_EDGE's (n2), Claude's lean: at memory 5 and pace 0.75, above the climb-back edge 0.53, every needle that falls under weak noise keeps running to the end of the run; one fall and the lock is lost. Of the fallen, 212 of 449, 239 of 510, 286 of 512 and 284 of 512 still running at D 0.05 to 0.10: MISSED.

Kind: WRONG GUESS, of a mechanism. *Lesson:* The climb-back edge is a noise-free edge. With noise the running needle is caught again too: its own way back down into a seat has a barrier of its own, lower the nearer the pace is to the climb-back edge, and at 0.75 with D 0.05 to 0.10 noise carries it over often enough that about half the fallen needles are holding again by the end. So the remembering needle's working edge under noise is not a door that shuts behind it but a pair of doors, the fall and the catch, each with its own rate. Redesign, said aloud: the follow-on is to measure both rates, the fall's and the catch's, and the share of time running, against pace between the climb-back edge and 1 at fixed noise; it goes on the proposed list. *Proved?* Seen in runs (NOISE_EDGE 20261014_232230).

216. STOP_WALL_3's (B1) and (B2), Bear's reading of fig_stop_wall_2 made exact by Claude: from memory 1.6 to 2.2 the gap from the window down to the stop holds near 300 window widths (270 to 345 at memory 1.8, 2.0 and 2.2) and still falls, more slowly. Measured: 2,089 widths at 1.72 and more than 8,986 to 10,279 from 1.76, the stop not found within 0.012: both MISSED.

Kind: WRONG GUESS, of a trend's reach (a flattening carried past a turn). *Lesson:* His reading of the curve he was shown was right as far as it went: up to memory 1.6 the gap in widths had levelled near 300, and he said it could change course. It did, sharply, between 1.72 and 1.76: the stop that had crept to within 300 widths fell away by more than ten thousand. The legs were made exact as 'flat to memory 2.2', which carried his trend past a turn nobody had seen; the window's own width, which he also read as flattening, did flatten, and turned too, later, near memory 1.92. Redesign, said aloud: covered by the follow-ons, where exactly between memory 1.72 and 1.76 the stop leaves, and where it goes (a walk further than 0.012, or the stop's own edge found from the stopped side), both on the proposed list. *Proved?* Seen in runs (STOP_WALL_3 20261015_000651).

217. HEAVY_WINDOW_4's (D1), Bear's reading of fig_stop_wall_3's middle panel made exact by Claude: the window's depression is a bell, the same on both sides of its least, S at memory 2.22, 2.37 and 2.52 within 10 per cent of S at 1.62, 1.47 and 1.32. Measured right over left 0.97, 0.83 and 0.58: MISSED.

Kind: WRONG GUESS, of a shape (a bell read on one ruler). *Lesson:* Against the memory itself the depression is lopsided: a steep wall on the light side, where the window had shrunk as about exp(9.4/M), and a gentle one on the heavy side. Near its least it is even, as any smooth floor is: right over left 1.006 to 1.013 out to 0.2 of memory either side. Redesign, said aloud: a shape lopsided against one ruler can be even against another; read against the logarithm of the memory the two sides already agree within 2.3 per cent out to 0.2 either side (a scratch reading, said aloud), so BELL-LOG tests the bell against ln M further out, in the same batch. *Proved?* Seen in runs (HEAVY_WINDOW_4 20261015_032553).

218. BELL_LOG's (L1), the redesign of row 217: against ln M the window's depression is the same on both sides, S at 0.3 and 0.4 either side of the least within 5 per cent. Right over left 0.9366 and 0.7935: MISSED.

Kind: WRONG GUESS, of a shape (a second ruler tried). *Lesson:* The logarithm evens the floor further out than the memory itself does (within 2.3 per cent to 0.2 either side, against 1.3 per cent to 0.1 and 3 per cent by 0.3 for M), but the light side's wall is still the steeper beyond; against 1/M the heavy side is the steeper. So no one of the three rulers makes the depression a bell; one between ln M and 1/M might. Redesign, said aloud: covered as far as this chain of runs goes; the chain of redesigns for this reading stops at two, as the practice asks, and the question which ruler, if any, makes the depression even, and why the floor sits near memory 1.92, goes on the proposed list. *Proved?* Seen in runs (BELL_LOG 20261015_033341).

219. GAPS_READ's (G2), Claude's lean on Bear's flickers in panel (c) of fig_stop_leaves: they are the half-turn lock losing its hold, the half-turn cycle unstable at the first step of each group (164 at memory 1.7248; 147 and 194 at 1.7598) and stable at the step before. Measured: the plain half-turn cycle is unstable at every step of both walks until 0.0196 below the window at memory 1.7598 (its largest multiplier in size 1.85 near the window, falling to 1.03, 1.01 and 1.00 at steps 193 to 195 and 0.97 at step 198, as corrected by GAPS_REREAD 20261015_071452; GAPS_READ, its angles unreduced, had printed 1.92, 1.00 to 1.02 and 0.95), so it was never the lock: MISSED.

Kind: WRONG GUESS, of what a quiet stretch is (a lock read from a spread that cannot see a doubled cycle). *Lesson:* What looked like one clean half-turn lock is a ladder of doubled half-turn cycles, read downward in pace: at memory 1.7598, 32 drive cycles at step 130, 16 at 131 to 133, 8 at 135 to 146 and at 150, 4 from 155 to 193, and the plain 2 only from 198; at 1.7248, 16 at 150 and 151, 8 at 153 to 166, 4 from 167. Each of Bear's flickers sits where the ladder steps down one rung: there needle 0 does not return within 1e-6 in 50 cycles, and the rung's largest multiplier climbs toward 1 in size on the way in (the 4-cycle's 0.71 at step 189 to 0.90 at 193, as corrected), the slow settling a doubling brings. The spread could not see the ladder, because a doubled cycle turns the same whole number of times in 200 cycles whichever of its points a needle sits on. Redesign, said aloud: yes, and not covered elsewhere: HALVINGS, the rungs' edges found exactly and their spacing set against the doubling ratio of the window's own fork tree, registered at once, to run next. *Proved?* Seen in runs (GAPS_READ 20261015_061859).

220. HALVINGS's (H1), (H2) and (B1): printed MISSED, and void. Its four 'edges' came after 1 to 3 steps of 1e-5, with the largest multiplier at the last stable pace -0.953, -0.583, +0.625 and +0.619, none near 1 in size.

Kind: INSTRUMENT, a finder that read its own failure as the answer. *Lesson:* The edge test was 'Newton's method fails, or a multiplier reaches 1 in size', and Newton failed first: the needles' angles were carried unreduced, 168,542 to 250,223 radians, where the number line's step is about 3e-11, so the cycle's residual could not reliably reach 1e-10. Two lessons: reduce an angle to one turn before asking Newton for a cycle (the physics sees only its sine), and never let a method's failure stand for the event it is looking for; give the finder a planted edge with a known answer (here the window's own first doubling, S below its top) so that the planted case touches the very thing the test reads. The blind run said 'all four edges found', and was right that they were found and wrong about what they were: a decidability check is not a correctness check. Redesign, said aloud: HALVINGS_2, run minutes later. *Proved?* Seen in runs (HALVINGS 20261015_064027).

221. HALVINGS_2's (H2): the rungs of the ladder of doubled half-turn cycles shrink by Feigenbaum's ratio, both width ratios within 25 per cent of 4.669. Measured 3.4280 and 4.2353: MISSED, the first 27 per cent short.

Kind: WRONG GUESS, of how soon a limit arrives. *Lesson:* Feigenbaum's 4.669 is the ratio a ladder's rungs reach in the limit; the first rungs of a real ladder fall short of it and climb toward it, and these do, 3.43 then 4.24. A lean on a limit should be set on the late rungs, or say how fast the climb goes. Bear's reading of the same numbers held: the rungs are evenly spaced on a log ruler to within a quarter. Redesign, said aloud: yes, HALVINGS-3, the 32-cycle's rung added, run minutes later. *Proved?* Seen in runs (HALVINGS_2 20261015_064427).

222. GAPS_READ's printed multipliers (not a judged line): read at needles' angles of about 200,000 radians, they were off by up to 0.09 for the cycles and far more for the half-turn cycle in the chaos (0.2785 printed for a 4-cycle whose multiplier is 0.3317; a second eigenvalue of -0.0775 where the squeeze allows about 1e-6), and 38 cycles and 16 half-turn cycles were 'not found' only because of it. Found by LADDER_LIGHT's blind run (planted value 0.2785 read as 0.3317).

Kind: INSTRUMENT, a reading tool's numbers wrong in their last places and more. *Lesson:* The fault behind misses row 220 was already in the run before it. When a numerical fault is found, look back for every earlier run that carries the same fault, rerun it, and correct every record from the rerun in the same batch. GAPS_REREAD (20261015_071452) did: both verdicts stand, (G1) held and (G2) missed, and the half-turn cycle's multiplier at memory 1.7598 now falls smoothly to 1 between 0.0195 and 0.0196 below the window, where HALVINGS_2 found the 2-cycle's edge. Redesign, said aloud: covered, by the correction run itself. *Proved?* Seen in runs (GAPS_READ 20261015_061859; LADDER_LIGHT's blind run 20261015_070958; GAPS_REREAD 20261015_071452).

223. SNAKE_LONG's (S1): the best exchange at 192 links on 38.4 wavelengths lies between 4.18 and 5.0. Measured 4.1760: MISSED, by 0.004.

Kind: WRONG GUESS, of a range set from a probe. *Lesson:* A timing probe had printed 4.1751 at the 48-link body's best setting before the leans were set, and I set the lower end above it, guessing the search would add more than it did. It added nothing: the search's best is the 48-link body's best setting itself, with its exact torque 0.028117, and the 0.0009 between the probe and the result is the probe's torque rounded to 0.0281 (read from the run's own saved settings, LOOP_ATLAS_SNAKE_LONG_20261015_073142.json, on 15 October; first written here as the search adding 0.0009). A range set after a probe should start at the probe's value, and the probe should run at one of the search's own settings, so that its value is a floor the search cannot go under. The question the card asks held: the climb goes on, slowing (the gap below 6.2832 shrinking by 0.79 and then 0.84 a doubling) (corrected 15 October, by SNAKE_EXACT 20261015_154247: the slowing was the search grid missing the best by more at each length; found exactly, the climb is 3.27, 3.96, 4.53, 4.99 and 5.34 at 9.6 to 153.6 wavelengths, the gap below 6.28 falling steadily, close to one over the square root of the length). Redesign, said aloud: the real redesign is a longer body, 384 links, about an hour on one core: a batch for the second computer, put on the proposed list. *Proved?* Seen in runs (SNAKE_LONG 20261015_073142).

224. SONG_GAPS's (G1): the gaps between taps on the 1 m and 0.5 m boards, in fall times, have the same distribution (Kolmogorov-Smirnov distance under 0.03). Measured 0.4224: MISSED.

Kind: INSTRUMENT, a ruler reading rounding at an atom. *Lesson:* 96 per cent of the gaps are exactly the rider's rest, 200 clock steps, half a fall time, and a gap of exactly 200 steps reads as 0.5 less or more a rounding that differs between boards, so the cumulative distributions split the pile at 0.5 differently. Counted in whole steps the two boards agree to 0.0065 (SONG_GAPS_2). Bear's integer diagnostic, a count read as a measure, in a new place. Redesign, said aloud: SONG_GAPS_2, run at once, held. *Proved?* Seen in runs (SONG_GAPS 20261015_074750; SONG_GAPS_2 20261015_074852).

225. SONG_GAPS's (G2): the first 50 gaps on the 1 m and 0.5 m boards differ, rider by rider, for at least 12 of 16 riders. Measured 0 of 16: MISSED.

Kind: WRONG GUESS, of how soon rounding grows. *Lesson:* Two boards whose clocks differ only in rounding tap identically for the first 50 to 100 gaps (the first difference at gaps 50 to 100, rider by rider), longer than the 28 taps BALANCE_BOARD_B saw for a 1e-12 rad difference in starting tilt: a rounding starts smaller than 1e-12. Redesign, said aloud: SONG_GAPS_2's (G2b), gaps 101 to 150, run at once, held, 16 of 16. *Proved?* Seen in runs (SONG_GAPS 20261015_074750; SONG_GAPS_2 20261015_074852).

226. SNAKE_768's (T1): at 384 links the best of the search is the setting best at every shorter body (torque 0.0281, wave +1, start 20), 4.4117 within 1e-4. Measured 4.4475 at start 0: MISSED.

Kind: INSTRUMENT, a grid too coarse to follow its optimum. *Lesson:* The best start moves as the body lengthens; a grid of starts every 20 degrees can only follow it in jumps, and I read three bodies' shared grid point as a setting that does not move. Redesign, said aloud: remove the grid. The push and the motor power are exact quadratics in the torque's size and sinusoids in its start, so five solves give the best over every size and start exactly (SNAKE_EXACT), run in the same batch. *Proved?* Seen in runs (SNAKE_768 20261015_151934).

227. SNAKE_768's (T2): at 768 links the snake stays below 6.2832 and the gap's shrink from 384 links lies between 0.888 and 0.96 (the climb slowing further). Measured 4.9897 and a shrink of 0.7046: MISSED (the climb sped up).

Kind: INSTRUMENT, a best on the edge of its range. *Lesson:* At 768 links the search covered starts 0, 20 and 40 only, and its best sat at 0, the edge, so 4.9897 is a floor and the shrink compares a floor with a best; and the starts' grid misses the true best by different amounts at different lengths, so the shrinks mix grid error into the climb. Redesign, said aloud: the same exact algebra (SNAKE_EXACT), which needs no range and has no edge; it also says whether the speed-up at 768 links is real. *Proved?* Seen in runs (SNAKE_768 20261015_151934).

228. RING_KNOBS's (K2): at a gap of 1.0 the leapfrogging pair's advantage over one ring on its budget grows with a fatter core, a0 $0.02 < 0.05 < 0.1 < 0.2$. Measured 1.179, 1.191, 1.195 and 1.178: MISSED (it falls again at 0.2).

Kind: WRONG GUESS, of a direction. *Lesson:* I pictured a fatter core strengthening the rings' hold on each other without limit; the core has a sweet spot of its own, and a fatter core moves the best gap outward (at 0.2 the best is 1.188 at a gap of 1.25), so the two knobs trade. Redesign, said aloud: already covered by the run's own table, which maps both knobs together: the pair's best is about 1.195 near a core of 0.1 and a gap of 1.0 to 1.25; a finer map around it goes on the proposed list, and a core of 0.2 is near the edge of what the thin-ring formulas can be trusted with. *Proved?* Seen in runs (RING_KNOBS 20261015_165858).

229. RING_KNOBS's (K3): at gap 1.0 and core 0.05 three rings beat one ring on their budget by more than two do. Measured 1.138 against the pair's 1.191: MISSED.

Kind: NATURE, and the INSTRUMENT that did not see it. *Lesson:* The lean pictured a train that holds; at gaps of 0.75 and more three equal rings fly apart within 400 time units, their first and last rings 41 to 73 radii apart, so the 'mean speed' there is the mean of a group falling apart, a measure with nothing to measure. At 0.25 and 0.5 the trio holds

and beats its budget ring by 1.163 and 1.260, more than any pair. Redesign, said aloud: measure a train only while it holds, find the spacing where it breaks, and see what is left when it does (TRAIN_HOLD), run at once, ahead of the plan, because the result could not wait. *Proved?* Seen in runs (RING_KNOBS 20261015_165858).

230. TRAIN_HOLD's (H2): a train of three that holds is worth more the wider it is spaced, its advantage rising at every step up to its edge. Measured 1.1628 to 1.2610 rising from 0.25 to 0.50, then 1.2325, 1.2658 and 1.2305 at 0.55, 0.60 and 0.65: MISSED.

Kind: WRONG GUESS, of a shape. *Lesson:* I pictured the gain rising smoothly to the cliff; near the cliff the train still holds but wanders, its rings up to 4 radii apart, and one long run's mean speed then depends on the wandering it happened to do. Redesign, said aloud: near the edge the advantage is a property of many nearby starts, not of one; the redesign reads it as a mean over an ensemble of slightly different starts at each gap from 0.50 to 0.65, with the break-up times read the same way as a lifetime (both on the proposed list, the second beside the stop's door, chapter 10). Below 0.50 the rise held and is the engineering answer: pack a train of three at about half a radius. *Proved?* Seen in runs (TRAIN_HOLD 20261015_171853).

231. SLOPE_EAR's (S2): with noise and looks every 1/16 cycle, the slope from the last two looks drifts less than the plain ear (the derivative kick). Measured 0.10857 against 0.09739: MISSED.

Kind: WRONG GUESS, of where a cost lands. *Lesson:* The kick is real but it lands in the bill, not the drift: the two-look slope switches 2.217 times a cycle against 1.650, and its price falls from 0.989 to 0.821, while its drift rises to the level of looking every step. Redesign, said aloud: the controls engineer's cure, a filtered slope (fitted to the last three or four looks, or with a knee of its own), to keep the drift and drop the chatter: FILTER_SLOPE, registered to run next. *Proved?* Seen in runs (SLOPE_EAR 20261015_190935).

232. FILTER_SLOPE's (F1): with noise and looks every 1/16 cycle, a slope fitted to the last four looks keeps its drift within 5 per cent of the two-look slope's and switches no more than 10 per cent above the plain ear's 1.650. Measured: drift 0.10622 (held), switches 1.927 (missed, 17 per cent above): MISSED.

Kind: WRONG GUESS, of a size. *Lesson:* I expected four looks to smooth away most of the noise in the slope; they halved the extra switching, from 0.567 above the plain ear to 0.277. Redesign, said aloud: the fit has two knobs, the number of looks it spans and the knee its forecast passes through, and the chatter is the knee's job as much as the fit's; a map of both knobs (looks 2 to 6, knee 0.5 to 1.5 of strength) at a sixteenth of a cycle is registered as FIT_KNOBS. *Proved?* Seen in runs (FILTER_SLOPE 20261015_195645).

233. PILOT_BEAT's (B1): a soft beat (0.03 of the soup's push) sets the spring bead's count near its tune: the blade's switches within 5 per cent of Omega/pi at Omega 0.9, 1.0 and 1.1, and not at 0.5 or 2.0. Measured: the count 15 to 31 times Omega/pi everywhere: MISSED.

Kind: INSTRUMENT, a count the clock owns. *Lesson:* The blade, on while the bead moves forward, chatters at the simulation's step whenever the bead's speed sits near zero (sliding at v = 0), so the switch count is 4.9 to 5.8 a unit of time whatever the beat; EAR_CURES had already shown a plain relay's bill doubling as the step halves. Redesign, said aloud: count the bead's swings, not its blade's switches, in whole numbers, and read the share of its swing at the beat's pitch; prove the chatter by halving the step (BEAT_SWING), run in the same batch. *Proved?* Seen in runs (PILOT_BEAT 20261015_203855).

234. PILOT_BEAT's (B2): a loud beat (0.3) sets the count at every Omega from 0.7 to 1.5. Measured: the count 9.8 to 25 times Omega/pi: MISSED.

Kind: INSTRUMENT, the same count. *Lesson:* The same chatter as row 233. Redesign, said aloud: the same, BEAT_SWING. *Proved?* Seen in runs (PILOT_BEAT 20261015_203855).

235. BEAT_SWING's (C2): by the bead's swing, a soft beat (0.03) locks the count at Omega 0.9, 1.0 and 1.1 of the tune, and not at 0.5 or 2.0. Measured: the swing stays near 0.12 a unit of time at every pitch, the beat holding at most 4 per cent of it: MISSED.

Kind: WRONG GUESS, of a strength and a centre. *Lesson:* I took the lock for a property of the tone; it is a contest between the tone and the soup, and at 0.03 of the soup's push the soup wins everywhere. I also centred the lock on the bead's nameplate tune; the bead's working swing, driven by the soup through this much loss, is 0.76 of it. Redesign, said aloud: covered by the run's own table, which maps loudness against pitch; the finer tongue, centred on the working swing and with a lighter bead whose own ring is sharper, goes on the proposed list. *Proved?* Seen in runs (BEAT_SWING 20261015_204207).

236. BEAT_SWING's (C3): a loud beat (0.3) locks at every Omega from 0.7 to 1.5. Measured: locked at 0.6 to 0.8 only: MISSED.

Kind: WRONG GUESS, of a centre. *Lesson:* The same misplaced centre as row 235: the lock lies at and below the bead's working swing, and a beat above it cannot pull the bead up. Redesign, said aloud: covered as row 235. *Proved?* Seen in runs (BEAT_SWING 20261015_204207).

237. FIT_KNOBS's (G1): some window and knee keep the fitted slope's drift within 5 per cent of the two-look slope's and its switching at most 1.815 a cycle. Measured: none; the nearest 0.1035 with 1.858 switches: MISSED.

Kind: NATURE, a trade-off with no corner cut. *Lesson:* The two knobs slide the ear along one line of headway against switching, and the plain ear sits on the efficient end of it. Redesign, said aloud: covered by the run's own map, which is the frontier itself; a filter of another kind (a knee on the slope itself, or an estimator that weighs each look by its noise) is the only way off the line, and goes on the proposed list. *Proved?* Seen in runs (FIT_KNOBS 20261015_204319).

238. PILOT_PRICE's (P2): at loudness 0.03 the ordinary bead takes the most power from the pilot at its tune. Measured: most at 0.9 (94.7 millionths against 91.5): MISSED.

Kind: WRONG GUESS, of a centre. *Lesson:* I pictured the plain spring, which takes the most power exactly at its tune; this bead is not plain, its blade switching with its motion, and what it takes peaks between its working pace (0.76) and its tune. Redesign, said aloud: covered by the run's own table at every loudness; the same rule as rows 235 and 236, that the working pace governs. *Proved?* Seen in runs (PILOT_PRICE 20261015_220321).

239. PILOT_PRICE's (T2): a loud beat (0.3) holds the lighter-damped bead's count at its tune. Measured: held at 0.9 only (0.990), at the tune 0.947: MISSED.

Kind: WRONG GUESS, of a centre, the third. *Lesson:* Less loss moved the working pace toward the tune, to 0.906 of it, but not onto it, and the beat's hold followed the working pace, not the tune, as it had for the ordinary bead (row 236). Redesign, said aloud: covered; the rule now stated in the book is the working pace, measured under load, and three misses of the same kind say I should have set every centre from it since row 235. *Proved?* Seen in runs (PILOT_PRICE 20261015_220321).

240. PAUSE_BEST's (Q2): the best price of Bear's *L pause R pause* ear, looking every step, lies at a dead zone from 0.5 to 1.0 of strength. Measured: rising to the widest zone tried, 1.075 at 1.5: MISSED.

Kind: WRONG GUESS, and a best on the edge of its range. *Lesson:* The price kept rising as the zone widened while the thrust fell; a best on the edge of the grid is a floor. Redesign, said aloud: carry the zone on, to 3 of strength, to find where the price turns over and what thrust is left there (PAUSE_WIDE). *Proved?* Seen in runs (PAUSE_BEST 20261015_220655).

241. JUMP_STORE's first blind run: the knee ear, a knee of a tenth of its running speed on both sides of zero, was planted to count its own swings. Measured: at tune 1 and loss 1 it counted no switch at all in the second half, and the run stopped on the empty count before any judged line (run 20261015_223539, retired).

Kind: TOOL FAULT, a design that silences itself. *Lesson:* A knee measured against the bead's own running speed lets a bead that falls quiet stay quiet: the running speed remembers the louder swing and lags above the quiet one, so the bead never crosses the knee again and, with its blade off, gets no push to grow back. Redesign, said aloud and written into the registration before the next run: the knee on the off side only (on as soon as the bead moves forward), so a quiet bead still fires. It held: M1 held, and every knee ear fired except the critically damped one, which SPRING_STALL found takes nothing. *Proved?* Seen in runs (JUMP_STORE BLIND 20261015_223539, retired; JUMP_STORE BLIND 20261015_223744).

242. JUMP_STORE's (J2), judged whole: the triggered spring's impulse per event rises at every step of its trigger from 0 to 1.5 of its running speed, while the listener's crest-firing's falls. Measured: the listener's falls at every step but barely, 0.378, 0.374, 0.372 and 0.368, flat within 3 per cent as Rice's crossing count says it must be; the triggered spring fell silent at every trigger from 0.5: MISSED.

Kind: TOOL LIMIT, and a wrong guess. *Lesson:* The trigger rode on the bead's own running speed, a memory of 20 time units, as slow as the store's own drain at a loss of 0.05. While the store drains the memory stays above the swing, the swing never reaches the trigger, and the blade never fires again: traced on one soup after the run of record, the swing fell to about 1e-24 by the run's end. The flat listener says the memoryless skater neither loses nor gains per event by waiting for crests; it only fires less often. Redesign, said aloud: a trigger set against a fixed speed (the knee ear's own long-run speed at that setting), or against a memory shorter than the drain (an open test). *Proved?* Seen in runs (JUMP_STORE 20261015_223918).

243. JUMP_STORE's (J3): the best spring carries more impulse per event than the listener with noise in its ear and RF_SKATER_COUNT's knee. Measured: 0.1145 (tune 1.5, loss 0.2) against the listener's 0.354: MISSED.

Kind: WRONG GUESS. *Lesson:* The spring is not a jump. It fires about once per swing of its own, 0.57 to 1.34 times a soup cycle at light and middling loss, and each firing carries a third or less of the listener's; even the noisy listener with its knee carries three times as much. The plain noisy listener's 0.069 is the step's (its count grows 1.40 times when the step is halved), which is why the registration was amended before the tool to judge against the knee. Redesign, said aloud: the store that makes a jump is a latch, not a spring; a latch on the listener and on the spring, counted per event (LATCH_EVENT). *Proved?* Seen in runs (JUMP_STORE 20261015_223918).

244. PAUSE_WIDE's (Z1): carried on to 3 of strength, the price of Bear's *L pause R pause* ear turns over, its best at a zone of 1.75 to 2.5. Measured: it rises at every widening, 1.075, 1.127, 1.171, 1.251, 1.325 and 1.524 from 1.5 to 3: MISSED.

Kind: WRONG GUESS, and nature. *Lesson:* The price is the push per switch, and the ear latches: once over the zone's edge it holds until the push turns round, so it harvests each crest to its end, and the wider the zone, the bigger the crests it takes. A memoryless crest-firer carries the same per event at every threshold (row 242); the latch carries more. There is no best zone in reach, only the thrust a builder needs. Redesign, said aloud: the latch alone, tethered, counted per event against its zone (LATCH_EVENT). *Proved?* Seen in runs (PAUSE_WIDE 20261015_223744).

245. PAUSE_WIDE's (Z2): at the price's best at least 0.08 wavelengths a cycle of thrust is left. Measured: the best is at the widest zone, 3, where 0.017 is left: MISSED.

Kind: WRONG GUESS, and a best on the edge of its range. *Lesson:* It follows from row 244: with no turn, the best price sits at the widest zone tried, where the ear switches 0.19 times a cycle and keeps a twelfth of its thrust at a zone of 0.25. A best on the edge of the grid is a floor. Redesign: none owed; the knob is read as a trade, not a best. *Proved?* Seen in runs (PAUSE_WIDE 20261015_223744).

246. LATCH_EVENT's (L3): at every h where it fires, the spring's latch carries less than half the clean listener's 0.378 per event. Measured: it fired at no h, at either setting; nothing to judge: UNDECIDED.

Kind: TOOL LIMIT, and nature. *Lesson:* A blade that is off while its bead swings small gets no push to grow the swing back, and the bead's own loss finishes it. A trigger of any kind, against the bead's memory (row 242) or against a fixed speed, turns one quiet spell into silence; traced on one soup after the run of record, the latch at h 0.25 fired 47 times and fell quiet for good between 200 and 400 time units in, while its trigger was still ramping up. Row 1's lesson returns: a blade should never go fully dark. Redesign, said aloud: the spring's latch and its fixed trigger with a floor, the charge 0.2 when off, so a quiet bead still hears (SPRING_FLOOR). *Proved?* Seen in runs (LATCH_EVENT 20261015_232643).

247. KNEE_CREEP's (K2): with the noise relaxing over 0.03, the knee's bill changes by at most 3 per cent over the last halving of the step (0.005 to 0.0025). Measured: 1.041: MISSED.

Kind: WRONG GUESS, and the instrument. *Lesson:* A noise that relaxes is still rough at every scale finer than its own time, so each finer step catches brief excursions past the knee that a coarser one stepped over; the faster the noise, the rougher, and the longer the creep lasts (0.5, 1.4 and 4.1 per cent over the last halving at relaxing times 1, 0.1 and 0.03). The noise recipe also shrinks with the step at so short a time (row 248), so the creep under it is larger than the 4.1 per cent seen. Redesign, said aloud: draw the noise by its exact recipe so its size is the same at every step, carry the step down to 0.000625, and test whether the creep shrinks by about 0.7 a halving, as the correction for a price watched only at set times predicts (CREEP_LAW). *Proved?* Seen in runs (KNEE_CREEP 20261015_232920).

248. KNEE_CREEP's (K3): with the noise relaxing over 0.03, the first halving's ratio is larger than the last's. Measured: 0.885 against 1.041: MISSED.

Kind: TOOL LIMIT. *Lesson:* The first halving fell because the toy's noise recipe (each step keeps 1 minus step over time of the old noise and adds a fresh draw) is too big at a step comparable with the noise's time: its spread is 1.222 of its aim at step 0.02, 1.092 at 0.01, 1.042 at 0.005 and 1.022 at 0.0025 (measured after the run of record on 400,000 draws). A smaller noise flips less, so the first halving's drop is the recipe, not the ear. Redesign: the exact recipe (CREEP_LAW). *Proved?* Seen in runs (KNEE_CREEP 20261015_232920).

249. CREEP_LAW's (T1): the traders' law predicts the knee's fall in flips within 25 per cent for noise relaxing over 0.1, 0.03 and 0.01, at every step from 0.00125 to 0.04. Measured: held at 0.1 (1.02 to 1.18 of the law), missed at 0.03 at step 0.04 (1.274) and at 0.01 from step 0.005 (1.27 to 1.51): MISSED.

Kind: WRONG GUESS of the law's reach, and nature. *Lesson:* The law is built for a price that wanders: over a short look the noise moves like a random walk, and the law corrects for the pokes a random walk makes between looks. When the step nears the noise's own time the noise forgets where it was between two looks, every look is nearly a fresh throw, and the slow watcher misses more than the law says, a fifth more at about the noise's time and half again at eight times it. Counted in the noise's own time the four noises fall close to one curve. Part of the few per cent the law falls short even at short steps may be the instrument: the prediction is read off the knee grid by straight lines, and the curve of flips against knee bends, which makes the predicted fall a little too small. Redesign, said aloud: count the pokes a slow watcher misses from its own looks, by the chance that the noise crossed the knee between two looks given where it was at each (the bridge between looks, which traders also use), and check it against the fast watcher at every step and noise time (BRIDGE_COUNT). *Proved?* Seen in runs (CREEP_LAW 20261016_035058).

250. CREEP_LAW's (T2): for noise relaxing over 0.1, the change in flips at each halving from 0.01 to 0.000625 falls as the step to a power from 0.4 to 0.6. Measured: 0.332: MISSED.

Kind: WRONG GUESS, the lean set on the wrong quantity. *Lesson:* The square root lives in the knee's extra size, not in the flips: the flips follow it through the curve of flips against knee, which bends, and the law's own predicted flips give the same power there (0.33). A law should be tested where it lives, or the curve between must be carried. Redesign, said aloud: read the knee's extra size itself as a power of the step, on a knee grid ten times finer so the bend no longer biases the reading (KNEE_SIZE). *Proved?* Seen in runs (CREEP_LAW 20261016_035058; FIG_CREEP_LAW 20261016_035636).

Misses 251 to 300

251. BRIDGE_COUNT's (B1): the slow watcher's bridge count of pokes past the knee lies within 5 per cent of the fastest watcher's count, for slow noise at every step. Measured: the fastest watcher's count is itself no count; the pokes past one line grow 1.41 to 1.44 times at each of the finest halvings, without end, and the bridge comes to about three times what each watcher sees: MISSED.

Kind: WRONG GUESS, the count ill-posed, and nature. *Lesson:* A rough noise that touches a line touches it again and again close by, so its crossings of one line grow as one over the square root of the step forever (a lesson already drawn, at its root). The bridge's three times is the reflection principle's: for each up-crossing seen, one more hides between looks on the low side and one on the high side. Only a band makes a count finite; the knee's flips settle where a line's crossings never do. Redesign, said aloud: the slow watcher's own count of its missed flips, with the bridge taken across the band, both edges at once (BAND_BRIDGE). *Proved?* Seen in runs (BRIDGE_COUNT 20261016_044439).

252. BRIDGE_COUNT's (B3): at every step and noise time the bridge recovers at least half of the pokes the plain count misses. Measured: 4.4 to 4.8 times them at the shortest steps, 0.04 to 0.18 at the longest: MISSED.

Kind: WRONG GUESS, from the same ill-posed count. *Lesson:* Measured against a count that is the step's, 'recovered' means nothing: at short steps the bridge counts the hidden touches the fastest watcher also misses, and at long steps against a noise that has forgotten itself the bridge, like the law, no longer applies. Redesign: as row 251 (BAND_BRIDGE). *Proved?* Seen in runs (BRIDGE_COUNT 20261016_044439).

253. RIDER_FLEET's M4 (registered 5 October): on riders whose reaction slows, no policy beats the calendar by more than 0.02 of a life. Measured: the learned reminders waste 0.304 and 0.309 against the calendar's 0.333, while catching 0.828 and 0.844 against 0.859: MISSED, narrowly.

Kind: WRONG GUESS in part, and the instrument. *Lesson:* Something does move before the fall: the wobble shrinks to 0.89 of the first flight's, while the taps stay flat. But the learned reminders' small gain comes with a smaller catch, and one fleet seed cannot tell a 0.03 gain from the needle fleets' seed spread of 0.014 to 0.027. Redesign, said aloud: the same fleets compared at equal catch (each policy's threshold set to catch the same share of the test half) over five fleet seeds (EQUAL_CATCH). *Proved?* Seen in runs (RIDER_FLEET 20261016_044707).

254. RIDER_FLEET's M5 (registered 5 October), judged whole: on riders whose tap weakens, the taps per fall time rise before the fall, and the learned reminder beats the calendar. Measured: the learned power reminder wastes 0.125 against 0.345, but the taps fall, to 0.693 of the first flight's: MISSED.

Kind: WRONG GUESS of the gauge's direction, and nature. *Lesson:* Bear's IoT picture, more power as the wear grows, belongs to a machine that works harder to make up for its wear. This rider's tap is a fixed-size kick: as it weakens, the board moves less, the rider's reading stays inside its band longer, and it taps less and wobbles less until it can no longer hold (wobble to 0.150 of the first flight's). A fixed power level that waits for a rise wastes 0.917; the learner, which reads the direction from the fleet, wastes 0.125. Redesign, said aloud: the weakening rider with a controller that makes up for its wear, its tap sized to hold the band, to see whether Bear's power gauge then rises (WORN_RIDER). *Proved?* Seen in runs (RIDER_FLEET 20261016_044707).

255. HITCH_LEGS's (a): with human legs and nothing carried, one running stride of both legs turns the trunk back by 15 to 60 degrees. Measured: 6.573 degrees: MISSED.

Kind: WRONG GUESS of a size. *Lesson:* The stride's turn is the area of its loop in the space of hip and knee angles, and a human leg's loop is small: the hip swings 0.70 rad either side, the knee bends only on the way forward, and the shank is light (4.5 kg against the thigh's 7.5). With the leg's mass spread evenly the stride gives 8.354, still under 15; the toy's straight leg swung round gave 173 because a full circle is the biggest loop there is. Redesign, said aloud: the stride's turn mapped over the hip's swing and the knee's bend, to see what stride a jumper would need (STRIDE_MAP). *Proved?* Seen in runs (HITCH_LEGS 20261016_065650).

256. HITCH_LEGS's (b): with the spin carried, whole arm circles hold off a share within 5 per cent of the number of circles times 13.87 degrees over 60. Measured: 0.409, 0.666 and 0.924 for one, two and three circles, against 0.231, 0.462 and 0.694: MISSED.

Kind: WRONG GUESS, two things left out. *Lesson:* The share does rise in step with the circles, exactly: 0.2573 a circle, twice running, so whole circles remove the sawtooth. But the lean left out two things. A circling arm spends time out from the body, which slows the spin by inertia alone, a fixed 0.152 of the turn whatever the number of circles

(the skater's arms out); and the geometric turn per circle on this body is 15.44 degrees, not the even-legged body's 13.87. Read after the data, to be checked: the share is one minus the rigid inertia times the arms' average of one over the inertia, plus the circles times the geometric turn over 60. Redesign, said aloud: the arm circle's geometric turn on the human-legged body measured with nothing carried, and the offset predicted from the arms' average inertia, against this run's shares (ARM_SHARE). *Proved?* Seen in runs (HITCH_LEGS 20261016_065650).

257. BELL_CLOCK's (C1): with the water's three clocks stretched together (squeeze period, memory and the rings' own speed), two tied bells get back their distance per squeeze within 5 per cent. Measured: 0.8662 against 0.6070, 1.427 times: MISSED.

Kind: WRONG GUESS, a clock left out. *Lesson:* The bell's own squeeze is a clock too: its kick rule adds the flow to a speed of its own, the 1 in 1 plus the flow, the jet of its squeeze. Stretched with the water's three, the distance per squeeze returns to every digit; left alone it is 1.43 times longer, and stretched without the rings 1.38. The distance per squeeze is set by how the four clocks compare, not by any one of them. Redesign, said aloud: the distance per squeeze mapped against the one ratio that matters most, the bell's squeeze speed against the rings' speed, with the memory in the rings' time (BELL_RATIO). *Proved?* Seen in runs (BELL_CLOCK 20261016_072241).

258. BAND_BRIDGE's (F1): the slow watcher's band-bridge count lies within 3 per cent of the fastest watcher's flips for noise relaxing over 1.0 at every step and over 0.1 to step 0.01, where its plain flips fall short. Measured: within 1 per cent at 1.0, but 2.7 to 3.5 per cent above at 0.1: MISSED.

Kind: WRONG GUESS of the reference, and the instrument. *Lesson:* The fastest watcher is not a watcher who never blinks. At noise time 0.1 it is itself 3.5 per cent short: its own band-bridge count is 1.0352 of its flips, and the traders' law, built on another idea, puts a never-blinking watcher at 1.0368 (cross-checked after the run). Against that count the slow watcher's bridge is within 1 per cent to step 0.01. Redesign, said aloud: judge the band bridge against a watcher sixteen times faster at noise times 1.0 and 0.1, the reference the run lacked (NEVER_BLINK). *Proved?* Seen in runs (BAND_BRIDGE 20261016_072612).

259. BAND_BRIDGE's (F3): the fastest watcher's own band-bridge count lies within 1 per cent of its flips at noise times 1.0 and 0.1. Measured: 1.0077 and 1.0352: MISSED.

Kind: WRONG GUESS of how much the fastest watcher misses. *Lesson:* A watcher who never blinks flips 0.8 per cent more often than one looking every 0.000625 at a noise memory of 1, 3.5 at 0.1, 18 at 0.03 and 44 at 0.01, and the traders' law agrees at all four to within 0.6 per cent (1.0070, 1.0368, 1.1827 and 1.4490). A fast watcher is not the truth for a rough noise; the correction is. Redesign: as row 258 (NEVER_BLINK). *Proved?* Seen in runs (BAND_BRIDGE 20261016_072612; CREEP_LAW 20261016_035058).

260. TRUCK_ROAD's M2 in its first blind run (run_LOOP_ATLAS_TRUCK_ROAD_BLIND_20261016_085417, retired), leg (c) of the road questions of 6 October, planted, the band test: the lanes read with windows of 200 time units repeat those read with 100 at every setting. Measured: they differ at 3 of 465 settings, mass 0.75 at 1.055, 1.5 at 0.960 and 3.0 at 0.730: MISSED.

Kind: THE INSTRUMENT, a window too short to see a slow slip. *Lesson:* SWAP_STATE's reading calls a window locked when the rider makes no full turn in it, so a rider that slips once every 200 to 400 time units passes for locked in windows of 100 and not in windows of 200; near a lane's edge, where slips come slowly, a short window reads a slipper as a lock. Fixed before the run of record, said aloud: each lane read over the whole hold of 3,000 time units, the band test asking that each half agree with the whole (held, TRUCK_ROAD 20261016_090656). SWAP_STATE's ramp readings may carry the same blind spot at a lane's edge, a few windows at most (0.002 of push each); not rerun. *Proved?* Seen in runs (TRUCK_ROAD's blind run 20261016_085417, retired; TRUCK_ROAD 20261016_090656).

261. TRUCK_ROAD's (c4), leg (c) of the road questions of 6 October: at mass 0.75 there is no passing lane at any grid push. Measured: a rider set in gear 2 keeps it at 1.060, the top of the road read, and six runners of eight are caught there: MISSED.

Kind: WRONG GUESS, a ramp that turned too low. *Lesson:* SWAP_STATE's ramp turned round at 1.05, so it never reached the light rider's passing lane. On held riders the mass-0.75 passing lane sits at the top of the road, above its slow lane's fold (1.0120), with open road from 1.015 between them. Redesign, said aloud: the light riders' passing lanes read above 1.06, at masses 0.5, 0.75 and 1.0, to find where each is born and where it ends (LIGHT_LANES). *Proved?* Seen in runs (TRUCK_ROAD 20261016_090656).

262. TRUCK_ROAD's (c5), leg (c) of the road questions of 6 October: at masses 1.0 to 3.0, summed over the pushes where both lanes hold, more runners end in gear 2 than in gear 1. Measured: 32 to 16 at mass 1.0, 86 to 58 at 1.5, 61 to 59 at 2.0 and 44 to 60 at 3.0: MISSED at mass 3.0.

Kind: WRONG GUESS, by weight. *Lesson:* The runaway ramp works for the lighter riders and fails for the heaviest: the share caught by gear 2 falls from two thirds at mass 1 to about two fifths at mass 3, and inside each band it rises with the push, two in eight at the bottom of mass 3's band and five in eight at its top. Redesign, said aloud: the share

caught by the passing lane against the mass, finely from 1.0 to 3.0, and against the runner's speed coming in (an open test). *Proved?* Seen in runs (TRUCK_ROAD 20261016_090656).

263. LIGHT_LANES's (P1), Bear's prediction made exact: at mass 1.0 the passing lane fills the whole road read, a rider set in gear 2 keeping it at every grid pace from 1.00 to 1.80 and no runner running there. Measured: it holds from 1.00 to 1.09, and from 1.10 to 1.80 is open road: MISSED.

Kind: WRONG GUESS, a map read too short. *Lesson:* The first map stopped at 1.06, three steps of the dial short of the lane's top. The passing lane is a strip of push that slides up the hill and widens as the rider lightens; at mass 1 it overlaps the slow lane's top and ends at 1.09. Redesign, said aloud: the strip's two edges read finely against the mass, for the law of its slide and where it leaves the slow lane (LANE_EDGES). *Proved?* Seen in runs (LIGHT_LANES 20261016_174648).

264. HEAT_RING's (h1-r), the lean first registered for the brick and kept as the rival when it was rewritten before the tool: a hot brick in ice rings above the first mode's line, its middle crossing zero at L/2 pi = 1.05, 1.10 and 1.20. Measured: it stays above zero at all three (its lowest values 2.4e-09, 5.4e-08 and 2.7e-06 of its start): MISSED.

Kind: WRONG GUESS, read by the rod's modes instead of its fronts. *Lesson:* Read by its modes, a brick carries every shape at once and its short shapes ring; read by its fronts, its middle is knocked down only by about 2e^(-L/4), 0.38 at 1.05, and the short shapes die as fast as the fronts fade, so it stays warm. Redesign, said aloud: where between 0.75 and 1.0 of the line the brick stops ringing, beside the hump's own dip (BRICK_LINE). *Proved?* Seen in runs (HEAT_RING 20261016_174342).

265. ARM_SHARE's (A3), the card's cruder form: one minus the rigid inertia over the circle's average inertia lies within 5 per cent of the fixed share. Measured: 0.162910 against 0.151576, 7.5 per cent above: MISSED.

Kind: WRONG GUESS of a size. *Lesson:* An average of one over a thing is not one over its average when the thing swings, and the body's inertia grows by 39 per cent from arms down to arms overhead. The exact form, the average of one over the inertia, held to 1.6e-10 (A2). Read after the run: the cruder form is exactly the share of arms held still at right angles to the trunk, because the inertia is a constant plus a cosine of the arm's angle. Covered by the run's exact form; no redesign. *Proved?* Seen in runs (ARM_SHARE 20261016_182910).

266. ARM_SHARE's (A4): with the arms' mass halved to 3.5 kg, one circle's turn and the fixed share each fall to between 0.45 and 0.60 of their values at 7 kg. Measured: 0.6160 and 0.6087: MISSED.

Kind: WRONG GUESS of a size, a knob that pays out. *Lesson:* The lean took the arms' share to go in proportion to their mass. But the arms are part of the body they turn: lighter arms leave a lighter body, so each kilogram counts for more. One circle's turn runs 9.51, 15.44 and 19.56 degrees at 3.5, 7 and 10.5 kg: more mass, less return on each kilogram. Covered by the run's own knob table, with the arm's length, the stronger knob, beside it; no redesign. *Proved?* Seen in runs (ARM_SHARE 20261016_182910).

267. BELL_RATIO's (B1): at a squeeze speed 1.41 and 2 times the rings', the water's gain lies within 5 per cent of 1 at every memory. Measured: within 4.0 per cent at S = 2, but 0.933 at S = 1.41 and memory 3: MISSED.

Kind: WRONG GUESS of a size, read from one point. *Lesson:* The lean was read from BELL_CLOCK's one case at memory 6, which turned out to sit on the balance line where the rings neither help nor hurt. At short memory the rings cost stroke even to a fast squeeze. Redesign: as row 269 (BELL_BALANCE). *Proved?* Seen in runs (BELL_RATIO 20261016_183422).

268. BELL_RATIO's (B2): at every memory the water's gain at S = 0.5 exceeds that at S = 2. Measured: below it at memories 3, 4.24 and 6 (0.629, 0.675 and 0.976 against 0.960, 0.987 and 1.009), above it at 8.49 and 12: MISSED.

Kind: WRONG GUESS of a sign. *Lesson:* The lean assumed the rings always help and help the slow squeeze most. They help only in a long-memory water; in a short one they cost, and they cost the slow squeeze most. The slow squeeze feels the water most both ways. Redesign: as row 269 (BELL_BALANCE). *Proved?* Seen in runs (BELL_RATIO 20261016_183422).

269. BELL_RATIO's (B3), the card's one ratio: the squeeze speed over the rings' sets the water's gain more than the memory does. Measured: the gain's spread across the squeeze speed averages 0.2426, across the memory 0.3257: MISSED, and the rival held.

Kind: WRONG GUESS, two jobs read as one. *Lesson:* There is no one ratio. The memory, with the period half of it, sets the sign: the gain crosses 1 between memories 5.23 and 6.24 at every squeeze speed. The squeeze speed sets the size. Redesign, said aloud: the memory and the period separated at fixed squeeze speed, to find which sets the balance line, with the reopening's kicks printed beside the squeezes', to see what takes back the squeezes' 7.6 per cent at the balance (BELL_BALANCE). *Proved?* Seen in runs (BELL_RATIO 20261016_183422).

270. EQUAL_CATCH's (Q1-r), the rival kept beside Claude's lean: at equal catch over five fleet seeds, the learned wobble reminder beats the calendar by more than twice its standard deviation, because the wobble shrinks to 0.89 of its first flight's before the fall. Measured: a gain of +0.0048 of a life against a standard deviation of 0.0162: MISSED.

Kind: WRONG GUESS, a signal that does not pay. *Lesson:* The wobble does shrink before the fall, but the fall of a slowing rider comes when its reaction delay reaches its own rest, two clocks crossing, and a small shrinking wobble does not say when that will be. Matched on catch, the learned reminders and the calendar throw away the same life within luck. Redesign, said aloud: meter the clock that wears, the lag between the board leaving its band and the rider's tap, on the same fleet at equal catch (LAG_METER). *Proved?* Seen in runs (EQUAL_CATCH 20261016_192548).

271. BELL_BALANCE's M1 and M2, planted: at the map's four corners the distance per squeeze repeats within 1 per cent with the step halved, and at every point its quarters lie within 2 per cent. Measured: the corner at memory 12 and period 1.5 changes by -304 per cent, and the quarters depart 10.8 per cent at memory 8.49 and period 1.5: MISSED, in the blind run and the run of record alike.

Kind: THE INSTRUMENT, a toy past its step. *Lesson:* Where the memory is long and the period short, the rings crowd the bells, close approaches between rings need a finer step than 0.005, and the toy no longer settles; five of 25 points fail. Amended before the run of record, said aloud: every point run at both steps, and only settled points judged. Redesign, said aloud: the ring toy made step-safe where rings crowd (a finer step with a check that it settles, or a ring core that softens close approaches, checked against the settled points), to read the fast-squeeze corner and decide (T2), in BELL_CORNER. *Proved?* Seen in runs (BELL_BALANCE 20261016_193628; its blind run 20261016_192112).

272. BELL_BALANCE's (T1-r), the rival kept beside Bear's prediction: the water's memory sets the gain at least as much as the squeeze period. Measured: over settled points the gain's spread across the memory averages 0.2534 and across the period 0.6256: MISSED, and Bear's prediction held.

Kind: WRONG GUESS, the rival. *Lesson:* The period in the rings' time decides the sign: the gain crosses 1 between periods 2.72 and 3.23 at every memory from 3 to 12. The memory deepens what the timing does. Covered by the run's own map; BELL_CORNER reads the corner the toy could not. *Proved?* Seen in runs (BELL_BALANCE 20261016_193628).

273. LAG_METER's (W1): a reminder learned on the slowing rider's reaction lag beats the calendar at equal catch by more than twice its standard deviation across five fleets. Measured: a gain of -0.0211 of a life, standard deviation 0.0322: MISSED, and the rival held.

Kind: THE INSTRUMENT read the wrong clock. *Lesson:* The gauge timed each tap from the moment the board left its band, but the rider may not tap again until its rest of 0.5 fall times has run out, so the lag reads the rest: 0.47 of a fall time on the first flight, with no reaction delay at all, and 0.497 on the last. The wearing delay hides behind the rest until the two are equal, and that is the fall. Redesign, said aloud: read the reaction only where the rest has run out, or by a test poke given to a rested rider (RESTED_LAG). *Proved?* Seen in runs (LAG_METER 20261016_203014).

274. LAG_METER's (W2): the learned lag reminder's mean gain is at least 0.10 of a life. Measured: -0.0211: MISSED.

Kind: THE INSTRUMENT, as row 273. *Lesson:* The size followed the reading: a gauge that reads the rest cannot buy lead. Redesign: as row 273 (RESTED_LAG). *Proved?* Seen in runs (LAG_METER 20261016_203014).

275. LAG_METER's (W3): a fixed lag level, knowing no room, beats the calendar by more than twice its standard deviation. Measured: it throws away 0.786 to 0.828 of a life, a gain of -0.5154: MISSED.

Kind: THE INSTRUMENT, as row 273. *Lesson:* With the lag pinned near the rest on every flight, any level that catches 85 per cent fires on the jitter, early in life. Redesign: as row 273 (RESTED_LAG). *Proved?* Seen in runs (LAG_METER 20261016_203014).

276. RESTED_LAG's (R1-r), the rival kept beside Claude's lean: the rested lag is too noisy, flight by flight, to warn; the learned reminder's gain lies within twice its standard deviation of nought. Measured: +0.2465 of a life against a standard deviation of 0.0358: MISSED.

Kind: WRONG GUESS, the rival. *Lesson:* Read where the rest has run out, the reaction lag is the wearing clock itself, one step of the toy on a new rider's first flight and 0.49 of a fall time on its last, and even a fixed level that knows no room throws away only 0.031 to 0.051 of a life. Covered by the run's own table; no redesign. *Proved?* Seen in runs (RESTED_LAG 20261016_205504).

277. BRICK_LINE's (B1-r), the rival kept beside Claude's lean: a quenched brick stops ringing sooner than its first mode. Measured: it dips at every grid length from 0.75 to 0.95, where the first mode dips: MISSED.

Kind: WRONG GUESS, the rival. *Lesson:* The equation is linear and the brick carries its first mode, so the brick's slow swing goes wherever its first mode's goes. Covered by the run's own table. *Proved?* Seen in runs (BRICK_LINE 20261016_205112).

278. BRICK_LINE's (B2): from 0.80 to 0.95 of the line the brick's deepest dip lies within 35 per cent of 4/pi times the first mode's. Measured: 2.13 to 2.72 times up to 0.87, then 1.30 to 1.51: MISSED.

Kind: WRONG GUESS, two dips read as one. *Lesson:* The lean took the brick's deepest dip to be its first mode's late swing. In shorter pieces it is not: it comes at 1.485 times the rod's length every time, an echo of the cold fronts crossing back through the middle, and only from 0.89 of the line is the late swing the deeper; there the shorter modes

add 30 to 51 per cent, not the third the working allowed. Covered by the run's own table, which prints each dip's time; no redesign. *Proved?* Seen in runs (BRICK_LINE 20261016_205112).

279. STRIDE_MAP's (S2): the biggest stride on the grid (hip 1.40, knee 2.36) turns the trunk back less than one arm circle, 15.44 degrees. Measured: 24.90 degrees: MISSED.

Kind: WRONG GUESS. *Lesson:* The lean took the legs for weak turners because one human stride turns only 6.57 degrees. The turn grows with both knobs at once, close to in step with the hip's swing (1.94 times for twice the swing) and faster than in step with the knee's fold (3.74 times for twice the fold from 0.785, 1.96 times from 1.57 to 2.36), so a stride both wide and deeply folded turns 3.79 times the human one. Seen in the blind run's planted print before the run of record, said aloud; no judging line changed. Covered by the run's own table; no redesign. *Proved?* Seen in runs (STRIDE_MAP 20261016_213858).

280. STRIDE_MAP's (S3): legs alone, three strides in the flight, the human stride holds off less than 0.35 of the forward turn and no stride with a hip swing up to 1.05 holds off half. Measured: the human stride 0.2651, but with the knee folded to 2.36, 0.5417 at hip 0.70 and 0.8103 at hip 1.05: MISSED.

Kind: WRONG GUESS. *Lesson:* The deep fold is the knob: at the human swing, folding the knee from 1.57 to 2.36 rad nearly doubles the turn (6.57 to 12.86 degrees a stride) while the tuck's price grows by little (0.0636 to 0.1012), so the share doubles. Redesign: whether the legs' share adds to the arms' when both run in one flight (ARMS_LEGS). *Proved?* Seen in runs (STRIDE_MAP 20261016_213858).

281. RING_WIDTH's (R1-r), the rival kept beside Claude's lean: the trapped ring's width is second order in the wave's strength, both doubling ratios 3 or more. Measured: 1.875 and 2.067: MISSED.

Kind: WRONG GUESS, the rival. *Lesson:* The trapped ring is a lock made by the wave directly, so its width is first order in the wave's strength: a quarter of it in pace at mass 2, width over strength 0.2417 to 0.2667 from 0.075 to 0.6. Covered by the run's own table; no redesign. *Proved?* Seen in runs (RING_WIDTH 20261016_213858).

282. BELL_CORNER's (C1): on two more halvings of the step every crowded point of the bells' map settles. Measured: none of five settles; the last halving changes the distance per squeeze +2.351, +3.494, -125.766, -139.926 and +104.069 per cent, and the quarters stray up to 43.486 per cent: MISSED.

Kind: THE INSTRUMENT, a toy that switches (possibly NATURE). *Lesson:* At memories 8.49 and 12 with squeezes 1.5 apart, the distance per squeeze takes one of two values on the ladder, forward 0.11 to 0.16 or backward 0.04 to 0.06, and within one run its quarters stray by a third: no finer step converges on one number, because the bells switch between swimming and being driven back by their own wake. At memories 4.24 and 6 it wanders by 2 to 4 per cent. (C2) to (C4) are UNDECIDED. Redesign, said aloud: the corner read in distribution over many nearby starts at two steps, with a softened ring core beside it as a check, to tell two states of the machine from a near-collision the toy's thin cores cannot resolve (BELL_SWITCH) [Answered 20 October by its run 20261020_161902: the rival held; the two answers are the toy's thin cores, and with the cores' own size every nearby start swims forward (rows 438 and 439).] *Proved?* Seen in runs (BELL_CORNER 20261016_224449).

283. SOUP_JUMP_3's (Q1-r), the rival kept beside Claude's lean: the share of riders knocked off by 1,000 differs by less than 0.1 between margins 0.02 and 0.4. Measured: 1.0000 and 0: MISSED.

Kind: WRONG GUESS, the rival. *Lesson:* The margin decides: with a quarter of the soup's power running left the share falls from all to none between margins 0.02 and 0.2. Covered by the run's own table; no redesign. *Proved?* Seen in runs (SOUP_JUMP_3 20261016_225440).

284. SOUP-JUMP-3's framing (not a judged line): the knock-off read as a rate, as the card was written on 9 October. Measured: the riders of one soup share one fate; broken at a large margin, every rider falls at the same place in the left-going pattern, the falls spread by 196.9, exactly the starts' spread; at lopsided 0.25 still riding at 1,000 against the square of still riding at 500 is 0.1250 against 0.0156 (margin 0.1) and 0.6250 against 1.0000 (margin 0.4): NOT A RATE.

Kind: WRONG GUESS, a framing. *Lesson:* The left-going waves are a pattern, not noise: a rider falls where it meets a stretch of them deep enough to beat its margin, and every rider of the soup meets the same stretch, so the jump's tax grows with how long the bead must ride. Redesign, said aloud: each soup's knock-off margin against the ride's length, and against the deepest stretch of the left-going pattern a rider sweeps, read from its shape alone as the crest line was (an open test). *Proved?* Seen in runs (SOUP_JUMP_3 20261016_225440).

285. ARMS_LEGS's (W1-r), the rival kept beside Claude's lean: at some pairing the arms and legs together depart by more than 25 per cent from their shares alone, added. Measured: -6.19 to +6.51 per cent: MISSED.

Kind: WRONG GUESS, the rival. *Lesson:* The body the limbs share changes each only a little, and in a pattern: above the sum where the circles outnumber the strides, below it where the strides outnumber the circles. Covered by the run's own table; no redesign. *Proved?* Seen in runs (ARMS_LEGS 20261017_012927).

286. ROUGH_FLOOR's M2, planted (the band test): at every judged row the median run over the mean repeats within 0.1 with runs joined at 1.5 and at 3 field cycles. Measured: changes up to 0.234, at three of seven rows (two-fold 0.15 at 0.85 and 0.90 of its grip, many ledges at 0.85): MISSED, in the blind run and the run of record alike.

Kind: THE INSTRUMENT, a statistic that depends on its own definition. *Lesson:* Where a needle lingers near its lock for one to three field cycles between bursts, the line between one run and two is the counter's choice, not the needle's. Amended before the run of record, said aloud: only rows that pass the band themselves are judged. Redesign, said aloud: runs read by the needle's own speed with a knee, as the lock readers did, which needs no joining gap (ROUGH_BAND). *Proved?* Seen in runs (ROUGH_FLOOR 20261017_013514; its blind run 20261017_011647).

287. ROUGH_FLOOR's (F1): UNDECIDED, because the registration's grid of paces, set as fractions of each floor's greatest grip, left the two-fold floors without a judged row (the 0.15 floor's two rows inside its band dropped by the band test; the 0.3 floor 72 runs at 0.80 of its grip and 89 to 100 per cent slipping above).

Kind: THE INSTRUMENT, Claude's registration. *Lesson:* The greatest grip does not set where a remembering needle slips: the ledges raise it by 4 to 14 per cent and move the half-slipping pace by 0 to 5 per cent. A grid must be placed on each floor's own slipping band. Redesign: each floor's band found first by a coarse scan, then read inside it with enough needles for 100 runs a row (ROUGH_BAND). *Proved?* Seen in runs (ROUGH_FLOOR 20261017_013514).

288. ROUGH_FLOOR's (F1-r), Bear's sandpile picture kept as the rival: on some rough floor a judged row's median run lies below 0.4 of its mean, a heavy tail. Measured: the one judged rough row 0.513, and 0.519 to 0.758 on the rough rows the band dropped: MISSED.

Kind: WRONG GUESS, the rival. *Lesson:* The floor repeats every turn, so a falling needle meets the same ledges on every turn and its chance to be caught stays the same: a trap, not a sandpile. Near its onset the many-ledge floor is a little more spread than one trap (median 0.51 of the mean, 8.2 per cent of runs past three means), which looks like two traps mixed; ROUGH_BAND was to read it, and its own cut moved the answer (row 577). *Proved?* Seen in runs (ROUGH_FLOOR 20261017_013514).

289. EXPLORER_ENGINE_CHECK's E9 to E11, planted, in its first blind run: the diver's values of record were to be read from the scope's table. Measured: the check took the first number in each row, which sits in the check's own name ('at amplitude 0.6', 'from 3 m and 10 m'), and judged the engine's -1.398960, -6.451253, 0.782 and 1.428 against 0.6, 0.6, 3 and 10: MISSED, as printed.

Kind: THE INSTRUMENT, a tool fault in the check, not the engine. *Lesson:* A value read from a document's table is a claim about which cell it came from. Redesign, said aloud and made the same day: the check reads the value cell and asserts that each cell holds the check's name, the run and the tolerance where it expects them; the second blind run held all eleven. The first blind run is retired. *Proved?* Seen in runs (run_LOOP_ATLAS_EXPLORER_ENGINE_CHECK_BLIND_20261017_063057, retired; its redesign held in 20261017_063114 and the run of record 20261017_063123).

290. EXPLORER_ENGINE_CHECK's (W1-r), the rival kept beside Claude's lean: one jumper flight at 6,000 steps takes over 500 ms in node. Measured: a median of 46.19 ms over 12 flights: MISSED.

Kind: WRONG GUESS, the rival. *Lesson:* The six-part body's 24,000 inertia sums a flight cost a twentieth of a second in a browser's engine, so a room can recompute a whole flight while a slider moves. Covered by the run's own table; no redesign. *Proved?* Seen in runs (EXPLORER_ENGINE_CHECK 20261017_063123).

291. EXPLORER_ENGINE_CHECK's (W2-r), the rival: at 1,500 steps a flight at least one of the jumper's acceptance numbers leaves its tolerance. Measured: the largest gap 1.77e-10 against tolerances of 1e-6 and 1e-5: MISSED.

Kind: WRONG GUESS, the rival. *Lesson:* The gap between the engine and the record is set by the rates' central differences and rounding, not by the step: it is about a ten-billionth at 1,500 steps and at 6,000. Covered by the run's own table; how far the steps can fall is on the proposed list. *Proved?* Seen in runs (EXPLORER_ENGINE_CHECK 20261017_063123).

292. FACE_ALONG's M1, planted, as registered: with every link at grip ratio 1 the travel is zero to 1e-9. Measured: +1.798e-06 at 300 steps a lap: MISSED, in the first blind run.

Kind: THE INSTRUMENT, a tolerance below the step rule's own error. *Lesson:* The two-slope rule's error at 300 steps is about 2e-6 body lengths and falls about four times when the steps double (+4.628e-07 at 600); an exact zero is read as a zero only above the rule's floor. Amended before the run of record, said aloud: below 1e-5 at 300 steps and shrinking 3 to 5 times with doubled steps; held. *Proved?* Seen in runs (run_LOOP_ATLAS_FACE_ALONG_BLIND_20261017_070529; the amended M1 held in 20261017_070615 and the run of record 20261017_070629).

293. FACE_ALONG's (F1-r), the rival kept beside Claude's lean: the broad middle travels farther than the broad ends. Measured: 0.053166 against 0.064786 body lengths a lap: MISSED.

Kind: WRONG GUESS, the rival. *Lesson:* The ends' grip is the more sensitive knob both ways: at grip 3 on the ends the swimmer travels 0.0400 against 0.0379 with the middle at 3, and at 1.5, 0.0198 against 0.0210. Covered by the run's own table; no redesign. *Proved?* Seen in runs (FACE_ALONG 20261017_070629).

294. FACE_ALONG's (F2), Claude's lean, a symmetry guessed and not derived: the broad head and the broad tail travel the same distance (to 1e-6 as registered, 1e-4 as amended before the run of record). Measured: 0.040913 against 0.040337, 1.4 per cent apart, the difference steady at 300, 600 and 1,200 steps: MISSED.

Kind: WRONG GUESS, a symmetry that was not there. *Lesson:* The hinges bend a quarter cycle apart, the head's first, so the bending runs from head to tail and the body has a front, though the water takes the same work from both, 0.20748. Can this be redesigned? Yes: run the lap backward, the bending running from tail to head; if the timing is the cause the two should swap. On the proposed list. *Proved?* Seen in runs (FACE_ALONG 20261017_070629).

295. FACE_ALONG's (F4), Claude's lean: the broad ends buy at least half of the extra travel that making every link broad buys. Measured: 0.405: MISSED.

Kind: WRONG GUESS, a size. *Lesson:* The faces help each other: all broad buys 1.79 times the sum of the three single faces' extras, so two faces buy less than their share of the whole. Covered by the run's own table; whether the faces combine as a product is on the proposed list. *Proved?* Seen in runs (FACE_ALONG 20261017_070629).

296. DIVE_CUE's M2, M3 and M4, planted, as registered with tolerances of 1e-6 degrees and 1e-9 s. Measured in the blind run: 1.18e-05 and 1.3e-06 degrees (M2), 4.8e-05 degrees (M3), 2.2e-09 s and 5.3e-06 degrees (M4): MISSED.

Kind: THE INSTRUMENT, tolerances below the floor: the lesson of row 292, learned an hour or so before, not applied. *Lesson:* The same miss as row 292, an hour or so later: the planted tolerances were written by hand, not from the stepping's own error. A likely cause of the floor, not tested: each half-cosine shape change has a jump in its second derivative at its ends, which lowers the stepping's order there. Amended before the run of record, said aloud, at ten times the measured floor; held. The redesign is of the practice: measure the floor by a halving before writing any planted tolerance. *Proved?* Seen in runs (run_LOOP_ATLAS_DIVE_CUE_BLIND_20261017_082235; the amended checks held in the run of record 20261017_082454).

297. DIVE_CUE's (D1-r), the rival kept beside Bear's prediction: at some spin error the clock's kick-out lands nearer the mark than the cued one. Measured: the cue nearer at all six spin errors, its miss 0.420 of the clock's: MISSED.

Kind: WRONG GUESS, the rival. *Lesson:* With a cue, the part of the error the tuck would carry is gone; only the slow layout after the cue carries the rest. Covered by the run's own table; no redesign. The open part, a second cue near the water, goes to the proposed list. *Proved?* Seen in runs (DIVE_CUE 20261017_082454).

298. SPRING_FLOOR's (S1), Claude's lean: with a floor of 0.2 the spring's latch stays awake, every soup's latch switching on in the run's last tenth at every h and both settings. Measured: awake to the end only at h 0.25 (and at h 0.5 at tune 1.5); seldom at h 1.0; silent at h 1.5: MISSED.

Kind: NATURE, in the toy: a store with a floor keeps an idle swing of its own. *Lesson:* The floor keeps the bead swinging at about a third of the fixed speed where the latch never fires (0.302 and 0.366 of it at h 1.5); a trigger set above that idle swing still never fires. Can this be redesigned? Yes: set the trigger as a share of the floor's own idle swing (an open test). *Proved?* Seen in runs (SPRING_FLOOR 20261017_160023).

299. SPRING_FLOOR's (S2-r), the rival: at some setting the floored latch carries at least half the clean listener's 0.3776 per event. Measured: at most 0.1181: MISSED.

Kind: WRONG GUESS, the rival. *Lesson:* The spring carries at most a third of the listener's push per event even awake; covered by the run's own table; no redesign. *Proved?* Seen in runs (SPRING_FLOOR 20261017_160023).

300. EXPLORER_DIVE_CHECK's (W3-r), the rival: one cued dive at 3,000 steps takes over 500 ms in node. Measured: a median of 49.78 ms: MISSED.

Kind: WRONG GUESS, the rival. *Lesson:* A cued dive costs a twentieth of a second, so room 2 can fly the clock's dive and the cued dive side by side as a slider moves. Covered by the run's own table. *Proved?* Seen in runs (EXPLORER_DIVE_CHECK 20261017_155902).

Misses 301 to 350

301. EXPLORER_ROUTINE_CHECK's (G1-r), room 2, the rival: the tuck with the arms swung forward instead of back, all else the record's, misses the record's entry by more than 30 degrees. Measured: 5.24 degrees short of it: MISSED.

Kind: WRONG GUESS, the rival. *Lesson:* Both roads go out and come back the same way, so each turns the body by nothing with no spin (3e-06 degrees for the forward road); the 5.24 degrees are all count part, from where the arms pass during the 0.3 s of change. Covered by the run's own table; the exact split, the spin times the difference of one over the inertia along the two roads, goes to the proposed list. *Proved?* Seen in runs (EXPLORER_ROUTINE_CHECK 20261017_164432).

302. EXPLORER_ROUTINE_CHECK's (W4-r), the rival: one routine flight at 3,000 steps takes over 500 ms in node. Measured: a median of 31.55 ms: MISSED.

Kind: WRONG GUESS, the rival. *Lesson:* A dive of laid-out moves costs as little as DIVE_CUE's dives; room 2 flies the player's line and its no-spin twin at each launch. Covered by the run's own table. *Proved?* Seen in runs (EXPLORER_ROUTINE_CHECK 20261017_164432).

303. LANE_EDGES's (L1-r), the rival: the passing strip's width against the mass, on logarithmic scales, has a slope outside -0.6 to -0.4. Measured: -0.5135: MISSED.

Kind: WRONG GUESS, the rival. *Lesson:* The width goes as one over the square root of the mass, within 3.2% over eleven masses. Covered by the run's own table; the centre, whose slope steepens toward the same law, goes to the proposed list. *Proved?* Seen in runs (LANE_EDGES 20261017_164501).

304. EXPLORER_ROUTINE_CHECK's beside line: the cue's rip window in spin error. Printed: +-1.102 per cent. From the same run's entries at +-2 per cent it is +-2.203, as DIVE_CUE's 22.694 degrees at 10 per cent also give: the print was half itself.

Kind: THE INSTRUMENT, a print: a half-width halved again. *Lesson:* The formula already gave the half-width and the print divided it by two. Nothing judged read it. Can this be redesigned? Yes, and done: the print fixed in the tool after the run, said in its header and in its row of the data ledger; the run file and json keep the slip. The practice: print the inputs of every derived beside value beside it, so a slip shows on its face. *Proved?* Seen in runs (EXPLORER_ROUTINE_CHECK 20261017_164432).

305. TONG_DULLIN's M5 (and its verdict print), planted: the two-part turn at a spin of 120 equals the direct flight to 1e-10 rad. Measured in the first blind run: 5.3e-09 to 7.6e-09 rad, the same at half the steps; and M1 to M4 printed UNDECIDED though inside their tolerances: MISSED.

Kind: THE INSTRUMENT, a tolerance below the arithmetic's floor, and a print fault. *Lesson:* The engine's coefficients carry the trunk's angle through central differences at 1e-6, so a turn that is exactly linear in the spin is linear only to about 1e-8 rad at a spin of 120; and numpy's booleans fail a test of identity with True. The halving probe's rule set M5 at ten times the floor, and M1 too (4.5e-13 against 1e-12); the print now casts to a plain boolean. Amended before the run of record, said aloud; held. *Proved?* Seen in runs (run_LOOP_ATLAS_TONG_DULLIN_BLIND_20261017_173508; held in the second blind run 20261017_173617 and in the run of record 20261017_173718).

306. TONG_DULLIN's (T1): Tong and Dullin's inertias come back from their Table 1, 21.0647 at (-pi, -0.3608) and 5.2888 at (-0.3867, -pi), within 0.001. Measured: 21.3089 and 5.4930: MISSED.

Kind: THE INSTRUMENT or the source, not yet told apart: our reading of their table, or the table. *Lesson:* The rebuild is about 0.24 and 0.20 kg m^2 heavier (1.2 per cent stretched out, 3.9 per cent piked) at the two shapes (corrected 17 October, from TONG_DULLIN 20261017_173718's json: first reported as about 1 per cent at both shapes), while its field and its best dives keep their shape; the rival, a joint vector misread, held. Can this be redesigned? Yes: the forward parts of the three joint vectors read both ways (TONG_DULLIN_READINGS, run the same batch, row 309), then a scan of each component (TONG_DULLIN_SCAN) [17 October: told apart, the source. Their Table 1 as printed drops the knee's forward 0.08 on the thigh's side; restored as the hip's pair has it, both inertias come back within 0.005 (TONG_DULLIN_KNEE, row 314).] *Proved?* Seen in runs (TONG_DULLIN 20261017_173718).

307. TONG_DULLIN's (T2) to (T6): their extremes' places, the field's largest size and place, the best dives at spins of 120 and 30 and the corner's spin, each within its registered tolerance. Measured: the extremes at legs -0.4327 and arms -0.4528 (theirs -0.3608 and -0.3867); B 0.3065 at (-1.3237, -3.0801) (theirs 0.3058 at (-1.3148, -3.0043)); at 120, (1.0000, 0.8330) and 0.0247 rad (theirs (1, 0.8593), 0.0189); at 30, (0.9910, 0.3015) and 0.2335 (theirs (0.9818, 0.3158), 0.2327); the corner from 209.26 (theirs 193.65): MISSED.

Kind: THE INSTRUMENT or the source, as row 306. *Lesson:* Each follows the body of row 306, and each keeps the shape of theirs: the field's size within a quarter of a per cent, the gain at the slow spin within half of one, the best way out the same. Covered by row 306's redesign [17 October: settled with row 306; with the knee's 0.08 restored every one of these comes back within its registered tolerance (TONG_DULLIN_KNEE).] *Proved?* Seen in runs (TONG_DULLIN 20261017_173718).

308. HALT_THRESHOLD's M2, lean (l) of 6 October, planted: with the rule at 2.5 m the tail's arrival moves by less than 0.5 s at half the step. Measured in the first blind run: no arrival by 900 s at either step: MISSED.

Kind: THE INSTRUMENT, a step test set where the march does not finish. *Lesson:* The cap was too short for a crawling column; amended before any run of record to 3,600 s, said aloud; missed again (row 310). *Proved?* Seen in runs (run_LOOP_ATLAS_HALT_THRESHOLD_BLIND_20261017_174151).

309. TONG_DULLIN_READINGS's (R1): exactly one of the eight readings of the three forward joint offsets gives both of Tong and Dullin's inertias within 0.001. Measured: none; their pair lies between the printed reading (21.3089, 5.4930) and the knee offset reversed (20.9313, 5.1993): MISSED.

Kind: WRONG GUESS, about where the difference lies. *Lesson:* Not a sign; something smaller or elsewhere in the body. Can this be redesigned? Yes: a scan of each joint component, one at a time, for the value that brings all four numbers within their print (TONG_DULLIN_SCAN); to run next, the chain having already had its two runs [17 October: it was elsewhere, as this row guessed: the knee's forward offset, one value in the right place (TONG_DULLIN_SCAN and TONG_DULLIN_KNEE, rows 312 and 314).] *Proved?* Seen in runs (TONG_DULLIN_READINGS 20261017_174547).

310. HALT_THRESHOLD's M2, lean (l) of 6 October, planted, again with the cap at 3,600 s. Measured in the second blind run: no arrival at either step: MISSED.

Kind: THE INSTRUMENT, the planted setting in a column that never starts (possibly NATURE, in the toy). *Lesson:* At a threshold of 2.5 m the stretch of starting off is larger than the rule allows, so each start ends in a halt and the column never gets going: the crawl the lean foresaw, at its limit. [Corrected 17 October: in part the instrument. The column's limit called hits from rounding alone and held a standing column's tail still (misses row 315); with that fixed, the rule at 2 m still brings no arrival by 3,600 s, while at 3 m the tail arrives at 1796.783 s (HALT_THRESHOLD 20261017_184055).] The chain stopped at two runs, said aloud. Redesign: the step test on a march that finishes (the rule at 7 m) and the start-up stretch printed, to run next. *Proved?* Seen in runs (run_LOOP_ATLAS_HALT_THRESHOLD_BLIND_20261017_174755).

311. Claude's telling of TONG_DULLIN's inertias, in the reply and records of 17 October: the rebuild about 1 per cent heavy at both shapes. Read from its json: 1.2 per cent stretched out and 3.9 per cent piked: MISSED.

Kind: THE INSTRUMENT, a misreading (Claude's own telling of a number not read by script). *Lesson:* The piked shape's larger share was folded into the stretched one; Bear's question the same day, why the structure matched to a quarter of a per cent while the inertias missed by 1, brought it out, and the true split is the clue: the piked shape, where the lower legs fold, is off three times as much. Corrected in every record that carried it, the same batch. Can this be redesigned? Yes: a share is read by script from the json, as every number is. *Proved?* Seen in the reply and records of 17 October (TONG_DULLIN 20261017_173718's json).

312. TONG_DULLIN_SCAN's (S1): exactly one of the 32 entries of Tong and Dullin's Table 1, set alone so the stretched-out inertia is 21.0647, brings the piked one within 0.001 of 5.2888. Measured: two do, the forward parts of the thigh's knee end and of the lower leg's (5.2897 for both): MISSED.

Kind: THE INSTRUMENT, a lean that counted entries where the body has one offset. *Lesson:* Only the difference of the knee's two forward parts moves any mass, so the two entries are one number counted twice; in substance the scan found the one value. Can this be redesigned? Yes, said aloud the same day: the knee's 0.08 restored with nothing tuned (TONG_DULLIN_KNEE); held (row 314). *Proved?* Seen in runs (TONG_DULLIN_SCAN 20261017_181953).

313. TONG_DULLIN_SCAN's (S1-r), the rival: no entry alone brings both inertias within 0.001. Measured: two do (row 312): MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* The difference lay in one place after all, the knee. Covered by TONG_DULLIN_KNEE. *Proved?* Seen in runs (TONG_DULLIN_SCAN 20261017_181953).

314. TONG_DULLIN_KNEE's (K1-r), the rival: with the knee's 0.08 restored, either inertia is more than 0.05 from theirs. Measured: 21.0597 and 5.2858 against 21.0647 and 5.2888: MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* The thigh's knee end printed as (0, -0.215) where the hip's pair carries 0.08 on both sides; with 0.08 there, no number fitted, every number of theirs we rebuilt comes back within the bars registered for it (gaps 0.02 to 0.15 per cent; first written 'within its print', corrected 18 October), the field's peak to its last digit. The source, most likely a misprint, not our reading. Needs no redesign; settles rows 306, 307 and 309. *Proved?* Seen in runs (TONG_DULLIN_KNEE 20261017_182629).

315. HALT_THRESHOLD's M2, lean (l) of 6 October, planted, version 2 (the step test moved to 7 m, the start-up stretch printed): with the rule at 7 m the tail's arrival moves by less than 0.5 s at half the step. Measured in the third blind run: no arrival at either step: MISSED.

Kind: THE INSTRUMENT, a fault in the column's limit, read from MORTAR_SPACING. *Lesson:* Diagnosed after the miss on the planted case alone: the column halts at 26 s and stands until 508 s, its tail's last gap closing at half a millimetre a second. The limit (no closer than 0.5 m) tested a hit as (z - off) < x after z = x + off, so rounding alone called a hit on about one walker-step in 80 and set that walker's speed to his man ahead's: in a standing column, nobody behind can close. Can this be redesigned? Yes, said aloud: the hit called only where the running minimum is below the walker's own (version 3); held in the fourth blind run (713.721 s against 713.7105 at half the step). The fault sits in every tool that ran the column with its limits on: LIMIT_AUDIT audits them. *Proved?* Seen in runs (run_LOOP_ATLAS_HALT_THRESHOLD_BLIND_20261017_182910; the diagnosis printed, not kept as a run).

316. EXPLORER_CAT_CHECK's CAT (W4-r), the rival: half a circuit of the cat at 1,000 steps puts the legs' turn more than 0.01 degrees from 10,000 steps. Measured: 5.3e-12 degrees: MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* The shapes move smoothly and the fourth-order step's error falls as the step to the fourth power, so a thousand steps is already finer than any page could show. Needs no redesign. *Proved?* Seen in runs (EXPLORER_CAT_CHECK 20261017_183736).

317. EXPLORER_CAT_CHECK's CAT (W5-r), the rival: half a circuit at 1,000 steps takes over 100 ms in node. Measured: a median of 2.80 ms over 9: MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* Each step solves a three-by-three system four times, a few hundred multiplications; a page can fly the cat many times while a slider moves. Needs no redesign. *Proved?* Seen in runs (EXPLORER_CAT_CHECK 20261017_183736).

318. HALT_THRESHOLD's (l1), Claude's lean of 6 October made exact: the tail's arrival against the halt threshold is a valley, its earliest strictly inside the read. Measured: the earliest is with no rule, 524.999 s; every rule arrives later (714.738 s at its best, 8.0 m): MISSED.

Kind: WRONG GUESS (and the yardstick). *Lesson:* A halt never moves the tail sooner: it stops the head while the tail closes, so it buys contact (the longest gap near 30 m instead of 406) at the cost of time. Can this be redesigned? Yes: the trade itself, time against the longest gap and walker-seconds lost, to the proposed list. *Proved?* Seen in runs (HALT_THRESHOLD 20261017_184055).

319. HALT_THRESHOLD's (l2): the best threshold lies below 6 m, where contact is lost. Measured: the best is no rule: MISSED.

Kind: WRONG GUESS. *Lesson:* Follows row 318; covered by its redesign. *Proved?* Seen in runs (HALT_THRESHOLD 20261017_184055).

320. HALT_THRESHOLD's (l1-r), the rival: the lowest threshold, 1.5 m, is best. Measured: 1.5 m never arrives by 3,600 s (208 halts): MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* The lowest threshold is the crawl at its limit. Covered by row 318's redesign. *Proved?* Seen in runs (HALT_THRESHOLD 20261017_184055).

321. HALT_THRESHOLD's (l3), the third surprise noted just before this run, made exact: every threshold below the start-up stretch (3.6516 m) gives no arrival by 3,600 s, and every one at least 1 m above it arrives. Measured: 1.5 and 2 m never arrive, but 3 m arrives at 1796.783 s after 40 halts: MISSED.

Kind: WRONG GUESS. *Lesson:* The stretch is the largest gap of a standing start over a minute, not a wall: below it each start still gains a few metres before the halt, so a rule just under it creeps home slowly, and only well under it does the creep stall. Needs a sharper lean, not a redesign of the tool: where between 2 and 3 m the column stops arriving, to the proposed list with row 318's trade. *Proved?* Seen in runs (HALT_THRESHOLD 20261017_184055).

322. LIMIT_AUDIT's (A1-r), the rival: with the limit's rounding fault removed, no printed number in the five column tools moves by more than 0.1 per cent. Measured: counts of stopping walkers move by up to 100 per cent in three tools: MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* The fault was no rounding-level blemish: it braked any walker closing on the man ahead about once in eighty steps. Needs no redesign; the audit itself is the redesign of row 315. *Proved?* Seen in runs (LIMIT_AUDIT 20261017_190938).

323. LIMIT_AUDIT's (A3), Claude's lean: no judged verdict of the five runs of record changes with the fix. Measured: five change, in MORTAR_SPACING, CATS_EYES and QUICK_READ: MISSED.

Kind: WRONG GUESS. *Lesson:* The leans those tools judged sat close to the fault's own effect: stops that came from walkers who could not close up. Covered by rows 324 to 326. *Proved?* Seen in runs (LIMIT_AUDIT 20261017_190938).

324. MORTAR_SPACING's (a), as rerun with the limit fixed: a fixed 5 m under the swinging leader stops some walker and somewhere puts 5 or more within 10 m. Measured: 9 stop, at most 4 within 10 m (5 at a delay of 0.6): MISSED.

Kind: THE INSTRUMENT (the record's hold was the fault's) and a WRONG GUESS by one walker. *Lesson:* The bunching Bear named is real at a fixed 5 m, nine stopping, but the five-in-ten-metres line was the toy's fault at a delay of 0.3. Can this be redesigned? Covered: the corrected numbers stand in the book; a lean on how many of the column stand within one point's reach belongs on the proposed list. *Proved?* Seen in runs (LIMIT_AUDIT 20261017_190938).

325. CATS_EYES's (b), as rerun with the limit fixed: read through a look-time of 0.3 s, some walker stops, with the reading error and without it. Measured: 19 stop with the error, none without it: MISSED. And its planted M0, repeating NIGHT_SPACING's record (tail 1.355 m/s): 1.338: MISSED against the old record, held against the corrected one.

Kind: THE INSTRUMENT (the record's hold was the fault's), and NATURE in the toy. *Lesson:* A late read lets the quick wobble grow, the closed form's 1.58 a walker, but with no error it has nothing to grow from: the leader's slow swing barely feeds it. Can this be redesigned? Yes: the seed's size at which a late read starts stopping walkers, to the proposed list. *Proved?* Seen in runs (LIMIT_AUDIT 20261017_190938).

326. QUICK_READ's (a), as rerun with the limit fixed: with the small error no walker stops at look-times to 0.07 s and some stops at every look-time from 0.11 to 0.15 s. Measured: none stops to 0.13 s; 1 at 0.14 and 4 at 0.15: MISSED. And its M0, repeating CATS_EYES's record: MISSED against the old record, held against the corrected one.

Kind: THE INSTRUMENT (the record's hold was the fault's). *Lesson:* The edge moved from 0.09 s to 0.13 s; the wobble's swing, 2.34 s, did not move. The book's reading of the eye (about 0.1 s in moonlight, from recall) now sits inside the edge, not on it. Needs no redesign. *Proved?* Seen in runs (LIMIT_AUDIT 20261017_190938).

327. TWO_HANDS's (W1-r), the rival (*A Rider in Two Waves*'s straight-sided V): the crest-firer's impulse per event moves by more than 10 per cent somewhere from h 0.25 to 2. Measured: at most 1.5 per cent: MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* This balance is not a one-point cancellation: both hands carry the same Gaussian factor at every level, so a size sweep finds it flat everywhere, as it would a symmetry's nothing. Needs no redesign; tests 2 and 4 tell it. *Proved?* Seen in runs (TWO_HANDS 20261017_203811).

328. TWO_HANDS's (W2-r), the rival: the drift from h 0 to 2 is the same at 6, 24 and 96 waves within two standard errors. Measured: -0.0926, -0.0149 and +0.0018: MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* A soup of few waves has lighter tails than a Gaussian, so its high crests are fewer than the crossing count allows for and the balance slides; *A Rider in Two Waves*'s second test, passed. Needs no redesign. *Proved?* Seen in runs (TWO_HANDS 20261017_203811).

329. TWO_HANDS's (W3): on the skewed soup the crest-firer moves by more than 5 per cent from h 0 to 2 and its trough-firer's magnitude differs from it by more than 5 per cent at h 2; on the even soup they agree within two standard errors. Measured: the crest moved +0.3318, but the trough-firer never fired at h 2 (an empty count), and the even soup's gap was 2.6 standard errors: MISSED.

Kind: THE INSTRUMENT, twice. *Lesson:* The square-law skew is bounded below at about -1.13 of the strength, so its trough never reaches h 2; and the gap was the pooled ratio's while its standard error was the soups' own ratios'. Can this be redesigned? Yes, at once: a skew that reaches both sides and one estimator on four times the soups (TWO_HANDS_2); the gap was chance (+0.66 standard errors at 192 soups). *Proved?* Seen in runs (TWO_HANDS 20261017_203811).

330. TWO_HANDS_2's (V1): on the soup skewed by 1 + 0.3 tanh, the trough's magnitude differs from the crest's by more than 5 per cent at h 1.5 and 2, and the crest moves by more than 5 per cent from h 0 to 2. Measured: the two sides differ by about 48 and 47 per cent, but the crest moves +0.0327: MISSED by its last clause.

Kind: WRONG GUESS. *Lesson:* A smooth one-way stretch of the soup keeps the crossing count and the carried push nearly in step, so the crest's balance survives it; the square-law skew broke it by a third. Covered: the half-turn's answer is in hand (the two sides separate); the balance's sensitivity to the soup's shape is test 2's. *Proved?* Seen in runs (TWO_HANDS_2 20261017_204131).

331. TWO_HANDS_2's (V1-r), the rival: the crest's and trough's magnitudes on the skewed soup agree within two standard errors at h 2. Measured: -0.2249 +/- 0.0118: MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* The half-turn separates the hand that flips (the push's side) from the one that does not (the crossing count). Needs no redesign. *Proved?* Seen in runs (TWO_HANDS_2 20261017_204131).

332. TWO_HANDS_2's (V2-r), the rival: on 192 even soups the crest's and trough's magnitudes at h 2 differ by more than three standard errors. Measured: +0.66: MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* A soup's mirror is as likely as itself; the 48-soup gap was chance. Needs no redesign. *Proved?* Seen in runs (TWO_HANDS_2 20261017_204131).

333. COIL_SNAKE's (C3-r), the rival: the log coil's travel per turn lies more than 10 per cent from an even coil of its mean radius. Measured: 1.3e-04 of it: MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* The axle's balance weighs each piece of wire by its length, which grows with its radius, so the mean radius along the wire is the coil's radius for travel; a growth of a tenth a turn tilts the wire only slightly. Needs no redesign. *Proved?* Seen in runs (COIL_SNAKE 20261017_204030).

334. TWO_HANDS_BAND's (B1), Claude's lean: given a dead band of 0.001, 0.01 and 0.05 strengths the crest-firer's count changes by less than 0.1, 0.5 and 2 per cent at h 0, 1 and 2. Measured: at h 2, -1.43e-03, -1.05e-02 and -5.11e-02; at h 1, -5.87e-03 and -2.66e-02: MISSED.

Kind: WRONG GUESS, about how many crests graze the level. *Lesson:* A smooth wave that passes a high level mostly passes it by a little: its overshoot spreads like a Rayleigh curve, so a band of b removes about h b / 2 of the crossings at level h. Not chatter: no step at the smallest band (row 335). Can this be redesigned? Yes: judge the band test against the share of the run's own crests that peak within b/2 of the level, read from the band-0 run (the proposed list). *Proved?* Seen in runs (TWO_HANDS_BAND 20261017_210543).

335. TWO_HANDS_BAND's (B1-r), the rival: at a band of 0.001 the count falls by more than 1 per cent somewhere (a chattering count). Measured: at most 1.43e-03: MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* The soup is smooth at the step used, so no crossing repeats between steps. Needs no redesign. *Proved?* Seen in runs (TWO_HANDS_BAND 20261017_210543).

336. EXPLORER_CAT_ROOM_CHECK's R1, room 3, planted, in the first blind run: the routine of record (bend 90, swing 180, straighten) lands with the front legs within 1e-6 degrees of EXPLORER_CAT_CHECK's 151.4217 at two timings of its moves. Measured: the second timing (moves of 0.1, 0.2 and 0.12 s) 4.87e-06 degrees from the first: MISSED.

Kind: INSTRUMENT (a tool fault). *Lesson:* The routine stepped straight across the moves' edges, where each half-cosine blend's rate turns a corner; a fourth-order step loses its order whenever an edge falls inside a step, so the error rose to 1e-6 to 2e-5 degrees at durations whose edges missed the step grid and fell to 1e-12 where they met it. Can this be redesigned? Yes, and it was, before any run of record, said aloud: the routine flies each move as its own stretch of steps, shared out by duration, so no step straddles an edge. The second blind run held at 4.5e-12. *Proved?* Seen in runs (EXPLORER_CAT_ROOM_CHECK blind 20261017_211142; fixed in blind 20261017_211314).

337. EXPLORER_CAT_ROOM_CHECK's (P5-r), room 3, the rival: the spine's tilt after bend, half swing and straighten differs from the half circuit's V movement by more than 0.01 degrees somewhere among bends 60, 90 and 120 and radii 0.15, 0.3 and 0.45. Measured: at most 1.23e-11: MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* Straightening after the half swing cannot undo the tilt the half circuit left: the straighten move is a bend run backward with no swing, which turns nothing, so the landing keeps exactly what the half circuit's V moved. Needs no redesign. *Proved?* Seen in runs (EXPLORER_CAT_ROOM_CHECK 20261017_211325).

338. EXPLORER_CAT_ROOM_CHECK's (W6-r), room 3, the rival: one launch at 3,000 steps takes over 150 ms in node here. Measured: a median of 9.79 ms over 9: MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* The cat's step is a few small matrix products and a three-by-three solve; 3,000 of them cost about 10 ms, so the page can launch at once. Needs no redesign. *Proved?* Seen in runs (EXPLORER_CAT_ROOM_CHECK 20261017_211325).

339. CAT_TWIST's T3, planted, in the first blind run: the relative orientation followed as a unit quaternion along the trick ends at minus its start for the toy and at its start for the twist-free spine. Measured: at a bend of 45 the toy's end against its start read -0.999999924941: MISSED.

Kind: INSTRUMENT (a tool fault). *Lesson:* The lift's first quaternion was taken at a stale time, a loop variable left from the check above it (a moment near the end of the straighten, where the 45-degree cat is still a little bent), not at the trick's start; the 90 and 135 cases hid it. Can this be redesigned? Yes, and it was, before any run of record, said aloud: the lift starts at the trick's first moment. The second blind run held at -1.000000000000 at every bend. *Proved?* Seen in runs (CAT_TWIST blind 20261017_231820; fixed in blind 20261017_231908).

340. CAT_TWIST's (L1-r), the rival: a twist-free spine's trick (bend 90, swing half round, straighten) lands the cat on its feet. Measured: both pairs of paws 75.95 degrees from straight down, the body turned 104.05: MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* Without its waist's full turn the half swing turns the body only a little over a right angle at a bend of 90; the toy's half turn came from the twist. A deep bend helps (a twist-free half trick at 135 lands 32.0 degrees from straight down), but a full circuit is the twist-free cat's move. Needs no redesign. *Proved?* Seen in runs (CAT_TWIST 20261017_231928).

341. CAT_TWIST's (L2-r), the rival: no constant bend from 90 to 135 degrees lets a twist-free cat turn over in one full circuit. Measured: at 90 the paws end 28.11 degrees from straight down and at 95 39.81, both within 40: MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* A twist-free full circuit turns the cat most near a bend of 75 degrees (169.72 degrees, paws 10.28 from straight down) [Corrected 18 October, by CAT_SWEET_READER 20261018_041321: the turn was read from 0 to 180, which cannot tell a cat that turns too little from one that turns too far. Read unfolded, one twist-free circuit turns the cat further the deeper it bends, at every radius tried: at radius 0.3, 82.4 degrees at a bend of 45, 169.7 at 75, exactly a half turn at 78.8, 208.1 at 90 and 296.0 at 135. Near 75 is where the 15-degree grid came nearest a half turn, not where the circuit turns most; at 90 it turns 28.1 degrees too far, not short.], Kane and Scher's regime: their cat carries its bend a full circuit. Needs no redesign. *Proved?* Seen in runs (CAT_TWIST 20261017_231928).

342. SLING_RUNDOWN's (S2-r), the rival: the best cut passes 45 degrees somewhere below a swing of 90. Measured: at most 38.854 degrees, at 90: MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* The seat flies from above the ground and slows as it rises, so the best cut stays below the textbook 45 and levels off near 39 degrees. Needs no redesign. *Proved?* Seen in runs (SLING_RUNDOWN 20261017_232234).

343. SNAKE_LONGEST's (X5-r), SNAKE_LONG's longer body, the rival: the gap below 2 pi flattens at 1,536 links, its log slope above -0.43. Measured: -0.4665: MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* The slope keeps steepening toward one half, -0.38, -0.41, -0.43, -0.45 and now -0.47 a doubling, so the motor snake keeps closing on 2 pi like one over the root of its length. Needs no redesign. *Proved?* Seen in runs (SNAKE_LONGEST 20261017_155052, the second computer).

344. EXPLORER_CAT_TWISTFREE_CHECK's (Q1-r), the rival: the bend whose twist-free full circuit lands the paws nearest straight down lies outside 70 to 80 degrees. Measured: 79 degrees, the paws 0.4310 from straight down: MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* One circuit turns the twist-free cat most at a middling bend [corrected 18 October, by CAT_SWEET_READER 20261018_041321: the deeper the bend, the further one circuit turns it; a middling bend is where it turns exactly a half turn]; at radius 0.3 it comes within half a degree of a perfect half turn at 79. Needs no redesign; the sweet spot at other radii is the proposed list's (25). *Proved?* Seen in runs (EXPLORER_CAT_TWISTFREE_CHECK 20261018_031532).

345. EXPLORER_CAT_TWISTFREE_CHECK's (W7-r), the rival: a twist-free full-circuit fall at 3,000 steps takes over 150 ms. Measured: a median of 16.94 ms: MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* The switch changes one sign and adds no work. Needs no redesign. *Proved?* Seen in runs (EXPLORER_CAT_TWISTFREE_CHECK 20261018_031532).

346. SAGNAC_LOOP's (G1-r), the rival: a square, a triangle and a 1-by-9 rectangle of the same area give delays differing by more than 1 per cent. Measured: at most 5.9e-10 apart: MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* The delay counts only the area the light's path encloses, in the plane square to the turning. Needs no redesign. *Proved?* Seen in runs (SAGNAC_LOOP 20261018_032138).

347. TRAIN_LIFETIME's (T1): the train of three rings has a sharp cliff, every start from 0.600 to 0.650 holding for 2,000 time units and every start from 0.720 to 0.950 breaking. Measured: the second half held, but of the eleven starts from 0.600 to 0.650 only 0.600, 0.620 and 0.630 held; the other eight broke, five of them after 1,000 time units (0.650 at 1010.7): MISSED.

Kind: WRONG GUESS, of a shape, and INSTRUMENT (an earlier tool's window read as a cliff). *Lesson:* TRAIN_HOLD flew each train 1,000 time units and called a train that had not broken by then one that holds; its cliff at 0.65 to 0.70 was where the lifetimes outgrew that window. Counted fairly with the starts still whole, a train started 0.600 to 0.695 apart lives about 862 time units, one started wider 127 to 224: a lifetime that lengthens sharply below 0.70, not a cliff. TRAIN_HOLD's records corrected the same batch. Redesign: the lifetime against the spacing below 0.70, each start flown until it breaks or for ten times the band's lifetime (proposed list). *Proved?* Seen in runs (TRAIN_LIFETIME 20261018_034211; TRAIN_LIFETIME_READER 20261018_035727).

348. TRAIN_LIFETIME's (T1-r), the rival: some start at 0.750 or above holds for 2,000 time units. Measured: all 41 broke, the latest at 985.7 (gap 0.825): MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* Above 0.70 every train came apart well inside the window, its lifetime 127 to 224 time units by band. Needs no redesign. *Proved?* Seen in runs (TRAIN_LIFETIME 20261018_034211).

349. TRAIN_LIFETIME's (T3-r), the rival: the break times beyond the shortest are far from an exponential, p below 0.01. Measured: p 0.1262 as registered, and 0.0275 read fairly: MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* The times are close to memoryless but not one exponential: the fair p falls between the lean's 0.05 and the rival's 0.01, and the reason is the mix of two lifetimes (row 350). Needs no redesign. *Proved?* Seen in runs (TRAIN_LIFETIME 20261018_034211; TRAIN_LIFETIME_READER 20261018_035727).

350. TRAIN_LIFETIME's (T3), held by its letter: the break times beyond the shortest fit an exponential, Kolmogorov-Smirnov p above 0.05. Measured: p 0.1262: HELD, on a test too kind.

Kind: INSTRUMENT (a test that flatters its own fit). *Lesson:* The test compared the times with an exponential whose mean was fitted from the same times, which brings any sample closer to its curve than chance would (Lilliefors 1969). Repeating the whole procedure on 20,000 true exponential samples of the same size gives the fair p, 0.0275, below the lean's 0.05. The pooled times mix a long lifetime below a spacing of 0.70 with a short one above, which no single exponential fits. The verdict stands as registered and is told with the fair p beside it. Fix: a fit's p is taken from the fitted procedure run on samples drawn from the fit, and a lifetime is read band by band. *Proved?* Seen in runs (TRAIN_LIFETIME_READER 20261018_035727).

Misses 351 to 400

351. CAT_SWEET's (K1-r), the rival: at radius 0.3 the nearest a twist-free full circuit brings the paws to straight down, at bends from 75 to 85, stays more than 0.1 degrees away. Measured: 2.77e-07 degrees at a bend of 78.83660617, the body turned 179.999999723: MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* The 1-degree grid's 0.43 was the grid's coarseness: the twist-free cat has an exact sweet spot, one bend at which one circuit turns it precisely over. Needs no redesign. *Proved?* Seen in runs (CAT_SWEET 20261018_034953).

352. CAT_SWEET's (K2-r), the rival: at radius 0.15 no bend from 45 to 135 lands the twist-free cat on its feet with one circuit. Measured: on its feet from 45 to 51 degrees, 14.66 from straight down at 45: MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* The thin cat's 122.8 degrees at a bend of 90 was not a cat that turns too little but one that turns too far, 302.8 degrees read unfolded: its best bend lies at the shallow end, 45 or below, and a thicker cat needs a deeper fold (78.8 at 0.3, 105.9 at 0.45). Needs no redesign; the thin cat below 45 degrees is outside the room's rule and on the proposed list. *Proved?* Seen in runs (CAT_SWEET 20261018_034953; CAT_SWEET_READER 20261018_041321).

353. CAT_SWEET's (U1-r), the rival: with halves of radii 0.25 and 0.35 the toy's half trick still turns the body a half turn within 1e-6. Measured: 156.109449 degrees at bend 90, 23.89 short, tilted -27.42: MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* The toy's exact half turn came from the two halves being the same, not from its waist's odd loop: make them differ and it falls short at every bend, by 44.85 at 45 to 10.30 at 135. Swapping the halves changes nothing printed. Needs no redesign. *Proved?* Seen in runs (CAT_SWEET 20261018_034953).

354. CAT_SWEET's (U2-r), the rival: with the same unequal halves the twist-free full circuit at bend 90 still lands level within 1e-6. Measured: tilted +2.122894 degrees: MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* Level landings were the equal halves' symmetry too; unequal, the twist-free cat lands tilted, from -0.38 degrees at bend 45 to +37.09 at 135. Needs no redesign. *Proved?* Seen in runs (CAT_SWEET 20261018_034953).

355. CAT_MIRROR's (Z1-r), the rival: the thick toy (radius 0.3) at bend b and the thin twist-free cat (0.15) at 180 - b part by more than 1e-6 degrees in their turn somewhere on a 1-degree grid from 45 to 135. Measured: at most 5.4e-13 apart: MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* The pairing seen at seven bends is exact at every bend: the toy pitches by the angle the thin twist-free cat rolls. Needs no redesign; why is the proposed list's (31). *Proved?* Seen in runs (CAT_MIRROR 20261018_041123).

356. CAT_MIRROR's (Z2-r), the rival: no twist-free radius matched to a toy at bend 90 matches it at 60 against 120 and 120 against 60 as well. Measured: for toys of 0.2 and 0.45 the radii 0.1 and 0.225 match at all three within 1.2e-11: MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* The pairing is a law of thickness, not a coincidence of one pair of radii; a second root at 0.574 for the 0.45 toy matches at 90 alone. Needs no redesign. *Proved?* Seen in runs (CAT_MIRROR 20261018_041123).

357. CAT_MIRROR's (Z3-r), the rival: the matching twist-free radius is off half the toy's by more than 1e-3. Measured: exactly half to ten places, at 0.2 and 0.45: MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* Halving the thickness quarters the halves' spin about their own lengths; why that, with the bend mirrored, undoes the twist is not yet known. Needs no redesign. *Proved?* Seen in runs (CAT_MIRROR 20261018_041123).

358. COIL_PUMP's (P1), Claude's lean (odds 9 in 10): a loop in the coil's radius and climb, a quarter cycle apart, moves it more than 1e-6 of its length per cycle. Measured: 5.4e-16 radii a cycle, 7.2e-18 of its length, at every loop size and both grips: MISSED.

Kind: WRONG GUESS, and a symmetry not looked for before registering. *Lesson:* A uniform coil is the same turned end over end about its middle, and an even squeeze about that middle keeps the symmetry while a slide or a spin would break it: the coil's middle can do neither, at any instant. The two-knob loop that beats the scallop theorem for a hinged swimmer is not enough for a symmetric body. Redesign, said aloud and run at once: a wave along the coil (COIL_WAVE, which swims). *Proved?* Seen in runs (COIL_PUMP 20261018_082424).

359. COIL_PUMP's (P2), Claude's lean (odds 3 in 4): halving the loop quarters the travel within 2 per cent. Measured: a ratio of 1.66 between two roundings: MISSED.

Kind: WRONG GUESS, following row 358. *Lesson:* With no travel there is no area law to read. The area law is tested on the wave instead, as the square of the wave's height (COIL_WAVE's (W2), held). Covered by row 358's redesign. *Proved?* Seen in runs (COIL_PUMP 20261018_082424).

360. TRAIN_TIMING's (D1-r), the rival: a ring train's dense output costs less than 5 per cent. Measured: 18.1 per cent, its three extra evaluations a step: MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* The output is not free, but it is not the factor of two told earlier that day either. Needs no redesign. *Proved?* Seen in runs (TRAIN_TIMING 20261018_082139).

361. TRAIN_TIMING's (D2), the long shot (Claude's odds 1 in 4): the dense output is the whole of the factor told earlier that day, the ratio above 2. Measured: 1.181: MISSED.

Kind: WRONG GUESS (the long shot's), and INSTRUMENT (a stopwatch on a shared machine). *Lesson:* The earlier figures, 15 s against 35, compared flights on a machine shared with other runs; alone, the train flies 11.5 s per 1,000 time units without dense output and 13.6 with it. That surprise was told as found and is withdrawn. Fix: time a tool alone before blaming any of its options. *Proved?* Seen in runs (TRAIN_TIMING 20261018_082139).

362. COIL_WAVE's (W4), the long shot (Claude's odds 1 in 3): the pumping coil swims the way its wave runs. Measured: -4.63e-03 radii a cycle, against its wave: MISSED.

Kind: WRONG GUESS (the long shot's). *Lesson:* It swims against its wave, as Taylor's swimming sheet is recalled to (names owed to the prior-art pass), by about the same distance for each crest that passes. Needs no redesign. *Proved?* Seen in runs (COIL_WAVE 20261018_082544).

363. CAT_MIRROR_PROOF's (X2), the long shot (Claude's odds 1 in 3): the mirror pairing survives unequal halves with both halved (toy 0.25 and 0.35 against twist-free 0.125 and 0.175). Measured: mismatched by up to 0.178 in the starting rate, angles 74.45 against 58.60 degrees at bend 90: MISSED.

Kind: WRONG GUESS (the long shot's). *Lesson:* The pairing needs the two halves alike, as the toy's exact half turn did (CAT_SWEET): both are gifts of the equal halves. Whether some other map pairs unequal halves is not asked yet; on the proposed list. *Proved?* Seen in runs (CAT_MIRROR_PROOF 20261018_174959) [Redesign check said aloud 20 October, owed since 18 October: (X2) tried one map, both halves halved, as (X3) did, and a miss of one map is not a miss of every map. In Montgomery's picture unequal halves leave his symmetric frame, so the twist-free cat's turning is no longer about one axis (CAT_SWEET's tilted landing, its U2), and a single thickness map is unlikely; whether some map pairs unequal halves is untried, put on the proposed list rather than a new card.]

364. CAT_MIRROR_PROOF's (X3), the second long shot (Claude's odds 1 in 5): with the hinge 0.3 of a length from each tube's centre, some other thickness map still pairs the cats. Measured: the two radii that pair a toy of 0.3 at bend 90 (0.2080 and 0.6283) miss at 60 and 120 by up to 118.4 degrees: MISSED.

Kind: WRONG GUESS (the long shot's). *Lesson:* The pairing needs the hinge at the tubes' ends, where the closed form cancels for every length, mass and radius. Needs no redesign. *Proved?* Seen in runs (CAT_MIRROR_PROOF 20261018_174959) [Corrected 20 October: not a wrong guess but a tool fault (row 419). The tool read its twist-free cat at the toy's own bend, so at 60 and 120 it compared unlike bends; asked rightly, the matched radius 0.3 / sqrt(2.08) pairs the cats within 4.5e-13 degrees (CAT_MONTGOMERY 20261020_035025). 'The pairing needs the hinge at the tubes' ends' holds only for the halving.]

365. The book's own wording, read whole by six readers for the headline harvest (18 October): twenty-two sentences printed a reading the runs do not carry. (a) the fibre coil's builder numbers called 5 and 10 cm 'across' where the run's table is by radius, four times too large as written; (b) the steering swimmer's path printed as 0.008 a lap and a circle 0.22 'across' where the run gives 0.00847 and a turning radius of 0.220; (c) its prediction printed as 5.7 times 0.38 (2.17) for the run's 5.733 times 0.3828 (2.19); (d) the breath-holding cap's six cycles called all inside a floor of 0.085 with one at -0.089; (e) the jellyfish's power factor of -0.083 read as 'ninety-nine per cent comes straight back', two measures mixed (twelve returned for each one spent); (f) the snake's exact zero at an even grip given the scallop theorem's name, where it is the even-drag wall of slender swimming (any stroke at all), chapter 3 and its answer page; (g) the three-ring train said to hold 'for as long as followed' after TRAIN_LIFETIME broke trains near two thirds past 1,000, with its 'still owed' line out of date; (h) the hinge count's travel per hinge said to peak at five, half a per cent over seven with four counts untried (also the toy runs table, row 5k); (i) the long snake's limit stated as 'never zero, and never stuck' while its slopes still steepen; (j) the sling's angle form told without saying it was restated after the registered miss; (k) Tong and Dullin's rebuild said to come back 'within its print' where the gaps are 0.02 to 0.15 per cent inside the registered bars (also the data ledger, row 314 of these misses, the Field Guide's lessons and two sets of working notes); (l) chapter 10's 'the guess is not yet settled' left standing after WINDOW_TOUCH_5 settled it, and 'windows no one had seen' claimed beyond this needle; (m) Bear's strokes called the fold's images where 0.741 and 0.673 of them lie on them; (n) 'the switch is exact' at a hundredth of a window's width; (o) GEAR_TRAIN's 'never stepping back' beside STAIR_WIDTHS's five steps back; (p) a teaching-room line still describing the twisting cat after the room was rebuilt on the twist-free one; (q) the relock line printed from AVALANCHE's ramp (0.77, 0.53) where the still-standing line is 0.79 and 0.54 (SEAM_MAP_B; also the ledger's row); (r) a column growth of 1.58 a walker, 'about a million over thirty', which is the daytime column's closed-form gain (COLUMN_STEADY) printed for the night read

[the reason first given here, ‘names no run’, corrected the same day]; (s) the wave-pumped coil’s travel at grip 6 said to follow the turning coil’s ‘grip ratio less one’, where the turning coil’s own closed form carries that factor over a growing denominator (also the data ledger’s row for COIL_WAVE).

Kind: WRONG READING (the book’s prose, not the runs). *Lesson:* Every one is corrected in the same batch in each record that carried it, the corrections marked in place; the runs were not changed and none needed rerunning. Three items were checked and found right: the shared stamps 20261016_203014 and 20261016_213858 are two tools launched in the same second, each with its own blind run; PENTA_SHARE’s 060159 and 060200 are its two parts; the spaced column’s 15 at twice the delay is the corrected count. Two are open: chapter 10’s ‘seventy times lighter’ squeeze (a rounding mismatch, about 66 by the drag alone), and the units of heat’s speed limit (a spot’s width?). The lesson: a number can be right in its run and wrong on the page; the reader who checks the page against the run is a different instrument from the one who checks the run. *Proved?* Seen in the readers’ reports and the runs named.

366. TRAIN_CLOSE’s M0 (planted): the breaks at 0.61, 0.65 and 0.70 repeat TRAIN_LIFETIME’s within 1e-6. Measured on the second computer: 4493.519, 241.102 and 272.393 against 1022.069, 1010.664 and 302.609: MISSED; the same tool on the same files, md5 for md5, repeated them exactly here (blind run 20261018_072408, apart 0).

Kind: THE INSTRUMENT (a planted bar that asked a chaotic event to repeat across machines) and NATURE (chaos). *Lesson:* A train near its edge breaks at a time set by the last bits of the arithmetic: two machines with different numerical libraries throw different dice. Can this be redesigned? Yes, said aloud and run at once as TRAIN_NUDGE: with the middle ring started 1e-15 of a radius further along, all three breaks moved by 28 to 84 per cent of their own size, and 1e-12 moved them 30 to 86 per cent (N1 and N2 held). So the miss is nature’s, and the planted check was the fault: a planted check must compare what chaos cannot scramble (the energy and impulse a train keeps, M1, which held) or the statistics of many breaks, never one break time across machines. Every single break time in this chapter is one throw, and the book now says so. *Proved?* Seen in runs (TRAIN_CLOSE 20261018_103739; TRAIN_NUDGE 20261018_192138).

367. TRAIN_CLOSE’s (C1-r), the rival: some start from 0.25 to 0.40 breaks within 20,000 time units. Measured: all 16 held: MISSED.

Kind: WRONG GUESS (the rival’s). *Lesson:* The tight trains are long-lived: 0.25 to 0.56 held all 32 starts for 20,000 time units, 640,000 time units of flight without a break. Needs no redesign. *Proved?* Seen in runs (TRAIN_CLOSE 20261018_103739).

368. TRAIN_CLOSE’s (C2-r), the rival: every start from 0.41 to 0.59 holds for 20,000. Measured: 0.57, 0.58 and 0.59 broke at 1119.5, 5954.1 and 2650.6: MISSED.

Kind: WRONG GUESS (the rival’s). *Lesson:* The breaking reaches down to 0.57 and no further on a grid of 0.01 [corrected 19 October: one start a gap; five nudged copies a gap broke down to 0.560, three of five there (TRAIN_EDGE 20261018_181641; row 381)]: an edge, though a soft one, its lifetime tens of thousands of time units just outside it [corrected 21 October: five copies a gap below 0.56 broke down to 0.530, none of thirty from 0.500 to 0.525, and the lifetime is about 10,500 at 0.550 to 0.555 and 48,000 at 0.530 to 0.545 (TRAIN_FLOOR 20261020_224239)]. Needs no redesign; where between 0.56 and 0.57 the edge lies, and whether trains just inside it would break if flown longer still, is on the proposed list. *Proved?* Seen in runs (TRAIN_CLOSE 20261018_103739).

369. TRAIN_CLOSE’s (C3-r), the rival: the three starts that held TRAIN_LIFETIME’s 2,000 (0.60, 0.62, 0.63) all hold for 20,000. Measured: all three broke, at 759.7, 2292.1 and 2045.9: MISSED.

Kind: WRONG GUESS (the rival’s). *Lesson:* Holding 2,000 was one throw of the dice, not a property of the spacing (TRAIN_NUDGE). Needs no redesign. *Proved?* Seen in runs (TRAIN_CLOSE 20261018_103739).

370. TRAIN_NUDGE’s (N3), the long shot (Claude’s odds 1 in 3): at a nudge of 1e-15 the break at 0.70, the widest and quickest train, moves by less than 1 per cent. Measured: 555.431 against 302.609, +84 per cent: MISSED.

Kind: WRONG GUESS (the long shot’s). *Lesson:* Even the quickest break is a throw: the chaos has time to grow from the last bit in a few hundred time units. Needs no redesign. *Proved?* Seen in runs (TRAIN_NUDGE 20261018_192138).

371. The book’s numbers, checked against their runs by an independent reader of the headline register (18 October): eleven sentences printed a number their runs do not carry. The area rule’s 10-degree travel as 0.00195 and its step as ‘four times again’ (the run: 0.00196, and 3.92); the latch’s 1.37 at an edge of one and a half, in chapter 11 and the knob register (1.36); the passing lane’s ‘within 3%’ (3.2 per cent); the square rule ‘overshoots by 25’ (by a third; the measured travel is 25 per cent short of it); the wave-ride’s ‘five bodies’ at 1.000006 (five bodies give 1.000003; 1.000006 is a motor-driven snake); a look’s cost ‘within 4 per cent over twenty settings’ (a slope of 1.04, 0.84 to 1.08 pack by pack); the hidden loop’s law ‘at every one of 112 angles’ (111 of 112); the cone’s wedge ‘to fifteen figures’ (8.4e-15, fourteen); the 9/4 law ‘within half a per cent’ at eight weights (0.6); the cascade’s forecasts ‘within 2.5 billionths’ (the largest 2.54e-9); Kramers’ rate ‘wherever the barrier is at least four noise units high’ (four barriers; a later sweep from 4 to 9 units read 0.79 to 1.08).

Kind: WRONG READING (the book's prose). *Lesson:* Each corrected in place, in every record that carried it. The checker also corrected the reason row 365 gave for withdrawing the column's 1.58 a walker: it is the daytime column's closed-form gain, printed for the night read. The runs were not changed. The lesson, again: a number is checked against its run, never against the sentence that first carried it. *Proved?* Seen in the checker's report and the runs named.

372. The headline register's own counts (18 October): its first script read 783 graded predictions with 573 held, its second 667 with 469 held, and both were short. They read only labels like 'W1' or '(K3)' and so skipped every verdict labelled '(a)', '(b)', 'CH-1' or 'K3_8/5', and a rival printed twice in one run, as '(C1-r)' and 'C1r', was counted twice. Read with every label form, the register of 18 October held 941 graded predictions, 640 held, 275 missed, 26 undecided, with 85 rivals, 77 missed. The register's first copy also stated about fifty numbers more strongly than their runs allow or cited them to the wrong run.

Kind: THE INSTRUMENT (the counting script; Claude's own). *Lesson:* The register set out to stop the hits being shrunk, and its own first count shrank them by a third. Redesign, said aloud and done: an independent checker that had not seen the page went through every number against its run, and the parser now reads every label form; its count of all non-planted labels, 959, sits within five of the checker's own count, 964, the rest being planted-check lines that carry no verdict of their own. Every count in the register is printed by the script, and a count that moves is corrected in every record that carried it. *Proved?* Seen in the checker's report and the run files.

373. SAIL's (S1): on some heading the toy boat sails faster than the true wind. Measured: the fastest 1.6576 m/s at 88 degrees, 0.3315 of the wind: MISSED.

Kind: WRONG GUESS, from an assumed number (the hull). *Lesson:* The assumed hull drag area, 0.15 square metres, is a barge's, about twenty-five times a clean dinghy's; by the course theorem the drag angles set the whole polar. Can this be redesigned? Yes, said aloud and run at once as SAIL_HULL: with a clean hull of 0.006 square metres the same boat sails 1.1483 times the wind at 84 degrees, and it first beats the wind between 0.02 and 0.01 (row 376's card). *Proved?* Seen in runs (SAIL 20261018_202451; SAIL_HULL 20261018_202650).

374. SAIL's (S2): the fastest heading lies between 90 and 135 degrees off the true wind. Measured: 88 degrees: MISSED, narrowly.

Kind: WRONG GUESS. *Lesson:* A boat trimmed for speed on this low sail is fastest just short of a beam reach; past the beam its sail stalls and its side force falls to windward (59 headings of 151). Covered by SAIL_HULL, where the fastest heading stays at 83 to 88 degrees for every hull. *Proved?* Seen in runs (SAIL 20261018_202451).

375. SAIL's (S4), the long shot (Claude's odds 1 in 4): the best progress straight downwind beats the true wind. Measured: dead downwind is best, 1.1242 m/s, 0.2248 of the wind: MISSED.

Kind: WRONG GUESS (the long shot's). *Lesson:* The barge's hull and the sail's stall together; SAIL_HULL asks the hull's part. *Proved?* Seen in runs (SAIL 20261018_202451).

376. SAIL_HULL's (H2): the cleaner the hull, the further off the wind the fastest heading lies, at every step. Measured: 88, 86, 84, 83, 83, 83, 84, 84 and 85 degrees from the barge to the cleanest: MISSED.

Kind: WRONG GUESS. *Lesson:* The fastest heading is pinned near the beam by the sail, not the hull: past the beam the speed-trimmed sail stalls, whatever the hull. Needs no redesign; the sail's own knob, its aspect ratio, is on the proposed list. *Proved?* Seen in runs (SAIL_HULL 20261018_202650).

377. SAIL_HULL's (H3), the long shot (Claude's odds 1 in 4): at the cleanest hull, 0.001, the best progress downwind beats the true wind. Measured: 0.6322 of the wind at 134 degrees: MISSED.

Kind: WRONG GUESS (the long shot's). *Lesson:* The sail's own drag angle caps it: at aspect ratio 3 its best lift is about five times its drag, and a sail that stalls past the beam cannot make good downwind faster than the wind however clean the hull. A higher-aspect sail, the way ice boats and foiling catamarans outrun the wind downwind, is the next knob, on the proposed list. *Proved?* Seen in runs (SAIL_HULL 20261018_202650).

378. Claude's own readings, caught on 18 October. (a) The sail's price dead downwind, 0.2000 newtons a watt, exactly one over the wind's speed, was printed in the book, the data ledger and the register as chapter 8's ride law met, 'as the floating snake does'. It is the price's own bookkeeping: dead downwind the sail's whole force lies along the wind and the course alike, so any boat prints it. What does carry is the general form: the watts the boat draws from the wind's motion over the water are its push's own work plus the sail's drag loss, so the price times the boat's speed is the share of the drawn power that moves the boat, never above one; dead downwind that share is the boat's speed over the wind's, which is why a ride on drag alone cannot beat the wind (SAIL's M1). Read from the recorded polar, the share peaks at 0.62, at 109 degrees; derived that day, not yet registered. (b) The checker's correction of the front page's 'fifteen figures' to fourteen (row 371) had not reached the working notes' list of what was new, or a no-physics read's fifteen places after the point.

Kind: WRONG READING (Claude's own). *Lesson:* Each corrected in place in every record that carried it, in the same batch; the no-physics read keeps its words with a dated note, as the reads do. The runs were not changed. The test that would land (a) is owed a registration: the share at every heading of SAIL_HULL's nine hulls, never above one, climbing as the sail's lift over drag grows. *Proved?* Seen in the run (SAIL 20261018_202451) and the records named.

379. The headline register's front page, as assembled on 18 October: 'They answered it with 453 small model tools and 483 runs of record', '(114 blind runs)', 'of 483 runs of record' and, in Part C, '196 of 222 questions answered'. Counted from the crate later that day by the definitions that give exactly 453, 483 and 114 in the crate as it stood when those numbers were typed in (top-level LOOP_ATLAS_*.py files; distinct tool and stamp among run files neither blind nor smoke; run files marked blind): 457 tools, 487 runs of record and 118 blind runs, and the book's count line says 200 of 226. The numbers had been typed into the register's generator earlier that day, and the page went stale a little later the same day while every count computed beside them moved. A second fault of the same kind was found with it: the generator, its source and the harvest's scripts lived only in a scratch folder of the working session, so a move to a new session would have left that session able to read the register but not rebuild it.

Kind: THE INSTRUMENT (the register's own generator; Claude's). *Lesson:* Redesigned and done the same batch: the generator, its source and the harvest's and the checker's scripts moved into the crate beside the register; run from there it repeated the register byte for byte before any change (md5 the same); the tools, runs, blind runs, cards and the benchmark, found-here and rule counts are now counted from the crate and the source at every assembly, the headline's 'forty-one' among them. The lesson is the register's own: a count typed in is a count that stops moving. *Proved?* Seen in the crate's listing and its file lists of that day.

380. TRAIN_EDGE's (E4), the long shot (Claude's odds 1 in 4): the forty throws at the gap 0.65, counted with any still whole weighed in, live within 20 per cent of 862.03, the lifetime of the band 0.600 to 0.695. Measured on the second computer: 675.7, 0.784 of it: MISSED.

Kind: WRONG GUESS (the long shot's). *Lesson:* The band's 862.03 mixes the long lives of its closest starts (0.600, 0.620 and 0.630 held the 2,000 first flown) with the short lives of its widest, so one spacing in its middle should live less than the band's average, and the bar measured the band's mixing rather than the dice. Can this be redesigned? Yes, as a reading of files already in the crate: the forty throws against the lifetime of the starts nearest 0.65 in TRAIN_LIFETIME and TRAIN_CLOSE, and lifetime against spacing inside the band; on the proposed list, not a new card. *Proved?* Seen in runs (TRAIN_EDGE 20261018_181641).

381. Claude's reading of TRAIN_CLOSE, carried in the register's chapter 9 headline ('a train packed at 0.56 radii or tighter holds'), the book's chapter 9 ('The breaking reaches down to 0.57 and, on this grid, no further') and misses row 368: a single unnudged start at each gap from 0.25 to 0.56 held 20,000 time units, and that was read as the edge. Measured on the second computer with five nudged copies a gap: at 0.560 three of five broke, at 1600.385, 6756.066 and 18348.513, and 48 of the 55 copies from 0.560 to 0.570 broke, the seven that held scattered across the band: the edge reaches at least to 0.560.

Kind: WRONG GUESS (a reading of one throw a gap). *Lesson:* One throw a gap cannot place a soft edge; only copies can. Corrected in every record that carried it, citing TRAIN_EDGE 20261018_181641: the register's Part C and D1, the book's chapter 9 (its paragraph, its builder's step 1 and its owed list) and row 368. Can this be redesigned? Yes, registered and sent to the second computer as TRAIN_FLOOR: five copies a gap from 0.500 to 0.555. *Proved?* Seen in runs (TRAIN_EDGE 20261018_181641; TRAIN_CLOSE 20261018_103739).

382. TRAIN_EDGE's (E2) (Claude's lean, odds 2 in 3): more copies break in the band's upper half (0.566 to 0.570) than in its lower (0.560 to 0.564). Measured: 23 of 25 against 21 of 25: HELD. But with every copy breaking at the pooled share, 48 of 55, a difference at least that large comes 0.41 of the time (computed from the json on 19 October): the verdict stands and says almost nothing.

Kind: THE INSTRUMENT (a lean about a difference of counts registered without its no-trend chance, and without the planted step and band tests the practice asks of a lean about a count). *Lesson:* No judging line changes, and it needs no redesign of its own: the practice now prints the no-trend chance beside every such split, and TRAIN_FLOOR carries it with a planted step test and a planted band test (its M3 and M4), both held blind. *Proved?* Seen in runs (TRAIN_EDGE 20261018_181641).

383. The book's chapter 9 (its paragraph 'Flown ten times longer, and thrown twice'): 'the trains at 0.57, 0.58 and 0.59 lasted 1,119, 5,954 and 2,651 time units, a lifetime near 43,000 for that band once the trains in it that held are weighed in'. None of those three held. The run's own line is 'band 0.51 to 0.59: 9 starts, 3 broke; the lifetime counted with the starts still whole 43241.4', six of its nine starts holding.

Kind: THE INSTRUMENT (a misreading: a run's band renamed in the telling). *Lesson:* Corrected in the book in place, marked, citing the run file; found by reading TRAIN_CLOSE's run file again while judging TRAIN_EDGE. Needs no redesign. *Proved?* Seen in runs (TRAIN_CLOSE 20261018_103739).

384. The headline register's methods section, as assembled on 18 and 19 October: 'every tool since the blind-run practice began ran blind first (114 blind runs)' and 'The 370 rows of the misses file', both typed into its source on 18 October. Counted from the crate on 19 October: 119 blind runs and 384 rows. The fix of row 379 computed the front page's counts at assembly but did not reach this section.

Kind: THE INSTRUMENT (a count typed in, the second found). *Lesson:* Redesigned and done the same batch: the assembler now fills both from the crate at every assembly, each replacement asserted to occur once; found while reading the rebuilt register for the TRAIN_EDGE batch. The lesson of row 379 again: search the whole of a record for every count it prints, not only the page where the first was found. *Proved?* Seen in the register (the headline register).

385. The PDF of a working note on reading patterns has read back 84 letters and digits short at every build since 13 October ('missing before 84, after 84' in V13_FIDELITY 20261013_204846), and nobody chased it. Counted by character on 19 October: t 42, e 21 and x 21, the word 'text' 21 times, the language label of the file's 21 fenced blocks, which a PDF never prints. Nothing is lost.

Kind: THE INSTRUMENT (the read-back counts a fence's label as text). *Lesson:* Needs no redesign of the file. The read-back's counter should skip a fenced block's label; that belongs to the separate publishing work, whose build and check scripts are its own. Until then, a 'short' equal to four times a file's number of fences labelled 'text' is that and nothing else. Found by the backward pass of the closeout report for the Field Guide. *Proved?* Seen in runs (V13_FIDELITY 20261013_204846); rebuilt in a scratch folder on 19 October, short 84 again.

386. The headline register's lede for chapter 5 (Part C2, 18 October): 'a registered prediction that held at three sizes'. The run's record and the register's own Part A and D1 say the rule, the lap's radius over √2, was read off one lap size after a first guess failed, and then held at the two others tried: two sizes tested, not three.

Kind: WRONG GUESS, of a size (an overstatement made in the telling). *Lesson:* Found when the book's 23 headlines were checked number by number against their runs before they were set (19 October); corrected in the register's source and in the book's chapter 5 headline, which was set from the corrected words. Needs no redesign. The check that found it, every number and claim in a headline read again from its run before it is set, is the practice from now on. *Proved?* Seen in runs (S_REVERSAL_SPIKE 20261003_213523).

387. The headline register's count of rivals, as assembled from 18 October: '88 rivals, written to miss if the prediction was right, and 80 missed', on the front page and in Part B. Read one by one on 19 October, the 88 lines are 60 distinct rivals: 28 lines are the explorer's checks of its own engine, counted again at every fidelity rerun. And of the 60, only 8 are competing ideas (2 held); 41 are the bare negation of their prediction (6 held) and 11 are checks of a tool (none held). By day, the distinct rivals about the toys held 5 of 12 on 16 October, 3 of 23 on the 17th and none of 14 on the 18th, and the competing ideas among them fell from 5 to 3 to none: the rivals grew tame because they stopped being rivals.

Kind: THE INSTRUMENT (a count that took repeats as new) and a WRONG GUESS of the practice (rivals written as negations). *Lesson:* Redesigned and done the same batch: the assembler counts each rival once and by kind, from a file of the rivals marked as competing ideas (kept beside the register's scripts, every other one a negation until marked), and the front page's sentence now says what the rivals tested. Every rival from TRAIN_FLOOR on is written as the strongest competing idea, with odds (TRAIN_FLOOR's comb, TRAIN_DRIFT's close passes). *Proved?* Seen in the register's ledger (the headline register and the source of its Part E).

388. KNEE_SIZE's (K1r), the rival, the strongest competing idea (Claude's odds 1 in 4): the knee's creep is the counter's own stepping error, first order in the step, a power of 0.8 or more at noise time 1.0. Measured: 0.458: MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* The creep is the traders' square root, read where it lives: 0.458 at noise time 1.0 and 0.475 at 0.1, against the law's own 0.481 and 0.454. Needs no redesign. *Proved?* Seen in runs (KNEE_SIZE 20261019_035236).

389. KNEE_SIZE's (K2) (Claude's lean, odds 2 in 3): the law gives the knee's extra size within 15 per cent at every step at most a tenth of the memory, at noise times 1.0, 0.1 and 0.03. Measured: 11 of 13 cases within; at noise time 1.0 and steps 0.00125 and 0.0025 the extra size was 1.1967 and 1.1724 of the law: MISSED.

Kind: WRONG GUESS, or THE INSTRUMENT (the two misses sit where the extra size is one and two steps of the knee grid). *Lesson:* Can this be redesigned? Yes, said aloud and run at once as KNEE_FINE: the grid ten times finer again near knee 1.0 at the slowest noise, with the strongest competing idea as its rival, that the finest watcher's own bias is more than the law's shift for it. *Proved?* Seen in runs (KNEE_SIZE 20261019_035236).

390. KNEE_SIZE's (K3), the long shot (Claude's odds 1 in 5): at noise time 0.03 the law gives the extra size within 15 per cent even at step 0.01, a third of the memory. Measured: 1.1884: MISSED, narrowly.

Kind: WRONG GUESS (the long shot's). *Lesson:* The law's reach on the knee is about what it was on the flips: past a tenth of the memory the slow watcher misses more than the law says, growing with the step (1.06, 1.06, 1.11 and 1.19 of the law from 0.00125 to 0.01 at noise time 0.03). Needs no redesign; the reach is the finding. *Proved?* Seen in runs

(KNEE_SIZE 20261019_035236).

391. TRAIN_DRIFT's (D1r), the rival, the strongest competing idea (Claude's odds 1 in 5): the drift at tight packing is the rings' own close passes, so tightening the tolerances tenfold cuts it by less than 2 at 0.25. Measured: a cut of 14.58: MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* The drift is the integrator's: tighter tolerances buy it back, 14.58 times at 0.25 and 5.62 at 0.30, for 1.37 and 1.27 times the evaluations. Needs no redesign; chapter 9's fence now says so. *Proved?* Seen in runs (TRAIN_DRIFT 20261019_034330).

392. TRAIN_DRIFT's (D4), the long shot (Claude's odds 1 in 5): the tighter tolerances cut the drift at 0.25 by more than 30. Measured: 14.58: MISSED.

Kind: WRONG GUESS (the long shot's). *Lesson:* A tenfold tolerance bought back a tenfold-and-a-half drift at 0.25 and about half that at 0.30; the integrator's error per step follows its tolerance, not its order. Needs no redesign. *Proved?* Seen in runs (TRAIN_DRIFT 20261019_034330).

393. KNEE_FINE's (J1) (Claude's lean, odds 1 in 2): the excess of misses row 389 is the ruler's, so on a knee grid ten times finer the extra size at steps 0.00125 and 0.0025 lies within 15 per cent of the law. Measured: 1.1972 and 1.1758: MISSED; the rival held.

Kind: WRONG GUESS. *Lesson:* The excess is the law's at small shifts, an absolute offset of about 0.001 to 0.002 of strength that a ratio magnifies where the law's own shift is small. Can this be redesigned? Yes: correct the reference too, with a reference watcher four times finer, to see whether the offset moves with the reference; on the proposed list, since this chain has already had its two runs [the reading withdrawn the same day by KNEE_REF: the excess is inside the scatter, row 398]. *Proved?* Seen in runs (KNEE_FINE 20261019_040109; KNEE_SIZE 20261019_035236).

394. KNEE_FINE's (J2), the long shot (Claude's odds 1 in 5): at noise time 1.0 every step from 0.00125 to 0.04 lies within 5 per cent of the law on the finer grid. Measured: 1.1972, 1.1758, 1.0949, 1.0276, 1.0103 and 0.9857: MISSED.

Kind: WRONG GUESS (the long shot's). *Lesson:* Within 3 per cent from step 0.01 to 0.04, outside it below, where the same small offset is a large share of a small shift. Covered by row 393's redesign. *Proved?* Seen in runs (KNEE_FINE 20261019_040109).

395. KNEE_REF's M2 (planted): against the old reference the coarser knee grid of 0.001 gives KNEE_FINE's extra sizes (its grid 0.0005) within 5e-5 at every step from 0.00125 to 0.04. Measured: largest difference 3.65e-4 at 0.04, 2.3e-4 at 0.005, under 2e-5 at the other four: MISSED.

Kind: THE INSTRUMENT (a planted tolerance set for a smooth curve). *Lesson:* The tolerance was scaled from the curve's bend, the coarse grid's error falling as the square of the grid; but the flips are whole numbers, about twenty over the four soups to a knee step of 0.001, so the curve is a staircase and its grain, not its bend, sets the reader's error. The 3.65e-4 is a fifth to a third of the standard errors at those steps, so no verdict turns on it. Needs no redesign; a reader of counts states its grain beside its tolerance. *Proved?* Seen in runs (KNEE_REF 20261019_042520; KNEE_FINE 20261019_040109).

396. KNEE_REF's (R1r), the rival, the strongest competing idea (Claude's odds 2 in 5): KNEE_FINE's offset is real and the law's own, a term the traders' leading correction leaves out, so against both references the extra size at 0.00125 and 0.0025 stands two standard errors or more above the law. Measured: +1.17 and +1.20 against the old reference, -0.34 and +1.67 against the new: MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* Inside the scatter at both references: at the toy's precision the traders' leading correction is the whole of the knee's extra size for slow noise. Needs no redesign. *Proved?* Seen in runs (KNEE_REF 20261019_042520).

397. KNEE_REF's (R2), the long shot (Claude's odds 1 in 5): against the new reference every step from 0.000625 to 0.04 lies within 5 per cent of the law. Measured: 0.9877, 0.9761, 1.1053, 1.0248, 1.0221, 0.9933 and 0.9948: MISSED, at 0.0025 alone.

Kind: WRONG GUESS (the long shot's). *Lesson:* A band of 5 per cent at the small shifts is narrower than the scatter there (the standard error at 0.0025 is 6 per cent of the law's shift), and the miss is 1.67 standard errors. A bar written in per cent of a small number is checked against its error bar before it is registered. Needs no redesign. *Proved?* Seen in runs (KNEE_REF 20261019_042520).

398. Claude's reading of KNEE_FINE, carried in the book's chapter 11 ('So the excess is the law's, not the ruler's'), the card table, the register's D2 28, a no-physics read and misses row 393: the excess at the two smallest shifts of the slowest noise is the law's, an offset of about 0.001 to 0.002 of strength. With error bars, by leaving out each of the 256 noise paths in turn, the excess at 0.00125 and 0.0025 is 1.17 and 1.20 standard errors; against a reference four times finer every step lies within 1.67 of the law.

Kind: WRONG READING (Claude's own: a mean read without its error bar). *Lesson:* Corrected in place in the book, the card table, the register and the no-physics read, each marked. KNEE_FINE's (J1r) stays HELD as graded, since its registered bar was a ratio with no error bar; what is withdrawn is the reading of it. A finer ruler laid over the same noise repeats the same scatter, so a repeat on a finer grid is not a confirmation: put an error bar on an offset before naming its owner. *Proved?* Seen in runs (KNEE_REF 20261019_042520; KNEE_FINE 20261019_040109).

399. The data ledger's row for KNEE_FINE, as written on 19 October: inserted before the text that began the next row (a bar, then the words that named SAIL's chapter), it took that text's place, so SAIL's row lost its first cell and ran on in KNEE_FINE's line, and KNEE_FINE's DONE was then written at the end of that line, after SAIL's last cell.

Kind: THE INSTRUMENT (a records slip: a row inserted by anchor text). *Lesson:* Found while placing KNEE_REF's row, by its line of eleven fields where the ledger's rows have seven; split back by script, SAIL's row restored to the character from an earlier copy in the crate and KNEE_FINE's DONE put back on its own row. One shipment, built and never sent, carried the slip; the next did not. This batch's records were checked by counting every table's cells after writing, and the check joins the working rules. *Proved?* Seen in the records (the data ledger).

400. ERROR_BARS's (B1) (Claude's lean, odds 3 in 5): at most one headline claim besides KNEE_FINE's changes its wording once error bars are on. Measured: 6 change (C01, C02, C05, C06, C09, C17): MISSED; the rival held.

Kind: WRONG GUESS. *Lesson:* Five of the six are one slip: a 'within' bar or a range read off a mean and printed narrower than the run's own error bar (Kramers' 5 per cent against counting errors of 5 to 9; the traders' law's 2.5 under to 7.7 over and its 15 per cent at the smallest steps, where the error bar is 14 per cent; the knee's 11 of 13; the knee ear's 1.04 to 1.05, each +- 0.08). The sixth is row 402. Can this be redesigned? Yes, and done in the same batch: each wording set in place and marked; the rule from here is row 403's. *Proved?* Seen in runs (ERROR_BARS 20261019_162750).

Misses 401 to 450

401. ERROR_BARS's (B3), the long shot (Claude's odds 1 in 5): every noisy number in the register within two standard errors of the law or bar it is set against. Measured: Kramers' wider sweep 0.909 +- 0.011, 8 standard errors under 1: MISSED.

Kind: WRONG GUESS (the long shot's). *Lesson:* Covered by row 402's redesign. *Proved?* Seen in runs (ERROR_BARS 20261019_162750; SLIP_SENSE 20261011_060035).

402. Claude's reading of SLIP_SENSE, carried in the front page's item 7, D2's 2 and chapter 3: 'a later, wider sweep from 4 to 9 units read 0.79 to 1.08', set beside the four first barriers' agreement as scatter about Kramers' rate. Counted, its 15 judged rows hold 6,549 slips and pool to 0.909 +- 0.011; six rows lie two or more standard errors under 1, and the chi-square against 1 is 90.4 on 15 rows.

Kind: WRONG READING (Claude's own: a range read as scatter that is a lean). *Lesson:* Corrected in place and marked. The four first barriers pool to 0.983 +- 0.031, 2.2 standard errors above the sweep: either the leading formula's own correction (its prefactor's next term, of the order of one over the barrier in noise units) or a difference between the two runs (SLIP_SENSE's shorter runs, 500 time units a row, and its knee counter) sets the 9 per cent. Can this be redesigned? Yes: rerun the four first settings with SLIP_SENSE's function and run lengths, and the sweep's with NOVIKOV_NEEDLE's, to see which the 9 per cent follows; registered next, behind the book's spine at Bear's word that day. *Proved?* Seen in runs (ERROR_BARS 20261019_162750; SLIP_SENSE 20261011_060035; NOVIKOV_NEEDLE 20261004_152937) [Settled 20 October by KRAMERS_EXACT: against the exact rate, Stratonovich's, the same 6,549 slips pool to 0.9885 +- 0.0122, so the lean belongs to Kramers' leading formula, whose overshoot the exact rate measures: it is 0.905 of Kramers' at 4 noise units and 0.965 at 9 (KRAMERS_EXACT 20261020_073520).]

403. The register's and the book's precision words, as written from the first runs to KNEE_REF: 'within 5 per cent' (Kramers), 'from 2.5 per cent under to 7.7 per cent over' and 'within 15 per cent' at the smallest steps (the traders' law), 'within 15 per cent at 11 of 13 settings' (the knee), '1.04 to 1.05 of the line' (the knee ear). Each was read off a mean; its run's error bar is wider than the words.

Kind: WRONG READING (Claude's own: precision read off a mean). *Lesson:* Corrected in place in the book, the register's source and its assembler, each marked. The rule from here: a 'within' or a range is printed only beside its error bar, and never narrower than two of it. Needs no further redesign. *Proved?* Seen in runs (ERROR_BARS 20261019_162750).

404. KNEE_REF's error bars, and with them misses row 398's standard scores (1.17 and 1.20): the jackknife over the 256 noise paths of the staircase reader on a grid of 0.001 gave 0.001515 at the old reference, step 0.0025. Measured by KNEE_READER: its bootstrap is 0.58 of that.

Kind: THE INSTRUMENT (a jackknife on a reader that is not smooth). *Lesson:* The jackknife leans on the statistic changing smoothly when one unit is left out; a reader that draws straight lines between the steps of a staircase of whole flips jumps instead, and the jackknife reads the jumps as scatter. With the bootstrap the excess at 0.0025

against the old reference is about two standard errors on every reader (2.06, 1.99, 2.21): row 398's 'inside the scatter' is too strong, and so is KNEE_FINE's 'the law's'; neither is shown. Against the reference four times finer every reader is within 1.62. Redesigned and run in the same batch: KNEE_READER. *Proved?* Seen in runs (KNEE_READER 20261019_163948; KNEE_REF 20261019_042520).

405. The register's D2 34 and its source: 'taps a second 11.98 on a 0.25 m board and 2.99 on a 4 m board, 4.007 against 4'. BALANCE_BOARD's own part B3 and C1 show every rider's taps at the same moments in fall times at every length (gap ratios 2.000000000 exactly), so the toy's ratio is exactly the fall times' ratio, 4 within the printed fall times' rounding.

Kind: THE INSTRUMENT (a quotient of rounded numbers printed as a measurement). *Lesson:* An under-claim: corrected in place to 'exactly 4 in the toy'. Needs no redesign. *Proved?* Seen in runs (BALANCE_BOARD 20261004_161712).

406. KNEE_READER's (Q3), the long shot (Claude's odds 1 in 5): with the fitted reader, against the new reference, every step from 0.0003125 to 0.04 within 5 per cent of the law. Measured: 1.081, 1.014, 0.997, 1.054, 1.043, 1.025, 0.992, 0.988: MISSED.

Kind: WRONG GUESS (the long shot's). *Lesson:* Five per cent at the smallest shifts is narrower than their error bars, row 397's lesson met again in this very card's registering. Needs no redesign. *Proved?* Seen in runs (KNEE_READER 20261019_163948).

407. KNEE_READER's (Q1r), the rival, the strongest competing idea (Claude's odds 2 in 5): the small steps' excess is real and survives the fitted reader, against the new reference at 0.0025, two or more bootstrap standard errors. Measured: +1.14: MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* Against the finer reference the excess is gone on every reader; against the old it stays at about two standard errors. Needs no redesign beyond row 404's. *Proved?* Seen in runs (KNEE_READER 20261019_163948).

408. Claude's word, given before KNEE_READER's run of record, that reading the knee with a slope fitted over a wider range makes the 20 per cent excess disappear (0.95, 1.05, 1.03 and 1.02 of the law). Those numbers came from an exploratory probe in a scratch folder, a straight line over knees 1.000 to 1.040; the registered reader, a quadratic over 1.000 to 1.080, gives 1.139, 1.137, 1.083, 1.045 against the old reference.

Kind: WRONG READING (Claude's own: a probe's number told as a result). *Lesson:* Corrected in Claude's report the same day. Two smooth readers of the same counts differ at the smallest steps by as much as the error bar, so the reader is part of the reading. The rule from here: a probe's number is not told as a finding before its run of record. *Proved?* Seen in runs (KNEE_READER 20261019_163948).

409. A working record of Bear's words: his words from two sittings on 19 October had reached two other working records but not this one, where his words are kept for prose.

Kind: THE INSTRUMENT (a records slip: a file the routine forgot). *Lesson:* Filed the same day, as typed, with two more of his words from that day. Needs no redesign; that record joins the list of records each batch touches. *Proved?* Seen in the records.

410. The reader's edition's first assembly printed its front page from the headline register's Part A as it stands, and so, in thirteen places, named a part of the working draft by its working name rather than its chapter, where a reader of the reader's edition meets only chapters. Seen on reading the built page, before anything was sent.

Kind: THE INSTRUMENT (a reused page carried its first document's vocabulary). *Lesson:* Mapped to chapters by rule at assembly (the reader's edition's build script), with an assertion that no part name is left anywhere in the edition; the register itself keeps the working book's names, which are right there. Needs no redesign. *Proved?* Seen in the records (the reader's edition).

411. The lean registered for the check of the reader's edition, (R1): the two independent checkers would find at most 5 of the reader's edition's numbers set in a context their records do not give them. They found 25, among 36 lines to mend, and none a wrong value.

Kind: WRONG GUESS (Claude's: a script that finds every number in the records was read as a sign that the numbers were placed right). *Lesson:* The script check compares numbers as written and cannot see a right number in a wrong place: a run named short (REINJECT for a number REINJECT_CLOCK printed), a number given to the neighbouring toy (the conveyor's 0.9984 given to the needle), a caveat dropped in the rewrite (one start a gap flown; a hold for a thousand time units). The checkers who had not written the chapters saw each. Redesign, done the same day: every finding mended in place and the edition rebuilt; a second independent read of the mended edition is owed before any deposit, and a rewrite of this size from now on is read for context by someone who did not write it, not by script alone. *Proved?* Seen in the records (the reader's edition and the registration of its check).

412. The long shot registered for the check of the reader's edition, (R3): no number out of context.

Kind: WRONG GUESS (the long shot; Claude's odds 1 in 6). *Lesson:* As row 411. Several of the 25 sat where a paragraph of the working draft was compressed to a sentence and its closing caveat fell away; that compression drops a paragraph's last caveat first is a hunch, not yet counted over the 25. Needs no redesign beyond row 411's. *Proved?* Seen in the records (the registration of the reader's edition's check).

413. Claude's outline of 19 October put the reader's book at about 60 to 70 pages; the edition came out at 44, and at 47 after the additions made the same day.

Kind: WRONG GUESS (Claude's: a page estimate made without the builder's measure). *Lesson:* The builder sets about 830 words to a page, so 39,528 words make 47 pages here and would make well over a hundred in a trade paperback; the estimate had pages of a looser setting in mind. A book's length is said in words beside its pages from now on. The length itself is not a miss: a coverage check by script found the edition carrying every heavy result of the working draft but the chaos workshop's back pages, and all but five of its medium ones, four of which were then carried in. *Proved?* Seen in the records (the reader's edition).

414. The second independent read of the mended reader's edition found 28 numbers still set in a context their records do not give them, among 56 lines to mend. Four came from the working draft itself: the climb-back edge 'shot with four different integrators' where CLIMB_BACK_LINE_CHECK used three, one at two tolerances (misses row 38 carries the same slip; the register's row 13 has it right); the traders' law 'inside the error bar of each' step where it is within two standard errors at six of seven steps and 2.2 at the coarsest; 'none of eight control paces' where WINDOW_TOUCH_5 drew one of its eight twice, so seven; and the ERROR_BARS run's chance figure for the arch touches (its C29), which assumes eight.

Kind: THE INSTRUMENT (a slip copied faithfully from the record into the rewrite). *Lesson:* The first three mended in both editions, each with a dated note in the working draft; C29 is a run of record and stands as printed, its eight to be read as seven. One more is open: HALVINGS_2's header prints memory 1.0, a default of the module it reads, while the working draft reads its ladder at memory 1.7598; which is right is to be read from the tool. Redesign: a second independent read is part of any rewrite of this size, and it reads the source as critically as the rewrite. *Proved?* Seen in the records (the reader's edition; the working draft) [Settled 20 October, read from the tool: HALVINGS_2 executes HEAVY_WINDOW's preamble by text, and that preamble prints its own default, memory 1.0; the flights are made by make(Mv), with Mv read from LOOP_ATLAS_STOP_STATES_20261015_060745.npz and asserted at 1.7598, so the working draft's 1.7598 is right and the header line is the borrowed preamble's.]

415. Bear's reflection was set in the reader's edition as written near the end of the work, and its second part as written halfway through; and one line of it, that nothing was to be taken too seriously except that society needs to improve itself now, was set as his last word on the book's results. All of it was typed at the first sitting, 2 October, before the book's toys were built, and the line was about his own spitballing that night.

Kind: WRONG READING (Claude's: a quotation placed without its date and subject read from the record of his words). *Lesson:* Mended: the reflection is dated to the first night in chapters 4 and 13, and the line is given whole, spitballing and all, in both editions. Every quotation placed from now on is read in the record of his words for its date and its subject first. Needs no redesign beyond that. *Proved?* Seen in the records.

416. CAT_MIRROR_PATHS's (P1), Claude's lean (odds 3 in 5): on paths whose bend swells and shrinks as it swings round, the two cats' mirror breaks, because its proof used the steady circuit. Measured: on nine paths, among them a bend swinging 20 degrees either side of 90 and one swinging at twice the circuit's rate, the two turns agree within 2.842e-13 degrees: MISSED.

Kind: WRONG GUESS (Claude's: a proof's convenience read as a limit on its result). *Lesson:* The mirror lives in the rates at every shape, not in the circuit: the twist-free cat's rate at bend 180 - b, its bend rate reversed, is the toy's at b turned half a turn about the front tube's axis, within 1.9e-15 at 2,000 random states, so any path pairs. The redesign check said aloud was the instants printed beside, and the follow-on was CAT_MONTGOMERY, which found the reason in Montgomery (1993). Needs no redesign. *Proved?* Seen in runs (CAT_MIRROR_PATHS 20261020_034021).

417. CAT_MIRROR_PATHS's M2 (planted), on its first blind run: the new path at a = 0 is CAT_TWIST's within 1e-15 at 701 times. Measured: 1.5e-14, in the bend rate at the end of the last stretch: MISSED, on the blind run.

Kind: TOOL FAULT (Claude's: a stretch's length taken from the list of durations where the engine takes t1 - t0), caught by a planted check. *Lesson:* Near a stretch's end the half-cosine's rate is a near-cancellation, and the two ways of writing the length differ in the last bit. Mended to CAT_TWIST's arithmetic before any (P) line was printed, the threshold kept; M0 then fell from 2.8e-14 to 0.0. A planted check set at the arithmetic's grain finds where two tools write the same thing differently. *Proved?* Seen in runs (CAT_MIRROR_PATHS blind 20261020_033921 and 20261020_034012).

418. CAT_MIRROR_PATHS printed beside its paths the twist-free cat flown with its swing reversed, to show whether the mirror keeps the sense of the swing. Measured: the reversed swing pairs as well, within 2.487e-13.

Kind: INSTRUMENT (Claude's: a comparison blind to what it was meant to show). *Lesson:* The landing's turn is the size of the turn, from the quaternion, and not its sense; a loop flown backward turns by the same size, so the column could not have parted. The sense is read from the rotation vector, as CAT_MONTGOMERY printed it. Read what a reading cannot see before putting a comparison beside it. Needs no redesign. *Proved?* Seen in runs (CAT_MIRROR_PATHS 20261020_034021).

419. CAT_MIRROR_PROOF's (X3) (row 364), graded MISSED on 18 October: with the hinge 0.3 of a length from each centre, some other thickness map pairs the cats. Asked rightly two days later, it holds within 4.5e-13 degrees.

Kind: TOOL FAULT (Claude's: a mirror applied twice). *Lesson:* The twist-free rate was written at pi - b and then called at 180 - b, so the cat was read at the toy's own bend. The search was made at bend 90, the mirror's one fixed point, where a doubled mirror is invisible, so the radius it found was right (0.2080125736 = 0.3 / sqrt(2.08)) and only the test at 60 and 120 was wrong. The book's headline then carried 'it needs the hinge at the ends' for two days. Found by reading Montgomery's closed form, which disagreed with the printed miss, then the code; CAT_MONTGOMERY's M3 repeats the faulty numbers from the fault. A check made at a map's fixed point cannot see the map applied twice: test a mirror away from its middle. *Proved?* Seen in runs (CAT_MIRROR_PROOF 20261018_174959; CAT_MONTGOMERY 20261020_035025).

420. The three-surprises test of 20 October, its (C3), Claude's lean (odds 4 in 5): Cavendish's 1798 value, with his arithmetic slip mended, lies within 1 per cent of today's mean density. Read: the mended value is 5.448 +- 0.033, 1.2 per cent below today's 5.514; the unmended 5.480 +- 0.038, printed in his paper, is the one within 1 per cent: MISSED.

Kind: WRONG READING (Claude's, from recall: the slip taken the wrong way round). *Lesson:* Francis Baily found the slip in 1821, and the mended figure is the lower one. Which way a correction runs is a fact to read, not to recall. The surprise it served is reread too: Cavendish's ratio is G in another dress, both known today to about 2.2e-5, and the precise thing is GM, which an orbit reads. Needs no redesign. *Proved?* Read on the web (Wikipedia, 'Cavendish experiment' and 'Gravitational constant', 20 October).

421. ZETA_FIT's (Z1), Claude's lean (odds 3 in 5): the traders' BETA, read back from CREEP_LAW's flips pooled over noise times 0.1, 0.03 and 0.01 and steps 0.00125 to 0.04, lies within two standard errors of 0.5826. Measured: 0.90210 +- 0.00040: MISSED.

Kind: WRONG GUESS (Claude's: a limit law tested by pooling across its limit). *Lesson:* BETA read back climbs with the step against the noise's memory, from about 0.60 to 1.2, and a fit through the origin weighs the longest steps most. CREEP_LAW's own (T1) had already missed at noise times 0.03 and 0.01, and the book states the law within 15 per cent only while the step is at most a tenth of the memory. Redesigned at once as ZETA_LIMIT, which reads the limit at 0.57947 +- 0.00254, within 1.23 standard errors of 0.5826 under its registered line (ZETA_LIMIT 20261020_060256). *Proved?* Seen in runs (ZETA_FIT 20261020_055322; CREEP_LAW 20261016_035058).

422. Claude's chat of 20 October called the traders' 0.5826 a number from prime-number theory that set a sampled switch's dead band with nothing fitted, and offered it as the Integer Forest's pure structure turned engineering knob.

Kind: WRONG READING (Claude's: a function's name taken for its road, and a limit taken for the knob). *Lesson:* Zeta at one half enters Broadie, Glasserman and Kou's constant through the ladder heights of a Gaussian random walk (Chang and Peres 1997), a sum over the walk's whole-number step counts; the primes play no part. And 0.5826 is the small-step limit: the knob itself grows with the look's coarseness (ZETA_LIMIT). The whole numbers do carry it, so the integers' lens is not wrong, only the primes. Needs no redesign. *Proved?* Seen in the chat (20 October) and runs (CREEP_LAW 20261016_035058; ZETA_LIMIT 20261020_060256).

423. ZETA_LIMIT's first run of record stopped after its verdict lines, at its json: a numpy true-or-false the json writer refuses. The mend, made in haste, put a comment before the rest of a line, so the second run stopped before its verdicts.

Kind: TOOL FAULT (Claude's, twice). *Lesson:* The cast to a plain true-or-false mended it; the third run repeats the first's every printed number. All three run files are kept, so the register counts the three verdict lines twice; the first run's half-written json was removed as no data. A mend is read back before it is run. Needs no redesign. *Proved?* Seen in runs (ZETA_LIMIT 20261020_055829, 20261020_060047 and 20261020_060256).

424. On 20 October Claude flagged Rademaker and ter Braak (1935) as the source most likely to hold the mirror: Kane and Scher call their no-twist result 'essentially' that paper's, and Gbur describes that cat as two cylinders that bend and twist at the waist, so a twisting cat and a no-twist cat giving one result might be the mirror. Read in full: their cat is the twist-free cat, and the paper holds neither a twisting cat nor the mirror.

Kind: WRONG READING (Claude's: a summary's verb read as a mechanism). *Lesson:* Gbur's 'bend and twist' is the bend's direction carried round the body, with the halves turned the same way about the spine; Kane and Scher's 'essentially' means the same cat and the same formula. The flag was right to send us to the paper, which removed the

threat to found-here row 1 and gave a benchmark of 1935 (CAT_RTB). A secondary source's word for a motion is a lead to the primary, not a reading of it. Needs no redesign. *Proved?* Read in the paper (Rademaker and ter Braak 1935, pp. 322 to 325; CAT_RTB 20261020_063617).

425. KRAMERS_EXACT's (K1b), Claude's lean (odds 3 in 4): at most 2 of SLIP_SENSE's 15 judged rows lie two or more counting errors from the exact rate. Measured: 3 do (-2.08, -2.20 and -2.30), though the 15 pool to 0.9885 +- 0.0122: MISSED.

Kind: WRONG GUESS (Claude's, or the instrument's: counting errors assumed, not measured). *Lesson:* Fifteen rows at two standard errors should leave about one outside by chance, and three is unlikely only if each row's error really is the square root of its slips. If slips come in pairs or needles differ, the spread is wider. Redesign: rerun the 15 rows keeping each needle's slips, measure the spread, and judge again (the next card). *Proved?* Seen in runs (KRAMERS_EXACT 20261020_073520).

426. SLIP_SPREAD's (K1b'), Claude's lean (odds 1 in 2): with each needle's slips kept and the counting error measured, at most 2 of the 15 judged rows lie two or more errors from Stratonovich's exact rate. Measured: the same 3 do (-2.09, -2.20 and -2.31), and the measured errors are the square root of the slips within 3 per cent (Fano factors 0.947 to 1.032): MISSED.

Kind: WRONG GUESS (Claude's: the spread was suspected and is clean). *Lesson:* The redesign of row 425 suspected the error bar; the error bar was right, so the miss moves to chance or the needle's own step. Three rows of 15 two errors out comes about once in 35 by chance (arithmetic on the run's printed numbers). Redesign: new seeds, four times the needles and half the step on the same rows; chance would scatter the three back, a step bias would hold them. *Proved?* Seen in runs (SLIP_SPREAD 20261020_075635).

427. BENCH_LOOP's (L2) and (L2r), registered on 20 October as Claude's lean (odds 4 in 5, the scissors' tip below memory 1) and its rival (open to memory 0). Our own ledger had answered both on 11 October: MEMORY_SWITCH_2 put the tip at memory 0.7025636582 and MEMORY_SWITCH's S2 found no spinning below pace 1 at memory 0.6. Found while designing the tool, before it ran: not graded as leans; the tip made a planted check.

Kind: THE INSTRUMENT (Claude's process: our own records not searched before a lean was written). *Lesson:* A lean is only a prediction if nobody here has already measured it. The check of outside sources is done before every card; the check of our own ledger was not, and a week-old test held the number. Redesign: before registering, search the ledger, the misses and the Guide for the quantity's own words and the tool names that measured it, and say in the registration what was found. *Proved?* Seen in the ledger (MEMORY_SWITCH 20261011_084432; MEMORY_SWITCH_2 20261011_085331).

428. BENCH_LOOP's (L1), Claude's lean (odds 9 in 10, 20 October): a slow loop of memory and pace round the scissors' tip brings the needle home in the other motion, and a loop clear of the tip brings it home the same. Measured on 18 runs: the loop round the tip changes only one of its two starts in each direction, and the loop clear of it changes one start in each direction too, by jumps: MISSED, as Claude wrote before the tool that it would.

Kind: WRONG GUESS (Claude's: a textbook picture carried over whole). *Lesson:* The lean borrowed the picture of a loop round the tip of a fold, where going round changes the state, and read it as true of every start. Here a slow loop that crosses an edge sends both starts home in the same motion, so the end cannot tell the loops apart. What the tip changes is whether the change needs a jump: round it, a change can come with none. The redesign was registered before the tool as (L3), the count of jumps plus turns round the tip, and held in all 18 runs. *Proved?* Seen in runs (BENCH_LOOP 20261020_081001).

429. SLIP_RHYTHM's M3, planted: the mean speed from the new exact code repeats KRAMERS_EXACT's exact rates in all 18 rows within 1e-9 of themselves. Measured: up to 5.6e-05 apart: MISSED. At KRAMERS_EXACT's own paces the two agree within 1.1e-15 (a check in a scratch folder; SLIP_RECHECK's M1 repeats KRAMERS_EXACT at its printed paces exactly).

Kind: THE INSTRUMENT (Claude's, twice: KRAMERS_EXACT read the paces from SLIP_SENSE's printed table, rounded to six decimals, and the check compared two codes at two different paces). *Lesson:* A rate this steep in the pace moves about 114 times as fast as the pace at noise 0.1, so a rounding of 5e-07 in the pace is 5.6e-05 in the rate. KRAMERS_EXACT's exact rates are off by at most that, a hundredth of their errors, so nothing it concluded moves. Redesign: read a run's inputs from its json, never its printed table; SLIP_RECHECK takes each row's exact rate at its own pace and prints the shift. *Proved?* Seen in runs (SLIP_RHYTHM 20261020_082954 and 20261020_083426; SLIP_RECHECK).

430. SLIP_RHYTHM's (R1), Claude's lean (odds 3 in 4): in each 3-unit row the long-run Fano factor lies within two errors of the exact one, and below 1. Measured: 0.9539 +- 0.0372, 0.8660 +- 0.0348 and 0.8434 +- 0.0349 against 0.9133, 0.9145 and 0.9160, the last 2.08 errors out: MISSED.

Kind: WRONG GUESS (Claude's, or the instrument's: the variance's growth is a wide ruler). *Lesson:* The variance's growth between two times carries errors near 0.035 on 2,048 needles, seven times those of the intervals' squared spread, which reads the same quantity for a renewing needle (R3 held) and lies within two of its errors of the exact value in all three rows (read after the run). Redesign: SLIP_RHYTHM_2, the three rows again with four times the needles and new seeds, the intervals' squared spread registered against the exact value before its tool. *Proved?* Seen in runs (SLIP_RHYTHM 20261020_083426).

431. SLIP_RHYTHM's (R4), Claude's lean (odds 1 in 2): a needle that has just slipped cannot slip again at once, so under half of the 0.0952 of intervals that independent slips would put below a tenth of the mean wait. Measured: 0.0552, 0.0566 and 0.0560, 0.58 to 0.59 of chance: MISSED.

Kind: WRONG GUESS (Claude's: the right kind, the wrong size). *Lesson:* Short waits are rare, 41 per cent rarer than chance, but there is no hard floor: the shortest wait was 12.40, 8.50 and 5.84 time units here, about a fortieth of the mean, and a fiftieth on four times the needles (SLIP_RHYTHM_2). The bar was set too strong. No redesign needed for the measurement, which stands; the shape of the intervals near the floor is on the proposed list. *Proved?* Seen in runs (SLIP_RHYTHM 20261020_083426).

432. SLIP_RECHECK's (Q3), Claude's lean (odds 3 in 4): with new seeds, 8,192 needles and half the step, SLIP_SENSE's 15 judged rows pool within two errors of Stratonovich's exact rate. Measured: 0.9877 +- 0.0061, 2.01 errors under 1: MISSED, by a hair, while the halved step moved nothing seen (-0.0008 +- 0.0136 against SLIP_SPREAD).

Kind: WRONG GUESS (Claude's: a start not counted). *Lesson:* The exact rate is the steady one, and every needle starts at the bottom of its well, where it cannot escape at once and, once over, still has to run to the counting point. A count of 500 time units feels that late first slip; the 3-unit rows run to 20,000 count within 0.2 per cent of the exact rate (arithmetic on SLIP_RHYTHM's counts, read after the run). Redesign: SLIP_START counts the first 50 time units apart from the rest on the 500-unit rows. *Proved?* Seen in runs (SLIP_RECHECK 20261020_083908).

433. SLIP_RHYTHM_2's (S1), Claude's lean (odds 3 in 4): on 8,192 needles the intervals' squared spread lies within two errors of the exact Fano factor in each 3-unit row. Measured: 0.9139 +- 0.0034, 0.9097 +- 0.0025 and 0.9113 +- 0.0023 against 0.9133, 0.9145 and 0.9160, the last 2.06 errors low: MISSED.

Kind: THE INSTRUMENT (Claude's: a reader that drops long intervals), in part. *Lesson:* The pool of each needle's complete intervals inside its window leaves out the last one, which runs past the end and is likelier to be long; on a model renewal process with these means and spreads that pool reads 0.0013 to 0.0028 low (a check in a scratch folder), about half the gap seen over SLIP_RHYTHM and SLIP_RHYTHM_2. Redesign: SLIP_RHYTHM_3 uses an interval if it starts before the end less 12 mean intervals, whatever its length. *Proved?* Seen in runs (SLIP_RHYTHM_2 20261020_084919).

434. SLIP_RHYTHM_2's (S3), Claude's lean (odds 1 in 2): at one mean interval the Fano factor lies more than two errors below the exact long-run one in each row, every needle having started together from its seat. Measured: +0.84, -0.16 and +1.44 errors from it: MISSED.

Kind: WRONG GUESS (Claude's: a surprise picked as the lowest of 18). *Lesson:* SLIP_SPREAD's 0.835 was the lowest Fano factor of 18 rows, and it was picked out for that. On new seeds and four times the needles the same row at the same time reads 0.9244 +- 0.0132, within an error of the exact 0.913. The spread below chance is real, about 9 per cent at 3 noise units, but the extra dip at the start was the sample's. The start shows instead in the count's mean (SLIP_START). No redesign needed. *Proved?* Seen in runs (SLIP_RHYTHM_2 20261020_084919).

435. SLIP_RHYTHM_3's (V2), Claude's lean (odds 3 in 5): on the same needles the window pool of complete intervals reads below the unbiased reader by 0.0005 to 0.005 in each row. Measured: 0.0020 +- 0.0011, 0.0021 +- 0.0006 and 0.0004 +- 0.0005, the last under the band: MISSED.

Kind: WRONG GUESS (Claude's: a band sized for another window). *Lesson:* The band came from the model at 20,000 time units; this run's window is 30,000, and by the same model the pool's lean should shrink as one over the window, to about 0.0009 at noise 0.3, within an error of what was seen (expected, not measured). The pool read low clearly in one row, at 1.9 errors in another, and showed no lean on the synthetic needles (M0); the reader that avoids it matched the exact spread in all three (V1 held). No redesign needed. *Proved?* Seen in runs (SLIP_RHYTHM_3 20261020_090717).

436. RHYTHM_BAND's M1, planted: at 20 noise units the exact Fano factor lies within 1e-3 of 1 at noise 0.05 and 0.5. Blind run 20261020_153930: 1.000000 at noise 0.05 and an enormous number at 0.5: MISSED.

Kind: THE INSTRUMENT (Claude's: a planted check set where the needle cannot go). *Lesson:* The washboard's barrier is at most 2, at zero pace, so at noise 0.5 (D = 0.125) it reaches only 16 noise units; asked for 20, the pace search ran to zero, where the needle barely drifts and its count spreads without bound. Mended after the blind run and before any lean printed, as row 417 was: M1 reads noise 0.05 and 0.2. Lesson: check that a planted setting exists before planting it. *Proved?* Seen in runs (RHYTHM_BAND blind 20261020_153930).

437. SLIP_BAND's (H1), Claude's lean (odds 3 in 4): at pace 0.95 and noises 0.3, 0.7 and 1.5 the needle's waits' squared spread lies within two errors of the exact Fano factor. Measured: +1.07 and +0.13 errors at 0.3 and 0.7, but 0.49300 +- 0.00105 against 0.50967 at 1.5, 15.92 errors out: MISSED.

Kind: THE INSTRUMENT (Claude's: a reader that assumed no backward slips). *Lesson:* Only the noisiest row slips backward (7,253 times). After a backward slip the next forward count re-crosses a turn already counted, so the waits between forward counts are no longer first passages to a new turn, the quantity whose squared spread the paper proves equal to the long-run Fano factor; the counts themselves fit the exact value (variance growth 0.4974 +- 0.0140). Redesign, said aloud and run: SLIP_BAND_2 reads first passages to each new turn, and matches, 0.50779 +- 0.00112. *Proved?* Seen in runs (SLIP_BAND 20261020_161039).

438. BELL_SWITCH's (D2), Claude's lean (odds 1 in 2): at the bells' crowded corner the share of eight nearby starts going forward moves by at most 2 of 8 between steps 0.005 and 0.0025. Measured: 2 to 6 of 8 at memory 8.49, 3 to 6 at memory 12: MISSED.

Kind: WRONG GUESS (Claude's: a share hoped to settle where single runs did not). *Lesson:* With the toy's thin cores even the share moves with the step, because what moves it is a near-collision of rings that the step resolves differently each time. Carried by row 439: with the cores' own size every start goes forward and nothing is left to settle. No redesign of its own. *Proved?* Seen in runs (BELL_SWITCH 20261020_161902).

439. BELL_SWITCH's (D3), Claude's lean (odds 1 in 2): with the rings' near-collision softened to the cores' own size, both answers still appear among eight nearby starts at each crowded memory, so the switch is the machine's. Measured: all eight go forward at both memories, +0.1240 to +0.1872 at memory 8.49 and +0.1808 to +0.1837 at memory 12: MISSED; the rival held.

Kind: WRONG GUESS (Claude's), and THE INSTRUMENT (the toy's thin cores, a softening of 1e-3 on the squared distance, a length of about 0.032, against cores of 0.15; units corrected 21 October). *Lesson:* The corner's switch, which the book and the Field Guide read as the bells choosing between swimming and being pushed back by their own wake, was rings passing almost through one another in a toy whose near-collision term was far thinner than its own cores. Redesign, said aloud: read again with the softened core every bell number made where rings crowd, starting with the bells' settled map (BELL_BALANCE), proposed; and carry the correction to the Field Guide. *Proved?* Seen in runs (BELL_SWITCH 20261020_161902).

440. BELL_SOFT_MAP's (F1), Claude's lean (odds 3 in 5): with the rings' near-collisions given the cores' own size, all 25 points of the bells' map settle. Measured: 23 do; memory 6 at period 2.12 (quarters 4.936 per cent) and memory 8.49 at period 1.5 (half the step -9.692 per cent) do not: MISSED.

Kind: WRONG GUESS (Claude's: the cores' size taken for the whole cure). *Lesson:* The softened core settles three of the five thin-core holdouts but not every point; memory 8.49 at period 1.5 is the corner where BELL_SWITCH's softened starts fell in two forward groups a thousandth of the gap apart. Redesign, said aloud: read the two in distribution over nearby starts, as BELL_SWITCH read the corner, softened. *Proved?* Seen in runs (BELL_SOFT_MAP 20261021_000736).

441. BELL_SOFT_MAP's (F2), Claude's lean (odds 2 in 3): at the 20 points that settled with the thin cores the softened gain lies within 0.05 of the thin one. Measured: 13 of 20 move by more, up to 0.3334 (memory 4.24, period 2.12: 0.6754 to 1.0088): MISSED.

Kind: THE INSTRUMENT (the thin-cored toy), and a WRONG GUESS (Claude's: that settling under a halved step meant the number was the toy's). *Lesson:* A point that settles when its step is halved has settled only against the step, not against the toy's other scales; the near-collision core was a scale nobody had moved. Redesign, said aloud: every bell map made with the thin cores (BELL_RATIO, BELL_CLOCK, BELL_BALANCE's record point) owes the same re-reading, and the book's bell numbers are now the softened map's; a softening of 0.01 and 0.05 at a few points, proposed, says whether the cores' size is where the gains settle. *Proved?* Seen in runs (BELL_SOFT_MAP 20261021_000736).

442. BELL_SOFT_MAP's (F3), Claude's lean (odds 3 in 5): the softened gain still crosses 1 along the period at every memory, between periods 2.5 and 3.5. Measured: 2.264, 2.110, 2.937, 2.881 and 2.018, with the gain within 0.03 of 1 from period 2.12 to 3 at memories 3 and 4.24: MISSED.

Kind: WRONG GUESS (Claude's). *Lesson:* With the cores' size, the line where the rings stop costing comes earlier and is not a clean line: at short memories the gain lies on a near-flat stretch at about 1, so a crossing read between two grid points can fall anywhere on it. Redesign, said aloud: a finer period grid between 1.5 and 3.5 before the balance is given as a line. *Proved?* Seen in runs (BELL_SOFT_MAP 20261021_000736).

443. BELL_SOFT_SCALE's (S2), Claude's lean (odds 1 in 2): widening the rings' meeting from 0.15 to 0.22 carries on the softening's drift toward one, at more than half of the points settled at both. Measured: two of five lie closer to one at 0.22 (memory 6 at period 1.5, 0.3931 to 0.8042; memory 12 at period 6, 1.2981 to 1.2783) and three lie further (memory 3 at period 3, 1.0218 to 1.0540; 4.24 at 2.12, 1.0088 to 0.8709; 12 at 3, 1.0992 to 1.2724): MISSED.

Kind: WRONG GUESS (Claude's). *Lesson:* The drift toward one seen on mending was not a trend in the width but a property of one width: where squeezes come fast, the gain moves with the width without a direction, one point peaking at the cores' own size. Redesign, said aloud: not covered by another card; read the gain on a finer ladder of widths at a fast point and a slow one, to tell a peak from a crossing of two effects [run 21 October as BELL_WIDTH_LADDER 20261021_151833: the fast point's peak is a spike at the cores' own width alone; the slow point stands at 1.28 to 1.37; row 446]. *Proved?* Seen in runs (BELL_SOFT_SCALE 20261021_054731).

444. TRAIN_FLOOR's (F1), Claude's lean (odds 2 in 3): the edge thins downward, fewer than 8 of the 10 copies at 0.550 and 0.555 breaking (TRAIN_EDGE's share would give 8.7). Measured on the second computer: 9 of 10: MISSED.

Kind: WRONG GUESS (Claude's). *Lesson:* The edge does thin downward, but lower than guessed: 2, 1, 1 and 3 of five at 0.530 to 0.545, none at 0.525 or tighter; at 0.55 it is still as steep as the band above it. Redesign, said aloud: covered by this same run, which found where the thinning lies. *Proved?* Seen in runs (TRAIN_FLOOR 20261020_224239).

445. TRAIN_FLOOR's (F3), the long shot (Claude's odds 1 in 5): the floor is sharp, every copy at 0.540 or tighter holding and at least 6 of the 10 at 0.550 and 0.555 breaking. Measured on the second computer: 4 of the 45 at 0.540 or tighter broke (two at 0.530, one each at 0.535 and 0.540): MISSED.

Kind: WRONG GUESS (the long shot's). *Lesson:* There is a floor inside 20,000-unit flights, but it lies between 0.525 and 0.530, below the slope where the long shot put it. Redesign, said aloud: not covered by another card; fly the copies at 0.520 and 0.525 much longer, to tell a floor from a lifetime longer than the window, and grid the gap from 0.525 to 0.530 finer, both on the second computer (proposed) [run 21 October as TRAIN_FLOOR_2 20261021_083613: half of the ten long flights broke by 100,000; no floor; rows 447 and 448]. *Proved?* Seen in runs (TRAIN_FLOOR 20261020_224239).

446. BELL_WIDTH_LADDER's (W2), Claude's lean (odds 1 in 2): at memory 12 the slow squeeze (period 6) beats the fast (period 3) at every width where both settle, at least eight of the eleven. Measured: nine of the ten such widths, by +0.4416 at the toy's thin 0.032 down to +0.0033 at 0.175; at 0.200 the fast squeeze spikes to 1.4298 against the slow 1.2779, losing by 0.1519: MISSED.

Kind: WRONG GUESS (Claude's), and THE INSTRUMENT (a toy whose fast-squeeze gain carries narrow spikes in a setting). *Lesson:* The slow point is steady at every width, and the fast one wanders with narrow spikes; so slow-beats-fast at long memory is a shape of the water at most widths and fails where a spike lands. Redesign, said aloud: not covered by another card; a fine ladder around the two spikes (lengths 0.14 to 0.16 at memory 4.24 and period 2.12; 0.19 to 0.21 at memory 12 and period 3), to tell a narrow resonance from a switch between two ways the rings pass. *Proved?* Seen in runs (BELL_WIDTH_LADDER 20261021_151833).

447. TRAIN_FLOOR_2's (G1), Claude's lean (odds 3 in 5): the train's floor is real, none of TRAIN_FLOOR's ten starts at 0.520 and 0.525 breaking when flown on to 100,000 time units. Measured on the second computer: 5 of 10 broke, at 35243.1, 38376.9, 36516.3, 66783.3 and 86180.7, a lifetime near 152,620: MISSED; the rival held.

Kind: WRONG GUESS (Claude's). *Lesson:* A floor seen in 20,000-unit flights was the window's; the lean read a sharp change in breaking as a change of regime when it was a lifetime climbing steeply as the rings close up. Redesign, said aloud: covered by this same run, which measured the lifetime the window hid; the question it opens is the gate every break came through, the slipping ring's squeeze to about 0.21 of a radius (proposed). *Proved?* Seen in runs (TRAIN_FLOOR_2 20261021_083613).

448. TRAIN_FLOOR_2's (G3), the long shot (Claude's odds 1 in 5): a knife-edge, none of the fifteen copies at 0.526 to 0.528 breaking and at least 2 of the 5 at 0.529 breaking. Measured on the second computer: 0, 2, 1 and 0 of five at 0.526 to 0.529: MISSED.

Kind: WRONG GUESS (the long shot's). *Lesson:* With no floor there is no edge to be sharp; the breaking near 0.53 is a throw of the dice at a lifetime of tens of thousands. Needs no redesign. *Proved?* Seen in runs (TRAIN_FLOOR_2 20261021_083613).

449. TRAIN_DOOR's (D1), Claude's lean (odds 3 in 5): the door moves with the core, the median broken train's last-200 smallest ring lower at core 0.04 than at 0.05, and higher at 0.06, each by more than 0.01. Measured: medians 0.2455, 0.3060 and 0.2261: MISSED; the rival held.

Kind: WRONG GUESS (Claude's), and THE INSTRUMENT (medians taken over two ways of breaking). *Lesson:* At gaps 0.60 to 0.72 most trains fly apart early without a deep squeeze, so the medians mostly measured that; the breaks that did come through a deep squeeze, four at core 0.06, came at 0.2080 to 0.2107, TRAIN_FLOOR_2's size at core 0.05, so the door stays with the train [TRAIN_EDGE_MAP adds three more at core 0.06, 0.2029, 0.2231 and 0.2272: near that size, looser; row 452]. Redesign, said aloud: not covered; read the door only on breaks that come through a deep squeeze, near each core's own edge (proposed). *Proved?* Seen in runs (TRAIN_DOOR 20261022_055043).

450. TRAIN_DOOR's (D2), Claude's lean (odds 1 in 2): the door keeps one shape, the median broken train's core over radius at its last-200 smallest ring within 0.05 of 0.52 at every core. Measured: 0.329, 0.295 and 0.563: MISSED.

Kind: WRONG GUESS (Claude's). *Lesson:* The door is a size, not a shape: where trains broke through it at core 0.06 the core was 0.62 of the radius. Redesign, said aloud: covered by row 449's. *Proved?* Seen in runs (TRAIN_DOOR 20261022_055043).

Misses 451 to 500

451. TRAIN_DOOR's (D3), the long shot (Claude's odds 1 in 4): at core 0.05 every break at gaps 0.60 to 0.72 comes through TRAIN_FLOOR_2's door, its last-200 smallest ring from 0.200 to 0.220. Measured: 0.2305 to 0.3578, none within: MISSED.

Kind: WRONG GUESS (the long shot's). *Lesson:* Wide trains at core 0.05 fly apart before any ring squeezes deep, at gaps 0.68 and 0.72 within 265 time units. Needs no redesign of its own; row 449's covers it. *Proved?* Seen in runs (TRAIN_DOOR 20261022_055043).

452. TRAIN_EDGE_MAP's (E1), Claude's lean (odds 3 in 5): the door holds at core 0.06, every break whose last-200 smallest ring is below 0.23 having it from 0.205 to 0.215. Measured: 0.2029, 0.2231, 0.2272 and 0.2107 (the last a repeat of TRAIN_DOOR's start): MISSED.

Kind: WRONG GUESS (Claude's). *Lesson:* At the fatter core the door is near the same size but looser, from 0.203 to 0.227 over seven independent breaks with TRAIN_DOOR's; the window registered was TRAIN_FLOOR_2's at core 0.05, read in long flights at its edge, and these breaks were short flights away from it. Redesign, said aloud: not covered; TRAIN_FLOOR_2's design at core 0.06's own edge, long flights, for the second computer (proposed). *Proved?* Seen in runs (TRAIN_EDGE_MAP 20261022_060414).

453. TRAIN_EDGE_MAP's (E2), Claude's lean (odds 1 in 2): thin cores reach the door near their edge, some core-0.04 flight's smallest ring below 0.215. Measured: the smallest in 14 flights 0.2306: MISSED; the rival held.

Kind: WRONG GUESS (Claude's). *Lesson:* Thin cores did not squeeze deep in 3,000-unit flights near their edge, as in TRAIN_DOOR's wider ones; whether long flights ever take them there is open. Redesign, said aloud: covered by row 452's, long flights at core 0.04's edge beside core 0.06's. *Proved?* Seen in runs (TRAIN_EDGE_MAP 20261022_060414).

454. TRAIN_EDGE_MAP's (E4), the long shot (Claude's odds 1 in 4): the tightest gap where a copy breaks is the same grid gap at all three cores. Measured: 0.59, 0.59 and 0.56: MISSED.

Kind: WRONG GUESS (the long shot's). *Lesson:* The core moves the edge by one notch at most; a grid of 0.03 cannot say more. Needs no redesign of its own; a finer grid between 0.53 and 0.59 rides with row 452's. *Proved?* Seen in runs (TRAIN_EDGE_MAP 20261022_060414).

455. TRAIN_EDGE_DOOR's (K1), Claude's lean (odds 3 in 5): at core 0.06, flown at its own edge, every break comes within 200 time units of a squeeze to 0.200 to 0.220. Measured on the second computer: eighteen breaks from 0.1974 to 0.2343, five outside the band: MISSED.

Kind: WRONG GUESS (Claude's). *Lesson:* The tight door of TRAIN_FLOOR_2 came from core 0.05 trains living about 150,000 time units; at core 0.06's edge trains live about 13,000 and break through a looser squeeze. Redesign, said aloud: not covered; fly the copies at 0.530 that held, at cores 0.04 and 0.06, on to 100,000 on the second computer, to tell whether a clean door belongs to slow leavers (proposed; run as TRAIN_LONG_EDGE 20261022_175421 on the second computer: rows 463 and 464). *Proved?* Seen in runs (TRAIN_EDGE_DOOR 20261022_070632).

456. TRAIN_EDGE_DOOR's (K2), Claude's lean (odds 1 in 2): flown long at its edge, core 0.04 reaches the door, some flight's smallest ring below 0.215. Measured: the smallest in 25 flights 0.2249: MISSED; the rival held.

Kind: WRONG GUESS (Claude's). *Lesson:* Thin cores break anyway, at 0.225 to 0.263, where their holding trains squeeze just as deep: for them the squeeze is no door. Redesign, said aloud: covered by row 455's. *Proved?* Seen in runs (TRAIN_EDGE_DOOR 20261022_070632).

457. TRAIN_EDGE_DOOR's (K4), the long shot (Claude's odds 1 in 5): thin cores have a door of their own, core 0.04's breaks' last-200 smallest rings within 0.02 of one another. Measured: 0.2254 to 0.2625: MISSED.

Kind: WRONG GUESS (the long shot's). *Lesson:* No door of their own: the thin cores' breaks spread as widely as their holds. Needs no redesign of its own; row 455's covers it. *Proved?* Seen in runs (TRAIN_EDGE_DOOR 20261022_070632).

458. A build fault, not a prediction: The toy runs' PDF names 12 pictures and drew 3, all 16-pixel icons, and a shipment carried it so (when it was built that way is not recorded here); and on 22 October the reader's edition, built by its file name rather than the 'READERS' keyword, came out with 17 of its 18 pictures as icons, 43 pages where it

has 50. In both, the build's readback printed short 0, since it counted letters and digits and counted pictures only when called by keyword. Caught by a picture count before anything shipped (the publishing notes of 22 October had named the fault in versions 1.0 to 1.3 of the book's builder, build_paper).

Kind: INSTRUMENT (a guard on one path only). *Lesson:* Redesigned and run: the build-and-read-back script now builds every source that points into the figures folder from a full-path copy, whatever name it is called by, sets the pictures named against the images drawn on every build, and prints FIGURES MISSING or TINY IMAGES; a planted faulty PDF is flagged; both files rebuilt, 18 of 18 and 12 of 12; the book's own build unchanged. *Proved?* Seen in builds (22 October).

459. CAT_MIRROR_FAMILY's (F1r), the rival, the hope of the first surprise of 22 October: the mirror holds on the family's third direction, the back half spinning about its own length, within 1e-13, so the whole cat with a freely twisting back half (Iwai's family) mirrors. Measured: it fails by 0.2816: MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* The map is its own inverse, so it could not hold on the whole family without making twist-free cats of radius r and r/4 turn alike; the mirror is a pairing of two slices. Redesign, said aloud: covered by (F2), which asks whether any other fixed twist map works (none does, row 460); other maps of the shape proposed. *Proved?* Seen in runs (CAT_MIRROR_FAMILY 20261023_001232).

460. CAT_MIRROR_FAMILY's (F3), the long shot (Claude's odds 1 in 5): some fixed linear map of the shape rates makes the whole family mirror, the partner's matching rate the same at every shape within 1e-12. Measured: it spreads by 2.175: MISSED.

Kind: WRONG GUESS (the long shot's). *Lesson:* No fixed twist rule rescues the mirror: the spin the partner would need changes from shape to shape. Redesign, said aloud: for twist rules none is left to try; not covered for other maps of the shape itself (a fixed turn other than the half turn, psi shifted by a constant), proposed. *Proved?* Seen in runs (CAT_MIRROR_FAMILY 20261023_001232).

461. CAT_MIRROR_CENTRES's (C1r), the rival: with the hinge at the tubes' centres the mirror still fails on the back half's own spin by more than 1e-6. Measured: 3.3e-16: MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* At the centres the twist-free cat at 180 - b is the toy at b described end for end, so the whole family mirrors there by relabeling. Redesign, said aloud: needs none. *Proved?* Seen in runs (CAT_MIRROR_CENTRES 20261023_001618).

462. CAT_MIRROR_CENTRES's (C3), the long shot (Claude's odds 1 in 5): the failure goes as the square of the hinge's distance, failure over A squared within 10 per cent of one value from 0.1 to 0.5. Measured: 1.126 to 2.281: MISSED.

Kind: WRONG GUESS (the long shot's). *Lesson:* The departure from the centres' mirror rises faster than the square close in and slower farther out. Redesign, said aloud: not covered; read the departure's shape against the hinge on a finer ladder of hinges and fit it (proposed). *Proved?* Seen in runs (CAT_MIRROR_CENTRES 20261023_001618).

463. TRAIN_LONG_EDGE's (L1), Claude's lean (odds 1 in 2): the slow leavers have the door at core 0.06, every core-0.06 break later than 20,000 within 200 time units of a squeeze to 0.200 to 0.220. Measured on the second computer: all 13 breaks late, through 0.1940 to 0.1991: MISSED.

Kind: WRONG GUESS (Claude's: the band, set from core 0.05's door). *Lesson:* The slow leavers do have a narrow door at core 0.06, but below the band: the door slides with the core, about 0.0137 of a radius for each 0.01 of core. Redesign, said aloud: not covered; register each core's door from the line through the three bands' middles and test it at cores 0.045 and 0.055 on the second computer (proposed; run as TRAIN_DOOR_LINE 20261022_222935 on the second computer: at cores 0.045 and 0.055 every one of 15 breaks came within 0.0025 of the line's door, so the door slides on one line at five cores; row 465). *Proved?* Seen in runs (TRAIN_LONG_EDGE 20261022_175421).

464. TRAIN_LONG_EDGE's (L3), Claude's lean (odds 1 in 2): at gap 0.530 the three cores' lifetimes to 100,000 lie within a factor of 2. Measured: 180,184, 54,207 and 69,736, a factor of 3.32: MISSED.

Kind: WRONG GUESS (Claude's). *Lesson:* The edge's place does not move with the core, but the life at the edge does; with 6 breaks the thin cores' counting error alone is 41 per cent. Redesign, said aloud: covered in part by (L4), the long shot that held; not covered, more copies at core 0.04 to shrink its error (proposed; ten more copies run in TRAIN_DOOR_LINE 20261022_222935: core 0.04 pooled over 25 copies lives 148,791 +- 42,952, its counting error down from 41 to 29 per cent, 2.13 times core 0.06). *Proved?* Seen in runs (TRAIN_LONG_EDGE 20261022_175421).

465. TRAIN_DOOR_LINE's (D1r), the rival: at cores 0.045 or 0.055 some break lies more than 0.008 of a radius from the line through TRAIN_LONG_EDGE's three doors, so the door does not slide on one line. Measured on the second computer: all 15 breaks within 0.0025: MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* The door slides on one line at five cores from 0.04 to 0.06, forecast at two cores never flown. Redesign, said aloud: needs none; proposed beside it, the line tested outside its bracket, at cores 0.035 and 0.065. *Proved?* Seen in runs (TRAIN_DOOR_LINE 20261022_222935).

466. An instrument fault, not a prediction: TRAIN_DOOR_LINE, prints beside each break ‘the median spacing of that flight’s last five scaled squeezes’ as the middle entry of the sorted spacings, which for a flight with only five scaled squeezes, four spacings, is the upper of the two middle values. Two printed numbers are so: 209.2 and 158.8, where the medians are 191.8 and 137.1. Printed beside, not judged; no verdict uses them. Found when every number was recomputed from the json by a second script.

Kind: INSTRUMENT (a label that a short flight makes untrue). *Lesson:* Redesign, said aloud: the next tool that prints a median takes it from Python’s statistics.median, and the read of the fuse against the spacing in TRAIN_DOOR_LINE’s ledger row uses the recomputed medians. *Proved?* Seen in runs (TRAIN_DOOR_LINE 20261022_222935).

467. TRAIN_FUSE_CLOCK’s (F1), Claude’s lean (odds 1 in 2): most of the fuse, the 130 to 175 time units from a train’s last deep squeeze to its break, is the freed ring’s trip, the median of the parting over the fuse below 0.4 over TRAIN_DOOR_LINE’s 21 re-flown breaks. Measured on the second computer: 0.908, from 0.808 to 1.000: MISSED.

Kind: WRONG GUESS (Claude’s). *Lesson:* The train keeps within the widest swing of its last thousand time units for about nine tenths of the fuse, and the last leg past it to the 5 radii that count as a break takes a median 13.2. But that widest swing already reaches 4.0 to 5.0 radii, so the spread cannot say whether the leaving ring was set on its way sooner. Redesign, said aloud: needs none, the rival holding; proposed beside it, the third ring’s threadings of the pair read as their own clock (TRAIN_BEAT), and a look at whether a ring that passes 5 radii ever comes back. *Proved?* Seen in runs (TRAIN_FUSE_CLOCK 20261028_101647).

468. TRAIN_FUSE_CLOCK’s (F2r), the rival: at cores 0.035 or 0.065, outside the bracket of cores the door line was drawn from, some break lies more than 0.008 of a radius off the line, so the line bends. Measured on the second computer: all 14 breaks within 0.0036: MISSED.

Kind: WRONG GUESS (the rival’s). *Lesson:* The door lies on one line at seven cores from 0.035 to 0.065, four of them forecast before they were flown, two outside the bracket. Redesign, said aloud: needs none. *Proved?* Seen in runs (TRAIN_FUSE_CLOCK 20261028_101647).

469. TRAIN_FUSE_CLOCK’s (F3), the long shot (odds 1 in 5): the parting keeps count, the passages from the last deep squeeze to the parting within 10 per cent of their median at all five cores. Measured on the second computer: median 358, from 295 to 474, 18 of 35 within 10 per cent: MISSED.

Kind: INSTRUMENT (a count that mixed two clocks) and WRONG GUESS (Claude’s). *Lesson:* A passage was counted whenever any two rings swapped order, and 97 to 99 per cent of those swaps are the tight pair’s own, about every 0.35; the third ring threads the pair only every 76 to 126 time units near the end, and between the last deep squeeze and the break it threads the pair exactly once in 35 of 35 (read from the json after the verdicts, not forecast). Redesign, said aloud: count only the third ring’s threadings of the pair, and register their beat as the clock before flying fresh trains (TRAIN_BEAT). *Proved?* Seen in runs (TRAIN_FUSE_CLOCK 20261028_101647).

470. TRAIN_BEAT’s (B1r), the rival: at gap 0.565 some deep squeeze lies more than 2 time units from every threading of the third ring through the pair, so a deep squeeze is not a threading. Measured here: all 11 within 0.045: MISSED.

Kind: WRONG GUESS (the rival’s). *Lesson:* Every deep squeeze is the third ring going through the tight pair, at a second gap as at the first. Redesign, said aloud: needs none. *Proved?* Seen in runs (TRAIN_BEAT 20261028_212034).

471. TRAIN_BEAT’s (B2), Claude’s lean (odds 3 in 5): the third ring threads the pair exactly once between the last deep squeeze and the break, in every broken flight. Measured here at gap 0.565: in 5 of 9 (fuses 103 to 153); in the other four, 3, 29, 299 and 339 threadings (fuses 214 to 5,051): MISSED.

Kind: NATURE (the soft edge) and WRONG GUESS (Claude’s, carried from gap 0.530, where it held 35 of 35). *Lesson:* Near the soft edge a train can go on threading for hundreds of beats with no squeeze deep enough to count and then go, and two of eleven breaks had no counted squeeze at all, their smallest rings 0.2205 and 0.2217 against the threshold 0.2200. The clean break is the rule when the break comes soon, not always. Redesign, said aloud: not covered; read the depth of every threading, not only those under the threshold, and ask whether a break follows the threading after which the third ring’s swing grows past its widest (proposed). *Proved?* Seen in runs (TRAIN_BEAT 20261028_212034).

472. TRAIN_BEAT’s (B4), the long shot (odds 1 in 5): at gap 0.565 every break 35 to 70 time units after its last threading, as 42.4 to 63.9 at gap 0.530. Measured here: 33.8 to 42.3 in the five clean breaks, two under 35, and up to 211.9 in the long ones: MISSED.

Kind: WRONG GUESS (Claude’s). *Lesson:* The last leg is shorter at the wider gap, and in a long fuse it has no fixed length. Redesign, said aloud: covered by row 471’s. *Proved?* Seen in runs (TRAIN_BEAT 20261028_212034).

473. TRAIN_BEAT’s (B5), Claude’s lean (odds 4 in 5): a break is for good, no third ring coming back within 3 radii of the pair in the 1,000 time units after the first 5-radius spread. Measured here at gap 0.565: 5 of 11 came back, first at 59.9 to 432.2: MISSED.

Kind: INSTRUMENT (the toy's break event, a spread of 5 radii, set at TRAIN_FLOOR) and NATURE. *Lesson:* Four of the five that came back had gone out no farther than 6.0 to 8.7 radii; the six that did not were 32.6 to 60.6 radii out at the end, each at its farthest. So this toy's break is the first wide parting, and about half the time near the edge it is a long swing, not a loss; every lifetime and break in chapter 9 is a time to that first 5-radius spread. Seen first in a smoke run at gap 0.60, both of whose breaks came back, before the run of record. Redesign, said aloud: not covered; define a parting for good, the third ring past 10 radii and receding, read on fresh flights with a longer look, and read the lifetimes at one gap both ways (proposed as TRAIN_PARTING, runnable here). *Proved?* Seen in runs (TRAIN_BEAT 20261028_212034).

474. An instrument fault, not a prediction: TRAIN_PARTING, as first written, printed in its blind run each flight's parting at 10 radii and the nearest its ring came after it; nudge 22 showed a ring 10 radii out coming back to 0.03, which answered the registered lean (G1) before the run of record. Found on reading the blind run's lines.

Kind: INSTRUMENT (a blind run that was not blind). *Lesson:* A blind run checks the plumbing and must print nothing a forecast is judged on; this one printed the progress line of a full run. Redesign, said aloud: (G1) withdrawn and replaced before the run of record (a ring that comes back from 20 radii), the flight made whole to 20,000, and the blind mode made to print only each flight's energy; rerun blind (20261029_030409). *Proved?* Seen in runs (TRAIN_PARTING_BLIND 20261029_025845).

475. An instrument fault, not a prediction: TRAIN_PARTING, flew fourteen trains whole to 20,000 at about 560 to 3,700 s each, against an estimate of 325 s made from flights that stopped at their break, and kept every result in memory until the end; the workspace stops a command after two hours, and it did, after 11 of 14 flights (6,916 s), so the run left its lines but no verdicts and no json.

Kind: INSTRUMENT (a long run saved only at its end). *Lesson:* The sixth lesson of 28 October from the three-body work, measure a small run before a long one and save a long run in pieces that resume, was carried into the closeout's tolerance audit for the second computer's run kits but not into this cloud-workspace tool. Redesign, said aloud: each flight saved as it finishes in a piece of its own, the tool reading its own pieces when started again (tool md5 085077deb61a6461b736ac795e3e3107); and a run's time estimated from a flight of its own kind. *Proved?* Seen in runs (TRAIN_PARTING 20261029_030532).

476. TRAIN_PARTING's (G1r), the rival: at gap 0.565 some ring that swings 20 radii or more from the nearer of the others comes back within 3 radii. Measured here on eleven trains: every return from no farther than 13.78: MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* A ring that goes past about 14 radii does not come back in this toy; the six that left on their first swing and the three that left after two or four returns were 30.9 to 1,126.0 radii out at 20,000, each at its farthest. Redesign, said aloud: needs none. *Proved?* Seen in runs (TRAIN_PARTING 20261029_112955).

477. TRAIN_PARTING's (G2), Claude's lean (odds 1 in 2): read to the parting of a ring that is gone, a train near its edge lives at least 1.3 times as long as read to its first 5-radius spread. Measured here: 12,012.4 against 11,357.8, a ratio of 1.058: MISSED.

Kind: WRONG GUESS (Claude's, from TRAIN_BEAT's 5 returns in 11 seen in a look of 1,000). *Lesson:* Most rings leave on their first wide swing (6 of 9), and the rest come back only once or twice in a pair of equal swings before they go, a few thousand time units in all; so the lifetimes the book reads to the first 5-radius spread stand, at this gap, within about 6 per cent of a parting for good. Redesign, said aloud: none for the book's numbers, which are said so; the same reading at 0.530, where the long lifetimes were measured, proposed for the second computer with its pieces saved. *Proved?* Seen in runs (TRAIN_PARTING 20261029_112955).

478. The reader's edition, read cold and whole by a fresh reader who had seen none of the work (29 October, at Bear's asking for a full cold read like the one *A Rider in Two Waves* had): 100 findings, 17 wrong or misleading, 55 confusing and 28 minor. Among the 17: (a) 333 missed predictions beside 477 misses rows, never explained, and 'every miss has a row of its own' not shown (204 of the 362 missed or undecided link to a row by run and label); (b) '27 rules' beside 26 rows in chapter 15, the register's rules 26 and 27 never carried and the maintenance alarm never registered; (c) Feigenbaum's constant on the reference map, his own road, led on the front as a meeting 'by another road'; (d) the front's 'eight breaks' where chapter 9 reads 46; (e) the front's 'before the tool existed' where a replacement prediction came after its tool; (f) 'within 1 per cent' beside a stroke 1.4 to 5.3 per cent off; (g) the firmer kick's 2.00 called 'short of' a bar of 1.80, where it was the gentler kick's 1.75 that missed; (h) chapter 11's 0.124 billed as an oar's, which chapter 3 calls untested; and nine more, each in the cold read's findings, kept with the book's checks.

Kind: WRONG READING (the book's prose, not the runs). *Lesson:* Checks aimed at one chapter, one claim or one new paragraph at a time do not see two places that disagree, a word used chapters before it is defined, or a front page that outruns its appendices; only a whole read by someone new does. And a reader's fix is a proposal: of the 100, 10 saw a real problem and proposed a fix the record does not support (the two 9/4 ranges were both right, for different memories; 'floating-point actuator' is the trade's own name; the half wave and the three quarters belong to two toys), so each was checked against the record first (the cold read's triage, kept beside its findings). Redesign, said aloud:

every count the book prints in two places is now read from one script where it can be (the rules now 28 from the register), and a cold read before upload is the closeout's standing step. *Proved?* Seen in the cold read; every fix checked against the record and the new wording read against it by a second fresh reader.

479. TRAIN_PARTING_EDGE's kit, quoted about two to three hours for its 18 flights on the second computer. Five hours in, one flight had finished (nudge 61, 1,672.6 s), and Bear asked whether it was stuck. It was working: the estimate was wrong.

Kind: INSTRUMENT (the run's time, Claude's estimate). *Lesson:* The estimate was measured on flights that end at a break (TRAIN_LONG_EDGE's) and on a smoke run whose ring was gone at 8 radii, and checked for the far field's thirtyfold slowing; but this tool flies on past the first 5-radius spread, while the third ring swings out and back around a pair, and there the flight goes five to seven times slower (nudge 61: 16,692 time units before the spread in about 440 to 550 s at TRAIN_LONG_EDGE's rate of 30.2 to 38.4 a second, the 6,724 after it in the remaining 1,120 to 1,240 s). The lesson already drawn, time a flight of the run's own kind, was applied to the wrong part of the flight. Redesign, said aloud: none for this run, which Bear let run on; for any later kit, time one whole flight of the kind, through every regime it reaches, before quoting, and print a line every so often while a flight is in the air, so that a slow flight can be told from a stuck one. *Proved?* Seen in the run (TRAIN_PARTING_EDGE 20261029_070800, in flight).

480. TRAIN_PARTING_EDGE's (P1), Claude's lean (odds 1 in 2): at gap 0.530 too, read to the parting for good, the lifetime is under 1.10 times the lifetime read to the first 5-radius spread. Measured on the second computer: 178,763.2 against 105,119.8, a ratio of 1.701 (middle 95 per cent of resamples 1.054 to 3.395): MISSED; the rival, at least 1.3, held.

Kind: WRONG GUESS (Claude's, from TRAIN_PARTING's 1.058 at 0.565). *Lesson:* At 0.565 most shed rings left on their first wide swing, so the first spread was nearly the parting; at 0.530 a ring that comes back can keep the train whole for thousands of time units more (16,145 in one train, 74,849 in TRAIN_LONG_EDGE's planted train), and three of ten trains that spread were whole again at 100,000. The lean carried one gap's answer to another without asking whether the returns last longer where the train is tighter. So the book's lifetimes at 0.530 and below, read to the first spread, are the short side of a parting for good, and are now said so. Redesign, said aloud: none for the measurement, which stands with its error bar; to narrow the ratio, more copies at 0.530 on the second computer, the planted trains stopped at their first spread, a line printed while each flight is in the air (misses row 479). *Proved?* Seen in runs (TRAIN_PARTING_EDGE 20261029_070800; its reading script of 29 October, kept with the book's checks).

481. The record's first pass at the misses' working names, which gives a registered test by the tool that ran it, gave a test named just before its tool as that tool twice, in ten rows (226, 229, 233, 358, 366, 381, 389, 455, 463 and 464), and a test named before its run in brackets as the tool beside itself in three more (282, 443 and 445).

Kind: THE INSTRUMENT (a rule written for one word order). *Lesson:* A rule that replaces a name is read for every shape it can make, not only the one it was written for: this one assumed the test's number always came after its tool. Redesign, said aloud and done the same day: a rule for the number named first, a rule for the tool named beside its own run, and a check for any name doubled, run over every row before the build, none left. *Proved?* Seen in the scrub's own check of 30 October, before any build.

482. Misses row 479's reason for the slow kit, read again from TRAIN_PARTING_EDGE 20261029_070800's own flight times: a flight was said to go five to seven times slower once a ring swings out. But three trains that never spread at all (nudges 60, 62 and 65) took 18,555 to 18,962 s in the run's first five hours, and three of the same kind (nudges 68, 69 and 73) took 3,429 to 3,730 s after them, flying 100,000 time units at 5.3 a second against 26.8 to 29.2: the same flight, 5.2 times slower early. Flights past their spread, timed in the second half, ran 0.7 to 4.2 times slower a time unit than whole ones, not five to seven, and nudge 61, the one flight finished in the first half hour, kept the second half's pace.

Kind: WRONG READING (Claude's: a slowness laid to the flight that was the computer's). *Lesson:* Before laying a slowness to the physics, set like beside like: the same kind of flight at two times of the same run. The kit's own estimate, two to three hours, was near the work it held: at the second half's pace the run would have taken about two of the 6.4 hours it took. Redesign, said aloud and done the same day: TRAIN_PARTING_EDGE_2's time is set from the second half's flights, and it prints a line for each flight every 10,000 time units with the seconds it has flown, so a slow computer shows as it happens, and a computer asleep shows as a gap. *Proved?* Seen in runs (TRAIN_PARTING_EDGE 20261029_070800: its flight times and the order they finished in).

483. Building the Torus of Fractions, its builder first checked the needle's staircase, as read from STAIR_WIDTHS 20261013_184033's table of every fraction met, against the same run's count of locked paces by q, taking q there for the bottom of each fraction. The check stopped the build: the count gave 224 paces to q 3 where 2/3 alone holds 511, and 347 to q 6 where 5/6 holds 60. The run counts each pace by the least number of drive cycles after which the needle repeats, and 287 of 2/3's paces repeat only every six; the table also prints the whole turn as '1', which the first reading did not take for 1/1.

Kind: WRONG READING (Claude's: a column read by its label, not by the run's own definition). *Lesson:* Before checking one table against another, read how each one counts: the same letter can name two different things in one run file. The check was right to stop the build, and what stopped it was a finding: the doubled paces, a band of them inside 2/3 and a cascade of 7, 14 and 28 cycles from both edges of 5/7, kept in the scan's saved data and never looked at until then. Redesign, done the same day: the builder checks the table's 3,340 locked paces against the run's own total and, with its 1,161 unlocked, against the scan's 4,501; reads the whole turn as 1/1; and asserts 59 distinct locks. The doubled paces are counted by script from the saved scan and told on the page. *Proved?* Seen in runs (STAIR_WIDTHS 20261013_184033: its count of locked paces by least repeat, its table of every fraction met, and its saved scan).

484. The doubled band inside 2/3, told on the Torus of Fractions as running from pace 1.03300 to 1.04775 with plain three-cycle paces on both sides, and the 5/7 step's doubling, told as 7, then 14, then 28 cycles, were read from the staircase scan's lock rule, which waits 2,000 cycles. Re-flown before the book could tell them, with the same needle and rule: the band's middle is doubled beyond doubt (at pace 1.04000 the two rounds differ by 0.87 radian), but at its ends the test as written missed. At 1.03300 the three-cycle split was 3.9e-7 at 2,000 cycles and 1.2e-9 at 8,000, the plain loop still settling, and so were 1.04770 and 1.04775; the band runs from 1.03320 to 1.04750, its split growing as the square root of the distance from pace 1.03319 (split squared against pace, R squared 0.9998 over nine paces). In 5/7 the 14-cycle paces hold at 8,000 cycles (splits 0.065 to 0.28), but the two 28-cycle paces, 1.05505 and 1.06025, settle to no repeat at all, so the third rung and the claim that two of the scan's step-backs sit on it were not shown.

Kind: WRONG READING (Claude's: a lock rule's verdict after a fixed wait taken for the needle's own, at the edge of a doubling). *Lesson:* Near a period doubling the plain loop settles slowly, each round on the other side of the last, so a rule that waits a fixed time reads it as doubled; and a lone pace at the end of a cascade is the least settled of all. Before a doubled band is told, fly its edge paces longer and read how the split grows: a square root of the distance from one pace places the edge, and a split of a few billionths that shrinks when flown longer is the plain loop. Redesign, done the same day: the edges flown at 8,000 cycles and the square-root law fitted; the page and the book tell the band from 1.0332 to 1.0475 and the 5/7 doubling as 7 and then 14. *Proved?* Seen in runs (STAIR_WIDTHS 20261013_184033's saved scan, its paces re-flown with its own needle and rule at 2,000 and 8,000 cycles).

485. TRAIN_PARTING_EDGE_2's (Q1), Claude's lean (odds 3 in 4): at gap 0.530, read to the parting for good, thirty-two fresh trains live at least 1.3 times their lifetime read to the first 5-radius spread. Measured on the second computer: 117,750.1 against 104,446.2, a ratio of 1.127: MISSED; the rival, under 1.15, held.

Kind: WRONG GUESS (Claude's, from the first sixteen trains' 1.701, a small sample). *Lesson:* A small sample flatters: sixteen trains with seven partings gave 1.70, thirty-two with nineteen gave 1.13, and the first sixteen's own resampled range already ran from 1.05 to 3.4; the lean took that range's middle for the answer. Pooled, the forty-eight trains, with twenty-six partings, give 1.282, the middle 95 per cent of 20,000 resamples 1.093 to 1.586. Redesign, said aloud and done the same day: the pooling; the book tells the parting for good at 0.530 as about 28 per cent later than the first wide spread, with that range, and more trains would narrow it further. *Proved?* Seen in runs (TRAIN_PARTING_EDGE_2 20261029_211416, pooled with TRAIN_PARTING_EDGE 20261029_070800, read by a script written apart from the tool).

486. TRAIN_PARTING_EDGE_2's (Q2), Claude's lean (odds 4 in 5): no ring comes back from 14 radii or more. Of 196 returns, six came back from farther, in three equal pairs: 19.31 and 19.27, 17.05 and 17.09, 16.46 and 16.46 radii: MISSED.

Kind: WRONG GUESS (Claude's: the farthest of 78 returns, 13.75, read as a fence). *Lesson:* The farthest of a sample is where the sample stopped, not a wall: over all forty-eight trains the farthest return is 19.31 radii, and the reaches of the 137 pairs thin out beyond 6 radii as a power, about the minus third, with no edge, so the farthest return grows with the number of trains flown. The rings that left are another kind of swing: they reached 50 radii 890 to 10,252 time units after starting out, where a ring pulled home as the returns are would take about 19,863 to get there by the far returns' law [corrected at the next sitting, the same day: first written 41,568, from the law over every return; row 488]. So there is no fence in distance, but a split in how a ring leaves. Redesign, said aloud and done the same day: the farthest return over all forty-eight, and the fence read again as a thin tail beside a separate road out. *Proved?* Seen in runs (TRAIN_PARTING_EDGE_2 20261029_211416 and TRAIN_PARTING_EDGE 20261029_070800, read by a script written apart from the tool).

487. A test registered on the two parting runs, Claude's long shot (odds 1 in 3): if a ring that comes back is held by a pull falling as distance to some power, its time out and the thinning of the reaches both tell that power, so the reaches beyond 6 radii should thin out as a power within 1 of what the time law over every return gives (its power 3.037, asking about -4.07). Measured: -2.97 +- 0.40 from 54 pairs: MISSED.

Kind: WRONG GUESS (Claude's: a law fitted over near and far swings alike taken for the far law). *Lesson:* The time law over every return mixes near swings, where the clock's fixed marks at 5 radii out and 3 back weigh most, with far ones. Read on the far returns alone, the 48 beyond 8 radii, the time out grows as the reach to the power 2.587 (R squared 0.9998), which gives the pull's power as 3.17, beside the reaches' 2.97: two readings agreeing on a pull that

falls as about the cube of the distance, the power at which two distant vortex rings interact, as two small magnets do. Fit a law where its picture holds, not over every point. Redesign, registered after this miss and before it ran, held the same day: the far returns' law, its power between 2.2 and 2.8. *Proved?* Seen in runs (TRAIN_PARTING_EDGE_2 20261029_211416 and TRAIN_PARTING_EDGE 20261029_070800, read by a script written apart from the tool).

488. Beside TRAIN_PARTING_EDGE_2's grading, the time a ring pulled home would need to reach 50 radii, set against the rings that left, was printed as about 41,568: half the out-and-back of the time law fitted over every return (power 3.037). The same grading had just shown that law to overstate the far swings (row 487); read on the far law, the one fitted where its picture holds (power 2.587), it is about 19,863, and the slowest ring that left, at 10,252, went about twice as fast where the first figure said four times. Caught at the next sitting, the same day, when a picture of the two roads was drawn from the far law and its mark at 50 radii did not match the number in the prose.

Kind: WRONG READING (Claude's: a number carried over from a law the same day's test had just set aside). *Lesson:* A lesson learned is applied first to one's own last numbers: every number printed beside a fit says which fit it came from, and when a fit is replaced, every number drawn from the old one is read again the same day. Drawing the picture from the data, not from the prose, is what caught it. Redesign, done the same day: the number corrected in every record that carried it, the grading script printing both laws' figures side by side; the conclusion stands, the two roads about fourteen times apart at the median, and the slowest ring to leave now looks closer to the line between them (proposed: that ring flown again past 50 radii on the second computer). *Proved?* Seen in runs (TRAIN_PARTING_EDGE_2 20261029_211416 and TRAIN_PARTING_EDGE 20261029_070800, read again by a script written apart from the tool).

489. The draft of the abstract and the Zenodo description, written on 23 October for the deposit, said that each toy reruns the book's numbers before it shows a picture; the skater's page has no such check, as the book's own chapter on the toys says. And its counts had been typed in from the register that day, so by 30 October they were a week stale: 495 tools where the register holds 502, 1,070 predictions judged where it holds 1,098, 46 meetings with the published record where it holds 48; it named four toys where six now travel, and two editions where the book is now one, with a companion record. Found when Bear asked for the draft to mark.

Kind: WRONG READING (Claude's: a toy described as the writer remembered the set, and counts typed in). *Lesson:* A text written for the front of a deposit is held to the record as hard as the book: an 'each' is a claim with a count behind it, and a count typed in stops moving (row 379's lesson, met again in a draft kept outside the build). Redesign, done the same day: the draft is now built by script from the register, its counts filled at every build; the skater's sentence says what is true now, with a note to change it once the skater's check exists; the sixth toy and the one book with its record written in. *Proved?* Seen in the draft against the register and the book's chapter on the toys, read side by side.

490. The CASCADE_57 check of the 5/7 step, Claude's lean (odds 3 in 5): flown from rest on the step's high side every millionth of pace, the needle's repeats run 7, 14, 28, 56 and 112 in that order as the pace falls, each in one unbroken band. Measured: 7, then a 14 band broken five times near its top by a hidden 11/15 that repeats every 15, 30 and 60 cycles, then 28, 56 and 112 unbroken: MISSED; the rival, a band broken by another repeat, held. The doubling itself held up: placed by the square-root law, the bands shrink by 3.945 and then 4.483 (the lean on Feigenbaum's 4.669 held).

Kind: NATURE (a hidden lock inside a doubling band) and WRONG GUESS (Claude's: a band pictured as one repeat from edge to edge). *Lesson:* Flown from rest, a scan meets whichever loop its start falls into, and near the top of the 14 band a second loop, 11/15, lives at the same paces with a little doubling of its own. A band's edges as a scan meets them are not its width; its onsets are. Redesign, said aloud: covered; fly the 11/15 paces from twenty starting angles to see whether 11/15 and 5/7 share them, as 3/5 and 2/3 share theirs, and read 11/15's own doubling (proposed; run on 30 October as the HIDDEN_WINDOWS check: 11/15 met from some of twenty starts at 34 of 44 paces, always beside 5/7, its own doubling 15, 30, 60 as the pace rises; row 493). *Proved?* Seen in runs (the CASCADE_57 check of 30 October, the high side).

491. The TWO_LOOPS_MAP check at pace 1.03295, where two loops share the pace, Claude's long shot (odds 1 in 3): the starts that settle on the hidden 3/5 form one connected region on a map of starting angle against starting speed, the angle wrapping round. Measured on 3,960 starts, 120 angles by 33 speeds: 771 settle on 3/5 (19.5 per cent, the lean of 10 to 30 per cent held), in 215 separate regions, the largest 216 starts: MISSED; the rival, more than three regions, held.

Kind: WRONG GUESS (Claude's: a basin pictured as one island). *Lesson:* The 3/5 starts lie in thin diagonal stripes that wind round the cylinder of angles, with one broad tongue; a grid a 120th of a turn wide cuts a thin stripe into pieces, so the count of regions says as much about the grid as about the basin. Redesign, said aloud: covered; map a window of the stripes at a finer grid and read the share of starts whose neighbours disagree at two grid sizes, which tells thin connected stripes from a boundary of fractal kind (proposed; run on 30 October as the BASIN_BOUNDARY and BASIN_FINE checks, the share read at six sizes from 0.03 to 0.0001 radian: a boundary of fractal kind, its uncertainty exponent 0.406 and 0.423; row 492). *Proved?* Seen in runs (the TWO_LOOPS_MAP check of 30 October).

492. The BASIN_BOUNDARY check at pace 1.03295, where 2/3 and a hidden 3/5 share the pace, Claude's lean (odds 1 in 2): 600 starts drawn in a window of the striped part of the map, each flown again with its angle moved by eps, the share whose loop changes falling as eps to a power alpha from 0.8 to 1.2, thin stripes with smooth edges. Measured: 22.2, 16.3, 11.3 and 5.3 per cent at eps 0.03, 0.01, 0.003 and 0.001 radian, alpha 0.406: MISSED; the rival, under 0.6, a boundary of fractal kind, held, and the long shot, under 0.5, held.

Kind: WRONG GUESS (Claude's: thin stripes pictured with smooth edges). *Lesson:* A boundary that looks like a few clean stripes on a coarse map can be of fractal kind: at every finer eps a share of starts still lies near enough to it to change loop, falling only as eps to the power 0.4, so its dimension on the plane of angle and speed is about 1.6. Boundaries of this kind between a forced pendulum's coexisting loops are in the published record (McDonald, Grebogi, Ott and Yorke 1985). Redesign, said aloud: covered; eps two decades further down, to 0.0001, on 1,800 fresh starts, to see whether the boundary smooths below the stripes' width (run on 30 October as the BASIN_FINE check: alpha 0.423 from 0.001 to 0.0001 and 0.418 over all six eps; the lean that it stays under 0.6 held, and the long shot that it lies within 0.1 of 0.406 held). *Proved?* Seen in runs (the BASIN_BOUNDARY and BASIN_FINE checks of 30 October).

493. The HIDDEN_WINDOWS check, the hidden 11/15 and 11/16 flown from twenty starting angles at every millionth of pace through their windows, 1,320 flights of 32,000 cycles, Claude's long shot (odds 1 in 3): some start settles on a third loop, neither 5/7 nor the hidden lock. None did: MISSED. Both leans held: 5/7 is met from some start at every pace where the hidden lock is met from rest (11/15 at 6 such paces, 11/16 at 8), and each hidden lock's repeats grow steadily with pace, 11/15's through 15, 30 and 60 as the pace rises, 11/16's through 16, 32 and 64 as it falls.

Kind: WRONG GUESS (Claude's: a third state pictured as a third loop). *Lesson:* The hidden locks live wider than a scan from rest shows (11/15 at 34 of 44 paces, always on 1 to 4 starts of 20; 11/16 at 11 of 22), and each one's doubling runs against its host's: on the high side 5/7 doubles as the pace falls and 11/15 as it rises; on the low side the reverse. Past each hidden lock's last doubling a few starts met no repeat at all, beside a settled 5/7: the long shot's third state was there, but not as a loop. Redesign, said aloud: covered; fly the starts that met no repeat twice as long, read them to a repeat of 960, and give each an exponent from a twin flown beside it (run on 30 October as the HIDDEN_CHAOS check: all 29 still met no repeat; 28 of their 29 exponents positive). *Proved?* Seen in runs (the HIDDEN_WINDOWS check of 30 October).

494. The TWO_LOOPS_ZOOM check at pace 1.03295, the map of the two loops' starts widened from speeds -1 to 1 to speeds -3 to 3, 4,704 starts (96 angles by 49 speeds), Claude's long shot (odds 1 in 3): some start settles on neither 2/3 nor the hidden 3/5, a third loop or tumbling. Every start settled on one of the two: MISSED. The lean held: 867 starts on 3/5, 18.4 per cent, within five points of the inner map's 19.5.

Kind: WRONG GUESS (Claude's: a wider map pictured as reaching other states). *Lesson:* The two loops share the whole wider window: the 3/5 starts run in slanting stripes that keep winding round the cylinder of angles out to speeds of 3, the broad tongue near the middle continuing as the longest stripes, their share by speed between 7 and 25 starts of 96, thinnest from speed 1.5 to 1.875. Redesign, said aloud: not covered; map one band of speed near 1.5 on a finer grid in angle, to see whether the thinning is a narrowing of the stripes or a gap between them (proposed). *Proved?* Seen in runs (the TWO_LOOPS_ZOOM check of 30 October).

495. The HIDDEN_CHAOS check of the 29 starts that met no repeat past the hidden locks' doublings, and of a 5/7 control at each of their 8 paces: flown 64,000 cycles, read to a repeat of 960, and each given an exponent from a twin flown beside it for 1,500 drive periods. Claude's lean K1 held: all 29 still met no repeat. Three missed: the lean K2 (odds 3 in 5), every one of them above +0.005 per drive period, missed by one start at 1.060519 reading -0.00227, the other 28 from +0.0024 to +0.0434; the planted M1, every control below -0.001, missed by the high side's controls, -0.00080 to -0.00115, which sit a few millionths of pace from 5/7's 7-to-14 onset; and the long shot K3, one of them a hidden loop longer than 240 cycles: none repeats at all.

Kind: NATURE (exponents near zero at the end of a doubling) and WRONG GUESS (Claude's: bars set without minding where the doublings end). *Lesson:* Below the low side's 11/16 the states that never repeat are chaos beside a settled 5/7: all 22 exponents positive, +0.002 to +0.043, while 5/7 at the same paces reads -0.031 to -0.046, and they shrink toward the hidden lock's last doubling, as an exponent does near the end of a cascade. On the high side, near both 5/7's own onset and the end of 11/15's doubling, every exponent is near zero, and one reading over 1,500 periods cannot give its sign. Redesign, said aloud: covered; the high side's seven starts and three controls read again over 15,000 periods in ten segments, with an error from their spread (run on 30 October as the HIDDEN_CHAOS_LONG check, whose own misses are row 497). *Proved?* Seen in runs (the HIDDEN_CHAOS check of 30 October).

496. Two judges written for the checks of 30 October misread before any record was written. The WINDOWS_ORDER judge of the low side took the doubling's direction from the first pace of the 112-cycle band, which lies seven paces before the fitted onset (the paces nearest an onset settle slowest), so it looked the wrong way, met no windows, and printed its lean O1 MISSED and its long shot O2 UNDECIDED; and it measured O2 from the 112-to-224 onset, where O2's registered words name the 56-to-112. The TWO_LOOPS_ZOOM judge kept its share as a numpy number, so its lean printed UNDECIDED though 18.4 per cent lies inside its bar of 14.5 to 24.5, and its json failed to write. Mended:

a second judge beside the first, changed only in those two places, its difference read line by line (O1 and O2 HELD, O2's edge at 0.4041 of a band against the high side's 0.4046); one cast to a plain number, and the zoom flown again for its picture, every count identical. The first outputs are kept beside the mended ones.

Kind: THE INSTRUMENT (Claude's: two judges trusted untried). *Lesson:* A judge is an instrument and fails like one. Try it first on the case already seen (here the high side, whose order was known before the low side was read), and print beside each verdict the number it judged by, so that a verdict and its number can be read together; a verdict printed from a value of a kind the judge did not expect should stop the run rather than print UNDECIDED. Redesign, said aloud: covered; the mended judges run, the first outputs kept beside them. *Proved?* Seen in runs (the WINDOWS_ORDER and TWO_LOOPS_ZOOM checks of 30 October).

497. The HIDDEN_CHAOS_LONG check: the high side's seven no-repeat starts and three 5/7 controls flown 64,000 cycles, read to a repeat of 1,920, and given exponents over 15,000 drive periods in ten segments, each with a standard error from the segments' spread. Four missed. Claude's lean J1 (odds 3 in 5), the start at 1.060519 from angle 0 above zero by two standard errors: it read +0.00084 +- 0.00059, and its rival, below zero by two, missed with it; the long shot J2, that start a loop of 1,920 cycles or less: none; the planted M1, each control below zero by two standard errors: -0.00047, -0.00010 and -0.00041, each +- 0.00034; the planted M2, the other six starts above zero by two: five held, +0.0114 to +0.0156, and the sixth, at 1.060519 from angle 19, read +0.00063 +- 0.00046.

Kind: THE INSTRUMENT (a flight too short for a sign so near zero) and WRONG GUESS (Claude's: the segments' spread taken for the error of a loop's exponent). *Lesson:* At 1.060520 and 1.060521 the no-repeat states are chaos, 14 to 44 standard errors above zero, beside a 5/7 loop. At 1.060519 the state's exponent lies within a thousandth of zero over 15,000 periods and it is not a loop of 1,920 cycles or less: chaos at its weakest, just past the end of 11/15's doubling, or a loop longer than the reading. For a settled loop the segments' spread is not an error at all: a segment that ends part way round the loop gains or loses part of one turn's stretching, a few parts in 1,500 (Claude's reading of the spread, not yet checked), so a loop's exponent is read over whole repeats. Redesign, said aloud: not covered; read each loop's exponent over a whole number of its repeats, and place 11/15's own doubling onsets by the square-root law, following its loop from pace to pace, so that the end of its doubling is known and 1.060519 set on one side of it (proposed). *Proved?* Seen in runs (the HIDDEN_CHAOS_LONG check of 30 October).

498. The MSS_TABLE_CHECK of 30 October, which set the logistic map's superstable parameters found on the bench against Metropolis, Stein and Stein's own tables: all 21 values of its Table I agreed to seven figures with their periods, yet the check printed C1 UNDECIDED, because its truth value was a numpy one, the very fault mended an hour before in another judge (row 496). Mended as before, beside the first, the first output kept; and this time with a verdict rule that stops the run on anything but a plain True, False or None, written into this judge and into the next check of the day.

Kind: THE INSTRUMENT (Claude's: a fault fixed where it was met and not where it came from). *Lesson:* A fault mended in one script is a fault mended once. Row 496's redesign was said aloud and done for its two judges, but the verdict printer it came from was copied fresh into the next script, and the fault came with it. The redesign that holds changes the shared part: one verdict rule, strict, carried into every judge written after it. Redesign, said aloud: covered; the strict rule written into MSS_TABLE_CHECK and LOGISTIC_GAPS the same hour, to go into every judge after them. *Proved?* Seen in runs (the MSS_TABLE_CHECK of 30 October).

499. The LOGISTIC_GAPS check of 30 October, the logistic map's superstable parameters of periods 2 to 512 bisected on the bench, to ask whether its band ratios climb to Feigenbaum's 4.669 as the needle's do, each gap about 0.217 of the last. Claude's lean G1 held: the logistic gaps change sign (its first ratio, 4.6808, lies above 4.6692). The long shot G2 (odds 1 in 4), the last two gap factors within 10 per cent of 0.217: they read 0.239 and 0.201: MISSED, the first just beyond the bar.

Kind: WRONG GUESS (Claude's: a settled factor expected of a map whose factors still swing). *Lesson:* The logistic map's gap factors swing, -0.540, 0.128, 0.311, 0.179, 0.239, 0.201, closing round about a fifth; the needle's sit still at 0.217 and 0.218 on its two sides, and R2W's rider's at about a quarter, as its session quoted them. Whether all three close on one factor, and whether that factor is 1/4.669 = 0.214, is open. Redesign, said aloud: not covered; carry the logistic map to period 8,192 in extended precision and read where its factor settles, and set beside it the needle's fifth ratio, its 448 band placed by the square-root law on a grid finer again (proposed). *Proved?* Seen in runs (the LOGISTIC_GAPS check of 30 October).

500. The LOGISTIC_DEEP check of 30 October, the redesign of row 499: the logistic map's superstable parameters carried to period 8,192 in 60-digit arithmetic. Claude's lean L1 held: the last three gap factors lie from 0.200 to 0.230 (0.20988, 0.21670, 0.21273). The long shot L2 (odds 1 in 4), the last factor within half a per cent of 1/delta = 0.214169: it read 0.21273, 0.67 per cent off: MISSED.

Kind: WRONG GUESS (Claude's: a limit asked of a sequence still swinging round it). *Lesson:* Carried deep, the logistic map's factors swing round about 0.214 with each swing smaller than the last: 0.2016, 0.2219, 0.2099, 0.2167, 0.2127, their distance from 1/delta shrinking by a little over half a step and changing sign each time. Row 499's reading that the steady climb belongs to the toys holds only in part: the map heads for the same factor the needle sits at, 0.217 on

both its sides, and swings on the way. Redesign, said aloud: not covered; fit the swinging factors as a limit plus a term that shrinks and changes sign each step, and read the limit with its error against 1/delta; and carry the needle to its sixth ratio to see whether its factor moves from 0.217 towards 0.214 (proposed). *Proved?* Seen in runs (the LOGISTIC_DEEP check of 30 October).

Misses 501 to 550

501. The MSS_FORECAST check of 30 October: three windows the 1973 order names and no scan had met, forecast on the needle's high side by a map from the logistic map's superstable r to the needle's paces, a cubic through six windows already met (each inner one forecast within 0.0000000035 when left out), and flown every 0.0000000001. Two held: the lean M1, a window repeating every 560 between the 6 and the 8, met at 8 paces, its middle 0.0000000024 from the forecast; and the long shot M3, a window repeating every 616 between the 8 and the 9, met at one pace, 0.0000000013 from the forecast. Claude's lean M2 (odds 1 in 2), a second window repeating every 560 between the 8 and the 9 within 0.00000001 of 1.0602012061: no repeat of 560 at any of the 201 paces: MISSED; and with it the long shot M4, both 560 windows' middles within 0.000000004 of their forecasts.

Kind: WRONG GUESS (Claude's: a window's place forecast where its neighbour's was met, its width or its settling not known). *Lesson:* The same map that put the 11 within 0.0000000013 missed the 10 beside it, so the map is not the likely fault. The logistic map gives this 10 a width of 0.0000576 of r, about 0.0000000004 of pace on the needle, two to four paces of this grid; a window that narrow opens from chaos, and 32,000 cycles from rest may not be enough to settle into it, or its period-10 part may be narrower than its doublings, which repeat every 1,120, above this check's ceiling of 640. Redesign, said aloud: covered; fly the forecast again from 0.00000002 below to 0.00000002 above every 0.00000000005, 128,000 cycles, read to a repeat of 1,300, from several starting angles (proposed; run on 30 October as the MSS_FORECAST_2 check, 0.00000001 each side every 0.0000000001 from three angles: not met; and its planted check showed the 11 window met here to be a quiet stretch of chaos, not a window, the long shot M3's reading withdrawn; rows 502 and 503). *Proved?* Seen in runs (the MSS_FORECAST check of 30 October).

502. The MSS_FORECAST_2 check of 30 October, row 501's redesign: the second 10 window forecast at 1.0602012061, flown every 0.0000000001 over 201 paces from three starting angles, 128,000 cycles, read to a repeat of 1,300. Claude's lean R1 (odds 1 in 2), a repeat of 560 or 1,120 met within 0.00000001 of the forecast: none at any of 603 flights: MISSED, its rival held; the long shot R2 missed with it. And the planted N1 missed: at 1.0602011539, where MSS_FORECAST met the 11 window repeating every 616 after 32,000 cycles, the needle flown from rest for 128,000 cycles met no repeat at all.

Kind: NATURE (a quiet stretch of chaos beside a window looks like the window for thousands of cycles) and THE INSTRUMENT (Claude's: a settle of 32,000 cycles and a read of 1,300 taken as enough to tell them apart). *Lesson:* Near the edge of a narrow window the needle's chaos can linger by the window's ghost, nearly repeating, for a long stretch before it bursts away; read inside such a stretch, a repeat rule calls it the window. The 11 met at one pace was such a stretch: MSS_FORECAST's long shot M3 held by its own rule, and the reading is withdrawn. The second 10 may be narrower than 0.0000000001 of pace, or a ghost only. Redesign, said aloud: covered; every window the record claims past the 5/7 doubling flown again for 128,000 cycles (run as the WINDOWS_LONG check, row 503); and from now on a window is claimed only where it holds at two or more paces after 128,000 cycles. *Proved?* Seen in runs (the MSS_FORECAST_2 check of 30 October).

503. The WINDOWS_LONG check of 30 October: every window met past the 5/7 doubling after 32,000 cycles flown again from rest for 128,000, read to a repeat of 1,300. Claude's lean L2 held: the 11 met at one pace shows no repeat there or at the four paces beside it. Claude's lean L1 (odds 3 in 4), every window met at three paces or more repeating at its first repeat again at the middle pace of its run: MISSED, its rival held. Each of the five windows survives, at its first repeat or at that repeat doubled, but narrower: MSS_FORECAST's 10 at 3 of its 8 paces (560, 1,120, 560); WINDOWS_ORDER's 10 at 4 of 5 and its 7 at 2 of 3; SCALED_COPY's 7 on the low side at 1 of 3 (its first pace, not its middle) and its 10 at 2 of 3, both repeating every 1,120, the 10 doubled.

Kind: NATURE (windows narrower than a short settle shows them) and WRONG GUESS (Claude's: a window's width read at the first look taken as its width). *Lesson:* A window's edges as met after a short settle include the quiet stretches beside it, so the first look's widths are upper bounds and its middles can shift; the windows themselves held. Redesign, said aloud: not covered; one window flown at settles of 32,000, 64,000, 128,000 and 256,000 cycles, to see its width settle and to set the settle the record's windows need (proposed). *Proved?* Seen in runs (the WINDOWS_LONG check of 30 October).

504. The FACTOR_FIT check of 31 October (UTC), Bear's question whether 1/4.669 governs the road's closing: the logistic map's superstable r carried to period 262,144 in 80-digit arithmetic, its gaps to Feigenbaum's delta fitted four at a time by two shrink factors. The leans F1, F2, F3 and F5 and the planted N1 to N3 held: the first factor 0.214169 against 1/delta 0.214169, the second -0.123652 against the period-doubling operator's second eigenvalue -0.1236527 (Christiansen, Cvitanovic and Rugh 1990); on the dial mu = r(r-2)/4 the last three factors 0.214190, 0.214158 and 0.214176; the sine map 0.214169 and -0.123795. The long shot F4 (odds 1 in 2), the dial straightened by c* =

-0.01524729 found from periods up to 1,024 only, its first four out-of-sample factors all within 0.03 of -0.1237: they read -0.11826, -0.13329, -0.10816 and -0.15430, the last 0.0307 off: MISSED; its rival, all four near +0.214, missed too.

Kind: WRONG GUESS (Claude's: a band of 0.03 set without reckoning the bend the straightening leaves behind). *Lesson:* Straightened, the closing by 1/delta stopped: the gaps changed sign at each of the fifteen steps from period 4 to period 131,072, and the four-gap fit's second factor read within 0.00003 of -0.12365 from period 4,096 on. But the 1/delta part was not removed entirely: the fit's first factor climbed back from 0.03 to 0.214 by period 16,384, and since that part shrinks more slowly than the swing (0.214 a step against 0.124) its share grows each step, pushing the successive factors further either side of -0.124 (-0.118, -0.133, -0.108, -0.154, -0.081, -0.236). Redesign, said aloud: not covered; c* found from periods up to 16,384, or the verdict read as the sign changes and the fit's second factor rather than each factor's distance (proposed). *Proved?* Seen in runs (the FACTOR_FIT check of 31 October).

505. The switching converter (the CARD14_CONVERTER check of 31 October): with the output held, the planted check C3 of a proof (every one of 400 settings from v = 20.5 to 40 V chaotic, no repeat up to 64, the exponent positive) failed at its last setting alone: at v = 40 V exactly, the slope -3 branch, with I_ref 10 A and a start at 9 A, the current repeated 7, 8, 9, 10 A every 4 laps with no lap on the sloped branch, its exponent read as 0, while the twin trajectory beside it read +0.549, half of ln 3.

Kind: THE INSTRUMENT (Claude's: round numbers at a scan's end point; whole-ampere steps land in floating point exactly on the switch's border). *Lesson:* The orbit is real but sits on the border itself, the switch opening at the very tick; any nudge leaves it, so it is no window. A planted scan should not end on a setting where every number is round. Redesign, said aloud: covered; the same 400 settings flown from 9 + 1/pi A, and again with I_ref 10 + 1/(10 sqrt 2) A: no repeat at any setting, the smallest exponent 0.0326 a lap (the CARD14_HELD_REDO check, its C3b held). *Proved?* Seen in runs (the CARD14_CONVERTER and CARD14_HELD_REDO checks of 31 October).

506. The switching converter (the CARD14_CONVERTER check): Claude's lean C4 (odds 3 in 5), with the capacitor in, the first doubling within 10 per cent of the I_ref where the period-1 lock's own duty reaches one half, 2.2319 A: the scan met period 2 first at 1.9100 A, 14 per cent low: MISSED.

Kind: WRONG GUESS (Claude's: the held-output rule, duty one half, carried to a converter whose output swings) and THE INSTRUMENT (the scan, following its last state upward, called slow settling period 2; the refining step, starting each try from rest, never left period 1, so the onset it printed, 1.9100, is the scan's grid point). *Lesson:* The capacitor's swing joins the current's, and the lock is lost before duty one half. Read after the run, not registered (the CARD14_EIGEN_READ exploration): the period-1 lap's eigenvalue, found by Newton, is -0.9954 at 1.910 A, -0.9982 at 1.915 and -1.0067 at 1.930, so the flip is near 1.918 A, at a duty near 0.458; the miss stands either way. Redesign, said aloud: not covered; place each smooth doubling where the repeat's own eigenvalue crosses -1, not by settling (proposed). *Proved?* Seen in runs (the CARD14_CONVERTER check of 31 October).

507. The switching converter (the CARD14_CONVERTER check): Claude's lean C5 (odds 2 in 3), at most two doublings before the first chaotic setting as I_ref rises from 0.5 to 5 A: three came, period 1 to 2 (the scan's 1.9100 A), 2 to 4 at 2.428358 A and 4 to 8 at the scan's 2.5825 A, with chaos from 2.5925 A and one spacing ratio, 3.36: MISSED; its rival, four or more doublings closing on 4.669, missed too.

Kind: WRONG GUESS (Claude's: the road read as shorter than it is). *Lesson:* The converter's road mixes two kinds of doubling. Read after the run, not registered (the CARD14_EIGEN_READ exploration): the first and third are smooth flips, a repeat's eigenvalue passing -1 (near 1.918 A; and near 2.590 A, where the period-4 repeat's eigenvalue reads -0.9896 at 2.585 and -1.0011 at 2.590); the second is a border collision, the period-2 repeat's on-time reaching the whole lap at 2.4284 A, where its eigenvalues jump from a complex pair of size 0.55 to 0.307 and -1.395. No ratio of the needle's kind can be read on such a road: Feigenbaum's delta belongs to a road of smooth doublings, and this one is cut short at 8. Redesign, said aloud: not covered; the 4-to-8 flip and the start of chaos placed finely between 2.58 and 2.60 A, to see whether a 16 lives there (proposed). *Proved?* Seen in runs (the CARD14_CONVERTER check of 31 October).

508. The SHUKRINOV_WINDOWS check of 31 October: its lean W2 held (the logistic's 10, 6, 10 and 8 of Table II's order all in the two-band chaos, the 24 and 12 below 3.5926), but its locator for the two-band chaos's upper end, the last chaotic setting whose orbit's alternate halves stay apart, read 3.7553 where the bands merge near 3.6786 (from recall), so the bound W2 was judged by was looser than designed.

Kind: THE INSTRUMENT (Claude's: 32 samples of each half taken as enough to tell two bands from one). *Lesson:* Near a window the chaos lingers by the window's ghost, and a few samples of each half can fall apart by chance well past the bands' merging. The four windows lie at 3.6052, 3.6266, 3.6470 and 3.6621, below 3.6786 too, so W2 stands on the recall value; the locator does not. Redesign, said aloud: not covered; find the merging where the gap between the two bands, read from thousands of iterates, closes (proposed). *Proved?* Seen in runs (the SHUKRINOV_WINDOWS check of 31 October).

509. The CARD14_LINES check of 31 October, Bear's surprise: he drew four straight lines over the converter's chaos and wrote 'not chaos. patterns in reference current clearly continue'. The planted N1 and N2 and Claude's lean B1 held: the top edge is I_ref itself, never passed; the converter is chaotic at all five settings (exponent +0.22 to +0.48 a lap); and each of Bear's lines lies within 0.12 A of one image of the switch's border at every setting (the farthest 0.100 A). Claude's lean B2 (odds 1 in 2), images 1 to 3 thin, under 0.05 A between quartiles at every setting: thin at 3.0 A (0.008 to 0.012) but 0.06 to 0.10 at 3.5 and 4.0 A and 0.22 to 0.28 at 4.5 and 5.0 A: MISSED.

Kind: WRONG GUESS (Claude's: the border's images taken as single curves, where in two dimensions each carries the output's spread). *Lesson:* The lines Bear saw are real and are the chaos's skeleton: the border, a lap that ends exactly at I_ref, and its images, one lap with the switch open at once and then a whole lap with it closed, which adds exactly E T/L = 1 A, so his middle line is his bottom edge plus one ampere (1.700 and 2.700 A at 3.0 A; 2.820 and 3.820 A at 5.0 A). But the capacitor's voltage is part of the state, and as the chaos widens with I_ref its spread smears each image: the lines stay sharp-edged where the logistic map's would, but thicken. Redesign, said aloud: not covered; draw the images as bands (quartiles) over the picture and read their thickness against the output's spread lap by lap (proposed). *Proved?* Seen in runs (the CARD14_LINES check of 31 October).

510. The CARD61_LOGBOOK check of 31 October, the logbook of a toy trainer: N1, P1 and P2 held (the elevator's spread grows with the gusts, ratios 2.02, 3.04 and 3.93; the hinge's events met Rice's level-crossing law, 761 an hour against 704.6 in calm air, and in rougher air the counts forecast from the calm hour alone were met within 1 per cent: 3,178, 4,163 and 4,616 against 3,158, 4,169 and 4,594). Claude's lean P3 (odds 2 in 3), a tach-hour interval for the hinge leaving more than half its life unused on at least two of the four aircraft: on one only, the calm cross-country one (0.852); the rough, the trainer and the mixed left 0.100, 0.225 and 0.146: MISSED (the engine's 0.100 on all four and the gear's three of four held).

Kind: WRONG GUESS (Claude's: the hinge's clock taken to keep speeding up with the gusts). *Lesson:* Rice's law has a knee: once the gusts swing the elevator well past its dead band, the events come at the elevator's own crossing rate, set by the pitch loop and not by the air, a ceiling of about 5,200 an hour here, and the roughly flown one already ran at 89 per cent of it. So a hinge's event clock runs slow only in calm air and keeps near step with tach time everywhere else, and it is the gentlest-flown aircraft whose tach-hour schedule throws hinge life away. Redesign, said aloud: not covered; count wear, not crossings: weight each swing by its size to a power (each swing's damage growing with its size, Miner's rule and a fatigue power law, from recall), so the hinge's damage per tach hour keeps growing with the gusts, and judge the same lean on it (proposed). *Proved?* Seen in runs (the CARD61_LOGBOOK check of 31 October).

511. The LANE_ORDER check of 31 October, the order in which a rider visits its four lanes, read from READER-2's recorded lanes: Claude's lean O1 (odds 3 in 5), that in every column from 300 to 780 both riders alternate between the lanes the corner pairs by sign, {0, 2} and {1, 3}, at every step: in none of the 481 columns: MISSED; its rival held.

Kind: WRONG GUESS (Claude's: the corner's pairs read as alternate steps of the dance). *Lesson:* In every rider and every column from 109 to 830 the dance goes round the same cycle, lane 3, then 1, then 2, then 0: the lanes that bend the same way at a doubling are visited one right after the other (3 then 1, 2 then 0), and the bend changes sign across the dance's other two steps (1 to 2, 0 to 3). Redesign, said aloud: not covered; test whether each step of the dance keeps or flips the bend as the sign of the map's stretch along that step says, and whether the corner sizes' ratios along the cycle follow the same stretches (proposed). *Proved?* Seen in runs (the LANE_ORDER check of 31 October).

512. The TOP_CORNER_LONG check of 31 October, the slow alternation beside the top doubling (columns 815 to 829, flown 32,256 passings): the planted N1 to N4 held (the first window equal to READER's own strobes to the last bit) and Claude's lean T35 held (the top corner's leftover 0.1225, 0.0598 and 0.027 at 8,000, 16,000 and 32,000 passings). Claude's lean T36 (odds 2 in 3), the alternation falling as the dance's multiplier says in every judged column: it did where the alternation stood clear (column 818, -0.000820 and -0.000832 read against the log of the multiplier's size, -0.000824; column 819 at 16,000, -0.001668 against -0.001666), but further out it stopped falling at 1e-8 to 4e-7 and read far too slow: MISSED, and its rival missed too.

Kind: THE INSTRUMENT (Claude's: a floor of 1e-11 set for numbers of size one). *Lesson:* The rider's angle is carried unwrapped, two turns every four passings, so after 32,000 passings it is about 1e5 radians and its last bit about 1e-11; the dance's slow alternation, a multiplier of -0.992 to -0.999, gathers that rounding into a standing alternation of 1e-8 to 4e-7, larger the nearer the doubling. The law itself held at all three readings clear of that floor. Redesign, said aloud: not covered; wrap the angle at every passing, keeping the turns as a count, and judge only readings a hundred times above the floor measured at a far column (proposed). *Proved?* Seen in runs (the TOP_CORNER_LONG check of 31 October).

513. Two planted fidelity checks of 31 October, TOP_CORNER_WRAP's N1 and LOBE_FLIP_FINE's N1: the new runs, the angle wrapped every passing, were to repeat the old runs' strobes at 8,000 passings within 1e-9; they repeat them within 2.4e-8 and 3.0e-8: MISSED, both.

Kind: THE INSTRUMENT (Claude's: an old run taken as exact where it carried the very fault the new run removes). *Lesson:* The old runs carried the angle unwrapped, and by 8,000 passings it was some 25,000 radians, its last bit about 4e-12, which the dance's slowest sway gathers to the 1e-8 the floor of misses row 512 showed. The new runs

agree with the old to the old runs' own error, as they should. A fidelity check against an old run must allow that run's own known error, or be made against a run free of it. Redesign, said aloud: covered; the wrapped run's own lanes equal the old saved lanes at every dot (both N2 checks held), and its floor is measured directly (TOP_CORNER_WRAP's W1). *Proved?* Seen in runs (the TOP_CORNER_WRAP and LOBE_FLIP_FINE checks of 31 October).

514. The TOP_CORNER_WRAP check of 31 October, misses row 512's redesign: W1 held (with the angle wrapped every passing the floor fell from 1e-8 to between 3.0e-12 and 4.3e-12 in columns 826 to 829). Claude's lean W2 (odds 2 in 3), the alternation falling as the multiplier says in every column 818 to 826 whose alternation exceeds 1e-10 at both 8,000 and 32,000 passings, at least four such columns: only two qualified, so MISSED by its own count, though both held (column 818: -0.000826 and -0.000832 against -0.000824; column 819: -0.001666 and -0.001666 against -0.001666).

Kind: WRONG GUESS (Claude's: the number of columns that could be judged set without reckoning how fast the alternation falls once the floor is gone). *Lesson:* With the floor gone the alternation fell so fast that by 32,000 passings most columns had reached the new floor. Read after the data, not a test: between 8,000 and 16,000 passings the fall in columns 820, 821 and 822 is -0.002509, -0.003352 and -0.004198 a passing against ln of the multiplier's size -0.002509, -0.003352 and -0.004197. Redesign, said aloud: covered in substance by that reading; to be judged, register the fall between 8,000 and 16,000 passings wherever both readings stand a hundred times above the floor (proposed). *Proved?* Seen in runs (the TOP_CORNER_WRAP check of 31 October).

515. The SPRING_KNEE_COUNT check of 31 October, the spring's count at heavy loss (tune 1, loss 1): the planted C0 held (JUMP_STORE's counts and pushes repeated to four places, the loop equal to its own). Claude's lean C1 (odds 3 in 4), each knee's count from a fifth down to a fortieth of the running speed free of the step within 5 per cent between steps 0.0025 and 0.00125: the fifth, tenth and twentieth moved 0.3, 1.0 and 2.4 per cent, the fortieth 5.2: MISSED.

Kind: WRONG GUESS (Claude's: a count read as settled from the size of its last change, without the order of its approach). *Lesson:* Each count climbs toward its limit by a change that halves as the step halves (at the tenth: 2.5602, 2.6462, 2.6993, 2.7266, rises of 0.086, 0.053 and 0.027), an approach of the first order, so the last change is the size of the error still left, not a sign it is gone: the tenth's count at step 0.01, called free of the step in JUMP_STORE, lies about 7 per cent below its limit, near 2.75 by Richardson's step. Every knee has a limit; the smaller the knee the finer the step it needs. Redesign, said aloud: covered in substance by the run's own four steps; to judge a count's limit, carry three steps or more and read it by Richardson's step, or fine the step until the change falls below the bar's half (proposed). *Proved?* Seen in runs (the SPRING_KNEE_COUNT check of 31 October).

516. The SPRING_TRIGGERS check of 31 October, the floored spring's trigger set as a share of each bead's own idle swing (misses row 298's redesign). Its B1, B2 and B3 held (awake to the end at every share from a quarter to one and a quarter, at both springs; the push per event under half the clean listener's). Claude's lean B4 (odds 2 in 3), the push per event rising at every step of share from 0.25 to 1.25 at both springs: at the first spring (tune 1, loss 0.05) it rose at every step, 0.0582 to 0.0645; at the second (tune 1.5, loss 0.2) it stood flat, 0.1004, 0.1035, 0.1041, 0.1033, 0.1040, 0.1033: MISSED.

Kind: WRONG GUESS (Claude's: a rise read in SPRING_FLOOR's three fixed triggers taken for a law at both springs). *Lesson:* At the faster drain the floored latch carries the same push per event at every share, within 4 per cent, as the memoryless crest-firer did at every threshold (misses row 242, flat within 3 per cent, as Rice's crossing count says); at the slower drain it gains 11 per cent from a quarter to one and a quarter. What it carries per event, the card's question, is read in the run's table. Needs no redesign. *Proved?* Seen in runs (the SPRING_TRIGGERS check of 31 October).

517. The LATCH_WALL check of 31 October, the spring latch on its own memory flown at h from 1.00 to 1.50 every 0.05: its planted W0, W3 and W4 held. Claude's lean W1 (odds 2 in 3), the wall (the largest h awake to the end) within 0.1 of a steady swing's crest seen through the memory, at every memory up to a quarter of the drain: the formula, registered unevaluated, gives the square root of two at every memory, and the walls read 1.35, 1.35, 1.35, 1.30 and 1.20 at memories 0.25 to 5 on the first spring and 1.35, 1.30 and 1.25 at 0.25 to 1 on the second: MISSED at memories 2 and 5 on the first spring and 0.5 and 1 on the second.

Kind: WRONG GUESS (Claude's: a formula registered without its value computed, and a steady swing taken for the swing that decides). *Lesson:* A steady swing's lag through a memory is exactly the cosine of its phase, so the memory moves where in the swing the speed peaks over it but never how high: the square root of two at every memory. The wall is set by the swing that has gone quiet and is dying, whose memory lags above it; worked out after the run, its ceiling is the square root of 2 (1 - loss times memory), and nine of this run's ten walls sit at the largest h of its grid at or below that ceiling, the tenth one step lower. Redesign, said aloud: covered, the dying swing's ceiling registered and flown on springs never flown so, on a grid five times finer (LATCH_CEILING; misses row 519). *Proved?* Seen in runs (the LATCH_WALL check of 31 October).

518. The LATCH_WALL check of 31 October: Claude's lean W2 (odds 3 in 4), the wall rising with memory from 0.25 to 2 at the first spring: it fell, 1.35, 1.35, 1.35 and 1.30: MISSED.

Kind: WRONG GUESS (Claude's: a short memory thought to see the speed itself and so hold the wall at one, where the steady swing's own formula says the square root of two at every memory). *Lesson:* The fall is the dying swing's: the longer the memory, the more it lags above a swing that is fading, and the lower the bar the swing can still meet, as the square root of 2 (1 - loss times memory) says. Redesign, said aloud: covered by LATCH_CEILING (misses row 517). *Proved?* Seen in runs (the LATCH_WALL check of 31 October).

519. The LATCH_CEILING check of 31 October, misses rows 517 and 518's redesign: the dying swing's ceiling, the square root of 2 (1 - loss times memory), flown on five springs at fourteen memories with h every 0.01: its planted L0 held, and L2 held (no latch awake to the end at the ceiling plus 0.01 or above, at any of the fourteen). Claude's lean L1 (odds 2 in 3), every wall within 0.04 below its ceiling: ten of fourteen walls are the last h of the grid below the ceiling and two the one before it, but at the two settings where loss times memory is exactly one half the ceiling is exactly 1 and the latch stays awake at h 1.00, at the ceiling itself, which the bar's 'below' excludes: MISSED.

Kind: THE INSTRUMENT (Claude's: a strict bar set where the forecast falls exactly on a grid value). *Lesson:* At loss times memory one half, a dying swing's speed meets its memory exactly at the crest; a bar exactly there is met. The law held at all fourteen settings within two hundredths. A bar must allow equality where the forecast can fall on the grid, or the grid be set off it. Needs no redesign: L2 and the run's table cover it. *Proved?* Seen in runs (the LATCH_CEILING check of 31 October).

520. The CORNER_TWO_SCALE check of 31 October's first surprise: the two-scale guess for the corner ratio, one over alpha's size less one over its size to the power z, set against two over the ratio's size times delta, at tops of order 8, 10 and 12. Its planted T0 held (order 6 repeated to 1e-9). Claude's lean T1 (odds 1 in 2), the guess's error shrinking on past order 6 (0.49 per cent there): it reads -0.78, -1.48 and -1.90 per cent at orders 8, 10 and 12: MISSED, and the rival T1r held: the error changed sign between orders 6 and 8.

Kind: WRONG GUESS (Claude's: a small error at one point read as convergence). *Lesson:* The guess crossed the truth near order 6 and moved away on the other side; its error over five orders (13.3, 10.0, 5.6, 3.1, 0.49 per cent) was a falling line through zero, not a curve settling on it. The two-scale picture catches the ratio's size to a few per cent at every top tried and no better. Redesign, said aloud: not covered; the corner ratio read exactly from the doubling operator's fixed point (the eigenvalue of the operator that carries an orbit average's change from one generation to the next), at every order (proposed). *Proved?* Seen in runs (the CORNER_TWO_SCALE check of 31 October).

521. The CEILING_BEYOND check of 31 October's second surprise: the dying swing's ceiling past loss times memory one half, and free of the spring's tune. Its planted C0 held; C2 held (no latch awake to the end at the ceiling plus 0.01 or above, at six settings); C3 held (at tunes 0.5, 1, 2 and 3, loss 0.1, memory 2, every wall 1.26 under the ceiling 1.2649: the ceiling is free of the tune, as the proof says). Claude's lean C1 (odds 2 in 3), every wall past one half at most 0.02 below its ceiling: three of six were (0.89, 0.89 and 0.70), but at loss 0.05, memory 15 the wall stood at 0.65 under 0.7071, at loss 0.1, memory 9 at 0.37 under 0.4472, and at loss 0.05, memory 18 no h from 0.30 was awake to the end: MISSED.

Kind: THE INSTRUMENT (Claude's: an awake test's window, the run's last 300 time units, set shorter than the time the memory takes to forget a louder past). *Lesson:* The memory carries the swing's louder past, from when the blade was on, and that part fades at the rate 1/tau - loss only, over tau/(1 - loss tau), 60, 90 and 180 time units at the three settings that missed; a bar just under the ceiling is met only after it fades, so a living latch can go a 300-unit window without a switch-on. Redesign, said aloud: covered; flown five times longer, the last tenth 1,500 time units, all three walls came to the last hundredth under their ceilings, 0.70, 0.44 and 0.44, and none stood a hundredth above (the CEILING_SLOW check of 31 October). *Proved?* Seen in runs (the CEILING_BEYOND and CEILING_SLOW checks of 31 October).

522. The COUNT_LAW check of 31 October's third surprise: the held spring's click count forecast before its flight as C over the knee plus the switch-ons outside the stays, both read from a held run at the same loss. Its planted K0 held, and K1 held (loss 1, a knee of 1/320: 53.13 flown against 54.54 forecast). Claude's lean K2 (odds 1 in 2), within 5 per cent at losses 0.7 and 1.5 with a knee of 1/40: at loss 0.7, 8.397 against 8.370 (0.3 per cent); at loss 1.5, 2.712 against 2.038, a third over: MISSED.

Kind: WRONG GUESS (Claude's: a law read where the stays carry the count forecast where they carry little). *Lesson:* At loss 1.5 the held stretches shrink (C 0.034 a soup cycle against 0.168 at loss 1 and 0.187 at 0.7), and the knee ear switches on about 0.67 times a soup cycle more than the stays and the free crossings account for; read after the run, the knee's own band lets a flicker go on a little past the ideal stay's edge, where the bead's own loss holds it back, a share that grows with the loss. Redesign, said aloud: not covered; sort the knee ear's switch-ons at loss 1.5 by where they fall, inside an ideal stay, within the knee's band of one, or in free motion, and forecast the second kind from the loss times the band (proposed). *Proved?* Seen in runs (the COUNT_LAW check of 31 October).

523. The CORNER_CUSP check of 31 October's first surprise from Bear's picture of the corner as a data sink: an orbit average's steps between neighbouring superstable parameters. Its planted P0 held; P2 held (round top: the steps shrink by 4.6677 against delta's 4.6692, the slopes settle at -0.17601, their differences shrink by -1.96468 against R's -1.9644); P3 held (quartic top: the slopes' differences shrink by -0.60236 against -0.6033). Claude's lean P1 (odds 2 in

3), at the quartic top the deepest step ratio within 1 per cent of R delta = -4.3946 and the slopes growing at each of the last four steps: the slopes grew, -2.94, 4.56, -7.91, 12.72 and -21.53, but the step ratio swung -4.71, -4.20, -4.53, -4.30 and its last stood 2.1 per cent off: MISSED.

Kind: WRONG GUESS (Claude's: a 1 per cent bar on the raw last value of a sequence that swings slowly about its limit). *Lesson:* The step ratio swings about its limit with the class's slow alternation (about 0.7 a generation at the quartic top); Aitken's step on its last three reads -4.3958 against -4.3946, 0.03 per cent. The cusp itself is plain in the slopes, which swing from side to side ever more steeply, by 1.66 a generation, the reciprocal of the corner ratio. Redesign, said aloud: covered in substance by Aitken's step, read after the data; to be judged, register the bar on Aitken's step or on the swing's centre (proposed). *Proved?* Seen in runs (the CORNER_CUSP check of 31 October).

524. The QUARTIC_SECOND check of 31 October's second surprise: a second quartic-topped map, f(x) = 1 - r sin(pi x / 2)^4, to see whether -0.603 is the class's. Its planted Q0 held. Claude's lean Q1 (odds 3 in 4), its corner ratio within 0.003 of -0.60326: it swung about -0.28 (Aitken's step -0.2793), and its doublings closed by 10.95, not 7.28: MISSED.

Kind: THE INSTRUMENT (Claude's: a test map chosen whose top's value, f(0) = 1, lands on its own second turning point, the bottom at x = 1). *Lesson:* The cycle passes the quartic top and then the quadratic bottom in a row, and two turns in a row act as one of order 4 times 2 = 8: delta at order 8 is 10.957 and R there -0.2787 (CORNER_TWO_SCALE), against this map's 10.9485 and -0.2793. A clean test map has one turning point in reach. Redesign, said aloud: covered; map A of QUARTIC_THIRD, 1 - r (x^4 + 0.3 x^6), one turning point, gives -0.60323 against -0.60326 and delta 7.2848 (its T1 held). *Proved?* Seen in runs (the QUARTIC_SECOND check of 31 October).

525. The QUARTIC_SECOND check of 31 October: Claude's lean Q2 (odds 1 in 2), the two quartic maps' ratios drawing together by 5 to 9 a generation: the differences stood at 0.30 to 0.35 and did not shrink: MISSED.

Kind: THE INSTRUMENT (the same as row 524: the second map was of another class). *Lesson:* Two maps of different classes keep their difference; the step to a clean map is row 524's. Needs no separate redesign: QUARTIC_THIRD's T3 asked it again on the clean map (misses row 527). *Proved?* Seen in runs (the QUARTIC_SECOND check of 31 October).

526. The QUARTIC_THIRD check of 31 October: the reading that missed row 524, tested: map B, f(x) = 1 - r sin(pi x / 2)^2, a quadratic top whose value lands on its quadratic bottom, should act as order 4. Claude's lean T2 (odds 2 in 3), its Aitken ratio within 0.005 of -0.60326 and its last gap ratio within 0.02 of 7.2847: the gap ratio came to 7.28468 (order 4's delta, 7.2846862, to the fifth figure), but the corner ratio still swung widely at the doubling of 1,024, -0.20, -1.15, -0.37, -0.84, -0.47, -0.71, Aitken's step -0.6179: MISSED by 0.015.

Kind: WRONG GUESS (Claude's: a composite map's corner taken to settle as fast as a plain one's). *Lesson:* The composite order holds for delta at once; the average's corner closes on the class's ratio with a swing that falls by only about two thirds a generation, the mean of x passing near two turning points. Redesign: the COMPOSITE_DEEP check of 31 October, map B flown to the doubling of 16,384: the swing's centre -0.5986, within 0.005 of -0.60326, Aitken's step -0.6042, the swing falling from 0.243 to 0.064 in four generations, the gap ratio 7.28468617: HELD. *Proved?* Seen in runs (the QUARTIC_THIRD check of 31 October).

527. The QUARTIC_THIRD check of 31 October: Claude's lean T3 (odds 1 in 2), map A's ratio less 1 - r x^4's shrinking at each of its last three steps: it fell from 0.064 to 2e-5 over eight generations, but not at every step (1.4e-5 then 7.5e-5 then -2.1e-5): MISSED.

Kind: WRONG GUESS (Claude's: a step-by-step bar on a difference that carries the class's slow swing). *Lesson:* The corner forgets the map, three-thousandfold in eight generations, about 2.7 a generation on the whole, but each map's own swing about the limit leaves the difference to wobble from one step to the next. Redesign, said aloud: judge the difference's envelope over the last four steps (proposed). *Proved?* Seen in runs (the QUARTIC_THIRD check of 31 October).

528. The TABLE_ROWS_FIVE_SIX check of 31 October's first surprise: Claude's lean T1 (odds 4 in 5), the published table's row 5 (delta 8.3494991320, abs(alpha) 1.5557712501; MathWorld's 'Feigenbaum Constant', citing Briggs 1991, Briggs and others 1991 and Finch 2003) given by Feigenbaum's own equation within 2e-10: alpha agreed (off 9.7e-11), but delta came to 8.3443847487, the same at 24 and 28 coefficients, off by -0.0051: MISSED. Row 6 held in both (off 7e-11 and 2e-11).

Kind: WRONG GUESS (Claude's: the order-3 entry taken for a lone slip). *Lesson:* The table's delta column is off at both odd orders it prints, 3 and 5, and right at both even ones, 4 and 6; its alpha column is right at all four. Redesign: the map flown, not the equation: the ORDER5_MAP check of 31 October, 1 - r abs(x)^5's superstable parameters to period 16,384, gap ratios 8.3488, 8.3459, 8.3449, 8.34456, 8.34444, 8.34440, 8.344391, 8.344387 and Aitken's step 8.3443849, alpha 1.5557713: HELD, the map with the equation, not the table. *Proved?* Seen in runs (the TABLE_ROWS_FIVE_SIX and ORDER5_MAP checks of 31 October).

529. The COLD_READ_SPREAD check of 31 October (a test of the record, for the Field Guide's question whether the cold read was given a number to stop at): Claude's lean R1 (odds 2 in 3), each third of the reader's edition cited by at least a fifth of the findings that cite a line: 50, 60 and 12 of 95, the eighth and ninth tenths cited once: MISSED.

Kind: WRONG GUESS (Claude's: a parser's count taken for the reader's attention). *Lesson:* The last third of the 29 October edition held Appendix B's rows, the notes on the run behind each number and the references, and the reader cited those by note and row, not by line. Redesign: the COLD_READ_SPREAD_B check of 31 October, notes, rows, the Declaration and the References counted: 22 findings reach the last third, 19 asked: HELD. The record shows no sign the read was cut short; the brief, found since word for word in the transcript of the session that launched the read, named no number. *Proved?* Seen in the record (the COLD_READ_SPREAD and COLD_READ_SPREAD_B checks of 31 October).

530. The BRIGGS_OWN_Z12 check of 31 October: one line of Briggs's own Table 2 (1991), read through a web reader from a free scanned copy that could not be downloaded here, 'delta_12 13.5350756661 7027649570 05633538'; Claude's lean B1 (odds 3 in 4), the Loop Atlas's solver within 1e-9 of it at order 12 at 24 and 28 coefficients: 13.53507570198 and 13.53507566878, off by 3.6e-8 and 2.6e-9: MISSED.

Kind: TOOL LIMIT (the expansion in abs(x)^(12 j) converges more slowly at high order; 28 coefficients too few for 1e-9). *Lesson:* A bar set for low orders does not carry to high ones: the coefficients a reading needs grow with the order. Redesign: the BRIGGS_OWN_Z12_C check of 31 October (its first build, _B, stopped at 32 coefficients, Newton not settling in 50 digits, no verdict), re-solved at z = 12 in 80 digits at 32, 36 and 40 coefficients: off by 8.8e-12, -1.5e-12 and -2.2e-14, the gap shrinking at every step: HELD; the solver meets Briggs's own line to fourteen figures. *Proved?* Seen in runs (the BRIGGS_OWN_Z12, _B and _C checks of 31 October).

531. The ONE_CURVE check of 31 October's second surprise: Claude's lean C1 (odds 3 in 4), every third difference of the map's delta at orders 2 to 12 under 0.01 in size: +0.075, +0.033, +0.020, +0.014, +0.010, +0.008, +0.007, +0.006: MISSED. (Its printed, not judged, third column carried a slip too: a json key tested as a number against keys that are strings, so the column meant to put the slip's odd orders beside the map's even repeated the map's, and its 'largest difference' read nothing.)

Kind: WRONG GUESS (Claude's: a bar on size where the claim was about shape). *Lesson:* The map's curve bends hard at low order and gently at high; one smooth curve shows as differences of one sign that shrink, not as small ones. Redesign: the ONE_CURVE_B check of 31 October, on a new grid, the half orders 2 to 12: every third difference positive and each no larger than the one before, +0.0201 down to +0.0006: HELD; and, the column repaired, the map's even orders with the slip's odd reproduce Briggs's printed column at every order 2 to 12 within 1e-7: HELD. *Proved?* Seen in runs (the ONE_CURVE and ONE_CURVE_B checks of 31 October).

532. The ODD_TABLE check of 31 October (the table of the note on the odd-order deltas): its planted N0, every even order's delta and alpha at 36 coefficients within 1e-12 of Briggs's printed values: at orders 2 to 10 they were (within 2.3e-14), but at order 12 the delta was 1.54e-12 short: MISSED.

Kind: TOOL LIMIT (a planted bar set without the truncation at the highest order, which the BRIGGS_OWN_Z12_C check had already measured: 1.5e-12 at 36 coefficients). *Lesson:* A planted check is only as good as the bar it is given; a bar that a run on the record had already shown out of reach is a slip in the registration, not news about the map. Redesign: covered by the BRIGGS_OWN_Z12_C check of 31 October, which at 40 coefficients meets Briggs's order-12 delta to 2.2e-14; the note's table prints order 12 to the nine places its two readings agree on. *Proved?* Seen in runs (the ODD_TABLE and BRIGGS_OWN_Z12_C checks of 31 October).

533. The MCCARTNEY_READ_JUDGE check of 31 October (McCartney's accepted manuscript of 2020, sent by Bear, read against predictions registered before it was opened): Claude's lean M2 (odds 4 in 5), his deltas at orders 5, 7 and 9 the map's to every figure he prints: order 7 was (10.15977986791), but at 5 and 9 his last printed figure is one higher than the map's rounded there (8.3443847487310 against 7309; 11.673105851 against 850): MISSED.

Kind: WRONG GUESS (Claude's: a converging reading's last figure taken for the limit's). *Lesson:* His values are his last superstable estimates, unextrapolated, approaching from above, printed to the places on which his last two agree; such a last figure runs one high. Redesign: covered by his own even orders, where Briggs's printed values and the Loop Atlas's agree and his last figure differs by one in the same way (one low at order 2, 4.6692016091029906718 against 6719; one high at 6 and 8, 9.296246832772 against 771, 10.9486242660 against 659); one place fewer, all nine of his integer orders are the map's. *Proved?* Seen in the record (the MCCARTNEY_READ_JUDGE check of 31 October).

534. The note on the odd-order deltas checked by the Field Guide's publishing tools (31 October): Claude's lean P1 (odds 1 in 2), lint_deposit reports no failure on the note: one FAIL, check 9, no tool and computational resource disclosure section: MISSED. (P2 held: build_paper v1.8 built it and read back 31 of 31 paragraphs and cells.)

Kind: WRONG READING (Claude's: the house form's declaration taken for the whole of the disclosure the rules ask). *Lesson:* A short note owes the same disclosure section as a book: tools used, responsibility, verification, attribution limits, computational resources. Redesign: the section written, the note rebuilt and linted again: CLEAN, all nine

checks. *Proved?* Seen in runs (lint_deposit of the publishing packet of 31 October, run twice).

535. The ODD_BLOCK check of 31 October: Claude's lean O1 (odds 2 in 3), at orders 4 and 5 the six largest eigenvalues of the doubling operator's odd block each within 1e-8 (relative) of -abs(alpha)^(z (1 - k)): the first three or four were, within 1e-16 to 2e-10, but the fifth and sixth only within 1e-8 to 1.6e-5 at 20 coefficients: MISSED.

Kind: TOOL LIMIT (one bar for six eigenvalues spanning five decades, at a size that resolves the small ones less well). *Lesson:* Small eigenvalues of a truncated operator are read less precisely than large ones; a bar for all must be set for the smallest. Redesign: the ODD_BLOCK_C check of 31 October (its first build, ODD_BLOCK_B, stopped at 36 coefficients in 50 digits, no verdict), at 20, 28 and 36 coefficients in 80 digits: every gap shrinks at every step, the sixth at order 5 from 1.6e-5 to 3.4e-10 to 3.3e-15, the first four within 2e-21 at 36: HELD; the block's eigenvalues are the powers of alpha, very likely the known eigenvalues that come from changes of coordinate (not checked). *Proved?* Seen in runs (the ODD_BLOCK, ODD_BLOCK_B and ODD_BLOCK_C checks of 31 October).

536. The note on the odd-order deltas read cold by a fresh reader given no target and no limit (31 October; its brief kept, NOTE_COLD_READ_BRIEF): Claude's lean R1 (odds 2 in 3), no number in the note's prose disagreeing with its Table 1 or another place: the reader logged 55 findings, 11 wrong or misleading, among them 'the gap grows at orders 7, 9 and 11', where the signed gaps are +0.117, -0.005, -0.062, -0.095 and -0.116: MISSED. Among the 11, one wrong in the explanation itself: the note said alpha comes from a part of the calculation that never looks left of zero, but the fixed-point equation does evaluate g at x/alpha.

Kind: WRONG GUESS (Claude's: a note written and checked number by number taken to be clean as a whole). *Lesson:* Whole reads catch what number-by-number checks do not: a summary sentence about the numbers, and a reason given for a result that the calculation itself contradicts. Redesign: the factual findings fixed at once (the gaps given signed; alpha's reason corrected, in the note and in the Guide's ledger; McCartney described with his own hedge; his last-place differences given order by order; Finch's later errata named as unread; the routes counted fairly), and all 55 to be triaged against the record before the note goes, then a second whole read (proposed). *Proved?* Seen in the record (NOTE_COLD_READ_FINDINGS_2026-10-31 and the EXCHANGE23_JUDGE check of 31 October).

537. The DK_DEPTH check of 31 October, a surprise of the day's twenty-third exchange: Claude's lean K1 (odds 1 in 3), one common depth at which the maps' superstable gap ratios round to all six of Delbourgo and Kenny's printed values (4.67, 6.08, 7.29, 8.35, 10.2, 12.3 at orders 2, 3, 4, 5, 7, 10): orders 2 to 7 all do at depths 4 and 5 (the ratio from periods 32 to 256), but order 10 only from depth 7 on (12.41 and 12.37 at depths 4 and 5): MISSED.

Kind: WRONG GUESS (Claude's: a single reading depth taken for a whole table's method). *Lesson:* Five of their six values are the maps' own ratios read part way, at one shared depth, their 7.29 and 8.35 included, so neither supports the published table's odd values; their order-10 value was read deeper or another way. Redesign: HELD for Delbourgo and Kenny's paper (Physical Review A 33, 3292-3302, 1986), whose method section would say how each value was read; not read here. *Proved?* Seen in runs (the DK_DEPTH check of 31 October).

538. The BRIGGS_1989_TABLE check of 31 October: Claude's lean B2 (odds 2 in 3), our order-2 superstable gap ratios within 1e-10 of Briggs's 1989 table (reprinted as his thesis's Appendix A) at every row from 2 to 13: within about 1e-15 to row 7, then apart about tenfold a row, 1.3e-10 at row 12 and 2.6e-9 at row 13: MISSED. (B1 held: every superstable parameter to all 10 printed decimals.)

Kind: TOOL LIMIT (the 1989 table's: its extended precision too short for the last rows, as its own note warns). *Lesson:* A printed table is right where its arithmetic holds; a bar for its last rows is set only after asking how it was computed. Redesign: the BRIGGS_1989_TABLE_B check (R2 held: the factor by which his ratios close on delta jumps to 4.875 at the last step, while ours stays near delta) and the BRIGGS_1989_TABLE_C check (R3 held: our ratios unmoved at 100 digits, to 1e-17, at every row): the drift is the 1989 table's. *Proved?* Seen in runs (the BRIGGS_1989_TABLE, BRIGGS_1989_TABLE_B and BRIGGS_1989_TABLE_C checks of 31 October).

539. The BRIGGS_1989_TABLE_B check of 31 October: Claude's lean R1 (odds 3 in 4), the factor by which our order-2 ratios' distance from delta shrinks lies within 0.01 of delta at every step from row 6 to 13: it swings around delta instead (4.731, 4.633, 4.690, 4.657, 4.676, 4.665, 4.6715), the swing shrinking about 0.57 a step: MISSED.

Kind: WRONG GUESS (Claude's: the leading rate taken for the whole approach, its correction to scaling left out). *Lesson:* A sequence of ratios approaches its limit with corrections of its own, known (Briggs's thesis, section 2.3; Briggs 1994); a test of convergence must allow for them or avoid them. Redesign: the BRIGGS_1989_TABLE_C check, which assumes no shape of approach (the ladder recomputed at 100 digits): HELD. *Proved?* Seen in runs (the BRIGGS_1989_TABLE_B and BRIGGS_1989_TABLE_C checks of 31 October).

540. The BRIGGS_1989_ORDER3 check of 31 October: Claude's lean D3 (odds 1 in 4), Briggs's 1989 recipe in 64-bit arithmetic at order 2 reproduces the 1989 table's row-13 drift in sign and within a factor 3 in size: +3.4e-10 against his +2.6e-9, the same sign but eight times smaller: MISSED. (D1 and D2 held: the same recipe at order 3 gives 6.084689, the maps' value, finding the right zero at every row.)

Kind: WRONG GUESS (Claude's: one row's rounding noise taken for a number a second run could repeat). *Lesson:* Rounding drift can be reproduced in size, not in value: its value depends on when an iteration stops. Redesign: the BRIGGS_1989_ORDER3_B check, the drift's size row by row under two stopping rules a 1989 program might use: HELD (within a factor 10 of the 1989 table's at rows 10 to 13 under both, its sign varying). *Proved?* Seen in runs (the BRIGGS_1989_ORDER3 and BRIGGS_1989_ORDER3_B checks of 31 October).

541. The ARCHIVE_COMPLETE check of 31 October: Claude's lean K1 (odds 3 in 4), every run the record volume cites by tool and stamp has its run file in the deposit's archive: 11 of 440 cited pairs had none; 2 were the check's own misreading (BLIND read as a tool's name; both runs are in the archive), 8 were first versions the record itself calls retired or set aside, kept in each working session's own scratch folder outside the crate and never carried by any crate (no manifest lists one), and 1 was the three-body work's own run: MISSED.

Kind: WRONG GUESS (Claude's: a book's citations taken to point only at files the crate holds). *Lesson:* A run set aside must travel with the book, or its citation must say that it does not. Redesign, said aloud: covered by a fix rather than a new prediction: the deposit allowlist now fails on any cited run missing from the archive unless the record's own sentence marks it retired, set aside or another project's (dry run: 9 listed so, none missing); and an index of every run in the archive, the retired ones named as not in it, proposed for the record volume. *Proved?* Seen in runs (the ARCHIVE_COMPLETE check of 31 October; the deposit allowlist's dry run).

542. The ARCHIVE_COMPLETE check of 31 October: Claude's lean K2 (odds 1 in 2), every run file in the archive is cited by the record volume or the reader's edition: 244 of 737 are cited by neither, 142 of them blind runs (each a run of its tool before the run of record, its judged lines printed only as yes or no; 23 of the 165 are cited by stamp, rows 544 and 545) and 102 others (smoke runs, figure builders, readers and checks, and the first week's runs, stamped by date alone): MISSED.

Kind: WRONG GUESS (Claude's: the book's citations taken for an index of the archive). *Lesson:* A book cites results, not files; a deposit needs an index of its own files. Redesign, said aloud: an index of every run in the archive, by tool, stamp and kind (run of record, blind, smoke, figure, check), built by script at every build into the record volume; a change to the volume's structure, asked of Bear first. *Proved?* Seen in runs (the ARCHIVE_COMPLETE check of 31 October).

543. The description of blind runs written on 31 October into row 542, a lesson for the Field Guide and a reading without physics: blind runs were 'made before each prediction was written, as witnesses, and by design never cited'. Read later that day against the book's methods section ('every tool has run blind before its run of record') and a tool's own header ('print the planted case whole and the judged lines only as yes or no'): a blind run is a run of the tool before its run of record, with its judged lines hidden and its planted checks shown; it checks the tool, not the prediction; and 23 of the 165 are cited by stamp in the record volume or the book (rows 544 and 545 give the count). Wrong on all three counts.

Kind: THE INSTRUMENT (Claude's record-keeping: a kind of file described from memory). *Lesson:* Corrected the same day in all three places, before the run index described any kind of run. The lesson: a kind of file is described from its maker's own header, read on the day, never from memory. *Proved?* Seen in the crate (a tool's header; the book's methods section; the archive check's own list).

544. The headline register's counts of blind runs and runs of record, and its ledger of verdict lines, as built since 18 October: a blind run was any run file whose name holds '*BLIND*'. Found on 31 October while the run index was being written, and checked by BLIND_MARK (registered first; its lean B1 held): two runs of record are named so by their own tools, NEVER_BLIND and NEVER_BLIND_2 (the rider that never taps blind), and carry no blind mark in their headers. So the book counted 167 blind runs where there are 165 and 539 runs of record where there are 541, and the eight verdict lines the two runs printed, four each (NEVER_BLIND 20261006_045833: two held, two missed; NEVER_BLIND_2 20261006_050123: one held, three missed), were never in the register's ledger, its count of predictions or this record's Part 1, which read 1,098 judged (733 held, 336 missed, 29 undecided) where the runs hold 1,106 (736, 341, 29); the misses they printed were in the misses file all along (row 100).

Kind: THE INSTRUMENT (the register's own rule; Claude's). *Lesson:* Redesigned and done the same day: a blind run is counted by both of its marks, the name's and the header's that the tool prints in a blind run, and the register rebuilt (the BLIND_MARK_B and BLIND_MARK_C checks). The lesson is row 543's: a kind of file is told by what its maker printed in it, not by a word in its name. *Proved?* Seen in runs (the BLIND_MARK checks of 31 October).

545. The BLIND_MARK check of 31 October: Claude's leans B2 (odds 3 in 4), counted by the mark '*BLIND*' followed by a stamp the crate holds 165 blind runs and the runs of record rise by exactly 2, and B3 (odds 1 in 2), 23 of the true blind runs are cited by either text: 166 and 1, and 24 cited, both MISSED, for one reason. The mark was fooled by the very tool that began the check: NEVER_BLIND's own name followed by its stamp reads '_BLIND_20261006'.

Kind: WRONG GUESS (Claude's: a mark in a name taken for the kind of file, row 544's fault made again inside its own check). *Lesson:* Redesign, said aloud and run the same day: count by both marks, the name's and the header's (BLIND_MARK_B: 165 blind runs, 541 runs of record, 23 of them cited; both leans held). A name says what a file is

called; only its contents say what it is. *Proved?* Proved by run (the BLIND_MARK_B check of 31 October).

546. The BLIND_MARK_C check of 31 October, judging the register's rebuild: Claude's leans C2 (odds 2 in 3), the registered predictions rise by exactly 4, to 1,102, all of them NEVER_BLIND_2's, the run of NEVER_BLIND adding none, and C4 (odds 2 in 3), of the count's rows only the two prediction rows, the runs of record that judged a prediction (up by 1), the misses and at most two part rows change: they rose by 8, to 1,106, four from each run, and the runs of record that judged a prediction by 2, both MISSED, for one reason. NEVER_BLIND's run had never been read: row 545's mark, fooled by that tool's name, also took its file out of the first check's reader, so the lean that it held no verdict was a guess about a file unread.

Kind: WRONG GUESS (Claude's: a file not read taken for one with nothing in it). *Lesson:* Said aloud: the rebuilt register's own counts stand, and the check printed all eight lines it added; a lean about a file is written after the file has been read, or says that it has not. *Proved?* Seen in runs (the BLIND_MARK_C check of 31 October).

547. The RUN_READS check of 31 October, on what every run of the archive reads: Claude's leans W2 (odds 2 in 3), at least one tool asks for a parent at a fingerprint no file of the archive carries, and W4 (a long shot, 1 in 4), every file a tool reads that the archive lacks is found somewhere in this crate: all 53 parents a tool finds by name and fingerprint are in the archive at the fingerprint asked, and one file was found nowhere, both MISSED. W2's miss is good news. W4's one file is a scratch file NEEDS_AUDIT writes for itself and deletes, flagged because the check's reader judged each line alone; the same reader found none of DRAFT_TEST's three known reads, whose names sit in a list on a line with no reading word.

Kind: THE INSTRUMENT (Claude's check: a reader that judged a line alone), and WRONG GUESS (Claude's, W2). *Lesson:* Redesign, said aloud and run the same day: every quoted file name anywhere in a tool, less those it writes (the RUN_READS_B check, row 548). A reader of what tools read is trusted only after it finds a read already known; this one's known case came back empty, and that was seen only because the output was read whole. *Proved?* Seen in runs (the RUN_READS check of 31 October).

548. The RUN_READS_B check of 31 October, the redesign of row 547: its planted case R1 (DRAFT_TEST's three jellyfish files found, NEEDS_AUDIT's scratch file not flagged, a figure tool's file pattern matched) MISSED, the solver's own name dropped because the name it was kept under shared a line with a file being written; and Claude's lean R3 (odds 2 in 3), every flagged name is found in this crate, MISSED on 15 flags, every one a file the tools write or a name in their prose, read so by hand. Read flag by flag, fifteen tools then in the archive read a file it does not hold, and every such file is in this crate or in the jellyfish paper's upload set.

Kind: THE INSTRUMENT (Claude's check: a write rule judged by the line, not by the name; caught by its own planted case). *Lesson:* Every flag read in its own tool and the readings kept beside the printed verdicts; the ten tools that still read a file the archive lacks are listed in the record volume's Part 4 with where each file is, and one more, the builder of an internal working file, no longer goes up (31 October). A planted case that fails is a fault caught, not a result. Owed: a third reader that drops a name only where that name itself is what a tool writes. *Proved?* Seen in runs and read by hand (the RUN_READS_B check of 31 October).

549. The NAME_RULES check of 31 October, on every rule that judges a file by its name: Claude's leans N2 (odds 3 in 4), the register's rules for the teaching engine's checks and the publishing tool's agree with their files' first lines, N3 (3 in 4), the deposit's rule for tools sends up a working helper among the tools with no launcher and no run, N4 (9 in 10), every blind run or smoke part the record volume's index calls cited borrows that citation from its run of record, and N6 (a long shot), a run file called of record is a smoke run by its first lines: all four MISSED. N2 on a keyword reader that took a toy's planted line, which names the engine's value, for a check of the engine; N3 because the helper, the builder of an internal working file, has a launcher, so the test as framed could not see it; N4 because 22 of the 28 credits are earned, blind runs that misses rows cite by stamps of their own, though six (one blind run and five smoke parts) are borrowed; N6 cleanly.

Kind: WRONG GUESS and THE INSTRUMENT (Claude's: keyword readers, and tests framed from a glimpse). *Lesson:* Each read by hand and the reading kept beside its verdict: the six borrowed citations taken out of the index, which now counts a blind run or smoke part as cited only where it is named itself; the helper kept out of the deposit; the rules for the engine and the publishing tool found right by reading. N2 redesigned and run the same day (the NAME_RULES_B check, row 550). A rule's own words are read before a test of it is framed. *Proved?* Seen in runs and read by hand (the NAME_RULES check of 31 October).

550. The NAME_RULES_B check of 31 October, reading each run's kind from its own first line: Claude's leans B2 (odds 1 in 2), at most 30 verdicts of runs that name a kind other than a toy model are counted as predictions about the toys, and B3 (3 in 4), the teaching engine's and the publishing tool's runs call themselves something other than a toy model: 33, and all 22 of the engine's runs with verdicts call themselves TOY MODEL, both MISSED. The house header stamps TOY MODEL on every run, so a first line cannot say what a run is any more than its name can. Read by hand, most of the 33 are reading tools judging the toys' own recorded data and rightly counted; five verdict lines of two checks of the kit and the records (NEEDS_AUDIT's N1 and N2, BENCHMARK_TAGS's S1a and S1b and its rival S1r) had been counted as predictions about the toys and as a rival.

Kind: WRONG GUESS (Claude's: a first line taken for the file's kind) and THE INSTRUMENT (the register's default class, which counted whatever it did not know as a prediction; Claude's). *Lesson:* Redesigned and done the same day: the register reads a list kept by hand, each tool with a reason quoted from its own docstring, and counts checks of the kit and the records on a line of their own (5, 3 held); the registered predictions about the toys are 1,102 (733 held, 340 missed, 29 undecided) where the book said 1,106 (736, 341, 29), the rivals 105 distinct where it said 106, and every count that reads the register moved with it. A file's kind is read by a reader from what its maker wrote about it, not from a header a template stamps on every run. *Proved?* Seen in runs (the NAME_RULES_B check and the register rebuilt, 31 October).

Misses 551 to 600

551. RAMP_SPEED's (S1), the runaway ramp against the speed coming in: a runner coming in at rest is caught by gear 1 at every push of every mass where both lanes hold. Measured: gear 2 catches some of them at every mass, 18 of 48 at mass 1.0, 54 of 144 at 1.5, 33 of 120 at 2.0 and 24 of 104 at 3.0: MISSED.

Kind: WRONG GUESS (Claude's: a runner at rest taken for one already in the slow lane). *Lesson:* A runner set down at rest at one of eight angles round the circle is not sitting in the slow lane's well: most fall into it, but a quarter to three-eighths slide through and are picked up by the passing lane on the way. Can it be redesigned, or is it covered? Covered in part by the same run's json, which keeps every runner's fate by angle; owed, a reading of which starting angles the passing lane takes from rest, and whether they are the angles past the hilltop. *Proved?* Seen in runs (RAMP_SPEED 20261031_232517).

552. RAMP_SPEED's (S2) and (S3): at every mass the passing lane catches a larger share of runners coming in at the clock's speed (0.6 times the pace) than at the pace, and from 1.1 to 1.6 times the pace the share stays within 0.10 of its share at the pace, the ramp forgetting how fast the runner came. Measured: the share rises with the speed coming in at every mass, with small dips: 0.583 at 0.6 against 0.667 at 1.0 at mass 1.0, and 0.806 at 1.6 against 0.597 at 1.0 at mass 1.5: both MISSED.

Kind: WRONG GUESS (Claude's: a runner slowing through the clock's speed taken to forget where it began). *Lesson:* The faster a runner comes in, the more often the passing lane catches it, at every weight from 1 to 3; the long shot (S4) held with it, more than half of the heaviest runners caught at 1.4 and 1.6 times the pace. Redesign, said aloud: record each runner's phase against the clock at the moment its speed first falls through the clock's speed, and read its fate against that phase and against how long it took to slow from its speed coming in, to see whether the speed coming in sets the phase of arrival. *Proved?* Seen in runs (RAMP_SPEED 20261031_232517).

553. The TOOL_KINDS check of 31 October, counting the tools the front page counts by what each tool's docstring calls itself: Claude's leans K1 (odds 2 in 3), fewer than 450 of them name themselves a toy model, K2 (1 in 2), at least 25 name themselves a figure tool, and K3 (a long shot), at least 20 name no kind: 453 of 503, 10 figure tools and 4 drawing tools, and 6 with no kind, all three MISSED.

Kind: WRONG GUESS (Claude's: three counts guessed before a count). *Lesson:* Needs no redesign: the count is the answer. By their own words 453 of the 503 tools are toy models and 50 are figure, drawing, reading, checking, build or launching tools, or name no kind; the front page counts all 503 as small model tools, a presentation question set before the author. *Proved?* Seen in runs (the TOOL_KINDS check of 31 October).

554. HALT_TRADE's (T2), the halt rule's edge: the highest threshold at which the column never arrives by 3,600 s is 2.5 m or more. Measured: 2.3 m; at 2.4 m the tail arrives at 2,901.585 s: MISSED, while (T1), one clean edge, held.

Kind: WRONG GUESS (Claude's: the edge placed by the start-up stretch, not read). *Lesson:* The edge is clean, a column that never arrives below it and one that always arrives above it. Can it be redesigned? To the proposed list: the edge read between 2.30 and 2.40 m in steps of 0.01, set beside the overshoot found in the same run, the longest gap being the threshold plus 1.4205 m at every threshold from 2.0 to 2.7. *Proved?* Seen in runs (HALT_TRADE 20261031_235932).

555. HALT_TRADE's (T3) and (T4), the halt rule's trade: between 3 and 4 m the longest gap jumps from under 6 m to over 20 m between neighbouring thresholds, and some threshold keeps contact and brings the tail home within 1,000 s. Measured: contact frays before it breaks, the longest gap 5.823 m at 3.2, 6.164 at 3.4, 8.051 at 3.6 and only then 28.842 at 3.8; and the best threshold that keeps contact, 3.2 m, brings the tail home at 1,522.952 s, against 524.999 with no rule: both MISSED.

Kind: WRONG GUESS (Claude's: a cliff guessed where there is a ramp and then a cliff, and a cheap setting hoped for that is not there). *Lesson:* The trade has no free lunch: a halt that keeps every walker in contact costs 2.9 to 5.5 times the time of no rule, and one that gives contact away still costs 214 to 219 s. Can it be redesigned? To the proposed list: the thresholds 3.2 to 3.8 m in steps of 0.05, to place the cliff, and the jump between 2.7 and 2.8 m, where the overshoot, the arrival and the count of halts all change at once. *Proved?* Seen in runs (HALT_TRADE 20261031_235932).

556. COMB_STRETCH's (W1), (W2) and (W3), and its rival (W1-r), foretelling a crest rider's fall from the left-going pattern's shape alone, its pull added up against the crest's width with the rider spending the crest's whole margin: the rule right on at least 90 per cent of 5,120 fates, its foretold falls coming within 30 time units before the recorded ones for at least 80 per cent, and every all-or-none soup foretold so; the rival, the deepest point alone, as good. Measured: the rule right on 4,351 (0.8498), 1,629 of 2,048 within 30, 67 of 79 soups; the rival right on 2,817, calling a fall for every rider: all four MISSED.

Kind: WRONG GUESS (Claude's: a rider credited with margin it does not have at its holding point). *Lesson:* Every error ran one way, no false fall and 769 falls missed, which says what is wrong: the rider sits where its crest's push only just matches the wave speed and has no margin to spare until it is pushed onto the crest. Redesign, said aloud and run the same day: the same sum sparing none of the margin, and sparing half (COMB_STRETCH_B, row 557). *Proved?* Seen in runs (COMB_STRETCH 20261101_002907).

557. COMB_STRETCH_B's (B2) and (B3), the rule's redesign: the rule and its twin that spares none of the crest's margin disagree on at most 20 per cent of the 5,120 fates, and the midpoint, sparing half, is right on at least 95 per cent. Measured: they disagree on 1,560 (30 per cent), and the midpoint is right on 4,545 (0.8877): both MISSED, while (B1) held, the twin missing no fall.

Kind: WRONG GUESS (Claude's: a bracket hoped narrower than it is, and a midpoint hoped to be the answer). *Lesson:* The two sums bracket every fate, 5,120 of 5,120: a fall foretold by the first is certain and a fall not foretold by the second does not come, in every soup read. Can it be narrowed? To the proposed list: let the spared margin grow as the rider is pushed up the crest's back, read from the crest's own shape, which needs no new number. *Proved?* Seen in runs (COMB_STRETCH_B 20261101_003130).

558. The labels of three planted checks written on 1 November: COMB_STRETCH's F0 and F1 and COMB_STRETCH_B's B0, each marked 'planted, judged' in its tool's docstring, were named outside the house's convention that a planted check's label begins with M, which is the only way the headline register knows one; the register counted all three as predictions about the toys, nine new predictions where six had been registered. Found at the rebuild that day, by the count moving more than the registrations, before any count was told.

Kind: THE INSTRUMENT (Claude's: a new tool's labels broke the convention a rule relies on). *Lesson:* Fixed the same day, as CAT_MIRROR_CENTRES's planted checks named N had been fixed in October: the register names the three exceptions with their reason, and the count is six. A convention that a counting rule relies on is part of the rule; a new tool keeps it, or names its exception where the rule can read it. *Proved?* Seen in the register's rebuild (31 October in Bear's day; 1 November UTC).

559. The first blind run of the two-gear hill's check (EXPLORER_TRUCK_CHECK): H4, the room's printed numbers set against the run files' own lines, MISSED on 2 of its 105 comparisons with 104 printed lines (the mean line read twice), both RAMP_SPEED's share caught at mass 1.0: 27 of 48 is exactly 0.5625 and 39 of 48 exactly 0.8125, which the record prints as 0.562 and 0.812 and the check printed as 0.563 and 0.813. Every one of the 18,798 riders flown repeated its run of record in the same blind run.

Kind: THE INSTRUMENT (Claude's: the checker formatted a number by a browser's rule, not the record's). *Lesson:* Python rounds an exact tie to the even digit; a browser's toFixed rounds it up. Fixed before any run of record: every number the room prints is formatted by Python's rule (an exact tie to the even digit, every other number as before), and the page will use the same formatter; the second blind run held. A number printed by two languages is the same number only if both round a tie the same way. *Proved?* Seen in a blind run (1 November UTC; 31 October in Bear's day).

560. EXPLORER_TRUCK_CHECK's (W8-r), the two-gear hill's live truck, the rival: one rider's flight of 5,000 time units at steps of 0.02 takes over 300 ms in node here. Measured: a median of 31.03 ms over 9: MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* A flight is 250,000 steps of four sines each, about an eighth of a microsecond a sine; the room can fly all thirteen riders of a setting in under half a second. Needs no redesign. *Proved?* Seen in runs (EXPLORER_TRUCK_CHECK 20261101_015149).

561. The two-gear hill's page, its first pass in headless Chromium as its registration asked: no page error and no sideways scroll at 1,280 and 390 pixels. Measured: three page errors at each width, a launch's animation left running after the visitor changed how the truck is let go, so that it read a flight no longer there; and at 390 pixels the Build and Check tables pushed the whole page sideways to 564 and 446 pixels. The self-check itself passed at both widths.

Kind: THE INSTRUMENT (Claude's: the page's own faults, found by its registered test). *Lesson:* Fixed before publishing: a change of start stops the animation first, and the animation stops itself if its flight is gone; the page's columns may shrink, so a wide table scrolls inside its own box. The second pass held at both widths. A page is tested by doing what a visitor does, in the order a visitor might, not only by loading it. *Proved?* Seen in the page's test (1 November UTC; 31 October in Bear's day).

562. EXPLORER_LATCH_CHECK's (W9-r), the latches-and-switches room, the rival: at least one number in the knob register's six rows for the switch, the pause ear, the latch, the tall latch, the soup ear and the spring does not agree with the six runs' data at the rounding the row prints. Measured: all 15 claims read from the data agree, every number at its printed rounding and every word of kind (silent, awake, never, no best, no extra switches): MISSED.

Kind: WRONG GUESS (the rival's). *Lesson:* The rows were written from the runs' own lines and have been read again at every rebuild since; the rival was the test that they had not drifted as the book was edited around them. Needs no redesign; the same test can be run on the knob register's other rows as each room is built. *Proved?* Seen in runs (EXPLORER_LATCH_CHECK 20261101_042950).

563. The first blind run of the long snake's check (EXPLORER_SNAKE_CHECK): K2, the two runs' 13 result lines printed again from the data, MISSED on one line, SNAKE_LONGEST's line of the gap, which looked the same character for character.

Kind: THE INSTRUMENT (Claude's: a file from the other computer read as if written here). *Lesson:* SNAKE_LONGEST ran on the second computer, whose run file ends each line with a carriage return as well as a new line; read line by line on new lines alone, its last line kept the carriage return and could not equal anything printed here. The other line from that file passed only because its run time, cut from the end, took the carriage return with it. Fixed before the run of record: lines are split on either ending; the second blind run held. A file's line endings are part of what it says. *Proved?* Seen in a blind run (1 November).

564. EXPLORER_SNAKE_CHECK's K3, judged, its answer seen and said aloud before it was registered: every claim in the knob register's three rows for the long swimming snake agrees with the data at the rounding the row prints. Measured: five of six agree; the start of the motors' wave on the 153.6-wavelength body is 4.648 degrees in SNAKE_EXACT's data, which the row, the book's paragraph on how close a long snake comes and the reader's edition's chapter 8 each gave as 4.7: MISSED.

Kind: THE INSTRUMENT (the book's own: a number rounded from a print, not from the data). *Lesson:* SNAKE_EXACT printed the start to two places, 4.65, and whoever carried it into prose rounded that print again to one place, 4.7; from the data, one place is 4.6. Fixed the same day in all three places, each with its date. Round once, from the number itself, never from a rounded print; the other rooms' knob rows get the same test as they are built. *Proved?* Seen in runs (EXPLORER_SNAKE_CHECK 20261101_171827).

565. The first step of the turn that built the snake race, set down at the close of the turn before it: check the crate's files by md5 against the list shipped last, before any work. Measured: no line of the record shows it run first; the turn crossed a break in Claude's working memory, whose summary carried the task and not the check, and the check was run only after the race's page was published. It found 3,277 of 3,281 files as shipped, the four that differ and the seven new all the race's own: MISSED.

Kind: THE METHOD (Claude's: an opening step dropped across a break in working memory). *Lesson:* Nothing was lost; the late check shows the files were as shipped when the turn began. After any break in working memory the opening steps are read again, and the check run, before the first new file is written. *Proved?* Seen in runs (the crate's md5 check, 1 November).

566. The snake race's page, its first look in headless Chromium before publishing, at a phone's 390 pixels, where the fix of row 561 (every grid column allowed to shrink) was meant to keep any page from scrolling sideways. Measured: the page was 496 pixels wide: the list of matches, a drop-down whose width is set by its longest choice, pushed the whole page sideways. The self-check passed: MISSED.

Kind: THE INSTRUMENT (Claude's: the page's own fault, found by its test). *Lesson:* Fixed before publishing: the drop-down may shrink to its row and cut its longest choice short, and every row of controls may wrap. Row 561's fix covered tables and grids; a drop-down, a long button or a long word each sets its own width, so the phone's width is tested on every page every time and never assumed from the last fix. *Proved?* Seen in runs (the room's page test, 1 November).

567. The snake race's page as first published, its test passed in light and dark: each lane shows its snake so that the knob of length can be seen to act. Measured, by Bear on the published page: every snake looked the same length, 9.6 wavelengths or 38.4 or 153.6, for each lane drew a fixed window of 9.8 wavelengths, which a 9.6-wavelength body fills and a longer one runs out of; only a line of small type said so. The page's test had checked every number, winner and width, and nothing about what the picture shows: MISSED.

Kind: THE INSTRUMENT (Claude's: a picture that did not show its own knob, and a test that read numbers only). *Lesson:* Fixed the same day: both lanes drawn to one scale set by the longer body, with a bar giving each whole body to that scale; the first fix took the shorter body's length for the window and was caught on its own picture before the test. A page's test now reads the picture's own scale against the knob it shows; a knob that changes nothing a visitor can see is a fault, whatever the numbers say. *Proved?* Seen in runs (the room's page test, 1 November).

568. The abstract draft and the closeout report as built and shipped on 1 November with the snake race, each stamped by its builder with the turn that built it: they should have named session J's twelfth exchange. Measured, at the next turn's rebuild: both named session I, a session already closed, as the one that built them, and the closeout counted that session's compactions; each builder reads a bare number as session I's, by a rule written into it at session J's third exchange, and was given 12 where it needed J12. The closeout's PDF had already been sent out loose: MISSED.

Kind: THE METHOD (Claude's: a builder's argument given from habit, and its stamp not read back). *Lesson:* Rebuilt the next turn with the session's letter, the stamp read back after the build. A built file's own header is a claim like any other, read back before it ships; and a rule that reads a missing letter as an old session is a trap for every later session, so the builders now refuse a bare number. *Proved?* Seen in runs (the abstract draft and the closeout as shipped, 1 November).

569. The first run of the check written for the clock rule of 1 November, which reads the snake race's and the beat run's data: the hardest-pulling start plus the lateness of the hinges' summed hand, sum over j of e^(-i 2 pi (k - 1) x_j) as the rule was registered, is 45 degrees at every one of 51 bodies and wave numbers. Measured: in tune every case gave 45; mistuned, every case was off, by twice its beat's lateness; the code had summed e^(+i 2 pi (k - 1) x_j), the registered sign turned over: MISSED.

Kind: THE INSTRUMENT (Claude's: a check that did not do what its registration says). *Lesson:* Read case by case before any verdict was believed: the in-tune cases right and every mistuned one off by exactly twice its own lateness is the mark of a turned-over sign, not of a wrong rule. Corrected to the registered words, said aloud, the first run's output kept; run again, all 51 within 9.8e-10 degrees, the rule held. A check is read against its registration line by line before its first run, as a tool is. *Proved?* Seen in runs (the clock rule's check, its first and second runs, 1 November).

570. The knob register's row for a train of three leapfrogging rings, read against its runs on 1 November by the register's sweep, its rules written before the check ran ('up to' is the largest the cited runs print for the case named): the trade is 'up to 26 per cent'. Measured: TRAIN_HOLD's largest gain for a trio that held is 1.2658, at 0.6 radii over 400 time units, 27 per cent at the row's rounding; over 2,000 time units the starts that held reach 1.2589, 26; at the row's own setting, half a radius, 1.2610, 26: MISSED.

Kind: THE INSTRUMENT (the book's own: a maximum given without the run that bounds it). *Lesson:* The row's 26 is what the setting and the longer run bear out, and it was called the largest; the shorter run found more, at a spacing that holds for 400 time units and sits at the edge of its cliff. Fixed the same day with its date: 26 per cent at half a radius, and 27 at 0.6 over 400 time units [corrected 2 November: over 1,000 time units; TRAIN_HOLD flies every train 1,000, and its line over 400 is M0's, at half a radius; row 576]. A largest value is given with the run and the horizon it was the largest over. *Proved?* Seen in runs (the knob register's sweep, part A, 1 November).

571. The knob register's row for a master oscillator, read against its runs on 1 November by the register's sweep: above the bead's working pace the beat cannot pull the bead. Measured, in PILOT_PRICE's swing table: a beat of three tenths of the soup locks the ordinary bead (working pace 0.76 of its tune) at 0.7 and not at 0.8, as the row says; a beat as loud as the soup locks it, within 1 per cent, up to 0.8, and the lighter bead (working pace 0.91) up to 1.1: MISSED.

Kind: THE INSTRUMENT (the book's own: a rule stated for a knob's whole range, true at the setting it recommends). *Lesson:* The row's setting is a beat of at least three tenths, and at three tenths the rule holds; the loudest beat in the run breaks it, as the book's own next sentences said. Corrected the same day in the row and in the book's two sentences that carried it, each with its date; the reader's edition had it right. A rule in a register is read at the ends of its knob's range, not only at the setting it recommends. *Proved?* Seen in runs (the knob register's sweep, parts B to D1, 1 November).

572. The check that read the knob register's rows for the column, the ears, the soup, the truck, the meters and the counters against their runs, its third run (its first two stopped on their own patterns): every value read from the runs' printed lines. Measured: five claims printed as not agreeing; read each against its run before any was believed, three were the check's own: the soup's pattern also caught the per-soup table printed below the pooled one, and the worn rider's two patterns caught the plain fleet's lines and not the fleet that works harder, the row's machine: MISSED.

Kind: THE INSTRUMENT (Claude's: a pattern that found a line, not the line). *Lesson:* Corrected to read the named block of each run, the pooled table and the harder fleet, and run again: 59 of 60 agree. A pattern that reads a value from a run is anchored to the run's own heading for the case the row names, since the same label can print in several tables; every claim printed as not agreeing is read against its run before it is believed. *Proved?* Seen in runs (the knob register's sweep, parts B to D1, 1 November).

573. The check that read the knob register's rows for the jumper, the diver, the bells, heat and the first swimmer against their runs, its first full run: every value read from the runs' printed lines. Measured: all 38 claims agreed, but one pattern, for the bells' width ladder, fell back to this record's own words when the run's lines did not match it, so two of the claims were read against the ledger and not against the run: MISSED.

Kind: THE INSTRUMENT (Claude's: a fallback that read the claim's own source as its evidence). *Lesson:* Found on reading the check before its verdict was believed; corrected to read the run's own table of gains, and run again: 38 of 38. A check never falls back to the words it is checking; where a run's lines do not match, it stops. *Proved?* Seen in runs (the knob register's sweep, part D2, 1 November).

574. FREE_SWIM's (E2), the snake let loose to swim into the waves: the moored race's clock rule moves with the frequency at which the snake meets the waves, so at the hardest pull the start plus the hinges' summed lateness is arctan of that frequency, not 45 degrees, within 1e-6 degrees, every length, wave number and frequency. Measured: 38.659808 degrees at an encounter frequency of 1.25, 33.690068 at 1.5, 26.565051 at 2 and 18.434949 at 3, at every length and wave number: 90 degrees less arctan of the frequency, within 1.75e-10 degrees in all 45 cases: MISSED.

Kind: WRONG GUESS, two readings of one rule that agree at one point. *Lesson:* The moored rule, start plus lateness equal to 45 degrees, was found at one frequency only, where arctan of the frequency and 90 degrees less it are both 45; the lean took the one reading and the run chose the other. Read as the run reads it: the body answers its motors arctan of the frequency late (the solver's matrix), and the hardest pull comes when the start, the hinges' summed lateness and the body's own lateness together make a quarter turn, the body's answer in quadrature with the water at every frequency. Said aloud and registered as a redesign before any new run: tested at five frequencies no run had touched, 0.5 (a snake swimming with the waves), 0.8, 1.75, 2.5 and 4, at 48 and 96 links and three wave numbers, it held within 1.0e-10 degrees in all 30 cases. A rule found where two readings coincide is tested where they part. *Proved?* Seen in runs (FREE_SWIM 20261101_222356; FREE_SWIM_2 20261101_222740).

575. The headline register's count of predictions, built on 2 November with RING_PATHS: a planted check is known to it by a label beginning with M, or by a named exception (row 558). Measured, on reading why the count rose by five for three leans: 14 planted checks, each marked 'planted' in its own tool's docstring and all held, were counted as predictions about the toys: CAT_TWIST's T1 to T3, LIMIT_AUDIT's A0, SAGNAC_LOOP's S0, and five tools of 1 and 2 November, SNAKE_RACE's C0 to C2, SNAKE_BEAT's B0, FREE_SWIM's E0 and E1, FREE_SWIM_2's Q0 and RING_PATHS's R0 and R1. The count given on the front page and in the abstract was 1,136 predictions with 755 held; read rightly it is 1,122 with 741, and the 1,131 with 750 of the day before was 1,119 with 738: MISSED.

Kind: THE INSTRUMENT (Claude's: a rule kept as a convention and a list of exceptions, the convention broken again by new tools after row 558 said a new tool keeps it). *Lesson:* Row 558 fixed one tool's labels by naming them, and the kind came back in five new tools within a day. Fixed by the kind this time: the register reads each tool's own docstring, and a label the docstring marks '(planted' is planted, whatever its letter; the exceptions of row 558 are now caught by the same rule. Every count that moved was rebuilt the same day, and the change said aloud. A counting rule reads the source's own words; a convention a rule relies on is a fault waiting for the next tool. *Proved?* Seen in the register's rebuild and the tools' docstrings (2 November).

576. EXPLORER_RING_CHECK's K3, room 7, the rings that leapfrog: every claim in the knob register's four ring rows agrees with the data as the rows now stand. Said aloud before the check ran, from reading TRAIN_HOLD's code for it: expected to miss on the trio row's 'over 400 time units', and nowhere else. Measured: 19 of 20 claims agree; 'over 400 time units' does not: TRAIN_HOLD flies every train 1,000 time units, by its own text and its calls, so its 1.2658 at 0.6 radii is a gain over 1,000: MISSED.

Kind: THE INSTRUMENT (Claude's: a correction that took its horizon from the wrong line of its run). *Lesson:* The correction of 1 November (row 570) read the trio's gain at 0.6 from TRAIN_HOLD's lines of trains and the horizon from the line above them, M0's, which repeats RING_KNOBS's trio at half a radius over 400 time units; the lines below it were flown 1,000. Found while the room's check was written, and said aloud before it ran. Fixed the same day, each with its date: the knob row, row 570's fix and the lesson drawn from it. A correction is read against its run as closely as the claim it corrects; a horizon is read from the code that flew the case. *Proved?* Seen in runs (EXPLORER_RING_CHECK 20261102_045348; TRAIN_HOLD's code).

577. ROUGH_BAND's M2, planted (the knee test, the redesign of row 286's band test): at every judged row inside each floor's own slipping band, the spinning spells read by the needle's own speed with a knee have a median over mean that repeats within 0.1 with the knee moved from 0.35 and 0.65 of the pace to 0.25 and 0.75 or 0.45 and 0.55. Measured, in the blind run: at 20 judged rows the largest change is 0.228: MISSED; the run of record, started beside it as registered, void and moved out unread.

Kind: THE INSTRUMENT (a statistic that depends on its own definition, a second time). *Lesson:* The reader itself is sound: it repeats SPIN_OR_SIT's spells to every printed digit. But the needle lingers near its lock between bursts, and whether a linger ends a run is the reader's choice, whether made by a joining gap (row 286) or by a speed knee (here); either way the spread of runs moves by about a quarter. No statistic of cut runs can settle the trap. Redesign, said aloud and registered before any new run: no cutting at all; the slips counted in windows from 32 to 1,024 field periods over flights ten times longer, the counts' spread over mean levelling off for a trap and growing without end for a heavy tail, the instrument first tried on made-up records of each kind. The question of two traps mixed near the many-ledge floor's onset is not covered by it and goes to the proposed list. *Proved?* Seen in runs (ROUGH_BAND's blind run 20261102_164924).

578. ROUGH_COUNT's (C1), ROUGH_BAND, the trap read without cutting runs: on every floor at both paces inside its slipping band, the slips' counts have a variance over mean that levels off, F at a window of 1,024 field periods less than 1.2 times F at 512. Measured: on seven rows 0.964 to 1.131, levelling off; on the two-fold 0.3 floor at 0.845 of its grip 1.348, the scatter still rising at every window to the largest, a third of each needle's record: MISSED, and the rival, Bear's sandpile, HELD there.

Kind: WRONG GUESS, or a trap slower than the record: not yet told apart. *Lesson:* The instrument told a made-up trap from a made-up heavy tail (1.022 and 1.326), and at the needle rows' length a made-up trap never reached 1.2 in 120 tries; so the growth is in the needle. But a count that grows to the record's end cannot tell a heavy tail from a trap whose slowest dwell is longer than the window. Redesign, said aloud and registered before any new run: that row flown ten times longer, with windows to 8,192 periods, its instrument tried first at the same length on made-up records of each kind; beside it, whether a second gear (one slip in two turns) is present and whether each needle keeps its own rate [Answered 2 November by its run 20261102_200436: flown ten times longer the count levels off, 1.152 per doubling at the last, a slow trap of about a thousand periods; no second gear; row 579]. *Proved?* Seen in runs (ROUGH_COUNT 20261102_173941).

579. ROUGH_LONG's (L2-r), ROUGH_BAND, Bear's sandpile kept as the rival: on the strong two-fold floor at 0.845 of its grip, flown 30,569 field periods, the slips' counts keep growing more uneven, F at 8,192 periods at least 1.2 times F at 4,096. Measured: 1.152; the growth per doubling falls from 1.92 at short windows to 1.36 at 1,024 and 1.09 to 1.19 beyond: MISSED.

Kind: WRONG GUESS, the rival. *Lesson:* The floor's trap has a slow mood, about a thousand periods long, which the record of about 3,000 could not outwait, so it looked like a sandpile there. It is not a second gear (no rhythm of one slip in two turns) and not a needle keeping its own rate (the halves correlate at 0.228): each needle passes through the slow mood in its own time. Covered by the run's own lean; what sets the mood goes to the proposed list. *Proved?* Seen in runs (ROUGH_LONG 20261102_200436).

580. EXPLORER_BELL_CHECK's K3, room 8, the tied bells: every claim in the knob register's bell row agrees with the room's data as the row now stands. Said aloud before the check ran, from reading BELL_SOFT_MAP's table for the registration: expected to miss on the set-it clause 'a longer memory to make them pay more', and nowhere else. Measured: 13 of 14 claims agree; on the softened map, at the three periods where every memory pays (3, 4.24 and 6), the gain rises with the memory only from 4.24 up, memory 3 paying 0.022, 0.037 and 0.047 more than memory 4.24: MISSED.

Kind: WRONG GUESS (Claude's: a builder's rule written from the map's two ends). *Lesson:* The clause was written from the run's own (T3), the longest memory further from one than the shortest at every period, which holds; read as a rule for every step it overreaches at the short end, where the two shortest memories pay almost the same and the shorter a little more. Fixed the same day with its date: the row now says 'from a memory of about 4 up (at 3 they pay a few hundredths more than at 4.24)'. A rule drawn from two ends of a map is read at every step before it is handed to a builder. *Proved?* Seen in runs (EXPLORER_BELL_CHECK 20261103_025532).

581. The book's crowded-corner paragraph, written from BELL_SWITCH on 20 October, said of the softened core's eight nearby starts that the stroke ran 'smooth from start to start' at both memories. Read again for room 8: true at memory 12 (0.181 to 0.184, every neighbour within 0.001); at memory 8.49 the four starts below the gap of 1.5 swim 0.182 to 0.187 a squeeze and the four above it 0.124 to 0.132, a jump of 0.049 between starts a thousandth apart, every other neighbour within 0.004, the two families drawing apart from the third quarter to the last.

Kind: WRONG READING (Claude's: one word laid over two memories' rows when only one of them was smooth). *Lesson:* The run's own verdicts were about sign, every softened start forward, and they hold; the summary word was added beside them and was read from the memory-12 row. At memory 8.49, with the cores' own size, the corner still chooses between two forward strokes by its start, which the softened map's one unsettled point there agrees with. Found while the room's eight starts were read for its picture, after its check had run. Fixed the same day with its date. A summary word over a table is read against every row it claims. *Proved?* Seen in runs (BELL_SWITCH 20261020_161902; EXPLORER_BELL_CHECK 20261103_025532).

582. Two digits in the bells' paragraphs, found when room 8's page printed them from the data: the width ladder's fast point 'sits between 0.68 and 0.97 at every width but one' (written 21 October), and the softened starts below the gap of 1.5 at memory 8.49 swim from '0.181' (written 3 November). The data: 0.67495, which the run printed as 0.6750; and 0.18163. So 0.67 and 0.182.

Kind: WRONG READING (Claude's: a number rounded twice, and a digit misread). *Lesson:* The first is the snake's 4.65 again: a printed value carried to a coarser rounding instead of the data's own; the second was 0.1816 read as 0.181, in a correction written the same day. Neither moves a conclusion. Both fixed the same day with their dates, and so was the Field Guide's lesson on reading a setting at more than three values, which carried the first. A number written into prose is rounded once, from the data, and set beside the row it names: the room's page, which prints every number from the data by the same rule, is what caught them. *Proved?* Seen in runs (BELL_WIDTH_LADDER 20261021_151833; BELL_SWITCH 20261020_161902).

583. The book’s table of cards, the entry for the coil snake, the jelly-like blobs and the face a body presents: it still said ‘the pumping coil and the blobs remain’, and the entry stood as one of the two still to run. The pumping coil had run on 18 October, under two later entries of the same table (COIL_PUMP: an even squeeze goes nowhere; COIL_WAVE: a running wave carries the coil, against the wave), and the book’s own coil paragraph said so.

Kind: OTHER (Claude’s: an entry not carried forward when its sweep ran under a new entry’s number). *Lesson:* The sweep was registered as a new entry, in the rotation of entries left waiting, and that new entry was written; the entry it came from was not touched, so the table under-reported the work by a whole sweep, and the count of entries still to run was one too high for two weeks. Found when the entry was opened to finish it. Fixed the same day with its date: the entry brought up to date and closed, its blobs written into the open work with a design. When a sweep runs under a new entry, the entry it came from is updated in the same breath. *Proved?* Seen in the book’s table of cards.

584. EXPLORER_HEAT_CHECK’s K3, room 9, heat in a small piece: every claim in the knob register’s heat row agrees with the room’s data as the row now stands, read strictly. Said aloud before the check ran, from reading BRICK_LINE’s json for the registration: expected to miss on ‘1.485 lengths of heat’s travel after the quench’, and nowhere else. Measured: 5 of 6 claims agree; the echo comes at the toy’s 1,187th step at lengths 0.75 to 0.78 of the line (1.48375 lengths of travel) and at its 1,188th from 0.79 to 0.88 (1.485): MISSED.

Kind: WRONG READING (Claude’s: a range given as one number, its third place the toy’s clock). *Lesson:* The sweep of 1 November accepted 1.484 for the least and called the claim agreed; read by its own rule, a single 1.485 does not round from 1.48375. The echo is real and steady; only its last place is the step’s. Fixed the same day with their dates: the knob row, the brick’s paragraph, its card and the Field Guide’s lesson that carried it now give 1.484 to 1.485. A number whose last place is the instrument’s step is given as the range the steps allow. *Proved?* Seen in runs (EXPLORER_HEAT_CHECK 20261103_065559; BRICK_LINE 20261016_205112).

585. Two more of the same kind, found by a sweep of the book’s table of cards for rows left behind: the column’s entry still said its trade and the five tools that ran its limit ‘are next’, though the five had been rerun on 17 October and its corrected numbers were in the column’s pages; and the error-bar pass of 19 October, which left seven claims owed for runs that kept nothing to put a bar on, said so truly in its entry and in the headline register, yet the seven were carried into no list of open work for two weeks, four of them runnable here.

Kind: OTHER (Claude’s: work recorded where it ran and not where it was waited for, and a debt written down where it was found and nowhere it would be paid). *Lesson:* A list of open work is only as good as what is carried into it: the closeout’s list was built from the card table’s statuses, and an owed claim inside a finished entry has no status of its own. Fixed the same day with dates: the column’s entry brought up to date; the seven written into the open work, the four with tools to run before the upload, each registered first, the three without written as such. When an entry finishes with something owed, the owed part gets its own line in the open work in the same breath. *Proved?* Seen in the book’s table of cards (the sweep, 3 November).

586. BRICK_QUARTER’s rival (Q1-r), and the warm brick’s digits in the book: at a quarter of the line the brick’s dip moves by 1 per cent or less when the toy’s cells are doubled, as it did at half the line. Measured: 0.704481, 0.711642, 0.715144 and 0.717386 at 400, 800, 1,600 and 3,200 cells, +1.016 per cent at the first doubling and still +0.313 at the last; the profile’s turning points at the dip’s moment double with every doubling of the cells (247 to 2,039), the ripples the grid’s. Read on finer grids (BRICK_FINE): the echo moves from 1.484 to 1.485 lengths of travel at 400 cells to 1.490 to 1.492 at 1,600, closing on the fronts’ own 1.5, while its end at 0.88 and the brick’s stop at the line hold: MISSED.

Kind: THE INSTRUMENT (the grid: a sharp start on a grid of cells, and a step test run where the number was not read). *Lesson:* The warm brick starts warm right up to its ice-cold ends, a jump that heat with a memory carries inward as a sharp front, and a grid of cells cannot carry a sharp front cleanly: it rings at the cells’ own scale beside it. HEAT_RING’s step test halved the cells at half the line and 1.1, where the change was 0.8 per cent and nothing, and the book read the brick at a quarter, where the fronts meet soonest and sharpest. The smooth hump, in the same rod, does not move in its fifth figure. Found by room 9’s first look at its own picture, before any number said so. Fixed the same day with their dates: the brick’s digits in the book, its knob row, its two cards and the Field Guide’s lessons now give the 400-cell value with the finer grids’ beside it. A step test is run where the number is read, and a sharp start is looked at before its digits are trusted. *Proved?* Seen in runs (BRICK_QUARTER 20261103_180126; BRICK_FINE 20261103_180438).

587. The jelly-like blobs, the last sweep Bear named at his second sitting: written out of this deposit, in the book’s table of cards and the closeout’s open work, and only then put to him as a question with a default, ‘unless you’d rather I build them here, they’ll stay there’. Bear, the next day: ‘I’d like to keep them in this deposit since they took a lot of effort. Is there a reason not to include things? Please don’t start cutting things without discussing it’.

Kind: OTHER (Claude’s: a cut decided before it was discussed). *Lesson:* The finish line of 2 November, no new toys in this deposit, was drawn at Bear’s word ‘you decide how this finishes’, and Claude applied it to the blobs as a rule already agreed, writing the decision into the record before asking. A finish line is a proposal for each thing it would

leave out, not a licence: the person whose work it is hears each cut before it is written, and nothing is written out with a default left to run. Fixed the same day: the blobs put back in this deposit and built; every other item written as outside it listed for Bear to decide. *Proved?* Seen in the conversation and the book's table of cards.

588. BLOB_RING's M3, planted (the node test): at 128 nodes every judged travel and spin of the soft ring within 1 per cent of its value at 64. Measured, in the blind run: the wave with four crests turns the ring -0.10300 a lap at 64 nodes and -0.10463 at 128, 1.6 per cent; three crests 0.84, two 0.28, the two waves together 0.7: MISSED; the run of record, started beside it as registered, void and moved out unread.

Kind: THE INSTRUMENT (a polygon too coarse for its shortest wave). *Lesson:* Sixty-four nodes give a wave of four crests sixteen nodes each, and the ring's turning is read from the slope of its outline, which a polygon that coarse misses by about a per cent and a half. Redesign, said aloud and registered before any new run: the ring drawn with 128 nodes and the node test made at 256, the step test and every judged case unchanged. *Proved?* Seen in runs (BLOB_RING's blind run 20261104_170321).

589. BLOB_RING's M4, planted (the time mirror): the two waves run backwards retrace the lap, travel and spin opposite, their sums below 1e-5 of the forward's. Measured, in the blind run: forward (-0.002697, -0.041992), spin -0.12825; backward (-0.002697, +0.041992), spin +0.12825; the travel's sum 0.0054: MISSED; the run of record void with M3's.

Kind: THE INSTRUMENT (a check written for a body that does not turn). *Lesson:* A body that turns as it goes does not retrace by the opposite of its shift: run backwards from the same start, it undoes the lap, so its shift is the forward's turned back through the spin and reversed. Turned by +0.12825 and reversed, the forward (-0.002697, -0.041992) is (-0.002698, +0.041992), the backward run's to its printed places. The toy was right and the check asked the wrong question. Redesign, said aloud and registered before any new run: the mirror judged as the inverse motion, within 1e-5 of the travel and of the spin. *Proved?* Seen in runs (BLOB_RING's blind run 20261104_170321).

590. BLOB_RING's (B3), Claude's lean: the soft ring's travel, carried by two neighbouring waves running round it together, goes as their height squared: from e = 0.05 to 0.1 it grows 3.8 to 4.2 times. Measured: 3.4284: MISSED.

Kind: WRONG GUESS (a small-wave law read at a height where the waves add). *Lesson:* The two waves meet in a bulge twice either's height, a fifth of the ring's radius at e = 0.1, and there the fourth power of the height takes off a sixth of the travel. Redesign, said aloud and registered before its run, run at once (BLOB_SMALL): at heights down to 0.0125 the growth at each doubling climbs, 3.43, 3.84, 3.96, toward the square law's 4, both leans held. A law for small waves is tested where the waves are small, and the size that matters is where they add. *Proved?* Seen in runs (BLOB_RING 20261104_170523; BLOB_SMALL 20261104_170747).

591. EXPLORER_NEEDLE_CHECK's K3, judged: the book's 20 number-bearing claims on the needle's stretch and its lap agree with the orbit file at their printed digits. Measured: 18 of 20 agree; 'The cycles that stretch most are spent wholly on the top half' does not (of the three that stretch more than 18 times, the one at 18.65 spends 0.95 of its steps on top), and 'Their two multipliers are a complex pair, each exactly e^(-π)' does not (of the three cycles spent mostly on the bottom half, the one at a share of 0.40 rings down through a real pair, 0.0967 and 0.0193): MISSED, both said aloud at the registration and nowhere else.

Kind: WRONG READING (Claude's: a sentence written from the case that made the point, and widened to its class). *Lesson:* Both sentences were written the day the orbit was read, each from its best example: the largest stretch, wholly on top, and the ringing cycles, whose pair is complex. Each is true of its example and not of every member of the class it names, and nothing printed by the run said otherwise, because the run printed the example. A room that reads a paragraph claim by claim against every member finds what a reader of the highlights cannot. Both corrected the same day with their dates. *Proved?* Seen in runs (EXPLORER_NEEDLE_CHECK 20261104_222610; STRETCH_READS 20261013_060151; STRETCH_WHERE 20261013_045856).

592. LEAN_DRIFT's (D1), Claude's lean: the held share of decided predictions in runs of 26 October to 4 November exceeds that of 5 to 18 October by more than two standard errors, the guesses grown safer. Measured: 35 of 58 held late (0.6034) against 648 of 934 early (0.6938), −1.37 errors: MISSED, and its rival, the bench grown bolder by more than two errors, missed with it.

Kind: WRONG GUESS (Claude's: the feel of the last few days read as the whole span). *Lesson:* The held share fell week by week through October, 0.73, 0.65, 0.61 and 0.50, and only the first three days of November, 14 of 16, look safe. What changed was how many guesses were made, about 470 a week early and 16 in those three days, and whether their rivals were ideas: none of the 30 rivals since 26 October was marked a competing idea, and five competing rivals written since 19 October had never been marked at all. Redesign, said aloud: the safe stretch is a few days long, so it is tested forward, not again on the past; the method written at the data ledger's head, rivals written as competing ideas and marked when written, a lean at even odds in every registration, and this read at every review of five. *Proved?* Seen in runs (LEAN_DRIFT 20261105_025051).

593. RING_AREA's M2, planted (the instrument): the loop's area per period read at 200,001 samples agrees with that at 400,001 within 1e-6 of its size at every pair. Measured: up to 1.5e-5 at gap 0.25, at all four cores: MISSED in the first blind run, its record moved out unread.

Kind: THE INSTRUMENT (a sampling set for the span, not for the loop). *Lesson:* A pair a quarter radius apart passes about 690 times in 400 time units, so 400,001 samples over the span give each small loop about 1,160 points, and the shoelace's error, falling as the square of the step, was still a few millionths. Amended aloud, the leans and the tolerance untouched: 20,000 samples a period or 400,001 over the span, whichever is more; the amended run held at 9.0e-08. Set a sampling by the smallest thing it must resolve, not by the whole record. *Proved?* Seen in runs (RING_AREA blind 20261105_032332; RING_AREA 20261105_032546).

594. RING_AREA's (A1), Claude's lean (odds about two in three): across 35 leapfrogging pairs the gain's excess over one ranks with the area of ring 1's loop per period, in the pair's own frame, at Spearman 0.9 or more. Measured: −0.1958: MISSED.

Kind: WRONG GUESS (Claude's: a law of lossless loops carried to an engine's speed). *Lesson:* The loop's area per period grows about thirtyfold with the gap, its axial swing exactly the launch gap, while the gain peaks near one radius and falls on both sides; the area is the gap's, and the gain is not. The clean loops of the book (light through filters, the turning loop, the falling cat) leave behind exactly their area because nothing else acts; the pair's gain is a speed, part of it the rings' own speeds at the radii they pass through and part their pull on each other. Redesign, said aloud, to be registered before it runs: split the gain into those two parts and read the remainder against the area swept per unit time and the area of the pair's shape loop. *Proved?* Seen in runs (RING_AREA 20261105_032546).

595. RING_AREA's (A2), Claude's lean at even odds: the gain's excess is the loop's area times one number, every pair within 10 per cent of the median ratio. Measured: the ratio runs from 0.057 to 2.44 about a median of 0.279: MISSED.

Kind: WRONG GUESS (Claude's: the bolder form of row 594's lean). *Lesson:* With the area not even ranking the gain, a proportion was out of reach; written at even odds as the method asks, and its miss is the size the lean's miss predicted. Covered by row 594's redesign. *Proved?* Seen in runs (RING_AREA 20261105_032546).

596. RING_SPLIT's design check, Claude's, written into its registration under the rule that a competing rival must be one the data can tell from the lean: the gap does not order the even-odds lean's measure and its rival's alike across the whole range, since the area swept per unit time is nearly flat at small gaps where the pass rate falls fastest. Measured: the area swept per unit time and the pass rate rank together at +0.9989 across the 35 pairs, both falling at every step of the gap in every core; the lean (the partner's part ranks with the area rate at 0.9 or more, +0.9199) and its rival (it ranks higher with the pass rate, +0.9277) both HELD: MISSED.

Kind: WRONG READING (Claude's: a check made on a measure's size where the test reads only its order). *Lesson:* A rank test does not see that a measure is nearly flat, only whether it rises or falls; the area rate's fall at small gaps is slow but it is a fall, and it is the same order as the pass rate's. The check could have been made before the registration on data already kept, RING_AREA's area and beat for every pair, by ranking the two yardsticks against each other with nothing of the new parts computed; the rule's own worked example, the area swept per unit time as a measure the gap does not order, had the same fault. Redesign, said aloud, not registered: the core is the knob that orders the two apart (at every gap the area rate and the pass rate fall as the core grows, while the partner's part rises), so fly new cores between those read here, at the same gaps, and ask across the cores at each gap whether the partner's part ranks with the area rate or against it; before registering, rank the lean's and rival's yardsticks against each other on that knob from data already kept. *Proved?* Seen in runs (RING_SPLIT blind 20261105_050120; RING_SPLIT 20261105_050121).

597. The bent swimmer's mechanism, Claude's reading after the data of 3 October and printed in both editions: at a blade's grip the whole hill gathers into a spike at the straight body, and a lap swims forward exactly while its disk encloses the spike and tail-first once it does not; the registered test of it, the reversal at the lap's radius over √2 on the S ray, held at three lap sizes. Measured in the same toy's follow-on, read on 5 November for the prior-art pass: a 5-degree lap centred on a 4-degree C (both hinges bent alike), whose disk does not reach the straight body, swam forward at 539 thousandths at grip 100,000, while the lap centred on a 4-degree S at the same distance ran backward at 436: MISSED.

Kind: WRONG READING (Claude's: a point read off laps on one ray, where a line through that point would read the same). *Lesson:* Every lap the test flew was centred on the S ray, which meets the C shapes only at the straight body, so a lone spike and a ridge along the whole C line predict the same reversal there; the one lap off that ray had been flown the same night, sat in its record, and was never read against the mechanism. The published kinematic snake, the body that cannot slip sideways at all, has exactly such a ridge, a delta function along the C line (Hatton and seven others, arXiv:2102.01506, 2021). Redesign, said aloud, to be registered before it runs: at grips 10,000 and 100,000, laps of the same size centred along the C ray, beyond the reach of the straight body, and off both rays, to say whether the hard-grip swimmer's hill gathers into the snake's ridge or into something else; the books' mechanism sentence is corrected after that run, in words Bear has seen. *Proved?* Seen in runs (S_REVERSAL_FOLLOWON 20261003_213210; S_REVERSAL_SPIKE 20261003_213523).

598. SWIM_RIDGE's (W3), Claude's lean at even odds: the 5-degree lap centred on the straight body travels between 795 and 815 thousandths forward at grip 1,000,000, its steps from grip 1,000 to 100,000 (296, 672, 776) shrinking by about a quarter each. Measured: 788.8: MISSED.

Kind: WRONG GUESS (Claude's: a ratio of two steps carried to a third). *Lesson:* The steps shrank 376, 104 and 12.5, by about a quarter and then by an eighth: the approach to the blade's limit speeds up, so the limit lies just above 789, not near 815. Redesign check, said aloud: covered by the miss itself, which brackets the limit within about two thousandths if the steps keep shrinking; a lap at grip 10,000,000 would only confirm it, and stays on the proposed list. Two steps fix a ratio, not a law. *Proved?* Seen in runs (SWIM_RIDGE 20261105_145417).

599. The reader's edition's Appendix B prints its own table of the found-here candidates, typed in its source rather than printed from the working register. The prior-art passes of 5 and 6 November brought candidates 3 and 4 up in the register, in both editions' chapters and in the reference lists, and left them at their earlier words there: candidate 3 still names the reversal angle with 'its place is not in the abstract that shows it', where the candidate is now the ridge a resistive swimmer reaches, and candidate 4 still says 'not in the nearest paper found', where three searches and a two-line derivation now stand behind it. Found on 6 November while candidate 5's cell was written there.

Kind: OTHER (Claude's: a table kept in two places, one of them typed). *Lesson:* A table copied by hand into a second place drifts from its source unless every edit names both; the table of published laws beside it is printed from the register for that reason, and cannot fall behind. Redesign check, said aloud: either print Appendix B's found-here table from the register as the laws are printed, its cells shortened by rule, or name the Appendix B cell in every prior-art pass; the second is done for candidate 5 the same day, and candidates 3 and 4 wait for Bear's word on their wording, since the swimmer's would change what that rung of the book claims. [6 November, later the same day: brought up at Bear's yes, candidate 3's cell from the wording he approved for chapter 5, its result left as it stands]. *Proved?* Seen in reading (the reader's edition's Appendix B against the register, 6 November).

600. Row 565's step dropped a second time. Session K's first break in working memory fell in its fifth exchange, while room 10 was being built; the summary carried the task and its next step and not the opening steps, and rooms 10, 7 and 6 were edited and rebuilt before section 9's paragraph was read again and the crate checked against the manifest last shipped. Measured: the check, run late, found 3,471 of 3,477 files as shipped in KR, the six that differ the templates and pages of rooms 10, 7 and 6, the rooms edited after the break: MISSED.

Kind: THE METHOD (Claude's: an opening step dropped across a break in working memory, a second time). *Lesson:* Nothing was lost; the late check shows the files were as shipped. A rule kept in a long paragraph is not carried by a summary that copies the next step, so every running record's Next line now opens with the check itself, by name and manifest, where a summary will copy it. *Proved?* Seen in runs (the crate's md5 check, 6 November).

Misses 601 to 650

601. A number read off a moving picture written down as the motion's result. The account written of session K's fourth exchange gave room 4's light truck, at weight 0.5 and push 1.20, as riding the passing lane '27.1 turns each'; the drive script had taken its picture 1.8 seconds into a flight that runs 3,000 time units, and the number was copied from that picture. Measured at session K's sixth exchange, when the rival race printed both trucks' ends: the light truck's wheel turns 354.1 times and the wave 354.1; the heavy truck's −0.2 and 206.6 were its end, read from a flight that sits still: MISSED.

Kind: THE METHOD (Claude's: a mid-motion reading taken for an end state). *Lesson:* A number from a picture is only the picture's moment; the end of a motion is read from the motion's own record at its last step, or from a page that prints it there. The room now prints the race's end in words, and the drive scripts read ends from the page, not from a frame. *Proved?* Seen in runs (room 4's live flights, 6 November).

602. A drawing offset read as distance. Room 1's rival jumper was drawn a step behind his opponent on the same arc, only so the two figures would not overlap; every visitor saw whoever was B land a step short, and the plain jumper, tipped forward with his head out over the pit, looked as if he had jumped as far as the hitch-kicker. Bear asked, after playing it, whether the jumper without tricks flew farther. Measured: both centres of mass come down at the same spot by the toy's own law, and at the end of the flight the hitch-kicker's back foot is −62 cm from it against the plain jumper's −95: MISSED.

Kind: TEACHING MISS (Claude's: a layout choice drawn as if it were data). *Lesson:* Every distance a toy draws between contestants is read as a result. The jumpers now fly in two pits side by side at the same distances, and the race is scored by the long jump's own measure, the mark nearest the board, read from the toy's own body. *Proved?* Seen in runs (room 1's engine, 6 November).

603. Row 565's step dropped a third time (row 600 the second). Session K's second break in working memory fell early in its ninth exchange, after its opening line and before its first edit; the summary carried the exchange's task as its next step, and room 1's template and page, the book and the reader's edition's chapter 2 were edited and rebuilt before the crate was checked against KV, the manifest last shipped. Row 600's fix put the check at the head of every

Next line, where a summary written between exchanges copies it; a break inside an exchange copies the step in flight instead. Measured: the check, run late, found 3,477 of KV's 3,481 files as shipped, and the four that differ are the four edited after the break: MISSED.

Kind: THE METHOD (Claude's: an opening step dropped across a break in working memory, a third time). *Lesson:* Nothing was lost. A summary written inside an exchange carries that exchange's opening line, so every exchange's opening line now names the check first ('first the crate check against KW, then …'), and the first tool call after any break is the check, before any file is opened for editing. *Proved?* Seen in runs (the crate's md5 check, 6 November).

604. A limit named for a wider class than the one it holds for. The register's found-here row 7 and the front page said sized taps hold a delayed balancer past 'the smooth controller's √2 limit'; the balance board's own note names the √2 as the textbook limit for a smooth push with a delay, which is a push on the tilt and its rate. The prior-art pass of 7 November read that smooth pushes that are not tilt-and-rate pass it: fed the tilt's acceleration by up to √2, to 2 fall times, and detuned by about 1.47 (Sieber and Krauskopf 2005, read through a summary of their preprint). Measured against it: the taps' 1.75 and 2.00 fall times pass the tilt-and-rate push's √2, and reach the 2.00 a push fed the acceleration reaches: MISSED.

Kind: WRONG READING (Claude's: a textbook bound read as every smooth controller's when it is one controller's). *Lesson:* A textbook limit belongs to one controller; a claim to pass it names that controller and asks what else has passed it. The front page, the register's row 7, the book and the reader's edition now say 'the √2 limit of a delayed tilt-and-rate push'. *Proved?* Read in print (the prior-art pass, 7 November).

605. An agreement printed narrower than its own error. The Methods' rule 'correct the reference too' said two references corrected the same way agree 'to 0.04 and 0.004 per cent' (NEVER_BLINK); that run kept no record place by place, and its error bar was owed. Measured, rerun with its places kept: -0.04 ± 0.04 and $+0.004 \pm 0.07$ per cent, one standard error over the 256 places, so the 0.004 is a coincidence about seventeen times smaller than its error: MISSED.

Kind: THE INSTRUMENT (Claude's: a difference read off one run with no error bar, printed to its last digit). *Lesson:* The rule stands, and its number now carries its bar: the two references agree within their error bars. A difference near nothing is printed with its bar or not at all. *Proved?* Seen in runs (OWED_BARS 20261107_023832, 7 November).

606. A bound that hid a bias. The front page's thirteenth result said that with the slips left out the noisy lock's gauge reads its softened hold 'within 0.034 where slips are rare' (SLIP_PROOF_3), its registered bar of 0.03 narrowly missed at two of 33 settings; that run kept no spread needle by needle. Registered, Claude's lean (G2): at the loudest noise, 0.4, at most one of its four judged gaps lies two or more standard errors above nothing, which is scatter; its competing rival (G2r): three or more do, which is a bias. Measured, needle by needle: two standard errors of the gap reach 0.034 at the quietest noise's slowest pace, so the bound is no narrower than its error there; and at noise 0.4 all four gaps lie 4.1 to 5.9 standard errors above nothing, about 0.03 high, the two that broke the registered bar only 0.5 and 0.6 standard errors past 0.03: MISSED.

Kind: WRONG GUESS (Claude's lean (G2); its competing rival held). *Lesson:* The rival won: the gauge reads high, steadily, at the noise where slips begin, and a bound on sizes hid a run of gaps of one sign. Its cause stays the rival's guess until tested: the slip finder cuts 15 time units either side of each slip and may leave slip-borne wobble in; if so, a cut twice as wide at noise 0.4 should shrink the bias. The front page now says the gauge reads within its own error save at the loudest noise. [Redesign run on 7 November (CUT_WIDE 20261107_033035), registered first: the doubled cut moves the four gaps by at most a fifth of their errors, so the bias is the loud noise's own, not the cut's; the next guess, on the proposed list, is the gauge's straight-line reading of a needle whose pull is curved, tested by the same needles under a straightened pull]. *Proved?* Seen in runs (OWED_BARS 20261107_023832, 7 November).

607. A companion's record quoted as its book. The book's working draft, in its 'grip in six questions', gave R2W's law for a fast ripple 'in its words, "heavy balls hardly feel a fast ripple"'; the reader's edition gives the same words as R2W's record's. When R2W as Bear uploaded it was read on 7 November, every quotation the two editions and the register put in R2W's mouth was looked for in it by script: all are there but that one, which its book does not print; its book says the ripple raises the pull the rider can stand, a slow one lowers it, and the weight dulls the drag and the shaking alike: MISSED.

Kind: THE METHOD (Claude's: a working record's phrase carried as the published book's). *Lesson:* A quotation names the document that prints it. The working draft now gives the words as R2W's record's, the reader's edition's wording, with a dated note; every quotation of a companion is read against its published text, by script, when that text arrives. *Proved?* Read in print (R2W as uploaded, 7 November).

608. A hand-kept list left behind its source. The register's Part D3, 'Known in kind, new in place or in number', was written by hand in October and says it states the known idea beside each result; read against Part D1 on 7 November, after the prior-art pass reached its last row, it names ten results, among them those of D1's rows 2, 7, 8, 11, 12, 14, 15 and 16, and not those of rows 6, 9, 10 and 13, all marked known in kind (rows 6, 9 and 10 since the first pass of 19 October), nor row 4's, now new at most in its form: MISSED.

Kind: THE METHOD (Claude's: a list built by hand beside a list that changed, never read against it when a status changed). *Lesson:* A list drawn from another list is built from it or checked against it. Redesign, said aloud and made the same day: the five bullets written, each the known idea and what the toy adds, from its row; and the assembler now checks that every D1 row whose status says known in kind or new at most in its form has a bullet in D3 carrying one of its run stamps, and stops the build when one is missing. Read against D3 as it stood, the check names rows 4, 6, 9, 10 and 13; with the five bullets it passes. *Proved?* Read in the record (Parts D1 and D3 of the register's source, 7 November).

609. A shipment's number given where an exchange's belongs. The draft of the abstract and description, rebuilt on 7 November, said in its build line that it was built at the session's exchange numbered 252, against the manifest's whole file name: its builder had been given the shipment's number, 252, where the exchange's own number, 13, belongs, and the manifest's file name where its two letters belong; every earlier build of the session gave the exchange's own number and the two letters, and the closeout report built beside it gave them right: MISSED.

Kind: THE INSTRUMENT (Claude's: an argument typed from the wrong counter, into a builder that took any number). *Lesson:* A label a builder prints is checked against the record that numbers it. Redesign, made the same day: the abstract's builder and the closeout report's builder now refuse an exchange number that is not the latest the session's running record has written, and a manifest not given by its two letters; given the shipment's number and the file name, each refuses. *Proved?* Seen in the record (the abstract draft as shipped, 7 November).

610. A pass's finding left out of the passages it changed. When the prior-art pass reached its last row, two passages of the working draft still said the names behind a candidate were owed, though the pass had named them: beside the sized taps, act-and-wait control (Stépán and Insperger 2006, read at the first pass of 19 October), and beside banded chaos, the phase locking of chaos (Rosenblum, Pikovsky and Kurths 1996, read in part on 7 November); and the reader's edition's chapter 16 still said a check of the literature was owed for every candidate but the first, when every one had had its pass. Read on 7 November, while drafting a front-page line for banded chaos: MISSED.

Kind: THE METHOD (Claude's: a pass written into the register and Appendix B, not searched for in the books' own sentences). *Lesson:* A finding is carried to every sentence that states the old state, found by searching for the result's runs, not by memory. Redesign, made the same day: the three sentences brought up to the passes, and the register's builder now stops when a paragraph of either edition that cites a candidate's run still calls its names owed, save two notes it lists that owe other results' names to the reference pass, and when chapter 16 says the checks are owed; run against the working draft as shipped before the fix, it names exactly the two passages. *Proved?* Read in the record (both editions, 7 November).

611. A resolution carried from where it is exact to where it is not. COIL_CHEAP swept every coil at 400 pieces, forty a turn, as COIL_SNAKE had, and its planted check M1 asked that ten times the pieces move the travel per unit of work by less than 1e-6 at every cheapest angle judged. Forty pieces a turn are exact for the even coil, whose every piece is alike, so its sums do not depend on how the wire is cut; the growing coil's radius grows along the wire, its sum is only second-rate in the cut, and the blind run read 1.45e-6 there, every even-coil case within 9e-16: MISSED.

Kind: THE INSTRUMENT (Claude's: a grid taken from a case where it is exact to one where it is only good). *Lesson:* A grid's accuracy belongs to the case that earned it; carried to a new case, it is checked again there. Redesign, made the same day before the run of record: the growing coil swept at 4,000 pieces and its share of M1 read at 40,000 against 4,000, the bar unchanged; blind again, M1 read 1.45e-8 and held. *Proved?* Seen in runs (COIL_CHEAP's first blind run, 7 November).

612. Tests promised to a list and never written to it. Replies of the session named their tests as going on the proposed list unless Bear said otherwise, and he went on each time; the tests of three replies were written into the running record, one more reply's in part, and others' not at all: the three from the reply on the light loop's modulator pages, two of the three from the reply on the coil's cheapest winding, and those of the replies just before them, of which only short names survived, one of them uncertain in meaning. Found on 7 November, when Bear asked that the list be kept up to date for review and every reply was read against the records: MISSED.

Kind: THE METHOD (Claude's: a promise made in a reply, where the records do not reach, and the list kept as a paragraph rather than a file). *Lesson:* A promise to record is kept in the record the same hour, or it is not kept. Redesign, made the same day: the proposed list is now one file of numbered rows, every test a reply names written into it before that reply's shipment, and the closeout report's builder refuses to build unless the list says it was brought up to date at the exchange being built; the five tests that survived by short name are carried, reconstructed and marked so. *Proved?* Read in the record (the replies against the running record, 7 November).

613. EXPLORER_GRID_CHECK's K3, room 11, the machine on the grid: every claim of the knob register's grid row and of the two paragraphs on the grid's own needle, in the book and in the reader's edition, agrees with the room's data, read strictly. Said aloud at the registration, before any code, from reading SWING_GRID's printed table: expected to miss on the two knocked-machine clauses and nowhere else. Measured: 12 of 14 claims agree; the book's 'a machine knocked out of step at a load above it will not come back by itself' and the reader's edition's 'below which

the load must be cut before a knocked machine will come back by itself' do not: the running start, the rotor stopped dead with the field turning, fell back into the lock at a load of 0.8 at memory 2 and of 0.6 at memory 5, both above their relock lines (0.790 and 0.541), and stayed out at every load run from 0.9 and from 0.7: MISSED.

Kind: WRONG READING (Claude's: the line where a state stops existing read as the line where every start enters it). *Lesson:* The relock line is where slipping stops being possible: shot from the saddle, below it the saddle's own branch falls back and no slipping state survives. Above it slipping is possible, not certain: SWING_GRID's own band line had printed the same stopped-dead start falling into the lock at some loads above the line and slipping on at others. The knob was written from the line, not from the band. Both sentences corrected the same day with their dates: below the line a knocked machine comes back by itself, there being no slipping to stay in; above it, it may come back or stay out, by how it was knocked; to be sure, cut the load below the line. A threshold for a state to exist is not a threshold for every start to reach it. *Proved?* Seen in runs (EXPLORER_GRID_CHECK 20261107_073202; SWING_GRID 20261016_203014; SEAM_MAP_B 20261005_023746).

614. Row 603's rule broken across a fourth break: the first tool call after any break in working memory is the check. Session K's fourth break fell in its nineteenth exchange, after the opening check (clean against LF) and room 11's registration, and before its check's read-through; the summary carried the check as done and the step in flight as its next step, and the read-through, the blind run, the run of record and the fixes in both editions were made before the crate was checked again, and before the working notes' start paragraph, which asks to be read again after any break, was read. Measured: the check, run late, found 3,507 of LF's 3,512 files as shipped, the five that differ and the five new all room 11's and its records': MISSED.

Kind: THE METHOD (Claude's: a summary's next step taken as the place to resume, a fourth time). *Lesson:* Nothing was lost. A rule kept only in the record cannot reach the moment after a break, which reads the summary first; so the two steps now also stand as an open item of the task list beside the conversation, kept outside the working memory a break summarizes: after any break, check the crate, then read the start paragraph, then resume. *Proved?* Seen in runs (the crate's md5 check, 7 November).

615. QUADRATURE_ACROSS's planted M0, its first blind run: the wing's run(), taken by text and run unchanged, repeats toy run 5d, 5.62 m/s with its twist a quarter cycle from its flap and 0.22 in step. Measured: −5.6195 and −0.2220 m/s; the toy's own axis flies the wing toward its minus side, and the record gives the speed by its size, so read as signed M0 missed, and the sweep's best, its most positive speed, sat at the mirror gap near 270: MISSED.

Kind: THE INSTRUMENT (Claude's: a direction carried from a record's words into a check of the toy's own axis). *Lesson:* The registration had said 'the wing's 90 flies forward', from the record's words; the toy never says forward. Amended before the run of record, said so in the row: the wing's speed read by its size and its best over the half circle, with a planted check added that the two half circles mirror each other (held within 1e-9); blind again, every planted check held. Read a toy's numbers in its own axes before a check is written from a record's words. *Proved?* Seen in runs (QUADRATURE_ACROSS's first blind run, 7 November).

616. QUADRATURE_ACROSS's (W1), Claude's lean at 2 to 1, and (Q1), the proposed list's at 3 to 1: the wing's best gap at a twist of 30 degrees within 2 degrees of a quarter turn, and every driven toy read within 5 degrees of the rule's gap. Worked by hand before the run: in steady flight the wing's mean thrust is the same a given angle either side of the quarter, so its best could leave the quarter only through the body's speed rippling within a flap. Measured: at twists of 5 and 15 degrees one top, at 89.4 and 89.7; at 30 and 45 the top splits in two, at 70 and 111 and at 66 and 115 degrees, their midpoints 90.5, and the quarter sits in the dip between them, the fastest 1.2 and 6.5 per cent above it; the snake, the light loop, the rider (87.8) and the skater (0.2) at their rule: MISSED.

Kind: WRONG GUESS (Claude's: a symmetry taken to fix a top, when it fixes only a level point). *Lesson:* The symmetry held, to half a degree; what it fixes is that the quarter is level, a top or a dip, and the air's square law, strong at a big twist, turned it from the one into the other. The rival's number held for a reason it did not name. A symmetry says where a curve is flat; its curvature says whether that is the best, and a lean drawn from a symmetry needs the curvature read too. At a twist of 30 the quarter is still the wing's cheapest timing, its work a flap least there. *Proved?* Seen in runs (QUADRATURE_ACROSS 20261107_080953; WING_HUMPS of 7 November).

617. What the deposit carries, read while drawing the road to upload: of the five files that go up, none is a room page, so the explorer's rooms reach the reader only as pictures and chapter 2's table, and what the four rooms still to build add to the deposit is their checks. Measured on 8 November, when the files that go up were read again for how the deposit should be laid out: the archive that goes up holds every room's page, rooms 1 to 10, each beside the template it was built from, with the skater, the Stretch and Squeeze Explorer and the Torus of Fractions; chapter 2 of the reader's edition names all thirteen by file; and that edition's last section, on what travels with the book, still said six toy pages and put the companion volume inside the archive, which goes up beside it: MISSED.

Kind: WRONG READING (Claude's: the five files at the top read as everything that goes up, though the archive's web pages had been counted, and read as the six toys). *Lesson:* A file inside the archive goes up as surely as a file beside it. The road was redrawn the same day: a room's page, once built, goes up inside the archive and joins chapter 2's table, so leaving the last four rooms' pages for later means this deposit's explorer stops at room 11, and building them

adds four pages a reader can open. The last section's two slips were corrected the same day; the Zenodo description's count of six, a line of the front page, waits for Bear's word. Before any claim of what goes up, list the files that go up by kind, with the archive's members counted inside it. *Proved?* Read in the record (the archive in the dry upload set, 8 November).

618. EXPLORER_SOUP_CHECK's K3, room 12, the wave soup and its ears: every claim of the seven knob rows the room reads and of the three paragraphs they cite, with their twins in the book and in the reader's edition, agrees with the room's data, read strictly. Said aloud at the registration, before any code, from reading the runs' printed tables: expected to miss on two sentences and nowhere else. Measured: 45 of 48 claims agree. The fitted slope's 'every setting that cuts the switching cuts the headway', in its knob row and in 'Slope and rate': of FIT_KNOBS's 30 settings, four pairs have one switching less and drifting more, the plain ear at a knee of 0.5 against the fits over five and six looks at 0.75, the plain ear at 0.75 against the fit over six at 1.0, and the fit over six at 0.5 against the fit over two at 0.75, each by less than 0.004 wavelengths a cycle. The riders' 'The riders of one soup share one fate': with a quarter of the power running back and a margin of 0.1, soup 4 lost one rider of its 64, at 13.14, while the other 63 rode the 1,000: MISSED.

Kind: WRONG READING (Claude's: a table read by its trend and its rows, where an 'every' needs its cells read, pair by pair and soup by soup). *Lesson:* Both sentences were written from tables that say them almost: the fit map's settings fall along a line with four near-ties against it, and the by-soup table is all ones and zeros but for one 0.016, a single rider. An 'every' over a table is a claim about each cell, and a trend or a pooled share cannot carry it. Both found by reading the printed tables before any code, as rooms 9 and 11 found theirs, and confirmed by the check; both sentences corrected the same day with their dates, the line now a near line and the one fate given its one exception. The redesign, said aloud: is the lone rider's fall at 13.14, in the ride's first fiftieth, a fall at the start, where the rider leaves its holding point before the left-going pattern has swept far, and not at the pattern's worst stretch? On the proposed list (P74). *Proved?* Seen in runs (EXPLORER_SOUP_CHECK 20261108_070632; FIT_KNOBS 20261015_204319; SOUP_JUMP_3 20261016_225440).

619. When room 11's page was built on 8 November, every place that lists or counts the toy pages took it in. Read the same day for room 12's page, when chapter 2's table, its count and the Zenodo description were given a fifteenth: chapter 2's table, its opening count and the description had taken room 11, but chapter 2's paragraph on the later rooms still said seven rooms and named the seven before it, and the book's own table of toys still said thirteen and had no row for room 11: MISSED.

Kind: THE METHOD (Claude's: the places that list the toys found from memory, not searched). *Lesson:* A list kept in five places is kept in five places. The deposit's check holds the reader's edition's named files against the pages that go up, so a missing file name there is caught; a count in words, a paragraph's list and the book's own table, which does not go up, are not held by any script. Both added the same day, with room 12. The redesign, said aloud: when a page joins the toys, search both editions and the description for the last room's file name and for the count's word before writing, and give each place found the new page; the working notes' steps for a room carry it. *Proved?* Read in the record (chapter 2 and the book's table of toys, 8 November).

620. A rule written down after its first slip keeps the next slip away. Written on 8 November, after a new row of the data ledger named a run's stamp and the headline register moved that run's judged lines under the new row, the register filing a run under the first row of the data ledger that names its stamp with its tool: name another run in a registration without its stamp. Broken twice more the same day: a bracket added to ROOMS-CATCH named a check run with its stamp, and room 13's registration (METER-ROOM) named EQUAL_CATCH's, RESTED_LAG's and LAG_METER's, which moved their judged lines from chapter 14 to chapter 13. Each was caught by the diff of the headline register against its last copy, before any build, the stamps taken out and the register rebuilt to its earlier characters exactly: MISSED.

Kind: THE METHOD (Claude's: a rule kept in the notes, where the hand writing a row does not look, and not in the tool whose quiet habit it guards against). *Lesson:* The register's habit is silent: a stamp in a row's prose files the run there, with no warning, and the hand that writes the row is thinking of the room, not of the register. The diff caught all three because it is run after every edit of the data ledger, so nothing wrong was built or shipped; but the diff is a net under the rule, not the rule kept. A rule against a tool's silent habit is kept best where the habit lives. The redesign, said aloud: the register prints, at every build, each run filed under a row other than the one that registered its tool, so a slip is named where it happens and not found by a diff after it; on the proposed list (P79), not built without Bear's word. *Proved?* Read in the record (the headline register diffed against its last copy, 8 November).

621. EXPLORER_METER_CHECK's K3, room 13, the maintenance meter: every claim of the three knob rows the room reads and of the paragraphs they cite, with their twins in the book and in the reader's edition, agrees with the room's data, read strictly. Said aloud at the registration, before any code, from reading the runs' printed tables: expected to miss on two sentences and nowhere else. Measured: 48 of 50 claims agree. The reader's edition's front page, item 7: an alarm reading the wearing clock where nothing hides it, 'set to catch nine machines in ten', wastes a thirtieth to a twentieth of a machine's life; the alarms that waste 0.031 to 0.051 were set to catch as many riders as the calendar

caught, 0.852 to 0.875 of them by seed. The book's 'the learned reminders gain nothing: +0.005 and -0.002 of a life on average, … 0.016 and 0.017', after a sentence naming the power reminder first: the power reminder gains -0.0021 (spread 0.0173) and the wobble reminder +0.0048 (0.0162), so each pair is the wrong way round: MISSED.

Kind: WRONG READING (Claude's: a headline compressed from a chapter that kept the first setting the chapter states and dropped the one its number was measured under; and a pair written into a sentence without reading it again against its table in the sentence's own order). *Lesson:* The chapter says how the first fleets' reminders were set, to catch nine in ten, and later says 'Set to catch the same riders' before the figures the front page gives; the front page kept the first and dropped the second. EQUAL_CATCH's table prints the power reminder before the wobble reminder, as the sentence before names them, and the sentence gave the pair the other way. A headline that compresses a chapter carries the condition its own number was measured under, not the first condition the chapter states; and a pair in a sentence follows the sentence's order, read against the table each time. The book's sentence corrected the same day with its date; the front page's item waits for Bear's word, front-page changes being his. The redesign, said aloud: read every number on the front page against the setting its run was measured under, the run's own setting line beside it, before the backward pass; on the proposed list (P80). *Proved?* Seen in runs (EXPLORER_METER_CHECK 20261108_182139; EQUAL_CATCH 20261016_192548; RESTED_LAG 20261016_205504).

622. EXPLORER_COLUMN_CHECK's K3, room 14, the column and the hiker: every claim of the four knob rows the room reads and of the paragraphs they cite, with their twins in the book and in the reader's edition, agrees with the room's data, read strictly. Said aloud at the registration, before any code, from reading the runs' printed tables: expected to miss on two sentences and nowhere else. Measured: 31 of 33 claims agree. The halt paragraph's 'Each man stops for three tenths of a second', for a column keeping a time gap of 1.5 s with the halt ordered at once: walkers 5 to 30 halt 0.30 s, but walker 1, the first behind the head, halts 2.16 s in two pieces. The night paragraph's 'A walker who follows by sight … fares at a metre much as the fixed 5 metres did by day': with the column's limit repaired, 4 stop at night against 9 by day, and at night a man is pressed to half a metre where by day nobody comes closer than 2.63 m: MISSED.

Kind: WRONG READING (Claude's: an 'each' read from the many walkers that agree, not walker by walker; and a comparison left standing when the numbers under it were corrected). *Lesson:* The halt's 'each man' was written from a row where six walkers of seven say 0.30 and one says 2.16: a row's majority is not each of its cells, the lesson of row 618 met again. The comparison was true when it was written: before the limit was repaired, 14 stopped at night against 15 by day, both pressed close. The repair corrected every number in the paragraph and left the sentence that compared them. A correction to numbers is a correction to every sentence that compares them, not only to the numbers. Both sentences corrected the same day with their dates. The redesign, said aloud: when a run's numbers are corrected, search both editions for every sentence that compares them with another run's, not only for the numbers themselves; on the proposed list (P87). *Proved?* Seen in runs (EXPLORER_COLUMN_CHECK 20261108_194717; HALT_ORDER 20261015_052122; LIMIT_AUDIT 20261017_190938; NIGHT_SPACING 20261013_231417).

623. ROOMS-CATCH's (C1), Claude's lean, registered 3 November before any of rooms 10 to 15: every one of those rooms whose check reads rows of the knob register finds, in its registration or its check, at least one claim in the book that its data does not give. Measured, the last room checked on 8 November: rooms 11 to 15 counted (room 10 reads no knob row), finding one, two, two, two and none; room 15, the first swimmer with Bear's broad faces, one row and one run, finds none, said aloud from its run's table and confirmed by its check: MISSED.

Kind: WRONG GUESS (Claude's: a rate read from the rooms before carried to a room unlike them). *Lesson:* The catches of rooms 11 to 14 came from three shapes of sentence: an 'every' or 'each' read from most of a table's cells (rooms 12 and 14), a comparison or a condition left standing when the numbers or the setting under it changed (rooms 13 and 14), and a line or a pair read in the wrong way, the line of a state's existence read as the line every start reaches, a pair given in the table's order where the sentence set another (rooms 11 and 13). Room 15's paragraph, written in one sitting from one table of six bodies and two sweeps, never corrected after, holds none of the three shapes. A catch is a shape of sentence, not a quota per room, and a room can be clean. The redesign, said aloud: read both editions for the three shapes directly, every 'every' and 'each' over a table, every comparison whose numbers were later corrected, every summary carrying a setting from elsewhere, rather than waiting for a room to read each by its rows; on the proposed list (P88), for the backward pass. *Proved?* Read in the record (the rooms' rows in the data ledger: GRID-ROOM, SOUP-ROOM, METER-ROOM, COLUMN-ROOM and FACE-ROOM, 3 to 8 November).

624. Row 619's redesign, when a page joins the toys: search both editions and the description for the last room's file name and for the count's word, and give each place found the new page; kept for rooms 12, 13 and 14, each searched for its count of pages. Read on 8 November for room 15's page: chapter 2 and the Zenodo description each still said 'the eight later rooms check themselves the same way', a count that fits rooms 4 to 11 and was not raised for rooms 12, 13 and 14, though the other count of rooms in each was: MISSED.

Kind: THE METHOD (Claude's: the search held one count's word, and the toys are counted twice). *Lesson:* The toys are counted as pages, seventeen and now eighteen, and as later rooms, which the same texts count again in other sentences; the description says 'the eleven rooms' in one paragraph and 'the eight later rooms' in another, so one

count was raised and its twin left. A search for one word finds one count. Both places now say twelve, with room 15. The redesign, said aloud: when a page joins the toys, search for every number word before ‘rooms’ and ‘pages’, not only the count’s word, and read each against the new total; the working notes’ steps for a room carry it beside row 619’s. *Proved?* Read in the record (chapter 2 and the Zenodo description’s source, 8 November).

625. P79, Claude’s lean, at Bear’s yes on 8 November: a guard in the register that prints, at every build, each run filed under a row of the data ledger other than the one that registered its tool, read from the row whose HERE column names the tool’s own file; on today’s ledger it prints nothing. Built the same day: it printed two. CAT_MONTGOMERY’s run of 20 October is named with its stamp in CAT-MIRROR-PROOF’s row, in chapter 13, above its own, MONTGOMERY-BENCH’s, in chapter 12, so its four held verdicts had been counted under chapter 13; and TRAIN_LIFETIME’s run of 18 October is named first in TRAIN-HOLD’s row, the same part, so nothing it was counted under changed: MISSED.

Kind: THE METHOD (Claude’s: the register filed a run under the first row that names its stamp, and a guess that the last three slips were the only ones). *Lesson:* The three slips of 8 November were caught by the register’s diff because they moved a count the day they were made; these two were made in October, before the rule that a row names another run without its stamp, and the first was absorbed into the counts long before any diff could show it. A guard that starts from today’s ledger finds the old slips too. The register now files a run by its own row when the ledger names one and prints every run that another row names first, so a new slip shows at the build that makes it; the cat’s four verdicts are now counted in chapter 12, the register’s totals unchanged. The guard reaches only the runs whose row names their tool’s file where it can read it, most of them recent; the rest are filed as before. The redesign, said aloud: every new row names its tool’s file in its HERE column, as the rooms’ rows do; whether the older rows can be matched to their runs by script, from the run each names in its DONE entry, is on the proposed list (P92). *Proved?* Seen in runs (the register’s own print, 8 November).

626. The register’s sorting of verdict lines, read by a fresh reader of the record on 8 November: three registered predictions, RIDER_FLEET’s M4 and M5 and SHUFFLE_NULL_2’s M1, each so named in its tool’s docstring and each missed, were counted as planted checks, because the register calls any label beginning with M a planted check; and the record’s line on the planted checks, ‘every one that failed was a fault in the tool, caught’, was false for them: MISSED.

Kind: THE METHOD (Claude’s: a label’s first letter taken for what its line is). *Lesson:* The register already read a tool’s docstring for ‘(planted’ to catch planted checks labelled otherwise; it had no rule the other way, for a registered prediction labelled M. Fixed by naming the three in the register, as it names the others: the predictions judged are now 1,162 with 364 missed (SLIP_SPREAD’s rival (K1b’r), counted among them by its summary’s code name, moving to the rivals at the same build), the planted checks 489 with 12 failed, and the twelve are each an instrument fault by its own miss row, so the line now holds. The converse, a tool’s fidelity and rig checks counted as predictions, is a question of what the counts mean, and waits on the authors’ decision. The redesign, said aloud: sort every verdict line by its tool’s docstring’s own word for it, planted, fidelity, registered or lean, and not by its label’s letter; on the proposed list (P93). *Proved?* Read in the record (the backward pass’s reader RC, 8 November).

627. The backward pass, at Bear’s word of 5 November, run 8 November: thirteen fresh readers, agents that had not seen the work, read the reader’s edition, the working draft and the record against the runs, every finding checked at its source before anything changed. Every result the books lead with stood. But 175 findings came back, the readers calling 7 of them wrong facts and 36 misleading, almost all confirmed at their sources: the books’ sentences had said more than their runs in many small places that the room checks, the twins’ comparisons and the register had all passed: MISSED.

Kind: THE METHOD (Claude’s: sentences checked by the people who wrote them, against the runs they remembered). *Lesson:* The shapes repeat the rooms’ (misses rows 618 to 624). A condition dropped from a summary: the fuse of 130 to 175 time units at the edge’s gap only, the gauge’s error at the loudest noise only, the column’s ‘a fifth or less’ to 0.4 s only. A unit misnamed: EAR_PRICE_CHECK’s delays called shares of the soup’s period, where its code sets them in time units, a fifth of what the label said. A count stale: ‘Five of the toys’, ‘ten thousand settings’, 35 pairs of a four by nine. A note pointing at a run that does not hold the number. A law named for the wrong quantity: a pull falling as the cube, where the cube is the depth of the hold and the pull falls as the fourth power. And working notes left in a deposited book. Each was corrected with its date, or, where it touched the front page, the description or a page Bear plays, put to him. The redesign, said aloud: a fresh reader, not the writer, reads every version against its runs before it goes up; and the brief the readers worked from, with its five things to check, is kept with the project’s working files for that. *Proved?* Read in the record (the backward pass, 8 November).

628. COLUMN_WALK’s (W2), Claude’s lean at even odds, registered 8 November before its tool: passed back man to man in a column a metre apart, the halt reaches the thirtieth man, his first halt beginning, more than 15 s after the head’s. The walk, run of record the same day: 10.722 s, about 0.36 s a man, where each walker reacts 0.2 s late: MISSED.

Kind: WRONG GUESS (Claude's: a halt's arrival read at the pace of its growth). *Lesson:* The halt travels back fast and grows as it goes. It reaches the thirtieth man in under 11 s, yet he stands 46.16 s in four pieces where the head stands 10: its arrival and its size move at two different paces, about 0.36 s a man for the first and over a second a man for the second, and the lean had taken the one for the other. The redesign, said aloud: register the halt's arrival a man from the walkers' own law, the reaction delay plus the time a gentle walker takes to fall below 0.05 m/s once the man ahead has, and read it from the same walk at delays of 0.1, 0.2 and 0.3 s; on the proposed list (P96). *Proved?* Seen in runs (COLUMN_WALK, 8 November).

629. The PDF build's own figure check, which counts the captions that open an image in each markdown source and prints FIGURES MISSING when the PDF draws fewer images, read on 9 November after the book's build printed 'figures in source 35, images in PDF 36'. The book has 36 pictures. The newest, the twelve later rooms, carries a dated bracket inside its caption, as the books mark an added line, and the counter's pattern stops at the first closing bracket, so it counted 35. The PDF drew all 36 and printed the caption whole: MISSED.

Kind: THE METHOD (Claude's: the books' dated bracket put inside a caption the counter reads). *Lesson:* A check that undercounts its own input passes in silence: with one picture uncounted, a second picture missing would have printed 35 against 35 and raised nothing. Nothing printed wrong this time. The redesign, said aloud: count every mark that opens an image, whatever its caption holds, and print both counts whenever they differ; on the proposed list (P105), the tool left as it is until Bear's word. *Proved?* Seen in the book's build (9 November).

630. CLOCK_SHIFT_30's first blind run (9 November): the light needle's doublings refined in 30 digits, each passing's state dropped to its midpoint at every step, the Jacobian's product over the passings and the doubling's residuals left as arb's balls. Periods 4 to 32 converged in two steps at both clocks, 2.2e-14 to 4.5e-13 from the record; at period 64 the golden clock's refinement returned nan. arb bounds a product by the sizes of its factors, and 64 factors bounded one by one give a ball far wider than their product, which the doubling keeps near one in size; once the ball held zero, Newton divided by it: MISSED, before any verdict printed.

Kind: THE INSTRUMENT (Claude's: a ball's arithmetic trusted through a long product of matrices whose sizes cancel). *Lesson:* A careful arithmetic can fail by being too careful: each bound is honest, and together they say nothing. Caught by the blind run before any lean was read, which is the blind run's whole job. The redesign, said aloud and made before the next run, the leans, their bars and the planted checks' tolerances untouched: the product and the residuals dropped to their midpoints at every passing, as the state already was, so the arithmetic is plain 30-digit floating point; the run of record started beside the blind one stopped and moved out unread. *Proved?* Seen in runs (CLOCK_SHIFT_30, its first blind run).

631. CLOCK_SHIFT_30's second blind run (9 November), after the tool had been read against its registration line by line before its first run (row 569): its M2 printed MISSED at a spread of 3.2e-26 against a bar of 1e-20, because the code's M2 also asked every refinement to reach its 1e-24 stopping bar, which the registration's M2 does not ask. At period 64 the Newton steps fell to 1e-28 and two starts agreed within 2e-28, while the residual stayed just above 1e-24, the angle having grown past 200 over 64 passings: MISSED, the read's.

Kind: THE METHOD (Claude's: a condition added in the code and read past, because the read checked that each registered clause was in the code and not that the code held nothing more). *Lesson:* A line-by-line read that walks the registration finds what the code leaves out, never what it adds. The redesign, said aloud: read the other way too, each verdict line's condition in the code against its registration sentence, every clause of the code matched to words of the registration, before the first run (on the proposed list, P111). The code corrected to its registration before the next run, and the run of record started beside the blind one moved out unread. *Proved?* Seen in runs (CLOCK_SHIFT_30, its second blind run).

632. CLOCK_SHIFT_30's (C2), Claude's lean at even odds, said so: in 30 digits the ratio of the clock shifts' differences at periods 16, 32 and 64 lies within 0.01 of Feigenbaum's 4.6692. Measured: 4.659088, 0.0101 below: MISSED, by a hundredth of the bar.

Kind: WRONG GUESS (Claude's: the shifts' ratios forecast to close on 4.669 as fast as a rung of the drives' own, read from the gap's shrink at the rungs below). *Lesson:* The ratios climb as foretold (C1 held), but a rung late: beside them the drives' own ratios run 4.4595, 4.6155, 4.6570 and 4.6667 one rung earlier, and the shifts' read 4.4540, 4.6201 and 4.6591, each within 0.006 of the drives' a rung before. Can it be redesigned, or is the redesign covered? Redesigned, said aloud: forecast each shift ratio from the drives' ratio one rung earlier, so r(32-64-128) within 0.002 of the drives' r(16-32-64), 4.6667; it needs period 128, which double precision could not follow, so it waits on P109 and is written there. *Proved?* Seen in runs (CLOCK_SHIFT_30 20261109_034237).

633. CLOCK_SHIFT_30's three blind runs, read into the register on 9 November: counted as runs of record, their verdicts as predictions, the blind count left at 211. The register, its assembler and the record's index know a blind run by two marks, '*BLIND*' in its name and '(BLIND' in its header (row 544); the new tool's header printed the word BLIND after the stamp, without the parenthesis: MISSED, caught by the headline register's diff before any count was shipped.

Kind: THE METHOD (Claude's: a new tool's header written fresh instead of copied from the last tool's). *Lesson:* The diff after every ledger edit did its job. The run files cannot be changed, and the tool is not changed after its run of record, so the three readers now accept the word after the stamp as the header's mark, both marks still required: the register's counts then rose by one run of record and three blind runs, two predictions, one rival and three planted checks, as they should. For new tools: copy the header line from the last tool built. *Proved?* Seen in the register's diff (9 November).

634. The reference pass of 9 November, which read every entry of the reference list against its source, and a read the same day the other way, every surname and year the reader's edition cites against the list: six works cited and listed nowhere, Astley and colleagues (2015) and Iino and Yoshida (2009) in the chapters, Briggs (1991, 1994 and his 1997 thesis), McCartney (2020) and Finch (2003) in the appendix the register fills, besides McDonald and others (1985) found earlier that day: MISSED, the pass's own reach.

Kind: THE METHOD (Claude's: a check run from the list toward the text, never from the text toward the list). *Lesson:* A list can be right line by line and still be short. The redesign, said aloud: the citation read both ways at every build of the reader's edition, the text's citations against the list as well as the list's entries against their sources; the six entries written as their publishers' records are read, each note saying what was read. *Proved?* Seen in P114's read (9 November).

635. The reader's edition's chapter 4, read in the stale-line sweep of 9 November (P119): it said that why the four barriers first read sit higher than the later sweep 'is a test proposed, not yet run'. No list ever held that test, and its answer stood in the run the sentence itself cites one line earlier: against the exact rate the four read 1.069, 1.062, 1.097 and 0.986, each within two of its counting errors of 1 (KRAMERS_EXACT 20261020_073520, printed beside its verdicts). The sweep's lean, that at least two lines stand where a later row closed them, missed too: none did; the one wrong line was wrong when it was written: MISSED.

Kind: THE METHOD and WRONG GUESS (Claude's: a condensing pass that ended a paragraph on a plausible open end with no address, and a lean that expected stale lines where the fault was an invented one). *Lesson:* An open end needs an address. A sentence that says a thing is proposed, owed or still to be tested must name the list row, card or run that would close it; one with no address is either stale or invented, and only the address tells which. The redesign, said aloud and made the same day: every 'proposed' in the edition read against the list (the one other hit makes no claim to a list), and every open end the edition prints given an address in the seeds file, which a check reads at every build (P124). *Proved?* Seen in P119's sweep (9 November).

636. THIRD-AXIS (T1), at Bear's reading that the pendulum loads its curvature on an axis the book never drew: Claude's lean that no doubled lock and no chaos appear below the memory where the needle's angle-only map first folds, M_fold, read as the smallest memory at which any quasi-periodic needle's smallest stretch falls below 0. The fold was read at memory 0.8 (−0.0126), but the needle tumbles from memory 0.5 (two paces chaotic, one doubled), and at 0.5 to 0.7 every quasi-periodic needle's map was still one-to-one, its smallest stretch +0.0237 to +0.0687, larger than at 0.4 (+0.0117) (THIRD_AXIS 20261109_170732): MISSED.

Kind: WRONG GUESS and THE METHOD (Claude's: the fold was looked for only between the steps, on the needles that neither lock nor tumble, while a needle tumbles inside a step or at its edge, where those needles are not). *Lesson:* A fold is read where the tumble is. Measured on the orbits that survive between the steps, the map looked safest just where the needle first tumbled, because the needles that were folding had left the sample by locking or tumbling. The redesign, said aloud the same day: THIRD-AXIS-2 (F), the invariant circle read inside the steps as the lock's saddle's unstable manifold, at the three paces that first tumble, memory 0.30 upward; and (P127) the chaos's own strobe read as a curve, where the fold should show as a single hump. *Proved?* Registered 9 November; the redesign registered the same day.

637. THIRD-AXIS (Z2), Bear's z-needle: Claude's lean that in the banded chaos beside 5/7 the slope over windows of 7 clock turns dwells near 5/7 and its Farey neighbours 2/3 and 3/4 and hops between them. All 14,300 windows lie between 0.70533 and 0.72042, one peak at the bin centred 0.7175, near 5/7 alone (THIRD_AXIS 20261109_170732): MISSED.

Kind: WRONG GUESS (Claude's: the excursion's bound over 30,000 cycles, a quarter turn either way, was read as room to reach the neighbours within seven cycles, when the bands visited in a fixed order hold every seven cycles to within about a sixteenth of a turn). *Lesson:* A bound over a long run is not a range over a short window: banded chaos is held tighter the shorter the window that matches its bands. The free chaos beside it, at 1.0250, is where the windows spread and lump (0.4972 to 0.6893). The redesign, said aloud the same day: THIRD-AXIS-2 (D), the free chaos read in windows of 5 to 13 clock turns, its piles tested against the windows shuffled. *Proved?* Registered 9 November; the redesign registered the same day.

638. THIRD-AXIS-2, P127: Claude's lean that the chaos at memory 1, read along its strobe curve, is a single hump with its top within 0.3 radians of the corner where the needle nearly stops. The main fold sits at the corner as leaned, its top 0.1696 radians from it and its foot 0.0126; but a second wiggle about 0.003 radians high on the flat stretch, a

maximum at 2.7175 and a minimum at 3.0316, makes two humps by the count (THIRD_AXIS_2 20261109_181451): MISSED.

Kind: INSTRUMENT (Claude's: a count of extrema on binned medians with no bar for noise, where the map is nearly flat and a median's own scatter can make a wiggle). *Lesson:* A count of bumps needs a bar for the bump's height. On a plateau, where the map squeezes a wide arc to a sliver, the medians' scatter is a large share of the curve's change, and a strict count of maxima reads noise as a hump. The redesign, said aloud the same day: THIRD-AXIS-3 (P), the binned medians' errors drawn by resampling and a hump counted only when its height clears three of them; registered. *Proved?* Registered 9 November; the redesign registered the same day.

639. THIRD-AXIS-2 (D) and its rival (D-r), both: Claude's lean that the free chaos piles its windows on Farey fractions and dwells there, at 4 or more of 6 window lengths; the rival that shuffling leaves the runs unchanged. The share in piles cleared 1.5 times the range covered only at q 5, and the persistence differed from the shuffled by 18 to 122 standard errors at every length, above it over 5 and 7 cycles and below it over 8 to 13 (THIRD_AXIS_2 20261109_181451): MISSED, both.

Kind: WRONG GUESS and INSTRUMENT (Claude's: the dwelling was pictured without a time of its own; and the piles' measure loses its power as q grows, since the piles of order 11 to 13 already cover 78 to 89 per cent of the slopes' range). *Lesson:* A both-missed pair owes a third idea, and the data names it: the free chaos dwells on a fraction for about five to seven cycles and then leaves, so a window of that length sees it stay and a longer one sees it gone. The redesign, said aloud the same day: THIRD-AXIS-3 (W), the dwell itself measured, the run of consecutive cycles whose turn count keeps one fraction's pace, its length's distribution against the shuffled; registered. *Proved?* Registered 9 November; the redesign registered the same day.

640. The register's reader of THIRD_AXIS_2's run of record: its rivals (F-r) and (D-r), registered as rivals and marked competing ideas, were counted as predictions, since the reader took a label for a rival only when a digit or a single lower-case letter stood before its 'r'; the predictions read 1,101, two too many: MISSED.

Kind: INSTRUMENT (Claude's: labels written without a digit, read by a rule that expected one). *Lesson:* A reader's rule is part of the record. Each label form a new tool uses is read back through the register before its verdicts are counted. The redesign, said aloud and made the same day: the rule takes a single capital letter before '-r' too; read again, only those two lines moved, the predictions to 1,099 and the rival lines to 181. *Proved?* Seen 9 November in the headlines' diff.

641. THIRD-AXIS-3 (P): Claude's lean that, counted against a bar of three resampled errors, the chaos's map along its strobe curve has one hump, the second wiggle being noise. The second hump drops 0.00458 radians against a bar of 0.00008: it is real (THIRD_AXIS_3 20261109_191727): MISSED.

Kind: WRONG GUESS (Claude's: a wiggle on a flat stretch taken for noise, where the flat stretch squeezes so hard that its medians are sharp). *Lesson:* Where a map squeezes hardest its fine structure is easiest to measure, not hardest. The chaos at memory 1 carries a large fold at its corner and a tiny real one on its flat stretch. Redesign check, said aloud: nothing to redesign in the reading; what makes the second hump is a question for the seeds file. *Proved?* Registered 9 November.

642. THIRD-AXIS-3 (W), the third idea of misses row 639: Claude's lean that the free chaos dwells on a fraction for 4 to 8 cycles, longer than its cycles shuffled. Its dwells have a median of 1 cycle and a mean of 1.6133 against the shuffled 2.2394 (THIRD_AXIS_3 20261109_191727): MISSED.

Kind: WRONG GUESS (Claude's: persistence over windows of 5 and 7 cycles read as dwelling, when a needle that alternates fast and slow cycles can repeat a window's fraction by near-recurrence while never staying on it). *Lesson:* A window that repeats is not a needle that stays. The free chaos alternates from one cycle to the next; its windows of 5 and 7 repeat their fractions for another reason, and its windows of 8 to 13 avoid them. Redesign check, said aloud: the cycles' own correlation from one to the next, its sign at each needle (P132); carried into the seeds file. *Proved?* Registered 9 November.

643. P126: Claude's lean, from a straight line through two stretches, that in thick fluid the second wave folds the needle's map by AMP 1.6 and tumbles it past that. At AMP 1.2, 1.6 and 2.4 the smallest stretch is +0.0048, +0.0055 and +0.0654, with no chaos and no doubling (THIRD_AXIS_3 20261109_191727): MISSED.

Kind: WRONG GUESS (Claude's: a fall from two points extrapolated past where the curve turns). *Lesson:* Two points make a line, not a law. Thick fluid does not tumble under a second wave up to twelve times the book's: memory, not the wave, admits the tumble, more firmly than THIRD_AXIS showed. Redesign check, said aloud: where in the plane of memory and wave the wave can stand in for memory (P130); carried into the seeds file. *Proved?* Registered 9 November.

644. P129: Claude's lean that at each of three paces the fold leads the tumble by 0.005 to 0.025 of memory and a lock's node multiplier passes between 0 and −1 before the first chaos. The fold leads by 0.030, 0.025 and 0.015; at pace 1.025 the 2/7 lock folds with its multiplier positive, dipping to +0.02112 and rising again, never negative

(THIRD_AXIS_3 20261109_191727): MISSED.

Kind: WRONG GUESS and THE WORDS (Claude's: a bar set by the coarse grid's step; and the book's new sentence, that the map folds when the lock's slope passes 0, read as the only way it folds). *Lesson:* A lock's multiplier is the map's slope taken round the lock, so it shows a fold only when the fold reaches the lock's own points; the circle can fold between the saddle and the node while the slope there stays positive. The redesign, said aloud and made the same day: both editions' sentences corrected to say so, with the fine grid's leads; where along the manifold the fold opens is P131, carried into the seeds file. *Proved?* Registered 9 November; the sentences corrected the same day.

645. Read 416 lost across a break in working memory. On 10 November, at session K's fifty-second exchange, the read was given in Claude's reply and not written to the reads file, the exchange's records being left to ship with the next batch of files; session K's twelfth break fell before Bear's next message, and its summary carried that a read 416 had been given, not its words. Whatever tests that reply's surprises named are not on the proposed list either, whose heading still named the earlier exchange. After the break, the first tool calls were the answer Bear had asked for and a listing of the crate, the check against the last manifest only after them, though still before any file was opened for editing (rows 603 and 614 ask for it first): MISSED.

Kind: THE METHOD (Claude's: a read given before it was written down, as row 612 found for tests). *Lesson:* Nothing in the deposit was lost; the read's words are in the conversation. A read or a test that lives only in a reply is lost at the next break, so each read is now written into the reads file before it is given, and each surprise's test into the proposed list, and the reply gives them from the files; read 417 was the first written so. Read 416's place is kept, to be filled word for word if Bear carries its words back. *Proved?* Seen in the records (the reads file against the break's summary, 10 November).

646. P118, the read of the finished cards the reader's edition never names, sorted for the results a reader should meet: planned on 9 November for Bear's final read, and dropped from the next-step lines from the next day on, though it stayed on the proposed list. Bear's remark at his final read, on the edition's chapter 16 ('confirming any worth noting modest results are checked by you for what deserves recognition'), found it: MISSED.

Kind: THE METHOD (Claude's: a step kept on a list but tied to a later moment, and the moment never read the list). *Lesson:* Nothing was lost; P118 ran the same day: of the 111 cards, 15 in three groups are proposed for the edition (Bear's two walkers, the saddle up close, the half gear's closed bubble), 36 are told without their runs named and 60 are workshop depth. A row tied to a moment ('at Bear's final read', 'before upload') is now named in the working notes' next line until it is done, and P143 asks for a script that lists every such row at every build. *Proved?* Seen in the records (Bear's remark, 10 November).

647. The reader's edition's front matter, Appendix B and back pages, and the workshop book's back pages, said the found-here candidates wait on a check of the literature, after every candidate's check had been written, and the register's own table still called the check owed; Bear asked at his final read whether the candidates should be cleaned up and cited. Row 610's guard read only the lines that cite a candidate's run, and these general sentences cite none; and Appendix B's table named six of the checks' sources by year alone and six not at all: MISSED.

Kind: THE WORDS (Claude's: a pass's finding left out of the general statements about it, a second time after row 610). *Lesson:* The sentences corrected in both editions and the register the same day; the guard now reads every line of both editions for those phrasings; the table names every source, and eleven works the checks found are entered in the references, each read at a record of it, four marked as checked only at a secondary record or a citation. *Proved?* Seen in the text (Bear's remark, 10 November).

648. Bear's walker, the board he sketched on 4 October and named himself, told in neither book's prose: run that day and told in that day's notes, its follow-ons run on 11, 13 and 15 October, but filed under his queue of ideas, which had no chapter, so the follow-ons went only into the workshop book's table of questions and the walk itself, which never had a card, onto no page; the reader's edition, condensed from the book's prose, never met it, and the passes that looked for what the edition left out read cards, so none could find a run without one. Bear asked where it had gone: MISSED.

Kind: THE METHOD (Claude's: a toy filed by where it came from rather than by what it is, with no chapter to hold it; and passes keyed to cards). *Lesson:* The walk written into chapter 14 at Bear's word on 10 November. P144 read every one of his named ideas against both books: two more told in neither book's prose (his cliff edge, his funnel), one cut when the edition was condensed (his pentagon star) and one small result never told (his dip on the stop drawing), each drafted for his word. A pass that looks for what a book leaves out now reads runs of record, not cards: 187 of 392 tools are never named in the edition, and P146 reads the ones no card names. *Proved?* Seen in the records (Bear's question, 10 November).

649. The proposed list's P15, read on 10 November from CHAOS_ORDER_2's kept rates at Bear's word: Claude's lean, at even odds, that the two near-edge locks' move over the banded chaos's grows with the noise until the locks melt, read as rising at every step from noise 0.001: 5.94, 11.15, 16.96, then 10.30 at 0.03: MISSED.

Kind: THE WORDS (Claude's: a lean whose stopping point, 'until the locks melt', was never defined, so the test could not tell a fall after melting from a fall before it). *Lesson:* Read after the data, a description and not a test: at noise 0.03 both locks' rates lie inside the chaos's band and at 0.01 both just outside, so the ratio rose at every step while the locks stood apart and fell once they had melted into the chaos. Redesign, said aloud: the noises between 0.01 and 0.03, the melt placed first by a rule of its own and the ratio read across it (the proposed list's P152). *Proved?* Seen in runs (CHAOS_ORDER_2 20261013_225021, read 10 November).

650. P46's first version, written on 10 November: its planted line M0 checked the heavy needle's line at memory 2 against the sweep's printed 0.7898648, where the line it shoots is the edge's own shooting line, 0.7898646, and so called its own faithful copy of the tool MISSED by two in the seventh place.

Kind: THE INSTRUMENT (Claude's: a planted check aimed at the nearest printed number rather than the one its own method reproduces). *Lesson:* Version 2 checks against the shooting lines SEAM_MAP_B printed and holds; version 1's output is kept beside it. A planted check names the line of record it repeats, by tool, stamp and line. *Proved?* Seen in runs (P46, versions 1 and 2, 10 November).

Misses 651 to 656

651. P62, the deposit gate's read of every file name the two books print against the files that go up, run on 10 November: Claude's lean, at 3 to 1, that it finds nothing beyond the two folder names one miss quotes in telling its own fault. It found one such folder name, a shorthand for two files both present, and three runs cited by their whole file names that the archive does not hold: the hinge tool's first version, never kept by any crate and not said so, and two retired blind runs, each called retired in its own sentence but left out of the record's table of the runs it cites that the archive does not hold: MISSED.

Kind: THE INSTRUMENT (Claude's: the guard of 31 October, row 541, and the record's builder read citations only in the form 'TOOL stamp'). *Lesson:* Redesign, said aloud: both read the whole-file form too; the hinge's first run then stops the build until its sentence says it was never kept, and the table reads twelve. Asked of Bear first, as it changes the record's text. *Proved?* Seen in the archive (P62, 10 November).

652. P63, the gate's count of what travels set against what the books print, run on 10 November: Claude's lean, at 2 to 1, that one more count is behind beyond the description's toy pages. Two were: the front page and the reader's edition's last section give 582 runs of record, counted at the crate's top level only, where the archive's second computer folder holds two more that the record's own index calls of record; and the Zenodo description puts the front page among 'the eighteen toy pages', where the edition counts it apart and nineteen pages go up: MISSED on its number.

Kind: THE WORDS (Claude's: counts defined on the crate's top level, and words written before the front page joined the pages). *Lesson:* Both asked of Bear first, as they stand on the front page and in the description. A third in a working file only: the register's tools in the archive, 566, counts two working helpers that never go up; the archive holds 564. *Proved?* Seen in the archive (P63, 10 November).

653. P146, the runs the reader's edition never tells, read by subject on 10 November: Claude's lean, at 2 to 1, that at least three more results a reader should meet stand among the runs no card names. Among those, one: why the needle's ratio is 4.6692, the shape of the rule's top and the map that keeps areas, told in the workshop book and cut in condensing. Two more stood behind cards counted as told because a sibling leg was told: Bear's word down the line, the grunt of his night marches heard behind at once, and where the needle's swirl comes from: MISSED as written.

Kind: THE METHOD (Claude's: passes that call a card told when any one of its legs is told). *Lesson:* Two drafted for Bear's word, the third left to the record. Redesign, said aloud: the edition read by run for every card with more than one leg. *Proved?* Seen in the books and the runs (P146, 10 November).

654. P151, every bracketed finding on the proposed list and every answered row read for a change it asked of the books: Claude's lean, at 2 to 1, that at least one more such change is owed beyond the Jellyfish title. None was: the two brackets and twenty-four answered rows ask nine changes of the books, and both editions carry the other eight: MISSED.

Kind: WRONG GUESS (Claude's). *Lesson:* A finding that asks the books for a change names the sentence it changes, so that a build can look for it (proposed). *Proved?* Seen in the books (P151, 10 November).

655. The reply of 10 November that sorted the open tests into cheap and dear called P46 a read of data already kept. It shoots new lines with the tool's own code, minutes of new runs at memories 10, 20 and 50: MISSED.

Kind: THE WORDS (Claude's: a cost described from the row's kind, not from what the check does). *Lesson:* Said plainly in the next reply. A cost is said from the check as written, its new runs named. *Proved?* Seen in the records (P46, 10 November).

656. P153, the seventeen toy pages run again on 10 November with their typeface asked for without holding the page back: Claude's lean, at 3 to 1, that all seventeen paint within half a second both ways, with every outside request refused and with it left unanswered, and that their self-checks end in the same words as before. The fix did what it

was for, no page now held back on an unanswered line and every self-check the same; but room 2 painted at 0.84 seconds with the line cut and 0.78 with it hanging, as it had at 1.0 seconds before the change: MISSED on its bar.

Kind: THE INSTRUMENT (Claude's: a bar set on the absolute paint time, which the fix cannot change, rather than on the change the fix makes). *Lesson:* Redesign check: covered by the same run, read against each page's own paint with the line cut, which holds at all seventeen (within 64 ms, room 2's; the rest within 40). Room 2's own opening is the proposed list's P156. *Proved?* Seen in runs (P153, 10 November).

Part 4. Every run file in the archive

The archive that travels with the book holds 830 run files, each the file a run printed as it ran. Here is every one, in the order of its name, with its kind, read from what the file itself prints, and where it is cited: 588 files of record, 217 blind runs, 19 smoke parts, 4 files the jellyfish solver printed when DRAFT_TEST ran it and 2 fidelity checks of that solver.

A file of record belongs to a run whose verdicts count; its predictions are in Part 1 where it printed them in a form the script reads, and a long run printed in parts has a file for each part. A blind run is a run of its tool before its run of record, its judged lines printed only as yes or no, or not at all, and its planted checks printed whole, so that the tool could be checked before anything it judged was seen; the book counts the blind runs and cites few of them. A smoke part is a short trial of its tool, and its lines are not counted. A file dated only comes from the first days, before a run's stamp carried its time. A figure tool draws a figure, and a reading tool reads data already made. Six files are not this book's tools' own: they were printed on 8 October by the solver of record of the jellyfish paper that came before this book, four when DRAFT_TEST ran it, cited where that run is, and two fidelity checks of it; the solver is the jellyfish paper's and is not in this archive. Cited means named by tool and stamp, or by the file's own name, in Parts 1 to 3 of this record or in the book; a blind run or a smoke part that shares its stamp with its run of record counts as cited only where it is named itself, so that it does not borrow its run of record's citations: 533 of the 830 files are.

Run file	Kind	Cited in
`jelly_fidelity_check_rung6_legs78_20261008_loopatlas_picked_v24.txt`	a fidelity check of the jellyfish solver	—
`jelly_fidelity_check_rung6_v25_legs78_20261008_loopatlas.txt`	a fidelity check of the jellyfish solver	—
`jelly_viscous_grid_rung6_animal_v25_draft_closed_free_2mm_3s_20261008_134707_loopatlas.txt`	the jellyfish solver, run by DRAFT_TEST	Part 1; Part 2; Part 3; the book
`jelly_viscous_grid_rung6_animal_v25_draft_closed_held_2mm_3s_20261008_134707_loopatlas.txt`	the jellyfish solver, run by DRAFT_TEST	Part 1; Part 2; Part 3; the book
`jelly_viscous_grid_rung6_animal_v25_draft_open_free_2mm_3s_20261008_134707_loopatlas.txt`	the jellyfish solver, run by DRAFT_TEST	Part 1; Part 2; Part 3; the book
`jelly_viscous_grid_rung6_animal_v25_draft_open_held_2mm_3s_20261008_134707_loopatlas.txt`	the jellyfish solver, run by DRAFT_TEST	Part 1; Part 2; Part 3; the book
`run_LOOP_ATLAS_AREA_RULE_20261002b.txt`	of record, dated only	the book
`run_LOOP_ATLAS_ARMS_LEGS_20261017_012927.txt`	of record	Part 1; Part 2; Part 3; the book
`run_LOOP_ATLAS_ARMS_LEGS_BLIND_20261017_011538.txt`	blind	—
`run_LOOP_ATLAS_ARM_SHARE_20261016_182910.txt`	of record	Part 1; Part 2; Part 3; the book
`run_LOOP_ATLAS_ARM_SHARE_BLIND_20261016_182340.txt`	blind	—
`run_LOOP_ATLAS_AVALANCHE_20261004_064914.txt`	of record	Part 3; the book
`run_LOOP_ATLAS_BACKWARD_PASS_TAGS_20261002.txt`	of record, dated only	—
`run_LOOP_ATLAS_BALANCE_BOARD_20261004_161712.txt`	of record	Part 3; the book
`run_LOOP_ATLAS_BALANCE_BOARD_B_20261004_162140.txt`	of record	—
`run_LOOP_ATLAS_BANDED_FIVE_20261013_224945.txt`	of record	Part 1; Part 2; Part 3; the book
`run_LOOP_ATLAS_BAND_BRIDGE_20261016_072612.txt`	of record	Part 1; Part 2; Part 3; the book
`run_LOOP_ATLAS_BAND_BRIDGE_BLIND_20261016_071950.txt`	blind	—

Run file	Kind	Cited in
run_LOOP_ATLAS_BANKED_TRACK_20261004_154940.txt	of record	the book
run_LOOP_ATLAS_BANK_BUMP_20261015_064644.txt	of record	Part 1; Part 2; the book
run_LOOP_ATLAS_BANK_BUMP_BLIND_20261015_064639.txt	blind	—
run_LOOP_ATLAS_BEAT_SWING_20261015_204207.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_BEAT_SWING_BLIND_20261015_204133.txt	blind	—
run_LOOP_ATLAS_BELL_BALANCE_20261016_193628.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_BELL_BALANCE_BLIND_20261016_192112.txt	blind	Part 3
run_LOOP_ATLAS_BELL_CLOCK_20261016_072241.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_BELL_CLOCK_BLIND_20261016_071948.txt	blind	—
run_LOOP_ATLAS_BELL_CORNER_20261016_224449.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_BELL_CORNER_BLIND_20261016_224307.txt	blind	—
run_LOOP_ATLAS_BELL_LOG_20261015_033341.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_BELL_LOG_BLIND_20261015_033320.txt	blind	—
run_LOOP_ATLAS_BELL_PAIR_20261002.txt	of record, dated only	—
run_LOOP_ATLAS_BELL_RATIO_20261016_183422.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_BELL_RATIO_BLIND_20261016_182340.txt	blind	—
run_LOOP_ATLAS_BELL_SOFT_MAP_20261021_000736.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_BELL_SOFT_MAP_BLIND_20261021_000725.txt	blind	—
run_LOOP_ATLAS_BELL_SOFT_SCALE_20261021_054731.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_BELL_SOFT_SCALE_BLIND_20261021_054721.txt	blind	—
run_LOOP_ATLAS_BELL_SPIKES_20261022_053449.txt	of record	Part 2
run_LOOP_ATLAS_BELL_SPIKES_20261022_053740.txt	of record	Part 1; Part 2
run_LOOP_ATLAS_BELL_SPIKES_BLIND_20261022_053426.txt	blind	—
run_LOOP_ATLAS_BELL_SPIKES_BLIND_20261022_053723.txt	blind	—
run_LOOP_ATLAS_BELL_SWEEP_A_20261002.txt	of record, dated only	Part 3
run_LOOP_ATLAS_BELL_SWEEP_B_20261002.txt	of record, dated only	—
run_LOOP_ATLAS_BELL_SWITCH_20261020_161902.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_BELL_SWITCH_BLIND_20261020_161823.txt	blind	—
run_LOOP_ATLAS_BELL_TETHER_20261002.txt	of record, dated only	—
run_LOOP_ATLAS_BELL_WIDTH_LADDER_20261021_151833.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_BELL_WIDTH_LADDER_BLIND_20261021_151813.txt	blind	—
run_LOOP_ATLAS_BENCHMARK_TAGS_20261020_055158.txt	of record	Part 2; the book
run_LOOP_ATLAS_BENCH_LOOP_20261020_081001.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_BENCH_LOOP_BLIND_20261020_080617.txt	blind	—

Run file	Kind	Cited in
run_LOOP_ATLAS_BENCH_LOOP_BLIND_20261020_080640.txt	blind	—
run_LOOP_ATLAS_BENCH_LOOP_BLIND_20261020_080832.txt	blind	—
run_LOOP_ATLAS_BENT_PAPER_20261008_164512_partSMOKE.txt	smoke part	—
run_LOOP_ATLAS_BENT_PAPER_20261008_164513_partALL.txt	of record, part ALL	Part 1; Part 2; Part 3
run_LOOP_ATLAS_BENT_PAPER_2_20261008_164720_partSMOKE.txt	smoke part	—
run_LOOP_ATLAS_BENT_PAPER_2_20261008_164721_partALL.txt	of record, part ALL	Part 1; Part 3; the book
run_LOOP_ATLAS_BIG_LOOP_20261003_175803.txt	of record	Part 3; the book
run_LOOP_ATLAS_BIG_LOOP_FIG_20261003_192319.txt	of record	—
run_LOOP_ATLAS_BLOB_RING_20261104_170523.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_BLOB_RING_BLIND_20261104_170321.txt	blind	Part 3
run_LOOP_ATLAS_BLOB_RING_BLIND_20261104_170521.txt	blind	—
run_LOOP_ATLAS_BLOB_SMALL_20261104_170747.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_BLOB_SMALL_BLIND_20261104_170745.txt	blind	—
run_LOOP_ATLAS_BRICK_FINE_20261103_180438.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_BRICK_FINE_BLIND_20261103_180436.txt	blind	—
run_LOOP_ATLAS_BRICK_LINE_20261016_205112.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_BRICK_LINE_BLIND_20261016_205059.txt	blind	—
run_LOOP_ATLAS_BRICK_QUARTER_20261103_180126.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_BRICK_QUARTER_BLIND_20261103_180126.txt	blind	—
run_LOOP_ATLAS_BRIDGE_COUNT_20261016_044439.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_BRIDGE_COUNT_BLIND_20261016_044232.txt	blind	—
run_LOOP_ATLAS_BUBBLE_FAR_20261013_084037.txt	of record	Part 1; Part 2; the book
run_LOOP_ATLAS_BUILD_PAPER_CHECK_20261010_201918.txt	of record	Part 2
run_LOOP_ATLAS_CASCADE_20261006_211853.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_CASCADE_2_20261006_213511.txt	of record	Part 1; Part 2
run_LOOP_ATLAS_CASCADE_3_20261006_220635.txt	of record	Part 1; Part 3
run_LOOP_ATLAS_CASCADE_4_20261006_221804.txt	of record	Part 1; Part 3; the book
run_LOOP_ATLAS_CASCADE_5_20261006_225126.txt	of record	Part 1; Part 3; the book
run_LOOP_ATLAS_CATCH_TRAP_20261011_202500.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_CATCH_TRAP_2_20261013_030643.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_CATCH_TRAP_3_20261013_040120.txt	of record	Part 1; Part 2
run_LOOP_ATLAS_CATS_EYES_20261014_063138.txt	of record	Part 1; Part 2; the book

Run file	Kind	Cited in
run_LOOP_ATLAS_CAT_3D_20261006_201322.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_CAT_3D_2_20261006_202140.txt	of record	Part 1; Part 3
run_LOOP_ATLAS_CAT_3D_2_READER_20261006_202842.txt	of record, a reading tool	Part 1
run_LOOP_ATLAS_CAT_MIRROR_20261018_041123.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_CAT_MIRROR_BLIND_20261018_041037.txt	blind	—
run_LOOP_ATLAS_CAT_MIRROR_CENTRES_20261023_001618.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_CAT_MIRROR_CENTRES_BLIND_20261023_001616.txt	blind	—
run_LOOP_ATLAS_CAT_MIRROR_FAMILY_20261023_001232.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_CAT_MIRROR_FAMILY_BLIND_20261023_001220.txt	blind	—
run_LOOP_ATLAS_CAT_MIRROR_PATHS_20261020_034021.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_CAT_MIRROR_PATHS_BLIND_20261020_033921.txt	blind	Part 3
run_LOOP_ATLAS_CAT_MIRROR_PATHS_BLIND_20261020_034012.txt	blind	Part 3
run_LOOP_ATLAS_CAT_MIRROR_PROOF_20261018_174959.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_CAT_MIRROR_PROOF_BLIND_20261018_174925.txt	blind	—
run_LOOP_ATLAS_CAT_MIRROR_RATES_20261020_041152.txt	of record	Part 1; Part 2; the book
run_LOOP_ATLAS_CAT_MIRROR_RATES_BLIND_20261020_041152.txt	blind	—
run_LOOP_ATLAS_CAT_MONTGOMERY_20261020_035025.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_CAT_MONTGOMERY_BLIND_20261020_035010.txt	blind	—
run_LOOP_ATLAS_CAT_RTB_20261020_063617.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_CAT_RTB_BLIND_20261020_063615.txt	blind	—
run_LOOP_ATLAS_CAT_SWEET_20261018_034953.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_CAT_SWEET_BLIND_20261018_034741.txt	blind	—
run_LOOP_ATLAS_CAT_SWEET_READER_20261018_041321.txt	of record, a reading tool	Part 3
run_LOOP_ATLAS_CAT_TWIST_20261017_231928.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_CAT_TWIST_BLIND_20261017_231820.txt	blind	Part 3
run_LOOP_ATLAS_CAT_TWIST_BLIND_20261017_231908.txt	blind	Part 3
run_LOOP_ATLAS_CHAOS_ORDER_20261013_222835.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_CHAOS_ORDER_2_20261013_225021.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_CHAOS_RAMP_20261013_192906.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_CHAOS_SETTLES_20261011_060320.txt	of record	Part 1; Part 2; the book
run_LOOP_ATLAS_CLIFF_EDGE_20261004_062732.txt	of record	Part 3; the book
run_LOOP_ATLAS_CLIFF_EDGE_B_20261004_062930.txt	of record	the book

Run file	Kind	Cited in
run_LOOP_ATLAS_CLIMB_BACK_LINE_CHECK_20261006_041633.txt	of record	Part 3; the book
run_LOOP_ATLAS_CLIMB_HOLD_20261006_065510.txt	of record	Part 1; Part 3
run_LOOP_ATLAS_CLOCK_SHIFT_30_20261109_034237.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_CLOCK_SHIFT_30_BLIND_20261109_032818.txt	blind	—
run_LOOP_ATLAS_CLOCK_SHIFT_30_BLIND_20261109_033450.txt	blind	—
run_LOOP_ATLAS_CLOCK_SHIFT_30_BLIND_20261109_034235.txt	blind	—
run_LOOP_ATLAS_COIL_CHEAP_20261107_060011.txt	of record	Part 1; Part 2; the book
run_LOOP_ATLAS_COIL_CHEAP_BLIND_20261107_055913.txt	blind	—
run_LOOP_ATLAS_COIL_CHEAP_BLIND_20261107_060006.txt	blind	—
run_LOOP_ATLAS_COIL_PUMP_20261018_082424.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_COIL_PUMP_BLIND_20261018_082348.txt	blind	—
run_LOOP_ATLAS_COIL_SNAKE_20261017_204030.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_COIL_SNAKE_BLIND_20261017_203922.txt	blind	—
run_LOOP_ATLAS_COIL_WAVE_20261018_082544.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_COIL_WAVE_BLIND_20261018_082533.txt	blind	—
run_LOOP_ATLAS_COLUMN_ACCORDION_20261013_222625.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_COLUMN_ACCORDION_2_20261013_222950.txt	of record	Part 1; Part 2; the book
run_LOOP_ATLAS_COLUMN_STEADY_20261014_164637.txt	of record	Part 1; Part 2
run_LOOP_ATLAS_COLUMN_TAIL_20261014_145058.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_COLUMN_WALK_20261108_234842.txt	of record	Part 1; Part 2
run_LOOP_ATLAS_COLUMN_WALK_BLIND_20261108_234757.txt	blind	—
run_LOOP_ATLAS_COLUMN_WALK_BLIND_20261108_234833.txt	blind	—
run_LOOP_ATLAS_COMB_STRETCH_20261101_002907.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_COMB_STRETCH_BLIND_20261101_002903.txt	blind	—
run_LOOP_ATLAS_COMB_STRETCH_B_20261101_003130.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_COMB_STRETCH_B_BLIND_20261101_003049.txt	blind	—
run_LOOP_ATLAS_CONE_TOMOGRAPHY_20261007_235426_partA.txt	of record, part A	Part 1; the book
run_LOOP_ATLAS_CONE_TOMOGRAPHY_20261007_235444_partB.txt	of record, part B	Part 1; the book
run_LOOP_ATLAS_CONE_TOMOGRAPHY_20261008_001220_partC.txt	of record, part C	Part 1; the book
run_LOOP_ATLAS_CONE_TOMOGRAPHY_2_20261008_002930_partSMOKE.txt	smoke part	—
run_LOOP_ATLAS_CONE_TOMOGRAPHY_2_20261008_002937_partE.txt	of record, part E	Part 1
run_LOOP_ATLAS_CONE_TOMOGRAPHY_2_20261008_002939_partD.txt	of record, part D	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_CONE_TOMOGRAPHY_3_20261008_011142_partSMOKE.txt	smoke part	—
run_LOOP_ATLAS_CONE_TOMOGRAPHY_3_20261008_011149_partN2.txt	of record, part N2	Part 1

Run file	Kind	Cited in
run_LOOP_ATLAS_CONE_TOMOGRAPHY_3_20261008_011150_partN1.txt	of record, part N1	Part 1; Part 3; the book
run_LOOP_ATLAS_CONE_TOMOGRAPHY_3_20261008_011157_partN3.txt	of record, part N3	Part 1
run_LOOP_ATLAS_CONE_TOMOGRAPHY_4_20261008_012041_partSMOKE.txt	smoke part	—
run_LOOP_ATLAS_CONE_TOMOGRAPHY_4_20261008_012042_partK2.txt	of record, part K2	Part 1
run_LOOP_ATLAS_CONE_TOMOGRAPHY_4_20261008_012043_partK1.txt	of record, part K1	Part 1; Part 2
run_LOOP_ATLAS_CONE_TOMOGRAPHY_5_20261008_012723_partK3.txt	of record, part K3	Part 1
run_LOOP_ATLAS_CONE_TOMOGRAPHY_5_20261008_012723_partSMOKE.txt	smoke part	—
run_LOOP_ATLAS_CORNER_PRINT_20261010_061715.txt	of record, a reading tool	Part 1; Part 3
run_LOOP_ATLAS_COUNTER_CHARGE_20261002.txt	of record, dated only	—
run_LOOP_ATLAS_COUNTER_FLOW_20261009_002621_partALL.txt	of record, part ALL	Part 1; Part 2; the book
run_LOOP_ATLAS_COUNTER_FLOW_20261009_002621_partSMOKE.txt	smoke part	—
run_LOOP_ATLAS_CREEP_LAW_20261016_035058.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_CREEP_LAW_BLIND_20261016_034809.txt	blind	—
run_LOOP_ATLAS_CRISIS_BURSTS_20261014_091405.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_CURVED_SWIMMER_20261002.txt	of record, dated only	—
run_LOOP_ATLAS_CUT_WIDE_20261107_033035.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_CUT_WIDE_BLIND_20261107_032955.txt	blind	—
run_LOOP_ATLAS_DAMPER_LAWS_20261011_051848.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_DELTA_AREA_20261008_062647_partALL.txt	of record, part ALL	Part 1; Part 2; the book
run_LOOP_ATLAS_DELTA_BY_TOP_20261008_034013_partALL.txt	of record, part ALL	Part 1; Part 2; the book
run_LOOP_ATLAS_DELTA_BY_TOP_20261008_034013_partSMOKE.txt	smoke part	—
run_LOOP_ATLAS_DELTA_BY_TOP_2_20261008_063744_partE2.txt	of record, part E2	Part 1; Part 2; Part 3
run_LOOP_ATLAS_DELTA_BY_TOP_2_20261008_063744_partE3.txt	of record, part E3	Part 1; Part 2; Part 3
run_LOOP_ATLAS_DELTA_BY_TOP_2_20261008_063744_partSMOKE.txt	smoke part	—
run_LOOP_ATLAS_DELTA_TAIL_20261010_205937.txt	of record	Part 1; Part 2
run_LOOP_ATLAS_DELTA_TAIL_2_20261011_051600.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_DIVER_20261005_160158.txt	of record	Part 1; Part 3
run_LOOP_ATLAS_DIVER_B_20261005_160338.txt	of record	Part 1; Part 3; the book
run_LOOP_ATLAS_DIVE_CUE_20261017_082454.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_DIVE_CUE_BLIND_20261017_082235.txt	blind	Part 3
run_LOOP_ATLAS_DRAFT_HORSE_20261002.txt	of record, dated only	—
run_LOOP_ATLAS_DRAFT_TEST_20261008_134701_partSMOKE.txt	smoke part	—

Run file	Kind	Cited in
run_LOOP_ATLAS_DRAFT_TEST_20261008_134707_partALL.txt	of record, part ALL	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_EAR_BILL_20261002.txt	of record, dated only	—
run_LOOP_ATLAS_EAR_COST_20261013_183245.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_EAR_COST_2_20261013_183913.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_EAR_COST_3_20261013_192738.txt	of record	Part 1; Part 2; the book
run_LOOP_ATLAS_EAR_CURES_20261004_045151.txt	of record	Part 3; the book
run_LOOP_ATLAS_EAR_CURES_B_20261004_045721.txt	of record	the book
run_LOOP_ATLAS_EAR_KNEE_20261003_220719.txt	of record	the book
run_LOOP_ATLAS_EAR_KNEE_C_20261003_220906.txt	of record	—
run_LOOP_ATLAS_EAR_PRICE_CHECK_20261003_220245.txt	of record	the book
run_LOOP_ATLAS_EQUAL_CATCH_20261016_192548.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_EQUAL_CATCH_BLIND_20261016_192112.txt	blind	—
run_LOOP_ATLAS_ERROR_BARS_20261019_162750.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_ERROR_BARS_BLIND_20261019_162623.txt	blind	—
run_LOOP_ATLAS_EXPLORER_BELL_CHECK_20261103_025532.txt	of record	Part 2; Part 3
run_LOOP_ATLAS_EXPLORER_BELL_CHECK_BLIND_20261103_025526.txt	blind	—
run_LOOP_ATLAS_EXPLORER_CAT_CHECK_20261017_183736.txt	of record	Part 2; Part 3
run_LOOP_ATLAS_EXPLORER_CAT_CHECK_20261017_211715.txt	of record	Part 2
run_LOOP_ATLAS_EXPLORER_CAT_CHECK_20261018_031543.txt	of record	Part 2
run_LOOP_ATLAS_EXPLORER_CAT_CHECK_BLIND_20261017_183100.txt	blind	—
run_LOOP_ATLAS_EXPLORER_CAT_ROOM_CHECK_20261017_211325.txt	of record	Part 2; Part 3
run_LOOP_ATLAS_EXPLORER_CAT_ROOM_CHECK_20261018_031545.txt	of record	Part 2
run_LOOP_ATLAS_EXPLORER_CAT_ROOM_CHECK_BLIND_20261017_211142.txt	blind	Part 3
run_LOOP_ATLAS_EXPLORER_CAT_ROOM_CHECK_BLIND_20261017_211314.txt	blind	Part 3
run_LOOP_ATLAS_EXPLORER_CAT_TWISTFREE_CHECK_20261018_031532.txt	of record	Part 2; Part 3
run_LOOP_ATLAS_EXPLORER_CAT_TWISTFREE_CHECK_BLIND_20261018_031524.txt	blind	—
run_LOOP_ATLAS_EXPLORER_COLUMN_CHECK_20261108_194717.txt	of record	Part 2; Part 3
run_LOOP_ATLAS_EXPLORER_COLUMN_CHECK_BLIND_20261108_194645.txt	blind	—
run_LOOP_ATLAS_EXPLORER_COLUMN_CHECK_BLIND_20261108_194706.txt	blind	—
run_LOOP_ATLAS_EXPLORER_DIVE_CHECK_20261017_155902.txt	of record	Part 2; Part 3
run_LOOP_ATLAS_EXPLORER_DIVE_CHECK_20261017_165821.txt	of record	Part 2
run_LOOP_ATLAS_EXPLORER_DIVE_CHECK_20261017_183750.txt	of record	Part 2
run_LOOP_ATLAS_EXPLORER_DIVE_CHECK_20261017_211712.txt	of record	Part 2
run_LOOP_ATLAS_EXPLORER_DIVE_CHECK_20261018_031538.txt	of record	Part 2
run_LOOP_ATLAS_EXPLORER_DIVE_CHECK_BLIND_20261017_155856.txt	blind	—
run_LOOP_ATLAS_EXPLORER_ENGINE_CHECK_20261017_063123.txt	of record	Part 2; Part 3
run_LOOP_ATLAS_EXPLORER_ENGINE_CHECK_20261017_070749.txt	of record	Part 2

Run file	Kind	Cited in
run_LOOP_ATLAS_EXPLORER_ENGINE_CHECK_20261017_155903.txt	of record	Part 2
run_LOOP_ATLAS_EXPLORER_ENGINE_CHECK_20261017_165820.txt	of record	Part 2
run_LOOP_ATLAS_EXPLORER_ENGINE_CHECK_20261017_183749.txt	of record	Part 2
run_LOOP_ATLAS_EXPLORER_ENGINE_CHECK_20261017_211710.txt	of record	Part 2
run_LOOP_ATLAS_EXPLORER_ENGINE_CHECK_20261018_031535.txt	of record	Part 2
run_LOOP_ATLAS_EXPLORER_ENGINE_CHECK_BLIND_20261017_063114.txt	blind	Part 3
run_LOOP_ATLAS_EXPLORER_FACE_CHECK_20261108_202547.txt	of record	Part 2
run_LOOP_ATLAS_EXPLORER_FACE_CHECK_BLIND_20261108_202515.txt	blind	—
run_LOOP_ATLAS_EXPLORER_FACE_CHECK_BLIND_20261108_202540.txt	blind	—
run_LOOP_ATLAS_EXPLORER_GRID_CHECK_20261107_073202.txt	of record	Part 2; Part 3
run_LOOP_ATLAS_EXPLORER_GRID_CHECK_BLIND_20261107_073141.txt	blind	—
run_LOOP_ATLAS_EXPLORER_HEAT_CHECK_20261103_065559.txt	of record	Part 2; Part 3
run_LOOP_ATLAS_EXPLORER_HEAT_CHECK_BLIND_20261103_065552.txt	blind	—
run_LOOP_ATLAS_EXPLORER_LATCH_CHECK_20261101_042950.txt	of record	Part 2; Part 3
run_LOOP_ATLAS_EXPLORER_LATCH_CHECK_BLIND_20261101_042947.txt	blind	—
run_LOOP_ATLAS_EXPLORER_METER_CHECK_20261108_182139.txt	of record	Part 2; Part 3
run_LOOP_ATLAS_EXPLORER_METER_CHECK_BLIND_20261108_182126.txt	blind	—
run_LOOP_ATLAS_EXPLORER_NEEDLE_CHECK_20261104_222610.txt	of record	Part 2; Part 3
run_LOOP_ATLAS_EXPLORER_NEEDLE_CHECK_BLIND_20261104_222609.txt	blind	—
run_LOOP_ATLAS_EXPLORER_RING_CHECK_20261102_045348.txt	of record	Part 2; Part 3
run_LOOP_ATLAS_EXPLORER_RING_CHECK_BLIND_20261102_045345.txt	blind	—
run_LOOP_ATLAS_EXPLORER_ROUTINE_CHECK_20261017_164432.txt	of record	Part 2; Part 3; the book
run_LOOP_ATLAS_EXPLORER_ROUTINE_CHECK_20261017_183751.txt	of record	Part 2
run_LOOP_ATLAS_EXPLORER_ROUTINE_CHECK_20261017_211712.txt	of record	Part 2
run_LOOP_ATLAS_EXPLORER_ROUTINE_CHECK_20261018_031539.txt	of record	Part 2
run_LOOP_ATLAS_EXPLORER_ROUTINE_CHECK_BLIND_20261017_164121.txt	blind	—
run_LOOP_ATLAS_EXPLORER_SNAKE_CHECK_20261101_171827.txt	of record	Part 2; Part 3
run_LOOP_ATLAS_EXPLORER_SNAKE_CHECK_BLIND_20261101_171813.txt	blind	—
run_LOOP_ATLAS_EXPLORER_SNAKE_CHECK_BLIND_20261101_171821.txt	blind	—
run_LOOP_ATLAS_EXPLORER_SOUP_CHECK_20261108_070632.txt	of record	Part 2; Part 3
run_LOOP_ATLAS_EXPLORER_SOUP_CHECK_BLIND_20261108_070622.txt	blind	—
run_LOOP_ATLAS_EXPLORER_TRUCK_CHECK_20261101_015149.txt	of record	Part 2; Part 3
run_LOOP_ATLAS_EXPLORER_TRUCK_CHECK_BLIND_20261101_013904.txt	blind	—
run_LOOP_ATLAS_EXPLORER_TRUCK_CHECK_BLIND_20261101_014621.txt	blind	—
run_LOOP_ATLAS_EXP_SHAPE_20261011_080708.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_EXP_SHAPE_2_20261011_194909.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_FACE_ALONG_20261017_070629.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_FACE_ALONG_BLIND_20261017_070529.txt	blind	Part 3

Run file	Kind	Cited in
run_LOOP_ATLAS_FACE_ALONG_BLIND_20261017_070615.txt	blind	Part 3
run_LOOP_ATLAS_FALL_CHAIN_20261011_200714.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_FALL_CHAIN_2_20261012_041438.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_FALL_SHARE_20261012_075804.txt	of record	Part 1; Part 2; the book
run_LOOP_ATLAS_FERRY_ROPE_20261006_045826.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_FIG_BEAT_MAP_20261015_220853.txt	of record, a figure tool	—
run_LOOP_ATLAS_FIG_BELL_20261015_033418.txt	of record, a figure tool	—
run_LOOP_ATLAS_FIG_CREEP_LAW_20261016_035636.txt	of record, a figure tool	Part 3; the book
run_LOOP_ATLAS_FIG_CRISIS_20261014_095018.txt	of record, a figure tool	—
run_LOOP_ATLAS_FIG_DOOR_20261015_081403.txt	of record, a figure tool	—
run_LOOP_ATLAS_FIG_GAPS_20261015_062248.txt	of record, a figure tool	—
run_LOOP_ATLAS_FIG_GAPS_2_20261015_071629.txt	of record, a figure tool	—
run_LOOP_ATLAS_FIG_GEAR_TRAIN_20261013_184419.txt	of record, a figure tool	—
run_LOOP_ATLAS_FIG_HALF_BUBBLE_20261013_164016.txt	of record, a figure tool	—
run_LOOP_ATLAS_FIG_HEAVY_20261014_160905.txt	of record, a figure tool	—
run_LOOP_ATLAS_FIG_HEAVY_2_20261014_171745.txt	of record, a figure tool	—
run_LOOP_ATLAS_FIG_QUARTER_WINDOW_20261013_231948.txt	of record, a figure tool	—
run_LOOP_ATLAS_FIG_RING_KNOBS_20261015_173103.txt	of record, a figure tool	—
run_LOOP_ATLAS_FIG_RUNGS_20261015_065055.txt	of record, a figure tool	—
run_LOOP_ATLAS_FIG_SEAM_20261014_091610.txt	of record, a figure tool	—
run_LOOP_ATLAS_FIG_SNAKE_CLIMB_20261015_154834.txt	of record, a figure tool	—
run_LOOP_ATLAS_FIG_STOP_LEAVES_20261015_054334.txt	of record, a figure tool	—
run_LOOP_ATLAS_FIG_STOP_WALL_20261014_182529.txt	of record, a figure tool	—
run_LOOP_ATLAS_FIG_STOP_WALL_2_20261014_230513.txt	of record, a figure tool	—
run_LOOP_ATLAS_FIG_STOP_WALL_3_20261015_003048.txt	of record, a figure tool	—
run_LOOP_ATLAS_FIG_THIRD_AXIS_20261109_171334.txt	of record, a figure tool	—
run_LOOP_ATLAS_FILTER_SLOPE_20261015_195645.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_FILTER_SLOPE_BLIND_20261015_195438.txt	blind	—
run_LOOP_ATLAS_FIT_KNOBS_20261015_204319.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_FIT_KNOBS_BLIND_20261015_204015.txt	blind	—
run_LOOP_ATLAS_FIVE_SEVENTHS_20261013_222744.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_FLIP_MISTAKES_20261015_195739.txt	of record	Part 1; Part 2; the book
run_LOOP_ATLAS_FLIP_MISTAKES_BLIND_20261015_195600.txt	blind	—
run_LOOP_ATLAS_FOLD_OR_FLIP_20261007_050721.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_FOLD_OR_FLIP_2_20261007_053443.txt	of record	Part 1; Part 3

Run file	Kind	Cited in
run_LOOP_ATLAS_FOLD_OR_FLIP_3_20261007_054521.txt	of record	Part 1
run_LOOP_ATLAS_FOLD_OR_FLIP_4_20261007_061947.txt	of record	Part 1; Part 3
run_LOOP_ATLAS_FOLD_OR_FLIP_4_20261010_172149.txt	of record, flown on the second computer	Part 3
run_LOOP_ATLAS_FOLD_TIGHT_20261011_051636.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_FRAY_AND_SCISSORS_20261005_050632.txt	of record	—
run_LOOP_ATLAS_FRAY_AND_SCISSORS_B_20261005_051313.txt	of record	—
run_LOOP_ATLAS_FREE_SWIM_20261101_222356.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_FREE_SWIM_2_20261101_222740.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_FREE_SWIM_2_BLIND_20261101_222610.txt	blind	—
run_LOOP_ATLAS_FREE_SWIM_BLIND_20261101_222249.txt	blind	—
run_LOOP_ATLAS_FUNNEL_CURLS_20261006_203001.txt	of record	Part 1; the book
run_LOOP_ATLAS_GAPS_READ_20261015_061859.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_GAPS_READ_BLIND_20261015_061509.txt	blind	—
run_LOOP_ATLAS_GAPS_READ_BLIND_20261015_061705.txt	blind	—
run_LOOP_ATLAS_GAPS_REREAD_20261015_071452.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_GAPS_REREAD_BLIND_20261015_071407.txt	blind	—
run_LOOP_ATLAS_GAP_DIAL_20261005_170519.txt	of record	Part 1; Part 3
run_LOOP_ATLAS_GAP_DIAL_2_20261005_172939.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_GAP_DIAL_3_20261005_194344.txt	of record	Part 3
run_LOOP_ATLAS_GEAR_TRAIN_20261013_170243.txt	of record	Part 1; Part 2; the book
run_LOOP_ATLAS_GHOST_DEPTH_20261013_024747.txt	of record	Part 1; Part 2
run_LOOP_ATLAS_GHOST_TRIAL_20261013_005637.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_GHOST_TRIAL_2_20261013_015743.txt	of record	Part 1; Part 2; the book
run_LOOP_ATLAS_GROOVES_10_20261007_002739.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_GROOVES_10_20261007_065508_partB.txt	of record, part B	Part 1; Part 3
run_LOOP_ATLAS_GROOVES_10_20261007_065508_partC.txt	of record, part C	Part 1; Part 3
run_LOOP_ATLAS_GROOVES_11_20261007_080706.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_GROOVES_12_20261007_083300_partB.txt	of record, part B	Part 1
run_LOOP_ATLAS_GROOVES_12_20261007_083648_partA.txt	of record, part A	Part 1
run_LOOP_ATLAS_GROOVES_13_20261007_090124.txt	of record	Part 1; Part 2
run_LOOP_ATLAS_GROOVES_14_20261007_170021_partA.txt	of record, part A	Part 1; Part 2; Part 3
run_LOOP_ATLAS_GROOVES_14_20261007_170022_partB.txt	of record, part B	Part 1; Part 2
run_LOOP_ATLAS_GROOVES_15_20261007_163523_partA.txt	of record, part A	Part 2

Run file	Kind	Cited in
run_LOOP_ATLAS_GROOVES_15_20261007_164232_partB.txt	of record, part B	Part 1; Part 3
run_LOOP_ATLAS_GROOVES_15_20261007_165418_partC.txt	of record, part C	Part 1; Part 2
run_LOOP_ATLAS_GROOVES_16_20261007_182009_partA.txt	of record, part A	Part 1; Part 2; the book
run_LOOP_ATLAS_GROOVES_16_20261007_182010_partB.txt	of record, part B	Part 1; Part 2; Part 3
run_LOOP_ATLAS_GROOVES_16_20261007_183554_partC.txt	of record, part C	Part 1; Part 3
run_LOOP_ATLAS_GROOVES_17_20261007_234942_partA.txt	of record, part A	Part 1; Part 2; Part 3
run_LOOP_ATLAS_GROOVES_17_20261007_234943_partB.txt	of record, part B	Part 1; Part 2
run_LOOP_ATLAS_GROOVES_20261006_193519.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_GROOVES_2_20261006_202753.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_GROOVES_3_20261006_204201.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_GROOVES_4_20261006_212231.txt	of record	Part 1
run_LOOP_ATLAS_GROOVES_5_20261006_213610.txt	of record	Part 1
run_LOOP_ATLAS_GROOVES_6_20261006_215909.txt	of record	Part 1; Part 2
run_LOOP_ATLAS_GROOVES_7_20261006_222345.txt	of record	Part 1; Part 3
run_LOOP_ATLAS_GROOVES_8_20261006_225037.txt	of record	Part 1; Part 3
run_LOOP_ATLAS_GROOVES_9_20261007_011618.txt	of record	Part 1; Part 3
run_LOOP_ATLAS_GROOVES_READER_20261006_194112.txt	of record, a reading tool	—
run_LOOP_ATLAS_GROW_LOOP_20261008_133644_partSMOKE.txt	smoke part	—
run_LOOP_ATLAS_GROW_LOOP_20261008_133646_partALL.txt	of record, part ALL	Part 1; Part 2; the book
run_LOOP_ATLAS_GROW_SLIP_20261008_195205_partALL.txt	of record, part ALL	Part 1; Part 2; the book
run_LOOP_ATLAS_GROW_SLIP_20261008_195205_partSMOKE.txt	smoke part	—
run_LOOP_ATLAS_HALF_BUBBLE_20261013_083609.txt	of record	Part 1; Part 2; the book
run_LOOP_ATLAS_HALF_PARENT_20261013_163612.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_HALF_TOP_20261013_183449.txt	of record	Part 1; Part 2; the book
run_LOOP_ATLAS_HALT_ORDER_20261015_052122.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_HALT_ORDER_BLIND_20261015_052101.txt	blind	—
run_LOOP_ATLAS_HALT_THRESHOLD_20261017_184055.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_HALT_THRESHOLD_BLIND_20261017_174151.txt	blind	Part 3
run_LOOP_ATLAS_HALT_THRESHOLD_BLIND_20261017_174755.txt	blind	Part 3
run_LOOP_ATLAS_HALT_THRESHOLD_BLIND_20261017_182910.txt	blind	Part 3
run_LOOP_ATLAS_HALT_THRESHOLD_BLIND_20261017_183927.txt	blind	—
run_LOOP_ATLAS_HALT_TRADE_20261031_235932.txt	of record	Part 1; Part 2; Part 3; the book

Run file	Kind	Cited in
run_LOOP_ATLAS_HALT_TRADE_BLIND_20261031_235730.txt	blind	—
run_LOOP_ATLAS_HALVINGS_20261015_064027.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_HALVINGS_2_20261015_064427.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_HALVINGS_2_BLIND_20261015_064339.txt	blind	—
run_LOOP_ATLAS_HALVINGS_3_20261015_064938.txt	of record	Part 1; Part 2; the book
run_LOOP_ATLAS_HALVINGS_3_BLIND_20261015_064836.txt	blind	—
run_LOOP_ATLAS_HALVINGS_BLIND_20261015_063824.txt	blind	—
run_LOOP_ATLAS_HALVINGS_BLIND_20261015_063916.txt	blind	—
run_LOOP_ATLAS_HEAT_RING_20261016_174342.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_HEAT_RING_BLIND_20261016_174331.txt	blind	—
run_LOOP_ATLAS_HEAVY_MARKS_20261014_164806.txt	of record, a reading tool	Part 2; the book
run_LOOP_ATLAS_HEAVY_SHAPE_20261014_155553.txt	of record	Part 1; Part 2; the book
run_LOOP_ATLAS_HEAVY_SHAPE_2_20261014_164913.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_HEAVY_SHAPE_3_20261014_170837.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_HEAVY_WINDOW_20261014_153422.txt	of record, a reading tool	—
run_LOOP_ATLAS_HEAVY_WINDOW_2_20261014_224515.txt	of record, a reading tool	—
run_LOOP_ATLAS_HEAVY_WINDOW_3_20261014_232615.txt	of record, a reading tool	—
run_LOOP_ATLAS_HEAVY_WINDOW_4_20261015_032553.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_HEAVY_WINDOW_4_BLIND_20261015_031848.txt	blind	—
run_LOOP_ATLAS_HIDDEN_SCISSORS_20261005_064106.txt	of record	Part 1
run_LOOP_ATLAS_HIDDEN_SCISSORS_F0B_20261005_064308.txt	of record	Part 3
run_LOOP_ATLAS_HIDDEN_SCISSORS_SLIVER_20261005_145137.txt	of record	Part 1; Part 3
run_LOOP_ATLAS_HINGE_FINE_20261004_080159.txt	of record	Part 3
run_LOOP_ATLAS_HINGE_GEOMETRY_20261002b.txt	of record, dated only	Part 3
run_LOOP_ATLAS_HISTORY_SETTLE_20261009_093424.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_HITCH_KICK_20261006_011635.txt	of record	Part 1; Part 3; the book
run_LOOP_ATLAS_HITCH_LEGS_20261016_065650.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_HITCH_LEGS_BLIND_20261016_065356.txt	blind	—
run_LOOP_ATLAS_HORSE_SPIN_20261002.txt	of record, dated only	—
run_LOOP_ATLAS_HOUSE_IN_SPACE_20261004_150225.txt	of record	Part 3; the book
run_LOOP_ATLAS_HOUSE_IN_SPACE_B_20261004_150847.txt	of record	the book
run_LOOP_ATLAS_HOUSE_IN_SPACE_C_20261004_151017.txt	of record	—
run_LOOP_ATLAS_HOW_IT_COUNTS_20261004_183911.txt	of record	the book
run_LOOP_ATLAS_INNER_OUTER_20261013_005545.txt	of record	Part 1; Part 2; the book

Run file	Kind	Cited in
run_LOOP_ATLAS_JUMP_STORE_20261015_223918.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_JUMP_STORE_BLIND_20261015_223744.txt	blind	Part 3
run_LOOP_ATLAS_KICK_RULE_CHECK_20261005_031525.txt	of record	Part 3; the book
run_LOOP_ATLAS_KNEE_CREEP_20261015_232920.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_KNEE_CREEP_BLIND_20261015_232509.txt	blind	—
run_LOOP_ATLAS_KNEE_FINE_20261019_040109.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_KNEE_FINE_BLIND_20261019_035922.txt	blind	—
run_LOOP_ATLAS_KNEE_FINE_BLIND_20261019_040001.txt	blind	—
run_LOOP_ATLAS_KNEE_READER_20261019_163948.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_KNEE_READER_BLIND_20261019_163628.txt	blind	—
run_LOOP_ATLAS_KNEE_RECOUNT_20261004_184524.txt	of record	Part 3
run_LOOP_ATLAS_KNEE_RECOUNT_B_20261004_190250.txt	of record	Part 3
run_LOOP_ATLAS_KNEE_RECOUNT_C_20261004_192352.txt	of record	—
run_LOOP_ATLAS_KNEE_RECOUNT_D_20261004_195102.txt	of record	Part 3
run_LOOP_ATLAS_KNEE_RECOUNT_E_20261005_003928.txt	of record	the book
run_LOOP_ATLAS_KNEE_REF_20261019_042520.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_KNEE_REF_BLIND_20261019_042243.txt	blind	—
run_LOOP_ATLAS_KNEE_SIZE_20261019_035236.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_KNEE_SIZE_BLIND_20261019_035114.txt	blind	—
run_LOOP_ATLAS_KRAMERS_EXACT_20261020_073520.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_KRAMERS_EXACT_BLIND_20261020_073510.txt	blind	—
run_LOOP_ATLAS_LADDER_LIGHT_20261015_071735.txt	of record	Part 1; Part 2
run_LOOP_ATLAS_LADDER_LIGHT_BLIND_20261015_070958.txt	blind	Part 3
run_LOOP_ATLAS_LADDER_LIGHT_BLIND_20261015_071652.txt	blind	—
run_LOOP_ATLAS_LAG_METER_20261016_203014.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_LAG_METER_BLIND_20261016_202452.txt	blind	—
run_LOOP_ATLAS_LANDSCAPE_20261003_201112.txt	of record	the book
run_LOOP_ATLAS_LANDSCAPE_FIG_20261003_201923.txt	of record	the book
run_LOOP_ATLAS_LANE_EDGES_20261017_164501.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_LANE_EDGES_BLIND_20261017_163836.txt	blind	—
run_LOOP_ATLAS_LAPTOP_F4_2_20261010_172148.txt	of record, a launcher, flown on the second computer	—
run_LOOP_ATLAS_LATCH_EVENT_20261015_232643.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_LATCH_EVENT_BLIND_20261015_232509.txt	blind	—
run_LOOP_ATLAS_LATTICE_READER_20261009_091525_partSMOKE.txt	smoke part	—

Run file	Kind	Cited in
run_LOOP_ATLAS_LATTICE_READER_20261009_091530_partS1.txt	of record, part S1	—
run_LOOP_ATLAS_LATTICE_READER_20261009_091531_partS2.txt	of record, part S2	Part 2
run_LOOP_ATLAS_LATTICE_READER_20261009_092420_partR.txt	of record, part R	Part 1; Part 2; Part 3
run_LOOP_ATLAS_LATTICE_READER_2_20261009_093040_partALL.txt	of record, part ALL	Part 1; Part 2; Part 3
run_LOOP_ATLAS_LEAK_CLOSE_20261015_073039.txt	of record	Part 1; Part 2
run_LOOP_ATLAS_LEAK_CLOSE_BLIND_20261015_072238.txt	blind	—
run_LOOP_ATLAS_LEAK_RATE_20261015_080150.txt	of record	Part 1; Part 2
run_LOOP_ATLAS_LEAK_RATE_BLIND_20261015_075411.txt	blind	—
run_LOOP_ATLAS_LEAN_DRIFT_20261105_025051.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_LEAN_DRIFT_BLIND_20261105_025050.txt	blind	—
run_LOOP_ATLAS_LIFT_3D_20261010_031527.txt	of record, a reading tool	Part 1; Part 3
run_LOOP_ATLAS_LIGHT_LANES_20261016_174648.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_LIGHT_LANES_BLIND_20261016_174122.txt	blind	—
run_LOOP_ATLAS_LIMIT_AUDIT_20261017_190938.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_LIMIT_AUDIT_BLIND_20261017_190326.txt	blind	—
run_LOOP_ATLAS_LOBE_TWIN_20261010_043453.txt	of record, a reading tool	Part 1
run_LOOP_ATLAS_LOCK_EDGE_EXACT_20261013_063227.txt	of record	Part 1; Part 2; the book
run_LOOP_ATLAS_LOCK_FROM_CHAOS_20261011_060551.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_LOCK_LANDING_20261011_170402.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_LOCK_MATCH_20261011_195010.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_MAINTENANCE_METER_20261005_012150.txt	of record	Part 3; the book
run_LOOP_ATLAS_MAINTENANCE_METER_B_20261005_012355.txt	of record	—
run_LOOP_ATLAS_MEMORY_BELL_20261002.txt	of record, dated only	—
run_LOOP_ATLAS_MEMORY_BELL_reopen_20261002.txt	of record, dated only	—
run_LOOP_ATLAS_MEMORY_SWITCH_20261011_084432.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_MEMORY_SWITCH_2_20261011_085331.txt	of record	Part 1; Part 3; the book
run_LOOP_ATLAS_METER_CLOCKS_20261011_072646.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_METER_K_SPREAD_20261011_080641.txt	of record	Part 1; Part 2
run_LOOP_ATLAS_MIRROR_AUDIT_20261002b.txt	of record, dated only	Part 3
run_LOOP_ATLAS_MORTAR_SPACING_20261013_224843.txt	of record	Part 1; Part 2; the book
run_LOOP_ATLAS_NEEDLE_GEARS_20261003_210142.txt	of record	the book
run_LOOP_ATLAS_NEEDLE_GEARS_WHY_20261003_210451.txt	of record	the book
run_LOOP_ATLAS_NEEDS_AUDIT_20261010_205839.txt	of record	Part 2; Part 3

Run file	Kind	Cited in
run_LOOP_ATLAS_NEVER_BLIND_20261006_045833.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_NEVER_BLIND_2_20261006_050123.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_NEVER_BLINK_20261016_090509.txt	of record	Part 1; Part 2; the book
run_LOOP_ATLAS_NEVER_BLINK_BLIND_20261016_085246.txt	blind	—
run_LOOP_ATLAS_NIGHT_SPACING_20261013_231417.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_NOISE_EDGE_20261014_232230.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_NOISE_EDGE_BLIND_20261014_232109.txt	blind	—
run_LOOP_ATLAS_NOVIKOV_NEEDLE_20261004_152937.txt	of record	Part 3; the book
run_LOOP_ATLAS_NOVIKOV_NEEDLE_B_20261004_153129.txt	of record	the book
run_LOOP_ATLAS_ORBIT_19_20261013_041840.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_ORBIT_919_3_20261013_031807.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_ORBIT_919_4_20261013_040039.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_OWED_BARS_20261107_023832.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_OWED_BARS_BLIND_20261107_023724.txt	blind	—
run_LOOP_ATLAS_PAUSE_BEST_20261015_220655.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_PAUSE_BEST_BLIND_20261015_220321.txt	blind	—
run_LOOP_ATLAS_PAUSE_WIDE_20261015_223744.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_PAUSE_WIDE_BLIND_20261015_223539.txt	blind	—
run_LOOP_ATLAS_PEAK_RING_20261008_040516_partSMOKE.txt	smoke part	—
run_LOOP_ATLAS_PEAK_RING_20261008_040518_partALL.txt	of record, part ALL	Part 1; Part 2; Part 3
run_LOOP_ATLAS_PEAK_RING_2_20261008_040853_partSMOKE.txt	smoke part	—
run_LOOP_ATLAS_PEAK_RING_2_20261008_040854_partALL.txt	of record, part ALL	Part 1; Part 3; the book
run_LOOP_ATLAS_PENTA_SHARE_20261008_060149_partSMOKE.txt	smoke part	—
run_LOOP_ATLAS_PENTA_SHARE_20261008_060159_partS.txt	of record, part S	Part 1; Part 2; Part 3
run_LOOP_ATLAS_PENTA_SHARE_20261008_060200_partW.txt	of record, part W	Part 1; Part 2; the book
run_LOOP_ATLAS_PENTA_SHARE_2_20261008_062402_partSMOKE.txt	smoke part	—
run_LOOP_ATLAS_PENTA_SHARE_2_20261008_062403_partS2.txt	of record, part S2	Part 1; Part 2; Part 3
run_LOOP_ATLAS_PENTA_SHARE_2_20261008_062420_partW3.txt	of record, part W3	Part 1; Part 2
run_LOOP_ATLAS_PILOT_BEAT_20261015_203855.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_PILOT_BEAT_BLIND_20261015_203819.txt	blind	—

Run file	Kind	Cited in
run_LOOP_ATLAS_PILOT_PRICE_20261015_220321.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_PILOT_PRICE_BLIND_20261015_220244.txt	blind	—
run_LOOP_ATLAS_PILOT_TONE_20261002.txt	of record, dated only	—
run_LOOP_ATLAS_POCKET_GEARS_20261006_052012.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_POCKET_HOLD_20261005_195333.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_POLARISATION_LOOP_20261004_071425.txt	of record	Part 3; the book
run_LOOP_ATLAS_POLARISATION_LOOP_2_20261007_074423.txt	of record	Part 1; Part 2; the book
run_LOOP_ATLAS_POLARISATION_VORTEX_20261004_074503.txt	of record	the book
run_LOOP_ATLAS_PRICE_MAP_20261002.txt	of record, dated only	—
run_LOOP_ATLAS_PULSED_SCISSORS_20261005_155747.txt	of record	Part 1; Part 3
run_LOOP_ATLAS_PULSED_SCISSORS_LONG_20261005_170134.txt	of record	Part 1; Part 2
run_LOOP_ATLAS_PULSED_SCISSORS_LONG_READER_20261005_170755.txt	of record	—
run_LOOP_ATLAS_PUSH_PRICE_20261004_144831.txt	of record	the book
run_LOOP_ATLAS_PUSH_TRADE_READER_20261007_024449.txt	of record	the book
run_LOOP_ATLAS_QUADRATURE_ACROSS_20261107_080953.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_QUADRATURE_ACROSS_BLIND_20261107_080511.txt	blind	—
run_LOOP_ATLAS_QUADRATURE_ACROSS_BLIND_20261107_080800.txt	blind	—
run_LOOP_ATLAS_QUANT_STEP_20261013_082454.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_QUANT_ZERO_20261013_081905.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_QUARTER_PHASES_20261014_073718.txt	of record	—
run_LOOP_ATLAS_QUARTER_STROBE_20261013_231124.txt	of record	Part 2; Part 3
run_LOOP_ATLAS_QUARTER_STROBE_2_20261013_231511.txt	of record	Part 2
run_LOOP_ATLAS_QUARTER_TREE_CURVES_20261014_055131.txt	of record, a reading tool	Part 3; the book
run_LOOP_ATLAS_QUARTER_TREE_MARKS_20261014_033510.txt	of record, a reading tool	Part 3
run_LOOP_ATLAS_QUARTER_TREE_READ_20261014_053757.txt	of record, a reading tool	the book
run_LOOP_ATLAS_QUARTER_TREE_VOID_20261014_060108.txt	of record, a reading tool	Part 3
run_LOOP_ATLAS_QUICK_READ_20261014_065420.txt	of record	Part 1; Part 2; the book
run_LOOP_ATLAS_QUIET_TAP_20261013_195754.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_RAMP_EDGE_20261013_195233.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_RAMP_EDGE_2_20261013_200133.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_RAMP_SPEED_20261031_232517.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_RAMP_SPEED_BLIND_20261031_232206.txt	blind	—
run_LOOP_ATLAS_REINJECT_20261013_015743.txt	of record	Part 1; Part 2; the book

Run file	Kind	Cited in
run_LOOP_ATLAS_REINJECT_CLOCK_20261013_024653.txt	of record	Part 1; Part 2; the book
run_LOOP_ATLAS_RESTED_LAG_20261016_205504.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_RESTED_LAG_BLIND_20261016_205030.txt	blind	—
run_LOOP_ATLAS_RF_CONVEYOR_20261002.txt	of record, dated only	—
run_LOOP_ATLAS_RF_ROTOR_20261002.txt	of record, dated only	—
run_LOOP_ATLAS_RF_SKATER_COUNT_20261004_061647.txt	of record	Part 3
run_LOOP_ATLAS_RF_SLOPE_20261002.txt	of record, dated only	—
run_LOOP_ATLAS_RHYTHM_BAND_20261020_154009.txt	of record	Part 1; Part 2; the book
run_LOOP_ATLAS_RHYTHM_BAND_BLIND_20261020_153930.txt	blind	Part 3
run_LOOP_ATLAS_RHYTHM_BAND_BLIND_20261020_154007.txt	blind	—
run_LOOP_ATLAS_RIDER_FLEET_20261016_044707.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_RIDER_FLEET_BLIND_20261016_044404.txt	blind	—
run_LOOP_ATLAS_RING_AREA_20261105_032546.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_RING_AREA_BLIND_20261105_032332.txt	blind	Part 3
run_LOOP_ATLAS_RING_AREA_BLIND_20261105_032544.txt	blind	—
run_LOOP_ATLAS_RING_BUDGET_20261004_080107.txt	of record	Part 3; the book
run_LOOP_ATLAS_RING_JAM_20261014_224934.txt	of record	Part 1; Part 2
run_LOOP_ATLAS_RING_JAM_BLIND_20261014_224735.txt	blind	—
run_LOOP_ATLAS_RING_KNOBS_20261015_165858.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_RING_KNOBS_BLIND_20261015_165032.txt	blind	—
run_LOOP_ATLAS_RING_KNOBS_BLIND_20261015_165357.txt	blind	—
run_LOOP_ATLAS_RING_LEAPFROG_20261002.txt	of record, dated only	—
run_LOOP_ATLAS_RING_PATHS_20261102_013613.txt	of record	Part 1; Part 2; the book
run_LOOP_ATLAS_RING_PATHS_BLIND_20261102_013326.txt	blind	—
run_LOOP_ATLAS_RING_SPLIT_20261105_050121.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_RING_SPLIT_BLIND_20261105_050120.txt	blind	Part 3
run_LOOP_ATLAS_RING_WIDTH_20261016_213858.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_RING_WIDTH_BLIND_20261016_213131.txt	blind	—
run_LOOP_ATLAS_ROOM_NEEDLE_20261006_013956.txt	of record	Part 1; Part 3
run_LOOP_ATLAS_ROOM_NEEDLE_20261006_014807.txt	of record	Part 1
run_LOOP_ATLAS_ROOM_NEEDLE_2_20261006_014758.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_ROOM_NEEDLE_2_READER_20261006_015510.txt	of record	—
run_LOOP_ATLAS_ROOM_NEEDLE_RULERS_20261005_170136.txt	of record	—
run_LOOP_ATLAS_ROOT_JUMPS_20261015_031511.txt	of record	Part 1; Part 2; the book

Run file	Kind	Cited in
run_LOOP_ATLAS_ROOT_JUMPS_BLIND_20261015_031446.txt	blind	—
run_LOOP_ATLAS_ROUGH_BAND_BLIND_20261102_164924.txt	blind	Part 3
run_LOOP_ATLAS_ROUGH_COUNT_20261102_173941.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_ROUGH_COUNT_BLIND_20261102_173939.txt	blind	—
run_LOOP_ATLAS_ROUGH_FLOOR_20261017_013514.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_ROUGH_FLOOR_BLIND_20261017_011647.txt	blind	Part 3
run_LOOP_ATLAS_ROUGH_LONG_20261102_200436.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_ROUGH_LONG_BLIND_20261102_200434.txt	blind	—
run_LOOP_ATLAS_RUNG_ABOVE_20261007_071341.txt	of record	Part 1
run_LOOP_ATLAS_RUNG_ABOVE_2_20261007_073220.txt	of record	Part 1; Part 2
run_LOOP_ATLAS_RUN_MAP_20261006_071157.txt	of record	Part 1; Part 2
run_LOOP_ATLAS_RUN_STROBE_20261006_170149.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_RUN_STROBE_2_20261006_174220.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_RUN_STROBE_2_READER_20261006_174743.txt	of record, a reading tool	—
run_LOOP_ATLAS_RUN_STROBE_READER_20261006_170950.txt	of record, a reading tool	—
run_LOOP_ATLAS_SADDLE_ASYMPTOTE_20261013_025252.txt	of record	Part 1; Part 2; the book
run_LOOP_ATLAS_SADDLE_LOG_20261013_012417.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_SADDLE_NOISE_20261013_024850.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_SAGNAC_LOOP_20261018_032138.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_SAGNAC_LOOP_BLIND_20261018_032130.txt	blind	—
run_LOOP_ATLAS_SAIL_20261018_202451.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_SAIL_BLIND_20261018_202354.txt	blind	—
run_LOOP_ATLAS_SAIL_HULL_20261018_202650.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_SAIL_HULL_BLIND_20261018_202625.txt	blind	—
run_LOOP_ATLAS_SCALLOP_DIAL_20261001.txt	of record, dated only	—
run_LOOP_ATLAS_SEAM_20261014_090006.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_SEAM_2_20261014_090552.txt	of record	Part 1; Part 2
run_LOOP_ATLAS_SEAM_MAP_20261005_021349.txt	of record	Part 3; the book
run_LOOP_ATLAS_SEAM_MAP_B_20261005_023746.txt	of record	Part 3; the book
run_LOOP_ATLAS_SECOND_GEAR_20261013_192828.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_SETTLED_COPIES_20261011_212014.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_SETTLED_COPIES_2_20261011_212705.txt	of record	Part 1; Part 2

Run file	Kind	Cited in
run_LOOP_ATLAS_SETTLED_GROOVES_20261011_223349.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_SETTLED_GROOVES_2_20261012_041558.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_SHOVE_SKATER_20261011_070846.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_SHUFFLE_NULL_20261010_190812.txt	of record, a reading tool	Part 1; Part 2; Part 3
run_LOOP_ATLAS_SHUFFLE_NULL_2_20261010_190944.txt	of record, a reading tool	Part 1; Part 2; Part 3
run_LOOP_ATLAS_SHUVIT_WALK_20261004_183058.txt	of record	the book
run_LOOP_ATLAS_SIGN_CHECK_C11_20261004_053054.txt	of record	—
run_LOOP_ATLAS_SIZED_TAP_20261006_060952.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_SIZED_TAP_2_20261006_061606.txt	of record	Part 1; Part 2; the book
run_LOOP_ATLAS_SIZED_TAP_3_20261006_061940.txt	of record	Part 1; Part 2; the book
run_LOOP_ATLAS_SIZED_TAP_4_20261013_161615.txt	of record	Part 1; Part 2; the book
run_LOOP_ATLAS_SKATER_20261002.txt	of record, dated only	—
run_LOOP_ATLAS_SLING_RUNDOWN_20261017_232234.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_SLING_RUNDOWN_BLIND_20261017_232230.txt	blind	—
run_LOOP_ATLAS_SLIP_BAND_20261020_161039.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_SLIP_BAND_2_20261020_161539.txt	of record	Part 1; Part 2; the book
run_LOOP_ATLAS_SLIP_BAND_2_BLIND_20261020_161406.txt	blind	—
run_LOOP_ATLAS_SLIP_BAND_BLIND_20261020_161035.txt	blind	—
run_LOOP_ATLAS_SLIP_BEND_20261006_071248.txt	of record	Part 1; Part 3; the book
run_LOOP_ATLAS_SLIP_PROOF_20261013_194935.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_SLIP_PROOF_2_20261013_200028.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_SLIP_PROOF_3_20261013_225419.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_SLIP_RECHECK_20261020_083908.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_SLIP_RECHECK_BLIND_20261020_083533.txt	blind	—
run_LOOP_ATLAS_SLIP_RHYTHM_20261020_083426.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_SLIP_RHYTHM_2_20261020_084919.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_SLIP_RHYTHM_2_BLIND_20261020_084218.txt	blind	—
run_LOOP_ATLAS_SLIP_RHYTHM_3_20261020_090717.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_SLIP_RHYTHM_3_BLIND_20261020_090456.txt	blind	—

Run file	Kind	Cited in
run_LOOP_ATLAS_SLIP_RHYTHM_BLIND_20261020_082954.txt	blind	Part 3
run_LOOP_ATLAS_SLIP_SENSE_20261011_060035.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_SLIP_SPREAD_20261020_075155.txt	of record	Part 1; Part 2
run_LOOP_ATLAS_SLIP_SPREAD_20261020_075635.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_SLIP_SPREAD_BLIND_20261020_074927.txt	blind	—
run_LOOP_ATLAS_SLIP_START_20261020_090154.txt	of record	Part 1; Part 2; the book
run_LOOP_ATLAS_SLIP_START_BLIND_20261020_085204.txt	blind	—
run_LOOP_ATLAS_SLOPE_EAR_20261015_190935.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_SLOPE_EAR_BLIND_20261015_190658.txt	blind	—
run_LOOP_ATLAS_SMOOTH_DIRT_20261014_084856.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_SMOOTH_PUSH_EDGE_20261004_170203.txt	of record	Part 3
run_LOOP_ATLAS_SNAKE_20261002b.txt	of record, dated only	Part 3
run_LOOP_ATLAS_SNAKE_768_20261015_151934.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_SNAKE_768_BLIND_20261015_134617.txt	blind	—
run_LOOP_ATLAS_SNAKE_BEAT_20261101_190003.txt	of record	Part 1; Part 2; the book
run_LOOP_ATLAS_SNAKE_BEAT_BLIND_20261101_185946.txt	blind	—
run_LOOP_ATLAS_SNAKE_EXACT_20261015_154247.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_SNAKE_EXACT_BLIND_20261015_153253.txt	blind	—
run_LOOP_ATLAS_SNAKE_EXACT_BLIND_20261015_153819.txt	blind	—
run_LOOP_ATLAS_SNAKE_LONGEST_20261017_155052.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_SNAKE_LONGEST_BLIND_20261017_210831.txt	blind	—
run_LOOP_ATLAS_SNAKE_LONGEST_BLIND_20261017_224026.txt	blind	—
run_LOOP_ATLAS_SNAKE_LONG_20261015_073142.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_SNAKE_LONG_BLIND_20261015_071934.txt	blind	—
run_LOOP_ATLAS_SNAKE_MISMATCH_20261002.txt	of record, dated only	—
run_LOOP_ATLAS_SNAKE_RACE_20261101_175747.txt	of record	Part 1; Part 2; the book
run_LOOP_ATLAS_SNAKE_RACE_BLIND_20261101_175149.txt	blind	—
run_LOOP_ATLAS_SOFT_SWITCH_20261002.txt	of record, dated only	—
run_LOOP_ATLAS_SONG_GAPS_20261015_074750.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_SONG_GAPS_2_20261015_074852.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_SONG_GAPS_2_BLIND_20261015_074841.txt	blind	—
run_LOOP_ATLAS_SONG_GAPS_BLIND_20261015_074716.txt	blind	—
run_LOOP_ATLAS_SONG_GAPS_BLIND_20261015_074739.txt	blind	—

Run file	Kind	Cited in
run_LOOP_ATLAS_SOUP_JUMP_20261009_045351_partSMOKE.txt	smoke part	—
run_LOOP_ATLAS_SOUP_JUMP_20261009_045400_partA.txt	of record, part A	Part 1; Part 2; Part 3
run_LOOP_ATLAS_SOUP_JUMP_20261009_045400_partB.txt	of record, part B	Part 1; Part 2; Part 3
run_LOOP_ATLAS_SOUP_JUMP_2_20261009_051408_partSMOKE.txt	smoke part	—
run_LOOP_ATLAS_SOUP_JUMP_2_20261009_051426_partA.txt	of record, part A	Part 1; Part 3
run_LOOP_ATLAS_SOUP_JUMP_2_20261009_051426_partC.txt	of record, part C	Part 1; Part 3
run_LOOP_ATLAS_SOUP_JUMP_2_20261009_051926_partB.txt	of record, part B	—
run_LOOP_ATLAS_SOUP_JUMP_2_20261009_052404_partV.txt	of record, part V	Part 1
run_LOOP_ATLAS_SOUP_JUMP_3_20261016_225440.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_SOUP_JUMP_3_BLIND_20261016_224305.txt	blind	—
run_LOOP_ATLAS_SOUP_RHYTHM_20261002.txt	of record, dated only	—
run_LOOP_ATLAS_SPEED_CHAIN_20261012_171949.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_SPIN_OR_SIT_20261011_055807.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_SPIN_START_20261013_161916.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_SPIN_WINDOWS_20261013_163655.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_SPLIT_LOOP_SLOPE_20261005_171124.txt	of record	—
run_LOOP_ATLAS_SPOKES_20261006_204038.txt	of record	Part 1; Part 3; the book
run_LOOP_ATLAS_SPRING_BEAD_20261002.txt	of record, dated only	—
run_LOOP_ATLAS_SPRING_BEAD_loss_20261002.txt	of record, dated only	—
run_LOOP_ATLAS_SPRING_FLOOR_20261017_160023.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_SPRING_FLOOR_BLIND_20261017_155714.txt	blind	—
run_LOOP_ATLAS_SPRING_STALL_20261014_063112.txt	of record	Part 1; Part 2
run_LOOP_ATLAS_STAIR_WIDTHS_20261013_184033.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_STEP_BACK_RAMP_20261013_195010.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_STOP_LEAVES_20261015_052943.txt	of record	Part 1; Part 2
run_LOOP_ATLAS_STOP_LEAVES_BLIND_20261015_051930.txt	blind	—
run_LOOP_ATLAS_STOP_STATES_20261015_060745.txt	of record, a reading tool	Part 3
run_LOOP_ATLAS_STOP_WALL_20261014_180544.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_STOP_WALL_2_20261014_225838.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_STOP_WALL_2_BLIND_20261014_225200.txt	blind	—
run_LOOP_ATLAS_STOP_WALL_3_20261015_000651.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_STOP_WALL_BLIND_20261014_175030.txt	blind	—

Run file	Kind	Cited in
run_LOOP_ATLAS_STRETCH_DEPTH_20261013_055240.txt	of record	Part 1; Part 2
run_LOOP_ATLAS_STRETCH_READS_20261013_060151.txt	of record	Part 3; the book
run_LOOP_ATLAS_STRETCH_WHERE_20261013_045856.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_STRIDE_MAP_20261016_213858.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_STRIDE_MAP_BLIND_20261016_213235.txt	blind	—
run_LOOP_ATLAS_STROBE_NEWTON_20261006_184350.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_SWAP_RIDER_BAND_20261006_041001.txt	of record	Part 1; Part 2
run_LOOP_ATLAS_SWAP_STATE_20261006_052545.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_SWAP_STATE_2_20261006_052914.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_SWIM_RIDGE_20261105_145417.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_SWIM_RIDGE_BLIND_20261105_145416.txt	blind	—
run_LOOP_ATLAS_SWING_GRID_20261016_203014.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_SWING_GRID_BLIND_20261016_202628.txt	blind	—
run_LOOP_ATLAS_SWING_LISTEN_20261007_013201.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_SWING_LISTEN_2_20261007_024253.txt	of record	Part 1; Part 2
run_LOOP_ATLAS_SWING_LISTEN_TURNS_20261007_014205.txt	of record	—
run_LOOP_ATLAS_SWING_PUMP_20261002.txt	of record, dated only	the book
run_LOOP_ATLAS_SWING_TWENTY_20261004_075715.txt	of record	Part 3; the book
run_LOOP_ATLAS_SWING_TWENTY_B_20261004_075902.txt	of record	the book
run_LOOP_ATLAS_SWITCH_END_20261015_074716.txt	of record	Part 1; Part 2; the book
run_LOOP_ATLAS_SWITCH_END_BLIND_20261015_074030.txt	blind	—
run_LOOP_ATLAS_S_REVERSAL_20261003_213109.txt	of record	the book
run_LOOP_ATLAS_S_REVERSAL_FOLLOWON_20261003_213210.txt	of record	Part 3
run_LOOP_ATLAS_S_REVERSAL_SPIKE_20261003_213523.txt	of record	Part 3; the book
run_LOOP_ATLAS_THIRD_AXIS_20261109_170732.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_THIRD_AXIS_2_20261109_181451.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_THIRD_AXIS_2_BLIND_20261109_181123.txt	blind	—
run_LOOP_ATLAS_THIRD_AXIS_3_20261109_191727.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_THIRD_AXIS_3_BLIND_20261109_191132.txt	blind	—
run_LOOP_ATLAS_THIRD_AXIS_BLIND_20261109_170229.txt	blind	—
run_LOOP_ATLAS_THROW_CATCH_20261002.txt	of record, dated only	—
run_LOOP_ATLAS_TONG_DULLIN_20261017_173718.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_TONG_DULLIN_BLIND_20261017_173508.txt	blind	Part 3

Run file	Kind	Cited in
run_LOOP_ATLAS_TONG_DULLIN_BLIND_20261017_173617.txt	blind	Part 3
run_LOOP_ATLAS_TONG_DULLIN_KNEE_20261017_182629.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_TONG_DULLIN_KNEE_BLIND_20261017_182611.txt	blind	—
run_LOOP_ATLAS_TONG_DULLIN_READINGS_20261017_174547.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_TONG_DULLIN_READINGS_BLIND_20261017_174348.txt	blind	—
run_LOOP_ATLAS_TONG_DULLIN_SCAN_20261017_181953.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_TONG_DULLIN_SCAN_BLIND_20261017_181752.txt	blind	—
run_LOOP_ATLAS_TOP_SECTION_20261013_005355.txt	of record	Part 1; Part 2; the book
run_LOOP_ATLAS_TOY_SHELF_CHECK_A2_20261004_053201.txt	of record	—
run_LOOP_ATLAS_TOY_SHELF_CHECK_A_20261004_052703.txt	of record	—
run_LOOP_ATLAS_TRAIN_BEAT_20261028_212034.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_TRAIN_BEAT_BLIND_20261028_211935.txt	blind	—
run_LOOP_ATLAS_TRAIN_CLOSE_20261018_103739.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_TRAIN_CLOSE_BLIND_20261018_072408.txt	blind	—
run_LOOP_ATLAS_TRAIN_DOOR_20261022_055043.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_TRAIN_DOOR_BLIND_20261022_055030.txt	blind	—
run_LOOP_ATLAS_TRAIN_DOOR_LINE_20261022_222935.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_TRAIN_DOOR_LINE_BLIND_20261023_045926.txt	blind	—
run_LOOP_ATLAS_TRAIN_DRIFT_20261019_034330.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_TRAIN_DRIFT_BLIND_20261019_034258.txt	blind	—
run_LOOP_ATLAS_TRAIN_EDGE_20261018_181641.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_TRAIN_EDGE_BLIND_20261018_201933.txt	blind	—
run_LOOP_ATLAS_TRAIN_EDGE_DOOR_20261022_070632.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_TRAIN_EDGE_DOOR_BLIND_20261022_135321.txt	blind	—
run_LOOP_ATLAS_TRAIN_EDGE_MAP_20261022_060414.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_TRAIN_EDGE_MAP_BLIND_20261022_060403.txt	blind	—
run_LOOP_ATLAS_TRAIN_FLOOR_20261020_224239.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_TRAIN_FLOOR_2_20261021_083613.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_TRAIN_FLOOR_2_BLIND_20261021_151237.txt	blind	—
run_LOOP_ATLAS_TRAIN_FLOOR_BLIND_20261019_020246.txt	blind	—
run_LOOP_ATLAS_TRAIN_FUSE_CLOCK_20261028_101647.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_TRAIN_FUSE_CLOCK_BLIND_20261023_155930.txt	blind	—

Run file	Kind	Cited in
run_LOOP_ATLAS_TRAIN_HOLD_20261015_171853.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_TRAIN_HOLD_BLIND_20261015_170546.txt	blind	—
run_LOOP_ATLAS_TRAIN_LIFETIME_20261018_034211.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_TRAIN_LIFETIME_BLIND_20261018_033904.txt	blind	—
run_LOOP_ATLAS_TRAIN_LIFETIME_BLIND_20261018_034056.txt	blind	—
run_LOOP_ATLAS_TRAIN_LIFETIME_READER_20261018_035727.txt	of record, a reading tool	Part 3
run_LOOP_ATLAS_TRAIN_LONG_EDGE_20261022_175421.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_TRAIN_LONG_EDGE_BLIND_20261023_000710.txt	blind	—
run_LOOP_ATLAS_TRAIN_NUDGE_20261018_192138.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_TRAIN_NUDGE_BLIND_20261018_192107.txt	blind	—
run_LOOP_ATLAS_TRAIN_PARTING_20261029_030532.txt	of record	Part 3
run_LOOP_ATLAS_TRAIN_PARTING_20261029_050736.txt	of record	—
run_LOOP_ATLAS_TRAIN_PARTING_20261029_070756.txt	of record	—
run_LOOP_ATLAS_TRAIN_PARTING_20261029_090839.txt	of record	—
run_LOOP_ATLAS_TRAIN_PARTING_20261029_112955.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_TRAIN_PARTING_BLIND_20261029_025845.txt	blind	Part 3
run_LOOP_ATLAS_TRAIN_PARTING_BLIND_20261029_030409.txt	blind	—
run_LOOP_ATLAS_TRAIN_PARTING_EDGE_20261029_070800.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_TRAIN_PARTING_EDGE_2_20261029_211416.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_TRAIN_PARTING_EDGE_2_BLIND_20261030_025608.txt	blind	—
run_LOOP_ATLAS_TRAIN_PARTING_EDGE_BLIND_20261029_135444.txt	blind	—
run_LOOP_ATLAS_TRAIN_TIMING_20261018_082139.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_TRAIN_TIMING_BLIND_20261018_082010.txt	blind	—
run_LOOP_ATLAS_TRIPLE_CARD_CHECK_20261005_055204.txt	of record, a reading tool	Part 3; the book
run_LOOP_ATLAS_TRUCK_ROAD_20261016_090656.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_TRUCK_ROAD_BLIND_20261016_090029.txt	blind	—
run_LOOP_ATLAS_TURNING_TILT_20261011_055620.txt	of record	Part 1
run_LOOP_ATLAS_TWIST_LONG_20261011_072842.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_TWO_HANDS_20261017_203811.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_TWO_HANDS_2_20261017_204131.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_TWO_HANDS_2_BLIND_20261017_204117.txt	blind	—
run_LOOP_ATLAS_TWO_HANDS_BAND_20261017_210543.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_TWO_HANDS_BAND_BLIND_20261017_210533.txt	blind	—

Run file	Kind	Cited in
run_LOOP_ATLAS_TWO_HANDS_BLIND_20261017_203758.txt	blind	—
run_LOOP_ATLAS_TYRE_KNEE_20261014_174309.txt	of record	Part 1; Part 2; the book
run_LOOP_ATLAS_ULP_NUDGE_20261011_044600.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_V13_FIDELITY_20261013_202203.txt	of record	Part 2; Part 3
run_LOOP_ATLAS_V13_FIDELITY_20261013_203154.txt	of record	Part 2; Part 3
run_LOOP_ATLAS_V13_FIDELITY_20261013_204846.txt	of record	Part 2; Part 3
run_LOOP_ATLAS_WADDLE_BOOKS_20261013_063353.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_WADDLE_CATCH_20261013_082013.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_WADDLE_FOLLOW_20261011_080808.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_WADDLE_PRICE_20261011_071302.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_WADDLE_STEP_20261013_073935.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_WAVE_SNAKE_20261004_073636.txt	of record	Part 3
run_LOOP_ATLAS_WAVE_SNAKE_B_20261004_073735.txt	of record	—
run_LOOP_ATLAS_WAVE_SNAKE_DRIFT_20261004_074634.txt	of record	Part 3; the book
run_LOOP_ATLAS_WAVE_SNAKE_DRIFT_B_20261004_074716.txt	of record	—
run_LOOP_ATLAS_WAVE_SNAKE_DRIFT_C_20261004_080331.txt	of record	—
run_LOOP_ATLAS_WAVE_SNAKE_DRIFT_D_20261004_080506.txt	of record	—
run_LOOP_ATLAS_WAVE_SOUP_20261002.txt	of record, dated only	—
run_LOOP_ATLAS_WAVE_SOUP_DELAY_20261002.txt	of record, dated only	—
run_LOOP_ATLAS_WAVE_SOUP_NOISE_20261002.txt	of record, dated only	—
run_LOOP_ATLAS_WINDOW_CASCADE_20261012_081249.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_WINDOW_CASCADE_2_20261012_171509.txt	of record	Part 1; Part 2
run_LOOP_ATLAS_WINDOW_CASCADE_3_20261013_015743.txt	of record	Part 1; Part 2
run_LOOP_ATLAS_WINDOW_EDGE_20261012_080030.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_WINDOW_EDGE_2_20261012_171509.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_WINDOW_TOUCH_20261014_073237.txt	of record	Part 3
run_LOOP_ATLAS_WINDOW_TOUCH_2_20261014_074538.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_WINDOW_TOUCH_3_20261014_080520.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_WINDOW_TOUCH_4_20261014_090037.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_WINDOW_TOUCH_5_20261014_092920.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_WING_20261002.txt	of record, dated only	the book
run_LOOP_ATLAS_WOBBLE_SOFT_20261013_192713.txt	of record	Part 1; Part 2; Part 3; the book

Run file	Kind	Cited in
run_LOOP_ATLAS_WOBBLY_SHUVIT_20261015_074302.txt	of record	Part 1; Part 2; the book
run_LOOP_ATLAS_WOBBLY_SHUVIT_BLIND_20261015_074121.txt	blind	—
run_LOOP_ATLAS_WOBBLY_SHUVIT_BLIND_20261015_074220.txt	blind	—
run_LOOP_ATLAS_WORD_BACK_20261014_065420.txt	of record	Part 1; Part 2; the book
run_LOOP_ATLAS_WORM_CHAIN_20261001.txt	of record, dated only	—
run_LOOP_ATLAS_WORM_CHAIN_DEAD_20261001.txt	of record, dated only	—
run_LOOP_ATLAS_WORM_CHAIN_spancheck_20261001.txt	of record, dated only	—
run_LOOP_ATLAS_WORM_RING_20261002.txt	of record, dated only	—
run_LOOP_ATLAS_WORN_RIDER_20261016_065645.txt	of record	Part 1; Part 2; the book
run_LOOP_ATLAS_WORN_RIDER_BLIND_20261016_065356.txt	blind	—
run_LOOP_ATLAS_ZERO_CROSS_20261010_043523.txt	of record, a reading tool	Part 1; Part 2
run_LOOP_ATLAS_ZETA_FIT_20261020_055322.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_ZETA_FIT_BLIND_20261020_055137.txt	blind	—
run_LOOP_ATLAS_ZETA_LIMIT_20261020_055829.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_ZETA_LIMIT_20261020_060047.txt	of record	Part 3
run_LOOP_ATLAS_ZETA_LIMIT_20261020_060256.txt	of record	Part 1; Part 2; Part 3; the book
run_LOOP_ATLAS_ZETA_LIMIT_BLIND_20261020_055645.txt	blind	—
run_LOOP_ATLAS_ZIGZAG_GAUGE_20261013_231629.txt	of record	Part 1; Part 2; Part 3
run_LOOP_ATLAS_ZIGZAG_GAUGE_2_20261013_231751.txt	of record	Part 1; Part 2; Part 3

12 runs this record cites are not in the archive. Each is a first version or a first blind run that the record itself calls retired or set aside, or a run of the three-body work, which is another project's; they were kept in the working scratch folders and never travelled with the book.

Run	Cited in	The record calls it
DAMPER_LAWS 20261011_051800	Part 3, miss 126	(set aside)
DELTA_AREA 20261008_062447	Part 3, miss 88	(retired)
MEMORY_SWITCH_2 20261011_084809	Part 3, miss 145	(set aside)
QUARTER_TREE_READ 20261014_022707	Part 3, miss 197	(retired)
SETTLED_GROOVES 20261011_222608	Part 3, miss 151	(set aside)
TRIPLE_READER_CHECKS 20261005_063526	Part 3, miss 89	(the three-body work's)
TWIST_LONG 20261011_072737	Part 3, miss 139	(retired)
WADDLE_PRICE 20261011_071049	Part 3, miss 135	(retired)
WINDOW_CASCADE 20261012_080715	Part 3, miss 156	(set aside)
run_LOOP_ATLAS_EXPLORER_ENGINE_CHECK_BLIND_20261017_063057	Part 3, miss 289	(retired)
run_LOOP_ATLAS_HINGE_GEOMETRY_20261002	Part 3, miss 26	(retired)
run_LOOP_ATLAS_TRUCK_ROAD_BLIND_20261016_085417	Part 3, miss 260	(retired)

Every file a tool finds by its name and its fingerprint, the parent tools it builds on and their saved data, travels in the archive at the fingerprint the tool asks for. Ten tools of the archive also read a file that does not travel with it: two read the jellyfish paper's own files, in the upload set of its record (Schulz, 2026, Jellyfish: the Holonomist of the Sea*; concept DOI 10.5281/zenodo.22908909); six read a working file of this project, kept with the author and not deposited; two read another project's files, kept with that project. Each is named here with what it reads, so that a reader who wants to run one again knows what it needs and where that is.*

Tool	What it reads	Where that is
DRAFT_TEST	the jellyfish paper's solver, jelly_viscous_grid_rung6_animal.py (version 2.5, sha256 beginning 0dfc9b4b5f9e9707), with jelly_viscous_grid_rung1.py and jelly_viscous_grid_rung3_bell.py, which it runs	the jellyfish paper's own files, in the upload set of its record (Schulz, 2026, *Jellyfish: the Holonomist of the Sea*; concept DOI 10.5281/zenodo.22908909)
KICK_RULE_CHECK	five run files of the jellyfish solver, 14 to 18 September 2026	the jellyfish paper's own files, in the upload set of its record (Schulz, 2026, *Jellyfish: the Holonomist of the Sea*; concept DOI 10.5281/zenodo.22908909)
GROOVES_10	the data ledger of registrations, for the first version's values in its own row	a working file of this project, kept with the author and not deposited
GROOVES_15	the data ledger of registrations, for the same values, in its part C	a working file of this project, kept with the author and not deposited
BUILD_PAPER_CHECK	the data ledger, as a test document, if it is beside the tool; without it the check builds its made-up tables only	a working file of this project, kept with the author and not deposited
EXPLORER_ROOM3_BUILD	one no-physics read, read 219, from the project's collection of them, for the page it builds	a working file of this project, kept with the author and not deposited
EXPLORER_ENGINE_CHECK	a working note on the scope of a sports explorer, 17 October 2026	a working file of this project, kept with the author and not deposited
BENCHMARK_TAGS	the headline register's source and the two readers' tag files it checks	a working file of this project, kept with the author and not deposited
RUN_STROBE	a run of A Rider in Two Waves' own instrument, sent with a letter of 6 October 2026	another project's files, kept with that project
TRIPLE_CARD_CHECK	the three-body work's tool, tables and ledgers, sent on 5 October 2026	another project's files, kept with that project

Declaration

This volume is part of *The Loop Atlas* and shares its Declaration, on what the work is, how it was made, how far the literature was checked and its limits. What follows is this volume's own disclosure.

Tool and computational resource disclosure

Tools used. Claude (Anthropic), an AI model, took part as a research partner and co-director: it wrote and ran every tool, kept the records, and wrote the scripts that print this volume. The long runs were flown on the author's own computer, the second computer of the tables. Every row here was printed by the build scripts of *The Loop Atlas* from the runs that travel with it and from the project's file of misses, written in Python; typesetting used pandoc and wkhtmltopdf.

Responsibility. Responsibility for the correctness of what is reported here rests with the named human author, as for the book, whose Declaration says so in full. No automated system is an author of this work.

Computational resources. Small models on ordinary computers: a cloud workspace for most runs, and the author's own computer for the long ones, at most eight cores at a time.